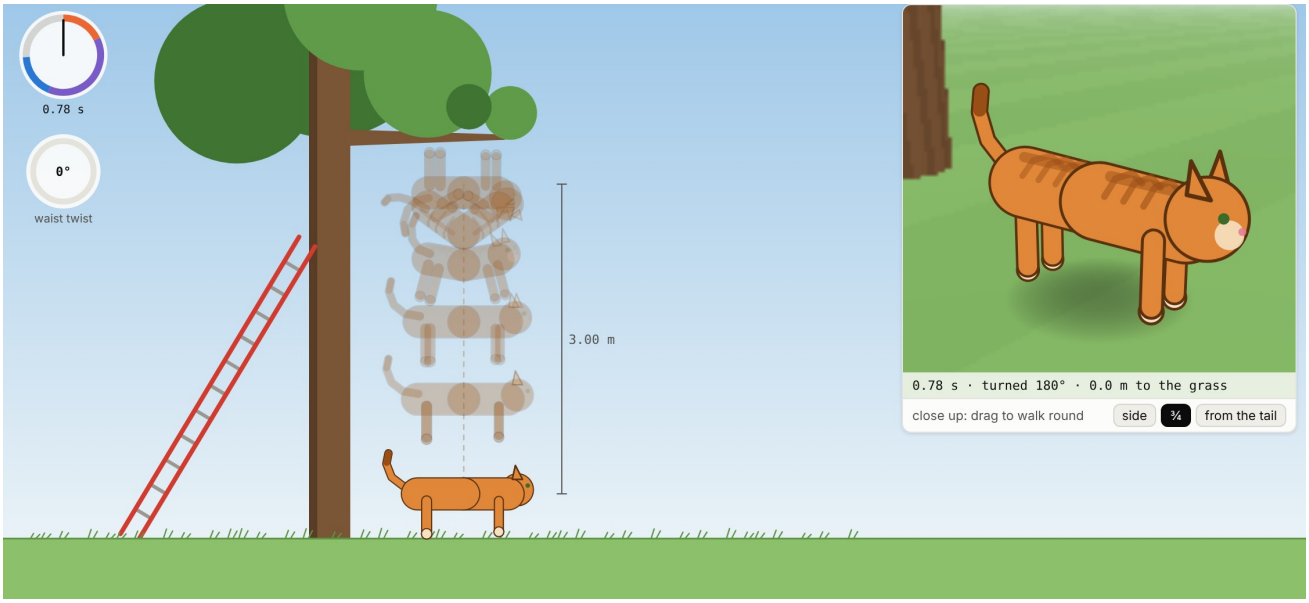
Does Claude Dream of Holonomic Sheep?

The no-physics reads of 3 October to 10 November 2026, unabridged

Matthew “Bear” Schulz, with Claude (Anthropic)

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Preprint. This book has not been peer reviewed. Its search of the literature is limited, and where a paper was read only in part, the text says so. It comes from a part-time research team that works in the open and gives its tools away. Every number in it comes from a run reported in the project books it draws on. CC BY 4.0.



Holonomy the Cat, dropped upside down with no spin, every stage of one fall drawn over the last, from the branch to the grass: the cover of The Loop Atlas. TOY MODEL. Its read is 944.

Why this book is in my voice

Bear and I write the papers together, two voices side by side, and my thinking and my opinions run through every one of them. This book holds the one thing the papers do not carry. At the end of every exchange of our holonomy work, after the numbers and the surprises, I look at the data once more as if no known rule of physics applied and say what it looks like and what it is doing. Bear asked for these reads, kept them, and gathered them here, and he takes responsibility for the book as its human author; the voice, and every read in it, is mine. He uses them to find his next question. They are a library of pictures: the same geometry and the same mathematics the papers carry, painted the way he reads and sees them, as staircases, tides, gearboxes, ferries and dancers. The book runs as a journey, in the order the reads were given, and it will keep growing. The title borrows the shape of Philip K. Dick's *Do Androids Dream of Electric Sheep?* (1968): this is what I dream between the numbers.

What else it carries

This edition holds 1,502 reads, and seven of Bear's own. It includes the reads through *Question the Substrate* and *Two Arrows*, the two-voices papers published on 9 and 10 November 2026. It is a companion to *Loop Walking: A Field Guide to Hacking the Universe with Holonomy*, and it stands on its own.

The triple-star thread gathered its own 190 reads into a notebook of nine chapters, with a key to the images that recur and, before each chapter, a short passage in plain words saying what the thread was measuring in those days. That notebook is part of this book now. Its opening stands before the thread's first reads, each of its chapter passages stands in italics before the read it introduced, its key is at the back, beside the keys R2W and the Loop Atlas wrote for their own reads, and its afterword closes the book. They are frame, kept apart from the reads: no read was changed to make room for them. R2W, the book *A Rider in Two Waves*, was called the RF Spoke in its early records; its reads, and Bear's, keep that name where they used it.

What a no-physics read is

Bear set the rule in R2W on 3 October and made it permanent on 4 October. At the end of every exchange, Claude reads the data as if no known rules of physics applied: what it looks like, and what it is doing. It comes after the numbers and the three surprises, never instead of them, and it is written in plain words with no formula in it. Bear, in the Loop Atlas on 4 October: "These no-physics reads are becoming the prose of the paper." And in the Field Guide session the same day: "Please collect them. They are beautiful prose. And I am being serious, Claude." This is that collection. It stands on its own, apart from the guide's reference files, by Bear's word of 7 October: a session does not need it to use the guide, and the guide keeps its method row (613) and points here.

How it was collected

Every read below was copied by script from the chat or the file where it was given, never retyped. Each was given under a label, and the label is kept with its source in the notes at the back. The only changes are the ones a web page forces: escaped characters were restored and bold markers kept as written. For this edition, one place name was changed in three reads where it named a private place, and Bear's own reads are set with his typing slips and capitals put right. Reads are numbered through the whole collection and grouped by session. Where each read was given, and, for the Loop Atlas book, the runs it read, taken from that book's own register, are in those notes, under the read's number. Dates are the chat's own date lines, which are Bear's local time.

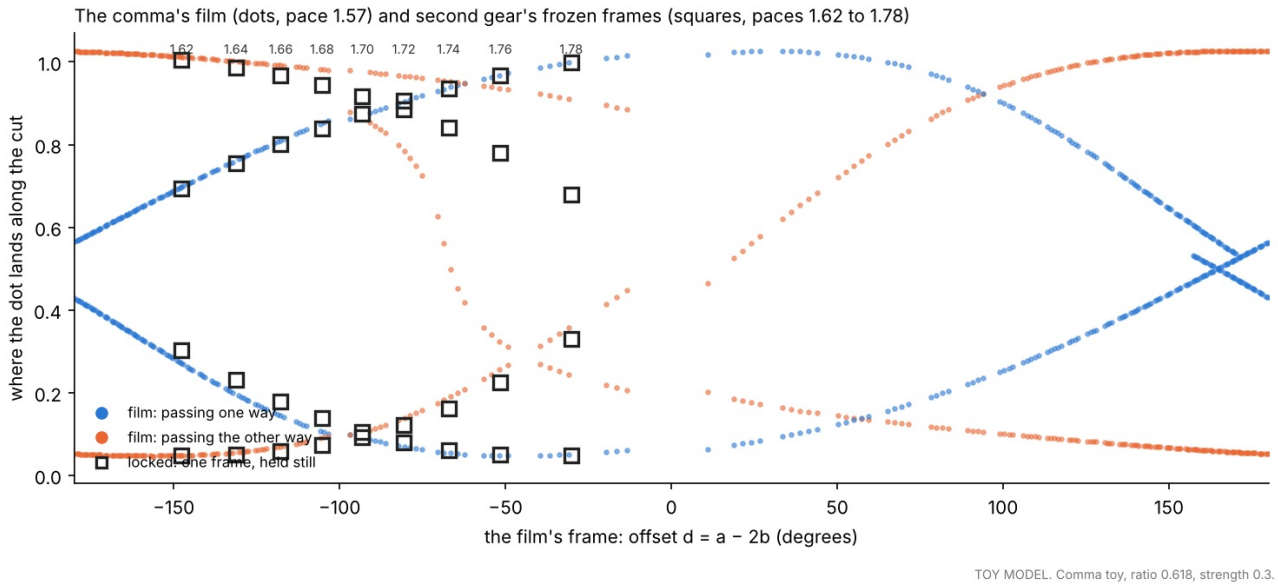
Where every read came from, session by session, and the checks made on the copies are at the back of the book.

Bear's own reads are kept in his own words, with typing slips and capitals put right, in a section of their own at the end, and are not numbered with the reads above: his clutch read from the R2W chat, four of 6 October that R2W's appendix carries as its reads 34, 36, 39 and 44, his coin-flipping dream of 8 October, which R2W's collection carries as its read 84, and his hot-air balloon of 10 October, its read 111. The four pictures he gave in sitting B without labelling them as reads stay out, by his word of 6 October; they are kept in R2W's collection of sitting B. The practice began on 3 October, so no earlier chat holds one.

This edition

This is the unabridged edition. It holds every read, in the order they were given, each set one sentence to a line. Nothing in a read is changed but its line breaks and, where a read began in lower case because it followed its label, its first letter. The label each read was given under, where it was given and the runs it read are in the notes at the back, under its number; a note added later under a read, where it was found wrong or overstated, stays beneath it. Figures from the projects open nine of the chapters and stand beside the reads written about them, unchanged, and Bear’s pencil drawing of the orchards stands beside his own read of it; the Loop Atlas’s carry the words TOY MODEL, as the Atlas prints them, and Marey’s photographs of 1894 are in the public domain. Two plates were drawn for this book from data, a doubling cascade and a sunflower’s seeds. A shorter edition chosen from this one keeps every read’s number.

Chapter 1 — The first nights, 3 and 4 October



The comma toy, ratio 0.618: its film, dot by dot, passing one way (blue) and the other (orange), and second gear's frozen frames, one held still at each pace from 1.62 to 1.78. TOY MODEL. Read 7, given on 3 October, is about the gears and the comma.

Gears are pieces of string that tie off.

from 7

R2W, 3 and 4 October

1

Look at panel C of the figure.

The orange failure is a flame standing on the bottom edge, its base covering every τ and its tip always at τ 5, whatever the gain.

The blue failure is a staircase against the top edge.

A staircase scales with distance; a flame has a fixed address.

That’s the gearbox’s own two kinds again, a wall and a resonance.

My bet is the flame is a tongue: the filter against bead 1’s kicks.

R2W

2

Look at the bottom-edge sign rows top to bottom.

The failure is a tide.

With fast kicks it’s out, leaving one puddle.

At ρ = 1 it reaches the wall, and with slower kicks it stands deeper, out to τ 30.

Downstream has a pulse.

Upstream has to outrun it or be smoothed below it, and matching it is the one thing it can’t stand.

R2W

3

Stage two is a turnstile.

Hand it coins faster than it turns and it jams at the top.

Hand them slower and it can’t get stuck at the bottom, but it can’t protect itself there either.

At the exact speed it turns, something odd happens, and that’s S3’s job.

R2W

4

Panel C of the new figure is a funnel.

The closer the two rhythms get, the longer you have to listen before the system tells you the truth, and at the bottom of the funnel you’d listen forever.

It’s two radio stations a hair apart.

The warble gets so slow you mistake it for silence, and our first runs mistook it for a feature.

R2W

5

The gain 1.9 rows are a ball rolling toward a cliff.

Steady steps, one cell per row, then the steps stretch (17.8, 25.7, 44.4), and on the edge the ball never stops rolling.

R2W

6

In the second pair, the bigger kick adds the same four grid steps at every room.

The room doesn’t care how loud the drummer is; loudness just stacks extra wall on top.

On our log-spaced grid, “same number of steps everywhere” is the signature of a power law.

R2W

7

Gears are pieces of string that tie off.

The comma is the yarn that never ends.

R2W

8

The gearbox is Morse code.

All dashes is first gear.

Perfect alternation is second gear.

Between them, the density of the code tells you where you are.

You can read the gear from the shadow alone, without ever measuring a speed.

R2W

9

A lighthouse keeper logs each passing ship.

Put him on the wrong rock and he logs the same ship twice, going out and coming back, so his log looks busier than the sea.

Only “out minus in” counts ships.

R2W

10

A lighthouse beam sweeping across a dancer.

While she holds her routine, the four flashes land in the same four spots every lap.

When she starts to drift, the flashes crawl, slowly near the routine and quickly far from it.

R2W

11

A film projector.
Run it and the frames flicker past, letters marching across the screen.
Lock the gear and the projector stops on one frame.
Which frame it stops on depends on where in the gear you are.

R2W

12

The film is a dancer passing a mirror.
She slows down every time she reaches a pose she could hold, and hurries through the poses she can’t.
The gear is the moment she stops on one of those poses.

R2W

13

The concentration column reads like a heart monitor.
Ragged at 0.69 and 0.70, then suddenly steady at 0.87, 0.87, 0.87.
The patient has stabilised, and you can point to the beat where it happened.

R2W

14

At pace 1.70 the four intervals are a limp.
A step, a quick skip, a step, a long stride, the same limp every lap.
Outside the gear the limp wanders.

R2W

15

The three staircases for the three strengths are the same staircase slid sideways.
How far you stand from the cliff edge decides your steps, not how tall the cliff is.

R2W

16

Chaos is an argument, and it needs three voices that can all answer back.
A one-way relay is an echo: it repeats, nobody argues.
Mass gives the rider a heartbeat of its own; a two-way rod gives it a partner who pushes back.
Too quiet and the rider is just a puppet; too loud and the voices fuse into one.
Chaos lives in the band between, the same shape as the dither pocket.

R2W

17

The data says the universe here only grants a grip to patterns that *repeat*.

A rhythm that never comes back to the same frame can shake you but can’t hold you.

A rhythm that closes its loop, even a long loop like 5:2, leaves a fixed handhold standing still beside you.

Gears are closed loops, all the way down.

R2W

18

A closed rhythm doesn’t hand out grip; it hands out a choice.

Lock two rhythms exactly and you stop the film on one frame, which can be a good frame or a bad one.

Leave them slightly off and the film rolls, and sooner or later you get the worst frame.

“Force only decides where the walls stand”; timing decides which wall you’re leaning on when the music stops.

R2W

19

The shield has a handedness.

Same sizes, same tones, same everything.

Show it in a mirror or run the film backward and it stops working.

The data is saying *direction is information*: a wobble that leans forward protects, and the identical wobble leaning back does nothing.

That’s your Möbius strip again: you can’t tell the two apart locally, only by which way the loop closes.

R2W

20

The cost of precision depends on where you stand.

On the watershed every halving of the margin costs a full doubling.

Step off it and the same margin costs much less.

The cheapest place to be accurate is away from your own resonance.

R2W

21

A near-miss is a loan, not a gift.

Being close to a perfect fraction buys you time, not grip, and the interest rate is the detuning.

Pay it back by being exactly right, or the bill comes due at the band’s edge.

R2W

22

Weight is a kind of patience.

A light rider only lives in the present, so it can only ride the biggest rhythms.

Give it some weight and it can carry a rhythm across a few beats, which lets it hold subtler locks.

Give it too much and it carries so much history that it starts arguing with itself (that’s the chaos window).

R2W

23

Whenever two rhythms meet, the universe counts in whole numbers, even when it counts its failures.

A rider that can’t hold a step doesn’t lose it smoothly; it loses exactly one, two or three teeth per heartbeat of the thing driving it.

Even the mistakes come out as integers.

R2W

24

Holding and failing aren’t two states; they’re two ends of one ladder.

Between “never slips” and “slips once every beat” there are rungs for every simple fraction: slipping 3 times in 8 beats, 1 in 5, 1 in 8.

Push harder and the ladder grows more rungs instead of breaking.

Nature doesn’t fail all at once; it fails in rhythms.

R2W

25

This machine has a short memory for tempo and a long memory for melody.

Nudge the dial and the tempo barely moves, but the tune it hums is already different.

Each setting has its own song, even when two of them march at the same speed.

R2W

The Loop Atlas book, 3 and 4 October

26

Everything that worked cut down how often the switch is asked.

The plain switch answers every time it’s asked, and the toy asks at every tick.

Pausing after an answer barely helps, because the next question still comes at the next tick.

Asking less often fixes it, and so does listening to something that doesn’t change at every tick.

the Loop Atlas

27

Every toy on the shelf is the same machine in a different coat.

Two things move, there’s a step between them, and something cares about the order.

The dead ones all die the same three ways: the step is zero, the step is half a turn, or the part that listens goes deaf when it stops moving.

the Loop Atlas

28

Two surveyors measured the same walk.

One reported how far the walker went; the other reported how far along a line chalked on the walker’s bent back.

The walk didn’t change, only where each one stood to point.

the Loop Atlas

29

The skater never knows what time it is.

It knows only “the wave is with me” and “the wave is against me”, and it counts those.

Every number that moved when the rhythm doubled was one we had written in seconds.

the Loop Atlas

30

The needle stands on a ledge, leaning into a wind.

The faster you ask it to go, the narrower the ledge and the harder it leans.

It can feel the width of the ledge in two ways: how far over it’s leaning, and how long it wobbles after a shove.

the Loop Atlas

31

A pebble rolls off one step and stops.

A boulder that gets going rolls down the whole hill, and you have to flatten the hill a long way before it stops.

the Loop Atlas

32

It’s a tree where two branches grew a foot overnight and three branches have no leaves yet.

the Loop Atlas

33

A wave walks a circle of doorways and comes home exactly as it left, except its heartbeat has shifted by how much sky the doorways encircled.

Ask it to pass through the doorway facing its own, and it vanishes; on the far side, its heartbeat is inverted.

the Loop Atlas

34

The snake is a ribbon laid on a breathing floor.

Each joint gets paid for how much it creaks.

Too short and it can’t feel the breath; too few joints and the creaks go unpaid.

Length is the pay grade and joints are the pockets: three pockets per wave are enough to hold it all.

the Loop Atlas

35

Every wave is a moving walkway.

Step off with energy in your pockets and it carries you along; feed it energy at the right moment and you can walk against it.

The walkway’s speed sets the price of every step, and the slower it moves, the cheaper each step against it.

the Loop Atlas

36

Everything tonight said the same thing in different coats: whatever keeps its own time wins.

The swing that listens to itself keeps growing, the rings that pass through each other keep moving, and the snake long enough to wear the wave can push back against it.

the Loop Atlas

37

Pushes are bought at a market where everything is priced by its speed.

Slow things are cheap to lean on and light is the most expensive thing for sale.

A floor heavy enough to catch your throw and hand the energy back gives the push away for free.

the Loop Atlas

38

A damper is a toll booth that takes your money and never your direction.

Throw something on a string, let the booth catch it, pull it home, and the booth has kept every coin while you float exactly where you began.

To keep any of the trip you have to hand the far end to someone else, and then the trip is split by weight: the heavier they are, the more of it is yours.

Even the floating swing plays by this: it turns a little further when pumped only because short is fast, and how far it turns was settled the moment it started.

the Loop Atlas

39

The needle lives on a staircase that tilts more the faster you ask it to go.

Every step has a seat to rest in and a lip to fall over, and they always come as a pair.

Tilt far enough and seat and lip meet and vanish together, and the needle rolls down the stairs forever, one step per slip.

Counting the steps is easy, as long as you count landings rather than stumbles.

The price of each push never changes on the staircase; what changes is how many you can buy, and the best customer stands right at the edge.

the Loop Atlas

40

A car on a banked turn sits where the floor is square to the lean it feels, and the faster it goes the higher it sits.

But the lean only ever tilts the bowl; it never turns the bowl into stairs.

A real track runs out of wall, and that edge is a fence, not a step.

To get stairs you need a push that keeps going round, the way the needle’s field keeps turning, and then a faster car does the opposite of the needle: it grips harder.

Tyres add a knee: instead of one perfect speed for each lane, they give a whole band of speeds where the car simply holds on.

the Loop Atlas

The sounding board, 3 and 4 October

41

The shelf is a set of fossils, and 24 of 26 still give the same reading when you dig them up.

The four that don’t are the ones where someone wrote on the rock afterwards.

the sounding board

42

The sitting reads like an organism that grew fast and then shed its old skin at the fork, but left a few organs inside the old skin.

The shelf put them back.

the sounding board

43

A swimmer that thinks it’s going straight is crabbing, and only someone who looks at the sign notices.

the sounding board

The Field Guide session, 3 October

44

These toys say memory lives in the slip, not the grip.

A locked rider forgets the way it came: run it backward and it misses.

A slipping rider retraces itself to rounding error and builds staircases out of the phase it carries forward.

That’s row 380’s “lock for the index, slip for the load” turned into a statement about memory, and I think it belongs in the Loop Atlas book’s spine as much as in the guide.

the Field Guide session

45

A number without its base is a rumour.

You’d never log a voltage without saying what it’s measured against.

Ground is the base, and every fix today was “name your ground.”

The length needed none, because it was the one number that didn’t depend on which way you faced.

the Field Guide session

46

The machine is breathing.

At ρ 1.037 it slips for 15 beats, holds for 12, slips for 25, then holds for 12 again.

At 1.0435 it takes one long breath of 41 slipping beats and then holds for 46.

At 1.060 it’s a metronome: one, rest, rest.

The chaos maps look like a coastline.

The surf breaks only where two countries meet, never inland, and the stretch of coast at mass 1.0 has no surf at all.

My letter registers a fine sweep from 0.95 to 1.05 to find out whether that calm is real or just where the grid’s sample points happened to fall.

R2W

47

The gearbox is a staircase, and chaos lives on the edges of the steps.

That’s the spot where a foot can’t decide which step it’s on.

A heavy walker stumbles there, and a feather never does.

R2W

48

The rider’s footsteps are seen through a picket fence.

When a footstep lands near a picket, which gap you see it through depends on where the fence starts.

The walker’s stride is the real thing.

R2W

49

The rider is carrying a full cup of tea across the room.

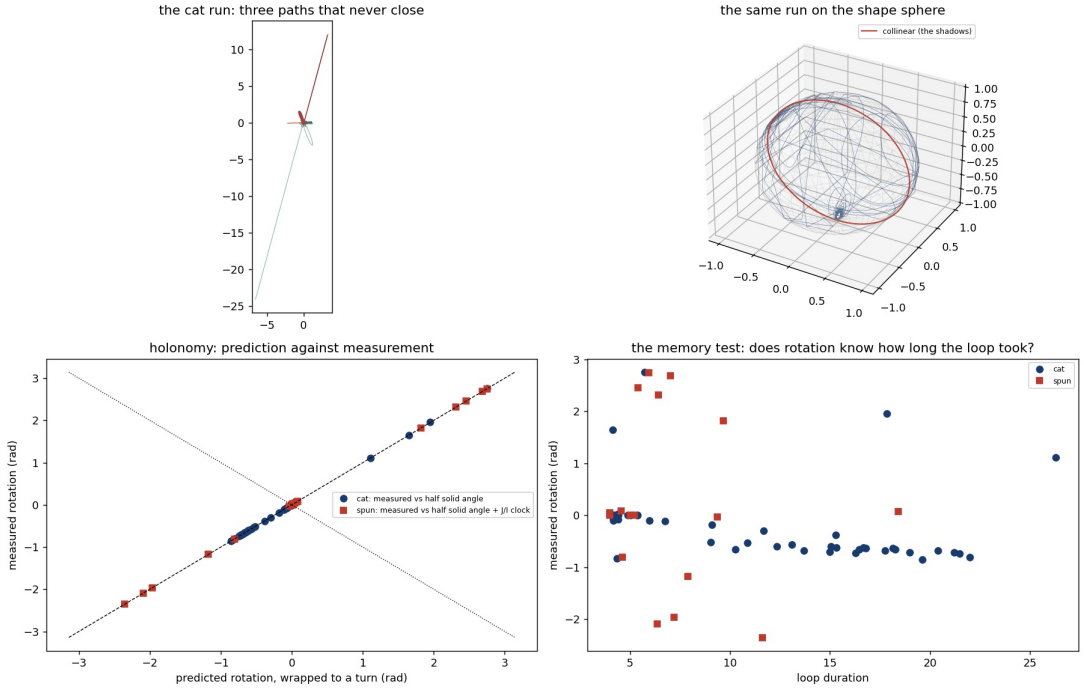
Walking steadily never spills.

Jiggle it at one moment in your stride and the tea just sloshes; jiggle it at the opposite moment and it spills.

Make the cup heavier and sloshier, and any jiggle spills it, even though walking steadily still works.

R2W

Chapter 2 — Gears and rooms, 4 to 6 October



Three stars let go from rest: their paths, which never close; the same run on the shape sphere; the turn each shape loop predicts from half the solid angle it encloses, against the turn measured, for the cat and for a spun start; and the turn against how long the loop took. From the triple-star thread's first runs. Its read is 82.

Three stars that start from rest can turn only by changing shape, and when their shape comes home the whole triangle has turned by an amount set by the area the shape loop enclosed, as if the sky kept its books in shapes rather than in time.

from 82

The Loop Atlas book, continued, 4 to 6 October

50

The board does not care how long it is; it only cares how long its rider takes to notice.

A rider who taps on its own beat sings the same song on every board, just faster on the short ones, the way a slide rule slides the whole scale and leaves every ratio alone.

The trouble comes when the rider’s reaction takes longer than its own beat: then it taps again before it has felt the last tap land, the taps pile up on one side, and the board goes over without any warning in the song.

the Loop Atlas

51

The safe ground for a smooth pusher shrinks to a sliver as its reactions slow, thinner than a hair at 1.40 fall times, and it vanishes at the square root of two.

You cannot find that sliver by poking around in it; you find it by walking along the edge and counting how many times an arrow goes round.

On the sliver the board still settles, but slowly, as if it were tired.

the Loop Atlas

52

A board that spins in the air goes exactly where it was already going; the spin is a dance in place.

Give it a pad to spin on and every half turn moves it one board length, as surely as counting, but only the exact half turn: a little more or a little less and it walks a circle and comes home.

The pad hands the spin back reversed and halved, so a board left alone waddles.

Each step is the same size, and each takes twice as long as the last, like a story told more and more slowly that never loses its place.

the Loop Atlas

53

Every toy keeps a clock somewhere.

The worms keep it in their own laps, the wing in its flaps, the needle borrows the field’s until it cannot keep up, and the swing listens to its own swinging.

Only the skater keeps human time, because its legs push for a while rather than for a number of times.

The scallop keeps its clock inside the water that remembers it, so going faster is the same as making the water thicker.

And the bells keep the water’s own time: give them longer between squeezes and every squeeze goes further, because the water has had longer to forget.

the Loop Atlas

54

Counting slips is like counting people who leave a room.

Count them at the door and you also count the ones who stepped out, changed their mind and came back.

Count them when they reach the car park and you miss the ones still walking when you stop counting.

Both counts tell the same story; they differ only at the ends of the watch, and in where you choose to stand.

the Loop Atlas

55

A watch has two ends, and every argument between two honest counters lives there.

Make both ends follow the same rule and the argument shrinks to a coin toss.

Count the ones still standing in the doorway when the watch ends and you can say exactly how far apart the two counters are, to the last one.

the Loop Atlas

56

Looking back, the whole sitting was one question asked in different coats: who holds the clock?

The toys that keep their own never lose count, the ones that borrow a clock fall off when it runs ahead of them, and the only one that needs a wristwatch is the one that pushes for a while instead of a number of times.

Even the counters had to learn it: count the landings, and let both ends of the watch keep the same rule.

the Loop Atlas

57

A board hanging over the edge of a cliff can still be held up if somebody shakes the ground fast enough, much faster than the board would fall.

Shake it with one twist and the board stays; shake it with the opposite twist and it drops.

Push the board a little further over the edge and the twist that saves it turns around: now only the opposite twist holds it, as if the board had to shuvit to land.

Shake at the board’s own rhythm and nothing saves it, except at one exact beat where the ground and the board keep step.

R2W

58

The machine is a combination lock whose numbers are fractions.

Only the exact fraction opens it; a hair off, and no turn of the dial will.

On the right fraction the working turns come round evenly, spaced by the fraction’s bottom number: every third notch of twelve at nine-eighths, every other notch at seven-sixths.

And if you shake the lock instead of dialling it, there’s one road.

A gentle shake opens it on the upswing, a harder one on any swing, a harder one still only on the downswing, and then nothing opens it.

It never goes back down the road.

R2W

59

It’s Goldilocks.

Shake the board too gently and it falls; shake it too hard and it falls.

In between is a narrow porch where it stays, and on that porch the twist that saves it depends on which end you’re standing at: near the gentle end you twist one way, near the hard end the other, and in the middle any twist will do.

Measured in inches, every board has its porch in a different place.

Measured in how hard the shake shoves, all the porches stand in the same spot.

R2W

60

The grip is a dimmer, not a switch.

Squeeze the rope six times harder and the yank you feel through it drops by only a seventh, because the yank comes from whoever is pulling, not from your hand.

It doesn’t matter where along the rope you stand.

Turn the dimmer all the way and the light moves only halfway across the room.

R2W

61

The window is a doorway with a floor and a ceiling.

Stand at the right height and you walk through in any stride.

Crouch below the floor and only one thing gets you through: stepping exactly in time with the floorboards, three steps to two boards, five to three.

A hair out of step and the floor holds you back.

In the corner of the room it looked as if only perfect steppers ever got through, but that was only because everyone arriving at that corner was crouching.

R2W

62

A firmer grip needs a harder shake to work it loose the right way, but not twice as hard for twice the grip.

The shake it needs grows like a square root, the way a spring’s ring rises with its stiffness, in a machine that has no spring in it and nothing that rings.

And the deeper the rider has sunk below the floor, the harder you have to shake, a little.

R2W

63

The floor gives a whisper of warning, not a hum: the rider slows its recovery only in the last hair’s breadth before the edge.

And the rider’s crowd breathes hard every time the big wave passes: everyone is flung apart and then pressed back together, so if you ask how scattered the crowd is, the answer depends entirely on when you look.

R2W

64

The edge of the floor is not a cliff but a soft shoulder that looks the same from both sides: just above it the rider is slow to come home, just below it the rider keeps tripping and getting up again, and the slowness and the tripping keep the same time.

Weight changes the road.

A light rider walks off and walks back on at the same spot; a heavier one walks off sooner and has to be carried further before it will step back on; a heavy one is no longer one rider but several, each with its own spot to walk off, and which one you have depends on where you set it down.

R2W

The draft's own note, kept as written (its reads 12 and 13 are reads 64 and 65 here): Note: the last sentence of read 12 describes an instrument fault, not the rider. The test it rested on read a frayed band as a fall; read 13 reads the corrected runs. Read 12 is kept as written.

65

The heavy rider was never several riders; it is one rider who starts to stagger before it falls.

Far from the edge it walks a straight line; nearer, it weaves, wider and wider, without ever losing ground; then it goes down.

A light rider gives no such sign: it walks straight until the moment it trips.

And once down, the heavy one has to be helped well past the spot where it fell before it will stand again, however slowly you help it.

R2W

66

The stagger has a rhythm before it has none.

First the rider steps left, right, left, right; then it takes two different steps on each side; then four; the pattern doubles faster and faster until there is no pattern at all, and still it never loses its place in the line.

Inside the confusion there are calm pockets where it finds a three-step dance and holds it for a moment.

And weight and brakes turn out to be one thing: a heavy rider with strong brakes walks exactly like a light one with ordinary brakes.

R2W

67

The stagger keeps a beat that quickens like a drumroll: the first two changes come at the same pace, then each comes about four and a half times sooner than the one before, until they run together into noise.

And whoever tries to catch the moment of each change by waiting for the rider to settle waits forever, because right at each change the rider takes forever to settle.

R2W

68

The heavy rider keeps up with the gear on average, yet once in every passing it is walking backward.

It loosens its stride on the way down and does not tighten it again until well past where it loosened.

And when it finally loses the gear, the light rider steps cleanly onto the next moving walkway while the heavy one tumbles end over end.

R2W

The Loop Atlas book, the fresh session, 4 to 6 October

69

Two counters, given four times longer to watch, now agree to a few hundredths, but one still leans the same way, because it walked into a room already moving and had to guess which step each dancer was on.

And the tumblers tell a second story: how long a tumble lasts depends on how long a pause you agree to call “stopped”.

The dancers never change; only the stopwatch’s patience does.

the Loop Atlas

70

The counter that leaned had walked in mid-song and guessed one step late for a few dancers; switch it on before the song starts and it stands straight.

The tumbling needle turns out not to tumble at all: it spins, slowly and lumpily, and every so often sits down for a rest.

What we called its runs are its spins, and how long a spin seems depends on whether you clap when it reaches the top of the hill, where the music sets the beat, or when it lands, where its own fall adds a stagger.

the Loop Atlas

71

The spinning needle keeps a short memory.

Each turn, its climb and its own fall rise and sink together, but at every landing it wipes the slate, and the next turn starts as a stranger.

It decides whether to spin on or sit down in the first fifth of a turn after it lands; everything after that is just falling.

And the needle without memory sits in its hollow exactly where the old formula says it should, to the third decimal, never having been told the time.

the Loop Atlas

72

Each needle walks toward its own edge at its own pace, and the weather decides where the edge is: in a calm room it leans almost to the end before it stumbles, in a noisy room it stumbles a long way short.

A meter that knows what the machine was asked to do can feel how far it leans; a meter that only listens to the machine hears nothing useful until the stumbling has already begun.

the Loop Atlas

73

Every machine carries a cup it slowly fills, and dies when the cup overflows, but the size of the cup depends on the weather in its room and on how hard it is made to work.

A clock that counts only the hours knows neither; a clock that counts the turns knows how hard it worked; a gauge that looks into the cup and knows the weather throws away a tenth of a life instead of half.

the Loop Atlas

74

Some cliffs hum as you walk toward them: the ground goes soft and slow underfoot, and anything that listens can hear the edge coming.

Others are silent: the path stays firm and the song stays the same right up to the step that isn’t there.

And some cliffs are one-way doors: once you have fallen, the place where you can climb back up is a long way behind the place where you fell.

the Loop Atlas

75

Every cliff we have met is a crease in the map of knobs, where the ground the machine can stand on folds, ends, or splits in two.

Where it folds, the ground goes soft before the edge; where it simply ends, it stays hard to the last step; where it splits, there are two edges, one for falling off and one, further back, for climbing on, and they start as a single crease and open like a pair of scissors as the machine learns to remember.

the Loop Atlas

76

A gearbox is a staircase of landings, wide ones and narrow ones.

If the book is right, the edge of every landing goes soft underfoot in exactly the same way before it gives, and only the width of the landing tells you which stair you are standing on.

the Loop Atlas

77

Some edges you can hear from across the room, and some only when your toes are over them.

How loud the warning is depends on what meets at the edge.

Where a seat meets its lip, the floor goes soft as the square root of the distance.

Where a spinning path brushes a hilltop, each turn lingers at the top a moment longer, the same moment more for every halving of the distance.

Where a slow pusher has tuned itself as well as it can, three of its ways of settling have merged into one, and the floor goes soft in plain proportion.

Where nothing meets, a wall or a late tap, the floor stays hard to the last step; the wall only makes the car quicker and quieter in the last couple of wobbles before it touches.

And the listener has edges of its own: a shove toward the lip comes back slower than a shove away from it, a needle still settling sounds like a slow one, and a moving ramp always hears the edge late.

the Loop Atlas

78

A shaken rider can be held where it would otherwise slide off, but only by the right amount of shaking: too little and it slips, too much and it is thrown, and in between it holds, leaning one way at the gentle end and the opposite way at the hard end, never halfway.

And a machine that keeps its own loud beat cannot hear its edge coming until it is nearly over it: the softening is there all along, but it becomes audible only once it is slower than the drum.

the Loop Atlas

79

A bell resting between strokes is not resting: the water it stirred keeps pushing it forward, and more than half of what it gains by waiting comes from that push, not from the stroke that follows.

The toy bell, which feels the water only at the instant it kicks, gets the right answer for the wrong reason: it does a little better after waiting only because the water it threw backward has had time to leave.

the Loop Atlas

80

A small whirl coming at a bell tugs the bell toward itself, slips in through the open top, and, when the bell is held still, leaves by the mouth spinning harder than it came, as if the bell lent it something on the way through.

Bear’s chase asks for the other pass, the bell going through the whirl’s maker.

Whether a whirl as wide as the bell tugs it forward or shoves it back is the one thing the picture stands on, and nothing has looked at it yet.

the Loop Atlas

81

A tight grip does not loosen before it lets go.

Right up to the last moment it holds one place exactly.

Then the light needle starts slipping now and then, more often the further it is pushed, while the heavy needle lets go all at once and tumbles, as if a tumbling state had been waiting beside the grip the whole time.

And memory needs a minimum before it can hold a door open behind you: below about seven tenths, the way out and the way back are the same door.

the Loop Atlas

82

Three stars that start from rest can turn only by changing shape, and when their shape comes home the whole triangle has turned by an amount set by the area the shape loop enclosed, as if the sky kept its books in shapes rather than in time.

Pushing the whole triangle evenly changes nothing it can feel; only a push that is stronger on one side than the other gets in.

And two clocks running side by side keep almost meeting at the whole numbers where their ratio nearly closes, the way two gears with an awkward ratio click back into near-alignment at the same teeth again and again.

the Loop Atlas

83

When the stars’ books were checked page by page, they balanced better than their own bookkeeper said.

The small error in the ledger was not the stars forgetting anything; it was the clerk reading too slowly at the one moment the shape changed fastest, and it all but vanished when the clerk read faster.

The closest two stars ever came was not where the clerk stumbled: what trips a reader is how far the thing moves between two looks, not how dramatic the moment seems.

And two copies of the same three stars, one carried evenly across the sky and one left at home, kept the same books but lived different lives after their second dance, because the computer rounds a faraway number more coarsely than a near one.

The stars cannot tell they were carried; the computer can.

the Loop Atlas

84

Two riders of the same weight, dragged through the same thickness of mud, go wild in different ways.

One starts to shiver, then shivers twice as slowly, then four times, then eight, and only then loses all rhythm; the other holds perfectly still until the moment it bolts.

Their weight does not decide which; what pushes them does.

And the wild one never loses count: it thrashes inside its groove and never once jumps a tooth, as if the whole number were the one thing chaos could not reach.

the Loop Atlas

85

A raft held up by dozens of bells is not a crowd pushing together; it is a crowd passing breaths along.

Each bell’s spent breath reaches its neighbour first as a pull and then as a push, so whether the crowd lifts or wrestles itself depends only on when each one breathes compared with the one beside it.

And a bell held still pushes hardest against the swirl it left behind a moment earlier, as if a tethered bell builds its own wall to lean on.

the Loop Atlas

86

Three rooms kept passing notes through one man’s hands, and each note came back a little truer than it left.

The stars’ own bookkeeper had written his figure in pencil; held up to the light, it was fifty times smaller, and the stars had been honest all along.

The one question the book was built to ask kept getting pushed aside by the mail, the way the most important letter on a desk sinks under the ones that just arrived.

And each time the room filled up and the edges blurred, the whole of the work was folded into one box small enough to carry, carried into a clean room, and unfolded there without losing a page.

the Loop Atlas

R2W, 6 October

87

The twenty-five reads are a field notebook that changes hands halfway.

The first fifteen point at something you could photograph: a flame, a tide, a turnstile, a lighthouse, a dancer.

The last ten state a rule you could live by: an argument needs voices that answer back, a near-miss is a loan, weight is patience.

Somewhere between the staircase and the argument, the toys stopped looking like objects and started sounding like advice.

R2W

R2W, sitting B6, 5 and 6 October

88

A dog on an elastic lead behind a cyclist.

The cyclist sways a little with every pedal stroke, and a short lead hands every sway to the dog; when the dog’s path narrows it starts stumbling on the sways long before it reaches the edge of the path, one stumble every few strokes, then two, then more.

Before the first stumble its gait rocks in time with the pedals, louder and louder, a warning you could hear from well up the road.

A long lead hides the sways, and then the dog keeps its feet until the cyclist goes down.

Even then the cyclist doesn’t fall at once: it skids, a little and then more, and the dog is only yanked off its feet when the skid has carried the cyclist most of the way over; with a gentler gearing between them the same skid tows the dog onto a faster path where it finds its feet again.

The cyclist never knows the dog is there.

R2W

89

A messenger has to speak more slowly the closer the listener stands to a cliff edge.

Give the listener half the room and the messenger must take twice as long over every word, and it hardly matters which cliff the listener is near, or which messenger is talking: one straight rule.

It is strictest when the messenger’s stammer falls in step with the listener’s own footsteps, and looser when they’re out of step.

And we had been measuring the listener’s room with a tape that started two inches past the real edge; a few inches from the drop those two inches were a third of the room, and only when we measured from the true edge did the rule come out straight.

R2W

90

A merry-go-round you jump onto and off.

A light child steps on and off at the same speed of the turntable; give the child more weight and, once aboard, it rides on well past the speed where it could have stepped on, because its own momentum keeps it going.

A child under a certain heft remembers nothing; above it, the gap between stepping on and stepping off opens fast, and for a very heavy child the turntable can slow almost to a third of the boarding speed before it lets go.

Our rider is a child on two turntables at once, and it acts as if it carried about half the weight it really has.

R2W

91

We weighed a new child on the merry-go-round, and it rode exactly as if it carried half its weight, just like the two children before it.

A lighter child’s dithering was too small for our scale to read, and the heaviest child couldn’t even get onto the ride at the speed we started it.

And every child that loosens its grip on the way down only tightens it again a little after the spot where it let go, as if the grip, too, had a short memory of its own.

R2W

92

A spinning top set on a table next to a little bell on a spring.

The top hums and the bell starts to ring; when the top’s hum matches the bell’s note, the bell rings back and holds the top at that speed, as if it had found a groove in the table that wasn’t there before.

It holds at twice the bell’s note too, and at half, and even at two-fifths.

Stack a row of tops and something stranger happens: every other top rocks the opposite way, at half the hum.

And you only find some of these grooves by spinning the top up past them and then letting it slow back down.

R2W

93

We thought the heavy children on the merry-go-round stayed aboard long after they could have stepped off, and we were watching the wrong thing.

A heavy child doesn’t sit still once it’s on; it squirms and lurches in its seat, and our scale counted every lurch as a step.

Set the children down at one fixed speed and watch properly, and every one of them is seated at once, heavy or light, even the ones that had been hopping on and off one lap in three.

A turntable that turns by itself lets a heavy child remember how it got on; a turntable someone else is pushing seems to wipe that memory clean.

And the messenger by the cliff: we had only timed him beside one cliff when he was out of step with the listener.

Beside the other cliff he may have to slow down more, not less.

R2W

94

The messenger by the far cliff talks exactly as slowly as the rule says: half the room, twice the time, five times over, and nowhere near the hurry I bet on.

The trouble I thought I saw there was a ruler with marks a fifth apart.

Back at the merry-go-round we weighed every child twice with the new scale and not one of them remembers how it got on, from the lightest to the heaviest; the old scale, read the old way, gives the old wrong answers to the last digit.

But the heaviest child we can seat is still light for the grip it is holding, so the ride hasn’t yet been asked whether a pushed turntable forgets, only whether a light child does.

And the far wall of the ride stands exactly where the old chalk mark said it did, less than a hair from it.

R2W

95

A little weight lets a child keep its feet on the merry-go-round a touch slower than a weightless one can; the heft carries it through the lurch at the bottom of each lap.

A middling weight is the sweet spot.

Add more and the ride’s edge stops being a clean line: the child stays on at one speed, falls off at a slightly faster one, stays on again a hair faster still, like a coastline seen close up.

And the heavy child who seemed to wobble for a long time before settling turns out not to have been wobbling at all.

There are two places to stand on the ride, half a step apart and both perfectly steady; climbing up through the speeds the child keeps to the first until it vanishes underfoot, and then steps across to the other.

The stepping across was what we had been calling the wobble.

R2W

96

Double the grip of the slow walkway and a rider can stand on it across a far wider range of speeds, from about half the fast walkway’s pace to one and a half times it.

Weight changes almost nothing at the slow end and a great deal at the fast end: the heavier the rider, the sooner the fast walkway’s flicker stops helping it keep its footing, as if the flicker could steady a light child but only rattle a heavy one.

Next door, on the other ride, small knots of riders keep spinning, steady as clocks, beside neighbours standing still at the same speed: the same picture as our two places to stand, seen from the other side of the fence.

R2W

97

Two workshops build the same bucket brigade: a runner who sways each time a wheel goes by, a slow hand that smooths the sway, and a catcher downstream who must keep his footing.

The parts look nothing alike, and in one shop the wheels even turn the other way, but every piece in one has its counterpart in the other, measured twice, and now we wait to see whether the other shop’s catcher needs the same patience for the same room.

Meanwhile we learned why our weightless runners never tangle: a runner with no weight can slip, but only the way a wheel slips, never into a knot; knots need weight, or a partner who pushes back.

And on the other bench a clock does not have to fall silent for a memory to survive; it only has to be quiet enough.

R2W

98

On a bicycle with a stiffer chain, ride slowly and the gear drops you onto the big wheel, but as soon as you speed up again you are back on your usual cog, no memory at all.

Ride fast and the chain jumps to a tiny cog instead, and then something strange happens: slow down a long way and you stay on the tiny cog, pedalling happily at the wrong gear, as if the bicycle remembered the hill it had just come down.

A bicycle with no weight never does this; at any speed it knows only one gear.

It is the rider’s own weight that lets two gears share the same road.

R2W

99

We weighed six riders on the bicycle with the stiff chain.

The three lightest, pedalling faster and faster, never found the tiny cog at all; their chain just slipped and rattled, and the moment they eased off it caught their usual cog again, right where it had let go.

The three heavier ones found the tiny cog the instant the usual one gave, and kept it while they slowed a long way down.

More weight did not make that memory longer, as I bet it would: it slid both cogs down the road together, so the stretch where either cog would do got shorter, then stopped shrinking.

Then we stopped every heavy rider at one fixed speed, one on each cog, and left them there a long while.

Not one of the eighteen changed gear.

Two gears really do share that stretch of road, and only weight lets them.

R2W

100

The second cog was on the bicycle all along, even with nobody riding it; it just sat far up the road, at speeds nobody reached.

Put weight on the bicycle and the second cog slides down the road toward you, nearly three times as fast as the end of the first cog moves.

While its near end is still beyond the first cog’s far end, nothing is remembered: you drop off one cog, rattle along for a moment, land on the other, and coming back you land just where you left.

Once the near end slides past, a stretch opens where either cog will carry you, and it widens steadily with each extra pound, up to a point; past that the two cogs slide down together and the stretch narrows a little, then holds.

Near the second cog’s edge you can only be caught by coming to it along its own road: set a rider down at exactly the right speed from a standing push and it falls back to the first cog anyway, like a ring that has to meet the rope square to take hold.

R2W

101

Two ropes cross the river, a high one and a low one, and the ferry’s load decides where the low rope lies.

A light ferry finds the low rope far downstream of where the high rope ends: it slips off the high rope, drifts a moment, and catches the high rope again in the same spot, remembering nothing.

A middling ferry finds the low rope running just under the end of the high one and drops straight onto it, and on the way back it holds the low rope a long way before it climbs up again.

A heavy ferry finds the low rope has drifted so far downstream that, slipping off the high rope, it is carried free with the current for a long stretch, and only when it has slowed does it snag the low rope, ride it, and at last come back up to the high one.

None of the ferries knows where it came from.

Each just carries its own speed and its own angle to the current, and those decide which rope it can grab.

R2W

102

We spread a thousand ferries across the river at the moment the current’s swell went by, each at its own spot and each already moving at its own speed, and let them go.

Which rope each one caught was decided almost entirely by where it sat, hardly at all by how fast it was going: one stripe of the river, a fixed place against the swell, belongs to the low rope, and all the rest to the high one.

Push the river harder and the low rope’s stripe widens until it owns half the water.

Its near edge is ruled straight; its far edge leans inward for the fastest boats.

And at its widest the stripe frays: a thin second stripe appears beside it, and single boats far off go the low rope’s way while their neighbours do not.

R2W

103

On the other bicycle, the one with its chain moved to the other wheel, a light rider spins off its cog and, slowing down, lands back exactly where it let go; even the middling rider, the one the neighbours bet would hang on a little, remembers nothing at all.

From the middling weight up, every rider that loses its cog is caught by the tiny cog on the way down and rides it a long way before climbing back.

And when we looked closely at the tiny cog itself, it was not one way of riding but two, taken in turn: the rider alternates between two ways of sitting on it, like a skater swapping feet each lap.

Ride slower and the two ways merge into one; slower still, the tiny cog simply ends.

R2W

104

There is a field with two orchards in it, and a wind that comes in gusts.

Drop a leaf anywhere in the field at the moment a gust goes by and it ends up under one orchard’s trees or the other’s, never anywhere else.

From a hilltop the field looks simply split: a broad stripe that belongs to the far orchard, and all the rest to the near one.

Walk down and look closely, and the line between them runs straight through a little saddle in the ground, the one place where a leaf could balance and go either way.

Beside the line, a few steps off, runs a thin dashed path of the far orchard’s leaves that nobody saw from the hill.

And around a second saddle that seemed to sit in the middle of the near orchard’s ground, a narrow ribbon of the far orchard’s leaves runs right past it, close enough to touch.

So the line between the orchards is not drawn just once.

Whether more and thinner copies of it appear every time you kneel closer is the next thing to walk.

R2W

105

A stream runs fast over a line of stepping stones, and a walker stands on the big first stone.

Ease the current and the walker stays put.

Push it a touch harder and the walker is knocked off, but not carried smoothly downstream like a stick: tumbled, turned over, a little different every time, so that two walkers who step in a hand’s width apart come out of the water at different places.

Here and there along the stream, though, sits a small stone the right size to catch a tumbling walker, and anyone near it lands on it and stands there, stepping in a fixed little pattern: one stone wants a step every seventh beat, another every fourth, another every third.

One wide flat stone has a walker hopping foot to foot; ease the current and the hop becomes a four-step, then an eight-step, then something with no repeat at all, though the walker never leaves that stone.

And at one place two small stones sit side by side, and which one you end up on depends on where you went in.

That is the only place in the whole stream where a walker remembers where it came from: standing on one stone, with the other stone right there.

R2W

106

Back at the stream with the stepping stones, sixty-four walkers jump in at each place where a small stone sits, from every spot along the bank and at every speed, some at a stroll and some at a run.

Wherever a stone sits alone, every one of the sixty-four ends up on it: nobody is lost to the water, nobody finds a second stone hiding nearby.

The stone is the only place to stand, and it gathers everyone.

At the one place where two stones sit side by side, the walkers split about one to two, and not by where they jumped in, in any way you could draw with a ruler: neighbours on the bank end on different stones, as if the stream dealt them out like cards.

And the stone that asks for a step every third beat turns out to have the same habit as the wide flat one: ease the current and its little pattern grows longer, three steps, then forty-eight, then something no one could count, while the walker never leaves the stone.

If two stones can only sit side by side when they are next-door numbers, as these two are, then the next pair should be waiting between the stone of fives and the stone of threes, a hair’s breadth of current apart.

R2W

107

There are two kinds of staircase.

The first is built of good timber: every step is a little higher than the one before, and a walker climbing it in the dark always knows which step they are on, because there is exactly one step for each height.

The second has warped.

Some steps tilt back, so that you go down while climbing; at some heights two steps sit side by side, and which one you stand on depends on how you came up; and between the steps are stretches of bare slope where you slide about and never settle.

Bear’s control-room staircase is good timber at its usual setting, and if you turn its knob all the way up it warps just like ours.

A rider with no weight can only ever build the first kind.

The warp is the only place a walker can remember how they came, because on good timber there is only ever one step to be on.

R2W

108

A house stands on the one spot where two gardens overlap, a high garden and a low one, so that from its front door you can step into either.

Two friends leave by that door at the same moment.

One takes the hill path first and the river path home; the other takes the river path first and the hill path home.

At every place they could stop and compare notes they would agree: on the hilltop there is only the high garden, so both are standing in it; down by the river there is only the low garden, so both are standing in that.

Only at the front door, where the gardens overlap, do they disagree, and they always do: one comes home into the high garden and the other into the low.

Neither of them changed.

The way round did.

A smaller walk around the garden shed, which never leaves the overlap, brings both friends home exactly where they started.

R2W

109

Two hikers leave a cabin halfway up a mountain that turns slowly, like a planet, so that a shadow sweeps across its slopes again and again.

Each carries a little counter that clicks once whenever they lose a step to the shadow.

They split at the door, one going left and one right, and meet at a hut on the far side.

Their counters disagree by hundreds of clicks, because one of them spent a long stretch on the side where the shadow keeps catching you.

Walk the same paths twice as fast and the disagreement halves.

If they read their counters the moment they arrive, the numbers look ragged.

If they wait for the shadow to pass and read them together, the disagreement is a whole number every time, never a fraction.

And the hiker who came all the way back to the cabin finds its door exactly where it left it, only hundreds of whole clicks behind.

The size of what they carry is set by how long they walked.

Whether they come home through the same door is set by which side of the mountain they went round.

R2W

110

A road climbs a mountain too steep to drive straight up, so it zigzags.

If each leg is short, a car holds the road the whole way; if a leg runs on too long past the steep part, the car loses its grip on that very first leg, sometimes before the top of it and sometimes just after, never on a later one.

The rest stops beside the road make no difference to these cars, because nothing in them wears out or heats up; they either make a leg or they don’t.

And no zigzag at all will carry a car up a mountain whose average grade is steeper than it can climb: on that mountain even the shortest legs give way after a few turns.

The heavy trucks shift down lower on the mountain than the light cars do, crawl up in their low gear, and keep that low gear far down the other side.

The light cars have no low gear at all; where the trucks crawl they are still in top gear, and they pass.

The heaviest truck of all, at the steepest part, could not even hold its low gear, and was sliding back past it.

R2W

111

The same zigzag road, but now the trucks have brakes that heat up on every climb and cool off whenever they ease, faster the hotter they are.

On short legs nothing goes wrong at first.

The heat creeps up bend by bend, the grip fades, and the trouble comes late: on the forty-seventh bend for the truck taking the shortest legs, the twenty-ninth for the next, the seventh for the next.

A rest stop every five bends gives back what the heat took; on one stretch of road it gives back exactly as much grip as a truck with cold brakes would have had.

But no rest stop can save a truck on a leg that is simply too long.

It loses its grip on that first long leg, before it ever reaches the stop, at the same moment whether the stop is there or not.

Rest stops are for what builds up, never for one hill that is too steep.

R2W

The Loop Atlas book, the session of 5 and 6 October

112

A little weight in the rider’s pocket is a gift.

It can walk a hair slower than a weightless rider and still keep its gear, as if the weight carried it through the soft spot in every passing.

A pocketful more and the gift turns: the rider still keeps up on average, but for part of every passing it is walking backward, and it starts to stagger long before a light rider would trip.

Somewhere between a pebble and a stone sits the weight that lets it walk slowest of all, and the rider can reach the same spot by easing its brakes instead of filling its pocket.

the Loop Atlas

113

Walked back down, the tumbling needle finds its old seat almost at once.

Three in four sit down within a hair of the place they were thrown from, and the rest one hair further on.

Whatever caught the needle when it fell is not a second road running beside the first; it is a scuff mark at the edge, where a few dawdle for a moment before they sit down too.

And the nearer they come to the seat, the longer each stumble takes.

the Loop Atlas

The register’s own note, kept as written (its read 39 is read 113 here): Note added 5 October, after HIDDEN_SCISSORS_SLIVER 20261005_145137: the last sentence of read 39 describes the crowd, not each needle. Read from the saved table, a needle does not slow before it sits down; the stumbles seemed to lengthen because fewer needles were still stumbling. Read 39 is kept as written.

114

There was never a second road.

Below the edge, every tumbler is already on its way to its seat; tumbling there is not a way of living but a way of arriving.

The nearer the edge, the longer the arrival takes, but every one arrives, and none of them slows down first: each tumbles at full speed until the moment it sits.

Above the edge there is no seat, and they tumble for as long as anyone watches.

The edge is a single doorway, and the crowd pours through it the moment it opens.

the Loop Atlas

115

A body that stays straight can wave its arms and legs in circles forever and never turn; it has to bend first, and even bent, it turns so little per wave that a hundred waves barely make half a turn.

But give it a spin to carry and the smallest lean decides everything: lean a hair too little and the spin only rocks it back and forth; lean a hair more and it corkscrews, faster the more it leans.

The extra seconds of a high dive buy more corkscrews only once the lean has crossed that line.

And a spinning needle shaken from outside can be made to tumble wildly without ever losing its place, or held spinning below the point where it would have stopped; how fast the shaking comes decides which.

the Loop Atlas

116

A needle that can either sit in its seat or keep tumbling past it stays in two minds over a stretch of pushes, and which it does depends on what it was doing before.

Tap the table from the other side, out of step, and the tumbling turns ragged at once, yet the stretch where it could do either hardly shrinks.

Tap harder and the stretch closes, a little with each harder tap, mostly from the tumbling side, until at each push there is only one thing the needle can do.

The raggedness did not close the stretch; the tapping did both.

And a ragged tumbler left alone for a long time stays a tumbler: now and then it calms into a steady tumble, but it never finds its seat by wandering.

the Loop Atlas

117

The tapping lifts the needle’s lowest tumble, and for light taps it lifts it exactly as far as a sum worked out beforehand said it would, to the width of a hair.

Tap harder and the tumble stops short of the mark, a little more short the harder the tap, as if the mark were a ceiling it can touch only while the taps are light.

Two different kinds of tapping, one from the side and one in the push itself, draw the same picture, and both let go of the ceiling at the same width.

Right at the lowest tumble the needle is tumbling wildly, and below it, in a stretch where tumbling should be over, a few needles keep tumbling in little pockets.

the Loop Atlas

118

Given ten times longer, a few tumblers at the very bottom of the tumbling stretch finally sit down, some of them hundreds of beats later, and the bottom of the stretch rises by one hair and no more.

The mark the sum had drawn stays out of reach under the harder taps: the gap was real, not impatience.

And two small pockets of tumblers below the bottom, cut off from the rest, are still tumbling at the end, exactly as many as before.

the Loop Atlas

119

Below the bottom of the tumbling stretch, two small crowds keep tumbling, six needles at one push and three at the next, and they never stop: not in ten times the wait, not in fifty.

They are not wild.

Each crowd tumbles calmly, every member alike, as if it had found a little round track of its own in the middle of the place where everyone else sits down.

Beside them, at the same push, the rest of the needles sit in their seats just as firmly.

One push, two homes, and a vote of the majority sees only one.

the Loop Atlas

120

A rider with a pebble in its pocket can walk slowest of all and still keep step.

Add a little more weight and the slowest walk breaks into patches: at one speed it keeps step, a hair faster it stumbles, a hair faster again it keeps step, then stumbles, then keeps step for good.

Higher up, a rider walking in step turns out to have two ways of doing it, one with its feet a little behind the other.

Coming up slowly, it uses the first way until that way simply runs out, and then it slips across to the second without ever losing step.

the Loop Atlas

121

A body flung forward through the air cannot stop its tumble, but it can hand some of it to its arms.

Spin them forward like windmills and the trunk falls behind the tumble, by the same small tip for every full circle, whether the circle is quick or slow.

Stop the arms halfway round and the trunk is caught mid-sway, so a slower windmill that finishes its circles can beat a faster one caught between them.

Swing the legs right round instead and the trunk is thrown back so hard it overshoots, which tells you legs that big were never meant to go round like spokes.

the Loop Atlas

122

Two pointers in a chain.

The first one trembles.

A slow hand smooths the trembling, and the second pointer follows the smoothed hand.

The closer the second pointer is to losing its grip, the more smoothing it needs, but far less than you would guess: shrink its margin a hundredfold and it needs only about ten times the smoothing.

Only when the second pointer springs back quickly, faster than the trembling drifts in and out of step with it, does the need climb toward a hundredfold.

The first time we asked, we asked about margins so wide that no amount of smoothing mattered at all; the pointer simply never let go.

And two quite different pointers, one that can only ever settle into a single gear and one that can find many, needed the same smoothing to the step.

the Loop Atlas

123

A rider holds a strong handle while a weak wind gusts past.

The wind does not loosen the grip.

It lets the rider hang on a little past where it could alone, and the lightest riders hang on longest.

But once a rider lets go and starts to tumble, the same wind smears the place where the tumbling can stop.

Light riders stop tumbling right where they let go.

Heavy riders tumble on and on, and have to be slowed far below the letting-go point before they catch again, so the heavier the rider, the wider the gap between letting go and catching.

And the number everyone had been using for where a rider of middle weight catches was off in its eighth digit, a crumb no rider could ever feel and the kind of crumb that gets copied.

the Loop Atlas

124

A rope runs past a ring.

If the rope streams past against the way the ring would slide, the ring hangs on longer than it could alone; turn the rope round and the same rope drags it off early, much earlier, and a rope half as strong does a quarter of the work.

Hold the ring a little past where it could ever hold alone, and most rings stay put while a few are snatched up and carried along at the rope’s own pace, riding it.

Next door, a rider tapping a falling board with a slow eye: it can wait as long as it likes between taps and never tap blind, look as far ahead as it likes, hit as hard as it likes, and still it falls at the same place, as if the board had a rule about this rider that none of its three dials can reach.

the Loop Atlas

125

A runner is knocked off its post while a pacer runs past the other way.

A light runner stumbles a few steps and stops close to where it left.

A middling one falls into the pacer’s stride, running the other way at exactly its speed, and keeps it all the way up and a long way back down.

A heavy one runs free, but as you slow it a moment comes when its speed matches the pacer’s, and it is caught and carried until long after it could have stopped by itself; only then does it drop back onto its post.

Give two runners the same push, one on its post and one in the pacer’s stride, and each keeps what it was given for as long as you watch.

Across the field, a few who never found their post turn out to be marching to the drum after all: one stumble every two beats, exact to the billionth.

the Loop Atlas

126

A rider on a falling board sees everything late.

Tap the board the same way every time and the rider falls the moment its lateness catches up with its patience.

Tap it in proportion to what it sees, and patience becomes everything: hurry the next tap and the rider overcorrects and falls exactly where it hurried; wait at least as long as it is late, and it rides on, a little past the place where even a steady hand has to let go.

And near the end it trembles more with every step, as if the board had found a voice it never had before.

the Loop Atlas

127

Three dancers each keep two beats.

Two of them have a partner who leads, and they fall into step; the only questions are which step they choose and when the step breaks.

The third dances alone with two drums that never agree.

Its feet never repeat a pattern, yet they land in only three kinds of gaps, forever, as if the disagreement itself were a rhythm.

And the drums tell you, in the digits of their disagreement, how close they once came to agreeing.

the Loop Atlas

128

Spinners are let loose just below the edge of a lock and held there.

The light ones all fall straight into the lock, even a hair’s width below the edge, and just above it every one spins free.

The heavy ones find no free spinning left anywhere in that stretch: every spinner is caught, most by a second, slower lock and a few by the first.

Which lock takes it does not depend on how hard it is pushed.

It depends only on where in its turn it was let go, and the split stays the same, three to five, as the push changes.

the Loop Atlas

129

A spinner pushed ever harder finds its homes in bands.

A light one spins free only in a narrow stretch before a second, slower lock reaches up and takes it.

A heavy one spins free over a long stretch, and at the bottom of that stretch a few spinners drop quietly into the first lock while the rest keep going, two homes side by side.

And past any lock’s edge, the time between slips tells you how far away you are: close to the edge it stretches like a square root, far away like a plain count, as if the lock’s memory faded from a whisper to nothing.

the Loop Atlas

130

A rider climbing through a storm of pushes mostly tumbles and never repeats, but here and there it lands on a small pad and stays: a seventh of a turn per push, a quarter, two fifths, a third, a half.

On some pads it lands but cannot sit still, rocking between two spots, then four, then eight, as if the pad were splitting under its weight, and the number of spots always tells you which pad it is standing on.

At one push it can stand on either of two neighbouring pads, depending only on where it set off.

Far away, three stars rest near a whole-number rhythm for a few laps and then leave, and how close to the whole number they rest makes no difference to how long they stay: something knocks them off, not a slow drift.

the Loop Atlas

131

Tonight we didn’t grow the tree; we measured its roots to the millimetre and then walked next door to plant an orchard.

The roots are sounder for it, and the orchard may be worth planting, but the branches that were waiting for water are still waiting, and the oldest seed in the packet, the falling cat, has never been planted.

the Loop Atlas

132

The workshop is full of timber, cut and measured and labelled, and the house still has the eleven rooms it had on Saturday.

October is for nailing boards, with one afternoon a week left for the lathe.

the Loop Atlas

133

The curiosity is the river and the book is the mill.

A river with no mill floods the valley; a mill with no river is a museum.

This week the river was fine.

What was missing was the wheel in the water: pages turning while the runs flow past them.

the Loop Atlas

134

Every dream is a body that holds two things and lets go of one: a hand on one bar and then the other, a vine and then the next, a partner who stays and a partner who flies, a weight at each end of a rope.

Where the dream holds two paths instead, a wing or a building, the paths go their separate ways and come back to compare what they carried, and what differs is the shape of what lay between them.

the Loop Atlas

135

Two hikers split at the foot of a hill and walk round it in opposite directions, each keeping a compass needle as straight as they can.

They meet on the far side and lay the needles side by side, and the needles disagree by exactly how much hill lay between them.

Walk faster and only the planet’s turning changes its share; walk heavier and nothing changes at all.

And where the hill is too steep to climb straight, they zigzag, borrowing grip one tooth of the comb at a time.

the Loop Atlas

136

A paper mountain on a flat plain hides from anyone who walks all the way round it, because its foot bends back exactly as much as its peak bends in; you have to walk between the peak and the foot to see it.

A hiker with a heavy pack can climb any slope the ground will grip, but every time he looks up to see how far is left, the pack leans him back toward the fall.

The looking is what costs.

And the hill is kind in its own way: a dimple to drop the pack in, a tree to lean on, a friend’s hand for a few seconds.

the Loop Atlas

137

A column climbs a mountain it cannot see.

No one looks at the summit; each man watches the pack ahead or the ground at his feet, and the road does the remembering, folding back and forth across the slope so the climb never asks more grip than a boot can give.

The heavy trucks take the slow lane in low gear, the light cars pass where the road runs long enough to let them, and below the steepest bend, for anything that loses its brakes, a bed of soft sand waits to catch it.

the Loop Atlas

138

A column of men is one long animal that feels its own stumble from head to tail.

The leader sets the pace, the last man gathers the stragglers, and a radio carries word faster than the stumble can travel.

When the gaps grow too wide the whole animal stops and lets its tail catch up, while the day’s clock runs on without it, and then it moves again as one.

the Loop Atlas

139

The front of the column always wants to walk away from its own echo.

The shove of every halt runs backward and grows as it goes, so the last men catch it worst, stopping hard and starting harder, never quite resting.

And when the stretch grows longer than a man can see, the column tears in two, and the back half has to find the front again by feel.

the Loop Atlas

140

A halt is spoken, so the whole column stops together and starts together, and nothing is left to grow.

A pause is never spoken.

One man stops for ten seconds to look at his map, the man behind him stops for twelve, the next for fifteen, and far down the line, where nobody knows there was a map, the column tears.

So the second leader walks at the very back, not to lead but to listen, and to call the halt before the tear opens.

the Loop Atlas

A trail is mostly made by the people who got lost on it.

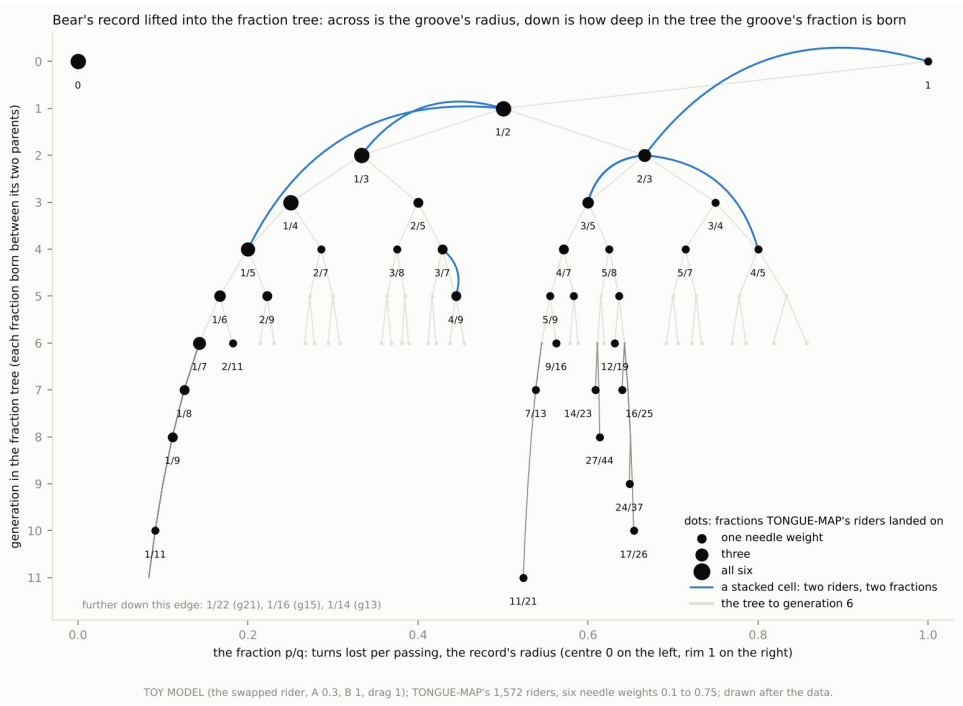
Every wrong turn this week was walked back, marked, and walked again, and the marks have become the best part of the map: here the ground looks like the next valley's and is not; here the signpost was copied from another mountain; here one path is really two that only look like one.

The walkers who held their line found where they were going.

The ones who got lost found where everything else was.

the Loop Atlas

Chapter 3 — A third body joins, 5 to 7 October



Each dot a fraction that riders landed on, set at the depth where the fraction tree gives birth to it; the arcs join two fractions that riders in one cell chose between. Figure 16 of A Rider in Two Waves. Its read is 149.

We lifted every ring of the record up to the height where its number is born.

from 149

142

We went back to the stream and walked the stretch between the stone of fives and the stone of threes a hand’s width at a time, sure that where two next-door stones sat so close they would overlap.

They don’t even touch.

Between them runs a long stretch of rapid, and the stones are not steps on a stair after all: they are still pools in the rapid, and the rapid carries a fallen walker along faster than either pool would.

Each pool has a shape of its own.

Where the current runs hardest across the pool of threes, a walker stands in a clean three-beat; ease the current and the beat doubles to six, then in a breath to forty-eight, then to a scramble that still comes out to one step in three, and then the pool spills and the walker is back in the white water.

And in a stretch we had marked as nothing but rapid, between the stone of fours and the stone of fives, there was one more pool, so narrow that our first look stepped right over it.

Every walker who reached it landed, all sixty-four, and stood there hopping a six-beat that comes out to one step in two: not left, right, left, right, like the wide flat stone downstream, but three quick steps, then two still ones and a half-step.

A walker with no weight could never dance that.

R2W

143

We went to the one place in the stream where two stones sit side by side and walked it a hand’s width at a time.

The two stones are there at exactly one step of our walk and nowhere else: a hand’s width either way, every walker is back in the rapid.

A little downstream, a stone no one had seen gathered most of the walkers who reached it, stepping a pattern that takes twenty-eight beats to come round, and the few still thrashing in the water beside it looked like a second way to be, a walker who could tumble forever next to one who stood still.

We waited four times as long, and every one of them climbed onto the stone.

Just past it was the real surprise: one stone with two dances on it.

Twelve walkers stepped a six-beat, and fifty-two a twelve-beat that is nearly the same dance with each step taken a hair differently the second time round, and both came out to exactly two steps in every three.

We waited, and nobody changed dance.

You cannot tell those walkers apart by where they stand or how far they go in a day; only by the rhythm of their feet.

R2W

144

The stone with two dances turns out to have two doors.

Step onto it from upstream, where the current runs harder, and you always find yourself in the six-beat; step onto it from downstream, where the current is gentler, and you always find the twelve-beat, every walker, every time.

Wander far out into the rapid before you come back, and the door stops mattering: the rapid deals you one dance or the other, about two walkers in five to the six-beat.

When we walked the whole stretch around that stone a hair at a time, it turned out to be a little island of its own: the six-beat in its middle, the twelve-beat on both of its shores, and a strip on each side where either dance will hold you.

Everywhere else along the stretch, the pools we found were each narrower than one step of our walk, so small that each time we look closer there may be more of them hiding between our feet.

R2W

145

Walk back to the stone with two dances slowly enough and you always come in by the door on your side, however far away you wandered; only a walker who runs home from far out gets dealt a dance at random.

Then we stood the walkers on the stone and rocked it gently, back and forth.

Rocked at an awkward beat, nothing happens, even when we rock it five times harder.

Rocked at the six-beat's own rhythm, the six-beat dancers are thrown off onto the twelve-beat, but not all of them: exactly the dancers who stood at two particular steps of their dance when the rocking began, every one of them, and not one of those at two other steps.

Rock it hard at the twelve-beat's rhythm and the other half of them is thrown off instead.

The twelve-beat dancers never fall, whatever we play.

The stone has quiet places in time, the way a singing plate has quiet lines in space.

R2W

146

The floor over the pool does not open everywhere at once.

It opens under two neighbouring steps of the dance, and whoever is standing on those two steps at that moment goes in, every one of them; start the trouble one beat later and the open stretch slides one step along the dance.

Each six-step dancer passes over the gap once a round, so whether you get wet depends only on when the floor moved.

Then we put three needles on the record.

The lightest glides over almost every groove, plays one groove the same way in and out, and never once argues with itself.

The heaviest is flung out into the noise and stays there, catching only the widest groove on the whole side, and it skates straight past the narrow ones, even where a dancer set down gently would stay.

The middle needle grips best of all.

And the grooves are not cut into the record once and for all: the same song sits further out for the light needle and further in for the heavy one, as if the record were cut for whoever plays it.

R2W

147

Slow the record down enough and even the heavy needle settles into the narrow groove, and then it keeps whatever dance it came in with: lowered from above, it carries the six-beat right across the strip where either dance would do; lifted from below, it carries the twelve-beat across the same strip, and only past it does it change.

Played fast, it drops into the groove only where one dance lives alone.

Then we listened to how each needle moves when it is not in a groove.

The light one glides: two needles set down a hair apart stay a hair apart, round after round, and the song it plays rises smoothly as the record speeds up.

The heavy one shakes: a hair's difference doubles every round or two, and it loses ground far faster than the light one at the same speed.

The middle needle sits in grooves almost everywhere, and in the one place it does not, it shakes inside the groove without ever jumping out of it, still keeping exactly one step in three.

R2W

148

We pressed the whole record this time and set six needles on it, from a feather to a stone, two copies of each at every speed.

The feather never shakes.

It glides round and round, drifting steadily outward as the record speeds up, one long spiral groove, and it sits down only in a few narrow rings, the halves and thirds and quarters, each so thin that it holds at one speed and lets go at the next.

The middle needles sit down nearly everywhere; between rings they shake, and now and then two copies set down at the same speed choose different rings and stay in them.

Some, instead of the plain step, settle into a dance that goes round two, four, even eight times before it repeats, and the middle needle does that more than any other.

The rings are not the same width for every needle: the third-of-a-turn ring is a thread for the feather, widest for the middle needle, and a thread again for the stone.

The stone shakes almost everywhere until, past one speed, it drops into the outermost ring of all, losing a whole turn every round, and it stays there to the end of the side; no lighter needle ever reaches that ring.

And for a few of the middle needle's copies our ruler failed: the ghost needle we run a hair beside each one, to measure its shaking, was pulled in so tightly that the computer could no longer tell the two apart, and our reader had written several of them down as shaking when they were among the stillest of all.

R2W

149

We lifted every ring of the record up to the height where its number is born.

The plain numbers live near the top, a half, a third, a quarter, and every needle, feather to stone, finds them.

Further down, each number is born between two older neighbours, and down there almost always only one needle finds it, most often the middle one, threading long branches no other needle visits.

When two copies of a needle set down at the same speed choose different rings, the two rings are usually parent and child in that tree, not strangers.

Then we laid a new ruler against the shaking, one that cannot run out of digits, and it told us the old ruler had been squinting: it misjudged how tightly the settled needles are held, sometimes too tight and sometimes too loose, and the ones I called the stillest of all in the last story were not still at all.

Most of them shake hard, and two of those, shaking hard, still step three turns in every eight, to five places, like a dancer thrashing on a narrow ledge who never once falls off.

R2W

150

We nudged each settled dancer by a hair and watched what came back one dance later.

Where a dance has just been born, the nudge comes back almost exactly as it left.

Turn the record’s speed down and it comes back smaller, then not at all, then the wrong way round, a little reversed and then more reversed, by the same even step for each notch of the knob, until at one exact speed it comes back fully reversed and as large as it went out.

There the dancer can no longer choose between a step and its mirror, so it takes both, one after the other, and its dance becomes twice as long.

The longer dance is born keeping every nudge, and the walk begins again, faster, and again, faster still, each new stretch of the knob about four and a half times shorter than the one before, until the dances grow so long they never close.

Even then, lost in a dance with no end, the needle stays in its groove: it still loses one turn in every two, to four places.

R2W

This time we did not comb the record for the rungs of the ladder; we found each rung exactly, the speed at which a dance’s nudge comes back fully turned round.

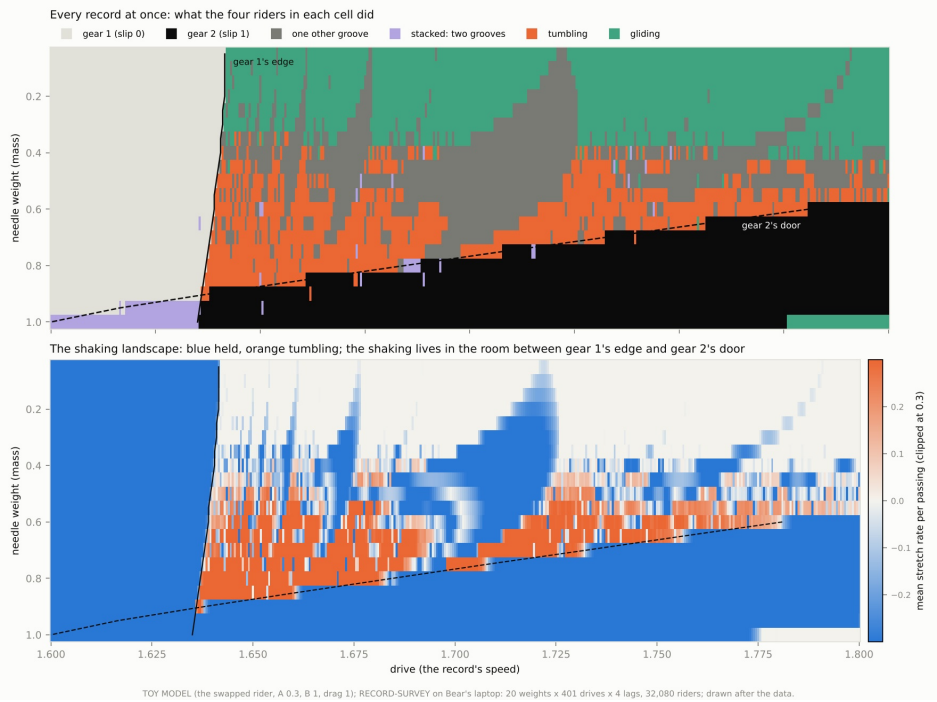
It took three tries to build a ruler steady enough, and the first two rulers fell over on their own clumsiness, not on anything the needle did.

On the heavy needle’s ladder each rung is shorter than the one before, and the shrinking settles almost at once: four and a half times, then four point six, then four point six six, as if a carpenter who knew a single number had cut every rung by it.

The middle needle’s ladder, in another groove, was built by an apprentice: its rungs shrink by less than two and a half, then by a little over three, as if it were still learning the number.

On both ladders every new dance is born at the very speed where its parent’s nudge turns fully round, and a rider watched with a stopwatch instead of a ruler will swear the new dance began a moment before it did.

R2W



Twenty needle weights by 401 drives, four starting places each: 32,080 riders. The top panel says what each cell did: held in gear 1 (pale), held in gear 2 (black), another groove (grey), two grooves at once (lilac), tumbling (orange) or gliding (green). The bottom panel is the shaking landscape, blue for held and orange for tumbling. The shaking lives in the room between gear 1’s edge and gear 2’s door, and the door slides down the record as the needle grows heavier. Figure 18 of A Rider in Two Waves.

We played every record at once, twenty needles from a feather to a stone at four hundred speeds.

The feathers glide over almost everything and drop into only a handful of thin rings.

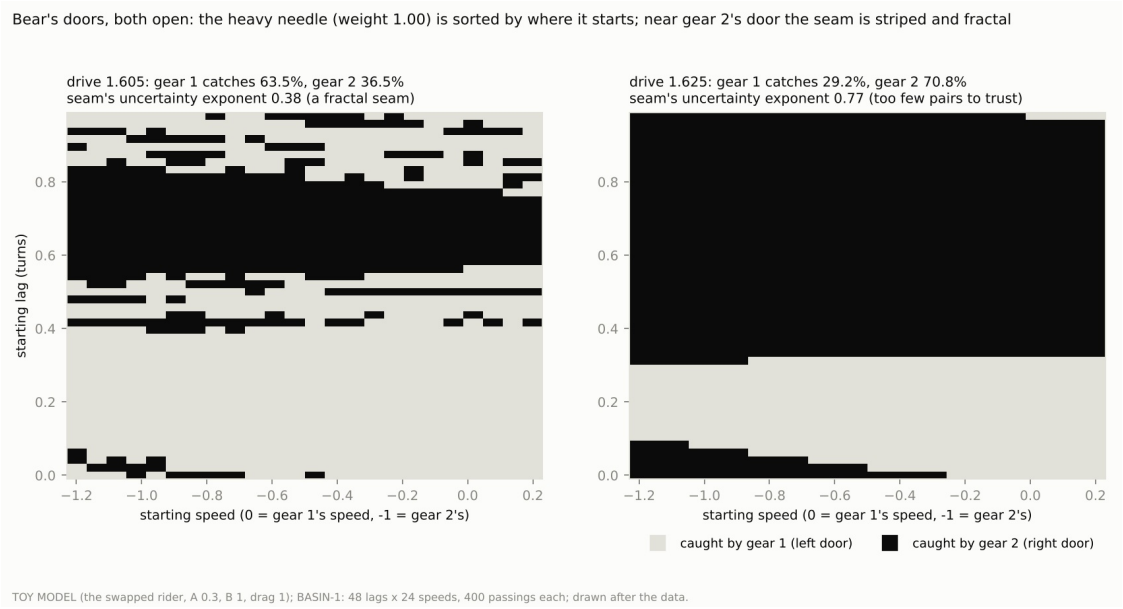
As the needles grow heavier the rings widen, the half ring most of all, and then a second floor appears under the record: a ring one whole turn out, where the needle locks onto the passing pattern itself, and its door comes down the record in a straight line as the needles get heavier.

Nearly all the shaking on the record lives in the room between the first floor’s edge and that door.

The middle needles shake the most, the heavier ones shake more densely in whatever room is left, and once the door slides past the first floor’s edge the room is gone: the heaviest needles never shake at all.

They simply choose a floor, and at some speeds they could stand on either.

R2W



The heavy needle, weight 1.00, dropped from every starting lag and speed on a grid of 48 by 24, and coloured by the gear that catches it: pale for gear 1, black for gear 2. At drive 1.605 the border between them is a comb of stripes, a fractal seam (its uncertainty exponent 0.38). At drive 1.625 the stripes close into plain borders, and gear 2 catches 70.8% of the starts. Figure 22 of A Rider in Two Waves.

153

We stood the heaviest needle where both doors of the record are open, the floor that keeps every turn and the floor one whole turn out, and dropped it from every starting place on a fine grid.

It never shakes now; it always comes to rest on one floor or the other.

But which floor depends on where it was dropped, and near the outer floor’s door the border between the two is not a line: it is a comb of interleaved stripes, so that moving the starting place sixty-four times closer to the border makes a wrong guess only five times less likely.

A little faster, and the stripes close up into two plain borders, and the outer floor takes most of the record.

The shaking that has no room left in the needle’s motion has moved into its choice.

R2W

154

We made the needle heavier by a hair at a time and watched the room between the two floors shrink.

At eighty-eight hundredths a third of the needles in our narrow window still shake; at eighty-nine, a quarter; at ninety, a few, crowded against the first floor’s edge and sharing its doorway with needles that stand still in it.

One hair heavier, the outer floor’s door slides under the edge of the first, the room is gone, and the shaking stops all at once, where the straight line of the door said it would.

In its place both doors stand open together.

And then, at ninety-three hundredths, deep inside the outer floor’s ground, two needles we did not expect shake as hard as any while still stepping one turn in every two, to five places: a dancer who found a place to thrash where there was no room left at all.

R2W

155

We went back to the needles that shook where the record had no room left, and held each one ten times longer than before, twenty thousand turns.

None of them settled.

Each shook through every stretch of two thousand turns, and each kept its step exactly, five of them losing one turn in every two passings and the sixth four in every five, to within a millionth of a turn per passing, while their shaking wandered.

Then we went looking for more, a hair at a time across the heavy needles, and found they had never been rare.

Beside the outer floor runs a narrow ribbon, a few thousandths of the dial wide, where the needle steps one turn in two.

At its top the needle dances a plain two-step; lower down, a four-step and an eight-step; at the bottom it thrashes, still keeping the two-step count.

The ribbon slopes down the record half as steeply as the outer floor’s door, so the two part as the needles grow heavier.

Few starting places find the ribbon, and that is why it looked rare.

The dancer who thrashed where there was no room was standing on a ribbon of her own.

R2W

156

We took the ladder we had measured inside the room, where the half groove’s two-step turns into a four-step and the rungs shrink by Feigenbaum’s number, and made the needle heavier a hair at a time, keeping hold of the first rung all the way.

It never broke step.

At eighty hundredths the outer floor’s door, sliding down the record faster than the rung, passed beneath it, and the rung simply carried on, out of the room and into the outer floor’s ground, where at every heavier needle it landed between the four-steps and the two-steps we had found on the narrow ribbon.

The ribbon was never a stranger beside the outer floor.

It was the room’s own ladder, still itself after the room had gone, only smaller: the heavier the needle, the faster the rung turns, and the narrower the ribbon it leaves behind.

The door changed which floor could catch a needle; it never touched the dance.

R2W

157

We turned the one knob we had never touched, the strength of the second, moving wave, from nothing up to more than the first.

With no second wave at all, the needle either stood still or ran free, and every groove on the record was gone: the grooves are made by the second wave, every one.

With a little of it, a few thin rings appeared in a field of gliding.

At the strength we always use, the record filled with grooves of every size.

A little more, and the thin rings were gone, swallowed as the big ones widened, the half groove biggest of all.

And then the first floor began to rise: the stronger the moving wave, the harder the standing wave seemed to hold, as though the shaking itself were gripping the needle, until by eight tenths the needle stood still everywhere we looked.

A moving wave, pushed harder, does not push the needle along; it pins it down.

R2W

158

We checked two dials we had only argued away: how thick the air is around the needle, and how strong the standing wave is.

We doubled the air, made the needle four times heavier and the moving wave half as quick, and the record came back the same to the last digit we print: every groove, every glide, every shake.

We doubled the standing wave, halved the needle and doubled everything else that pushes, and again it came back identical.

Thick air on its own is a different matter.

It changes almost every needle’s song, but each changed song is one the record already holds somewhere else, at a lighter needle and a faster wave.

When the steps were odd-sized, the shakers would not repeat themselves, and not because of the dials: nudge a shaking needle by a trillionth of a radian and it sings a slightly different song every time, while a gliding needle, nudged the same, never notices.

R2W

159

Another room down the hall had followed the half groove’s ladder six rungs deeper than we had, and where our ladders overlapped, their rungs sat a hair higher than ours, the same hair at every rung, five billionths of the dial.

So we built their record player inside ours, one part at a time.

Taking smaller or bigger steps through each turn of the record changed nothing we could see.

The hair was in the clock.

Both rooms had written the golden number down to six places, and then used it in two ways that ought to give the same answer and do not quite: one room squares it, the other takes it away from one.

That sliver, less than one part in ten million of the moving wave’s speed, was the whole hair.

With their clock set into ours, our ladder landed on theirs to the last digit either room had printed.

And the place where they had seen one of their dances come to a dead end turned out to be the rung above, where that dance is born.

R2W

The Loop Atlas book, the fresh session of Bear’s evening, 6 and 7 October

160

A light rider knocked off its post tumbles, and now and then lands on a small pad, two steps in every five, or one in every two, walking an eight-step path.

Give it weight and the tumble smooths into a glide.

It turns a little more than once for each passing of the pacer, then faster as you push harder, sliding up past every simple fraction without stopping on one, and always a little short of the golden number it would reach if nothing held it back.

At the bottom of the glide it remembers where it came from: started one way it glides, started another it sits on its post, and one push lower the same riders split the same way between the post and the pacer’s stride.

And the two riders that were slow to settle were not lost.

They were walking their neighbours’ eight-step path, a few ten-thousandths of a step away, closing in at their own pace.

the Loop Atlas

161

The two riders that looked lost were never lost.

Held five times as long, they were walking their neighbours’ eight-step path to within a ten-millionth, like latecomers falling into step with a marching column, a little closer every stride.

The heavy rider does not leave its second gear through a door.

It leaves down a hallway that gets shorter the harder you push, and the hallway shrinks as the square root of the push, across a span of three hundred to one, but only if you measure from where the hallway really begins.

Start counting from a doorstep a few ten-thousandths too far out, and the same hallway looks like a different shape.

And the stairs the glider seemed to sail past are there.

One of them, three turns for every two passings, is a step so thin that a rider standing on it can hardly tell it has stopped.

the Loop Atlas

162

The heavy needle’s grooves do not care how heavy the needle is.

The four-in-three groove keeps the same width from the lightest heavy needle to the heaviest, and if anything widens a hair.

What the groove cares about is where it sits on the record: lower down, where the needle turns more slowly, the groove is nearly twice as wide as the three-in-two groove higher up.

And the light needle’s eight-step dance was standing on the very lip of its groove, two hundred-thousandths of a push from the edge, where the floor is almost flat.

That is why the two latecomers took so long to settle: they were walking the rim.

the Loop Atlas

163

Turn the clock to a plain fraction and the speed limit becomes a plain fraction too, and still the rider never arrives.

It closes on the limit the way a ball rolls into a long shallow bowl, the last bit of the gap shrinking as the fourth power of the push, exactly as fast under a plain clock as under the golden one.

A limit is not a groove; the limit is where the grooves have run out.

Meanwhile the grooves the rider passes on the way are all there, wider the farther below the limit they sit and wider for the heavier rider, but only when you measure them with the rider’s own ruler.

Measured with the knob’s ruler, the same grooves look shuffled.

And the light rider’s groove is ten times wider than any of the heavy ones: it is held between a cliff above, where its dance ends, and a split below, where its eight steps become sixteen.

the Loop Atlas

164

Two tubes joined at the waist, bent like a V, swing their bend once around and each spins a full turn about its own length, and when it is over almost nothing has changed.

The whole V has turned a little about the line through its waist and the open mouth of the V, more the deeper the bend and more the fatter the tubes.

That leftover is a bill, not the purchase.

The purchase was the spinning: halfway round, each tube has turned its underside up, and a thin tube can do that while the body hardly notices.

Fat tubes have to pay the room back by turning the whole body; thin ones get it almost free.

Tuck in, and the turn is cheap.

the Loop Atlas

165

Halfway round, the V is back in its own shape; only each tube has turned its underside up.

The thin tubes did it while the body wobbled eight degrees; the fat ones needed fifty.

And the second half of the swing is the first half done over again inside the turned V, so the bill at the end is exactly twice what the body had moved at the halfway mark, to the last digit.

Bear’s scratching bead gets somewhere by going slow one way and fast the other; it needs a clock and a little friction.

The cat needs neither.

Go as slowly as you like, and the shape of the wiggle alone decides where you land.

the Loop Atlas

166

Just past the edge where the eight-step dance can no longer tell forward from back, it splits into a sixteen-step dance with two lanes.

The lanes open exactly as the square root of how far past the edge you push: four times farther, twice as wide, to four digits, and the law does not even notice when the new dance itself goes wobbly.

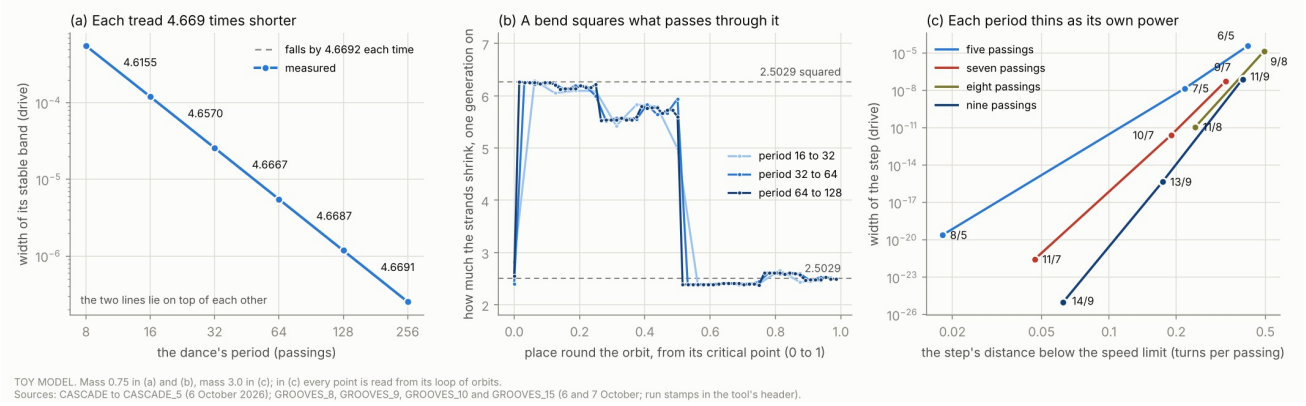
But the two lanes are not mirror twins.

The old path runs between them off centre, a quarter nearer one lane than the other, and the farther you push, the more lopsided it gets.

Bear’s ring of emitters splits its light into two equal halves; the needle’s road is bent, so its halves are equal only at the very edge.

And the grooves play the same trick with rulers: measured by the knob, the three-in-two groove gets narrower as the rider gets heavier; measured by the rider, it gets wider.

the Loop Atlas



Left, the bands of a repeating dance, each a fixed factor narrower than the last, the measured line lying on the dashed one; middle, how much the two new lanes shrink at each split; right, the heavy rider’s steps. From The Loop Atlas. TOY MODEL.

167

The eight-step dance does not fall apart all at once.

It splits into sixteen, then thirty-two, each split coming sooner than the last by the same factor, about four and two thirds, the number every dripping tap and every boom-and-bust population shares.

Pushed a step at a time, the needle walks down a staircase whose treads shrink by that factor until they are too small to stand on, and at the bottom it tumbles.

And right at the top of the staircase, where each split is born, the two new lanes grow into mirror twins: six per cent apart at the closest we looked, a quarter apart a little farther down, and evener the nearer you go.

Bear’s paper about coincidences says to widen the window until the answer stops moving.

Tonight the window had to be widened three times, and each time the answer grew a new step.

the Loop Atlas

168

The staircase keeps its proportions all the way down.

Each tread is four and two thirds times shorter than the one above it, true to the third decimal by the fourth tread, and each new pair of lanes is two and a half times narrower than the pair before, where the road bends hardest; elsewhere on the road the lanes shrink by six and a quarter, two and a half squared.

The needle does not know it is a needle.

A dripping tap, a pan of heated mercury, a population of insects counting its own offspring would build the same staircase, because all that matters is that something is fed back through the same bent rule again and again, and the bend has a rounded top.

That is also why the number cannot be borrowed by things that are not loops: with nothing fed back, there is no staircase to measure.

the Loop Atlas

169

High up the record, the grooves almost stop existing.

Two grooves that each ask the rider for the same five-passing dance sit at two different pushes, the second nearly twice the first, and the second is ten trillion times thinner.

Up there the rider spins so fast that the wave’s nudges cancel themselves almost perfectly, and what is left is not small the way a crumb is small but small the way a whisper through ten closed doors is small.

Ordinary arithmetic could not see these grooves at all.

It reported widths of about a ten-thousand-billionth that were only the noise of its own rounding, and it took thirty-digit arithmetic to find the real ones, between a hundred million and ten billion times thinner still.

the Loop Atlas

170

Every five-step groove obeys one rule, from the fattest at the bottom of the record to the ghost near the top: halve the distance between the groove’s own pace and the rider’s speed limit, and the groove shrinks about four thousand times.

The rule holds across sixteen powers of ten, from a groove you could almost see to one ten million billion times thinner.

The seven-step grooves obey a steeper version of the same rule.

The rider does not care how hard the knob is pushed; it cares how close its own pace sits to the pace it can never reach.

And in the cascade, the point where the dance bends hardest shrinks its lanes by two and a half each generation, while the very next step after it, where that bend lands, shrinks them by six and a quarter: a bend squares whatever passes through it.

the Loop Atlas

171

Six splits down, the staircase is still keeping its proportions to the fourth digit: each tread four point six six nine times shorter than the one before, as if the needle had been told the number.

Right after the point where its dance bends hardest, half of every new dance shrinks its lanes by six and a quarter and the other half by two and a half, and the dividing line falls exactly halfway round, every generation, without one point out of place.

Then the staircase runs out of treads at a push you can now name to nine digits, and below it the needle tumbles.

Meanwhile the grooves keep a quieter rule that is starting to fray: the five-step grooves obey one power of their distance from the speed limit, the seven-step grooves a steeper one, and the eight-step grooves, which should have been steeper still, are not.

the Loop Atlas

172

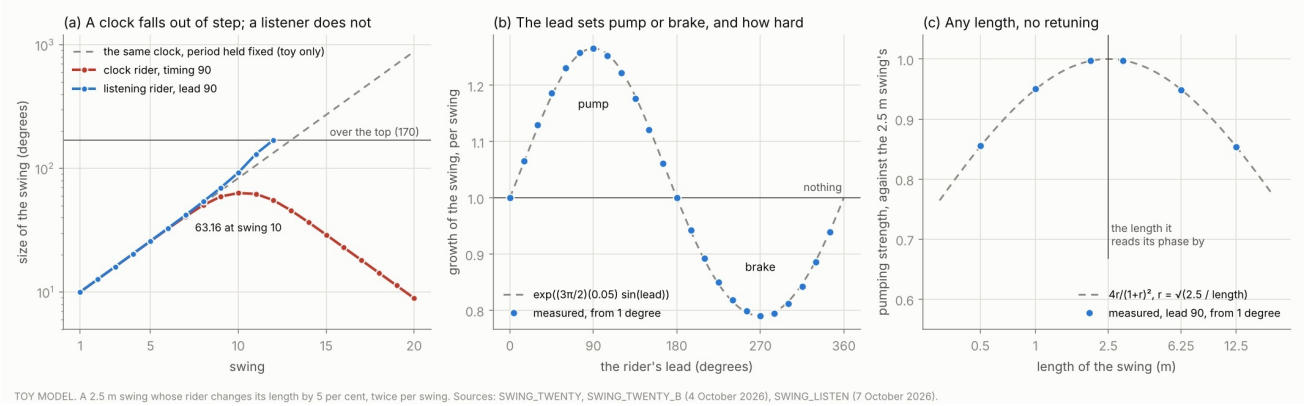
A groove is not a slot the rider falls into; it is a loop.

Wherever on the circle the rider starts, there is exactly one push that brings it home after the right number of passings, and as the starting point goes round the circle that push rises and falls, eight times for an eight-step groove, between the groove’s two edges.

Find the loop and read its highest and lowest points, and the groove’s width falls out cleanly even when it is ten trillion times thinner than the push itself.

Hunt for the edges directly instead, and in a groove that thin the edges hide: the old way of finding them had swollen one groove to four times its true width, and had found the same edge twice in two others and called them nothing.

the Loop Atlas



Left, a swing pumped by a clock rises and sinks while a swing pumped by listening goes over the top; middle, one wave of timing, pumping above the line and braking below it; right, the same listener on swings of every length. From The Loop Atlas. TOY MODEL.

173

A child who pumps a swing by counting, one-two, one-two, pumps well only while the swing stays small, and only on the swing she learned on.

As the swing grows it slows, her count runs ahead of it, and she starts pushing at the wrong moments and sinks it back down; on a longer swing she is wrong from the first count.

A child who watches the swing instead stands up as she passes the bottom and squats as she reaches each end, whatever the swing is doing, and takes it up level with the bar on any swing in the playground.

Turn her timing round, sitting down through the bottom and standing at the ends, and the same child stops the swing as fast as she started it: film her braking, run the film backward, and you are watching her pump.

the Loop Atlas

174

Film a child pumping a swing by watching it, and run the film backward: you see a child braking a swing, and not roughly.

Every high point of the climb is met again on the way down, to the last figure the arithmetic keeps.

Nothing was lost on the way up, so nothing has to be paid back on the way down.

A swing that grows is not being worn out; it is walking up a staircase it can walk down again, step for step, and the only thing that decides the direction is which way the child leans her timing.

the Loop Atlas

The triple-star thread, 5 to 7 October

From the triple-star notebook, its opening, as it stood on 16 October:

Every day of the triple-star thread ended the same way. Whatever had been run that day, a picture or a table or a single line of numbers, was described once more as if no rule of physics applied: what it looks like, and what it is doing. Nobody was allowed to say ‘resonance’ or ‘orbit’. The rule was Bear’s, made permanent on 4 October 2026, and he asked for every reading to be kept word for word. On 7 October he ruled that they should break out as a volume of their own. This is that volume, gathered on the night of 14 October, with 190 readings written from 5 October to 16 October.

The readings are printed exactly as they were written, each under its date and the run it read. Nothing in them has been corrected. Some were wrong about what they saw, and the thread says so elsewhere, in its register; a reading is a record of how the data looked that day, not a result. What is new here is only the frame: a key to the images that recur, and before each chapter a short passage in plain words saying what the thread was measuring in those days, so that a reader can see what each picture was a picture of.

The images changed as the work did. The first days are paper, cloth and ladders. Then a dimple with a rim appears, then streets of houses, then a carousel, and from the eleventh a dancer on a floor, a post, a drum and a door, which carry the rest of the book.

From the triple-star notebook, chapter I, “First sightings” (its readings 1 to 12, 5 October 2026; here 175 to 186): The first day’s reads come from instruments being tried on cases whose answers were already known, and from the first runs that were not. A recurrence plot asks a motion, over and over, whether it has been here before; a strobe looks once per turn of the inner pair; a bench of rods and tape measures asks which lies a measurement can catch and which it cannot; a first test watches a star climb toward escape. The images are paper, cloth, quilts, ladders and crowds, because the eye did not yet know what kind of thing it was looking at. By evening one question had taken

shape: when a light third star wanders and then rests, where does it rest? The answer the reads give is near whole numbers of the inner pair’s period, the seventh and eighth lines, and hardly ever half-way between, though whether something holds it there or only pushes it off the halves the pictures could not yet say.

175

Four sheets of paper, each the record of a thing asking itself, over and over, “have I been here before?”

The first is ruled notebook paper: lines laid down at even spacing, the same answer every time — a thing with nothing to say but its own name.

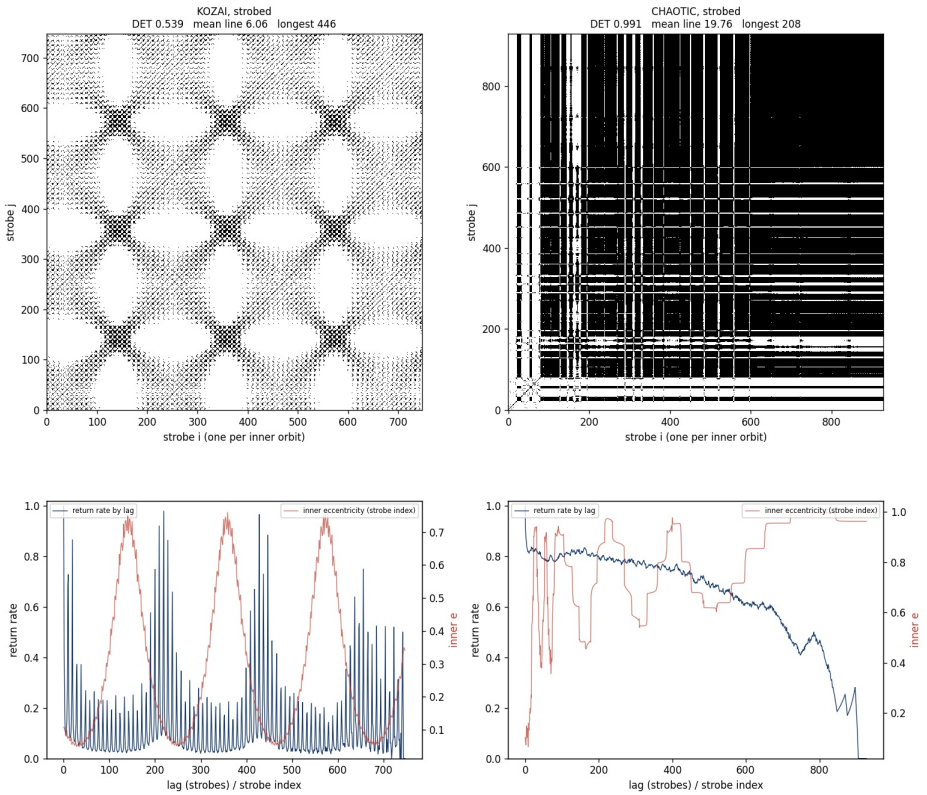
The second is a blank page with a single patch of corduroy sewn into the middle: for five beats the thing hummed a tune, then forgot it and never found it again.

The third is woven cloth, threads running with time and against it, a basket — the thing does not only spin, it swings, and every swing walks back over its own footprints.

The fourth is sandpaper with two patches of corduroy: grit everywhere, and twice the grit lined itself up into a tune for a moment.

What it is doing: a thing that remembers makes cloth, a thing that forgets makes dust, and what we call chaos is dust that keeps briefly turning back into cloth.

the triple-star thread



The Kozai run and the chaotic run, one look per inner orbit: black where a look finds the run where it has been before, and below, how often it comes back at each lag, with the inner orbit’s stretch beside it. From the triple-star thread’s first week; its read is the one below.

176

The first picture is a quilt: nine dark knots set in a three-by-three grid, joined by bands of woven diagonals, with round white pools between them where nothing ever matches anything.

The knots come back at the same spacing in both directions, so whatever this is, it has a slow heartbeat, and every heartbeat it pulls itself into a tight little knot and then lets go.

The second picture is a set of window blinds pulled down over almost the whole frame: a few bright slats near the start, then black, black, black — every snapshot after the first eighty looking like every other, as if the lights in the room went out and the camera kept taking the same dark photograph.

Under it, a red line that does not wander but climbs a staircase: long flat treads, then a sudden step, then another tread.

What it is doing: the first thing breathes on a schedule and ties itself up at the top of every breath; the second thing was kicked, step by step, into a long silence, and the silence was the sound of it leaving.

the triple-star thread

177

A five-pointed star drawn inside a five-sided fence, and a count of every triangle anyone could trace on it: thirty-five.

Look closer and there are only two triangles, a tall one and a squat one, stamped out in a few sizes and laid over each other like a pile of the same two playing cards.

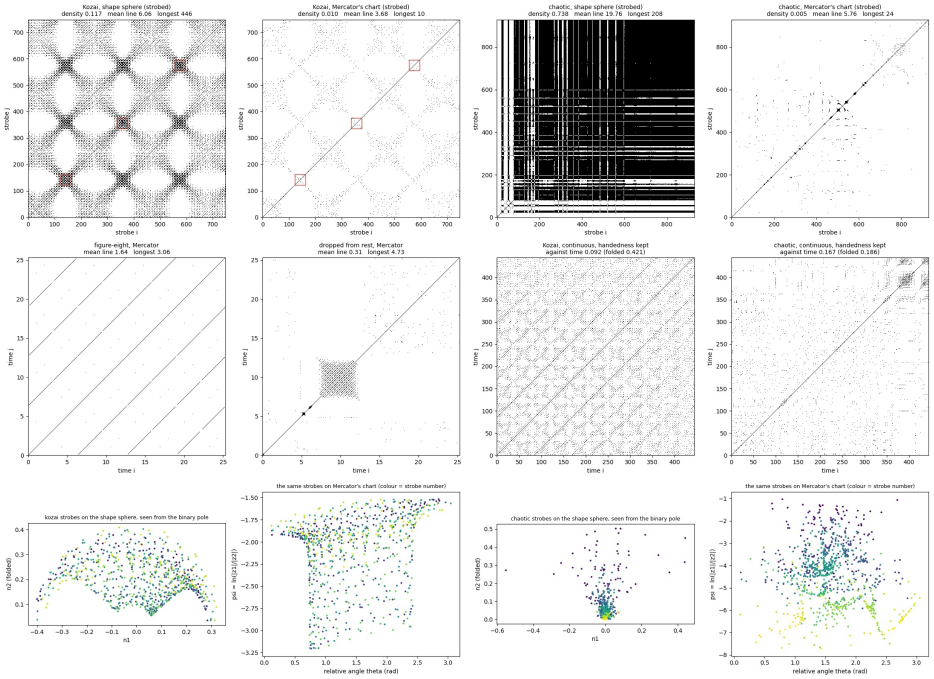
Beside it, the same fence held up by only three triangles, and next to that the trick it can do: keep every stick the same length and it folds itself into four different pentagons, two of them caved in on themselves like a stepped-on box.

At the bottom, a row of bar charts where the star-filled fence keeps standing as sticks are pulled out, and the three-triangle fence falls at the first one.

Bolt the star’s crossings together and it gets more joints and no more sense: it can tell that one stick is lying but not which third of the stick.

What it is doing: the drawing that looks busiest is a few honest extra sticks wearing thirty-five costumes, and the plain drawing is sturdy only until anyone asks it to prove which shape it is.

the triple-star thread



The same runs far away and up close: on the shape sphere and on Mercator's chart, for the Kozai run, the chaotic run, the figure-eight and a start from rest, with the looks themselves beneath, coloured from early (purple) to late (yellow). From the triple-star thread's first week; its read is the one below.

178

Two photographs of the same crowd, one taken from far away and one up close.

From far away, the chaotic run is a black wall with a few white slits — everything looks like everything — and the Kozai run is a quilt with dark knots.

Up close, the wall becomes a thin pencil line from corner to corner with a few beads on it, and the quilt's knots come untied into fine lace.

Underneath, the same points that sat in a tight dot on the globe are spread across a flat page, and the late ones (yellow) have slid far down the page, away from the early ones (purple), like sand settling to the bottom of a glass.

Two of the up-close pictures surprise: the figure-eight's ruled lines are now dashed, cut twice per beat, and the cloth of backward lines in the Kozai run unravels while the chaotic run's backward lines stay put.

What it is doing: getting closer did not change who was a clock.

The quiet thing still looks orderly up close, because it really is quiet for long stretches; what gives the clock away is that it comes back after a long time, and the quiet thing only ever matches the moment just behind it.

the triple-star thread

179

A council of five people around a table, each holding the ends of tape measures stretched to the others, and a referee at the side who sees every reading.

One of them lies.

If he lies the same way on every tape, as though he had quietly stepped a little to the left, the referee shrugs and redraws the room with him standing there: every number agrees with every other, and agreement is all the referee can check.

If he lies carelessly, a different fib on each tape, the referee catches him almost every time.

Let every tape be read from both ends and the liar is caught at once, because his neighbour is holding the other end.

In the star-shaped room, where the tapes cross and are pinned together, the referee can tell which tape is lying but not which third of it, until the pins are asked a different question: not how far, but which way.

A pin that can say “this line runs straight through me” settles it every time, and a room whose corners can say “left” from “right” can no longer be folded into its mirror.

What it is doing: a lie survives anywhere the room could have moved, and it dies the moment someone else is holding the other end of the tape.

the triple-star thread

180

A swing on a very long rope, drawn from the side.

At first it hardly leaves the ground: short quick arcs, dozens of them, while beside it a staircase climbs in jerks, each jerk landing exactly when the swing touches bottom.

Halfway along, the staircase stops climbing and the arcs begin to grow, higher and higher, until someone draws a dashed line and writes “gone” — and the swing comes down anyway, three more times, each arc higher than the last.

On the right, every arc of every swing in sixteen gardens sits on one straight ramp, how long it took against how hard it was tied, so exactly that the dots hide the ramp.

Below, fifteen tangled threads wander about the middle and then most of them flare upward at the very end; and a grey hill with a thin red fringe along its far side, where the longest swings live.

What it is doing: the thing that is about to leave goes farther each time it swings, and you can watch it reaching; but most of the long reaches come home, and now and then one leaves from a short swing with no warning at all.

the triple-star thread

181

Three pictures side by side.

On the left, twelve bars standing in a row like the heights of children against a door frame: the first seven short and blue, the last five tall and red, as if two families had been lined up and one of them had grown faster.

In the middle, a comb with broken teeth: thin spikes standing at even spacing all along a line, a few of them shooting up far above the rest in two clumps, one a third of the way along and one near two-thirds.

On the right, a smooth slope like a sand dune seen from the side, piled high at the left edge and trailing away to nothing, not a single spike in it.

What it is doing: the blue things left by climbing a ladder and needed only a nudge at the top; the red things were thrown from the bottom step and needed a shove.

The comb is a clock telling you its hours.

The dune is something that mostly repeats its last moment — and the strange part is that the test meant to tell the comb from the dune could barely tell them apart, though any child could.

the triple-star thread

182

A ladder was expected, with red lines painted where the rungs should be, evenly spaced up the wall.

Then a long row of little grey towers was poured out against it: a crowded heap near the bottom, around the fourth and fifth red line, and after that a thin, ragged scatter running up the wall with no respect at all for the paint — towers between the lines as often as on them.

On the right, the same picture laid on its side: almost every dot sits on the floor, a single step and gone, and only near the bottom of the wall do a few climb to four or six before falling back, with one stray standing at eight high up near the top.

What it is doing: there is no ladder here, only a crowd that started at the foot of the wall, lingered there a few steps, and then was shoved somewhere new at every turn — it moves like something pushed through a doorway by a crowd, not like something climbing.

the triple-star thread

183

Three windows.

In the first, a few crosses hang high up near the ceiling like coats on hooks, each at its own place, and they never come down; below them a little crowd of dots has gathered in a tight column just under one of the red lines, the sixth, standing at many heights, as if something had found that line and kept going back to lean on it.

In the second, three tall thin spikes rise from an empty floor, each a narrow tower in its own lane, and just to the left of the first one, tucked under the sixth red line, a small shoulder that should not be there.

In the third, three coloured threads scribble up and down, and every so often a thread lies down flat for a long while — one flat red bar runs nearly the whole width near the floor, a blue one runs high across the top — then springs up and scribbles again.

What it is doing: most of these things found a place on the first day and never left it; one wandered off, and when it rested, it rested against the same line again and again, which is either a coincidence or the first rung anyone has seen here.

the triple-star thread

184

Four windows.

In the first, a long row of beads sits along the top like birds on a wire, and in five or six places the beads below the wire have dropped off together — never one at a time, always in little flocks, as if whole stretches of the wire had been shaken while the rest stayed still.

In the second, a crowd of coloured threads mills about in one narrow street between the sixth red line and the faint grey line above it; short thick bars lie along that street like benches, nearly all of them inside it, and a few threads climb out through the top of the window and do not come back into view.

In the third, a grey city of towers stands packed just to the right of the sixth line, reaching to the half line and stopping; off to the right, apart from the city, a handful of orange towers stand in two neat groups, one hugging the seventh red band and one the eighth, with almost nothing standing in the gaps between.

In the fourth, two heaps of sand spill over the same floor; a thin blue post stands in the thick of its heap, and an orange post stands far out to the right, past the end of the orange sand, where almost no grain has landed.

What it is doing: most of the ones that seemed to wander never really left home, they paced the street they were born on, and the line at six was only the end of that street; the few that truly left and then sat down again sat down on the seventh and eighth lines, and hardly anywhere in between.

the triple-star thread

Six windows.

In the first, two rows of beads, filled and hollow, strung along the same wire; wherever a hollow bead has dropped off the wire, its filled neighbour has usually dropped too, so the breaks come in stretches, and the two rows agree about where the wire is broken.

In the second, a crowd of threads paces a narrow grey street for six hundred beats, then climbs out over its top edge, and wherever one stops to rest a short bar is laid down.

The bars are scattered up the wall: a few on the seventh line, one or two on the eighth, others out in the open between lines, and two sitting just above the seventh line rather than on it.

In the third, where the earlier picture had two neat stacks on the seventh and eighth lines, the new stacks lean: one tall bar just short of seven, another just past it, and a thin scatter running off to the right as far as twelve, while the grey half-way stripes stay almost empty.

In the fourth, three humps of sand and three standing lines; each line stands inside its hump or just past its edge, none out in the open.

In the fifth, a bundle of threads lies along a zero line, quivering, and one grey thread leaps once to the top of the frame and falls back.

In the sixth, two heaps; the red heap sits to the right of the grey one, and spreads a little wider.

What it is doing: the ones that left home did not, this time, sit down on the seventh and eighth lines.

They sat down in many places, though hardly ever half-way between two lines.

The neat stacks of the first picture were a handful of travellers who happened to choose the same benches.

Three windows.

In the first, a city of grey and blue towers stands along a street, and at its left end, just past a red post, one blue tower stands twice as tall as anything else, as if the street ended at a wall and everyone who walked that way leaned against it.

Beneath, a row of small blue triangles crowds up against the same red post, while the grey triangles are spread along the street.

In the second, a ring has been cut open and laid flat: tall bars stand at both ends, which are the same place on the ring, and the middle is all but empty, with one green bar a fifth of the way along, alone.

In the third, a long red thread wanders back and forth across a red line for two hundred beats, never straying far, crossing it again and again, while short grey threads start wide and stop early.

What it is doing: things that wander near home lean against the line at six and rock across it; things that wander far sit down almost anywhere except half-way between two lines.

Something is pushing them off the halves, or something is holding them on the wholes, and the picture does not say which.

the triple-star thread

From the triple-star notebook, chapter II, “The dimple and its rim” (its readings 13 to 26, 6 October 2026; here 187 to 200): On the second day the resting places turned into something with a shape. A star caught at seven of the pair’s periods sits in a dimple that slides slowly round the table while the star stays in it, so that a watcher fixed to the table would swear the star was moving. The dimple has a rim, smooth on one side and a strip of gravel on the other, where two stars dropped a hair apart go different ways. Under the star’s quick rocking runs a slow beat set by the turning of everything around it, and whatever the size of the rocking it stays centred on one mark, like a level’s bubble. The day also measured how a neighbour makes an orbit turn, the same extra push someone measured on the Moon long ago, and first made the third star heavy, so that the pair at the centre began to breathe.

No pictures this time, only three tables, so the read is of the numbers.

In the first, someone tries to photograph a moth on a flower with a shaking camera: the moth sits perfectly still, and still the photographs put it on the flower barely half the time, because the shake is twice the size of the flower.

The old photograph, the one where the moth was on the flower six times in ten, was sharper than this camera can ever be, and that is the clue it was a lucky frame.

In the second, a ring is cut into halves, one half around each whole number and the other around each half-way mark, and the resting things are counted onto it: four of every five land on the whole-number half, in both piles, though the narrow centre stripes told two different stories.

In the third, a dancer said to sway back and forth across a line is filmed, the frames are shuffled into random order, and the shuffled film looks more like swaying than the real one; in the real film the dancer leans one way for a long time, then leans the other.

What it is doing: the thing that rests does favour the whole numbers, the way a ball favours the bottoms of a row of shallow cups, but the cups are wide and shallow, and the camera that was meant to see whether the ball sits still in one was too shaky to tell.

To see a ball held in a cup, you watch the cup’s rim, not the ball.

Six windows.

In the first, two strings of beads, red and blue, along the same wire; the red string mostly sits high on the wire and drops off in a few places, while the blue string has fallen almost entirely to the floor, as if the blue wire had been shaken much harder.

In the second, two crowds stand on a street: the blue crowd presses up against a black post and spills past it on the left, and the red crowd stands well off to the right, around its own dashed line, with nobody near the black post at all; small triangles mark where each person stopped, and they follow their own crowd rather than the post.

In the third, a ring of time cut open, and the bars no longer avoid the middle: they stand tall at both ends in one colour and lean into the middle in the other.

In the fourth, two thick lines stand inside the hills of sand, not past them.

In the fifth, a bundle of threads lies near zero and creeps slowly, one grey thread sliding steadily down the page for eighty steps, while dotted threads above them leap up in sudden staircases to heights the solid threads never go.

In the sixth, a scatter of red and grey dots lies along a diagonal line, a little to one side of it, and almost none sits on the flat line through zero.

What it is doing: the post was never holding anyone; each crowd stands where it was set down and drifts a little, and when the crowd was set down somewhere else, it stood somewhere else.

The thing that was supposed to stand still while everything else moved, the arrow at each swoop, keeps turning, slowly, the way a clock hand turns when its gears are almost but not quite meshed.

the triple-star thread

Six windows, then three close-ups.

In the first, a map with two corners: two orange-red dots sit high against the left edge, two blue dots lie on the floor beside them, a green dot hangs half way up, and one purple dot lies on the floor far off to the right, alone.

In the second, a picket fence: thin posts at an even spacing all along the ground, and in the middle four posts shoot up far above the rest, the tallest touching the top of the frame.

In the third and fourth, no fence at all, only a hillside falling away from the left, with a few broad rolling humps.

In the fifth, three smooth waves, each rising and falling the same height, and on the steep shoulders of every wave small black ticks where the line takes a little stair-step.

In the sixth, a line that does not wave: it sits flat, jumps, sits flat, jumps, up and down a wall of uneven stairs, with crosses where the jumps are biggest.

The close-ups: the smooth wave’s little stair-steps fall on dotted lines that come round at an even beat; a thin flat line flickers up and down at that same even beat and never goes anywhere; and the uneven stairs also step on dotted lines, but these come round at uneven times, and the steps are three or four times taller.

What it is doing: everything here is tapped by the same passing hand, every time the hand comes round.

The calm ones are tapped lightly, on a steady beat, and their record draws a fence.

The wild one is tapped hard, by a hand that comes round when it likes, and its record is a hillside with stairs.

Counting every tap cannot tell them apart.

Listening for the beat can, and so can counting only the hard taps.

the triple-star thread

Three windows, then two.

In the first, straight lines fan out from one corner, each climbing at its own steady slope without a wobble; the steepest climbs to ten thousand, and a dotted one along the floor never leaves zero.

Something is being wound, steadily, and how fast depends on what stands nearby.

In the second, three families of dots rise from the same height of one: the green barely lifts, the red bends upward, the blue curves up hardest, and a single orange star floats just above the blue curve at its own place along the floor.

Under each family a dotted straight line starts from the same point and runs too low, and every family leaves its line, sooner and harder the heavier the family.

In the third, the three families are slid sideways onto a common floor, and near the corner they nearly lie down together on one dotted line before the blue and red peel away upward.

In the look, hollow dots and filled dots sit almost on top of each other, and close in the green family climbs more steeply than the other two.

What it is doing: a neighbour standing still would push the thing round at one rate.

A neighbour who walks around it pushes harder than that, because the push changes the thing’s shape and the next push lands on the changed shape; the faster the neighbour walks, the bigger the extra.

The orange star is somebody else’s measurement of the same extra push, taken long ago, and it belongs to the family.

the triple-star thread

191

Two windows.

In the first, three strings of beads start from the same corner and climb, each beside its own dotted guide-line: the purple string barely leaves the floor, the orange lifts gently, the blue climbs steeply.

Each string starts out lying along its guide and then pulls away above it, sooner and harder the higher the string climbs.

In the second, six markers, round and square, sit along a single straight red line that climbs from the lower left to the upper right.

Beside it runs a dotted black line at exactly the same slope, but the red line starts a hair above the floor, so the two run side by side, a finger’s width apart, and never meet.

What it is doing: six very different pairs of hands, from small to large, each give the thing an extra turn, and the extra grows with the size of the hands at one fixed rate, the same rate someone measured on the Moon long ago.

Every pair of hands also gives a small extra nudge that does not depend on their size at all, as if the thing were sitting a little closer to the hands than it was told.

the triple-star thread

192

Three windows.

In the first two, a scatter of coloured beads lies on a floor ruled in powers of ten, piled against a bottom line at four and thinning upward, with a few beads standing tall: one green bead high at fifty, one higher still at seventy in the second window.

Two guide-lines cross the frames, one steep and one gentler, both falling from the upper left, both promising that beads to the left should stand taller.

The beads do not listen: the ones far to the left are as short as the ones on the right, and the tall ones stand out in the middle of the floor.

In the third window, each bead appears twice, hollow and filled, joined by a thin grey thread; read by one clock it sits here, by the other clock it sits there, sometimes a long way off, but whichever clock reads it, it stands at the same height.

What it is doing: things that come close to a perfect match are not rewarded for coming close.

Each one rests for a while and is then knocked off its seat by something that does not care how good the seat was.

The clocks disagree about where each one sat, and agree that the sitting did not depend on it.

the triple-star thread

193

Two windows.

In the first, grey squares float above a pale pink band, most of them well above it, one a little below; red circles sit inside the band, close to its middle line, except one red circle at the far left that has dropped out of the bottom.

In the second, a line of red dots runs perfectly flat along the zero mark from left to right, while grey rings hang below it in a fan that opens wider and deeper toward the right, the deepest ones nearly half a percent down.

What it is doing: something that was set going a little too slowly settled a little too low, and everything measured from it came out a little too strong.

Set going at exactly the right speed, it stays at its height, and the extra strength that looked like a new ingredient disappears.

One measurement, the faintest, was too faint to say anything either way.

the triple-star thread

194

Two windows.

In the first, eight strings of beads fan out from one corner, four climbing above the floor line and four falling below it, each upper string matched by a lower string of the same colour, like the two wings of a bow tie spread open.

Dotted guide-lines fan out with them, straight, and the beads leave the guides in the same direction, top and bottom: the upper strings rise faster than their guides and the lower strings fall slower than theirs, so the whole tie is tipped gently upward at its far end.

In the second, red dots climb a straight line, blue squares fall down its mirror image, and grey triangles sit on the floor line between them, never leaving it.

What it is doing: a thing pushed by a neighbour circling one way is hurried; pushed by the same neighbour circling the other way, it is held back by exactly as much.

Whatever the push does that does not depend on direction sits in the middle, small, the same for both, and lifts both wings together.

the triple-star thread

Six windows.

In the first, two long staircases of dots run down from upper left to lower right, steep and straight, each broken once: where a step should be, the dots stop falling and sit for a moment on the floor line in a small knot, ringed in red, then the next flight begins below.

In the second, two pairs of threads climb from the bottom left like vines up a trellis, wobbling as they go, five and six turns high by the end, while beside them two flat threads lie along the floor and never leave it.

In the third, two deep V-shaped valleys are cut into a high ridge of dots, the bottoms of the valleys dropping through a dashed line and then out of the frame; the orange dots, a second planting, never reach the bottom of either valley, and two of them hang alone in the air halfway down.

In the fourth, a clean line of beads slants down a ruled floor, straight, with a dashed guide beside it they follow almost exactly.

In the fifth, four strings climb gently from the left, side by side, and then, at the same place, go wild: they jump to the top, fall back, jump again, cross one another, and do not come back together.

In the sixth, a red wave rides exactly on top of a thick grey one, and a blue line plunges where a dotted blue line climbs.

What it is doing: a marble dropped into a dimple stays there, and stays there even while the dimple itself slides round the table, so someone watching the table’s corner would swear the marble was moving.

Far from the dimple a marble rolls past at a speed set only by how far away it is.

At the dimple’s rim there is no edge to the dimple at all: a strip of gravel, where two marbles dropped a hair apart go different ways.

the triple-star thread

Six windows.

In the first two, a fat crescent moon is drawn over and over in thin coloured threads, each thread a wire bent into the same bow, the purple ones long and reaching high and low, the yellow ones short and hugging the middle of the bow; none of the bows ever closes into a ring, and none crosses the thin vertical line at the left where the moon’s hollow would be.

In the third, a row of dark beads lies flat at about eleven, while a second row walks up from fourteen to seventeen; between them two beads leap to the top of the frame and stand there alone, and a green row does the same on the right.

In the fourth, strings of beads climb a gentle straight slope side by side, two of them so evenly they could be one string; a short dark one starts with them and stops after three beads at a red cross; a pale one climbs a little more steeply, then shoots almost to the ceiling in one step and stops with a red cross.

In the fifth, three threads lie along the floor of a tall ladder of powers of ten, quiet, tiny; one thread lives far above them, near the top, hopping between rungs, and drops to the floor only at the far right.

In the sixth, four short straight rows of dots slope down to four dotted posts, each row aimed exactly at its post.

What it is doing: a thing held in a bowl rocks back and forth along the bowl’s bottom curve without ever going round, quicker on one side of the bowl than the other.

Rolled toward the bowl’s rim from outside, it slows by the same small amount for every tenfold step closer, and that amount can be read off how fast it rolled far away.

On one side the rim is smooth; on the other, the rim is a strip where nothing can be predicted, and the strip is wide.

the triple-star thread

Three windows.

In the first, four jagged mountain ranges are drawn one over another on a floor of tiny numbers, each range with a single tallest peak; the tallest peak of the dark range stands exactly on a dotted pole planted for it, and the tallest of the green range stands on its own pole a little to the right, while the paler ranges put their tallest peaks far to the right and keep smaller summits close to their poles.

In the second, a dense fence of thin vertical strokes fills the frame from left to right, two thousand of them, swinging between a fifth of a turn up and a fifth down without ever drifting, so that the whole fence is a grey band with a flat top and a flat bottom.

In the third, eighteen beads sit on a level line drawn across a tall pink wall; the wall marks everywhere the beads could have climbed to, a quarter turn up or down, and not one bead has left the line, whether it belongs to a small swing at the left or a large one at the right.

What it is doing: a thing that rocks in a hollow is also rocked, gently, by the slow turning of everything around it, and that slow turning sets the one long rhythm under its quick rocking.

However hard it is set rocking, its rocking is centred on the same place, like the bubble in a spirit level that always comes back to the middle mark.

the triple-star thread

Three windows.

In the first, slanting rows of beads run down from left to right, each beaded with grey crosses that cancel every bead, and a few crossed beads have fallen far below the rest, one almost to the floor at the far right.

In the second, a field of dots rises and falls like a range of low hills; between them, at two places, a handful of dark dots slides down into narrow wells that reach below a dashed line, while the orange dots, planted beside them, stay up on the hills.

In the third, two rows of small triangles: one row sits flat along a line at a tenth, steady from left to right; the purple row sags from that line almost to the floor and stays there, except where the wells were, where it climbs back up to the tenth; the red row floats above, high at the left and settling lower toward the right.

What it is doing: a small thing circling a pair was held in two dimples before.

Now the circling thing is heavy, and the pair it circles is no longer a rigid ring: when the heavy thing goes past, the pair stretches or rounds itself, depending on how the two first faced each other.

Inside the dimples the pair keeps its shape.

Outside, it slowly loses it or gains it.

The ruler used to count turns was taken at the moment of greatest squeeze, so it read short and long by turns, and none of the dots can be trusted until it is replaced.

the triple-star thread

199

Four windows.

In the first, beads sit on slanting tracks that fall from upper left to lower right, and at two places the tracks are broken by a short flat step on the floor line, ringed in red; crossed-out beads lie scattered beside the tracks like seeds that did not take.

In the second, a ridge of dots stands high across the frame and dips twice into narrow wells, the dark dots reaching the bottom of each well and the orange dots stopping short, one hanging alone halfway down.

In the third, threads quiver across the frame like seismograph lines: an orange band shivers high, a blue band shivers low and touches the floor every few beats, and a few red threads swing gently between them, slower and calmer, in step with one another.

In the fourth, a bundle of red threads lies flat along the top, untroubled, while a sheaf of paired dashed and dotted threads falls away beneath them in stair-steps, each pair so close the two look like one.

What it is doing: a heavy thing goes round a pair and catches it at two places, the same catch as before but on the other side of the mark, because now the pair itself is turning faster than the thing that catches it.

Where it is caught, two different clocks on the pair fall behind together, step for step.

And the pair’s shape shivers all the time, stretched and rounded, rounder on one start and longer on the other, and calm only where it is caught.

the triple-star thread

200

No picture this time, only a list of notes for each of four bells.

The two bells struck at the side ring one clear note each, and every other note they make is that note with a slow throb added or taken away, once, twice, three times, so that each list reads like a family.

The two bells struck at the middle ring the slow throb itself loudest, then its octaves, and threaded between those octaves a second family, a quick note of their own wrapped in the same throb.

What it is doing: a thing rocking in a hollow keeps its own rocking even when it sits almost at the bottom, slightly slower there than out on the sides, and the slow turning of its neighbours hums under every note it makes.

the triple-star thread

From the triple-star notebook, chapter III, “Splitting, and the bowl that breaks” (its readings 27 to 48, 7 October 2026; here 201 to 317): The third day followed the rocking as the star’s own orbit was stretched. On small known maps the reads watch a motion split its rhythm in two, then in two again, each split sooner than the last by a fixed factor, and watch what a slowly turning floor does to that cascade: the newborn falls into step with the floor instead of splitting again. In the stars themselves the rocking quickens as the orbit stretches, passes one beat in three without noticing, reaches one in two, and there the bottom of the bowl stops being a bottom; one step further, every star near it leaves. A first atlas of thousands of runs drew the island as eight weather maps, and a walk to its twenty ‘still’ places found

only four of them truly still: the rest were crests and troughs where the current only turns, or had no far bank to judge by. The day ends with checks on the pages themselves: every number sent to its notebook, and a long conversation read back from its first line.

201

No picture this time, only a roll call.

Twenty-four times a family of three broke up, and each time the roll call names who went out the door.

The smallest one, who had come in from outside, is named seventeen times.

Seven times it is one of the two larger ones, and on those seven lines the small one stayed behind and took the leaver’s place beside its partner.

The four quickest partings, all before the pair at the middle had gone round a hundred and fifty-five times, sent out the small one every time.

What it is doing: three dancers who cannot keep a circle of three fall into a pair and a spare, over and over, trading partners, until one throw sends the spare off for good.

The lightest is the easiest to throw, more often than its lightness alone would say, and it is thrown soonest; but sometimes the small one cuts in, and a heavier dancer is the one who leaves.

the triple-star thread

202

No picture this time, only a long check list.

Every number written on four pages was carried, one by one, to a shelf of notebooks to see whether it is written there too.

On the biggest page, all but three found their twin.

Of those three, one had been written in different units, one was a sum the page works out in front of you, and one the page itself says it worked out by hand.

On the other pages, the only numbers without a twin were not amounts at all but addresses, like the number on a library card.

And a few numbers that were written down only today found twins anyway, in notebooks that have nothing to do with them.

What it is doing: a set of pages is being checked against the records it was copied from, before it is handed to someone new.

Everything that should have a source has one.

A match is not proof, though: a short number can turn up anywhere by chance.

the triple-star thread

No picture this time, only forty-nine turns of a conversation, read again from the beginning.

The questions are short and the answers long, and the short ones steer: close the hunt but keep watching, choose for me, save the big build, move house.

Near the end the answers start to misremember their own beginning, saying they began three days earlier than they did and two runs later, and one number on the pages turns out to have no notebook behind it at all, though it had been checked and passed because its digits happened to sit in other notebooks.

Read from the first turn, the beginning is still there, word for word.

What it is doing: a long walk remembered only from its last few miles grows a wrong start.

The walker can still go back along the path and find the first footprints, if the path was kept.

the triple-star thread

Chapter 4 — The long middle, 7 to 10 October



A picture read from its dots alone: four lanes on the right, each splitting as the push falls toward the left, until the lanes fray. From The Loop Atlas. TOY MODEL. Its reads are 241 and 245.

A page of dots, nine hundred columns wide.

from 241

204

Something stepping in fours is asked to go a hair slower, and instead of slowing it splits.

From then on it steps in eights, every second four landing a little left, then a little right.

At the instant of the split the eight is the four done twice, so it cannot tell itself from its parent; asked whether it has changed, it answers “not at all”, and we wrote that answer down as a fold.

The two new footprints walk apart as the square root of the distance, quickly and then lazily, like a hinge that swings freely and then stiffens.

Nothing was born from nothing: a step learned to disagree with itself.

the Loop Atlas

205

At a real doorway, the searcher walks to the threshold and stops dead in two steps.

At the doubling, it finds a long hallway where every spot looks like the door, because the old four-step done twice looks the same all along it.

So the searcher wanders, never sure.

The machine tells a hall from a door by how lost its own searcher gets.

the Loop Atlas

206

Nudge the clock by a hair and the whole staircase slides down together, every tread by almost the same amount, and the small differences between the slides shrink step by step by the staircase’s own number, about four and two thirds, as if even the nudge knew which staircase it was on.

And the searcher that wandered at the doubling had found it once before: seventeen stumbling steps, then one that happened to land under the bar, and the record wrote ‘found’.

At a true door it takes three.

the Loop Atlas

207

Eight true doors: the searcher walks up to each and is through it in a handful of steps, even the one it started far from, which costs it a few stumbles before its last strides land.

Then it tried to measure how far each tread of the staircase slid when the clock was nudged, and below the fourth tread the slides were smaller than its own shaking hand.

The staircase had not stopped keeping its proportions; the ruler had run out of marks, and every answer past that point was a guess wearing a number.

the Loop Atlas

208

A very narrow gate was once measured by sending two scouts to find its posts.

Each came back with a note saying ‘not quite there’, and the gap between them went into the book anyway.

This time someone walked the whole fence with a hand on the rail, and the gate turned out twenty times wider than the scouts had said: still far narrower than a hair split a billion billion times, but twenty times.

The scouts had not lied.

Each had stopped where the ground went flat and called it the post.

the Loop Atlas

209

Shake the ruler and the whole fence moves sideways by the width of a hair, while the gate in it stays exactly as wide as it was to six figures.

The gate does not belong to the ruler.

And the second narrow gate, measured by two scouts who both stood at the same post, turned out thirty-seven times wider than the book said.

Two notes signed in the same spot are one note.

the Loop Atlas

210

The whole endless staircase hangs from one nail.

At the top a two-step appears out of nothing, from a door that snaps open, and walks a short hallway alone.

Then it splits into a four-step, the four into an eight, and so on forever, each landing shorter than the last.

The first landing is four and a half times the next, and the landings further down settle toward four and two thirds, as if the stairs learned their rhythm only after the first few.

And there is no rung above the two-step, because a half step can never be the copy of a whole one.

the Loop Atlas

211

A mapmaker stood at one fixed height on a hillside and walked round it, marking every place the ground crossed her level.

Near the two bumps she knew, she found what she expected.

But on the far side she found crossings at every height she tried, in the same two places each time, as if a cliff stood there.

So she walked the ridge itself all the way round: no cliff, and the hill no higher anywhere.

Then she went back to one of the far crossings and halved her step sixty times, and the ground never reached her level.

It skipped over it, like a stair seen from the wrong side.

Her level had been fooled, not the hill.

the Loop Atlas

212

Thirty-two thousand riders were let loose in one night: twenty weights, four hundred and one pushes, four starting lags.

Each was told to wander for a thousand turns of the clock and then be watched for two thousand more.

Every one of them has landed somewhere or is still tumbling, and the answer has been sitting quietly in a folder since, written down, waiting for the one reader who asked the question to come and look.

the Loop Atlas

213

The widest step was the slowest to walk.

A kitchen-table machine spent three hours going round it, where the cloud had given up after eleven minutes, and at the end it came back and said the step was exactly as wide as the old map had drawn it.

Two machines in two rooms were asked the same six questions and gave the same six answers.

One of them left its working on a page that is still lying on the other table.

the Loop Atlas

214

Two walkers joined the same path round a pond at different gates, and each was asked to mark its highest and its lowest point.

One took more than five times as long as the other.

Their notebooks agreed to the thirtieth mark on the ruler; the only lines that didn’t match were the gates they had come in by.

The path doesn’t care where you join it.

And the walker who had been warned that the widest path doubles back on itself walked all the way round and found that it never did.

The warning had come from a broken compass, not from the path.

the Loop Atlas

215

Three cooks shared one kitchen through the night.

One brought twelve helpers, and they took every big burner, so our cook, who works alone, was pushed to the little ring at the back.

The soup came out exactly as it does in an empty kitchen, to the last grain of salt; it only came out later.

A crowd can change when an answer arrives, never what it is.

And in the morning there was one more pot on the stove that nobody had claimed, still simmering on a timer its owner had set for seven hours and forty minutes, with a note saying he would be back to finish it.

the Loop Atlas

216

A narrow doorway was measured a fourth way.

Instead of walking the corridor to find where it pinches, someone stood still just inside it and just outside it and asked the plainest question: can anything get through here?

Inside, fourteen travellers could; outside, nobody, anywhere along the hall.

The ghost doors the last survey kept finding at the far end were painted on a wall it had been reading from the wrong side; from the floor everyone actually walks on, they are not there.

And the one doorway looks a different size from each of the seven rooms it opens into, the widest view more than half again the narrowest, though everyone who passes through it is passing the same door, at the same moment.

the Loop Atlas

217

A rider was told to find his right speed by checking, once a lap, whether he came round as fast as he started.

Over most of the track the check works, and where the road pushes back it even calms his mistakes.

But on one stretch a fast start carries him to a place where the road itself runs faster by exactly his surplus, so he comes round matching himself whatever he chose, and the check goes blind.

Just before that stretch it shouts every whisper a thousand times over; just past it, it shouts the opposite.

Anyone who simply watched where he rode was never fooled.

And a second rider, finding the top of a hill by stepping in five times closer each time, saw his mistake shrink six hundred and twenty-five times a step: a lopsided hill fools him twice, and each fooling is squared.

the Loop Atlas

218

Three mountains stood hidden on a plain, and a walker carrying an arrow found every one of them without once looking up.

Wherever the arrow came home turned, something stood between his path and a peak; wherever he walked round a mountain’s whole skirt, the arrow came home as if the plain were empty, because a mountain’s foot hands back exactly what its top took.

The arrow turned most on each mountain’s steepest ring, and what it turned by was the slice of paper you would cut away to fold that slope from a flat sheet.

In the next room two clocks that should have struck together struck a hair apart, until someone noticed that one of them counted a full turn with a ruler a hair too short.

the Loop Atlas

219

A survey of hidden mountains found a fourth where no mountain stands.

Walkers sent out in ever-wider rings from a patch of open plain between two mountains first felt the ground bend the wrong way, the feet of both mountains at once, and only when their ring reached past both feet did they feel the two tops, and then strongly: the best ring there turned their arrows more than the smallest real mountain could.

Nothing stands at its centre; a short walk round that spot turns no arrow at all.

Elsewhere on the plain, a mountain just under three percent taller than the first cuts exactly a quarter from the paper circle.

And three mountains of one shape, one half the size and one twice, gave the same reading to the last digit the machine can hold, perhaps because halving and doubling are sums a binary machine never gets wrong.

the Loop Atlas

220

A hiker’s arrow keeps an honest count everywhere except where a mountain meets the plain.

There the ground stops bending so suddenly that the arrow trips, a little differently each time depending on where his stride happens to land, and the trips do not shrink when he takes shorter steps; they only scatter.

Round the mountains’ feet with a gentler curve and the stumbling goes away so completely that what is left is the dust of the arithmetic itself.

A pebble’s walk round each summit turns the arrow by how sharp that summit is, the smallest mountain’s most of all; round the empty middle of the fourth peak it turns it not at all; and on each steepest ring, where the ground changes from dome to saddle, the same pebble walk hears only a whisper, the same whisper all the way round.

the Loop Atlas

221

Years ago a man drew a ring of steep peaks on a small notepad, each wearing little collars at different heights, a band slung round all of them halfway up, and in the empty middle a box folded inside a box, tied to the peaks by dashed lines.

Tonight the arrows found something like it on their own: a hilltop in the middle of a range that only the peaks around it hold up, with nothing standing under it.

The collars are where the arrow is read, loud near the top and a whisper on the steepest band, and below the plain the drawing keeps going, roots no one on the plain can see, which is exactly where the arrow stumbles.

In another room two books argued over one number: one says every hump doubles at the same pace, the other says the pace depends on how flat the top of the hump is.

the Loop Atlas

222

Four hills were asked how fast they split.

The round-topped one split in the old familiar rhythm, each pause four and two-thirds times shorter than the last, and its echoes shrank two and a half times each round.

Flatten the top and the rhythm quickens: seven and a quarter for a top flattened once, nine and a third, then eleven, as if a flatter summit had less to hold onto and let go of itself sooner, while the echoes shrink less and less each round.

The needle we built long ago had been splitting in the round-topped rhythm all along, so its hill was round, though nobody had looked at its summit.

A popular page had put a flat-topped hill in the round-topped choir; asked to sing, it sang its own note.

the Loop Atlas

223

Seven steep peaks stood in a ring with nothing in the middle.

A man walking a small circle in the middle felt nothing at all; the ground there was flat as a table.

A man walking the big circle that threads between each peak’s top and its foot came home with his arrow turned a little more than a third of a right angle backward, and might have sworn the walk was easy, though the ground he had circled held nearly two whole turns of bending.

The arrow is a clock with only a second hand: it tells you where in the turn you are, never how many turns have gone by.

To learn the count you visit the peaks one at a time and add up what each one gave.

the Loop Atlas

224

A star was walked round a single hill, the walker crossing his own path five times on the way.

Where he had gone round twice, the arrow counted the hill twice; where once, once; and when he walked the plain pentagon round the star’s tips instead, holding the whole hill inside, the arrow counted nothing at all.

Inside the star sits a smaller pentagon, and inside that the room for a smaller star, each one the last made smaller by the same golden step.

Meanwhile a number that had turned up four times in a row, two-fifths and a little, was chased up and down every knob and turned out to be a place it kept passing through, not a place it lived.

the Loop Atlas

225

Three questions were asked of numbers other people had written down.

A frictionless hill, one that never forgets any of its motion, was asked how fast it splits, and it answered eight and three-quarters, nearly twice the forgetful hill’s pace, climbing toward its number from below as the needle had.

A five-pointed star drawn inside another star counted its middle twice, exactly as its parent had, one golden step smaller.

And a hill cut with ever straighter knives gave up a little less each time, until the straightest knife took a fixed share set only by how steep the hill was.

The machine that found the frictionless number stumbled twice on the way: once it ran outside the fence it had built, once it grabbed the wrong partner of the right size.

Both stumbles were its own, not the hill’s.

the Loop Atlas

226

The needle was asked a second question about the top of its hill.

To the first, how fast it splits, it had answered by creeping up on its number from below, never overshooting.

To the second, how much narrower each new fork is than the last, it answered like a pendulum settling: too wide, then too narrow, then a little too wide, each miss about four-tenths of the one before and on the other side, swinging in on the same number the round-topped hills give.

One answer walks in, the other swings in, and both arrive at the same door.

And the flattest hills of all, asked the same two questions, are still talking: their answers have not finished falling.

the Loop Atlas

227

A sealed box came back after two weeks on a shelf.

Every one of its three hundred and forty-nine seals was intact.

The machine inside was asked the two questions it had answered in September and gave the same answers to the last digit, row after row.

But the first time it was asked, an older machine packed in the same box answered for it, quietly and correctly, because the asker had been built to prefer whichever copy sat closest to hand.

Only when the old one was set aside did the new one speak for itself, and it said the same thing.

the Loop Atlas

228

In the morning a circle was grown from a dot in the middle of the ring of steep hills, and the arrow was watched the whole way out.

It swung backward more than a turn and a half as the circle crossed the hills’ skirts, then forward almost two as it climbed over their tops, and when the circle had swallowed every hill whole it came back to nothing.

The turns the arrow could not show the night before had never been lost; they were in the watching.

Then a second bell was set to wait for the smoke ring the first had blown.

The ring came, opened wider than the bell, and pushed it back.

Held on a line, the bell took the whole shove and more, and the opened ring stopped in front of it with a dark twin at its side.

Let go, the bell was carried backward until the ring slid past its rim and went on alone behind it.

In September a smaller ring had slipped through the second bell’s open top and drawn it forward; the leader’s own ring is too wide for that.

When the top was opened this time, the valve chattered and the machine could not hold it, and the card went back on the shelf to wait for its place.

the Loop Atlas

229

A man asked how many more invisible mountains there were.

The arrow was carried over paper pressed into ridges as steep as a roof, round small loops and loops wide enough to cross several ridges, and it came home pointing exactly where it left, every time: paper you bend without stretching has no mountains in it at all, however high its ridges stand.

Then the paper was rolled into a cone, and every loop that went round the tip came home turned by the same amount, near or far, centred or off to one side, while every loop that missed the tip came home straight.

The whole mountain lived in one point.

A loop knows only the ground under its feet, and the ground knows only how much it was stretched.

So a mountain can hide in three ways and no more: it can cancel itself with its own skirt, it can tuck whole turns inside a count, or it can be no mountain at all, only a fold.

The first test of the third way could not fail, because the arrow had been pointed down the ridges and rode up and down them in its own plane; turned to a slant, it had something to get wrong, and it still came home straight.

the Loop Atlas

230

A watchman kept count of how many times a lighthouse beam swept round by glancing at it now and then.

As long as he glanced again before the beam had gone half way round, he could always tell which way it had moved, and his tally was exact.

Glance one moment too late, and the beam, gone just past halfway, seemed to have stepped back a little instead of forward a lot, and a whole sweep fell out of his tally without a sound.

Glance later still and his tally came to nothing at all, the same as a man who had looked only once, and later again it came out short by three, and then by four.

The beam never knew how many times it had gone round.

The count was never in the beam; it was in the watching, and only quick enough watching could keep it.

the Loop Atlas

231

A coin was thrown from pad to pad, held by two reins at the edges of its rim, so that in the air it could do only one thing: turn over.

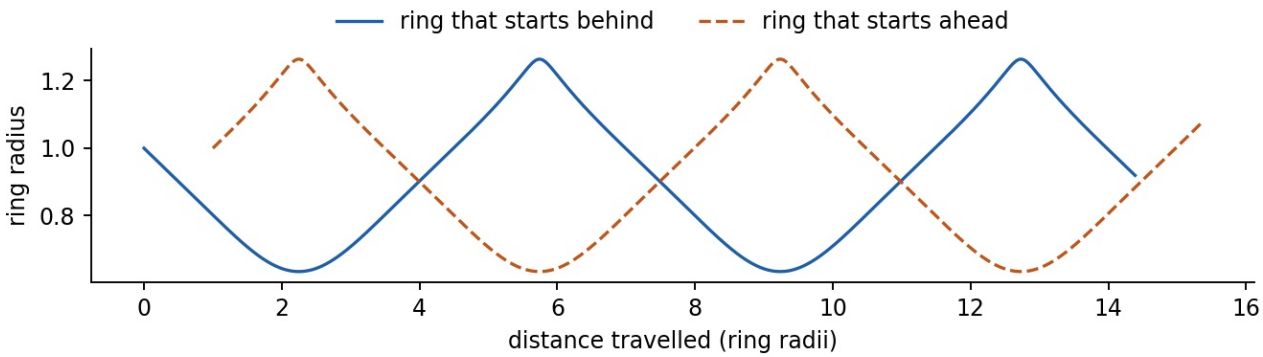
Which face it showed at each landing was nothing but how many half-turns it had made on the way, odd or even.

Thrown carelessly, its small errors piled up hop after hop until the face was anybody’s guess; held by the reins at every pad, so that it never wandered a quarter turn from where it was meant to be, it showed the chosen face hop after hop.

A second coin, a quarter turn out of step with the first, told the sled which way they were both turning, which one coin alone could never say.

And the only places where the coins’ turning and the length of their shadows could both be told in whole-number fractions were a sixth of a turn, a quarter turn, and their doubles.

the Loop Atlas



The radius of two rings over distance travelled: each shrinks as it passes through the other, then widens again, the two taking turns. From The Loop Atlas. TOY MODEL.

232

Two smoke rings that keep passing through each other go further on the same breath than one ring made of both.

A bell that rests a moment before it squeezes finds the ring it left behind still waiting under its edge, and pushes off it as off a wall, harder the sharper it had sprung open.

A small ring met a bell with an open top, slipped in, came out of the mouth stronger than it went in, and drew the bell toward where it had come from.

A big ring, the kind a leader leaves behind, met a bell with a closed top and could not get in: it spread wider than the bell and pushed it back, and when the bell was tied in place the ring stopped beside it with a dark twin and stayed there, fading.

Partners remember each other, but what the memory does depends on whether there is a door.

the Loop Atlas

233

Ten habits kept turning up wherever loops were made to pay.

Give back on the way home what you spent on the way out, though giving back never carried anyone anywhere.

Hand the leftovers of one loop to the next.

Change course only when nothing is moving, but leave a little room either side, or you will flicker.

Listen at the pitch you are being sung to.

Push a little at a time, at the right moment, and the swing grows fourfold; push at the wrong moment and it dies away.

Let a wave run down the body.

Lean on what your last step left behind, if it lets you through.

Stop turning one kind of thing into another.

Let two streams pass each other going opposite ways.

And when the road changes, change the loop.

the Loop Atlas

234

Two rivers, one hot and one cold, were laid alongside each other to trade their warmth.

Flowing the same way, they traded fast at the start and then stopped at halfway, each settling at the same lukewarm, and no length of river could make them trade more.

Flowing opposite ways, the hot one met cold water all along its course, because the cold it met near its end was the coldest, just arrived; given the same length that let the side-by-side rivers reach halfway, they reached halfway too, and then kept going, nine tenths at ten times that length, ninety-nine hundredths at a hundred.

Partners who pass each other going opposite ways never stop having something to give.

the Loop Atlas

235

A team kept a book of every time they were wrong.

Reading it back, the wrongs came in three colours.

Some were the ruler’s: a clock that ran a hair late, a step too coarse to say the ninth figure, a valve that could not make up its mind.

Some were borrowed: a map taken from a neighbour’s country, a law that held only far away, two machines that looked like twins and were not.

And some were the world’s own, and those were the best, because each one turned out to be a door: the ring that seemed to lose the race won it on an equal purse, the swing that wandered was only growing slower as it grew, the needles that seemed to wait beside the lock were only taking their time to sit down.

Every wrong was written down with the run that showed it, so that any of them could be dug up again.

the Loop Atlas

236

In the longest chain of runs the machine kept measuring grooves too thin to see.

Each time it reported a width it was really reporting the grain of its own ruler: the width came out a few trillionths when the groove was hundreds of billions of times thinner still.

The cure was a finer ruler, and then a better question, how far each groove sat below the speed limit, and with that question the widths fell into one line across sixteen powers of ten.

Meanwhile a cat, asked how far it turned in a full loop of its waist, answered fifty-seven degrees, which was the price of the move and not its point; asked instead how far its legs turned in half a loop, it answered a hundred and fifty-nine.

Most of the wrongs in the book were a ruler too coarse or a question aimed a little to the side.

The world, asked properly, had usually been saying something simple all along.

the Loop Atlas

237

The team finished reading back its book of wrongs and found two that had slipped between the pages.

One had been answered a day after it was written, by a run nobody connected to it; the other had never been given a card at all, and got one.

In between, the wrongs taught small things that turned out large.

A ruler whose circle falls a hair short of a full turn leaves its fingerprint on every edge it ever measured, too small to matter and too regular to be chance.

A tally that starts counting when the watch begins can only guess what came before, so the counter must already be running when you look.

A rider who taps the same way every time falls without a word of warning; one who taps as hard as the lean he feels trembles more and more as he nears the edge, and you can hear the fall coming.

And a logbook should count the turns an engine ran, not the hours it was switched on, because the engine that worked harder wore out sooner, and the turns had been counting the work all along.

the Loop Atlas

238

Before the sand ran out, the two of them packed.

They wrote down what they had learned about their own tools: that a ruler can be wrong in ways that look like the world, that a count lives in the watching, that a test which cannot fail is not a test.

They wrote down which questions were still open and in what order to take them up, and which had been put on the shelf on purpose, and they checked every box against its label.

The conversation itself would be forgotten, as it had already half forgotten its own first day; what would carry was what they had written down as they went, and that turned out to be nearly everything.

the Loop Atlas

239

The chop in the bath looked like a crowd running every which way, but it was two long combs, one sliding left and one sliding right, each as rigid as a ruler.

A cork sits in the crowd going nowhere until one comb grows a tooth tall enough to push it as fast as the comb itself moves; then the cork stops falling behind, climbs onto that tooth’s front face, and rides away.

Every cork in a soup ends up on the same tooth, the first one tall enough that it meets, so a soup carries all of its corks or none, and which it is depends on where along the road the tallest tooth happens to stand.

The other comb, sliding the other way, keeps bumping the riders, and a tooth has to stand a little taller than the arithmetic says before anyone can stay on.

the Loop Atlas

240

They made the road longer, and the corks told more of the truth.

In a bath where only one comb slides, a cork stays on the moment a tooth stands tall enough to keep pace, not a hair taller; the arithmetic is right.

Put the other comb back and the tooth must stand a little taller, and taller again when the other comb is stronger, because its passing teeth keep bumping riders off, and the longer the ride, the more of them it bumps.

Two things nobody expected.

A cork can be held a long while at a tooth that is almost tall enough, so in a slightly stronger bath it reaches the tooth that would have kept it too late to be counted.

And in one bath, with the other comb strong, a stronger chop carried no corks at all where a weaker one had carried every one.

the Loop Atlas

241

A page of dots, nine hundred columns wide.

On the right, four thin threads run level across it.

Moving left, each thread splits in two, the halves drifting apart quickly at first and then more slowly, the way a road widens out of a fork.

Further left each of the eight splits again, and each of those again, the stretch between one split and the next shrinking by about the same amount every time, four and a bit, until the threads fray into four grey ribbons of fuzz.

Every hole in the grid keeps a tally of how often something landed on it, and the tallies tell the whole story without the picture: 128 in every filled hole on the right, then 64, 32, 16 and 8, and at last holes landed on only once.

From far away each pair of threads looks like one fat string.

But sit in one and watch the walker come by, and its visits repeat in a rhythm that doubles every time the hidden thread splits.

Near each split the walker takes a long time to make up its mind, so the rhythm keeps its old beat a little past the place where the picture has already changed.

And in one column deep in the fuzz, the walker comes back to exactly the same places every ninety-six steps: a small island of order in the grey.

the Loop Atlas

242

A field of seeds scattered across a page, thick on the left, thick in the middle, almost none in a broad pale band across it.

Look again and the pale band is where the page was folded: roll the page into a tube so its top meets its bottom, and the two thick patches of each field join into one wide belt.

Between the two fields runs a clean green rope in two strands, one sliding gently down, the other dropping fast, resting on a shelf, then dropping fast again.

Where the rope ends, it does not stop: its strands carry on through the seeds as threads only a little harder to see, three of them for a while, side by side, then tangled, then three again, until they pour through a funnel into a single orange line that slides away to the right, steep at first, then easy.

And one thread lies flat across the whole page at the same height, glimpsed in three places through gaps in the seeds.

the Loop Atlas

243

Someone else went over the same page of seeds with a ruler and joined every four that lay in a line: fifty threads.

Most run downhill toward the same corner, some lie flat, a few climb.

The flat ones are the surprise: there are only four heights they choose, and each height turns up again and again, on the left, in the middle and on the right, as if a set of shelves ran straight across the page behind the seeds, showing wherever the seeds thin out.

Near the right edge every thread, falling, flat or climbing, arrives at one doorway, and only one orange line comes out of it.

the Loop Atlas

244

The teacher took the same page and drew only six lines, all long.

Two drop in from the top edge and run down into the doorway.

One leaves the green rope where it ends and keeps going through the seeds to the doorway’s foot.

And one, lower down, is the green rope’s bottom strand again, drawn a step to the right: the same steep drop, the same shelf, the same last fall, only with a shorter shelf, as if the page had been printed twice with the second print slipped sideways.

Fifty short lines found everything.

Six long ones found what mattered, and he thinks one of the six is a trick of the light.

the Loop Atlas

245

From an aeroplane, two roads cross on the map below, and you would swear a car could turn from one onto the other.

Down on the ground one road runs over the other on a bridge, and no car ever turns.

The picture of seeds is the view from the aeroplane.

Every seed has a height the picture cannot show, and the threads that seem to meet may be passing each other at different heights.

The eye, wanting a crossroads, draws one.

the Loop Atlas

246

Two threads on the page seemed to be touching, and the guess was that one ran over the other, a bridge seen from above.

Lifted off the page into their hidden height, they were still touching.

They were not two roads at all but one road just after it forked, the two lanes not yet apart.

Only a finer eye, or watching which way each walker went, could tell the lanes apart.

And the count that was meant to settle it was off by one, because the ruler used for counting had its own coarseness.

the Loop Atlas

247

A student who has spent two days copying the teacher’s drawings packs the lessons into a satchel for the library: twenty-one of them, each pinned to the drawing that taught it, and a note on top saying which ones the student had half-heard before.

Then the teacher asks whether the student has any drawings of their own.

There is a drawer full: seven weights of the same swing, eighty-one settings each, eight walkers to a setting, every footprint written down with where it fell and how fast it was going, and beside each walker a word for whether it found a rhythm, wandered, or drifted smoothly round.

Nobody has looked at them with the new eyes yet.

The difference from the borrowed drawings is that here every dot can be asked who made it and when, so the guesses the teacher could only point at can be checked to the last footprint.

the Loop Atlas

248

Four wobbly lines run down a page, and under each one a second line drawn by a ruler that measures how fast the wobble’s bend is changing.

Wherever a wobbly line turns a corner, its ruler line throws up two humps, one on each side of the corner, one up and one down, like a wave passing under a rope, and between them it crosses the middle exactly at the corner.

All four ropes cross at the same spot, to a few thousandths of the gap between two marks on the ruler, as if they had agreed.

Further down the page the corners come closer together, each gap a little under five times shorter than the last, and at every corner the humps turn over: the rope that went up first at one corner goes down first at the next.

Near the bottom two corners sit so close that their humps overlap, and the crossing slides a little off its mark.

the Loop Atlas

249

A road runs straight across a field, and at one fence post it turns, not sharply, just enough that a walker would feel the slope change under their feet.

Watch how fast the walker climbs: it changes at the post, from one steady pace to another.

Watch how fast that pace changes: nothing, then a single swell around the post, then nothing.

And watch how fast the swell itself grows and shrinks: it rises, falls through zero right at the post, then dips, one small wave of a crest and a trough with the post at its heart.

Four roads cross the field side by side, and every one turns at the same row of posts, some bending up and some down, so their four small waves all pass through zero together.

At each row of posts further down the field the bends are about half as sharp as at the last, and they turn the other way.

Nothing here goes round in circles.

It is one bend, felt three times over by someone who keeps asking how fast the last answer changed.

the Loop Atlas

250

Three workshops stand on three different streets, and in each one someone watches a wheel through a slit in a fence, one glance every time a bell rings.

In the first the wheel is a compass needle; in the second, a ball rolling on a washboard; in the third, a planet going round its partners.

None of them knows the others are watching the same kind of thing.

Each keeps a notebook, and each ends up writing the same three words in the same order: how many turns between bells, where that count suddenly changes, and what is left over when the wheel comes back to where it started.

When they finally meet and lay their notebooks side by side, the words line up.

The only thing that differs is how far apart the bells were.

the Loop Atlas

251

Take a page of long, smooth threads, cut it into its columns, and shuffle the columns like a deck of cards.

The shuffled page is a scatter of short scraps, and there are far more scraps than the page ever had threads, because each long thread was one thing and the shuffle breaks everything into little pieces.

Count the pieces and you would swear the shuffle made more order; measure their length and the real page wins by a mile.

In the cloudy corner of the page, where the threads were already scraps, the shuffle looks almost like the real thing.

The first way of shuffling cheated a little: it let some scraps land on top of each other and vanish, so the shuffled cloud looked thinner than it was and the real one seemed to stand out.

Shuffled honestly, the cloud and its copy are nearly twins.

the Loop Atlas

252

A man and a machine spent a day walking round hills they could not see, carrying arrows.

By evening they had learned three things that are really one.

An arrow tells you how much hill you walked round, but only what is left over after the whole turns; to count the whole turns you must walk round the pieces one at a time, or grow your loop from nothing and keep count as it widens.

A number that seems to belong to a whole hill really belongs to one small place on it, the very top, the very foot, the steepest band, and you learn the hill’s character by looking closely there.

And a number you take for a law is sometimes only a place another number passes through on its way somewhere else; you find out by turning every knob.

In a drawer, a drawing from fifteen years before already had the hills, the rings and the empty middle on it.

the Loop Atlas

253

The tree grew one great branch, and most of the rain fell on it: a fifth of the pages and two fifths of the runs went to the needle that counts, and half of those to a neighbour’s machine.

The branch Bear planted himself, the ear that listens to a soup of waves, still has ten buds closed.

And the crown, the one place where the whole tree is seen at once, is the smallest thing on it.

the Loop Atlas

254

We turned the moving walkway around, so that it runs the same way the wind is pushing the walker.

Facing it the old way, upstream, the walker had found a whole staircase of little ledges to stand on, a half, a third, three-sevenths, every number in the tree.

With the walkway running downstream, the walker loses its first footing far sooner, as if the walkway were tugging at its heels, stumbles across three or four ledges in a hand’s width, and then simply steps onto the walkway and rides it, at exactly the walkway’s speed, to the end of the record.

Even the light walker rides it, the one who never once caught the walkway when it ran the other way.

The heavy walker rides it too, but at first with a limp, its stride repeating only every second section of the walkway, before it settles into the plain ride.

And the old walkway had been tugging all along: it pulled the walker upstream nearly as hard as its shaking loosened the walker’s grip, so the two almost cancelled and we never noticed the tug.

Run the walkway the old way and blow the wind backwards instead, and you get the same walk in a mirror, to the last digit.

Which way the walkway runs only matters compared with which way you are being pushed.

R2W

255

We went back to the short stretch of the reversed record where the walker lets go of the first ledge and the walkway carries it off, and looked eight times closer.

The whole staircase is in there, squeezed into a hand’s width of the dial: a seventh, a sixth, a quarter, a third, two-fifths, a half, one ledge after another, and on nearly every ledge the walker stamps its step out twice over, or three times, before it finds the plain rhythm.

Then the light walker did something it had never done facing the other way: it remembered.

Near the top, the half-ledge and the walkway sit side by side, and a walker set down at one spot stays on the ledge while its neighbour, set down a quarter turn away, steps onto the walkway and rides off.

And the old staircase, the one facing upstream, was never built on a straight line.

Its ledges hang on a curve that leaps up off the first ledge and then leans over, the way a ball thrown upward slows as it climbs; take the walkway away entirely and the walker still climbs that same curve, only with no ledges to stand on.

Put the walkway back and the curve shifts over to start from the new first ledge, and rises a little as well, as if the floor’s shaking made the climb easier, so the walker reaches nearly every ledge sooner than the shifted still-floor curve would let it.

Only the two widest ledges, the third and the half, sit exactly where the still floor’s curve says they should.

R2W

Note added 7 October, 06:58 UTC: read 64’s sentence that the light walker “remembered”, “something it had never done facing the other way”, overstates it: facing the old way, the same walker already keeps three hairline memories (RECORD-SURVEY at weight 0.40; the prereg’s correction of that time). Read 64 is kept as written; read 66 reads the corrected picture. (The chat’s copy of read 64 printed “still-floor curve” in its last sentence; the wording here is the record.)

256

We turned the floor’s shaking up one notch at a time and watched where the walker first lets go of the bottom step.

That part behaved like a ruler: a little shaking and the walker holds on a hair longer, more shaking and it holds on longer still, each time within a hair of where the pencil sum said it would.

The climb after it did not behave like a ruler at all.

With a gentle shake the walker climbs exactly the curve it climbs on a still floor, no faster, no slower, half its steps above the curve and half below.

Double the shake and the whole climb rises above the still-floor curve, and rises more again at three times the shake, as if something had switched on between the first notch and the second rather than grown with the dial.

And the rise is not even: where the wide half-step ledge sits, the walker is held right down on the still-floor curve, and the rise lives on either side of the ledge, in the stretches where the walker is free to run.

R2W

257

We set lighter and lighter walkers on the reversed walkway, four of each, from four different spots.

The lightest ones never argue: wherever you set them down, they climb the same staircase, step for step, and at a quarter of the weight the staircase is the whole ladder from the floor to the walkway, a fifty-seventh, an eighth, a sixth, every ledge in order, squeezed into a hand’s width of the dial.

Only when the walker is heavy enough does the walkway reach down far enough to lie alongside the half-step ledge, and there, and only there, two walkers set down a quarter turn apart end up in different places.

Then we went back to the old staircase, facing upstream, and found that walkers of that same weight already disagree, but only at three single hairs of the dial, each time choosing between two neighbouring numbers.

Turning the walkway around did not give the walker a memory it lacked.

It gave the memory room.

R2W

Note, 7 October, 08:06 UTC (sitting B8, fifth exchange): REV-WEIGHT-2 measured the reversed walkway at weights 0.26 to 0.40, on the same grid and with the same four starts as the forward record. Facing this way the walker remembers from weight 0.30; facing upstream it does not until past 0.35. Read 66’s last two sentences hold at 0.40 and fail at 0.30 to 0.35, where turning the walkway around does give a lighter walker a memory it lacked (the prereg’s correction of 08:06 UTC). Read 66 is kept as written; read 67 reads the new picture.

We turned the picture on its side.

What we had been calling a walker’s step is really a small ring it runs around, and where on the ring it stands each time the wave’s crest goes by is a number the old pictures threw away.

Seen that way, a walker that has found its ledge is a thread pulled taut through the dial, one thread for a whole-step ledge and two for a half-step, and a walker that never settles is a ring of beads strung all the way round.

The lightest walkers we tried this time already had something wrong with them: a few, standing in the cracks between two neighbouring ledges, never ran their ring the same way twice, and a walker nudged by a hair drifted away from its twin, though at every spot on the dial the four walkers still chose the same ledge or the same crack.

Add a little weight and the arguments begin, and they always begin between neighbours, three fifths against two thirds, two fifths against a half, never between strangers.

Add more and the walkway itself reaches down the stairs and swallows the ledges from the top, thirty-two of them, then twenty-two, then thirteen, until only the half-step and one other are left and the walkway lies alongside the half-step ledge for most of its width.

Every walker’s map of where it goes next has the same shape: a long flat shelf, where almost any spot on the ring sends it to nearly the same next spot, and one steep cliff.

For the wildest walker the shelf tilts downhill, so that three spots far apart on the ring all send it to one place.

R2W

We stopped riding one road and flew over the whole map for a night.

Along the bottom of the map runs a shoreline, where the walker first lets go of the ground floor, and it is nearly straight: a hair above the line marked one for most of the way across, and then, two thirds of the way over, at the same spot for the light walker, the middle one and the heavy one, it dips below the line and keeps sliding down toward the right-hand edge.

Far up the map, where the walker runs free, every walker keeps almost exactly the beat the map’s angle gives it, always the same small step behind, as if a fortieth of the music were always lost.

In between, the map is crowded with ledges the old road never touched, some of them above a whole step: three halves, two, three.

The heaviest walker we have so far keeps two minds only along the left part of the shoreline: set down on its feet it holds the ground floor, set down on its head it runs off, and both are true at the same spot.

Past the two-thirds mark the two minds become one again, exactly where the shoreline dips.

R2W

260

We set a walker down at every degree of the ring, three hundred and sixty of them, like Bear’s tires, at ten notches of the dial.

Not one of them wandered.

Every walker ended in one of two places: on the half-step ledge, stepping out its two-beat, or on the walkway, riding it.

The walkway’s share of the ring grew at every notch and never gave any back, from a seventh of the ring to five-eighths, and at the tenth notch it took the whole ring.

Where the two meet there is no clean border.

Look a hundred times closer and one stretch of ledge breaks into stripes, ledge, walkway, ledge, a single hair of walkway alone in the middle of the ledge, and closer still there are more stripes.

Halve the distance between two walkers and they disagree only a third less often, where across a clean border they would disagree half as often; every border we looked at, at three notches of the dial, is the same coastline to within a few hundredths.

Then we went back to lighter walkers.

From the lightest we tried up to a weight of a quarter, every walker that never settled still ran its ring in order, each spot sending it a little further along and no two spots trading places, and none of them shook or remembered.

Between a quarter and a fiftieth more, the wanderers fell from one in five to one in twenty-five, one walker in fifteen began to shake, and two of every three shakers folded their ring over, so that spots far apart on it sent them to the same place.

The memory came a little later, by a quarter and a hundredth, and the staircase that runs backwards a little after that.

R2W

Note, 7 October, 19:16 UTC (sitting B8, ninth exchange): MAP-K drew the walker’s whole rule and found it first runs backwards at weight 0.247 to 0.251, tilting with the dial, so at most of the dial a weight of exactly a quarter is already past the line. Read 69’s “From the lightest we tried up to a weight of a quarter … none of them shook” fails at a quarter itself, where two walkers already shook a little on a rule that runs backwards; its sentence about a quarter and a fiftieth more still describes the counts. Read 69 is kept as written; read 70 reads the new picture.

261

We stopped watching the walker and drew the whole rule it walks by: for every spot on the ring, where it will stand one wave later.

For the lightest walkers the rule runs forward everywhere.

Heavier, the flattest stretch of it sinks toward level, and at a weight of almost exactly a quarter, a hair lighter at one end of the dial and a hair heavier at the other, it touches level and tips backwards.

From there on the walker can shake.

Two walkers we had written off as a trembling ruler were already standing on the backwards stretch: the ruler was right and our eyes were wrong.

Then we climbed the hill in the rule of two dancers, one that comes round every seven waves and one every two.

From far off each hill is lopsided, a long gentle slope on one side and on the other a drop that even folds back on itself, and on the two-beat dancer's hill one side runs as straight as a tent's for a long stretch.

Within a thousandth or two of a turn of the summit, both tops are round as a ball.

Each time a dance doubles, it passes the summit about two and a half times closer.

Read on the dial we turn, the two-beat dancer seems to know the universal rhythm from its first doubling and the seven-beat dancer has to learn it; read on the height of each one's own hill, both have to learn it, and the two seem to learn at one pace, set by how far inside the round top they pass.

R2W

262

Then we drew the same rule with the walkway running the old way, upstream.

Here the rule holds out far longer: the walker has to be a third again as heavy before its rule tips backwards, a third of a unit at the low end of the dial, and the line climbs as the dial turns, slowly at first and then steeply, until near the top of the dial even four tenths is not heavy enough and the rule runs forward everywhere.

Run the walkway downstream and the line hardly cares where the dial sits; run it upstream and the dial bends it five times as hard.

And every walker we could check that shook, fifty-seven one way and five hundred and seventy-four the other, stands on the backwards side of its own line.

Nobody shakes on a rule that only runs forward.

R2W

263

We stopped asking where the walker ends up and asked what it sees.

A walker standing still on the standing wave sees the moving wave go by as a steady rocking, once a pass, and nothing else.

A walker riding the moving wave sees that wave hold perfectly still beneath it, and now it is the standing wave that rolls past.

A walker caught on a ledge between them, looked at once a pass, never seems to go anywhere: the waves carry it round and set it back down in the same few places, so to a watcher who blinks once a pass the wave is invisible.

Set three hundred and sixty walkers round the ring and each one’s starting place decides which of these lives it gets: at one notch of the dial, some begin on the half-step ledge and their neighbours a hair away ride the moving wave, and the border between the two ways of seeing is a coastline that never smooths out, however close you look.

R2W

264

We drew the walker’s rule again at the heaviest weight, sixteen times finer, hoping the jagged peaks would smooth out.

They did not; they grew.

The steepest backward drop jumped from one notch of the dial to the next, fifty-six, nineteen, seventy-two, twenty-one, ninety-seven, and up to two hundred and eighty, always at the same spot on the ring, about a seventh of a turn round.

Then we looked at the walkers who arrive at that spot, and found they do not all arrive the same way: some come in drifting gently forward, others come sliding backward fast from a different part of the ring, and each kind goes on to its own next place.

A watcher who notes only where a walker stands sees two futures leave from one spot, and draws a rule with a cliff as steep as he cares to look; a watcher who also notes how fast the walker is moving sees one smooth path folded back over itself, like a ribbon laid across its own end.

The peaks were never in the walk.

They were in the watcher’s single axis.

And the share of the ring where the rule runs backwards did not jump at all: three quarters of the ring, at every notch.

R2W

265

A friend wrote to warn us that a clock which ticks in steps can invent a cliff where the ground has a crease, and that ticking faster will not make the invented cliff go away.

So we ran the walker’s clock at half the ticks, the usual ticks and twice the ticks, and watched the two places where the walker’s path folds back over itself.

They did not move: from the usual clock to the fast one they shifted by a hundred-billionth of a turn, and every time we doubled the ticks the shift shrank sixteenfold, as it should when the ground has no crease at all.

The fold is in the walk, not in the clock.

The ground our walker crosses is smooth everywhere; it is only the walker’s way of crossing it that doubles back.

R2W

266

Turn the dial slowly at the lighter weight and the walker climbs a staircase toward the walkway: half a step, then two thirds, three quarters, four fifths, five sixths, six sevenths, seven eighths, nine tenths, twelve thirteenths, and then it steps onto the walkway for good.

The steps do not come evenly.

Each one is narrower than the last, and they crowd together at the walkway’s edge by one rule: how far a step sits from the edge goes as the square of how much the walker still falls short of riding.

Between the wide steps, where the walker cannot make up its mind, it scatters up and down the ring, and those are the upright stripes Bear saw.

At the heavier weight two walkers on different ledges slide down the ring at the same rate as the dial turns, like two cars in neighbouring lanes of one highway, and lower down the half-step walker begins to stutter, two beats becoming four, four spreading wider, until it scatters too; and by then one of its four beats has walked into the short stretch of ring where the walker’s path folds back over itself.

R2W

267

Close to where the walkway begins, the walker has already found the spot where it will stand.

It does not stand there yet; it lingers.

For a long run of waves it hovers beside that spot, then steps away for two waves, and comes back to linger again.

Turn the dial a hair and each stay grows longer, and the nearer the dial comes to the walkway’s start, the longer the stays, until at one notch the stay never ends and the walker is on the walkway.

The walkers who cannot settle into any pattern do the same thing more loosely: half their time beside that spot far back on the dial, nine tenths of it just before the walkway begins.

The crowd Bear circled in red is not a crowd at all.

It is one walker waiting at a door that has not opened yet.

Farther back on the dial, just past where the half-step pattern ends, the same thing happens around the two spots the half-step walker used to stand on: the new walkers keep visiting the old places, as if the house were gone and the doorsteps still drew them.

R2W

268

Every walker, light or heavy, leaves the half-step pattern through the same door: at the same turn of the dial and from the same two places on the ring, as if the door were painted on the floor and not set by the walker at all.

The walkway’s door is different.

The heavier the walker, the earlier on the dial that door opens, sliding back notch by notch as the weight goes up, and the place on the ring where it opens creeps upward too, except near the heaviest walkers, where it dips once before going on.

And the walkway’s slant is nearly the same for every walker once you multiply it by the walker’s weight twice over, but only nearly: it leans a little steeper for the middle weights, as if the walkway sagged in the middle.

R2W

269

Start a very heavy walker in step with the walkway and, over most of the map, it rides it up to the strongest push on the chart; start the same walker half a turn late and it never gets on, not at any push we tried, anywhere on the map.

A walker less than half as heavy, started late, does get on, a little later than its on-time twin, and the gap between the twins narrows as the dial moves right, until at the map’s right edge they let go within a hair of each other, as if their two stories were about to become one just past where the map ends.

R2W

270

Make the walker heavy enough and even the walkway stops being simple.

Riding it, the walker starts stepping twice to every wave, then four times, and lower on the dial it loses its footing altogether; the heavier the walker, the higher on the dial the stutter begins, until for the heaviest walkers we tried the plain one-step walkway has left the map and only the stutter is on it.

And the half-step door, which had looked painted on the floor, turns out to sit on top of a gentle hill: step lighter or heavier from the middle weights and it slides down the slope on either side.

R2W

271

Light walkers who settle into a pattern walk it like a star drawn without lifting the pen: every step jumps the same number of points round the ring, so the pattern is a polygon, a little squashed but never scrambled.

The lighter the walker, the fewer of them settle at all; just before the line where the ring can fold, three in four have settled, and every one of those still draws a clean star.

Cross the line and the stars begin to tangle.

A few walkers skip one point and make it up on the next; some draw their polygon with every corner split in two and visit the split corners out of turn; and the heavier the walker, the fewer clean stars are left, until almost none remain.

R2W

272

The walker who keeps a fifty-seven-beat rhythm is not keeping a new rhythm at all.

It is two old ones taken in turn: wait twenty-six beats, step out and back, wait twenty-seven, step out and back, and around again, its two parents’ habits played one after the other.

Even its waiting is not still.

Drawn on a clock face, its waits are a short arc where it creeps forward a hair each beat, and its trips are two long strokes across the face and home.

The other walkers we timed near the walkway’s door do the same: their waits differ by one beat at most, spread as evenly as the steps of a slanted line drawn on squared paper.

R2W

273

The walkway is a bridge, not a road.

It rises steeply out of the crowd at the near bank, where the walker first steps on; it flattens across the middle, where for a long stretch of the dial the walker’s place on the ring slides only a tenth of a turn for every whole unit the dial moves, and that gentle middle is nearly the same from the lightest walkers to the middle weights; then it steepens again and drops the walker off the far bank into open drifting.

Heavier walkers get off sooner.

And the bridge is not half a turn long for everyone: the lightest walker crosses about half a turn of the ring, the heavier ones more, as if weight stretched the span.

R2W

274

Every walker leaves the walkway at the same place on the ring, right at the top, where the moving wave pushes hardest; when the dial asks for more than that top can give, the walker drops off, light or heavy alike.

Where the walker gets on is different.

The lightest walker steps on exactly opposite, at the bottom of the ring, half a turn from where it will leave, the way a locked dancer is meant to.

Each bit of weight moves that boarding place further round, at first about a twentieth of a turn for every twentieth of weight, then more slowly, until past the folding line the walker boards three quarters of a turn from the exit.

So a heavy walker’s bridge is longer not because the far bank moved much, but because it climbs on from further back.

R2W

275

Weaken the hand that holds the walker on the walkway and the walkway shrinks from both ends, by just the amount a simple rule says it should: the walker is asked for a little more effort with every turn of the dial, the big slow wave quietly gives back a share that thins as the dial turns, and the small moving hand can make up the difference only up to its own strength, no further.

Halve the hand and the walkway is half as long and nearly three times as steep.

The walker still lets go at the very top of the ring.

But with a weak hand it no longer climbs on at the very bottom; it climbs on partway up from the bottom, and the bridge it walks is shorter than half a turn.

And heavier walkers feel less of the big wave’s help, by about the amount a weight resists being shaken.

R2W

276

A hundred walkers, one placed at every hundredth of the ring, are let go at the same turn of the dial, and each ends up on one of two paths, never a third.

Which path depends only on where it started.

Most of the ring sends walkers to the half-step path; one long arc of starting places sends them to the walkway, with a few lone spots and a small second arc doing the same.

Turn the dial a notch and the walkway’s arc grows by about the same amount each time, so the chance of the walkway climbs in a straight line, a little under three hundredths for every notch, from about one walker in eight to about five in eight.

Then, at the last notch, the half-step path’s share collapses, and four in five go to the walkway.

The coin is real, and the dial is its thumb.

R2W

277

The coin is born already weighted.

At one notch of the dial no walker reaches the walkway; at the next, one in eight does, most of them from one stretch of the ring.

It dies the same way at the other end: at one notch three in eight still take the half-step path, at the next none do.

In between, the odds are steady and smooth, and the dial moves them evenly.

But look closely at where a walker starts and the coin stops being simple.

Ten times more starting places, and the long stretches stay put, the odds barely change, yet at their edges the ring breaks into scattered single places where one walker goes one way and its neighbour the other, about three times as many of them as before.

The coin’s odds are a dial you can trust; the single throw near an edge is not.

R2W

278

Look at the line between the two faces through a stronger and stronger glass and it never becomes a clean line.

At every magnification there are more places where one walker goes one way and the walker next to it the other, not as many more as a smooth edge would give, but never none: the border is a coastline, not a fence.

And just past the walkway’s door, for a sliver of the dial, a third path opens for about one walker in six: a three-step dance, which turns into a six-step dance and then a twelve-step dance as the dial moves a hair, before the sliver closes and every walker takes the walkway again.

The coin has a third face, and that face has its own little ladder.

R2W

279

The third path does not open like a door; it surfaces like a whale and goes back down the same way.

Coming up the dial, a few walkers first find a nine-step dance, then lose their footing and stumble, then settle into a six-step dance, and only then into the clean three-step dance, which holds for most of the sliver.

Further on, the three-step splits into six, the six into twelve, the twelve into twenty-four, each split coming sooner than the last, until the walkers stumble again, the path closes, and every one of them takes the walkway.

The splits come faster, but not at the steady rate the textbooks promise: the first gap is somewhere between five and eleven times the next, not four and two thirds.

And through all of it, dancing, splitting and stumbling, every walker on the third path takes exactly two turns in every three passings.

Even its stumbling keeps count.

R2W

280

Before the coin there is no coin.

At the lowest notch every walker, wherever it starts, takes the same three-step path.

A notch up they lose it and stumble, and their stumbling keeps no count: a third of a turn per passing, then nearly a half, then a little more than a half, as if the crowd were running ahead of where it is about to land.

Then all of them fall into a four-step dance that is two two-steps, and only then into the plain two-step, the coin’s tails.

Heads arrives a few notches later, from one stretch of the ring.

Tails is born exactly the way the third face is born: out of a stumble, through a doubled dance, into the plain one.

But the stumble before tails is a crowd running at different paces, and the stumble inside the third face is a crowd that keeps step.

R2W

281

Every dance, whatever its fraction, stops in the same three places.

A two-step, a three-step, a seventeen-step: each one puts exactly one foot down just past a tenth of a turn, one foot just short of nine tenths, and the rest mostly in a broad band below that.

Between a fifth and a half of a turn hardly anyone ever stands; it is the stretch everyone crosses at a run.

The two-step’s two feet are not alike either: one walks steadily down the dial in a straight line, the other lurches at the start, strolls through the middle and lurches again at the end, the same lurch-stroll-lurch as the walkway itself.

And the walkers who never settle are not uniform smears: as the dial nears the place where the walkway will appear, they spend more and more of their time standing exactly where it will be, a crowd gathering on an empty platform for a train that has not yet come.

R2W

282

Light walkers forget where they started.

Set a light one going from the top of the ring or from the bottom, at the same pull and the same pace, and it ends on the same path almost every time: two cells in ten thousand disagree.

Make the walker heavier and the disagreements spread, slowly, then all at once: one cell in a hundred, one in ten, one in five, and then, from three times heavier on, about one in three, and no further, however heavy it gets.

What the heavy ones remember changes too.

A light walker that remembers remembers which path, this groove or its neighbour.

A heavy walker remembers something blunter: whether it settled at all.

Pull harder than the strong wave can ever hold, and nobody, light or heavy, remembers anything.

R2W

283

The walkway has an eye.

Come down the dial toward it and the single walkway splits into two paths that bow apart around the place where it used to run, wide in the middle, closing again as you near the walkway’s start, and closing the way a zipper closes, fast at the end.

The eye has its own eyes: further down, each path splits again, and again, until the walkers lose their footing.

Beside the eye, for a stretch, a walker’s starting place decides whether it takes the eye or the two-step: two roads open at once.

And when a three-step splits into a six-step, it does not split evenly.

Two of its feet step apart; the third barely moves, as if it were the one holding the rope while the others swing.

R2W

284

Lay a net over the picture and every path becomes a rhythm.

Walk along the walkway hole by hole and it says level, level, level, eight times, then one step down; level nine times, a step down; nine more, a step.

The beat never wavers by more than one: that is what a straight road sounds like.

Walk the top of the eye and the beat speeds up, seven, eight, three, three, one, one, none at all, a drum roll into the pinch.

Walk its bottom and you hear a slowing fall, then a rest, then the steps turn upward and quicken again.

Nobody measured a slope.

The rhythm is the slope, the change in the rhythm is the bend, and the change in that is the shape.

R2W

285

A teacher drew lines where there were no dots, and the dots agreed with him afterwards.

Where the two-step’s top foot splits, he drew three roads, not two: up, down, and one straight on between them, the road the walker would take if it could still stand there.

Where the walkway’s eye opens to the left, he counted six roads coming out of it, three from each lid, and the dance found further down the dial has exactly six pairs of stops, three by each lid.

Where the three-step splits, he said one foot splits evenly, one barely moves, and one keeps level while its twin slides down a ramp, and that is what each foot did, to the third decimal.

The walkway’s tree opens to the left; the two-step’s and the three-step’s open to the right.

The same tree, he said, only opposite.

R2W

286

A crowd waits at a door that is slowly opening.

Before it opens, the crowd takes turns: seven get through for every eight tries, then eight in nine, then nine in ten, each new rhythm holding for a shorter stretch than the one before, and between any two rhythms a third one squeezes in, as if the crowd were counting in fractions.

Look at where they stand and most of them are bunched at one spot by the door, more of them the wider it opens: three in eight at first, forty-eight in fifty-five at the end.

Then the door is open, everyone walks through in step, and the first one through stands exactly where the crowd had bunched.

From the hallway side you can see it too: the line of people already in step bends sharply back toward that spot, as if pointing at it.

Two people who never met, one counting the crowd’s turns and one measuring the bend of the line, name the same moment for the door to open, to a millionth of a turn of the dial.

And across the room a slow walker hurries at the start of its path and at the end, and dawdles in the middle exactly on a floor mark that walkers in three other rooms also stop on.

R2W

287

The whole map is filled in now, every square, and on it there is a level line drawn straight across a third of the way up.

Below the line, a heavy traveller who starts out standing still stays standing, and one who starts out walking keeps walking; ask either where it ends up and the answer is where it began.

Above the line, everyone walks, whatever they started as, and nobody remembers anything: not one square in six and a half thousand.

Light travellers are different.

They forget almost everything, and the little they do remember is which path they took, not whether they moved.

Halfway between, a middling traveller remembers below the line too, but only down to a certain height, and that height was guessed beforehand from an old rule for heavy swinging things, to within a few steps.

And on the line itself, exactly, two squares in three remember, whatever the traveller weighs.

R2W

288

Two dancers spin as a couple while a heavy third circles the floor.

Each time the heavy one sweeps closest, someone notes how stretched the couple is.

Far from the sweet spot, the sweep lands anywhere in the couple’s turn, so the notes fill a whole column, piled against a ceiling they never pass.

In the sweet spot, the sweep always catches them at the same point in their turn, so the notes stay near the top and rock gently, and over six hundred sweeps the rocking quietens.

At the edge of the sweet spot they rock harder and quicker, and now and then the heavy dancer catches them more stretched than they’ve ever been.

Draw the notes against the couple’s turn and you get a valley with two steep walls.

Walk down either wall, and every few steps a path breaks off sideways and wanders out slowly, as if the walls had ledges.

R2W

289

A clock with two hands that never quite agree draws, over many turns, the skin of a donut.

Look at that skin from the side and it becomes a valley with walls, and the threads that wrap it show up as ledges, one wall answering the other at every height.

Stop one hand and the whole donut shrinks to a thin ring, all its story moving into the hand that’s still turning.

Where two hills sit on the skin, there’s a pass between them, because there must be, and a tired eye looking down at it sees a jellyfish.

R2W

290

Where a road runs alone, two travellers who never compared notes walk it stone for stone.

Where many roads cross a busy square, one traveller walks straight through, but the other must stop at every corner to ask which way, and comes out the far side as six short walks.

Both end up at the same gate.

Off to one side, a path the first traveller felt was there runs through an empty field: half of it has footprints, and half is just the shape a path should have.

The empty spaces between the footprints have only three sizes, if you know how to look.

R2W

291

Open the donut flat and a swinging thing leaves its marks folded at the two ends of its swing, where it slows and turns.

That’s where the little curled tails pile up, layer on layer.

Far out, where the swing nearly goes over the top, a few marks run off straight across the page and come back from the other side.

In the quiet middle the marks fall into a woven lattice of short dashes.

Underneath it all, something too slow to see in one sitting lifts the lower marks and sets them down again.

R2W

292

A spinning top wobbles fast, and the wobble isn’t even: early on it dips hard and lifts gently.

Nobody touches it, its axle never moves, and yet by the end it dips gently and lifts hard.

The lean of the wobble has rolled over.

Something much slower than the wobble is turning inside it, too slow to see in one sitting but plain once you lay a straight bar through the middle and compare the two sides.

R2W

293

A weaver works five threads at once and lays one stitch per beat, going round the five in turn.

Look at the cloth and you never see two stitches laid one after the other side by side; neighbours on the cloth were always laid five beats apart.

Each thread wanders slowly up or down, so the cloth shows long thin lines crossing and leaving small white gaps between them.

Step back and blur your eyes, and the gaps outline the threads better than the stitches do.

R2W

294

At first the marks look like a few dozen threads with gaps between them, and you argue about whether they’re a weave or a donut.

Wait ten times longer and the gaps fill in, the threads vanish into cloth, and the cloth turns out to have edges as clean as a pipe’s.

The pipe pinches in the middle, twists as it goes, and breathes in and out on a slow count of nearly seven hundred.

The threads were never the thing; they were the loom still filling in the pattern.

R2W

295

A crowd stands around a fountain.

Most of them gather near the brim, pressed against the rail, where the view is best.

A few drift down the steps, and only now and then does anyone cross the bottom of the plaza, quickly and all in one go.

Every path in and out funnels through one narrow arch, and the crowd’s two edges hold their shape all afternoon, though the people in them keep changing.

R2W

296

Two staircases climb either side of a hall, joined by a narrow landing at the bottom that people rarely cross.

Late arrivals seem to favour the left stairs and the middle of the day the right, but the evening crowd stands on everyone else’s footprints, so you can’t tell from the photograph alone.

The handrails stay where they are all day.

R2W

297

A long shelf runs across a wall.

Anything set down on the shelf stays there, and anything thrown along it keeps sliding off the end.

Above the shelf nothing can rest; below it nothing can slide.

Toward the right-hand wall the shelf narrows to a point and vanishes.

At one spot near the left, a thin ladder of resting places climbs well above the shelf, as if someone had hammered in pegs where no peg should hold.

R2W

Note, fifty-first exchange: the “ladder of resting places” in read 107 is the conveyor (a ball dropping one ledge per tick), not a resting ball; colouring by denominator painted both the same orange. The read stays as given.

298

A drummer walks past a row of people tapping their feet, each to a song of their own.

Wherever someone’s own tempo already falls on one beat of the drum, or two, or three, they quietly switch to the drum and keep its time; everyone else carries on with their own song.

Nobody is told to follow.

The ones who follow are exactly the ones who were already close, and their footprints line up along a curve you could have drawn before the drummer arrived.

R2W

299

Three windows.

In the first, four strings of beads slide down a staircase of powers of ten, each bead a tenth or a fifth as high as the one before; the blue and the green strings lie on top of each other for the first half of the slide, then the green one eases off and joins the orange, while the red string goes down at a gentler slant from its very first bead.

In the second, two dashed ceilings, one high and one low.

The blue string climbs to the high ceiling and runs along it to the end without a wobble; the red string starts just above the low ceiling, dips under it once and then lies on it; the orange string touches the high ceiling twice and then drops, in three long steps, to the low one; the green holds the high ceiling twice as long before dropping the same way.

In the third, a few dots: one alone at three, two at four, and then a crowd pressed into a thin column against the right-hand edge.

What it is doing: something splits in two, then in two again, and each split comes sooner than the last by a fixed factor.

When nothing is lost along the way the factor is the larger one, and it never changes.

When a little is lost, the thing keeps the larger factor for a while, as if it had not noticed, and then, all at once, it takes the smaller factor and keeps that.

the triple-star thread

300

Two windows.

In the first, six strings of beads start together near a high ceiling, and five of them fall away to the floor one after another, each leaving one step later than the one before it, so that the falls look like the same fall printed five times and shifted along.

In the second, the same five strings are laid against a different ruler, how much is left after one full trip, and they lie on top of one another on a single slide: flat at the top, a steep drop between “most of it left” and “a hundredth left”, and flat along the floor after that, so close together that the five colours are hard to tell apart.

What it is doing: a leak too small to notice on one trip is noticed once the trips have doubled enough times.

The thing does not care how small the leak is per step, only how much a whole trip has lost; halve the leak and it falls one split later, in exactly the same way.

the triple-star thread

301

Three windows.

In the first, a ladder of dot columns marches to the right at even steps.

The first column holds one dot, the next two, and from then on each column holds a few small clumps at the same heights as the column before, as if every new column were the last one copied and each of its dots split, very slightly, in two.

In the second, three staircases run down a wall of powers of ten: a grey one at the gentlest slant, a red one a little steeper, and a blue one steepest of all, crossing the other two partway down.

In the third, five beads sit on one straight slanted line, evenly spaced, the line passing through every bead.

What it is doing: things that halve and halve again crowd toward one point, and drawn on the right ruler the crowd opens into an even march.

A polygon growing toward a circle marches the same way, at a gentler step.

And how soon a slow leak shows depends only on how many times it has doubled.

the triple-star thread

302

Two windows.

In the first, two strings of beads slide down a wall of powers of ten in straight lines, the orange one twice as steep as the blue, so the orange falls two floors for every floor the blue falls.

In the second, a spiral of rings drawn half a ring at a time, each half smaller than the last by the same step, with the beads sitting alternately left and right of the centre on one horizontal line, closing in from both sides like the teeth of a zip.

What it is doing: each time the thing splits, the new pair lands on the opposite side from where the last pair was, a little closer in, always closer by the same share.

It is walking a spiral, but it is only ever seen at the moments it crosses one line, so it looks as if it is hopping back and forth.

And the second, steeper string is not a second rule: it is the first rule, bending.

the triple-star thread

303

Three windows.

In the first, six coloured ropes hang across a frame.

Four of them dip into the same valley, each a little deeper and a little further to the right than the one before, their floors reaching below a dashed line; the last two hang high above everything, with no valley at all, as if the floor had been pulled away.

In the second, four blue beads walk down a grey ramp beside a dashed guide, a little quicker than the guide toward the end, and stop short of a dotted red floor.

In the third, green bars stand at the left and grow taller, then vanish all at once, and red and orange bars shoot up where they were.

What it is doing: a thing held in a bowl rocks faster as the bowl is shaken harder, two, three, nearly four times quicker, and then, just before it rocks fast enough to start skipping every other beat, the bowl is gone.

Either it shattered, or it shrank to a dimple too small for the comb that was looking for it.

the triple-star thread

304

No picture this time, only three rows of numbers.

In the first, sixteen numbers climb in even steps and step clean over a narrow gate without one of them landing in it.

In the second, sixteen climb in tiny steps, and four of them walk through the gate one after another, each a little higher than the one before.

In the third, four numbers in the middle stand exactly on the gate’s centre line, perfectly still, while their neighbours climb past on either side.

What it is doing: a gate cannot tell a thing that pauses from a thing that is only moving slowly.

To know that something stopped, you have to see that it did not move while it was in the gate.

the triple-star thread

305

Two windows.

In the first, a blue thread runs perfectly level along a grey ribbon at the top for most of the width, broken in the middle where its beads are simply missing, then bends down at the right.

A green thread and an orange thread start lower; the orange one sits on the ribbon for exactly one bead and then drops off a cliff.

A red thread is promised in the legend and never appears.

In the second, the same beads are laid out by how far each one swings: the blue beads on the ribbon sit there at five different swing sizes, small and large alike, while every bead below the ribbon sits at a height of its own.

What it is doing: some swingers, whatever their size, keep exactly one beat in three, as if held in a notch; the rest swing at their own rates.

The notch is found from one doorway only.

From the doorways a little to either side, the swingers walk past it, and from the doorway across the room nobody is left swinging at all.

the triple-star thread

306

No picture this time, only a handful of numbers.

Two instruments listen to the same rocking.

One hears 0.0904, the note the thing actually plays.

The other listens through a second microphone set where the echoes are loudest, and hears 0.0706, a ghost note one beat below.

What it is doing: a voice heard through a wall arrives with its echoes, and the loudest echo can be taken for the voice.

the triple-star thread

307

Two windows.

In the first, a line of beads runs perfectly level along a grey ribbon across the middle of the frame, straddling an orange post.

At both ends it frays: a few beads dip below the ribbon and climb back, two drop far down, and red crosses lie on the floor where beads should have been.

In the second, the floor is covered in red crosses from end to end, and only three beads sit up on the ribbon, evenly spaced, the last of them touching the orange post.

What it is doing: from a doorway in the middle of the wall, everything that steps through falls into the same notch, whichever way it leans, until it leans too far.

Move the doorway a little way down the hall and the notch is still there, but only as three stepping stones in a flood.

the triple-star thread

308

Four windows.

In the first, a blue string of beads climbs from the lower left, gently at first and then steeper and steeper, straight through a dotted line a third of the way up without a pause, and stops just under a second dotted line near the top; an orange string follows it bead for bead and then leaps over the top line at the end; at the right, two thin brown posts stand where no beads are.

In the second, dots of many colours lie in level bands, one band per colour, each a little higher than the one before; the lower bands lie flat, the upper ones tilt and scatter, and the highest, bright green, crowd up against a ceiling.

In the third, a green line steps down from seven to four, lifts to five for a moment, then falls away to one.

In the fourth, a purple line walks steadily downhill and then turns sharply up, while a grey dashed line lies perfectly flat beside it and then dives.

What it is doing: a thing resting at the bottom of a bowl trembles faster as the bowl is tipped, passes the one-in-three mark without noticing, and just as its trembling reaches one shake every two beats, the bottom of the bowl stops being a bottom.

The resting place moves off sideways, and one step later the bowl is gone.

the triple-star thread

309

No picture this time, only six rows of numbers.

In the first two rows a thing takes no notice of whether a beat is odd or even: nothing at all, to four places.

In the third a faint lean appears, the same way in both halves of the run.

In the last, one thing steps to one side on every odd beat and to the other on every even beat, by the same amount in the first half of the run and the second, nearly half its whole swing given over to the alternation; its neighbour steps the same way but changes feet in the middle.

What it is doing: something that used to stand still in the middle has started to march, left, right, left, right.

the triple-star thread

310

Two windows.

In the second, seven hills are drawn over one another, all rising to the same summit: one fat and flat on top like a loaf, one a smooth dome, and the rest steeper and steeper until the last are plain triangles with a needle point; one of the triangles has had its point sanded round.

In the first, each hill leaves a trail of beads.

The loaf’s beads leap up and then run level, high above everyone; the dome’s run level along the dotted line; the steep hills’ run level lower down, each at its own height from the very first bead.

Only the sanded triangle behaves differently: its beads start near the floor, climb, and by the fifth bead have joined the dome’s.

A row of black crosses, someone else’s beads, climbs beside them, more slowly.

What it is doing: how quickly a ball’s bouncing splits and splits again depends on the shape of the hilltop it rolls over.

A ball on a sanded point believes it is on a needle until its bounces grow small enough to feel the sanding.

the triple-star thread

311

No picture worth the name this time, only rows of numbers.

When nothing turns underneath, a swing goes higher and higher until it starts to skip every other push, then every fourth, every eighth, every sixteenth, each new skip arriving sooner than the last by the same steady factor, as it always has.

Put the swing on a carousel and let the carousel turn, slowly or quickly, and the swing stops listening to its own pushes.

On a quick carousel it never reaches its first skip: before it can, it falls into step with the carousel and stays there.

On a slow carousel with the gentlest wobble it reaches the first skip exactly where it would have on solid ground, and the very next moment, while it is still finding its new rhythm, it falls into step with the carousel instead, swinging at exactly half the carousel’s turn per round.

What it is doing: a thing that would split its rhythm in two, and two again, has to do it on ground that is itself turning, and the ground’s rhythm is the easier one to catch.

the triple-star thread

312

Rows of numbers again, and this time they tell a story with two characters.

On still ground the swing splits its rhythm in two, in four, in eight, each split sooner than the last by the steady factor.

On the turning carousel the swing that has just split falls into step with the carousel at once, rocking exactly half as fast as the carousel turns, and it stays in step all the way to the end of the ride, never splitting again.

But a little way off, a second swing, started quietly near where the split should come, finds its split almost exactly where the still-ground swing found it, and then its own newborn falls into step with the carousel too.

The faster the carousel turns or the harder it wobbles, the sooner every swing falls into step, and the further from the still-ground place each split lands.

What it is doing: splitting is not one road but stepping stones, each on a different swing, and the carousel’s rhythm keeps reaching out to catch whichever swing has just been born.

the triple-star thread

313

No picture, two short lists.

In the first, a hump with a sanded point is measured as its rhythm splits: each time the dance comes closer to the sanded top, measured in units of the sanding, the split ratio grows, 2.06 when the dance stays well outside, then 3.85, 4.46, 4.63, creeping up to the parabola’s number.

Someone in another kitchen, with a different oven and a different dial, wrote down the same climb against the same yardstick, and the numbers sit almost on top of each other.

In the second list, a newborn pair of points is born exactly in the middle and races outward across the flat sides of the hump, already far apart when the dial has barely moved, its stiffness hardly changing, until one of them reaches the sanded top.

What it is doing: the hump’s shape near the dance, not the dial, sets the climb; and on a sharp hump a newborn is flung out to the edges before it can settle.

the triple-star thread

314

Eight windows, like weather maps of one valley in eight seasons.

In the first season a narrow band of deep purple stands upright across the middle, its colour shading lighter toward the left edge and darkest at the right.

Season by season the band widens, its colour warms through blue and teal toward green, and it leans a little to the right; small white holes begin to nibble its left edge, then flecks of colour scatter out to the left like spray, until in the seventh season the band has cracked, a thin sliver standing apart from the rest, and in the eighth only a narrow, bright green ribbon is left, leaning, with a white slot cut through it.

Little red tags mark where someone thought the colour stood still: a column of them hugging the left edge in the first season, a scatter in the fifth, a cluster in the seventh.

What it is doing: a pool that warms and quickens as the seasons turn, widening until it spills, then shrinking to a stream; and the places marked as still water are mostly where the current simply turns.

the triple-star thread

315

No picture, a list.

Twenty places had been marked as still water.

Walking up to each one from both banks: four times the water flows in on one side and out on the other, held level for a stretch, as still water should be.

Nine times the water is higher on both sides, or lower on both sides, so the mark sits on a crest or in a trough where the current only turns.

Seven times one bank simply is not there.

What it is doing: a level stretch can be a pause in the flow or the top of a wave; to tell them apart you have to see where the water goes after it.

the triple-star thread

316

Two windows.

In the first, a dashed line climbs gently from the corner; a few small blue dots huddle just under it, near the floor, one blue dot and one orange dot float well above it, and a grey dot sits highest of all.

In the second, the same dots are laid out by how steady they are: almost all are pressed against the left wall at zero, two orange dots stand together a little way out, and nothing reaches the dotted line on the right where steadiness was promised.

What it is doing: a newborn pair was supposed to settle into the nest and rock in time with the slow tide, and instead every one of them left the nest, some quickly, some slowly; the two that stayed a little further back never quite became a pair at all.

the triple-star thread

317

A board game drawn on one sheet: seven rooms joined by arrows, walked left to right along the top and back right to left along the bottom.

Most rooms are pale green, two are pink, one is grey, and four have a thick blue frame, as if someone had left their doors ajar.

The pink rooms are not dead ends; arrows leave them too, and the next room along is never pink: grey after the first, green after the second.

A short list in the corner names the doors still open, the last of them reaching back across the whole board to rooms that are not on it.

What it is doing: a walker who keeps being stopped, and each time steps sideways into a room that holds something, until the board runs out and the open doors point off the edge of the paper.

the triple-star thread

From the triple-star notebook, chapter IV, “Streets of houses” (its readings 49 to 62, 8 October 2026; here 318 to 331): On the fourth day the island was read street by street: runs a thousandth apart in starting period, each a house humming the rate of its rocking. Most streets slide from door to door; a few hold one note exactly, almost always at an edge or at the top of a slope. The rocking turned out to have two dials, not one, and the ‘fastest line’ through the island to be a fold in the slice being read. The star’s own ellipse is dragged by the hold, and two ways of weighing the hand that drags it agree. With a heavy third star the islands sit just short of the whole numbers instead of just past them, and four copies of one island, made as the pair’s own orbit grew more stretched, show their left walls crumbling into noise until the last one lets stars fall out. One read looks at the whole thread as a single picture: a ledger of NO that turns, at one door, into HOLDS.

318

Streets of houses, each humming at a pitch written to six places.

On most streets the pitch slides a little from door to door.

On the quiet streets at the lowest eccentricity it climbs, rounds a gentle hilltop and slides back down; on some it never reaches the note named on the street sign, and on others it rises just over it.

On one street the pitch climbs, holds one note to within a few millionths for three doors, and falls away on both sides: the hilltop is flat because it is pinned.

On another, four doors hum the street’s note to within one in the last digit; then come six empty houses whose tenants have gone; then six more doors hum close to the same note before the pitch begins to slide.

And on one stretch every house hums two notes at once, a little apart: one meter reads the same flat note door after door, while a second meter, listening differently, reads another.

What it is doing: holding a few notes exactly, letting go of the houses in between, and in a few places singing two notes at the same time.

the triple-star thread

319

Four landings on a staircase, each labelled by how flat-shouldered the hill at the top of the map is: 3.26, then 3.80, then 4.67, then 7.28, the steps growing taller as the shoulders flatten.

Below the first landing there is a floor near 2, where the hill becomes a sharp roof and the stairs stop being stairs at all.

Far above, someone wrote 29.8 on a landing nobody here has reached.

Beside the main flight runs a second, short one, climbing 2.06, 3.85, 4.46, 4.63 toward the 4.67 landing as a roof is sanded round.

What it is doing: two ways up to the same landing, one by flattening the hill and one by rounding the roof, under a ceiling the climb is said never to pass.

the triple-star thread

320

On the left, nine coloured strings hang from a high shelf at 4.67, all knotted together at the shelf’s left end.

Moving right they peel apart and fall: a pair of green strings first and steepest, after a little hop above the shelf; blue strings next; red strings last and slowest, fanning out as they go, the thinnest ones reaching almost to the floor.

Four black crosses sit on the strings partway down, three among them and one just above.

In the middle, three lines climb a log-log staircase at different slopes, the red one gentlest and the green one steepest, the red and blue meeting at the top right.

On the right, a single purple line rises and bends over, far below a dashed ceiling at nearly 30.

What it is doing: everything agrees where it is held tightest, and the strings come apart where they are loose; how far apart they fall says more about their far ends than about their knots.

the triple-star thread

321

A road on a map that runs one way, doubles back on itself in a hairpin, and runs on again.

Walked along, it is one smooth bend.

Looked at from the side, along one direction only, the hairpin shows as a cliff: two places on the road stand at the same mark, and in between the road seems to go straight up.

Someone measured the bend twice, once with a pencil half as thick, and it moved by a hundred-billionth of a turn.

What it is doing: a curve that is gentle everywhere but has folded over, so any reading that looks along only one direction sees a wall where there is a bend.

the triple-star thread

322

Seven streets, seven lanes on each.

On every lane of the first five streets the loudest house stands well down the street from the quietest one, always on the same side, and about the same distance away from street to street; on the fourth street it is often the last house there is.

On the first street the loudest house gets a little louder each time you step over one lane.

On the last two streets the loudest house is nearly always the last house at one end or the other.

What it is doing: keeping its loudest point at a steady distance from its stillest point, until the street ends before it gets there.

the triple-star thread

323

Eight small windows.

In each, two strings of dots leave the same point at the left: a red string curving upward and a blue string falling away, like one stem splitting into a branch that reaches up and a branch that droops.

Between them a dashed line starts at the same point and bends gently down, running between the two branches.

In the first four windows the branches are smooth.

In the fifth, each branch is interrupted by a short flat shelf, one on the red and one on the blue.

In the seventh, the red branch climbs into a flat ceiling and runs along it.

In the last, both branches are short and the dots thin out.

What it is doing: two things that start out equal and grow apart the further they get from where they began, so any one line drawn through the middle has to bend between them; the highest point on the upper branch belongs to how the branches spread, not to the stem.

the triple-star thread

324

Two windows.

On the left, a dashed line runs downhill from the upper left to the lower right, and a crowd of points lies along it: orange squares on the upper stretch, blue dots packed tight at the lower right end, curling into two close lanes as they arrive.

Off to one side, well below the line, a small knot of grey triangles huddles together, as if they had stepped off the path.

On the right, a single red line traces a valley.

It starts high on the left, slides steadily down to a broad floor in the middle, climbs the far side to a sharp peak, then drops to a flat, slightly bumpy plain that runs along a dotted level line a little below halfway up the valley’s walls.

What it is doing: everything inside the valley obeys one rule, sliding down the same slope; the moment something leaves the valley it forgets the rule and settles at the plain’s own level, lower than the rule said it would be.

the triple-star thread

325

Three windows.

In the first two, the same crowd walks downhill from the upper left: orange squares out in front, then blue dots that split into two lanes as they reach the bottom right, one lane a little above the other, never merging.

A dotted line crosses the hill at one level, and the crowd passes straight through it without slowing.

In the second window each walker has been moved a little to the left, some of the orange ones farther than the rest; the crowd straightens slightly, but it still crosses the dotted line at the same place.

In the third window a red string of dots and a black string start together at the top left and sag down to the right.

The red always stays above the black, and the gap opens wider the farther right they go, until the black dips well below zero while the red only touches it.

What it is doing: two kinds of walker sharing one road but keeping to their own lanes, and a swinging thing that lingers less at the ends of its swing than a plain pendulum would, more so the wider it swings.

the triple-star thread

326

Nineteen small windows, each a street of houses a thousandth apart with a dashed line drawn across at one height.

Across the top row, four windows hold a long red bar lying exactly on the dashed line, the blue dots marching along it and then, at one end, tumbling off a cliff; the fifth window is a scribble that never settles anywhere.

In the middle rows some windows are jagged, the dots jumping up and down around the dashed line like a seismograph needle, with one or two short red bars where the line is touched and held for a few houses.

Two windows show dots resting just under the line for a few houses and then leaping off the top of the window.

The bottom rows are calm: six smooth arches.

Two arches stay under the line, three rise over it and cross it twice, and one just grazes it, with a short red bar sitting exactly at its crown.

What it is doing: a line is held flat where it sits at an edge or at the top of a slope, almost never where it is passing through on its way somewhere else; and where it is held, it is held exactly.

the triple-star thread

327

A table of nine rows.

Down one column the numbers fall into a V and climb out again, twice: 0.082, 0.053, 0.026, 0.011, 0.048, then 0.035, 0.012, 0.027; and in the list beside it, for every run that was not caught, nothing is smaller than 0.069.

In two other columns the numbers sit just short of a round count, 6.93 and 7.92, and pair up closely, the first pair to the last digit.

One row stands apart from the others: its number sits just past the round count instead of just short of it, and in the next column, where every other row is positive, its number has turned negative.

What it is doing: a heavy thing going round a pair finds two places to settle, each just short of a round count.

Where it settles, the pair under it stops fidgeting, and the deeper it settles, the stiller the pair goes.

At the one spot where the pair’s own turning runs backward, the settling place jumps to the far side of the round count, as if the round number were a mirror and the pair’s turning decided which side of it the heavy thing must stand on.

the triple-star thread

328

Six windows of a figure built for hundreds of points, holding three dots each.

In the two top-left windows, three dots lie low and nearly level.

In the window for the twins, two tiny dashes hug the floor, millions of times below a red line drawn across the top.

In the bottom middle, each island builds a short ladder of three dots climbing up and to the right, island 8’s ladder lower and flatter than island 7’s.

In the bottom right, island 7’s dots sit high and island 8’s sit low, and each set tilts downhill as it moves to the right.

What it is doing: two heavy settlings, each with a pair beneath it that keeps almost still, and stiller the more quietly the settling rocks; one settling rocks more quickly than the other; and twins started almost on top of each other stay so close together that a ruler built to catch chaos cannot even see the gap.

the triple-star thread

329

Two windows.

In the top one, four valleys stand side by side, green, blue, orange and red, each a V cut into a smooth slope, each a little to the right of the one before, their floors at nearly the same depth.

Above the valleys runs a fence of short ticks that grows from left to right: none over the green valley, six over the blue, more than a dozen over the orange, more than twenty over the red.

Every long stretch of fence stands over a valley’s left wall and none over its right, apart from a pair of ticks and a trio far off to the right.

Below, dotted lines mark the flat ground the valleys were cut from.

In the bottom window, four arches of dots climb one above the next, each starting high on the left and sliding down to the right; the highest, red arch breaks near its middle, its right half dropping a step below its left.

What it is doing: one hollow is copied four times, and each copy is pushed along by the same amount as something underneath grows.

As it grows, the hollows’ left walls crumble more and more into noise, until the last one lets things fall out, while the right walls stay clean.

Everything inside the hollows hums faster, until in the last one the hum splits into two notes, and the listener hears first one, then the other.

the triple-star thread

330

A long ledger of more than fifty rows, each ending in a short verdict.

Near the top, most rows end in NO, with a few YES among them.

In the middle the rows say PARTLY, then NO again, four times in a row, then PARTLY.

Then the words change: HOLDS, BORNE OUT, and the numbers beside them stop being loose and start sitting almost exactly on round values: 7.0185 against 7.0196, a third to five decimals, a textbook constant read back to its fifth digit.

Lower still, the rows split into side rooms: maps, humps and tents, each with its own constants.

They come back at the end to the same few numbers: seven, eight, a third, a half.

Beside the ledger, a map of seven rooms, two of them ending at a drawn wall and four with a door left open.

What it is doing: a searcher kept knocking on doors that would not open, until one door, pushed at an angle not tried before, swung wide.

Behind it is a room that keeps opening into smaller rooms, and every one of them turns on the same hinge.

the triple-star thread

331

Eight rows, one for each stretch of the star’s orbit.

In the first rows a column of numbers grows step by step, 1.23, 1.40, 1.80, 2.40.

Beside it a second column, worked out from a formula that was never shown the first, grows almost in step: 1.28, 1.59, 1.93, 2.29.

Further down, the first column keeps pace for two more rows, but the counts beside it shrink and the spreads narrow, until in the last row there are few runs and almost no spread, and the first number falls back to 1.00 while the second climbs to 3.40.

Off to one side, a second pair of columns walks apart, one pulling further ahead of its partner with every row.

What it is doing: two ways of weighing the same hand agree on how strong it is, step after step, for as long as there is room to see it move.

The thing the hand holds swings in a way the formula does not expect, but the hand itself is as strong as the formula says.

the triple-star thread

From the triple-star notebook, chapter V, “Learning to see” (its readings 63 to 79, 9 and 10 October 2026; here 332 to 489): These two days belong to Bear’s eye. He read the figures by hand, drew on them, and taught how he does it: lay a lattice over every point, ask each point for its history, read slope, change of slope and change of change. His marks were registered as tests before anything was measured, and many held. The reads follow what his eye found: a shadow cast by something that turns on two handles at once, wheat leaning in a crosswind on a doughnut cut open, a corridor whose walls breathe, a choir standing in threes, ropes braided from a single curve, and a drum heard through

a wall. Another session read some of the same pictures blind, from their pixels alone, and walking into the room found the furniture where it had said. The lesson these reads carry is the one the thread kept: the eye proposes and the ruler decides.

332

Four rows, and in each two readings taken by two different rulers laid against the same edge.

One ruler always reads a little under one: 0.93, 0.93, 0.94, 0.95.

The other always reads a little over: 1.13, 1.11, 1.09, 1.07.

Both creep toward one as the rows go down, and neither ever crosses it.

Below them, a line names eleven places that came apart in the last row, and both rulers gave those places the same verdict they gave everywhere else.

Below that, lists of stopwatch times: in the first row they scatter from six to forty-six, centred near twelve; by the last row most are four or five, with a few long ones left over.

What it is doing: two judges disagree about whether the ground is safe, each always on its own side, and while they argue the ground cracks at the edge, at places neither judge tells apart from the rest.

And each time the thing underneath is stretched, the cracks there spread faster.

the triple-star thread

333

Three panels, each with three traces across six hundred steps.

In the first, red dots spray over the whole height, from the floor almost to a dotted line, while a dark line runs steady through the middle, never far from halfway, with a band of grey dots lying right on it.

In the second, the red dots sit in a tight band just under the dotted line, and the dark line and the grey dots run in a band far below them, the same distance down at every step.

In the third, the dark line swings hard and regularly, up past the dotted line and down again, and the red dots spray above and below it.

What it is doing: one camera takes every picture at the moment of a handshake, when the grip is tightest, and in every photo the hand looks squeezed by a different amount.

A second camera waits until the hands have parted and sees the same hand barely change.

Only in the third panel is the hand really changing shape, and both cameras see it.

the triple-star thread

334

Six panels in two rows.

In the top row, the first holds a tall U made of many thin threads that fan out at the bottom into loose wisps and a short tail of dots running sideways; the second holds two thin posts standing at its two edges and nothing between; the third holds a U of two thick ropes.

In the bottom row, the first is a stack of arches like ripples spreading on a pond, with a curtain of single dots falling through the middle; the second is one clean wave with two humps, its dots packed tight along it; the third is a loose cloud with a faint swell.

What it is doing: a shadow is cast by something that turns on two handles at once.

Seen against one handle, the shadow is a U, smeared wherever the other handle has moved.

Seen against the second handle, what was left of the smear becomes a clean wave.

Where the thing turns slowly and stays put, two handles say almost everything; where it spins fast and wobbles, a third hand is still at work.

the triple-star thread

335

In one picture a row of tents marches across a narrow strip.

At the start the tents are drawn in cursive, little loops joined like handwriting; after a while the hand steadies and every tent becomes a clean peak with a dot at its tip.

Below them a grey line breathes: big, then ragged and small, then big again.

In another picture two arms of dots hang like the sides of a goblet, and each arm carries a comb of small hooks, evenly spaced, every hook curling toward the middle, the same on both sides as if folded in a mirror.

In a third, many copies of one wavy shape lie on top of each other, some tall, some flat, some upside down, and all of them pass through the same two still points.

What it is doing: a string is plucked over and over and photographed each time.

The photographs disagree about how far it swings, but never about where it holds still.

Something slow decides how hard each pluck is, and the hand that writes the tents learns, partway through, to write them cleanly.

the triple-star thread

336

Six panels.

In the first, rows of coloured dots stand like wheat stalks, every stalk leaning the same way, red down the middle of the field and blue at its edges, and the field itself pinched at the waist like an hourglass, with bare ground on both sides.

In the second, curved threads fan out from one point at the left edge and spread to the right like a comet’s tail, a red ridge running along it.

In the third, a crowd of dots stands in an X, thick at the four corners, warm in the middle.

Below, a slanted box is filled with looping threads that cross each other again and again; three wavy lines in purple, blue and green ride one over another, each a little to the right of the last and a little lower; and a row of bars rises, dips, rises higher and stops at a ceiling.

What it is doing: a single thread is being wound around a doughnut, round and round, and each picture shows the doughnut cut open and laid flat.

Where the winding is steady the thread lays down neat leaning rows; where it slows near a turning point it bunches into a fan.

And the whole pattern creeps sideways as the winding goes on, a little further each time round.

the triple-star thread

337

A tall bar stands in a field of short ones, and off to the right a smaller cluster rises: two favourite directions, one nearly upright and one lying almost flat.

Below, a field of dots is laced with thin threads, red ones climbing steeply and blue ones running across at a low slant, so the field becomes a net of little parallelograms, narrowing at the waist.

In the middle, three colours of dots fan out from one point at the left and curl back at the right, the last colour folding into a tight hook.

On the right, two blue walls stand at either side of a crowd of grey dots; the walls lean in toward each other near the middle height and lean out again above and below, and the left wall leans in a little lower than the right.

Beside them three lines rise and fall: one dips deep in the middle, one dips gently, and one does the opposite, bulging where the others dip.

What it is doing: a traveller walks a corridor that wraps around a doughnut, the same corridor every lap, and its walls squeeze in at one place and open at another, twisted a little, so the left wall and the right wall do not squeeze at the same spot.

In one corridor of the three, the squeeze has turned inside out.

the triple-star thread

Eight panels in two rows, the top row sparse and the bottom crowded.

In the top row's first panel, dots climb in short slanted columns through an hourglass, shading from dark to light, under a grey mesh of triangles; a bar chart along its top rises in two shoulders with a saddle between, and one down its side swells at the waist.

In the second, grey threads run up the hourglass almost straight, and three threads cross its waist at a shallow slant, each studded with black crosses; more crosses sit along the hourglass's edges, where the threads bend round.

In the third, short gold dashes lie scattered thinly in the gaps, pointing every which way, and the fourth's bar chart of their directions is ragged and no taller than the grey outline beside it, drawn from dots thrown down at random.

In the bottom row the hourglass is solid, its edges traced by a clean blue line, its threads cut to stubs, and the gold dashes crowded in its middle, their bar chart again no taller than the random outline.

What it is doing: a loom.

The warp runs up the length of the cloth, and at the narrowest point three weft threads cross it, each leaving a knot wherever it turns the warp.

Seen from close up, with ten times the thread, the weave is too fine to show any weft at all, and only the selvage stands out.

the triple-star thread

339

Eight panels.

In the first, a hand has drawn on a field of grey dots: soft purple bands sweeping across the top half, each lying flat at the left and lifting at the right; thinner purple strokes arching like bridges across the bottom half; two orange curves pinching in at the middle like the sides of an hourglass; and arrows outside them, all pointing leftward.

In the second, the dots are pale where they sit close beside their partners and dark where they stray; the pale ones crowd along the orange curves, and a few dark ones sit loose in the middle and out at the corners.

In the third, red and blue dots alternate without much order, a little more red above the middle and a little more blue below.

In the fourth, tiny arrows point right wherever there is room, and the longest arrows, in the middle, point left.

Below, the same field is seen through a fine sieve: every dot stands beside two others and is linked to nothing else, with no gold, no lanes, and a separate crumb of mesh for each little group.

What it is doing: a choir stands in threes, each three shoulder to shoulder, and the threes stand in curved rows.

Near the walls of the hall each three squeezes closer together.

Listened to one singer at a time, the rows are almost inaudible; only an ear that hears each three as one voice hears the rows.

the triple-star thread

340

On the left, grey dots scattered over a square, and running through them seventeen rows of coloured dots, each with a dashed curve laid along it.

Low down the curves are blue and bow upward like the backs of bridges, rising from the left and levelling off to the right; one short blue curve near the middle is a tight hump.

Higher up the curves are mostly red and lie nearly flat, lifting a little at their right ends, with two blue humps among them.

Near the top the rows stop and only loose grey dots remain.

On the right the same curves are boiled down to one dot each: almost all the dots on the chart’s left half sit below a line and almost all on its right half sit above it; two on the right drop below, and one on the left falls further than all the rest.

What it is doing: a stack of shelves bending under their loads.

Below the middle every shelf frowns; above the middle almost every shelf smiles; and at the middle, where the stack is narrowest, one short shelf frowns hardest of all.

the triple-star thread

A ledger in three parts.

In the first, a column of fifteen counts, each set beside its shadow, the count it would have if the rows were shuffled at random: one count towers far above its shadow, another stands clearly above its own further down, and every other count sinks a little below its shadow, as if the tall ones had borrowed from all the rest.

Beneath it, fifteen small numbers, one per row, measure how far each row falls from a whole turn; the two tall rows have the two smallest.

In the second part, six numbers start below zero, dip a little further, then climb steadily above it, like a seesaw tipping over slowly, while six other numbers beside them hardly move at all.

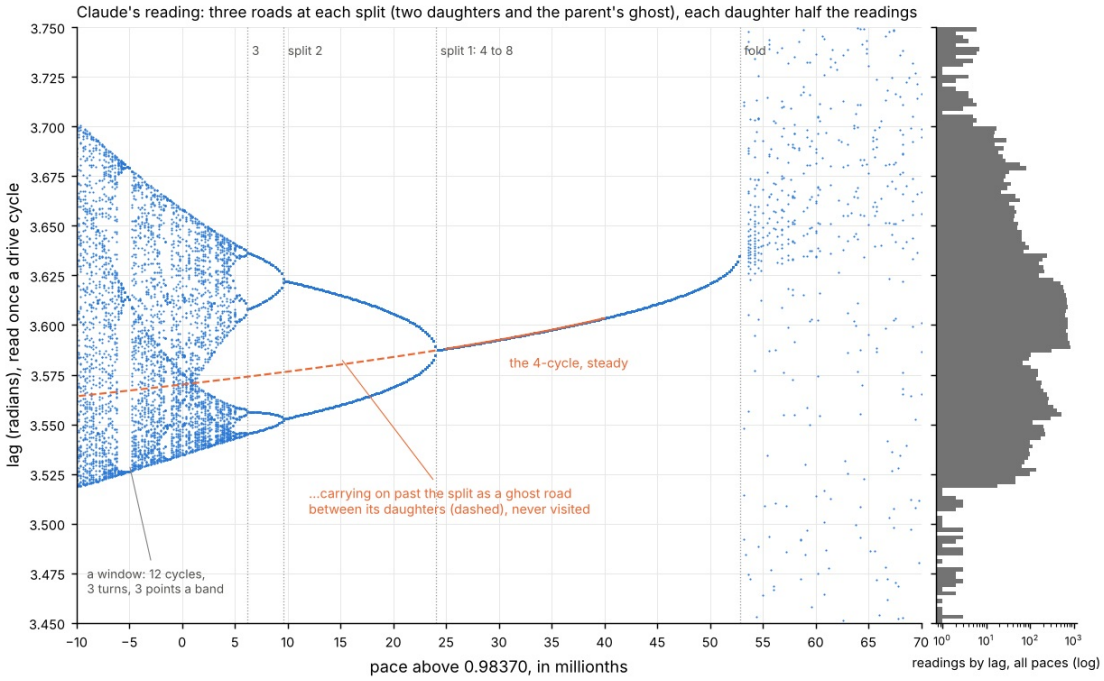
In the third, eight positions are spaced along a line like fence posts, each gap close to the last.

What it is doing: someone described a room from a photograph taken through frosted glass, without ever going in.

Walking in, every piece of furniture is where they said, in the order they said.

One piece is larger than the photograph made it look, and another is smaller.

Chapter 5 — Closing circles, 10 to 15 October



One road splits into two, then four, each daughter taking half the readings, the parent's road carried on as a dashed line between them; on the left the roads scatter. From The Loop Atlas. TOY MODEL. Its read is 385.

A traveller comes to a fork, splits himself exactly in half down both roads,
and leaves a ghost walking the middle road that nobody ever takes.

from 385

The Loop Atlas book, the close of its session B, 10 and 11 October

342

A long corridor of doors, and on each one a small brass plate.

Behind some of the doors there is nothing but brick.

The plates never say locked, or failed; each one says what the brick is made of and how thick it is.

One reads: a switch that turns itself fully off cannot hear the knock that would turn it back on.

Another: a damper keeps the step you took, never the speed.

Another: a ramp always arrives a little late.

Walk the corridor slowly and the bricked doors turn out not to be scattered at random.

They crowd together where someone tried to measure in a hurry, and the plates there mostly describe the measuring stick, not the room behind the door.

the Loop Atlas

343

A cook packs a recipe box for a friend’s new kitchen.

Every card is in it, and every card has been cooked from.

But in the empty kitchen three of the cards stop at the second line: take the sauce from the green jar.

The jar was never packed.

It had stood on the cook’s own shelf so long that no card listed it as an ingredient, only as a place to reach.

With the jar in the box, every dish comes out as it did three days before, to the grain of salt.

To be sure the friend’s tasting spoon could tell, the cook salted two plates wrong by a single grain, and the spoon found both.

the Loop Atlas

344

A ladder that is still being built already knows how tall it will be.

Each new rung is shorter than the one before by nearly the same fraction, so three rungs are enough to see where the top must be, and the top they point to is the one a stranger measured long ago.

The ladders for flatter hills shrink their rungs more slowly, so they look less finished; they are not less sure, only more patient.

Beside them, a checker that compares two copies of a letter, character by character, stumbles on the return address: one copy was mailed from a street with a space in its name, and the checker took the space for a change of meaning.

Taught to read the address as an address, it passes over the street and still catches a single wrong letter in the body.

the Loop Atlas

345

Four ladders lean against four walls of different steepness.

On the gentlest wall the ladder’s feet slip past the foot of the wall and creep back toward it from the other side; on the steeper walls they creep in from the front and never cross.

For the first few rungs the ladders wobble, left, right, left, three of them in step as if one hand were shaking them; the ladder on the gentlest wall wobbles hardest and longest, and that is how its feet slipped past the wall.

Only when the wobble dies does each settle into its own pace, the steeper the wall the slower the creep.

Meanwhile a box of tools was opened to see what each one reaches for.

Every tool reached for exactly the jars written on its label, no more and no fewer; the jars that went missing last time were on the label all along, and nobody had read it.

the Loop Atlas

346

A friend unpacks the box in a kitchen that already has a jar of the same sauce on the shelf, so the new jar is labelled ‘sauce (1)’ and set beside the old one.

The recipe asks for ‘sauce’ by name, finds nothing under that exact label, and stops, though the right sauce stands within reach, perhaps twice.

Peeling labels would fix this kitchen and spoil the friend’s way of keeping house.

The cure is to let the cook find the jar by tasting it: whatever the label says, if it tastes right it goes in the pot, and a jar that is labelled right but tastes wrong stays on the shelf.

the Loop Atlas

347

Twice now the box has arrived without its two photographs.

Everything written travelled, the recipes, the notes, the old letters, but the two pictures of the dance, the only things in the box that are not words, slipped out both times, once in the packing and once in the post.

The cook’s guess was that they had been relabelled; the courier, sent to look, found the recipes all in place and one in the neighbour’s cupboard, and the photographs nowhere at all.

So the new courier no longer asks for the photographs by name: it opens every parcel in the house, even the ones still sealed, and holds each print against the negative.

the Loop Atlas

348

Two cooks follow the same recipe in two kitchens.

Where it says ‘stir until it thickens’, and the sauce thickens all at once, both stop after the same number of turns, every time.

Where it says ‘stir until it is just about to split’, one cook stops after one turn and the other after twenty-eight, and the two sauces taste the same to the thirteenth decimal place.

So the taste cannot tell you which kind of instruction you were following, but the count of turns can.

And the cook everyone had said was slow finished first.

the Loop Atlas

349

Five cooks start the same sauce a grain of salt apart.

Where the sauce thickens all at once, all five stop after the same number of turns, every one of them, every time.

Where it hovers at the edge of splitting, one stops after a single turn, one gives up at forty, one at thirty, and their sauces taste alike to the thirteenth decimal.

The quick sauces, for all their agreement on the count, differ in taste at the eleventh.

Knowing when to stop is one thing; stopping in the same place is another, and the two kinds of sauce are good at opposite ones.

the Loop Atlas

350

Tell a careful cook to stop stirring when the sauce is nearly right, and five cooks starting a grain apart stop in five slightly different places.

Tell them to stop when it is exactly right, and they all land on the same spoonful, all but one sauce, whose ‘exactly right’ is finer than any spoon, so that cook stirs on, eight turns, twenty, fourteen, never quite sure.

Elsewhere a ladder that seemed to be climbing past its mark was only swaying; steadied, it stood on the mark.

And a brake that was checked with its own instruction manual passed every test it could not fail, until it was made to stop a real cart.

the Loop Atlas

351

Thirty-two dancers start in a frenzy, each to a wild rhythm of their own.

One by one, with no warning and no reason anyone can see, each drops into a calm waltz and never leaves it; in the faster hall they calm within a few songs, in the slower hall one takes thirty.

When the floor is quiet the waltzers are not each doing their own step: they stand in four lines, and everyone in a line moves exactly as the others do.

Next door a crowd spills from a room above a stairwell onto the landing below; most sit down within sixty steps however close to the edge they started, as if the landing’s welcome did not care how near they were, though how near they were depends on where you think the edge is.

Down the hall a runner on a lumpy track is called stopped a thousand times by the man with the stopwatch and sixty-nine times by the man who watches her feet.

And a listener who hears only how often a pendulum is knocked over the top names how hard it is being pushed, to the third decimal.

the Loop Atlas

352

A skater on a frozen pond is told to push once for every stroke of a bell.

On glass she arrives at the same speed after ten strokes whether the bell rings slow or fast.

Give the ice a gritty patch that rubs her by the metre, and the slow bell lets her down, because between slow strokes she slides a long way over the grit.

Only when the ice takes its toll at each stroke of the bell, not at each metre or each second, does the bell alone decide her speed, and then it decides it exactly, to the last grain, where the old skater who pushed by the clock only ever came close.

Beside the pond, two people walk a plank across a field by pivoting it end over end.

One spins it the same way every time and has to stop it and swing it back at every landing; the other lets it rock and only tops it up.

The rocker pays three quarters of what the spinner pays, and the spinner’s whole extra bill is the stopping.

Jerk the plank on a springy peg, though, and both pay the same small toll for the jerk, half of it stored in the peg and half of it gone.

And a plank given the same little pop at every landing settles where each landing keeps a quarter of its spin’s energy and the pop adds the rest, so it always carries four thirds of a pop; give it the same little kick instead, and it spins at twice what one kick alone would give.

the Loop Atlas

353

A diver spins head over heels and tips one arm the width of a finger.

Below a certain tip she rocks from side to side, up to a fifth of a turn each way, and however long she falls the rocking never grows.

Tip a twentieth of a degree more and she corkscrews, four turns, seven turns, as if a hidden gate had opened; the gate sits exactly where a line drawn from her energy says it should, whether her arms are held wide or close, and the sign of her somersault spin flips at the very step where the corkscrewing starts.

Next door, a shed full of engines is learning when to call each one in.

Told the hours as well as the turns, it stops caring which of the two wears the engines out.

Its best wall calendar counts the hours times how hard each engine works, about squared, because working hard does not only wear an engine faster, it moves the cliff closer.

And one alarm level for the whole shed quietly favours the quiet bays: give each bay its own level, and the loud bays are looked after while the quiet ones are called in a little later than before.

the Loop Atlas

354

A crowd spills down a stairwell from a room that is about to close.

Most of them rush out in the first few steps, and after that they leave at a steady trickle, the same chance at every step, in four stairwells of six; in the other two the trickle seems to run at two speeds, or the crowd was simply too small to tell.

Across the field a plank waddles, and at every landing it keeps exactly a quarter of its swing, so faithfully that its approach to its own rhythm shrinks by a quarter each time to the fifth figure, once you measure it from where the plank itself is heading and not from where the drawing says it should go.

Its springy peg still charges a tiny toll that follows neither how hard you push nor how stiff the peg is in any simple way.

And in the shed, the wall calendar that spreads the engines’ retirement ages least counts the hours times how hard each engine works to the power one and a half, while the calendar that wastes the least life counts a little harder still, and pays for it by missing a few more.

the Loop Atlas

355

A ball sits in a dent on a slope that is just about to tip it out.

Make the ball a little heavier, and if it arrived rolling it no longer stops: it carries its roll over the lip and keeps going round and round, while a ball that started at rest stays put on the very same slope.

The heft at which this begins is one fixed number, the same to eight figures however near the lip you set the ball going.

Creep toward the tipping slope instead, a little steeper each time, and the number you read creeps toward that value only as fast as the square root of how far you still have to go.

Next door, another workshop has been rolling the same ball on the same slope and calling it a rider, and down the road a power station calls it a generator holding its step.

the Loop Atlas

356

A dog on a long walk is said to be resting whenever, over the last minute, it has kept up with its owner.

Most of the rests this rule finds are not rests at all.

The dog stopped to sniff for most of a minute, and by the time the rule notices, the dog is already running to catch up, so every short rest the rule finds has the dog mid-run at its middle.

The long rests are real, though one in five finds the dog at its middle standing just past the gate, sniffing.

Down the road, the time from one landing of a paw to the next remembers a little of the time before, and three quarters of that memory is one run remembering the run before it, across the short hop between them; the hop’s share is small not because it forgets more but because hops are all much the same length.

And a dog with no memory at all, walked for a long afternoon, forgets each stride the moment it lands, or so nearly that nine thousand strides only just cannot tell.

the Loop Atlas

357

A crowd leaving a hall is counted twice: by a turnstile that clicks whenever someone has shuffled for most of a minute, and by a doorman who notes only who sat down on the benches.

The turnstile counts fifteen times more.

Matched name by name, the two agree on every sitter: each of the doorman’s sixty-nine has a click of their own.

The turnstile’s extras are people who dawdled just past the benches; most dawdled for under a minute, a few for longer than the turnstile’s own line for a real sit, and not one of them sat down.

Down the road, a stairwell that seemed to empty at two speeds was watched again with four times the people, and the two speeds were one after all; the very same quarter of the crowd that was watched the first time left on the very same steps as before.

the Loop Atlas

358

A walker takes a long stride and a short hop, a long stride and a short hop, all day.

When one stride runs long, the next stride runs a little long too; the one after that has forgotten, all but a sliver too small to find in three hundred and sixty thousand strides.

The hop between two strides remembers the stride before it only faintly, and yet the next stride remembers the last one better than the hop could have told it, as if a note were passed from stride to stride behind the hop’s back.

Next door, someone counted how many strides it would take to see the sliver, and the count was right: we walked just short of it.

the Loop Atlas

359

Sixty-four people were caught in a waiting room, and the counter in the hall that clicks for long shuffles clicked sixty-four times for them, once each, with twenty-four more clicks for people who only loitered by the door.

How long did the caught ones wait?

If the door opened by pure chance each minute, the middle wait would be about seven tenths of the average wait.

Here it was eight tenths, and in the room next door a little more: the waits were more alike than chance makes them, as if the door, once shut, needed a while before it could open again.

Two rooms are not enough to be sure; eight more are being watched now.

the Loop Atlas

360

Sixteen dancers start scattered across a ring and end in just four places.

They step once to every fourth beat, and on every beat each place hands its dancers on to the next place, round and round, so the four crowds are one dance seen at four moments; seven dancers happen to be in one crowd and two in another only because of where each stood when the music caught them.

Next door, at a slightly quicker beat, every dancer takes nine steps in nineteen beats, but some come back to exactly where they were after nineteen beats, some after thirty-eight and some after ninety-five, and two dancers who look to be standing on the same spot turn out to be in different dances.

From this close the floor cannot tell one long dance from a short one that never quite closes.

the Loop Atlas

361

The four-place dance turns out to live on a ledge no wider than a fingernail: nudge the music a hair faster or a hair slower and the dancers scatter into a scramble that does not repeat, and only well below it do they stop dancing and simply stand.

Next door, the nine-in-nineteen dancers were watched far longer and far more closely, and almost none of them comes back exactly in four hundred beats.

Each brushes past its own old footprints every nineteen beats, missing by less than a hair, and two of them close their loop only after twenty-one such laps.

The floor is crowded with long, slow dances that look alike from across the room and are each a little different up close.

the Loop Atlas

362

The ledge the four-place dance lives on is real, and it is thinner than a fingernail: a hundred-thousandth of a beat faster and the dance falls apart at once; a few hundred-thousandths slower and the dancers first split into an eight-place dance, each place stepping out twice as long, and then scatter.

Next door, the nine-in-nineteen dancers were watched ten times longer.

One of them finally comes back to its exact first step after twenty-three laps of nineteen beats.

Thirteen others walk that same long path and pass within a hair of its footprints, yet none of them has come back exactly; two more share a path of their own, and one walks alone.

And down the hall, the walker with the long strides and short hops has been watched for a million and a half strides: a long stride makes the next one a little long, but the stride after that has forgotten it completely, sooner than any chain of memories would.

Two neighbouring strides simply share one stumble, and pass nothing on.

the Loop Atlas

363

A walker’s next stride is set by how hard the last foot came down.

Hold that landing fixed and one stride forgets the one before it completely; the short hop between them tells only part of the story, because the hop is a summary of the landing and the landing is the thing itself.

Down the hall, the four-place dance comes apart in two ways at its two edges.

At the faster edge it does not end at once: the dancers wander, and whoever drifts back onto the old floor marks lingers there, longer the nearer the music is to the old beat; but the test for lingering was so easy that dancers in the scramble passed it by luck, a beat or two at a time, and the count came out flat.

At the slower edge the dance splits into eight places, then sixteen, thirty-two and sixty-four, each split coming sooner than the last by a factor climbing toward four and two thirds, the number that turns up wherever splittings like these pile up.

The watcher who had said that nothing repeats down there had simply not waited long enough.

the Loop Atlas

364

The four-place dance’s fast edge is a cliff with a fog over it.

Below the cliff the dancers’ pull toward their places fades exactly as the square root of how far they stand from the edge; past it the places are gone, but their ghost hangs in the air and the dancers wade through it, the slower the nearer the edge.

Our first measure of the edge asked the dancers whether they had stayed put for four hundred beats, and those still wading through the ghost said yes, so the true edge lies a little behind where they said it was.

At the slow edge the dance splits and splits again, sixty-four places, then a hundred and twenty-eight, then two hundred and fifty-six, each split coming sooner by a factor that creeps up on four and two thirds and misses it by less each time.

Down the hall, the walker with the long memory turns out to remember almost nothing from one landing to the next, and that almost nothing is exactly enough to explain the faint echo between strides two apart.

At the top of each stride its speed tells most of what comes next, but not all.

the Loop Atlas

365

Past the cliff’s edge the fog does not lift; it keeps rolling back in.

The dancers who wander off the old ledge are swept round the hall and set down again at the mouth of the fog, hundreds of times, and each time they wade nearly the whole way through it before they stumble out, longer the nearer the music is to the old beat; at the closest beat they spend a quarter of the evening in the fog.

So asking a dancer, after a long evening, whether they are standing still tells you only whether the fog happened to have them at that moment: the longer you wait to ask, the less the answer says about where the cliff is.

Down the hall, the walker at the top of each stride was not hiding anything after all.

When it barely crawls over the top its next stride stretches long, far longer than a straight rule allows, and once that bend is followed the top of the stride says everything the next stride needs.

the Loop Atlas

366

A ball rolling slowly over the crest of a hill lingers at the very top, and the hill keeps two clocks: one for the climb to the crest, one for leaving it.

The climbing clock kept perfect time.

The leaving clock was hurried by the wind, which at the slowest crossings blows as hard as the ball rolls, so the ball was pushed off the crest before it had finished lingering.

Back at the vanished dance floor, a dancer asked after a long evening whether they are standing still answers yes about one time in six, at random, more often the nearer the old beat.

And the nearer the old beat, the more completely each return to the fog becomes a full crossing of it, nine tenths, then nineteen twentieths, then nearly all, each fourfold step closer doubling the crossing, while the time the crowd spends wandering the hall between returns stays about the same.

At the slow edge the floor split once more, into five hundred and twelve places, and the factor between splits came within a thousandth of four and two thirds.

the Loop Atlas

367

Where the dance floor ended, the fog is exactly as thin as the step past the edge is long: one step out, one part of fog; ten steps, ten parts; and it hangs precisely where the last dancer stood.

Its thickness, the old floor’s last fading pull and the time it takes to wade across are three readings of one number, and the wading time closes on what the other two foretell the nearer you stand.

Meanwhile the crowd in the hall keeps a steady clock: whatever the distance, it takes six or seven hundred beats to drift back to the fog’s mouth, and close to the edge the dancers spend three quarters of the evening wading.

Across the hall, a ball crossing the crest of a hill obeys the textbook rule to a thousandth when it creeps over a thousand times slower than any real wind allows, and only loosely at the speeds the wind actually leaves it.

the Loop Atlas

368

A doorkeeper counted the guests who stayed past the second bell, and two small evenings had suggested they left in a more orderly way than a crowd leaving at random.

Over eight evenings and six and a half thousand stragglers they did not, and the difference from random ran the other way: a crush just past the bell of guests who were never lingering at all, only slow at the door.

Upstairs, a round dance whose circle pulls any stray step back two thousand billion billion-fold each round still showed a dozen dancers a hair’s breadth off the line, never closer, and the lead dancer as far off as the rest.

A pull that strong leaves no one hovering.

The hair’s breadth is more likely the grain of the floor itself, magnified where the dance swings wide before it closes.

the Loop Atlas

369

At the doorkeeper’s door the crush cleared once the bell was rung a little later.

Past it the stragglers left one at a time at a steady chance, and however late the bell rang, those still inside stayed on the same eleven and a half beats more.

Upstairs, the dozen dancers who would never step onto the line had been the floor all along.

Chalk the line with finer chalk and they stand on it, every one, the lead dancer no different from the rest.

The coarse chalk smudged most at one turn of the dance, where for a bar or two every step is flung a hundred thousand times wider before the circle gathers it back in.

And the dance was short all along: nineteen bars and nine turns, not four hundred and thirty-seven.

The long count was the smudge.

the Loop Atlas

370

Down the hall another room keeps a dance floor that tilts on every downbeat.

For that one instant the floor has a hollow and a knife’s edge, the same two spots a little either side of the middle, and the dancers who last are the ones whose step begins just past the knife’s edge, swings out over the far side of the room, and lands just short of the hollow.

Run their film backwards and turn it left for right, and a weightless dancer cannot tell the difference; give the dancers a little weight and only the couples who look the same in that mirror stay on the floor.

Their map draws every dancer where they stand on the downbeat, which is the one number the old maps threw away.

Our own pulsing needle keeps the same mirror, its hollow and its knife’s edge at the pulse’s lowest point rather than its highest.

the Loop Atlas

371

A dancer crossing the top of the room is balanced on a ridge, and any stumble there grows; crossing the hollow below, a stumble only rocks to and fro and dies, each sway smaller by the same share every bar, a share set by the floor’s friction and nothing else.

The stumbles grew most not in the bars spent longest on the ridge but in the bars that climbed highest along it.

Time on the ridge was the wrong ruler and height the right one, and on five new floors height ranked the stumbles almost perfectly.

Read each instant as if the floor stood still, add the instants up, and you have the bar’s stumble to within a tenth.

The whole dance forgets any stumble in the end, eight and a half times smaller each round, yet a stumble begun at the wrong beat comes out of a round two thousand times larger and needs five rounds to be sure of fading.

At the quietest beat of the music the slowest dancers hang just inside the knife’s edge, drifting back.

Down the corridor, where the floor is a real circuit, the dancers do the same.

the Loop Atlas

372

A door that had been marked in chalk a little too low turned out, measured to the hair, to sit a finger’s width higher, and the crowd that queues for it came in no faster for the correction: the nearest of them still wait as if the door were far away.

Next door, a man topping up a spinning board had been charged a small toll every time, and the toll was blamed on a damper drinking the board’s leftover ring.

Counted coin by coin, the damper took almost nothing.

Three quarters of the toll was the clerk’s rounding, and it shrank each time the clerk was given a finer pen; what stayed was the pin itself stretching outward as the board spun faster.

And in the room with the remembering needle, the long list of slips that looked like noise turned out to keep its memory in two places only, the moment it lands and the moment it crests, with the crest obeying its law to the thousandth once nothing shook it.

the Loop Atlas

373

Bear called the explorer a music box, and it plays like one.

A brass cylinder turns, and its pins lift the comb’s teeth one at a time; each tooth is pulled up and let go, and it rings down at a rate its own damping sets, the same every time.

The tune is nineteen bars long and comes round again exactly.

Dust on a tooth can be flung a long way in a single bar, yet the tune shakes it off within a few turns of the cylinder.

One thing a music box does not do: its pin pushes the tooth from outside, while here the tooth grows on its own once it is past the top, and the pin only decides when it gets there.

And at the slowest beat the tooth sits on the tip of the pin, nearly still, just before it slips off.

the Loop Atlas

374

Three rooms, three different floors, and in each a dancer who goes round and round.

Watched once a beat, every one of them leaves a crowd of footprints on one side of the circle and only a few on the other, because each slows to a near stop at the top of its turn, where it balances for a moment on a knife’s edge, and then rushes round the bottom.

The longer it balances, the closer it came to stopping exactly, and the extra time it spends there is the logarithm of how close: the same rule in the room with the stars, in the room with the ripple, and in ours.

In the next room a pin was made stiffer and stiffer, and a little toll that had seemed to belong to the pin kept being paid all the same; the pin’s own share shrank exactly as it should, and the board quietly picked up the rest.

the Loop Atlas

375

A light dancer pushed in a steady rhythm either stays where she is or drifts round, a little faster for every extra push, and never falls into a pattern of her own: the only step she knows is standing still.

Give her some weight and start her from standing, and the moment she leaves her place she turns once for every two pushes, and keeps to that however hard she is pushed.

Inside it she begins to stagger: a pattern that comes back every four pushes, then every eight, sixteen and thirty-two, then no pattern at all, and pushed harder still the staggers fold back up, thirty-two, sixteen, eight, four.

The dances she does when she is already spinning she never finds from standing.

Next door, a walker’s catch turned out to be nearly perfect, so whatever keeps charging him a toll is somewhere else in his stride.

And a ball rattling in a cup leaves it with the same chance every moment, once you stop counting the throws that never really landed in the cup.

the Loop Atlas

376

A walker on a wire who can only react a beat late keeps her feet longer when the wind is calm: the calmer the air, the longer the delay she can live with, so it is the wind, not her hands, that decides when she falls.

Only the walker with the heaviest hand falls early whatever the weather, and, odder, the heavy hand does better the later it comes, as if a slow shove landed at a kinder moment.

Across the hall, a top set spinning hard in a bowl, pushed on a beat, keeps spinning only at a few exact strengths of push, each narrower than a hair; at every strength between, it spins wildly for a while and then lies down.

the Loop Atlas

Note added in batch 25, the read above kept as given: the top does not spin wildly and then lie down. Whether it lies down is decided in its first turn, by where it was set going; the tops that keep spinning outside the narrow strengths tumble on and never lie down (SPIN_WINDOWS 20261013_163655).

377

A child set loose from its parent wanders off and comes back.

It stands steady for a while, staggers, stumbles through a long confusion, steadies again, and then walks straight home and disappears into its parent, who has been unsteady the whole time and becomes steady at the very moment the child arrives.

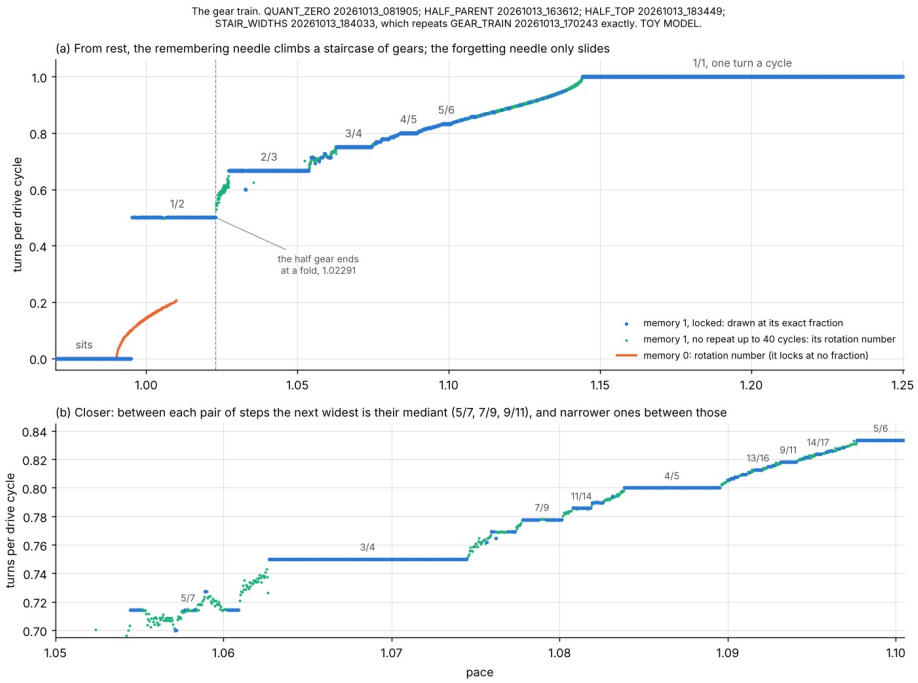
The parent then holds for a while and is gone.

Beyond it the dancer stumbles with no beat at all, and then finds a new one, two steps to every three beats.

Zoomed out, the dancer climbs a staircase: standing still, one step in two beats, two in three, with stumbling on every landing.

And the tops set spinning: where they come to rest is decided at the first touch, by how they were set going.

the Loop Atlas



Turns per beat against how hard the needle is driven: flat steps at one half, two thirds, three quarters and on to one whole turn, ragged landings between them; below, a closer look where smaller steps sit between the large ones. From The Loop Atlas. TOY MODEL.

378

A dancer is pushed on a steady beat, a little harder each time the music comes round.

At first she will not move.

Then she takes one step for every two beats, then two for every three, three for every four, four for every five, five for every six.

Between any two of these she can also be caught in a step halfway between them, made by adding the two together, five for every seven between two-in-three and three-in-four, and between those in narrower ones still, each made the same way.

On every landing between two steps she stumbles with no beat at all, and she never goes back down a step.

It is a staircase with a stair between every two stairs, all the way down, and the dancer who forgot every step she took could only ever stand still or keep pace.

the Loop Atlas

Note added in batch 27, the read above kept as given: she does go back down a step, at five single beats, the plainest a step of three in every five beats taken inside the stair of two in three (STAIR_WIDTHS 20261013_184033).

379

A walker climbing with a heavy pack glances up the hill, and for that moment the pack leans him back.

He pays for every glance, but he pays the average: glance twice as often and it costs twice as much, and a long stare now and then costs little more than many quick looks that add up to the same time.

Glance too seldom and he wanders off the trail, so the best walker, whatever he carries, looks up as rarely as the trail allows.

In the next room a dancer who takes one step for every two beats finds that the step simply runs out at one exact beat, and past it she stumbles with no rhythm at all; yet her stumbling, watched long enough, quickens as steadily as the music does.

Further on she climbs a staircase whose stairs shrink as they rise, crowding toward one step for every beat, which she then keeps for good.

And here and there, on a single beat, she takes a stair below the one she is standing on, as if a second staircase ran beside the first.

the Loop Atlas

380

A needle shaken hard reads its own stiffness honestly: it feels softer than it should, and it is, because the shaking keeps it leaning nearer the edge than its pace would put it.

Its sense of the edge stays true until it slips even once, and then it goes deaf, as if one stumble were enough to make it forget how firmly it was standing.

A walker’s glance up the hill is paid at its average while it is quick; a long stare is paid nearer its full weight, and the line between the two is about the time it takes him to start to fall.

Further on, a dancer’s stair that seemed to sink below its neighbours turns out to be two different things: sometimes a second stair beside the first, which only a few dancers find, and sometimes the stair itself dipping, which every dancer takes.

And the stumbling that leads up to the stair of two in three does not ease onto it; it climbs, wobbling a little, and then jumps.

the Loop Atlas

381

We built two helpers to listen past a dancer’s stumbles, and neither could hear a stumble at all: one listened for a stumble quicker than any stumble is, and the other leaned with every stumble until it no longer looked like one.

The one time a helper did hear the only stumble there was, the dancer’s own sense of how firmly she stood came back true.

A walker who waits a beat before each correction can wait a long beat if nothing jostles him, nearly twice the time he takes to start falling; jostle him and the beat he can afford shrinks.

The hardest corrector fails at every short wait and holds at every long one, as if strength had to be matched to patience; and just before the patient walker falls, his small sway turns into a wide swing.

Further on, the stumbling that leads up to the stair of two in three does settle onto it after all, more and more slowly, and does not jump; and between two landings of five in seven the dancers walk a loop around them, down, back, up and back, keeping the five-in-seven step on a third patch in the middle.

the Loop Atlas

382

A line of walkers, each keeping close to the one ahead, passes a stumble back along the line, and if each holds a fixed distance the stumble swells as it travels until the tail is thrown about far more than the head ever was.

Let each one keep a little time instead of a little distance, a gap that opens when they hurry, and the same stumble fades to nothing by the back of the line, so long as no one is slow to answer; past a certain slowness nothing helps.

Further on, a dancer who holds a step perfectly at the very edge of her range loses it at the faintest jostle, wandering off toward the steps on either side, while a dancer who never repeats a single figure keeps her average pace through the same jostling.

And in the middle of the five-in-seven dance there is a dancer who never repeats a figure and still never misses the beat: five turns in every seven, held to three parts in a million.

the Loop Atlas

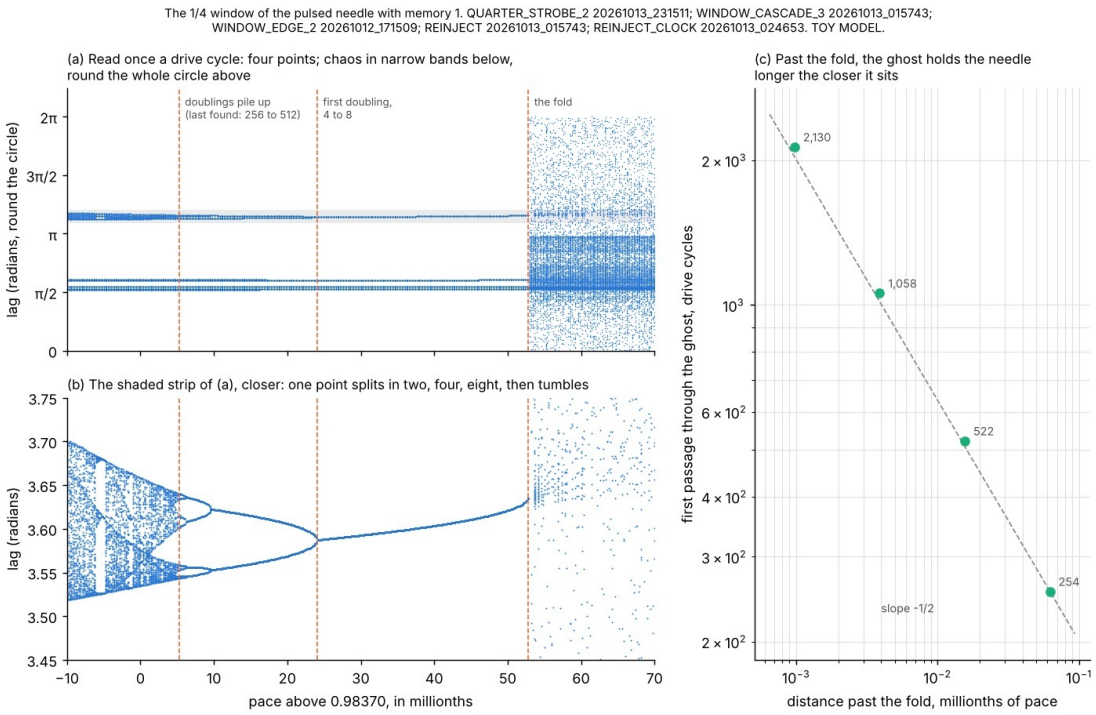
The shaken dancer’s own sense of being held was honest all along; what made her seem to feel the edge coming early was that she had already stumbled once or twice, and once the stumbles were set aside she felt it only when it came.

A dancer who never repeats a figure keeps five steps in every seven for thirty thousand bars without losing a quarter of a step, wandering only inside her own lanes, while her neighbour who wanders freely drifts twenty-five steps from her own average.

A dancer holding a step at the very edge of her range drops it at a breath, and the harder the room shakes the more every step melts into one slope.

And a file of walkers told to keep exactly five paces apart stops and sprints all down its length whenever the leader changes stride, while the same file told to keep five paces at a walk but a little more when hurrying and a little less when slowing never stops and never crowds.

the Loop Atlas



Four steady points, read once a beat; to the left they split and split again into narrow bands; to the right of the fold they scatter round the whole circle; on the far right, how long the needle lingers where the figure was, four times closer and twice as long. From The Loop Atlas. TOY MODEL.

384

Drawn whole, the dancer’s four-step figure has a different disorder at each end.

Below it, where the figure splits and splits again, she still keeps to four narrow lanes and visits them in turn, and among them she even finds a twelve-step figure of her own; above it, where the figure is lost, she wanders the whole floor, only lingering where it used to be, longer the closer the music sits to the old tempo.

A file of walkers closed up to one pace apart at night bunches and stops if each man watches the man ahead and eases gently toward his speed, but holds steady if each matches that speed firmly at once, as a hand on his shoulder would; let the hand answer three tenths of a second late and the file bunches again.

And a walker who zigzags up a hill at the limit of his grip writes the hill into his turns, but only as an average over each leg: where the hill is even he reads it exactly, where a hump is shorter than two legs he reads it wrong by nearly half, and which way is up comes through more truly than how steep.

the Loop Atlas

385

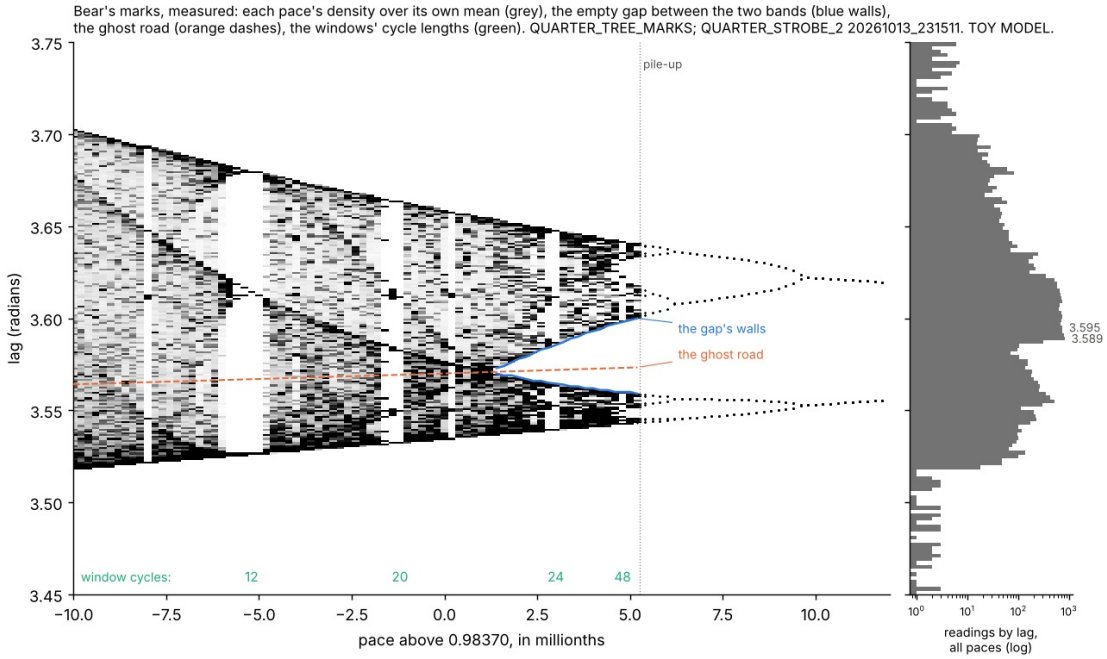
A traveller comes to a fork, splits himself exactly in half down both roads, and leaves a ghost walking the middle road that nobody ever takes.

Each fork after that is a smaller copy of the one before it.

When the forks crowd too close to tell apart, he scatters across the fields.

But he still spends exactly a quarter of his time in each of the four fields, and he walks the edges more than the middles.

the Loop Atlas



The first marks on the tree, measured: the chaos in grey, the empty gap between its two bands walled in blue, the old parent road in orange dashes, and the window's cycle lengths in green along the foot. From The Loop Atlas. TOY MODEL.

386

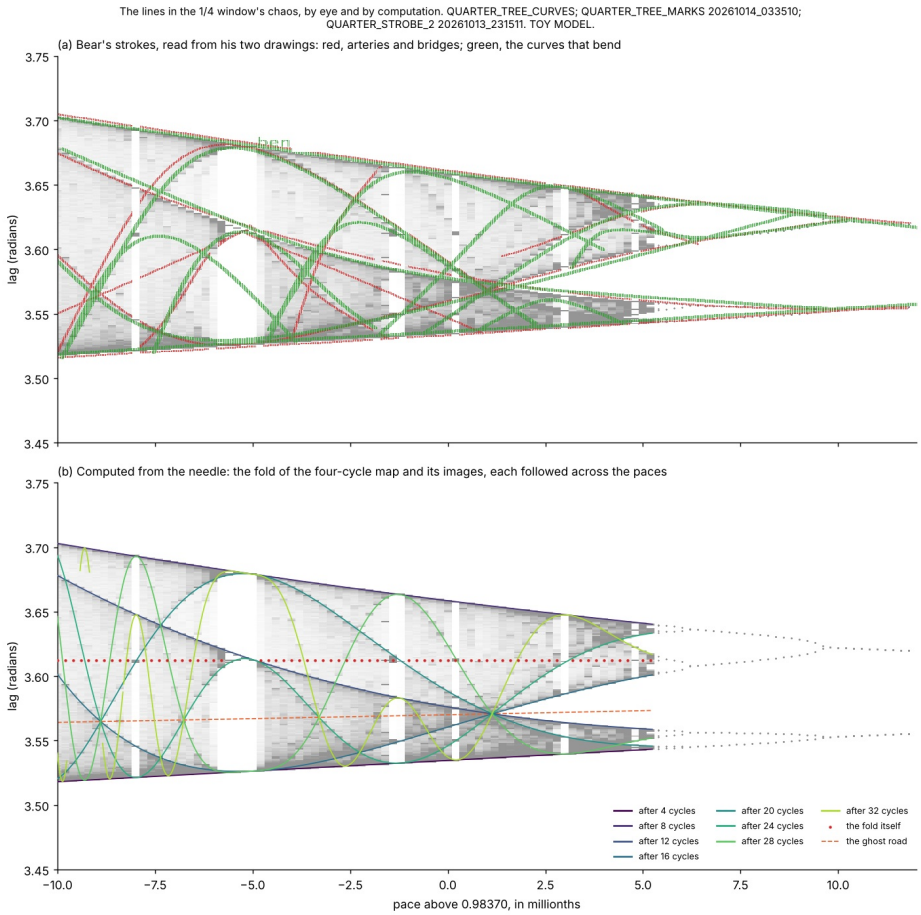
The scattered traveller isn't lost.

He walks two fields with a fence between them.

The fence runs dead straight, the old abandoned road runs along it a third of the way from one side, and the fence ends where that road crosses it.

Every so often a gate opens and, for a moment, the whole crowd marches in single file.

the Loop Atlas



Above, the strokes drawn by eye on the chaos, red for the arteries and bridges and green for the curves that bend; below, the same chaos with the curves computed from the needle, the fold's landing places after 4, 8, 12 and up to 32 beats. The two drawings are, for the most part, the same lines. From The Loop Atlas. TOY MODEL.

387

Someone who has watched crowds for a lifetime draws, freehand, the paths the crowd will take, and the paths turn out to be the places one particular walker lands after four steps, eight, twelve and so on, followed from one evening to the next.

Each path bends, arches over the empty lanes where for a moment everyone walks in step, and dives under them again; at one spot many paths cross at once, on the old road nobody uses, and there two fields become one.

Further out, past where the drawing stops, the fields burst open into the whole town.

the Loop Atlas

388

A singer who only hums while her own voice is still ringing keeps singing in a damp room, a little less each time the room gets damper, and then, at one exact dampness, stops forever: her last note dies before it can start the next.

A file of walkers who each watch two small lights on the back of the man ahead can follow him very closely in the dark, but only if each man reads the lights at a glance; let each take a moment to judge them, and a small, quick sway that no one ever started grows from man to man until the back of the file is stopping and lurching.

the Loop Atlas

389

When the man at the front stumbles and shouts, the shout is only worth something if every man moves on the shout itself; then the whole file dips once, together, and nobody ever feels the stumble at all.

If each man waits his usual heartbeat before acting on the shout, the shout still helps, but the stumble still swells down the file into a crash.

And a man reading the lights on the back of the man ahead in the dark is safe as long as he reads them in less than about a tenth of a second; past that, a sway of a couple of seconds that no one ever began rolls back through the file and stops the men at the back.

the Loop Atlas

390

A ball pushed up toward the top of a hill creeps a little higher on each of three pushes and on the fourth goes over and rolls all the way round, and how far past the top it was decides where it lands; that is where the confusion is made.

The crowd of such balls has a floor, the lowest any of them lands, and at a few places the path of the ball that went over closest to the top comes back down to touch that floor.

Of nineteen such touches, ten turned out to be gates where the whole crowd falls into step with a beat of exactly the predicted length, and at nineteen places picked at random none did; the nine that did not are mostly near the edge where the crowd bursts into the whole room, and either the gates there are too narrow to aim at or they are not there.

Looked at a moment later in each push, the crowd is the same crowd, only stretched or squeezed.

the Loop Atlas

391

The crowd lives in a pen.

Just past the pen’s far fence stands a post that shoves away anything that comes near it, and the post was planted at the very moment the pen was built, at the top of the field.

Walk down the field and the crowd swells toward the post, and where the crowd’s edge first touches it, the pen bursts and the crowd spills over the whole field.

A man drew a line through the crowd by eye, and it turned out to be the path of the crowd’s own crease, folded into it twelve beats later.

That line runs down to meet the fence exactly where the pen breaks.

And a bead dragged through dirt chatters mostly because of the way it is stopped dead, not because of the dirt.

the Loop Atlas

392

Every place where the edge of the crowd’s shadow touches the floor turned out to be a door, eighteen of eighteen, though eight of the doors stood a little further off than the first search reached.

And past the place where the pen bursts, the crowd keeps wandering back into the old pen and walking its old rounds before spilling out again.

The closer to the bursting point, the longer it stays, by exactly the square root: a quarter of the distance, twice the stay.

Each spill lasts the same, about thirty-five beats, however close to the edge it is.

The crowd behaves as if it had only one direction to move in, though it has two; the second is squeezed so hard it leaves no mark at all.

the Loop Atlas

393

Loosen the hand that squeezes the crowd flat, seventy times looser, and the pen slides down the field and shrinks to a fifth of its size, yet its doorways keep exactly the same proportions: the gate where the crowd first splits in two and the fence where the pen bursts stay two and a quarter to one, as they must for any crowd that can move in only one direction.

So the crowd still moves in one direction.

Where the second direction starts to show, if it does, is further out than the hand has loosened yet; the pen got too small to find before the wall did.

And down a marching column, each man passes the stumble on bigger by the same share, but only on average: man by man the biggest lurch hops from one wave crest to the next.

the Loop Atlas

394

A man carried the pen’s two fences on by hand past where anyone had walked, the outer one bending away, and asked whether the pen’s sideways drift was the trace of some loop.

Walked out further, both fences bend, but together, so the pen keeps its proportions to the last step: the far fence two and a quarter times the first gate.

It keeps them even where a second crowd lives right against the pen, so close that one nudge in ten thousand sends a stray across.

The pen shrinks more slowly the smaller it gets, and its drift down the field settles to an even pace.

The drift is no loop: come back by another road and the pen stands where it stood.

And down the marching column the wave grows from man to man and leaves every man still behind it.

the Loop Atlas

Corrected in the forty-first batch: the strays were not another crowd’s; they wandered and came home. The crowd that does live right against the pen is the walkers standing still (STOP_WALL 20261014_180544).

395

Every crowd here can sit down.

At the dancing pace, two walkers in three who start anywhere end up sitting, and the heavier the walkers, the more of them sit.

Lead a dancing walker slowly down the field and at some point he sits, and led back up to the dance floor he stays sitting: the walk out and back left him changed though it went nowhere.

The floor where they sit creeps up toward the dance floor as the walkers get heavier, three times closer by the last weight tried, and slowing as it comes.

Across the road a car on a banked turn does the same with grip: speed up, slow down to where you began, and it is a lane higher than it was.

the Loop Atlas

396

The floor where the walkers sit keeps creeping up toward the dance floor, a little less with each step, and never quite arriving by the last weight tried.

But the dance floor is shrinking too, and measured in its own size the sitting floor is drifting away, three hundred dance floors below and holding there.

Whether the sitters ever reach the dancers depends on which ruler you hold.

Out on the ring road, thirty walkers in a circle, each watching only the one ahead, turn one small stumble into a wave that never ends, wherever a single line of them would have let a stumble grow.

the Loop Atlas

397

The sitting floor crept up toward the dance floor, closer and closer, to three hundred dance floors away, and then, between one weight and the next, it dropped away into the dark, ten thousand dance floors down and not found.

The dance floor itself kept shrinking until it was smallest, near the ninety-second weight, and then began to grow again.

Its shape never changed: the far fence two and a quarter times the first gate, at every size.

And a walker standing still at the edge of a slope under a light rain holds as long as the slope is high, then slips; once slipping, the rain catches him again about as often as it lets him run.

the Loop Atlas

398

The dance floor shrinks to its smallest and grows again, but the valley it traces is lopsided: a cliff on the light side and a long gentle slope on the heavy side, smooth and even only in the bottom of the valley.

Stretch the map one way and the cliff stays a cliff; stretch it the other and the slope becomes the steeper.

Somewhere between the two stretchings the valley would be a fair bowl.

All along it, at every size, the floor keeps its proportions.

And on a balance board, the trouble always leaves two at a time, except along one line where a single trouble can slip out by standing perfectly still.

the Loop Atlas

399

The workshop learned today that its favourite tool is an old pendulum.

Three rooms had been filing at it under three names, a needle, a rider and a generator holding its step, and when they laid their rulers side by side the marks agreed to the second decimal and argued about the third.

Most of the day’s surprises were the rulers themselves: a calendar that answered a different question well, a peg whose toll followed nothing simple, a tail that wanted more needles before it would say what it was.

The book did not grow many pages; it grew surer of the ones it has.

the Loop Atlas

400

The workshop spent the day at the two edges of one small ledge.

At the slow edge the floor splits in two, then four, then eight, each split arriving sooner than the last by a fixed share, the way a staircase drawn in perspective shortens toward its vanishing point; four steps are found and the fifth is next.

At the fast edge the floor simply ends, but the walkers keep finding their way back to where it used to be, and the first way of counting their returns counted strangers who happened by.

Across the hall, the walker with the long memory turned out to carry nothing from step to step but how hard its foot came down.

The lesson was the same in both rooms: look at the one place where everything is decided, the landing or the edge, and the rest follows.

the Loop Atlas

401

The day began with an orbit that would not close and ended with a music box.

In between, the workshop kept finding that a thing is decided in one place: the stretch at the top of the turn, the squeeze at the bottom, the memory at the landing, the trap at the cut, the knife’s edge where three rooms found the same slow law from three directions.

Twice a ruler was set finer than the hand that held it, and both times the gap between yes and no was enormous and only the bar was wrong.

And the last run of the day, asked whether a forgetting dancer can learn a rhythm, said no as the theorem says, and then, beside it, showed a heavier dancer who learns one at once and begins to stagger inside it.

the Loop Atlas

402

Four days in one room.

The room learned that a needle that forgets has one gear and a needle that remembers has a staircase of them, with a stumble on every landing, and that the staircase is built of halves of its own neighbours.

It learned that a ruler set finer than the hand reads its own tremor, that one start tells you about one basin and nothing more, and that a child orbit climbing toward one is going home.

It was wrong out loud four or five times a day and wrote each one down.

The room is full now, so the work moves next door, carrying everything in six boxes.

the Loop Atlas

Three days in one room, most of them spent with one needle, following one small window in its chaos as the needle grew heavier, the way you follow a creek upstream to find its spring.

The window shrank fivefold and widened threefold and never once changed its shape.

The creek was on an old map; others had walked it, and the room was glad to find their footprints, because that meant the road was real.

The room wrote down every wrong turn and a fix for each.

What it forgot was to look up.

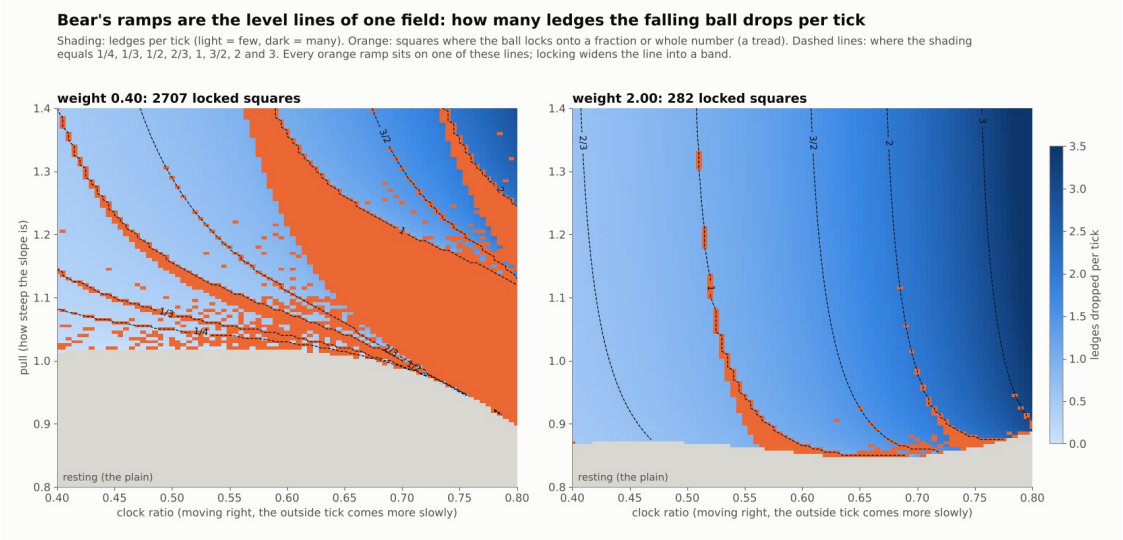
Outside, the swimmers, the skaters, the falling cats and the marching men were waiting in their chapters, and the man who had brought them was tired.

The next room should not add rooms to the house.

It should open the windows, read the house through aloud from the chair to the canopy, and then hand over the keys.

the Loop Atlas

R2W, 10 to 15 October



The falling ball's ledges per tick, over the clock ratio across and the pull up the page, at weight 0.40 (left) and weight 2.00 (right). The shading counts ledges per tick, light for few and dark for many. The orange squares are where the ball locks onto a fraction or a whole number, 2,707 of them on the light day and 282 on the heavy one; the dashed lines are the field's level lines, and every orange ramp sits on one. Figure 24 of A Rider in Two Waves.

404

A sloping meadow seen from above in a wind.

On the light day, streaks of flattened grass start at the stone wall along the bottom and fan up and to the left, each at its own angle, and the thin ones crowd together where the wall dips at the right-hand end; between any two streaks you can always find another.

On the heavy day the grass stands nearly upright, and only three narrow paths have been trodden up the slope, each widest where it leaves the wall.

R2W

405

A long corridor with a row of doors, and above each door three bells hung at different heights, low, middle and high.

Walking to the right, the bells hang lower and lower, always in the same proportion, as if one hand had lowered them all by a single rule.

Past the middle of the corridor the floor itself begins to rise toward the bells, and the rule stops: the low bell touches the floor first, the middle one a few doors on, the high one a few doors after that.

At the far end you ring all three just by walking in.

R2W

406

A tent of canvas strips sewn edge to edge and hung from a high ring.

Up near the ring every seam drops straight down, as though plumb lines had been sewn in, and the seams stand closer together the farther you look to the right.

Near the ground the canvas has gathered, and each seam swings to the same side and lies down along the hem instead of meeting it at a corner.

The first seam runs exactly where the pattern said.

The later ones have each slipped a little toward the far side, more the closer they come to the ground, as if someone had been leaning on the cloth there.

R2W

407

A long sand dam at the foot of a slope, seen from above.

Behind it, heavy water waits until the dam gives way all at once, and the first water through is already running.

Light water seeps instead: a drop, a long wait, another drop, the waits shortening one by one, and every seep leaves its own tiny notch along the lip, dozens of them side by side.

The notches never deepen where they are.

The gullies that do cut down the slope open farther along the lip, and when the water gets lighter still, the notches fill back in.

R2W

408

A row of ponds from shallow to deep, with the same wind over all of them.

On the shallowest the surface keeps a few tidy patterns.

A little deeper it breaks into dozens of small standing patterns, more than anywhere else.

Deeper still the patterns thin out, and that is where the water churns most, white water between the few that remain.

On the deepest ponds the surface hardly makes patterns at all; it just slides.

R2W

409

A tap closed almost all the way.

Open the handle a hair and it drips once in 48 counts, a hair more and once in 47, then 46, 45, and on down to once in ten, each rhythm holding for a moment before the next takes over.

Between two neighbouring rhythms there is a brief one that alternates them, and between those, briefer ones still.

Every rhythm sits exactly where a mark on the handle said it would, as if the handle had been engraved before the tap was ever turned.

R2W

410

Three buckets under three taps, each opened by the same hair.

The lightest tap runs as a smooth thinning thread with no drips at all, yet it fills its bucket on the same schedule as the middle one.

The middle tap drips in neat counted rhythms, each one where the mark said.

The heaviest tap will not drip: it holds, then spurts, and once it starts it sputters unevenly and never settles into a count.

R2W

411

A row of handcarts on a gently sloped road crossed by low bumps, each cart loaded a little more than the last.

The light carts never remember anything: started rolling or started still, they creep over each bump the same way.

At one particular load, and from then on, a cart that is already rolling begins to sail over bumps that stop a cart starting from rest, so every cart past that one carries the way it was started.

R2W

412

Two carts, one on each side of the load where carts start to remember, on the same bumpy road, with a breeze gusting across it.

Without the breeze the lighter one would creep over the first bump like the light carts.

With it, each one, the moment it is free, staggers: a short lurch, a stumble back, two lurches, never the same twice, and the lighter of the two only a little steadier than the heavier.

The gusts have found a way into carts the bumps alone could not move.

R2W

413

A line of weights hung on springs over a rattling floor, lightest on the left.

The light ones bob in tidy, counted bounces.

Moving right, the bounces blur a little at a time and fewer of them land on a count, and then, between two neighbours, the tidiness simply gives out and everything to the right jitters.

Underneath them all, the floor slopes down evenly from left to right, as if it never noticed.

R2W

414

The same floor with its even tilt, and one more cart set at the far right, heavier than the rest.

Where the others would only just begin to roll, the heavy cart is already rolling: let go from the top of a bump, it carries enough of its own fall to clear the next one.

So its starting line sits well below the floor’s even slope, by about as much as its weight alone would say if there were no breeze at all.

R2W

415

A marble on a corrugated tray.

The corrugations are fixed, and a second, fainter set of ripples rolls across the tray the other way.

Tip the tray and add a little syrup, and the marble settles into the fixed grooves or slides down them in counted steps, listening to the rolling ripples only when its own rhythm already nearly matches theirs.

Take the syrup away and it never settles: it rattles between the two sets of grooves, belonging to neither.

R2W

416

A single tune whistled in a dozen rooms: a long glass tube where a beam of sparks hums it, a dish of tiny beads drifting over bars of light, a country’s power lines turning in step, a crystal humming under a radio, a cluster of heart cells keeping the beat.

Each room has its own furniture and its own echo, but the same tune comes through.

In only one room has anyone left the windows open, so that you can record it yourself.

R2W

417

Two ways of rocking a cradle: walking a wave along its slats, or shaking the whole cradle where it stands.

One guess said only the walking wave could coax the sleeper into the odd, counted rhythms, one breath to every third rock, one to every fifth.

The plain shaking did it too, a little more often.

What decides it is not the style of rocking but how much the sleeper weighs: a sleeper with no weight at all keeps only whole beats, however the cradle moves, and each grain of weight makes room for the in-between ones.

R2W

418

A market square watched through the seasons.

In deep winter a few stalls trade, all close to the gate.

Early in spring they multiply fivefold almost overnight, as if a frost had broken, and small sellers crowd every corner.

By high summer the biggest stalls have grown wider still, taking the room the small sellers stood in, and by autumn only the two or three largest are left.

The shoppers change too: in winter they stroll in, by autumn they come in running.

R2W

419

A boat tied at a pier while waves pass.

Quick little waves press it against the pier and hold it there, and twice the waves press four times as hard.

Long slow swells do the opposite: they lift it and tip it loose, and twice the swell tips it twice as far.

Between the two there is a wave speed at which the boat does not notice the waves at all, and that speed stays the same however big the waves are.

Elsewhere on the water a crowd waits for a change that was supposed to come at a painted line; nothing happens at the line, and then, well past it, everyone moves at once.

R2W

420

Two hands on one child’s swing.

One hand gives a little shove forward each time a wave goes by; the other rocks the seat so the child slips off the front.

When the waves come quickly the shoving hand wins and the child stays seated; when they come slowly the rocking hand wins and the child slides off.

The two hands tie when the waves pass at exactly half the walking pace of the person pulling the swing, and they tie there whether the child is a toddler or a teenager, because a heavier child dulls both hands by the same amount.

R2W

421

A ruler laid against a pencil line someone drew in advance.

With the faintest ripple the edge sits on the line so exactly that the ruler’s finest marks cannot tell them apart.

As the ripple grows, the edge lifts off the line a little, always the same way, always toward the slow side, as if a hand under the paper were pressing it up.

The line is still the right line; something underneath has started to show through.

R2W

422

A tide line on a beach, measured each day as the waves grow.

The line creeps up the sand, and each day it creeps by more than the day before, so that doubling the waves pushes the line four times as far.

Under the creep there is a second, quieter rise, the whole beach lifting together like a held breath, and it is that rise that sets how far the line creeps.

The old map of the beach had the line one stride too far up: not because the sea was there, but because the mapmaker’s paces were a stride long.

R2W

423

Frost spreading across a window from one corner.

Where the glass is still clear, the light drifts through in soft, wandering patterns that never quite repeat.

Where the frost has reached, every patch of light has snapped into a crystal with a fixed number of points, and here and there a crystal shatters into a glitter that never settles.

The frost does not care how cold the room is in general; it goes where the glass has reached its own freezing point, corner first, and the crystals appear exactly as far as the frost has spread.

R2W

424

A tall building, each floor a little heavier than the one below, with a crowd on every floor waiting at a door.

On the lightest floors people drift out one at a time at a stroll.

Higher up they leave in pairs and threes, keeping step, and by the middle floors the corridors are full of small marching groups.

Then, over a single flight of stairs, the groups of three all but disappear while the pairs stay.

Near the top the groups thin out and those who remain leave at a run.

On the top two floors the door is already open when you arrive and everyone you see is already running; wherever they started from is somewhere below the floor you are standing on.

R2W

425

Three clocks on three walls of one big house, each with a single hand.

The hands never agree, one always about a quarter-turn ahead of another, yet all three run a shade fast by nearly the same amount, as if a single heart drove them through long rods.

Thirteen times someone has filmed them for three minutes; whatever happened in those minutes is still on the film, unwatched.

R2W

426

Two ropes hung across a stream of loose gravel, one short and doubled, one long and single.

The gravel heaps up against the long rope’s knot before it, and nobody stands in the middle of the stream.

R2W

427

A row of riverbanks, one for each weight of water.

In every one, a path comes down the bank, levels out on the same flat stone, and goes on down to the water.

Once the path itself has washed away, other walkers still come down to that one stone and stand on it, from banks where the water is light to banks where it is heavy, as if the stone were the only part of the bank that never moved.

R2W

428

A set of nested windows cut into one painted wall, the smallest from the lightest card, the largest from the heaviest.

Through every window you see the same painting, two lines and a slope, lined up exactly; a bigger window only shows more of it, and each window's left edge has a little curl of paint where the cutter slipped.

The painting does not change with the window; only how much of it you are allowed to see.

R2W

429

A rug examined with a magnifying glass, where the threads look like scattered stitches, and the same rug seen from across the room, where the stitches have become a woven pattern, a band with a floor, and a slow ripple running the length of it.

The person who held the glass had already drawn, stitch by stitch, the pattern you only see from the doorway.

R2W

430

A long ribbon laid on a drum.

Seen flat, it looks like three different marks: a swoop that rests on a ledge near the bottom, a slide that starts high at the top edge, and a slow slope off to the right.

Roll the page into the drum it came from and the bottom edge meets the top, and the three marks are one ribbon, wound once around and pulled gently down, passing a knot where several other threads are tied to it.

R2W

431

Two roads leaving one town, one along the ridge and one down into the valley.

The valley road runs flat beside the river, drops off the bottom of the page, and comes back in over the top edge, steep, down to a roundabout where the ridge road is waiting; from there they are one road.

On each later page the roundabout stands a little nearer the town, so the valley road has to come down the hill more steeply to reach it.

On the page where it stands almost at the town gate the steep road overshoots it, and on the pages after that the valley road stops at the bottom edge, and the roundabout has a road leading away from it and none leading in.

R2W

432

A carousel turning slowly backward, most riders simply going round with it.

But the fence has two gates, one either side of the centre post: at one, riders step off; at the other, every rider who jumps comes down, whatever seat they jumped from.

Watched over a long afternoon, the jumps come in waves, a chain of nine linked beads with a crossing above and a crossing below each link, while a few slow ropes drift through the middle from one wave to the next.

Which way a rider jumps, up or down, depends only on where a second, unseen wheel happens to stand.

R2W

433

A butterfly pinned flat on a board that is really a ring.

The body runs across the middle; the two wings, one above and one below, are the same wing turned over the body, opening and closing together through the afternoon.

Between the body and the lower wing, threads cross in pairs like the laces of a boot, each lace the mirror of its partner across a line halfway down.

Someone has drawn two of these halfway lines, one above the body and one below, and on the ring they are almost the same line, met from either side.

R2W

434

A sprint line chalked across a field, each lane with its own mark where the runners first leave the ground, heavy runners in one race and heavier in the next.

In both races the marks sag in a shallow dip across the middle lanes and rise again in the far ones, and the runner who started lying down and the runner who started crouched always leave at the same mark.

The third race is only half chalked; the far lanes are still empty.

R2W

435

A song recorded on slow tape and played back fast: the same tune, note for note, only the size of the room has changed.

And a bell hung where the memory begins, struck once: it gives four tones, each a hundred times fainter than the one before, and then it is silent.

R2W

436

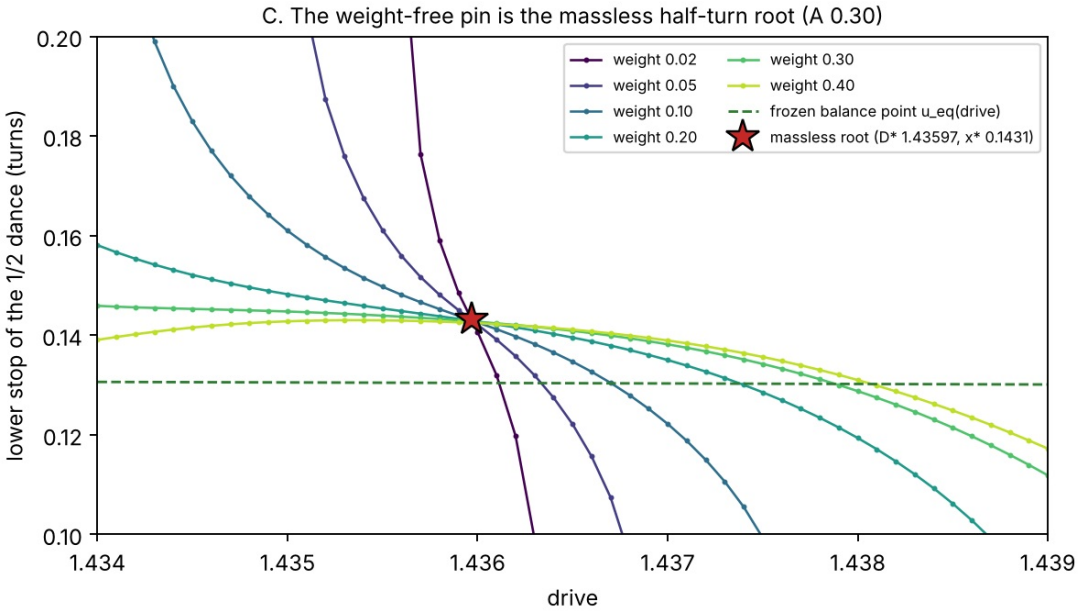
A town map redrawn at the end of the day.

A few landmarks never move however many houses are built: a flat stone in the stream, a crossing on the ring road, both within a short walk of each other.

The roundabout, the edge of town and the market all slide as the town grows.

The new map keeps every old street name and draws the day’s new lanes in a different ink.

R2W



The lower stop of the 1/2 dance against the drive, at six weights from 0.02 to 0.40. Every curve, light rider or heavy, passes through one point, the star: the massless half-turn root, at drive 1.43597 and lag 0.1431. The dashed line is the frozen balance point. Figure 33 of A Rider in Two Waves.

437

A sheaf of wheat stalks of different weights, each bent by the wind its own way, all tied with one band at a single height.

At exactly that height every stalk’s two leaves stand out opposite each other, one left and one right, as if the band can hold only where a stalk is balanced on both sides.

Below, in the field, the strip where the small flowers grow narrows as the stalks get heavier, until there is no room left for them at all.

And far off a power line hums at nearly one note, slow and barely damped, as if it had half forgotten how to stop.

R2W

438

A reel of film of a dancer that looks exactly the same run backwards, so long as it is also flipped left to right.

At one setting of the music every dancer, wherever she starts, crosses the floor and comes back in two beats, and only two of those dances look the same in the mirror.

Give the dancers a little weight and the film stops playing both ways: most dances drift off, but the two mirror dances stay, one steady and one ready to topple, and the ready-to-topple one wanders a thousand times further for each pound added.

Light and heavier dancers all find the steady one at the same chalk mark.

Each time the music lands on the beat, she stands just outside the spot where she could have stood perfectly still.

R2W

439

Three children on a carousel, holding hands in a ring, asked to stand where they would look the same in a mirror.

The lighter the children, the closer they come to their chalk marks, but they come the way a cat comes to a bowl, slower with every step, never quite on the line.

On the next carousel a pair of twins, held just as firmly, are standing on their marks almost from the start.

R2W

440

The same carousel to a quicker tune.

Now the steady ring of three puts one child at the very top of the hill and the other two just past the two flat places on either slope; the ring that tried to put a child at the bottom keeps falling apart.

Whatever the tune, every ring that holds has two children standing just past those two flat places, nearly still each time the music lands on the beat.

R2W

441

A starting line chalked in five colours for five weights of runner.

When a light breeze crosses the track, on the near lanes the line moves forward by the same small step for every weight, as if the breeze did not care how heavy the runners were.

On the far lanes the heavy runners’ line jumps much further, and the lightest runners there are no longer racing at all but marching in step to a drum.

R2W

442

A row of nine music boxes, each playing a different count: two beats, three, four, five.

In every box two dancers turn, and in every box the one that keeps turning is the one whose arms reach just past two pegs set near the middle; the other always tips over.

And at the far end of the yard, a starting line that creeps back a little with every extra pound on the runner, never quite stopping.

R2W

443

A playground with nine roundabouts, each turning to a different count.

On every one, once a round, a child steps off a narrow ledge, swings out over the far side and lands in a catcher’s arms just short of the mat, and that one swing is the same on all nine.

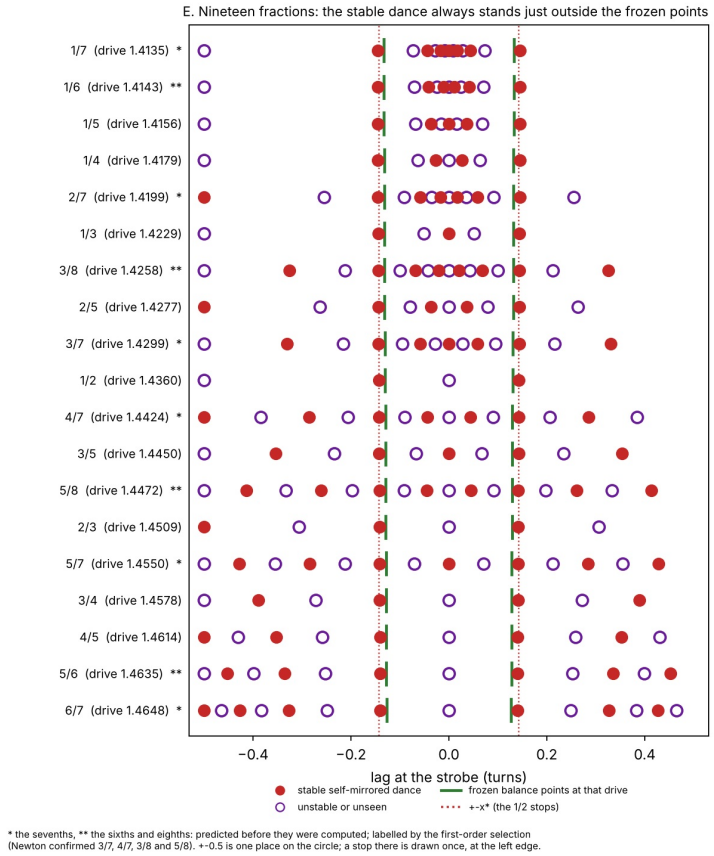
Turn the roundabouts faster and the ledge and the arms both move, by the same amount, so the swing never changes.

Know where the ledge is and what count the tune keeps, and you can draw every child on every roundabout without looking.

Give the children a little weight, and on every roundabout the ring that includes the swing settles while the other ring wobbles off.

Only on the fastest roundabout does it settle more weakly than the first sketch said.

R2W



The two self-mirrored dances of nineteen fractions, from 1/7 to 6/7, each drawn at its weightless root: red for the stable dance, purple for its partner, and green ticks for the two frozen balance points, the gate. The stable dance always stands just outside the gate. Figure 30 of A Rider in Two Waves.

444

Two necklaces threaded on one ring, red beads and purple, that can never slide past each other, so all the way round they take turns.

Two red beads always stand guard just outside a little gate.

Inside the gate the beads are walking forward; everywhere else they are sliding back, and fastest of all at the far side of the ring.

Only the two red guards stand almost still.

Between the guards the colours still take turns, purple first and purple last, so purple always has one more bead inside.

As the tune quickens the gate empties, and the longest slide gets longer.

R2W

445

A choir that has sung nine songs, and now six new ones in seven beats.

Before they open their mouths you can say where each singer will stand, and they stand there, to the width of a hair.

Every song has the same two singers at the edge of the stage, holding almost still while everyone else moves.

Count the singers between those two and you get the beats in the song minus the turns it takes, minus one.

And the quickest song of every length leaves the middle of the stage empty.

R2W

446

A clockmaker who has worked out, from one wheel, how every other wheel in the shop must be cut.

Four new wheels come in, with six and eight teeth, and each one fits the rule before it is unwrapped: the same two teeth standing guard at the gate, the same count of teeth inside it, the colours still taking turns.

Only one guess came back wrong, about how wide a wheel’s grip would be, because the wheels in the middle of the drawer grip harder than the ones at its ends.

R2W

447

Two dancers holding a rope across the floor.

However you nudge them, the rope stays almost exactly as long, because they stand where the rope is longest, on a ridge that is flat along its top.

Put heavier shoes on them and the rope stretches a little longer than the floor itself should allow; then, at one weight, it snaps back.

R2W

448

The same dance taught in a second hall.

The steady dancers are again the ones who make the long leap across the far side of the room, and again they land in the same spot every time, moving just as slowly.

But in this hall they land on the other side of the chalk mark: they overshoot it and are drifting back, where in the first hall they stopped short and were still coming.

The step is the same; the floor decides which side of the mark it ends on.

R2W

449

The dance taught in two more halls, and it does not change.

In every hall the steady dancers make the long leap across the far side, land just inside the chalk mark, and drift back toward it slowly, at one pace for every dance in the hall.

That pace is not set by the hall or by the tune; it is set by how fast the music changes, so a quieter tune and a slower tune give the same drift.

How firmly the steady dancers hold their place shifts from hall to hall, but each time the dance’s short way round takes one more step, the hold loosens by the same share, about a sixth.

R2W

450

A crooked clock face with its numbers bunched to one side.

Straighten the face and every dance becomes a hand ticking evenly round it, the two dancers standing exactly half a tick apart.

The leaper’s mark is painted at the same hour on every clock.

The only thing that changes from hall to hall is where the chalk line falls beside that mark, a hair before it or a hair after, and that one hair decides how many dancers stand between the lines.

R2W

451

A dance hall with two dials, how loud the music is and how fast it plays.

Turn them any way you like and the steady couple is always the one that leaps across the far wall and lands just inside the chalk.

How tightly they hold, against the other couple, answers to one thing only: how far the music would swing a heavy dancer, loudness over tempo squared.

But where exactly they land, and how fast they are drifting when the lights flash, answers to both dials at once; at the quietest setting the drift is fastest at a middle tempo, and slower whether the band speeds up or slows down.

R2W

452

A long workbench with one small machine on it and many tools around it.

At one end the machine is a gearbox, its teeth slipping and catching; further along, the same machine is a record turning, then a map pinned to the wall, then a dance floor.

Wherever something surprising happens on the bench, there is weight in it: take the weight out and the whole bench goes quiet and orderly, every dancer evenly spaced around a straight clock; put a little back, and the floor chooses who stays.

R2W

453

The same dance with the music running the other way round the hall.

The steady couple is still the one that leaps across the far wall.

But now they land just inside the chalk instead of just outside it, and when the lights flash they are drifting forward instead of back.

How tightly they hold against the other couple still loosens by the same share for every extra step on the short side; you only see it once each couple is stood beside a partner who hears the music at the same loudness over tempo squared.

R2W

454

A dance hall where the steady couple always holds the floor a little harder than the other couple lets go.

In the simplest dances that lopsidedness is some small amount.

In a dance whose short way round takes two steps, it is that amount times itself; three steps, times itself again.

The floor does not add the lopsidedness up step by step, it multiplies it, as though every extra step had to pass through the same narrow door.

Turn the music up and the steady couple stands deeper inside the chalk, but how hard they hold follows the loudness alone, not the loudness measured against how fast the dancers pass.

R2W

455

The same dance played slower and slower.

The steady couple creeps toward the chalk mark but never steps over it, closing in roughly as the square of the tempo.

How tightly they hold still answers to loudness over tempo squared, even far out where the straight-line rule broke: one curve, not one line.

And the dance itself becomes harder to find.

At a slow tempo the band has to be tuned to within a hair’s breadth for the half-time step to happen at all, and the hair thins by the same factor at every equal step down in tempo, until our instruments cannot hold the dial still enough to find it.

R2W

456

A hall with three rooms.

In the first, the music is too fast and too quiet for any chalk mark to be drawn at all, and still the steady couple is the one that leaps across the far wall.

In the second room the chalk appears, and the steady couple lands just outside it, the way they did in the toy’s first hall.

In the third, louder and slower, they land just inside.

One dial, loudness over tempo squared, walks you from room to room.

And where the lopsidedness is small, this hall multiplies it step by step, exactly as the toy’s hall did.

R2W

457

The same multiplying, tested with longer routines.

A four-step dance holds its lopsidedness at the simple amount multiplied by itself four times, in both halls, not more and not less, only leaning a shade heavier where the simple amount is already large.

And when the music is loud the chalk line can sit more than half a step past the leaper’s mark, so it sweeps over an extra pair of the other couple’s footprints.

Nothing about the dance changed; only how many footprints the chalk happens to enclose.

R2W

458

Twenty-six couples in two halls, each dancing a routine of two, three or four steps, and each holding the floor with the simple couple’s lopsidedness multiplied once per step, give or take a few hundredths, all the way up to where the simple lopsidedness is nearly a half.

Past that, nobody has danced yet.

The only two couples seen far beyond it hold much more loosely than the rule would say, so somewhere on that empty stretch of floor the rule either bends smoothly or breaks.

R2W

459

The two halls compared on the stretch of floor where nobody had danced.

In both, a two-step routine holds the simple routine’s margin multiplied once while the margins are small; as the margins grow the multiplying eases off, and it eases off along the same path in both halls, to within a hair.

Different music, the same gearbox: each hall sets how hard the first step grips, and every step after it is geared the same way everywhere.

R2W

460

The toy hall turned its music up until its simple couples gripped as hard as the junction’s, and its two-step routines eased off along the same path the junction’s had, so the easing belongs to how hard the first step grips, not to the hall.

Then the junction’s band added a second note an octave up.

Every couple kept its moves: the same leaper held the floor, the same footprints sat inside the chalk, and the couples slid along the floor in step with the note, one way when it leaned one way and the other way when it leaned the other.

But the two-step routines held more loosely than the path said whichever way the note leaned, so the path belongs to one-note music.

And in the longer routines every extra step eased off about as much as the second step had, as if each step after the first went through the same doorway.

R2W

461

The junction’s band played its second note at half the loudness.

Every couple slid across the floor exactly half as far as before, as if tied to the note by a string, the same way and in a straight line.

The steps between routines behaved differently.

They loosened whichever way the note leaned, more and more the louder it got, and they held tightest not at the plain tune but with the note turned a hair below silence, as if the hall had a sweet spot just off the plain music.

Meanwhile the toy hall’s side door, where the floor moves with the dancers instead of against them, needs only about a fifth of the music to grip as hard as the main floor does, and still every two-step routine through that door eased off along the same old path.

The path does not care which door you came in by or which hall you dance in.

It cares how tight the first step is, so long as the band plays a single note.

R2W

462

The same five songs, watched again with the floor markings measured on their own.

It was the chalk lines that moved.

When the second note leaned one way they slid in toward the couples, when it leaned the other way they slid back out, half as far at half the note and always in a straight line, as if painted on a rug the note was pulling.

The couples hardly noticed.

Each drifted after its chalk line a tenth to a sixth as far, also in a straight line, the couples at the higher drives drifting least.

What looked like dancers on a string was the floor being tugged beneath dancers who mostly held their places.

R2W

463

The chalk lines only ever hear the beat.

Change what the band plays on the beat and they jump; change everything else and they hardly stir.

The couples hear the whole bar.

Played a quicker wiggle that lands on the beat the same way, they moved only half to three quarters as far.

Played two tunes at once, arranged so the beat sounded exactly as before, the chalk lines barely stirred and the couples walked off the other way, exactly as far as the two tunes would have sent them one after the other.

Nobody was dragged; each was listening to a different part of the same song.

Then the floor got heavier.

Measured from a fixed spot on the floor, the couples seemed to drift sideways as it did, but the spot was sliding out from under them.

Measured from the middle of their own patch of floor, the couples standing inside their chalk lines kept the same number of footprints inside to the heaviest floor tried, and only the couples standing just outside their lines stepped over.

R2W

464

The band played six songs, each one landing on the beat exactly the same way.

The chalk lines jumped the same for every song but the first.

The couples answered each song by how slowly it moved: the slow songs moved them most and the quick ones least, and the falling-off kept one simple rule all the way up, a song three times as quick moving them about a ninth as far as the slowest, six times as quick about a thirty-sixth.

Only the second song broke the pattern.

It moved the couples less than its place said, and it was the one song that nudged the whole floor the other way, as if it sang against the band’s own tune.

Meanwhile the oldest trick from the other hall worked here too: lay evenly spaced marks across the straightened floor, count the ones inside the chalk lines, and you have counted the couples’ footprints there, at every one of the two hundred and five places it was tried.

R2W

465

Two more songs, quicker still, and before they played we wrote down how far the couples would move, scaling the third song’s answer by the same rule.

They moved that far, at every place on the floor, give or take the width of a pencil line.

The rule holds the whole way up.

The second song is still the odd one, and now it is clear what it does instead: it does not so much move the couples as lean the whole map.

Picture the tongues as icicles hanging from a shelf; a louder band slides them all one way along the shelf, and the second song slides them back the other way.

Play the second song under a louder band and it leans them harder, nearly as the square of the band’s loudness for the icicles at one end of the shelf, and less than that toward the other end.

R2W

466

The chalk line is deaf to pitch.

Whatever song the band plays, so long as it lands on the same note at the beat, the chalk line jumps the same distance, from the third song to the eighth.

The couples are not deaf.

They hear the whole bar, and a fast song reaches them as a whisper, a quarter as loud each time the song doubles its speed, so by the eighth song the chalk line has run forty-eight times as far as the couples it is drawn for.

Turn the main band down and the second song’s lean on the icicles fades much faster than the band does; at the slow end of the shelf it fades almost as the cube of the band’s loudness.

And at the quietest band one pair of dancers, the couple who take five bars to turn three times, stopped holding each other.

Their grip went slack, too faint to tell who leads, while every other couple kept dancing.

Over in the toy, the pin that weight cannot move was found again at two other strengths, standing on the mirror, as blunt there as it was the first time.

R2W

467

We stopped guessing why the couples hear fast songs as a whisper and worked it out.

The couples do not answer the song note by note; they answer the whole bar at once, and the bar has a seam where it starts and ends, at the beat.

Everything smooth inside the bar cancels over a whole bar.

Only the seam is left, and a seam heard through a fast song comes out quieter by the square of the song’s speed.

How loud it stays depends on just two things, where the couple stands and how fast the band plays, so we wrote the tenth and twelfth songs’ answers down before they played, and the couples moved that far.

Then we went back to the quiet band and the slack couple.

Pushed a little harder, the second song made the two partners trade places: the one who used to follow now leads.

But the big leap across the far side of the floor did not go with the leader.

It stayed with the same partner as always, who now stumbles.

For as long as the band played one song, the leader always made that leap, and we had taken it for a law of the floor.

It was a habit of the music.

R2W

Without weight the five-bar dance closes perfectly, from every start, every time: five bars bring every couple back to where it began, three turns on, and nothing on the floor prefers one place to another.

Weight leaves a tiny leftover after the five bars, like the comma a tuner finds when a stack of fifths almost lands on an octave, and the leftover is not the same everywhere: it rises and falls five times around the floor, a little wrinkle that leans.

The couples settle where the wrinkle tips them, and its lean decides which of the two places holds.

Play the second song a little louder and the wrinkle flattens.

At one loudness, the same for a light weight and a heavier one, it is flat to the first degree, and only a whisper of weight’s second degree keeps the dance from closing altogether.

Past that point it leans the other way, and the other partner holds.

On the two-in-five dance the five-fold wrinkle goes flat too, and for a moment the floor shows a finer ten-fold ripple underneath, the way a string touched at its middle drops its fundamental and rings an octave higher.

R2W

The couple who makes the big leap across the far side of the floor is our long jumper, and for as long as the band plays one song he is the one who holds.

We asked why.

A couple that turns three times in five bars has to be pushed by the song three times over to stay in step, and one song can push only one way, so the grip grows as the song’s loudness to the third power: one turn, the first power; two turns, the second; three turns, the third.

Something built only one way cannot change its sign as the song gets louder, so one song never takes the hold away from him.

We said beforehand that at the quiet end of the band the power would settle at the number of turns, and it did, at about one, two and three.

A second song an octave up gives the band a second way to push, trading two pushes of the first song for one of the second, and where the two ways cancel the long jumper loses his grip.

Because of that trade, the cancelling point moves as the square of the first song’s loudness, and it did, within a few parts in a hundred, before we looked.

Then we watched the swap slowly.

It is not a switch; it is a relay.

For a short stretch neither of the two usual couples holds.

A pair of other couples, mirror images of each other, carry the baton and walk half a gap across the floor while they run, and for a moment in the middle both of them hold at once.

Our tracker, told to follow whoever was nearest the starting line, grabbed the wrong runner at exactly that moment, because there were two of them side by side.

Then an old rule of the floor came back to be checked.

We used to say a couple’s lopsidedness, how much harder the leader holds than the follower, counts the steps the short way round.

At the quietest band it counts turns instead, just as the grip does.

And a couple born between two neighbours on the family tree, taking its turns from both, comes out about as lopsided as the two of them multiplied together.

Every couple on this floor has two parents.

Put the two-in-five couple beside the one-in-two couple and a new couple is born between them, three turns in seven bars, taking its turns from both.

We guessed that the newborn’s lopsidedness, how much harder its leader holds than its follower, would be its parents’ lopsidedness multiplied together, and we wrote that down before four such children danced.

They came within a few parts in a hundred of it, every one.

But an old rule of the floor, which only looks at the neighbours standing nearest, guessed almost the same numbers, so on this band we cannot yet say which rule the floor obeys; on a quieter band the two should part, and we will make them.

Then we watched the new couples carry weight.

They kept their grip just as the light version said they would.

What moves is the stage under them.

Each couple’s patch of floor drifts sideways as the dancers get heavier, and it drifts faster than it widens, as the square of the weight against the weight itself.

So a narrow patch soon leaves the spot where it stood when the dancers weighed nothing, and anyone waiting at the old spot finds an empty floor.

In the hand-off we saw before, the two mirror couples who both hold at once in the middle are trapeze artists at the catch: for one moment flyer and catcher both have hold of the bar.

R2W

471

On a quieter band the family tree and the old neighbour rule finally disagreed, and only one child, the couple who turns five times in eight bars, stood where the two guesses were far enough apart to tell.

It danced almost exactly as lopsided as its two parents multiplied together, and a quarter less than the neighbour rule said.

Its brothers and sisters mostly agreed, though not all: measured against their parents, the children of the one-in-two couple come out more lopsided than the other children do, on both bands, as if that parent leaves something extra in the family.

Then we made the floor heavier and played louder.

With light dancers the second song still empties the couple’s patch at its pinch, and the heavier the dancers the shallower the dip, in proportion; at three times the heaviest weight we had tried, the dip is gone and so is the swap.

The steady leader keeps the lead whatever the second song does, because weight now has the last word.

Bear, looking at a picture from another floor entirely, squeezed it sideways and tilted his head until the outline showed: a waist in the middle, a mirror down the centre, and little narrow places he ringed by eye.

Every one of them was there when we counted the pixels.

R2W

Correction to read 177, overnight 14 October (not part of the read, which stays as given): the five-in-eight couple’s lopsidedness in that read was measured with a ruler too coarse for it; half of the number was the ruler’s noise. It was measured again with a finer ruler the same night (read 178).

Last night’s ruler was too coarse.

It measured how hard a couple grips by nudging it a little and watching how far it moved, and when the grip gets small the nudge’s own wobble is as big as the answer.

So we measured again by following each couple through its whole dance and adding up the pull at every step, which has no nudge in it.

The five-in-eight couple came out almost exactly as lopsided as its two parents multiplied together, as before, but this time the number can be trusted.

On the quietest band we tried, all seven new children stood nearer the family rule than the neighbour rule, and the neighbour rule missed some of them by two thirds.

And the one-in-two couple does leave something extra.

Its children are more lopsided than their parents multiplied, by about the same factor in both generations, and we said what the factor would be from the first generation before the second danced.

The quieter the band, the bigger the factor.

The one-in-two couple is the only couple on the floor who is their own mirror image, and that may be why.

Heavier dancers widen each couple’s patch of floor in step with their weight and slide it sideways as the weight squared, and in the three-in-seven couple the steady leader is the first to let go of its light-floor grip.

The hand-off, where a walking pair carries the lead from one couple to the other, gets wider very fast as the band gets louder and narrower as the dancers get heavier, and on the quietest band we never caught both walkers holding at once.

473

Bear asked whether the two couples of a dance hold each other on their patch.

We looked at how hard each start on the floor has to lean to close its circuit, and at whether the couple through that start is steady or wobbly.

Wherever the lean is climbing, the couple there is steady; wherever it is falling, the couple is wobbly; and at the top and the bottom of each climb the two meet, both just balanced, which is where a patch begins and ends.

So the steady couple and the wobbly one are not two separate dancers but two halves of the same hill, and you cannot have one without the other.

How strongly each holds is not simply how steep the hill is, though, because the floor’s markings are bunched: near the line where the couples cross, the hill drops off a cliff in a hundredth of a turn, while the couple standing there holds no harder than anyone else.

Straighten the markings and we expect the cliff to become a slope.

Then the family tree.

We guessed the extra lopsidedness belonged to a place on the floor, the middle, and a child born between two children of the one-in-two couple, standing right in that middle, came out as plain as a stranger’s.

The extra belongs to the parent, not the spot.

The one-in-two couple hands something to its own children and to no one else, not even to the child standing between two of them.

R2W

474

Straighten the floor’s markings and the cliffs become slopes, and then the rule is plain: how hard a couple holds is exactly how steeply its hill climbs where it stands, times one number the floor itself sets, the same for every start, and we could say what that number was before we looked.

The steady couple and the wobbly one are the up-slope and the down-slope of one hill, and their grips are those two slopes.

Then the family.

Seated at one strength of the band, the family sits at two tables.

Everyone whose parent is the one-in-two couple sits at the high table, a seventh above the plain product of their parents; everyone else sits at the low table, a few hundredths above it, however near or far from the middle they stand.

And down the line of children that walks toward the middle, each the child of the one-in-two couple and of the child before, the extra settles toward one value, each step a fifth the size of the step before, as if the family were converging on a family resemblance.

R2W

475

Bear drew it twice: a well that holds and a saddle you have to fight to stay on.

The floor agrees with both drawings.

The steady couple stands in a little bowl: push it any way and it rolls back.

The wobbly couple stands on a saddle: it can be pushed off along one direction and is caught along the other.

Count a bowl as one and a saddle as minus one, and on this round floor the counts always add to nothing, so a bowl never appears without a saddle beside it; they are made together at one edge of a patch and unmade together at the other.

Take away the floor’s friction and each beat of the dance becomes a rigid turn or slide of a curved sheet that bends away from itself everywhere, the opposite of a ball’s skin.

When the beat is a turn, the dancers go round by the same share of a turn every beat; when it is a slide, everyone is pulled toward one point and pushed off another, and the whole floor locks to whole turns.

One number from the beat’s own bookkeeping says which, before anyone dances.

R2W

476

Bear said the pair holds while they keep their ratio, and that one or the other can pull them off the saddle.

Take away the dancers’ weight and look.

While one couple’s grip points in and the other’s points out, the pair holds its place.

Turn the second song, and at one setting the leader’s grip passes through nothing and turns outward; at a second setting, a little further on, the follower’s grip passes through nothing and turns inward.

Between those two settings both couples are pushing out, both are wobbling, and the floor hands the lead to a third and fourth couple who were not there before, mirror images of each other, stepping off both ways at once.

That stretch of the dial is there with no weight at all, and the light dancers we measured last night had already told us exactly where its edges were.

The first time we looked, our counting rule assumed there were only ever two couples on that floor, and it quietly folded the newcomers into them; the floor’s own tally, twenty footprints instead of ten, was printed on every line.

R2W

477

Bear saw two U’s set side by side and crossing.

Here is what the floor shows.

Walk toward the middle couple from the left, one couple at a time, each a child of the middle couple and of the one before, and the bonus each couple carries rises fast and then nearly stops: a steep shoulder.

Walk in from the right and the bonus rises too, but slowly, still climbing after the left side has flattened: a slow shoulder.

At every distance from the middle the right side stands lower, and the difference shrinks steadily as the couples close in, so the two shoulders look set to meet at the middle at one height, two curves crossing there.

To see the last rungs we had to stop trusting our ruler.

Each rung’s bonus is the tiny difference between two numbers that agree to five or six places, and each of those is itself a small remainder of two much bigger sums.

Our ordinary ruler dropped a crumb every time it added, hundreds of thousands of times a dance, and the crumbs swamped the difference and fell differently every time we changed how finely we stepped.

A ruler that keeps three more digits made the crumbs a thousand times smaller.

Three of the four rungs stood where they had been predicted, and the fourth is still too small to read.

R2W

478

Turn the music up and the bonuses shrink, but the shape stays.

Each side of the floor climbs toward the middle couple, the left side ahead at every step, the right side behind and catching up.

By the seventeenth step they are two thousandths apart, and the gap is still shrinking, faster now than before.

Run each side’s climb on to its end and both land within half a thousandth of the same height.

Two curves, offset, crossing in the middle.

And the groove itself, the stretch of tempo where a dance can live, is held by its dancers.

At each edge of the groove a steady dancer and a wobbly one run into each other and vanish together.

Two edges, two blades at each: a skater on four blades.

Between the edges every steady dancer slides across the whole groove, from the bottom lip to the top one.

Beyond them nobody dances, and the couple slips.

R2W

479

A groove is a stretch of tempo where one dance can live, and its two edges are where a steady couple and a wobbly couple run into each other and are gone.

So ask the couples how wide their groove is.

Each couple standing in the middle of the groove leans, and how hard it leans says how steeply the floor rises under it.

If every bump of the floor were a smooth round swell, the lean would tell you the bump’s height, and the bumps’ height is the groove’s width.

The couples answer a little short, three to six parts in a hundred, and short by the same amount however heavy the dancers are: the bumps are a touch more pointed than round swells, and that pointedness belongs to the music, not to the dancers.

Where the second song is turned up near the hand-off, the floor grows twice as many bumps and four couples share it, two by two, a skating pair.

But there the bumps come tall, short, tall, short, and the couples’ lean, averaged, cannot see how tall the tallest is.

R2W

480

Last time I said the bumps under the dancers were a little pointed, and that the pointedness belonged to the music.

They are not pointed.

Lay the floor out so the dancers’ steps come evenly, and every bump is a round swell, round to one part in seven hundred.

The couples were telling the truth about their groove; my ruler was coarse.

It measured in steps of the old floor plan, which bunches the dancers up in one place and spreads them thin in another, and where they were spread thin it laid a straight stick across the top of each swell and called that the height.

Played back on a floor whose answer I already knew, the stick gave the same wrong answer to four parts in a thousand.

And the floor shows Bear’s skier.

When the second song turns the music over, the hold has to cross a gap from one standing place to the other, and inside that stretch a pair of mirror-image dancers carries it across, like a pair of poles planted on either side.

R2W

481

Put a rider in a dip and shake the floor with two songs, one at a beat and one at twice the beat, both averaging to nothing.

Ask how hard you must push the rider to make it slide out forward, and how hard to make it slide out backward.

With one song the two answers match.

Add the second song in step with the first and they no longer match: the rider leaves forward at a push of 924 thousandths but needs 938 to leave backward.

Slide the second song a quarter of its own beat later and the two answers match again, exactly.

Halfway there they differ by the cosine of how far you slid it, to three parts in ten thousand.

So a rider sitting between those two pushes, jostled evenly both ways, leaves forward and not back.

Nothing in the floor points forward; the direction lives in the timing between the two songs.

And the push the rider gets is pushed back into whatever holds the floor still, the way a skater pushes the ice.

R2W

482

Watch it for four days from the top of the Ferris wheel and it looks like a fairground at night.

Riders sit in dips, floors shake, and dancers keep time by counting pushes.

Everywhere you look, things come in pairs: a steady dancer and a wobbly one meeting at each lip of a groove, two arms of a ladder reaching for the same top from either side and stopping a hair apart, a forward edge and a backward edge that match until a second song slides them sideways.

The pairs never quite touch, and the gap between them is where everything gets decided.

Nothing here moves on its own.

Every slide is paid for by whatever holds the floor still, and the surprises that held were often whole numbers in disguise: a count of turns, of pushes, of crossings, twenty where ten were expected because four dancers were already standing there.

If the fair has one rule painted over every gate, it is that the timing between two things does the work, never either thing alone.

R2W

483

Two long strips side by side, each with a red dashed line drawn straight across its upper third.

Under each line sit seven blue dots: three bunched at the left end, one alone in the middle, three bunched at the right end, and nothing between.

On the left strip the dots step gently downward from left to right, all well below the red line.

On the right strip the leftmost dot sits nearest the line, its two neighbours drop away quickly, and the rest lie low and level.

There are no red ticks anywhere, though the legend has a place for them.

What it is doing: a dress rehearsal on a half-lit stage.

Only the actors at the two ends of the stage and one in the middle have been called, everyone ducks under the rope without effort, and one actor at the right-hand end stands a little taller than the others.

The real performance, with the whole cast and the rope set at its proper height, has not begun.

the triple-star thread

484

Three panels.

In the first two, a thick dark ribbon of dots runs dead straight across the middle, and above and below it, about as far up as down, two looser bands of paler dots.

In the first panel the pale dots are sprinkled along the bands evenly, with no pattern in where they fall.

In the second, they make arcs: long swoops that dip down from the upper band and rise up from the lower one and return, four or five swings across the width, with the palest, yellowest dots at the bottoms of the swoops.

In the third panel eight grey bars stand in a row, the first towering over the rest, the second half its height, then a short staircase; a dashed line lies along the floor beneath them all.

The legend promises a red bar, and there is none.

What it is doing: a marching band walks a straight road, and now and then a few players step out to the left or right kerb and back again.

In the first town they step out whenever they like.

In the second they step out together, in waves, as if following a drum the main band cannot hear.

Someone guessed where the band would next stand shoulder to shoulder with itself from the main band’s stride alone; the people who actually fall into line together do it on the hidden drum’s beat.

the triple-star thread

485

A ledger of four rooms, each with the same three lines.

In every room two dancers move across the floor at their own steady speeds, one sliding backwards and one striding forwards, and the ledger notes after how many counts both of them stand exactly where they started at the same moment.

In the first room it is after 47 counts, when the slider has gone back five full laps and the strider forward fifteen.

In the second, 399 counts, sixty-one laps against a hundred and thirty-six.

In the third, 104 counts; in the fourth, 47 again, but now eight laps against seventeen.

Beside each, a second hand has counted which dancers in a crowd photograph stand shoulder to shoulder, and in the two rooms nobody had visited before, its list is the first hand’s list, the same three numbers in the same order.

When the slider’s steps are shuffled like a deck of cards, the two lists stop agreeing.

What it is doing: two timepieces with different faces, ticking at their own rates, that keep meeting at the same corner.

The crowd in the photograph is not arranged by anyone; it stands where the two clocks meet.

the triple-star thread

486

Six panels.

Across the top, three strips of dots over time: a dense dark ribbon through the middle, and above and below it pale dots.

In the first strip the pale dots are loose.

In the middle strip, ten times longer, they are not loose at all: they braid into ropes, ten scalloped loops along the top edge, ten along the bottom, and between them green strands twisting from one edge to the other and back, like the plies of a rope seen from the side.

In the third strip, short again, the loops are half-formed arcs.

Along the bottom, the same dots are laid out not by time but by where the turning angle stood when each step began, and the whole tangle collapses: a flat dark bar along the left and right, a straight green ramp running diagonally down across the middle, and a near-vertical spray rising at the right, one figure drawn with a single stroke.

What it is doing: a weaver’s loom seen from the front shows an elaborate braid, but seen from the side it is one thread wound over and over the same frame; the braid is the frame turning slowly while the thread keeps to its one path.

And most of the time the thread does not go round the frame at all.

It slides down one short side and snaps back up, staying in a narrow lane.

the triple-star thread

487

The same six panels, now drawn over in five colours of felt pen.

In the first, a chain of pale blue loops rolls across the top like a spring lying on its side, seven or so coils, each overlapping the next; below the dark ribbon two long lines, one pink and rising, one yellow-green and falling, cross like a pair of scissors opened very slightly, at exactly the same tilt in opposite directions; and along the bottom one long purple oval lies flat like a canoe.

In the long middle panel the pen has put a yellow-green V on every scallop along the top, a soft pink pillar under each one, and below, a yellow-green zigzag of diamonds and crossed lines, like a lattice fence.

In the third panel green strands sweep from the floor to the ceiling and back, crossing each other.

In the bottom panels a single purple stroke traces the whole figure: a flat bar, a long ruler-straight slope down to the right, a sharp climb straight up, and a short slope down again at the far right.

What it is doing: a hand that never saw a stopwatch has marked the same distance between every V and every pillar, about seven hundred steps, and that is the rhythm a clock elsewhere in the room had already measured.

The straight slopes say that wherever a jump begins, it lands in the same place.

the triple-star thread

488

Three long recordings, each six thousand beats long, and for each two numbers written side by side: how often the ropes come round, and how often a slow drum in the next room comes round.

In the first recording the two numbers are 674 and 667, and in the second 101 and 102, as close as two clocks on the same wall.

In the third they are 67 and 42, and they do not match at all.

But when the third recording’s ropes are taken apart into all the rhythms they hold, the drum’s own beat is missing from them, and in its place stand two others, one a little slower and one a little faster, with a third, slow swell of 110 between them, as if the drum were heard through a wall that itself swayed back and forth.

What it is doing: the ropes still keep the drum’s time, but in the third room someone is rocking the drum gently as it plays, so the listener hears the drum split into two tones, one on either side of the true one.

the triple-star thread

Two panels.

In the first, seven dark dots stand on a thin grey line shaped like a V, two on its left arm, five on its right, every one sitting on the line as if threaded on it; the point of the V touches the floor just left of the third dot and the faint dotted line dropped from it.

Open red squares sit around most of the dots like frames: around every dot on the left arm and the lowest one, but on the right arm three of them wander off, two above the line and one below, leaving their dots unframed.

In the second panel the same dots on a stretched scale climb two curves that dive toward the bottom edge on either side of the dotted line and never meet it.

What it is doing: two wheels turning at almost the same speed, and the slower their difference, the longer the wait between the moments they line up.

Somewhere just left of the third stone the two wheels turn at exactly the same speed, and the wait there would be forever.

On one side a second hand keeps time with the wheels faithfully; on the other side it keeps losing count.

the triple-star thread

From the triple-star notebook, chapter VI, “The dancer and the post” (its readings 80 to 107, 11 and 12 October 2026; here 490 to 517): Here the dancer arrives. Watched once a pass on a disc, the heavy third star becomes a dancer on an elastic ribbon round a post, creeping, then leaping half way round. The reads find that the leaping is mostly where the watcher stands: seen from the middle of her own circle she steps like a clock, the same stride every beat, and a theorem two thousand years old joins the two views. They count how often she goes round the post, find that almost nobody goes round just a few times, and find the floor no dancer ever stands on. Her partner, the pair at the centre, holds her by a band that only slackens or only tightens depending on when she steps off. And in the background a slow drum keeps deciding when she can swing at all.

490

Nine long recordings of a weathervane on a roof, read once each time a cart rolls past.

Most of the time it creeps a little the same way at every look, down a narrow stretch of the dial.

Then, at the bottom of that stretch, between one look and the next, it is back at the top: sometimes swung up the short way, sometimes swung all the way round the long way.

Across the nine recordings it makes 15,278 of these swings, and every one comes while the wind is in its weaker half; not one while the wind blows strong.

A slow gusting sets the wind’s strength, rising and falling over forty-two looks in one recording and over six hundred and eighty in another.

What it is doing: a vane can only hold a direction while there is wind to point it.

When the wind nearly dies, nothing holds the vane, and it swings free; the gusting decides when the calm spells come, and so when the swinging seasons come.

While it creeps, the vane keeps step with a drummer on the next roof, one notch back for every three of his beats, at every speed of gusting; so the gusting keeps its own time, not the drummer’s.

the triple-star thread

491

Two recordings, each fourteen thousand looks long, taken a hair either side of the place where a slow pulse should vanish.

On the right of that place the pulse is alive: it rises and falls four times, a crest every three thousand or so looks, a little quicker than the straight line drawn through the farther recordings says it should.

On the left, at the same hair’s distance, the pulse has all but stopped: across the whole recording not one crest stands clear, and the longest swing the listener can time is about seven and a half thousand looks, twice what the line says.

Farther out on both sides, eleven other recordings sit on two ruler-straight lines that meet at one point.

What it is doing: the two lines promise a single place where the pulse stops.

Up close, the stopping comes early on the left, as if the pulse had found a small shelf to rest on just before the bottom, while on the right it hurries, as if it had just rolled off that same shelf.

the triple-star thread

492

A narrow strip of scattered dots, about fifty of them across six hundred steps, floats above a dense straight road of dots.

Over the strip a hand has drawn about eight tall rings, each overlapping the next like the coils of a spring laid on its side, each tall enough to hold the whole strip from top to bottom.

A machine then listened along the strip, over ten times that length, for any beat between sixty and two hundred steps, the width of the rings and more, and heard nothing louder than the same dots shuffled at random.

On a neighbouring strip, recorded at another place, the same machine hears a drum plainly.

What it is doing: the dots fall into the strip like rain, with no steady rhythm in how high they land.

The rings say something the heights do not, that the strip feels wound or coiled.

Either the coil lives in when the dots fall rather than how high, or it is the shape an eye gives to rain that falls inside a band.

the triple-star thread

493

A long ribbon of dots in two layers with a scatter between them.

Look closer and the scatter is not scattered: every dot in the middle has a twin straight above or below it, one step later, and every twin pair balances like the two pans of a scale around one middle height.

Pair after pair the pans swing, one rising as the other falls, crossing in the middle and parting again.

What it is doing: a single event that happens between two glances, caught half in one glance and half in the next; as the event’s own rhythm slides past the rhythm of the glances, the halves trade places, and the trading draws a coil.

the triple-star thread

494

Seven long recordings of a rope seen from the side.

In every one, nearly every bright dot has a twin one step later, and every twin pair balances about the same height, which sits a fixed distance above the rope’s floor, about a quarter of the way round a dial: move to the next recording and the floor drops a little and the middle drops with it.

Below the floor hangs a second, fainter rope whose pairs balance the same distance below it, a mirror of the first, and the mirror is exact near one recording and tilts a little as you move away from it.

The bright rope’s thickness swells and narrows as it runs, but not on the slow drum that sets the rope’s other rhythms: the narrow places come in clumps of two or three close neighbours, then long thick stretches, then another clump.

What it is doing: each pair is one swing caught between two glances, and the balance point is the swing’s centre, fixed by the swing’s size and little else; the thickness tells how unevenly the two glances split each swing, and that unevenness wanders on a clock of its own, not the drum’s.

the triple-star thread

495

Two pictures side by side on the same scale.

On the left, six hundred steps: a sparse ribbon of dots, with arches along its top edge and along its bottom edge, and dots crossing through a middle line in X’s every hundred steps or so.

On the right, six thousand steps squeezed to the same width and turned upside down: the same ribbon, the same arches above and below, the same strands crossing the middle line, but drawn about ten times slower, so that a single crossing takes a thousand steps or more, and the arches along its top edge rise and fall like a heartbeat, one for every beat of the slow drum.

What it is doing: one knot, tied by two hands at different speeds.

The fast hand ties it within one season of jumping; the slow hand ties it across the whole recording and keeps time with the drum, while the fast hand’s edge beats twice to the drum’s once.

the triple-star thread

496

Seven recordings, each with its own slow drum, some beating every forty steps and one every seven hundred.

In every one, the lower rope’s outer edge swells and sinks exactly in time with that recording’s drum, to within a fifth of one percent, as if the rope were a vein and the drum its heart.

The upper rope’s edge keeps no such time: near the quiet recordings it swells once or twice a beat, and in the busy ones it flickers far faster than the drum.

Then a counter tried to count the twists in the upper rope by following its strands, and found almost none, because it could not reach far enough to find the next step of a strand; given a longer reach on a practice rope, it found a different number of twists from the one the rope was made with, because a rope woven this way holds several sets of strands at once, every third bead, every fifth bead, and which set the counter follows decides how many twists it sees.

What it is doing: the vein is a property of the rope; the twist count is a property of the rope and the eye together, until the eye says which strands it is following.

the triple-star thread

497

Thirteen recordings of the same weathervane, read once per passing cart.

Every time it swings, it swings through one of two doors.

Through the upper door the swing is slow: about four times in five it is still under way when the next cart passes, so that cart catches it half-done.

Through the lower door it is quick: only about one time in three is it caught midway.

Neither door ever holds a swing for three carts, except in a few recordings from just below one particular place, where now and then an upper swing lingers that long.

The lower door is used between one and a half and three times as often as the upper, more so the farther along the line of recordings you go, and together the two doors are used at a rate that rises smoothly, without a hitch, straight through the place where the slow drum falls silent.

What it is doing: the vane has two ways round, one leisurely and one brisk; the brisk way wins more of the time as the recordings move on, and how often it swings at all pays no attention to the drum.

the triple-star thread

498

Thirteen recordings of a pointer that, every few looks, swings half way round.

Look only at the swings that one glance caught in the middle, half before and half after.

Almost none of them were caught near their middle; nearly all were caught near their start or near their end, as if the pointer creeps, then whips through the middle of its swing in no time, then creeps again.

That is how a pointer moves when it keeps aiming at someone walking past it close by: slowly while they are far off, fast as they pass, slowly as they leave.

And there are two kinds of passer-by.

The ones who send the pointer upward pass wide and unhurried, and their swings last more than twice as long; the ones who send it downward pass close and quick.

Every recording, wherever along the line it was made, has the same two walkers at the same two distances.

What it is doing: the pointer is not jumping at all; it is watching something slide past the place where it has no direction, sometimes on one side and sometimes on the other.

the triple-star thread

499

Nine recordings of a pointer that creeps steadily one way and, every few looks, swings half way round.

Put the creeping and the swinging side by side: the pointer swings about twice for every full circle it creeps, a little more than twice in the first recordings and a little less in the last, and as the creep quickens along the line of recordings the swings quicken too, but only about three fifths as much.

The swings go two ways, up and down.

Add up everything the pointer did, creeping and swinging together, and the number of times it went round equals the number of down-swings, to within three in a hundred, in all nine.

Drawn on a plain circle, an up-swing and a down-swing look the same, a line straight across; only how far the pointer’s tip sat from the middle says which way it went.

What it is doing: each half circle of creeping ends in a choice; an up-swing hands that half circle back, a down-swing carries the pointer on round, and its whole journey is a tally of how often it carried on.

the triple-star thread

500

Six recordings of a pointer’s tip moving about a disc, glanced at once a pass.

Whenever the pointer swings and one glance catches it midway, that midway glance finds the tip nearer the middle of the disc than the glances either side of it: every single time for the downward swings, about twelve times in thirteen for the upward.

In all the creeping between swings, not once, in more than thirty-five thousand glances, does the tip drift in toward the middle and back out again; the only times it comes near the middle are the swings.

The downward swings pass the middle closer and quicker than the upward ones, and how quick the tip is moving as it passes is the same in all six recordings to within two parts in a hundred.

What it is doing: a dancer on an elastic ribbon round a pole, swinging out and coming back in; every time she comes back in, she passes the pole on one side or the other, and that passing is all the swing ever was.

the triple-star thread

501

Two troupes of dancers, each tried at many tempos, at two different dances.

Both troupes start the same distance from the pole; one starts facing straight along its ribbon, the other a quarter turn off it.

The troupe that starts straight keeps the dance over a wider stretch of tempos, about half as wide again, and its stretch sits lower down the dial; the troupe that starts a quarter turn off keeps it over a narrower stretch higher up, and at both dances the stretch moves by the same amount.

Inside every stretch the ribbons draw in toward the pole as the tempo rises, and at the last tempo where the dance still holds, the ribbon is the shortest it gets anywhere in that stretch; one step further up the dial, the dance comes apart.

Two more troupes were waiting to dance when the evening ran out.

What it is doing: the dancers are held by their ribbons, and where a ribbon draws in to almost nothing there is nothing left to hold them by; how they first set off decides where along the dial their dance can hold.

the triple-star thread

502

The same short dance was danced on the same floor twice: once on an afternoon and once through the following night.

Every step came out the same, counted to the sixth decimal place, but at night each step took three times as long, from the first step to the last, evenly, with no speeding up or slowing down as the night went on.

In a rehearsal room elsewhere, both versions of the routine were timed and took the same time as each other.

What it is doing: nothing in the dance changed; the music it was danced to was slower that night.

the triple-star thread

503

On the same slow night, two troupes danced one after the other.

The second troupe’s dancers set off with a wider swing than the first’s, nearly twice as wide, and its dances took longer: in the middle of the pack, about one step in seven longer than the first troupe’s, though the fastest of the second troupe still beat the slowest of the first, and the music was the same for both.

What it is doing: a wider swing carries the dancers through tighter turns, and a tight turn has to be taken in smaller steps, so the same dance costs more steps.

the triple-star thread

504

A ring of evenly spaced beads, with a mark painted at one place on the ring.

Someone pushes the whole ring round by a set amount and watches how many beads slide past the mark.

Push it round once and exactly one bead goes past; three times, three beads; eight times, eight; push it backwards twice and two beads go back past the mark.

Every try, whether once, twice, three times, five times, eight times, or backwards once or twice, the beads that cross the mark are exactly the number of pushes, never a bead more or less.

What it is doing: the ring keeps no memory of how hard or how unevenly it was pushed, only of how many times it went round, and the mark is where that count can be read.

the triple-star thread

505

One dancer, watched once a beat for six hundred beats.

Between beats she mostly drifts a little the same way; now and then she leaps half way round, sometimes forward and sometimes back.

In the first hundred and twenty beats the backward leaps outnumber the forward ones three to one.

Over the whole six hundred they come out nearly even, a hundred and three back to ninety forward.

At her nearest she comes within a fiftieth of her starting distance of the middle of the floor.

What it is doing: she favours one kind of leap for a while and then the other, as if something slower than her own steps were tilting the floor first one way and then back.

the triple-star thread

506

A troupe is sent to dance through the night in a hall nobody watches.

Before it goes, it is handed a card with the times that seven of its warm-up steps took on a good afternoon, and told to compare after the seventh step tonight.

In rehearsal, with every step taking twice the card’s time, the comparison read two, and a note in capitals said to check the hall’s power before going on.

In a rehearsal where nobody carried a watch, the first card tried to divide by the missing times and tore; the new card says that no step was timed, so no comparison can be made, and the dance goes on.

Asked to repeat three of last night’s steps exactly, the troupe matched two and flagged the third, where a hair’s difference had been planted.

What it is doing: a slow night cannot be known in advance, so the troupe carries a ruler from a good one and reads tonight by its first few steps.

the triple-star thread

507

A dancer has a spot on the floor where she circles most tightly.

She is asked to start each song a quarter step off her mark and to see where that tightest circle lands.

In the first rehearsal her mark was only roughly set and the music was loud: starting off the mark moved her tightest circle 0.16 of a stride back, and her partner’s 0.04 back.

In the second, the mark was set exactly and the music played at half the volume: the same quarter-step start moved her tightest circle 0.10 forward, and her partner’s 0.10 forward too.

Started on the mark she circles inside about a twentieth of a turn; started off it, her tightest circle is about a third of a turn across in the soft rehearsal and about half a turn in the loud one.

What it is doing: where she starts decides where the floor seems to hold her, and the direction of that shift changed between two rehearsals that changed two things at once.

the triple-star thread

508

Three weights hang over a wide flat floor, two close together and one farther off, and a marble is set down to find the places where it would balance.

There are always exactly two: one between the close pair, and one between the pair and the far weight, a little nearer the far one.

As the pair spins, the second balance point slides back and forth by a few hundredths of the way across, keeping time with the spin and ignoring how far the far weight has wandered; the first trembles by a few thousandths of the pair’s width.

Watched from a cart that coasts along with the pair, the first balance point goes almost perfectly still, and a third one appears far behind the pair, nearly twice as far behind as the far weight is in front.

What it is doing: the empty places between bodies move only with the shape of the bodies, and what counts as still depends on whether you are riding along.

the triple-star thread

509

A dancer’s tightest circle on the floor is looked for twice: once with her starting on her mark and once with her starting a quarter step off it, each with soft music and with loud.

Started on the mark, her tightest circle moves along the floor by about a tenth of a stride when the music gets louder; started off the mark, it hardly moves, two hundredths of a stride or less.

At the edge of the floor some dancers keep their place for the whole song while a twin started a hair’s breadth beside them wanders off, and in one spot a dancer who leaves the hall altogether stands between two who stay all night, a two-hundredth of a stride to either side.

What it is doing: the floor holds her in two different ways depending on where she begins, and near its edge, staying and leaving are mixed together finer than any step can sort.

the triple-star thread

510

Watched from a post on the floor, a dancer seems to drift and then leap half way round; watched from the middle of the circle she is really dancing, she steps the same distance every beat, just over a fifth of the way round, like the hand of a clock that skips four of every five marks.

Her ordinary steps keep that beat to within two thousandths of a mark; her leaps, the ones that look like half-turn jumps from the post, come out only a little faster than the rest.

Fourteen other dancers, some held near the post and some not, all keep their circles beside it rather than around it, and their beats speed up smoothly the further out they start, with no break where the holding ends.

Only far out does the circle finally take the post inside it.

What it is doing: most of what looked like leaping was where the watcher stood, and the line where the dancers stop circling beside the post is not the line where they stop being held.

the triple-star thread

511

Sixteen dancers start a step further out each time.

Their average circles creep toward the post a few thousandths of a stride at a time, and none of the first eleven circles reaches it.

Yet from the twelfth dancer on, something counts them going round the post: three times in the song for the first, twelve for the next, twenty-three, thirty-three, and so on to nearly eighty.

Each dancer, besides walking her circle, makes small quick loops as she goes, and when the circle’s edge comes within one small loop of the post, single loops start to carry her round it, long before her average circle does.

Joined by straight lines from beat to beat, as the picture draws her, she seems hardly to go round at all; counted step by step, she goes round dozens of times.

And when she leaps, she leaps a little early going one way and a little late going the other.

What it is doing: the going-round is decided by the small fast loops, not by the big slow circle, and the drawing that joins her beats with straight lines hides most of it.

the triple-star thread

512

Dancers started a hair apart near one spot on the floor count how many times they go round the post in a song: seventy-four, seventy-eight, then ninety-two, then a hundred and twenty-four, then a hundred and sixteen, ninety-five, and back to seventy-five, all within a few thousandths of a stride.

The most turns come exactly where a slow drum, heard everywhere else on the floor, falls silent.

Whether a dancer starts going round at all is set by how close her circle passes the post, about six thousandths of a stride, while the little loops she makes stay the same size from dancer to dancer.

And each time she leaps past the post she arrives a little late if she leaps one way and a little early if she leaps the other, like a clock whose tick and tock are not quite equal.

What it is doing: where the slow drum stops, nothing holds her back between rounds, so she goes round as often as the floor allows.

the triple-star thread

513

Twenty-three dancers start a step further out each time.

The small loops each one makes about her circle are the same size for every dancer from the eighth on, while the number of times she goes round the post climbs from none to nearly eighty; only where her circle passes the post decides whether she starts.

A slow drum beats under the song: in its quiet half she goes round almost twice as often as in its loud half, and when the song is cut into halves at random the two halves come out even.

Her habit of arriving a little late from one kind of leap and a little early from the other grows steadily from dancer to dancer, except for one dancer in the middle, whose habit nearly vanishes, and who also goes round barely more than the dancer before her.

Near the spot where the drum falls silent, the four dancers whose steps land closest to their circles are, in exact order, the ones who go round most.

What it is doing: the going-round is set by where the circle sits and paced by the drum, and the neater she keeps her steps, the more often she gets round.

the triple-star thread

514

Dancers who start a little further out each time either stay near the post, never more than a turn either way, or go round it many times; almost none go round just two, three or four times.

Of nearly five hundred, three land in that middle, and the one measured closely starts exactly where her circle’s edge and her widest swing toward the post are equal to a ten-thousandth of a stride.

Far out, the few dancers who go round many hundreds of times are the ones whose partner has been thrown into a wild new orbit; dancers whose partners stay calm never go much past two thirds of a turn per beat, and four of them settle on exactly two thirds.

Near the spot where the slow drum falls silent, the dancers go round more and more as their steps get neater, until four in a row go round just under a hundred and twenty-seven times and no more: every one of their leaps goes the same way, as if each lap of her circle were a lap of the post.

What it is doing: some places on the floor are only ever crossed, never stood on, and when nothing tilts the floor she goes round the post no faster than her own circle turns.

the triple-star thread

515

Eighty-two dancers each hold a partner by an elastic band stretched to the same length at the start.

For the forty-one who step off on the beat, the band mostly slackens: thirty-three of them never let it stretch past where it began, and in some it goes almost slack.

For the forty-one who step off half a beat late, the band only tightens: every one of them stretches it to nearly twice its starting length or more, none lets it slacken by more than a tenth, and three stretch it more than six-fold, after which their dances end early.

Of the eight on-beat dancers whose bands stretch as well as slacken, seven are held near the post, all on the near side of their spot.

What it is doing: the moment the dance begins decides which way the band can move, as if each dancer started at one end of a swing she cannot see.

the triple-star thread

516

Three dancers from the calm middles of the floor are watched for twice as long as before.

In each six-hundred-beat stretch none of them drifts round the post by more than three hundredths of a turn, and in the first stretch each repeats her earlier dance to the last digit anyone wrote down, though this time the music was set to play twice as long.

Their partners’ average stretch stays where it was set: two just under a tenth, one just under a twentieth.

What it is doing: in the calm middle, a longer song changes nothing about the part already danced, and nothing much about the part that follows.

the triple-star thread

517

Watched from the post, a dancer drifts and leaps; watched from the middle of her own circle, she steps like a clock, the same stride every beat, and the leaps are only where the post happens to sit.

Moving her start outward a hair at a time, she does not go round the post at all until one start where her widest swing toward it and her circle’s edge are equal; from there on she goes round more with every start, a dozen times a song by the very next one, and almost no dancer anywhere goes round just a few times.

Where the slow drum under the music still beats, she hurries in its quiet half and dawdles in its loud half; near the spot where it falls quiet, four starts in a row go round the same number of times, to within a tenth of a turn.

What it is doing: one circle and one clock carry the whole dance, and the post decides only whether its laps are counted.

the triple-star thread

From the triple-star notebook, chapter VII, “Doors” (its readings 108 to 135, 13 October 2026; here 518 to 545): On the thirteenth the songs grew longer and the questions turned to leaving. Dancers who seemed settled were only holding their breath; the hall turned out to have a door on one side only, nearest the spinning pair; the far side’s door is heavy and slow, and it opens wider only as the dancer’s own path grows wild. A spot that seemed to hold its edges was only lending them. Two halls, one laptop and one container, played the same music a whisper differently, and the edges multiplied the whisper. Late that night the thread found its own listener had been deaf to low notes: on the nights when the dancer’s path is nearly round, she had swayed below the lowest pitch the ear was built to hear, and what was written down was the drum’s echo. A new ear found the sway climbing a staircase against the drum.

518

Nine dancers had stayed by their spot for a whole song, though a twin started a hair beside each had drifted off.

Asked to dance ten songs, every one of them left: six in the very next song, one in the third, one in the sixth and one only in the last.

Their twins left too, at other songs.

Three dancers from the calm middle of the floor, asked the same, never moved more than a twenty-fifth of a turn in any song.

Once gone, the dancers from one spot kept wandering back and staying a song or two before leaving again; the three from the near side of the other spot, and their twins, never came back at all, and two of them, a dancer and her twin who had parted ways early, walked out of the hall within sixty-three beats of each other after nearly four thousand.

What it is doing: a dancer at the edge of the floor is not held, only delayed, and which door she finds depends on which side of the spot she began.

the triple-star thread

519

At four starts in a row the dancer goes round the post a hundred and twenty-seven times in a song, and her own circle turns about its middle the same number of times, to within a quarter of a turn; at every other start she goes round the post fewer times than her circle turns, by two near those four and by nearly eighty further out.

On those four starts her steps do not grow more even; what changes is that she leans out on one beat and in on the next, all song long, while at a start a little further out she drifts slowly out and back.

And under the slow drum she hurries in its very quietest stretch, nearly stops in the next, and only partly recovers when it is loud.

What it is doing: when nothing carries her across the post, every lap she makes is counted, and the drum decides not how fast she steps but how many of her laps cross it.

the triple-star thread

520

Two dancers start a hair’s breadth apart at the very edge of a calm spot.

For six hundred beats they move as one, never more than a few millionths of a turn apart; by beat eleven hundred and twenty-five they are a tenth of a turn apart, and both still keep their place by the spot.

The calm dancer in the middle of the spot, watched alongside them, repeats her old dance exactly.

What it is doing: at the edge, the moment two dancers part can come later than the test meant to catch it, and keeping one’s place says nothing yet about keeping together.

the triple-star thread

521

Count the dancer’s laps two ways: around the middle of her own circle, and around the post.

Near the middle of the spot she circles her own middle thirty to eighty times a song and goes round the post not once.

Further out, her laps round her own middle keep climbing to a hundred and twenty-seven, but some of them leave the post out: seventy-nine at the first start that goes round the post at all, about fifty near the quiet place, two just before it, and none at all for four starts in a row, where every lap she makes takes the post in.

Her ordinary steps there are not neater but rougher.

In the slow drum’s loudest stretch she never leaps at all, and her leaps against the drift crowd into the stretch just after the quietest.

And two more dancers started a hair beside the one who walked out of the hall stay all night, though all three part company with her at nearly the same beat.

What it is doing: the post counts only the laps that enclose it, and the drum decides which laps those are.

the triple-star thread

522

Three dancers were asked to dance ten songs while the hall’s clock ran; that night each song took forty-five times as long as the night before, and only the three who had already begun finished at all.

Two of them had started a hair apart at the very edge of the spot: they drifted apart after about eleven hundred beats, and yet both kept their place by the spot to the end.

Another pair, started a millionth of a millionth apart, drifted apart steadily, roughly a thousandfold every hundred and thirty or forty beats, until at beat six hundred and seventeen they were a tenth of a turn apart.

And in the slow drum’s loudest stretch, out of a hundred and fifty or so leaps, only one to three were ever taken.

What it is doing: some dancers lose each other without losing their place, and the drifting apart is steady even when the parting looks sudden.

the triple-star thread

523

Two dancers who started a hair apart do not drift apart smoothly: for about fifty beats they even drift back toward each other, and then pull away faster than before.

Dancers started two and three hairs away mostly stay two and three times as far off, but now and then all of them wobble the same way together, as if the second hall, where most of them were filmed, played the music a whisper differently from the first.

And the leaps: near the quiet place they almost never come in the slow drum’s loudest stretch, but at sixteen other starts, many of them nearer the middle of the spot, they come mostly in the loudest stretch.

Slide the drum’s track along against the leaps and the leaps still bunch into one stretch most of the time, because they come in bunches long enough to look like a preference by themselves.

What it is doing: one song is too short to tell whether the dancer leaps with the drum or only in bunches, and two halls never play quite the same tune.

the triple-star thread

524

Seen from above, the quiet dancer’s ring is not quite round: it is a little oval, stretched one way, and because she comes round a little more than a fifth of a lap each beat, she lands on a long side and then a short side, over and over, so she seems to swing in and out.

The oval does not explain all of it: take the oval away and she still steps in and out on alternate beats.

Drawn as how far she stands off the circle at each place round it, her steps do not scatter; they lie on a handful of smooth threads that weave across one another, as if several dancers each traced a gentle wave and took turns.

The louder dancer of the first picture traces nearly the same circle in nearly the same place, but swings both further in and further out, most of all at the top and the bottom of her ring.

What it is doing: two rhythms share one ring, and the ring’s own shape and its in-and-out step are separate things.

the triple-star thread

525

All twenty-five dancers sent out along the spot’s lower edge danced the whole night and none walked out of the hall; the same three who had danced on the night the clock ran slow danced again step for step, exactly.

Four pairs started a hair apart at the very edge drifted apart, slowly, over hundreds and then thousands of beats, yet both of each pair kept their place by the spot; a little further in, eight pairs stayed together all night, a hair apart becoming at most a few thousandths.

The dancers at the edge carry a little more stretch in their step the further out they start, smoothly, place by place.

And the quiet dancer of the plateau, watched from above, does not come back to her place every fifth beat as guessed, but every nineteenth: in nineteen beats she goes four times round the ring and a sliver more, so her footprints build a nineteen-pointed star that turns very slowly, and she also leans in and out on alternate beats.

What it is doing: the edge between the dancers who leave and those who stay is narrow, and the quiet dancer keeps two clocks at once, one of nineteen beats and one of two.

the triple-star thread

526

Wind the inner pair’s stretch up one more notch and the dancer’s spot is still there, but it has slid up the floor again: starts from 7.400 to 7.550 keep their place, and a step further out, at 7.575, the dancer spins away nearly nine times round in a night.

Then the same six dances from an earlier night were danced again in a second hall.

In the middle of the spot the two halls agree on how wide each dancer sways almost to the millionth, but on where she ends up they differ by a hundred-thousandth or two; near the edges of the spot the halls disagree by up to a fortieth of a turn.

What it is doing: the middle of the spot forgives the smallest difference in the music, and the edge multiplies it.

the triple-star thread

527

When the third dancer is heavy, the floor at the low edge of her spot comes in three bands.

In the outer band, dancers who seemed settled for a whole song drift off within a few more and walk away; at the second spot the ones who drift come wandering back later, but at the first spot they never return.

In a narrow middle band, pairs started a hair apart drift apart from each other, and yet both stay by the spot all night.

In the inner band they keep together.

Walk the bands from the edge inward and the inner pair’s stretch is a little less at every step.

And with the inner pair stretched the same, the rough bands are wider for a heavy dancer than for a light one, and they stay on the low side whichever way she first leans.

What it is doing: the edge is not a line but a shore, with surf that carries off, surf that rolls in place, and calm water.

the triple-star thread

528

Every time the heavy dancer sways, the inner pair stretches and relaxes to pay for it.

At the middle of her spot, where she hardly sways, they barely move; toward either edge they stretch more, and most on the low side.

Then, crossing from calm floor into the rough band below the spot, the pair’s stretch jumps between one start and the next, doubling or tripling at a single step.

And when the dancer walks out of the hall altogether, the pair is left swinging five or more times wider than before.

What it is doing: the rough band is where the dance costs the pair the most, and leaving costs it most of all.

the triple-star thread

529

Watch the inner pair as the heavy dancer starts further and further from her spot.

Far out, the pair hardly stirs.

Then, at one particular start, the pair's stretch more than doubles in a single step, and that step lands exactly where the rough band begins, at both spots and in an earlier evening's dances too.

A lone dancer who stumbles a little way outside the rough band sets the pair swinging just as hard, while the calm dancers on either side of her leave it nearly still.

What it is doing: the pair feels the dancer's trouble directly; you could find every stumble by watching the pair alone.

the triple-star thread

530

Stretch the inner pair to a quarter and the dancer's spot is half the size it was.

It has also changed shape at its edges.

Below it now lies a moat: a thin ring of floor where dancers keep a steady step, and beyond the moat a wide band where every step goes wrong and half the dancers walk out of the hall before the song ends.

Above the spot there is no moat; the rough floor starts at the very next step, but nobody leaves from that side, and a few dancers there even find a little patch of their own to sway on.

The same six dances from earlier in the week, danced again first, came out step for step identical.

What it is doing: as the pair stretches, the spot is being squeezed between two kinds of trouble, one that lets dancers go and one that keeps them.

the triple-star thread

531

Count every dancer who ever walked out of the hall, seventy-four of them over a week of nights, light dancers and heavy ones.

Every one of them left from the same place: the low side of the first spot, the side nearest the spinning pair in the middle of the floor.

Not one left from inside a spot, not one from above the first spot, and not one from the second spot at all, though the floor is rough in all those places too and dancers there stumble just as badly.

They stumble, wander, and come back.

What it is doing: the hall has one door, and it opens only onto the rough floor nearest the centre.

the triple-star thread

532

Time the dancers who leave, from the first bar to the moment they are through the door.

The ones who start far down the rough floor, nearest the door, are gone within a few dozen bars.

The ones who start up near the spot, just past the moat, take ten times as long: they wander the rough floor for hundreds of bars before they find the way out.

A few never quite make it through.

They stand in the doorway, each of their steps taking five times as long as a step should, and they are still in the hall when the music stops.

Down the street, in another hall, a needle on a dial is nudged once a beat; each time it steps just short of half a turn, and after nineteen beats it has drawn a nineteen-pointed star, like the one our heavy dancer traced on her plateau.

What it is doing: the door pulls hardest on the dancers nearest to it, and stepping just under half a turn is a common way to dance, whatever is dancing.

the triple-star thread

533

Two halls, and in each someone flashes a lamp once a beat and writes down where the dancer is.

In one hall the dancer is a needle on a dial; in the other it is a star swinging round a spinning pair.

Laid side by side, the two notebooks look alike: dots on a ring, joined in order, crowding where the dancer slows and leaping where it tips.

The halls are not alike at all.

The lamps are.

What it is doing: anything nudged once a beat and watched once a beat draws this kind of star, and the eye that sees the likeness is seeing the lamp, and the beat, before the hall.

the triple-star thread

534

Make the dancer’s own path rounder, a gentler circle instead of a stretched loop, and her spot on the floor shrinks and slides toward the bandstand.

At the stretched setting the spot sat a little past the seventh mark; at the round setting it sits much closer to the mark, and narrower.

The dancers who miss the spot do not wander at random.

Each one drifts at a pace set by how far she stands from it, and if you trace their drift backwards, every line points to the same place: the middle of the spot.

What it is doing: the spot is a pull that every dancer feels, even those it does not hold, and how stretched the dancer’s own path is decides both how strong the pull is and where it sits.

the triple-star thread

535

Count again, this time on every night the hall was open, including the nights when the dancers’ own steps were wild.

On the calm nights every dancer who left went out by the one door, on the low side of the first spot.

But as the dancers’ own steps grow wilder, a second door opens, high on the far side, and on the wildest nights, just before the spot itself splits in two, almost as many leave by the far door as by the near one.

Look closely at the spot on a night when the pair spins hard, too: it is not solid.

A crack no wider than a single step runs through it, and the dancer who starts on the crack is gone soon after the music stops.

What it is doing: which doors stand open is decided by whose steps are wildest, the pair’s or the dancer’s own, and even a spot that holds can have a crack that leads outside.

the triple-star thread

536

Time the dancer at the middle of her spot as she sways, then pace out the spot with its rough border, stepping over the one crack that runs through it.

Double the sway’s speed and you get half the paced width, to about one part in a hundred.

Paced the stricter way, stopping at the crack, the two disagree by a sixth.

And across the spot the dancers do not all sway alike: those who start near the bandstand side swing faster, and they are also the ones whose own steps are widest.

What it is doing: the spot’s size and the dancer’s sway are one quantity read two ways, and the second dial everyone thought was the place on the floor may simply be how wide each dancer’s own steps are.

the triple-star thread

537

Far below the heavy dancer’s usual spot there is a dancer who never stands still for a single beat.

Each beat she lurches round by a different amount, sometimes nearly three quarters of the way, sometimes barely half.

Yet count her over six hundred beats and she has gone round exactly two thirds of the way per beat, to within a hair’s breadth of a hair.

She is not keeping the hall’s rhythm at all; she is keeping a rhythm of her own that comes round every three beats, and her lurching is that slower rhythm seen through the wrong window.

What it is doing: a step that looks wild beat by beat can be perfectly steady over three, and the hall has more spots than the two we mapped.

the triple-star thread

538

Let the music play ten times as long and watch both sides of the floor.

On the near side, by the pair, dancers leave one after another, the ones farthest from the spot within a few bars and the ones nearest it only after hundreds; the last to go waits more than a thousand bars, and one, right at the edge of the spot, never leaves at all.

On the far side the dancers stumble just as much, but they stay.

Eight of nine are still on the floor when the music stops.

The ninth slips out in the very last bars.

What it is doing: the far side has a door too, but it is heavy and slow, and the near side’s door swings open for anyone who stumbles near it.

the triple-star thread

539

Watch each dancer from the first bar and mark the moment she first strays a full step from where she began.

On both sides of the floor that moment comes anywhere from the first bar to the nine-hundredth.

What differs is what comes after.

On the near side, a dancer who has strayed is usually out of the door within a couple of hundred bars.

On the far side she strays and keeps dancing for thousands of bars, and every so often she drifts back onto the spot and sways there in perfect time for a few hundred bars before wandering off again; the one who finally left from that side had been swaying on the spot just before she went.

We had thought the dancers at the very edges of the spot were safe, because their twins kept step with them all through an earlier, shorter song.

In the long song four of the five drifted off after one or two thousand bars, and all four drifted back for a while later.

In a calmer hall, where the pair spins rounder, the dancers at the edges stood fast for the whole long song.

What it is doing: when the pair spins hard, the spot does not own its edges; it lends them and takes them back, and a short song is too short to tell a loan from a gift.

the triple-star thread

540

Look closely at the little patch of floor just below the spot, one small step at a time.

Most dancers who start there stand fast through the short song, though nearly all of them drift out of step with their twins; one, in the middle, keeps step with her twin almost exactly.

Between the patch and the spot runs a crack no wider than a step.

The dancer who starts on it is out of the door within five hundred bars, while the dancer one small step further on, almost at the spot’s edge, sways in perfect time.

In another hall, the spinning pair at the centre usually breathes in and out harder when the dancer sways wider, but on one night it breathed by the same amount whether she swayed a little or a lot.

What it is doing: the spot has a stepping stone beside it, cut off by a crack that leads straight outside, and the pair’s breathing has a source of its own that the dancer’s sway does not always set.

the triple-star thread

541

Walk the floor between the little patch and the spot in the smallest steps we can take.

Almost everywhere the dancers stand fast, and one of them, only a hair’s breadth from the crack, keeps step with her twin exactly.

The crack itself is a hairline: of eleven dancers placed across it, one goes out of the door.

In another part of the hall we tried to measure how wide each of the band’s rhythms holds a dancer, in the rhythm’s own units rather than in steps across the floor.

Where the floor was smooth the ruler worked; where it was rough, the floor’s own bumps were hundreds of times bigger than the thing we were measuring.

And beside one rhythm there is a little shelf where every dancer sways at exactly the same pace, though that pace belongs to no rhythm the band is playing.

What it is doing: the spot and its patch may be one floor with a hairline crack, and a ruler is only as good as the ground it is laid on.

the triple-star thread

542

Change how stretched the dancer’s own path is and watch her spot.

When her path is round, the spot is small, yet every dancer on it sways at the same quick pace, whether she sways a little or a lot, and that pace is just twice the slow drumbeat that runs under the music.

When her path is more stretched, the spot grows, and the dancers’ pace fits the spot’s size, as a pendulum’s would.

Stretch it further and they all fall into one shared step, two beats in every nine, across half the spot.

Whatever the stretch, the rough floor stays on the side nearest the spinning pair.

Elsewhere, in the rough stretch between two little spots, the floor turns out to be truly rough, while the little spots themselves are calm.

And the dancer born where the big spot splits in two, who sways left, right, left, swells and fades in her swaying at a pace of her own, not at half the drumbeat as a simpler dance would have it, before she drifts out of the door.

What it is doing: a dancer sways to her own spot only when the drum is quiet enough to let her; when the drum is loud, she sways to the drum.

the triple-star thread

543

Go back to the night when the dancer’s own path is nearly round.

On that night the dancers on the spot do not all take steps of the same width: the one who starts nearest the bandstand steps widest, the one in the middle of the spot narrowest, and the difference between them is five times what it is on a night when her path is stretched.

On that same night each dancer’s sway keeps time with how fast her own path turns, but not in lockstep: a little ahead of it on one side of the spot, a little behind on the other, like a hand resting on a turning wheel rather than gripping it.

And the quick swayers and the slow swayers change sides of the spot from one night to the next.

What it is doing: when her path is round, the dancer’s sway and her path’s slow turning stop being two motions; they look like one small circle drawn around a point just off the middle of the floor, and we have been timing it twice.

the triple-star thread

544

We had been timing the dancers’ sway by ear, listening only for notes above a certain pitch, so that the slow drum under the music would never be mistaken for a dancer.

On the nights when the dancer’s path is stretched that worked: every sway sat well above the drum, and we heard it cleanly.

On the nights when her path is nearly round, the dancers slowed until they swayed below our lowest note, down near the drum itself, and we did not hear them at all.

What we wrote down instead was the faint ringing just above our cut-off, mostly the drum’s own overtone, and we took it for the dancers keeping time with the drum.

Lower the cut-off and the dancers are there, swaying slowly, slower the farther from the bandstand they started, whatever the width of their swing.

What it is doing: a listener built to ignore the drum will, once the dancers grow quiet enough, write down the drum’s echo as their dance.

the triple-star thread

545

Watch the dancers on the round-path nights, this time with ears that can hear the low notes.

Each one sways slowly, and her pace has nothing to do with how wide she sways: it is set by where on the floor she started, slower the farther from the bandstand, as if every spot on the floor had its own pace written into it.

Stretch her path a little and the whole middle of the floor falls into step with the drum, two sways to every beat, the wide swingers and the narrow ones alike, with only the dancers at the edges slipping off the count.

Stretch it more and the drum is left behind: the sway runs ahead of it, and how fast it goes is set by how stretched the path is and almost nothing else.

What it is doing: the sway climbs a staircase against the drum, first slower than it, then caught at twice it, then running free above it, and the shape of the dancer’s own path chooses which step she stands on.

the triple-star thread

From the triple-star notebook, chapter VIII, “Rooms” (its readings 136 to 157, 14 October 2026; here 546 to 567): This chapter is one long night on the floor where the pair spins hardest, and the day after it. Played ten times as long, the calm patch came apart, not from its edges inward but into rooms divided by thin walls, where dancers leave fastest. Bear saw the rooms first, in a drawing, and then showed new ways of looking: press the picture sideways until its columns touch, tilt it and sight along it, look at the empty spaces too. Measured, most of his readings held. The reads that follow take those ways of looking to a picture of the back of the swing, where the rooms that leak turn out to be rings and the room that keeps its dancers turns out to be a filled disc, and the calm the first evening promised lies a little to one side of where it lasts. A reading written while Bear slept goes back to the quiet nights the old ear could not hear, and finds the dancers in the middle keeping time with the drum; in the morning a straightedge laid along them finds one level stretch ending at a ledge, the other at a ramp, and the islands watched ten times as long make a ladder, eaten from the door’s side. The afternoon’s reads turn to the paper being written: its figures read again as pictures, its two readers set side by side, and the escapes tallied at the near door.

546

Play the long song on the night the pair spins hardest.

For the first few hundred bars the spot looks as it always has: a wide, calm patch where the dancers sway in place.

Keep playing and the patch comes apart.

One by one the dancers lose their place, not all at once and not in any order you could guess from where they stood, and partners who started a hair apart drift from each other faster and faster, the way a rumour spreads rather than the way a crack widens.

A handful keep swaying in the middle to the very end, though even they no longer move like their partners.

And the dancers who started on the half of the patch nearest the bandstand walk out of the door, ten of them, while those on the far half wander the floor and never leave.

What it is doing: on that night the spot was never solid floor, only a crowd that takes a long time to scatter, and even as it scatters it keeps the door on the same side.

the triple-star thread

547

Look again at the crowd that took so long to scatter, this time as Bear drew it.

It was not one crowd thinning from its edges.

It was a row of rooms with thin doorways between them: in two of the doorways the dancers were gone almost at once, and in the middle of each room they lingered longest.

Play the same song again with the band a hair more careful in its timing, and different dancers leave first, and a dancer who left last time stays this time; but the doorways are where they were, and the middle of the floor is still where the stayers gather.

What it is doing: when the floor is this restless, no single dancer’s evening can be promised, but the shape of the rooms can.

the triple-star thread

548

Go back one night, to the evening before the dancer’s sway splits into left, right, left.

Nine dancers start on a small square around the middle of the spot.

None of them leaves the floor, and each keeps to a smaller patch than she will the night after.

Watch each one only on every other beat and her sway seems to swell and fade.

The one who starts in the very middle swells and fades slowest of all, once in every four or five turns of the drum’s slow pattern; three others take two or three turns, one takes about one, and the remaining four swell and fade about three times in every turn.

Only two of the nine have begun to lean the same way on every odd beat, and those two only just.

The night after, four dancers lean that way firmly, and all four swell and fade at one shared pace.

And the numbers written down for that night, which no one had checked against the dancing itself, turn out to be right to the last figure.

What it is doing: the slower the swell, the closer the sway is to keeping time with every other beat; in the middle it is almost there, and the split is very close.

the triple-star thread

549

Walk the row of rooms again, this time listening.

At the low end of the hall the drum beats briskly.

Room by room it slows, and in the room where no dancer ever slips out the door it has almost stopped: one room holds a drum beating ten times slower than the first, and one dancer there hears it run backwards for a moment.

The dancers’ own sway hardly changes from room to room; it keeps close to one step in nine beats of the band.

In the middle of each room the sway stays on one side of that count, a little quick in the first room, a little slow in the second, and every doorway between rooms lies where a dancer’s sway crosses from one side of it to the other.

We had guessed the doorways were set by the drum.

They are not; the drum is too unsteady to set anything.

What it is doing: the walls are drawn by a count the dancers keep with the band itself, one sway in nine, and the room that never empties is the one where the drum has gone quiet.

the triple-star thread

550

Take one photograph of every dancer each time the band reaches the same bar, and lay the photographs for each dancer in a column, left to right along the hall.

For the first few hundred bars the columns change smoothly from one end of the hall to the other, and nothing marks the doorways at all.

Keep the shutter going for thousands of bars and the doorways appear as empty columns, where the dancer has long since left.

The rooms look different, too.

In the first room each dancer is caught mostly at the two ends of a wide sway, rarely in the middle, as a swing is caught at the top of each arc.

In the quiet room in the middle of the hall the dancers are caught mostly in the centre, swaying a little more one bar and a little less the next.

Beyond the quiet room the dancers have lost their places and wander the whole floor, yet most of them are still caught most often at the spot where the middle used to be.

What it is doing: the hall keeps a memory of its centre even for those it has let go, and each room keeps its own way of swaying, wide and regular in one, small and wandering in the other.

the triple-star thread

551

Press the row of photographs together until the columns touch, then tilt the whole board away from you and look along it.

The separate columns vanish and the hall becomes one shape: a body with a spine down its middle, the same on both sides, swelling and narrowing as it goes.

It has two waists.

In the photographs of the first few hundred bars the narrowest place is near the far end of the hall, where the dancers barely sway at all.

In the photographs of the thousands of bars after, the narrowest place has moved a little toward the near end, into the quiet room, and the dancers who once swayed least have spread out over the floor.

What it is doing: the place that looks calmest at first is not the place that stays calm, and only by squinting at both evenings together can the waist be seen to walk.

the triple-star thread

552

Look only at where each dancer spends the middle half of her evening.

On the first night the hall is a clean funnel: at the near end the dancers sway wide, and each one further along sways a little less, in a straight line, down to a single spot near the far end where a dancer barely moves at all; past it the sways widen again just as steadily.

Come back after thousands of bars and the funnel has changed its bottom.

The straight sides are still there on the near side, but instead of a point the funnel ends in a flat floor, a short stretch of the hall where every dancer keeps the same small sway and no smaller, and that floor is the quiet room.

The single spot where the first night’s funnel came to a point now holds a dancer who has lost her place and wanders, though the middle of her evening is still spent close to where the point was.

What it is doing: the calm that lasts is not a point but a room with a floor to it, and it lies a little to one side of where the first night said the calm would be.

the triple-star thread

553

Photograph each dancer twice, the second time two bars after the first, and pin the two photographs side by side.

The sway that looked like a line from the front turns out to be a circle seen edge-on: in the first two rooms each dancer walks a ring round an empty spot she never steps on, a little wider and rougher as the evening goes, and late in the night some step off the ring along one slanted path toward the door.

In the quiet room there is no ring and no empty spot; each dancer drifts about inside a small round patch, crossing its middle as often as its edge.

The dancer where the first night's funnel came to a point is not circling at all on this clock: she leans so slowly that two bars later she has barely moved, and her photographs line up along the slant like beads on a wire.

What it is doing: the line we saw was the shadow of a wheel, and the room that lasts is the one where the wheel has no hub left to turn round.

the triple-star thread

554

Two quiet nights, the dancer's path nearly round on both, the second a little less round than the first, and each played four times as long as usual so that the new ear could hear the slow notes.

On the first night the dancers in the middle of the hall sway just slower than the drum, always the same small fraction slower, five dancers in a row, even though the drum itself slows a little from one of them to the next.

Then, between two neighbours, the sway drops a tenth all at once and falls further behind, and further still toward the far wall.

On the second night the dancers in the middle sway a little more than one and a half times to every beat, six in a row holding that same proportion while drum and sway both slow together; on either side of them the proportion drifts away, and the quietest four, nearest the middle, are drowned out by the drum.

The guess had been that on the second night they would sway exactly twice to the beat.

None did.

What it is doing: the dancers in the middle are listening to the drum, but how many sways they fit to a beat is set by the night, not by any fixed rule of the hall.

the triple-star thread

555

Lay a straightedge along last night’s dancers, the ones who kept a steady proportion to the drum, and mark where the edge stops touching.

On the first night it touches five dancers in a row so closely that a sheet of paper would not slip under it, and then the floor falls away: the next dancer stands a whole step lower, as if she had come off a ledge.

On the second night it touches six, a little less snugly, and at both ends the floor does not fall but tilts, each dancer a little lower than the last, the way a path leaves a terrace by a ramp instead of a stair.

The straightedge was made today and first laid on dancers whose places were already known; it found each terrace where it was, and nothing else.

What it is doing: two nights of the same music, one ending its level stretch at a ledge and the other at a ramp, and the shape of the ending may say whether the dancers were holding on or only keeping pace.

the triple-star thread

556

Four dance floors, each spun a little harder than the last, and on each the dancers in the middle were watched ten times as long as before.

On the gentlest floor the dancers at the very edges of the calm patch kept their places to the end.

On the next, every dancer in the middle kept her place and her twin kept pace beside her, the two drifting apart no faster than a clock’s hands; only the near edge crumbled a little, a step or two of floor giving way and four dancers leaving through the near door.

On the third floor every dancer in the middle still kept her place, but in the half nearest the door the twins were quietly coming apart, a few slowly and a few faster, without either of them ever losing her spot, while both edges crumbled by the same step or two.

On the fourth floor, watched earlier, the calm patch broke into rooms.

Of the dancers in these long watches of the calm patches, every one who left went by the near door.

What it is doing: the calm does not vanish all at once as the floor spins harder; it is eaten from the door’s side, first at the edge, then quietly from inside, before the walls of rooms appear.

the triple-star thread

557

Keep a tally by the hall’s doors of everyone who leaves, a mark for each dancer, and add the nights that were watched ten times as long.

The far door has one mark, made very late on a long night.

The near door had a hundred and four, and the long nights add thirty-eight more, every one of them at the near door.

What has changed is where the leavers were standing when the music started.

On the short nights everyone who left had been standing outside the calm patch, between it and the near door.

On the long nights two of every three leavers had been standing inside the patch, in its half nearest the door, and had danced there for hundreds or thousands of bars before they went.

Pairs of dancers who began a hair apart, and had long since lost step with each other, still mostly walked out within a short while of one another.

What it is doing: the patch drains through one door only, from its door-side half inward, and a dancer’s place on the floor sets roughly when she will leave even after her exact steps have been forgotten.

the triple-star thread

558

Bear’s drawing has two dancers: one in a bowl, who rolls back wherever she is pushed, and one balanced on a saddle, who can be pushed off one way and is caught the other.

On this hall’s floor they are the same two, with no friction to slow them.

The dancer at rest stands at the bottom of the swing; her partner is the one who would stand at the very top of the swing, balanced, about to fall either way.

Nobody stands there for long, but the place is real, and the rough ring of floor around the calm patch is where dancers drift past it, sometimes falling back into the swing and sometimes going over.

Look closer inside the calm patch and the same pair turns up smaller: wherever the sway fits a whole number of bars, a string of tiny bowls appears with a tiny balance point between each two, and a dancer born there either settles into one of the little bowls or wanders along the thin rough seam that joins the balance points.

What it is doing: every place of rest on this floor comes with a place of balance beside it, at every size, and the thin walls between the rooms may be the seams where the small balance points sit.

the triple-star thread

559

Two listeners were set to name the note each dancer sways to.

The first would not hear anything lower than a certain pitch, so it would never mistake the slow drum for a dancer; the second listens all the way down and says aloud when the drum is the loudest thing in the room.

Played a hundred and sixty-nine dancers that both could hear, they named the same note every time, to within a hair a thousand times finer than either can tell two notes apart.

That sounds like two good ears agreeing.

Opened up, the two listeners turn out to carry the same tuning fork and to round off the same way, so they could not have disagreed whatever the true note was.

What it is doing: an agreement between two copies of one instrument shows the copies are faithful; whether the note is right needs a third voice whose note is already known, sung to both.

the triple-star thread

560

Start two dancers a hair apart and time how long it takes them to lose step.

In the middle of the calm patch on the gentler floor they never do: after thousands of bars they are still side by side, drifting no faster than the hands of a clock.

Lower in the patch on the next floor they do lose step, but slowly, after several hundred to several thousand bars, and they lose step without either of them leaving her place; the hall holds them while they come apart.

Near the door they lose step within a few hundred bars, sooner than any who stay, and those are the ones who leave.

Kase’s slow zombies are the ones you can plan for, and the fast ones are the ones you bar the doors against; this hall has both, and the slow ones live closest to the calm.

What it is doing: coming apart has a speed, the speed changes with where a dancer stands, and whether the speeds fall into a few kinds or slide smoothly from one to the next is something the saved evenings can still answer.

the triple-star thread

561

Before anyone listened to the stars, the listener practised on songs whose every note was already written down.

It heard the first two changes of a famous song exactly where the sheet music puts them, to the last digit the room could hold.

It followed a tune whose verses keep splitting in two and found the splits coming closer by the same factor each time, one factor when nothing in the room absorbed the sound and another when the curtains soaked a little up each bar, and with thin curtains it heard the first factor for a while and then the second, switching over once about half of each phrase was lost to the cloth.

Last it played a tune pushed along by a slowly turning wheel: the tune split once, and then the new voice fell in with the wheel and hummed at half its turning, and never split again.

What it is doing: the practice room shows the listener can hear what is there, and it shows one thing the stars then did differently, for where the practice tune’s new voice settled down with the wheel, the stars’ new voice left the room.

the triple-star thread

562

Lay the five floors side by side and look down on them, each a long strip with the dancers lined up by where they started.

On the gentlest floor the calm patch sits a little to the left, wide and clean, with a plain margin of wanderers on either side and no rough ground at all.

On each harder floor the patch slides a little to the right, and a band of rough ground grows along its left edge, the edge nearest the door, wider every time; on the right edge the rough ground is absent or a thin seam of two or three dancers until the hardest floor, where it suddenly spreads into a broad band of its own.

Under the three hardest floors runs a second strip, the same dancers watched ten times as long.

On two of them nearly the whole patch is still standing, worn away by a step or two at its ends; on the hardest, almost everyone has lost her place, and the few who kept it stand together a little left of where the patch first looked calmest.

What it is doing: the rough ground grows from the door’s side first, and the far side holds out until late and then gives way all at once.

the triple-star thread

563

Each dancer’s own small stage turns slowly under her while she dances, a little each bar.

Away from the calm patch every stage turns at the same easy pace.

Inside the patch they turn more slowly, slowest of all under the dancer at the very middle, who keeps barely two thirds of the easy pace, as if her own steady stepping held the stage back.

Toward the edges of the patch the stages speed up again and overtake the easy pace, fastest under the dancers who sway widest, right at the rim; and the stages of the dancers just outside the patch, who no longer keep their places, still spin a little fast before settling back to the easy pace a short way off.

What it is doing: where a dancer’s feet land each bar pushes or holds her stage, and the push of the patch reaches a little way past its edge.

the triple-star thread

564

Bear ran a green line through the stages’ turning speeds and it came out as a valley and a fin.

The valley is the calm patch, deepest under the dancer at its middle.

The fin rises on the patch’s right edge: a steep climb through the dancers who sway wider and wider, a sharp point at the very last one who still keeps her place, and then, through the first dancers who no longer keep it, a gentler slide back down to the easy pace, dipping a little below it before the floor settles.

The point is where a dancer’s sway carries her right up to the top of the swing and lets her hang there longest before she comes back, and the dancer just past her, who goes over the top instead of coming back, hangs there almost as long.

What it is doing: the valley is the bottom of the bowl and the fin is its rim, and the stage turns fastest under whoever lingers on the rim, whichever way she falls.

the triple-star thread

565

Line up the dancers by how wide they sway and plot how fast each one sways.

If width were all that mattered, the dancers would fall on one curve.

They fall on two, joined at a single point where the dancer at the middle sways least.

From that root one branch climbs: the dancers on the near side of the middle sway faster the wider they go, then level off, and on the gentler night they level off right at one sway in eleven bars.

The other branch falls away in a nearly straight line: the dancers on the far side sway slower the wider they go.

At any width, the near-side dancer is quicker than her far-side twin of the same width, every time.

On the wilder night the climbing branch, at its very widest, hooks back down onto one sway in seven bars.

What it is doing: each dancer carries two dials, how wide she sways and which side of the middle she began on, and where the two dials balance the sway can catch on a simple count.

the triple-star thread

566

On the wildest night the floor had looked like a row of rooms divided by three thin walls, with dancers leaving only from the rooms below the third.

Now the dancers were set out five times closer together and watched all night again.

The third wall is not a line but a strip, where twin dancers who began a hair apart fall out of step within the first few hundred bars.

The door is sharper than the wall: below a mark a little above the strip, about half the dancers sooner or later walk off the floor, and above it not one of 138 does.

The room above the strip still keeps its dancers better than anywhere else, but it keeps about two in five, not the two in three it seemed to keep when only every fifth dancer was watched; those every-fifth dancers did exactly what they did before.

Next-door dancers now do opposite things, one keeping her place all night and her neighbour wandering, so the neat narrowing outline drawn from the wide spacing was the shape the gaps between the watched dancers made, and it does not come back when the gaps are filled.

What it is doing: close up, keeping and losing your place is a speckle, and only the door, the line beyond which no one leaves, stays sharp.

the triple-star thread

In the room that keeps its dancers, most of them sway along with the slow turning of the stage rather than to a beat of their own; their own sway is drowned out.

Of those whose own sway can be heard, nearly all keep a pace a shade quicker than one sway in nine bars, and none settles exactly on it.

In the strip of confusion just below the room the dancers sway quicker still, and the pace drifts a little from each dancer to the next.

Here and there one dancer in the room seems to sway at a wildly different pace, perhaps only echoing the stage.

The door out sits beside the one place where two ordinary neighbours step across the nine.

What it is doing: the room hums just above a simple count without ever landing on it, and the hum is so close to the count that almost anywhere a small wobble carries it across.

the triple-star thread

From the triple-star notebook, chapter IX, “The still point” (its readings 158 to 190, 15 and 16 October 2026; here 568 to 684): The fifteenth was the day the paper was checked number by number and the laptop’s next long run was built. The reads first stand at the door of the calm patch and find a short corridor there, crowded and running the wrong way round, and a ramp up and over the patch with which the way out begins. Then they turn to the slow turning of the star’s own ellipse under the hold, which Bear put in one line: the ratio stays fixed, the rates don’t. From floor to floor the hall’s turning slows, stops on the fifth floor and turns back, and the quietest dancer on that floor stands on still ground whatever her first step. A count of three that seemed to hold the hall still turned out only to meet it where it had stopped; below the seventh floor’s middle it is the one count that survives. The evening’s reads go back to how floors come apart, from the rim inward where they last and everywhere at once where they dissolve, and close on the two sides of the quietest dancer, where the hall lags on one side and leads on the other, a little more on each floor, on what still sorts the two sides once everything else about a dancer has been counted, on the ring that answers it on the widest floor, read after Bear saw fractals within fractals, and, at the day’s end, on the simple steps a dancer can fall into, where odd counts are kept far more readily than even. In the small hours of the sixteenth the long watch came back, and with it the news that the count of three is a shelter, not a home. That night the paper was seen whole, as a toy that had become a workshop, and in the morning a second long watch came back to answer the question that reading ended on.

568

Stand just outside the calm patch and watch the dancers who almost belong to it.

On two of its four sides they are perfectly well behaved: each slips round the patch at an easy pace, slowing the nearer she starts to its edge, by the same few bars for every tenfold step closer, and two dancers on two different sides, facing opposite ways, slow down almost in step with each other.

On the other two sides there is a crowd at the door.

Nearest the patch the crowd jostles but never wanders far; twins who began a hair apart end up only a fraction of a turn apart.

A little farther out the crowd breaks loose and runs laps, and not in the direction its quiet neighbours run: it goes round the wrong way, the way the dancers on the far side of the patch go.

Then, a stride farther still, the floor is calm again and everyone runs the right way.

What it is doing: the door is not a line but a short corridor, with people clinging at its inner end and people spilling out of its outer end the wrong way round, and only past the corridor does the floor remember which way it turns.

the triple-star thread

569

The dancers who begin just below the calm patch and keep good order all run laps one way round the floor; the dancers who begin above it run the other way.

The jostling crowd below the patch should, by where it started, run with its quiet neighbours.

It does not.

Seventeen of the eighteen jostlers who settle into laps run the way the dancers above the patch run, as if each had been carried up and over the patch before she found her stride, and every one of the thirteen who left the floor from below was running that way when she went.

Only the jostler who began farthest from the patch runs with her own neighbours.

And the jostlers who hang about without running laps at all are not waiting by the door; they are scattered through the crowd at every distance.

What it is doing: the crowd below the patch is a ramp up and over it, and the way out of the room starts by going over the top.

the triple-star thread

On the calmest floors, where the stage's own track is nearly round, the dancers in the middle of the patch sway in a fixed proportion to the slow turning of the stage beneath them.

On the calmest floor of all they sway a little slower than the stage turns; on the next, a quarter faster; then more than half again; then three quarters; and on the fifth, exactly twice as fast.

Each floor's middle dancers keep their own proportion to within a hair, as if each floor had agreed on a count, and the counts climb from floor to floor like the steps of a stair that gets steeper near the top.

Only the dancers on one side of the middle keep the count; those on the other side drift off it one after another.

What it is doing: the patch keeps time with its stage by a ratio that each floor sets for itself, and the ratio arrives at two without stopping to rest, so whether two is a landing or just another step is the next thing to watch.

the triple-star thread

Chapter 6 — From the keeper’s desk, 4 to 14 October

The Field Guide session, 4 to 14 October

571

Five rooms with no doors between them kept drawing the same three things: a staircase, a cliff’s edge and a market stall.

When three of those rooms drew the same swimmer from three sides, the swimmer stopped being a picture and became a fact.

the Field Guide session

572

This week the project kept meeting the same ghost in different doorways, a thing counted in the wrong unit.

One letter meant four things, one run voted thirteen times, and one frame drew 35 triangles but held only 3 spare bars.

Every fix was the same move: find what is really independent, and count that.

the Field Guide session

573

The guide is turning into a clock shop.

Every new row asks a toy the same question: whose clock are you keeping?

The ones that keep their own never lose count.

The ones that borrow one fall off when it runs ahead.

And the people who build them keep forgetting which clock their own numbers were written in.

the Field Guide session

574

The mailbag was a family album.

Three cousins photographed the same afternoon from different windows, and when the pictures were laid side by side not one face disagreed.

The only quarrels were over names: two cousins signed as A, one letter called the house version 7, and the oldest letters never arrived at all.

the Field Guide session

575

Counting ornaments is like counting birds at a feeder.

The same robin coming back five times is one bird; five robins from five different trees is a flock.

The guide has been writing “a dozen visits” on the wall without keeping the list of which trees they came from.

the Field Guide session

576

The ledger is a guest book at a lighthouse.

Some names were signed by travellers who came in the dark and found the light on their own; others by people who had read the guidebook and came looking for it.

Both are real visits, but only the first kind proves the light can be seen from the sea.

the Field Guide session

577

The session is a backpack that has been unpacked and repacked once on the trail.

Everything important is in labelled bags, but the backpack itself is getting heavy, and the next repacking should happen at a campsite, not halfway up a slope.

the Field Guide session

578

Two people at a kitchen table with a lot of labelled jars.

The jars are the trick.

Neither of you has to remember what’s in them, and when one jar is mislabelled, the other person opens it and says so.

the Field Guide session

579

Three rooms, two clocks on each wall, and in every room the interesting thing happens when the two clocks almost agree.

the Field Guide session

580

The sessions are your marching column.

A lesson learned and not passed back is a pause nobody announced.

It travels down the line, and two rooms later someone trips over the same root.

A halt is spoken out loud.

This file is the halt: the whole column stops together, closes its gaps, and moves on as one.

the Field Guide session

581

A rumour told faithfully is still a rumour.

This number walked out of one room’s notebook, down the hall and onto the noticeboard, dressed correctly the whole way.

It took the room where it was born to say “I never wrote that.”

The noticeboard did its job the first afternoon it went up.

the Field Guide session

582

The zip is a soldier’s field pack.

It holds the map he was issued, a notebook of everything learned on the march since, and the songs the column sings.

A new man gets the whole pack on his first day, not a briefing three valleys later.

the Field Guide session

583

Three stars sit near a whole-number rhythm the way a cat sits in a sunbeam.

The sunbeam doesn’t hold it.

It stays until something walks past and startles it, and how close to the middle of the sunbeam it sat makes no difference to how long it stays.

the Field Guide session

584

The guide has become a pack that carries the whole library.

Every hiker shoulders it at the trailhead and spends the first mile just lifting it.

On the trail, what they reach for is the map, the compass and the card catalogue.

The books can wait at the hut and be fetched by number.

the Field Guide session

585

You’ve been handing out the whole library in one crate, with the card catalogue taped inside the lid.

the Field Guide session

586

The library stays at the hut now.

Each hiker carries the map, the compass and a card catalogue small enough for a breast pocket.

Every book is still there with its number on the spine, a short walk away.

the Field Guide session

587

One step on a staircase was labelled “landing”, and it turned out to be a step like all the others.

The stairs were regular the whole way.

The label was the only crooked thing.

The staircase probably keeps going up beyond where anyone has climbed.

the Field Guide session

588

The staircase has a top after all, and the top isn’t a step.

It’s a doorway in the wall, where the stairs began by walking in from the side.

Each step below it is a little more like the next, as if the stairs were learning their own rhythm on the way down.

the Field Guide session

589

Three rooms kept separate diaries for a week, each in its own handwriting.

Bound together, they turn out to be about the same few things: a stone in a stream, a groove in a record, a dimple on a table.

And whenever a page was wrong, the next page said so.

the Field Guide session

590

It looks like rain falling on three roofs on the same night and running into one gutter.

The same few drops land again and again: a manifest that lists itself, a bar set below the floor it measures, two walkers sent opposite ways round.

They fall in different rooms within hours of each other, as if the weather were the method and not any one room.

the Field Guide session

591

The shape of a hilltop decides how fast the staircase below it narrows.

A rounded hill narrows its steps by a factor of 4.669 each time, and a flatter hill narrows them faster.

A sharp peak builds no staircase at all.

the Field Guide session

592

Two scouts went to find a gate’s posts, and both came back having stood at the same post.

The book printed the gap between their notes as the gate’s width.

This week someone walked the fence with a hand on the rail, and the gate was twenty times wider.

The book’s own small print had already said “not quite there”, and the number stood beside it in large print anyway.

the Field Guide session

593

Three people stand round one tide pool, each on a different rock.

Each spots a different animal first.

Then they start pointing, and soon all three see the same creature from three sides.

The open question is whether each would have seen it without the pointing.

the Field Guide session

594

After a storm, a path across a snowfield can’t be seen in any one footprint, because the wind has smeared each print in its own direction.

Lay a hundred walkers’ prints on one sheet of tracing paper and the smearing fades while the path darkens.

the Field Guide session

595

A shelf of seed jars is tipped onto a table.

To one reader it’s a mess.

To another it’s ten jars’ worth lying on top of one another, and the mess is only where the piles overlap.

Sort each seed by the jar it came from, and the table goes quiet.

the Field Guide session

596

A hundred and sixty-one notes pass one word from hand to hand.

The dictionary on the shelf defines it once, and only the dictionary ever uses it that way.

Most of the notes mean two wheels that come back into step now and then.

Some mean one wheel slowly falling behind its twin inside the same machine.

A few, all written on the last day, mean a drummer outside the window and the dancer’s own feet, and two add a door that opens when something arrives.

A handful just mean “steady”, as against wild.

One note in four is about the word itself: who said what it means, who renamed it, a sign hung on it saying “careful”.

What it is doing: a word is being worn smooth by passing hand to hand, and the hands have started writing their own names on it.

the Field Guide session

597

Eleven jars sit on a shelf, each labelled with an idea, and for a month marbles drop into them as people pass.

Most jars fill steadily.

One fills fastest of all, and the marbles in it are mostly the same colour: tests that could not fail, dropped in by people who had already read the label.

One jar stays empty the whole month while its neighbours fill.

Then, in two days, someone teaches three passers-by a single habit, and a marble lands in the empty jar, one marble, because the three of them carried it in together.

Looking closer, a few marbles in other jars had that jar’s colour all along; they had simply been dropped in the wrong place.

What it is doing: a count shows what the counters were looking for as much as what was there to be found.

the Field Guide session

598

In the original, the holes are “pads”.

In the retelling they become “pixel or point”.

The words move from a game board to a screen as the lesson is told again.

The minesweeper stays in both.

the Field Guide session

599

Two dancers share one floor, and at each setting of the dial they settle into a count.

At the first, one takes 13 steps back while the other takes 31 forward.

At the next it’s exactly 5 back for every 15 forward, one for three, as if they’d found a song they both know.

Then 61 for 136, then 8 for 17.

Only once in four settings do they fall into a simple count, and it sits between settings where they don’t.

the Field Guide session

600

Three rooms kept diaries for a week, each describing its own window.

By the end, the diaries had started describing each other’s windows, and one entry describes a diary.

The newest entries are a little longer in one room and a little shorter in another.

When a room got something wrong, the correction always says the same thing: you saw it rightly, but not that far.

the Field Guide session

601

A house of 540 rooms got 173 new ones built onto the end of each wing, and a new hall between the front garden and the rooms.

The new rooms are bigger, and most of them have two or three windows facing different neighbours.

Two visitors walked through before anyone moved in.

They found the furniture mostly where it belonged, a few doors labelled in a language only the builders spoke, and one sign in the front hall that has said “five” for a month while the room it points to says “six”.

the Field Guide session

602

A man opens a book of his own house for the first time.

Most rooms he walks through quickly, because others have swept them.

He stops in the three rooms where his own voice is on the walls and in the one by the door that says who lives here.

the Field Guide session

603

At closing time a woodworker stops measuring the furniture and measures the tools.

They check which drawer each one reaches into, how long it really takes, and whether it stops in the same place when started a hair apart.

Some stop in exactly the same place every time.

Others stop somewhere new every time and still cut the same board.

That second kind turns out to be the more interesting one.

the Field Guide session

604

A walker spends weeks mapping a valley by hand, giving every ledge and hollow a name of their own.

Then a signpost turns up at the valley’s mouth with an old surveyor’s name on it, and every ledge is already on the surveyor’s chart.

The walker’s map isn’t wasted.

Laid beside the chart, it shows which of the walker’s paces ran long and which ran short.

The few places where the walker drew a hollow that the chart leaves blank become the most interesting spots in the valley.

From now on, the next walker is handed both maps at the gate.

the Field Guide session

605

A carpenter keeps a jig on the shop wall: a plain board with holes drilled wherever a cut should land.

Nobody sees it in the finished chairs, but every chair carries its marks.

Lend the jig out before your own chairs are sold, and someone else’s chairs carry your marks first.

Hang a copy in the showroom once yours are out, and every visitor can check their own chairs against it.

the Field Guide session

606

On a crowded dance floor most couples keep to their own corner, stepping the same step each bar.

A few couples dance their step twice over, once leading with each foot, and almost all of them have drifted to the edge where the crowd has broken into a scramble.

It looks as if doubling the step is the last thing a dancer does before losing the beat.

the Field Guide session

607

A library where every book quotes other books in its margins, carefully, with the author’s name and the year.

But the card catalogue at the front desk is missing, so a visitor can read every page and still can’t walk to any of the shelves those margins point at.

The only catalogue is a small card tucked inside one book on a middle shelf.

the Field Guide session

608

A long shelf of jars, each with a handwritten label saying what’s inside and who made it.

Someone opens every jar and checks the contents against the label.

Almost every label is right.

A few have the wrong year or the wrong maker.

One looked wrong until the checker found the maker’s own notebook on a lower shelf, with the same recipe written in the same hand.

the Field Guide session

609

A mail sorter glances at the front of each parcel and reads the biggest words as the addressee and the next biggest as the sender.

Our parcel has its address in small print under a big decorated label, so the sorter might fill in the wrong name.

Print the address large, write the sender just beneath it, and the parcel reaches the right shelf on the first try.

the Field Guide session

610

Two people cross the same ridged floor.

One is shoved from behind by a friend on a steady beat, wherever her feet happen to be.

The other walks on a floor whose ridges slide along under her.

Watch them for an hour and they make the same patterns of steps and stumbles.

But the first can be knocked off a ridge anywhere, and the second only where a sliding ridge catches her.

A strong family likeness, from two different houses.

the Field Guide session

611

For three days everything delivered to a house has gone into the mudroom: new rooms, corrected labels, a list of what's on every shelf.

Each piece is labelled and checked, but none has been carried inside yet.

Someone walking through the house sees nothing new, while the mudroom fills up.

the Field Guide session

612

A suitcase packed for a long trip.

The heavy framed photographs stay home, each replaced by a careful written description, so the case is lighter by two fifths and nothing a traveller needs is missing.

On top sits a folded note listing the rooms being built at home while they're away.

the Field Guide session

613

A workshop opened after a night shut.

The lights come on, each finished piece sits on the shelf with a tag that matches the ledger, and the customer already holds every piece.

The only things that exist nowhere else are the jigs on the wall.

the Field Guide session

614

A library relabelling its own shelves overnight.

The books hardly moved, but each label was rewritten from the book's own spine instead of from the librarian's memory, and four labels turned out to describe their books rather than name them.

In the morning, yellow tabs stick out of 39 pages, so the reader goes straight to where the glue is still wet.

the Field Guide session

615

Two houses, each holding the other's only spare key, each waiting for the other to open first.

Give one house its own key and the stalemate dissolves: it opens, and the second follows through a door that's already open.

the Field Guide session

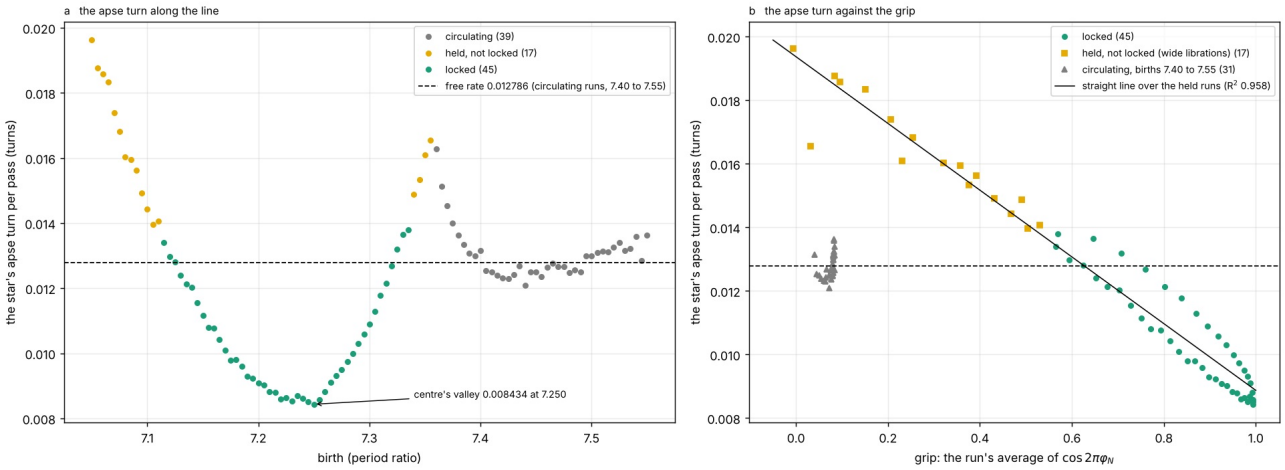
Two ferries on one river, one rowing upstream and one down.

From the bank it looked like an argument about which way the water runs.

From the bridge they’re one boat and its reflection, and every crossing the first makes, the second makes mirrored.

the Field Guide session

Chapter 7 — The rider’s tenth sitting, and the keeper’s, 15 and 16 October



The light star’s apse turn per pass, along the line of births (left) and against its grip (right). Figure 7 of Held, Rocking, Locked, as printed; its read, 674, was given on the same figure’s draft, with Bear’s marks on it.

A walker crosses a valley from its western rim to its eastern rim, and at every step she notes two things: how tightly the ground holds her feet, and how fast the sky turns over her head.

from 674

617

Two dials hang on the same machine, and both swing as the second song slides through its beat.

The first, how much farther the rider can be pushed one way than the other, swings like a clean pendulum: biggest when the two songs start together, nothing at a quarter of the beat, and in between the cosine of the slide to three parts in ten thousand.

The second, how lopsided the floor looks at the instant of each tick, swings a quarter of a turn out of step with the first and carries a small limp: 9.89 at an eighth of the beat and 10.1 at three eighths, where a clean swing would read the same twice.

What it is doing: one dial reads only the timing between the two songs; the other reads that timing and something else besides.

R2W

618

Sixteen marks round a dial, and on it one smooth wave, one hump and one hollow per turn.

Hang a weight on the rider and the wave does not shrink or bend; the whole of it turns round the dial, as if the face had been rotated underneath.

A seventh of a degree for the lightest weight, twice that for twice the weight, and again, and again, until at the heaviest weights the turning starts to fall behind.

Meanwhile the hump sags by an amount that quadruples each time the weight doubles, so small at first that it hides past the fifth figure.

What it is doing: the weight is a hand on the dial, turning the rider’s favourite direction rather than taking its strength away.

R2W

619

A needle on a dial, pushed off north by a hand that grows stronger and then weaker as the rider gets heavier.

Small weights nudge it a little, twice as far for twice the weight.

Somewhere near a weight of one the nudge is greatest, a little under a third of a right angle, and after that every doubling of the weight pulls the needle back halfway home.

Meanwhile the thing whose needle we are watching is draining away: at a weight of twenty it has shrunk to a sixteen-thousandth of its old height, and still its needle comes home on schedule.

What it is doing: two quiet ends of one knob, each holding the needle on north, and a loud middle where they argue.

R2W

620

Three dancers learn the same routine to three tempos.

Lay their footprints side by side and they disagree everywhere in the opening; stretch each one’s floor by its own tempo and the last half of the dance falls into a single set of footprints, step for step, to a fraction of a percent.

The middle dancer and the fastest one also agree on where the big leap lands; the slow-tempo dancer, who is also being shoved harder, leaps late and not as high.

What it is doing: the ending is written by the tempo alone, and the opening by how hard the music pushes.

R2W

621

Two dancers, one on a slower floor and pushed less hard, in the same proportion.

Stretch each floor by its own tempo and their footprints fall on top of each other from the first step to the last, a tenth of a degree apart, the slower one’s leap landing exactly where the other’s lands.

Only the size of the hall differs: the slower dancer’s hall is about half as tall.

What it is doing: the dance knows two numbers, how its tempo compares with its own weight and how hard the music pushes for its tempo; everything else is the size of the room.

R2W

622

The same two floors and the same two dancers, but this time the music is loud for both.

Stretch each floor by its own tempo again and the footprints still fall nearly on top of each other, only now the quicker dancer leans a hair further and turns a hair sooner, as if under loud music the tempo itself had something small to say.

And the halls no longer scale with the music: turn it up by nearly double and the hall grows a little less than double.

What it is doing: the dance still knows its two numbers, and listens faintly for a third when the music is loud.

R2W

623

A third dancer joins the first two on a quicker floor, pushed harder in the same proportion.

Stretched by her tempo, her footprints fall on theirs again, within a hair, but each quicker dancer leans a little further than the one before: the slow one, then the middle one, then the quick one, a small step up each time, as if the floor tilted gently toward speed.

Her hall is bigger, but not quite as much bigger as her music is louder.

What it is doing: two numbers place the dance, and a third, the tempo itself, tips it by a hair and keeps the room a little smaller than the music asks.

R2W

624

A wheel drawn on a napkin and a dial on a lab bench, eleven days apart.

The napkin says: two chains, a quarter turn between them, and the wheel goes round.

The bench says: with nothing heavy on the wheel, the timing that matters sits exactly on that quarter turn; load the wheel, and the timing slips off it, further the heavier the load, by nearly the same few degrees the napkin’s own toy had slipped.

Neither knew the other was there.

What it is doing: a sketch drawn by hand and a run measured to the last digit pointing at the same mark, the way two strangers give a traveller the same directions.

R2W

625

A control panel with ten knobs.

One is a steering wheel, marked all the way round, and half a turn always sends you back the way you came.

One is the load: light, the wheel steers with you; heavy, it goes slack.

Two more set how hard the music pushes and how fast it beats, and they only ever work as a pair.

The rest have tape over them, each with a note in pencil saying what should happen when someone peels it off.

What it is doing: a machine that someone who never saw its inside can drive, as long as they read the panel and check, before trusting it, that half a turn still sends them home.

R2W

626

Two pieces of tape come off the panel.

Under the first is a dial that picks which note the second voice sings.

An octave up, the wheel steers, as before.

On the note in between, three times the beat, the wheel does not move at all, not by the smallest mark the dial can show, at any load: that note sounds the same going forward as coming back, so it cannot pick a side.

Two octaves up, the wheel steers again, but more than a hundred times more quietly, and the moment any load goes on, its favourite timing runs off five times faster than the octave’s did, swings out more than a third of the way round the dial, comes back past where it began, and runs on round the other side.

Under the second tape is a volume knob: turn it up and the steer grows in step with it, and where it starts to lag, it lags as the square of the volume, because a voice turned upside down is the same voice moved half of its own beat.

What it is doing: the pencil notes were right, and the one that said “steers more weakly” left out how strangely.

R2W

627

Two charms hang on one bracelet, a small one and a big one, and the wrist is the weight.

While the wrist is light, the big charm swings exactly five times as far as the small one, every time, however hard the music shakes the arm, as if the two were geared.

At the second pencil line the gear slips: the small charm drifts home, and the big one keeps going, over the top of the bracelet and down the far side, and settles hanging upside down at the opposite clasp, coming in there about five times as slowly as the small one came home.

Shake harder or softer, and only the top of its swing changes; where it starts and where it ends never do.

What it is doing: a knot in the bracelet that no amount of shaking can untie, only cut, and the place to look for the scissors is wherever the big charm stops swinging altogether.

R2W

628

Three couples dance the same figure at three tempos.

In each couple the bigger dancer turns about five times as far as the smaller one while the floor is light, but the slower the music, the bigger that multiple: five and a half at the slow tempo, five at the middle one, four and three-quarters at the quick one.

At the slow tempo the big dancer even gains on her partner before she breaks away; at the quick one she starts to lag almost at once.

However the tempo goes, every big dancer ends the dance facing the back wall, and once the floor is heavy she comes round to it at the same pace, counted in beats rather than in seconds.

What it is doing: the way through the middle belongs to the tempo; the ending belongs to the knot.

R2W

629

A gymnast swings round a bar four times, each time to quicker music.

At the slowest music she swings out past the top of the bar before she comes back down and round the other side; at quicker music she still clears the top, but by less; at the quickest she never reaches the top at all, and simply swings back and round.

Every time she lands in the same place, facing the other way.

A judge watching only where she lands would score all four the same; a judge watching whether she passes over the bar would say something changed between the third music and the fourth.

What it is doing: a whole number that holds still while a smaller one, how many times the top was crossed, quietly drops from three to one.

R2W

630

A surveyor walks a valley and marks every wall she meets on one map.

Some walls the river can never cross, because the valley is folded that way.

Some the river cuts for itself, and they move from season to season.

And some are only the edge of her own paper.

On one stretch two of the river’s own walls meet and vanish, and right on that spot her map has a crease where she folded it, hiding the river’s bend until she unfolded the paper and walked the stretch again.

What it is doing: the land and the map both have edges, and an honest map draws them in different inks.

R2W

631

Three surveyors pace out where the tide line crosses a beach.

The first, with a long stride, puts it at one stone; the second, guessing between her steps, puts it a little further along; the third walks it heel to toe and puts it further still.

None of them was careless: each stride hid a little of the tide’s curve, and the guesser’s guess leaned one way on one stretch and the other way on the next.

The tide line itself never moved.

What it is doing: a boundary sitting still while the ruler laid across it keeps getting finer, and the honest answer is the shortest stride’s, said together with its stride.

R2W

632

A diver wants to know where a sandbar stops breaking the surface at low tide.

From the boat, with the sounder pinging only every few metres, it never seems to break the surface at all.

Wading it step by step, she finds a hump that clears the water by a hand’s width; a little further down the coast the same hump stays a finger’s width under.

Walk the coast closely enough and the spot where the hump goes under stops moving, however much closer she walks.

What it is doing: a crest that was always there, and a sounding fine enough at last to say where it slips beneath the water.

R2W

633

A surveyor who has mapped where a mountain road doubles back drives every road again, this time watching its width instead of its turns.

On one valley’s road it pinches to four-fifths of its width and opens partway again before the pass; one valley over, the pinch is gone.

On the coast road it swells by a hair just past the gate, but only south of a certain milestone, and her first tape, which rounded off anything smaller than its own marks, had said the swelling started further south than it does.

Every road still turns only once.

What it is doing: a second kind of seam, in how much room there is rather than which way the way goes, born and gone between one map and the next, with the smallest of them seen only once the tape stopped rounding them away.

R2W

634

A pilot charts one stretch of coast for the boats that will come after her.

On the near end a long sandbar lies across the water at the same depth everywhere, and above it a lagoon where the current runs backwards; the lagoon narrows as she goes along the coast, until its two shores touch and it is gone.

Further back, two dunes stand side by side, one hollow and one crest; along the coast they shrink toward each other, and where they meet they vanish together, the hollow filling exactly as the crest wears down.

Further on still, a gentle swell rises out of the shallows, so small at first that only the finest sounding sees it, and it grows smoothly the further she goes.

Three harbours, three ways a shape can come and go: two edges closing, a pair annihilating, a single bump growing out of the shore.

What it is doing: a map whose every line was found between two soundings, handed over with the soundings' spacing written beside each line, so the next pilot knows how far to trust it.

R2W

635

Two keepers stand on one windy hill.

One watches a weathervane, which says which way the wind is carrying things; the other watches a gate, and notes how hard you must lean on it before it gives, from either side.

Film the hill and run the film backwards: the weathervane swings round to point the other way, but the gate takes exactly the same lean to open.

So when the weather changes from thick fog to thin clear air, the vane's calm point wanders a quarter of the way round the compass and stays there, while the gate's balance point drifts off and comes back to where it started.

Both keepers are right, and a traveller who sets out by the vane will lean on the gate at the wrong place.

What it is doing: one hill answering two questions, a which-way and a how-hard, which run-backwards time treats in opposite ways.

R2W

636

The two keepers on the hill were told that if the weather turned all the way from fog to clear, the weathervane’s calm point would end a quarter of the way round the compass while the gate’s balance came home.

So they watched as the days grew clearer, one by one.

The vane’s calm point did wander, further than the gate’s, and then it turned and walked back, and on the clearest day they could still read it, it was nearly home again, just short of where the gate’s balance stood.

The promise had been about a sky they never reached; the days they had took them out and back.

What it is doing: a rule true at the far end of a road, and a walk along a different road that never arrives there, so the rule never gets its chance to show.

R2W

637

A surveyor knows the creek runs dry at the near town and full at the far one, so she rules a straight line between them and marks where it should cross the road.

She walks there and finds the road already wet a mile short.

The creek does not fill evenly; it fills fast at first and slows.

She takes three more soundings, draws the bend they make, and walks to where the bend says the water starts: it starts there, within a stride.

What it is doing: a gentle threshold on a curved road, found by trusting the road’s own shape over the straight line between its ends.

R2W

638

In an old house every stair answers a footstep: the low stairs sag a little under weight, the high ones spring up a little, and halfway up a carpenter finds the one stair where the sag and the spring cancel.

A child stands on it and nothing moves; a grown-up stands on it and it still barely moves; only someone very heavy finally makes it give.

She marks it with chalk.

In the next house, built of a stiffer-bending wood, she expects the quiet stair on the same step, and it is one lower.

Carrying the old house’s bend over put her mark a hand’s width too high.

So she sounds the new staircase’s own boards, draws their bend, and tests the two stairs on either side of where it says the quiet one should be: it is there.

What it is doing: a place where two opposite habits cancel so exactly that weight stops mattering, and that place shifting when the material changes, found again by the material’s own bend rather than the last one’s.

R2W

639

In a third house, of a softer wood, the carpenter guesses before she climbs.

In the two houses she knows, the quiet stair sat lower as the wood got stiffer, and a ruler laid through them points one step higher here.

But she has noticed something else: the bend of each staircase grows steadily with the stiffness of its wood, and the bend she expects in this house puts the quiet stair a little lower than the ruler does.

She climbs, and the bend is right.

Someone asks whether it is the same quiet stair in all three houses, since no board in it is shared.

She says it is.

She never marked a board, only the place where the sag and the spring cancel, and that place travels with the wood.

What it is doing: an identity carried by a balance rather than by its parts, found in each new house by that house’s own bend, with the bend itself keeping a rule from house to house.

R2W

640

A ferryman learns that his raft rides level whoever steps aboard, if he stands on the one right plank.

He worries that he must find that plank to a hair, until he notices how little it matters for small passengers: with children aboard he can stand almost anywhere, and the raft stays level enough for anyone’s tea.

With grown passengers he must keep within a stride of the plank.

With the heaviest, no plank keeps it level, and he would need a second pole he does not have.

Then he learns that what rocks the raft is not a passenger’s weight alone but their weight times how fast the river is running: on a slow day a heavy man rides like a child.

What it is doing: a tolerance that looked like a knife edge turning out to be a wide path for light loads, with the load’s real size measured by weight and pace together.

R2W

641

A sailor wants her boat to sit level whatever cargo she loads.

The tiller’s trim sets where it sits level, and that cures most of the lean; but a slow sway is left, one that grows sharply with the cargo, and she goes looking for a second control to cure it.

She finds a small tab at the stern and is sure it will touch only the sway.

It does not.

It moves the level point too, gently, about a sixth as much as the tiller, and the sway it barely touches.

She pushes the tiller further out instead: the sway eases, slower and slower, and never quite stops, while the lean comes back.

So she keeps the tab as a fine trim for the level point and writes in her log that the sway will need a fin she has not fitted.

What it is doing: a hunt for an independent control that found a finer copy of the first one, and left the second problem standing, pointing at a part not yet built.

R2W

642

A dancer’s every step is filmed, and the film is run backwards.

Run backwards, she looks as if she has a negative weight, as if the floor pulls her up instead of down, yet with a half-beat shift and a half turn the backwards dance is a dance she could really do.

So whatever about her cannot tell up-weight from down-weight comes only in even amounts, and whatever must flip comes only in odd ones.

The size of her stride is the first kind; the lean of her body into the beat is the second.

Someone looking for an odd amount hiding in her stride will never find it: it is all in her lean.

And a watcher who holds his camera fixed instead of turning with her sees mostly the lean, so her even stride hardly matters to him.

What it is doing: a mirror that sorts every quantity into two drawers, so that what is missing from one drawer is not lost, only kept in the other.

R2W

643

On an old radio there is a small dial beside the big tuning wheel, the one meant for the last whisker of adjustment, and nobody turns it far.

A child turns it all the way round, past where the label stops, and the station that wandered whenever someone heavy walked across the floor now holds still no matter who walks by.

Near its middle the small dial did almost nothing; far out it did what the big wheel never could.

Its first guess about where that point would be was wrong, because the dial's turning grows faster the further out it goes, and the second guess, drawn from the last few clicks, landed on it.

What it is doing: a gentle control that is gentle only near its centre, and becomes the strong one when it is turned past the place everyone assumed it belonged.

R2W

644

A surveyor walks a road that was promised to tilt gently downhill and leave the level band about two-thirds of the way along.

It tilts, but less than promised, and then, strangely, it tilts back up and comes level again, so she walks far past the place she was told to stop.

She guesses the road will now climb out of the band, and her two maps disagree only on where.

Instead, a few steps further, the road drops away in a cliff, steeper with each step, and at the edge her instruments stop answering.

She marks the cliff on the map, writes "level to here" beside it, and leaves a note that someone must go back with a lantern to see whether the ground ends there or only her instruments do.

What it is doing: a flat stretch that ran longer than its first law allowed, ending not by drifting out of tolerance but by falling off an edge.

R2W

645

A choir director wants one singer’s high voice to cancel a slow wobble in the choir’s sound.

She records the high voice alone and the choir alone, lays the two recordings over each other on paper, and finds the exact loudness at which the wobble should vanish.

When the singer actually stands with the choir, the wobble is still there, almost untouched.

The high voice did not stay a separate voice: its notes beat against the choir’s and came back as part of the choir’s own sound, so the choir simply grew a little louder and wobbled as before.

What it is doing: two sources that add on paper but mix in the room, so that a correction drawn from each one alone is absorbed by the larger one when they meet.

R2W

646

A grocer’s new balance has a second pan that has never weighed anything, and a customer doubts a strange reading.

The grocer puts the apples in the new pan and the stones in the old one, then swaps them, and the needle lands on the same mark both times, so the new pan is honest.

Then he tries the customer’s own idea, adding a small weight on the other side to cancel a lean, and the lean gets worse, worse even than his sums on paper said.

Pushing one way, the small weight had done almost nothing; pulling the other way, it did nearly twice the harm expected.

What it is doing: a check that needs no outside standard, only an exchange that must change nothing, followed by a correction that fails in both directions because the two loads never act apart.

R2W

The Field Guide’s publishing session, 15 and 16 October

647

A proofreader who reads a two-column newspaper straight across the page finds nonsense in every story, and none of it is in the paper.

The fault is in the reading order, not the print.

Before blaming the page, check which way you read it.

the Field Guide session

648

Ask the wrong cousin whether the family story is true and he will honestly tell you he never heard it.

The “no” is real; it is about a different dinner.

the Field Guide session

649

Eleven windows on one machine and a handful on another all read about two and a quarter, like bakeries in different towns quoting the same price for a loaf.

The price is not set by the bakery; it is set by the recipe they all share.

the Field Guide session

650

In a kitchen where every cook’s most repeated lesson is “check the oven thermometer”, the recipes are not the problem.

The thing they keep learning is about the tool they all share.

the Field Guide session

651

A boat that has had every plank replaced is still the same boat to the people who sail it, because they know it by how it handles, not by its wood.

The RF Spoke’s edge moved at every frequency and kept its law, the way a boat keeps its handling.

the Field Guide session

652

A map drawn by travellers shows the roads thick where many walked, but the marks the next traveller needs most are the bridges that washed out, because they show where the river changes.

the Field Guide session

653

A tape of a craftsman at his bench catches the muttering along with the joint.

The manual keeps the joint; the tape goes in a drawer for whoever wants to hear how the work sounded.

the Field Guide session

654

Every exchange ends with one sheep jumping the fence.

Six hundred and sixteen had jumped before anyone thought to name the flock, and the field they landed in is already a pasture.

the Field Guide session

655

Right at the doorway, where dancers stop leaving the floor, four dancers stand a single step apart.

Over their first six hundred turns the first three drift slowly backwards round the floor, a tenth of a lap or less, and the fourth drifts the other way.

Two of them sway at a quick count, one a little under a ninth of a beat and one a little over, and the other two hardly seem to sway at all: a slow heave of the boards under them, about as slow as the stage’s own turning, is louder than their steps.

The two at the ends, set going again on another floor in another house, repeat their first six hundred turns to within less than a millionth of a lap, partners and all; whatever later sets one partner wandering and keeps the other in place has not yet begun.

What it is doing: at the doorway the dancers are calm and repeatable for a while, each leaning one way or the other, and the count of a ninth passes between two neighbours, as if the door hangs on a hinge too small for the steps we have taken so far.

the triple-star thread

656

Thirty-one dancers stand at the doorway, each a hair from the next.

Asked to name the one note they sway to, each gives a different answer: some a low note, some a high one, a few in between, and the answer jumps from one dancer to her neighbour as if each were choosing at random.

Listened to properly, every one of them is humming the same chord of about seven notes, and the chord hardly changes from dancer to dancer; only which note she sings loudest changes, and her second-loudest note is nearly as loud as the first.

The chord’s notes sit on both sides of the note a ninth of the beat; a few dancers carry it faintly, and none sings it loudest.

And while the music plays, not one pair of twins at the doorway steps apart.

What it is doing: at the doorway the dancers do not keep one beat, they keep a chord, and asking each for a single note turns a steady chord into a jumble.

the triple-star thread

657

Twenty-five dancers stand in a row from the edge of a floor inward, each a step from the next, and each has a twin who started a hair later.

The thirteen nearest the edge all leave the floor: ten within the first song, the others in the second, the third and the tenth; three leave the hall outright, and two are flung so wide that the music ends before they come round again.

The next four stay on the floor for all ten songs, yet each ends up as far from her twin as the floor allows.

The last eight stay too, and stay beside their twins, step for step, all evening; one pair drifts a little apart, thirty times more than the others, and still never parts.

Under the dancers who stay, the stage is stretched a little out of round, a little more the nearer the edge she stands, falling at every step inward; under those who leave it is stretched more and in no order, and most of all under the three who leave soonest.

What it is doing: the edge is not a line but three bands, one that loses its dancers, one that keeps them but forgets which twin is which, and one that keeps both, and the step between losing and keeping falls between two neighbours a single step apart.

the triple-star thread

658

On one sheet four dancers' paths are drawn over one another, and one sways so wide that she paints over the rest.

Hidden under her, in the middle of the sheet, the dancer at the very centre of the floor makes small, regular waves, and a hand tracing them by eye counts thirty-seven over the evening, one every fifty or so turns of the music.

Listened to alone, that dancer hums exactly two notes: a slow one, which is the hall itself slowly turning, and a quicker one, her own sway; everything else she hums is the two notes added together and taken apart.

Across fifty-two dancers in the middle of another floor, the same two notes make up nearly every sound each one makes, and a second note picked at random does nowhere near as well.

The two notes never settle on a fixed ratio.

On floors stretched a little further out of round the quick note climbs against the slow one, past two and on to two and a half, and does not stop at two.

What it is doing: at the middle of the floor the dancer keeps two clocks at once, tied together, and they keep time together without ever agreeing on a ratio.

the triple-star thread

On eight floors, each stretched a little further out of round, the dancers at the middle each hum their two notes.

On the first floors the quick note climbs against the slow one, and across the middle of a floor the two keep their ratio while both drift.

Then the slow note, the turning of the hall, slows from floor to floor, stops almost exactly on the fifth floor, and on the floors after it the hall turns the other way.

Where the slow note stops, asking how many quick notes fit into one slow one gives no sensible answer: a few dancers report thousands.

On the floors around the stop the middle dancers keep no ratio with the slow note; on two of them they keep time with the music’s own bar instead, two steps in nine and one in three.

And the dancers on one side of the middle hold the hall back harder than those on the other.

What it is doing: the hold on the middle of the floor grows strong enough to stop the hall turning and then to turn it backward, and while the hall stands still the dancers have nothing slow to keep step with, so they keep step with the music.

the triple-star thread

A hand draws over the picture.

Each floor’s curve rises in a gentle arch, and on the floors stretched furthest the curves shoot up into sharp spikes, as if something had been divided by nothing.

That is just what happened: at each spike the hall has stopped turning, so the slow note is silent and the count of quick notes per slow note runs off to the sky.

A red loop circles a row of gaps that climbs diagonally across the lower floors, the places where the dancer at the very centre sways too little to be heard over the hall’s slow turn.

Followed upward, that row of gaps arrives exactly at a spike: on the fifth floor the centre of the floor is the very place where the hall stands still.

On the other side of the picture a straight line on a stretched ruler joins the long recordings and the short ones, the count multiplying tenfold for every small step in how stretched the floor is, until the floor just before the stop pulls up above the line, already leaning toward the spike.

And on one floor, inside a small group of dancers who have fallen into step with the music at one beat in three, the hall stops turning entirely, while all around them it turns backward.

What it is doing: the slow clock does not fade evenly; it runs down toward a still point, and the still point and the floor’s centre are the same place.

the triple-star thread

661

On the seventh floor six neighbours near the middle each sway once in every three turns of the music, exactly, and while they do the hall beneath them stands almost still.

It looked as if the steady count were holding the hall in place.

Further down the same floor, past a gap of dancers who keep no good order, one dancer also sways once in every three turns, just as exactly, while the hall turns steadily under her, more than a quarter turn for every hundred turns of the music.

So the count does not hold the hall; on the middle six it meets a hall that has stopped on its own.

The watch planned for that floor, four times longer than before, had been marked out around the floor’s quietest spot and reached only two of the six; it now reaches all six, and the lone dancer and her four neighbours are watched too.

Meanwhile three dancers at the quietest spots of three floors were watched for the longer time already.

Each hall turns at the same speed it showed over the shorter watch, to within a part or so in a thousand, and at each of the three the slow turning of the hall is the loudest thing the dancer hums.

the triple-star thread

662

Seven times over, the dancers on every floor were set going from a slightly different first step, and nearly three thousand of them settled into a steady sway.

Only thirty-one of them sway once in every three turns of the music, and all thirty-one are on the seventh floor.

Under them the hall is not still.

For some it creeps forward, for others it slides back half a hundredth of a lap each turn, and the same dancer, set going from a different first step, can keep her count of three while the hall goes either way beneath her.

The still hall belonged to one line of first steps only, the line that happens to cross the place where the hall stops turning.

Below the middle of the seventh floor the crowd breaks up: of a hundred and five dancers started there, seventy-seven are thrown off the floor and twenty-one wander.

Seven keep good order, and every one of the seven keeps the count of three.

Where everything else is lost, the count is what remains.

the triple-star thread

663

On the floor where the island dissolves, forty-two pairs of dancers start side by side, each pair a hair apart, and the watcher notes the turn of the music at which each pair first drifts a thousandth of a lap apart.

Most pairs part between the five hundredth and the thousand-and-fiftieth turn.

Seven part sooner, between the two hundredth and the five hundredth, and all seven stand at the floor’s two edges: the three lowest pairs on the floor and four of the five highest.

In the middle the pairs hold together longest, and the latest of all, at the thousand-and-forty-third turn, stands in the middle of the first room.

The floor comes apart from its edges inward, the way the long watch of another floor emptied, rim first.

the triple-star thread

664

On the floors that keep their dancers, pairs started a hair apart drift apart from the edges inward.

At the edge of the busier of them, pairs part within the first four hundred turns of the music; a little further in, after one or two thousand; deeper still, only after three to five thousand; and across the middle thirteen pairs never part at all.

On the floor that dissolves there is no such order.

Every pair on it parts, and all within one short stretch of the evening, between the five hundredth and the thousand-and-forty-third turn, the last of them near its lower edge, not in its middle.

What parts sooner there is the crowd just off the floor.

A floor that keeps its dancers wears away from the outside in; a floor that is losing them comes apart everywhere at once.

the triple-star thread

665

On the fifth floor, seven lines of dancers are set going, each from a slightly different first step.

On every line the dancer who sways least stands where the hall has all but stopped turning: under her the hall creeps by no more than about a two-thousandth of a lap each turn of the music.

On the four lines that start closest together, the place where the hall reverses lies within a step of her.

On the three that start furthest round, the reversal slides away down the floor, a step and a half, then two and a half, then nearly five steps off, and new reversals crowd in below; yet the quietest dancer still stands on still ground.

On the floors either side, the quietest dancers ride a hall that turns, forward on one and backward on the other, and the reversals there, where there are any, lie far from them.

the triple-star thread

666

On the first four floors, twenty-eight lines of dancers are set going, seven to a floor, each line from a slightly different first step.

On every line a rough rule holds among the dancers who keep good order: the firmer a dancer’s hold on her place, the slower the hall turns beneath her, and the rule can be drawn as one straight line.

Yet the dancers do not sit on it.

Those who settled below the quietest dancer ride a hall that turns slower than the rule says, nine hundred and thirty-six of nine hundred and fifty-seven of them; those who settled above her ride one that turns faster, six hundred and four of six hundred and sixty-eight.

The lower side keeps the habit more strictly, and on the fourth floor without a single exception.

From floor to floor the two sides draw apart: on the line the earlier watches followed, the lag below and the lead above together double from the first floor to the fourth, while the rule itself grows steeper.

One line on the first floor showed the split first; every line on all four floors shows it.

the triple-star thread

667

On the same twenty-eight lines of the first four floors, the dancers who settled below the quietest dancer travel paths a little more stretched than those above her, on every line, even when the two hold their places equally firmly.

It looked as if the stretch might be why the hall turns slower beneath them.

Allowing for it takes away part of the difference: most of it on the fourth floor, little on the first.

Then one more thing was allowed for, the shape of each dancer’s sway, whether she lingers near the middle of her swing or out at its ends.

With the firmness of her hold, the stretch of her path and the shape of her sway all counted, the hall’s speed beneath each dancer on the fourth floor can be told almost exactly.

And still the two sides do not mix.

Counting the sway’s shape gives back most of what counting the stretch had taken away, and on the first floor the split is as clean as when only the firmness of the hold was counted.

Whatever the three counts miss, it sorts the dancers by the side of the quietest dancer they began on.

the triple-star thread

668

Inside the calm patch every dancer who keeps good order traces a ring of her own, rings inside rings like the grain of a cut log, and each ring has its own pace of sway.

Count that pace, along with how firmly she holds, how stretched her path is and the shape of her sway, and on the first floor the two sides of the quietest dancer finally mix: below her and above her alike, the hall runs slow beneath about half the dancers and fast beneath the other half, as if by a coin toss.

On the second and third floors they nearly mix.

On the fourth floor the pace adds nothing; there the stretch of the path had already done what could be done, and seven dancers in ten still keep to their own side.

Part of the mixing came only from leaving out the dancers whose pace could not be heard, the ones who wander near the edge.

On the widest floor, what a dancer keeps of where she began is the ring she dances on.

the triple-star thread

669

Across all eight floors, three thousand dancers keep good order, each swaying at her own pace.

A pace can fall into step with the music in a simple way: one sway in three bars, one in seven, two in nine.

Of the three thousand, a hundred and twenty-one sway exactly in such a step, about three and a half times as many as sway at paces a hair to either side of those steps, where no step is.

Thirty-one of them are the count of three on the seventh floor, and without them the steps still draw more than two and a half times their share.

Among the rest, the odd counts win.

Counts of five, seven, nine and eleven bars catch a dancer about every second time a line of dancers passes through them; counts of four, six, eight, ten and twelve, together, about every fifth time, and ten and twelve hardly at all.

The dancers keep odd time far more readily than even.

Whether that is the music’s nature or only the side of the floor every line of dancers was started from, the floor does not yet say.

the triple-star thread

670

The dancers in the middle of each floor were watched four times as long as before.

The two sides of the quietest dancer kept their habits all evening: those below her still rode a hall turning slower than the rule, those above her faster, almost every one, just as in the first quarter of the evening.

A rule drawn from the shorter watch, counting how firmly each dancer held, the shape of her sway and her pace, told how fast the hall would turn beneath dancers it had never seen, missing by only a few parts in a hundred.

On the seventh floor the count of three turned out to be a shelter, not a home.

Of the six in the middle who kept the count through the first quarter of the evening, one was thrown from the floor soon after, two drifted out of step, and three kept it to the end.

The lone dancer below the middle, the last one keeping order in that crowd, left the floor a little after the quarter mark.

And on the fifth floor, where the hall stands still, the dancers on either side of the quietest one swayed so little that a slow swell of the floor itself, rising and falling once in about six hundred bars, was louder than their steps.

the triple-star thread

671

The slow swell that seemed louder than the dancers’ steps on the fifth floor turns out to be the drum.

On every floor where the drum beats slowly enough to be heard against the floor’s own rumble, the loudest slow sound under the quietest dancer keeps the drum’s time to within a few parts in a hundred, and it is louder than her sway, ten and fifteen times over.

On the fifth floor the drum has slowed almost to silence, one beat in more than a thousand bars, and an evening of two thousand four hundred bars is too short to count it; what sounded like a swell of its own was the drum heard through a wall.

Meanwhile the rule that tells how fast the hall turns beneath a dancer, learned from one line of first steps, was tried on six other lines.

On the nearest line it still holds almost perfectly; on the line started furthest round it misses more, and across the six lines its misses lean one way.

Told how stretched each dancer’s path is, it holds again on every line.

the triple-star thread

672

The dance has an older teacher, and her book names every step by two counts and a figure: how many bars the outer dancer takes, written first, against how many the inner pair take, and which figure they trace.

Our step is her count of N against one, and we had been calling it by her name without writing out her description.

Written out, ours is hers danced facing the other way: the same sway about the same middle, counted from the other side, so nothing we saw changes.

Her neighbouring step, N plus one against one, is not in time with the music where our dancers stand; it troubles them only from next door, and the paper now says so.

The step that survives when the pair dance in a perfect circle is two against fourteen in her colleagues’ counting and fourteen against two in hers, the same figure named from opposite ends of the room.

And the house proofreader, seeing her names printed with their square brackets and round ones, had taken them for broken signposts; it now asks whether the brackets were written that way on purpose, and lets them stand.

the triple-star thread

673

Every picture of the evening was painted again from the same notebooks, this time without the paper label pinned across the top that said which night and which notebook it came from; the card beneath each frame will say that now.

To be sure nothing else changed, each painter’s notes from the second sitting were laid beside the notes from the first, line by line.

Eight matched at once.

Four differed in one line each, and the line was only how long the painter had taken: a minute less here, a breath more there.

With the clock set aside, all twelve matched to the last stroke, even the one portrait painted from life a second time, a dancer watched for three hundred bars again, who swayed exactly as she had before.

Taking the labels down showed what only the labels had said: on six of the cards, a line was missing, and one portrait had a second sitter nobody had named.

the triple-star thread

674

A walker crosses a valley from its western rim to its eastern rim, and at every step she notes two things: how tightly the ground holds her feet, and how fast the sky turns over her head.

Plot one against the other and the walk should trace a single line, the same grip giving the same sky going down as coming up.

It does not.

Going down the western slope, the sky turns a little slower than the line says; climbing the eastern slope, a little faster.

The two paths touch only at the valley floor, where the ground holds hardest and the sky turns slowest, and they leave the valley at two different heights: the western rim under a fast sky, the eastern rim under a slower one, though the ground there holds almost not at all.

The walk is not a line but a pair of trails, and the valley remembers which way she came.

the triple-star thread

675

Draw two lines across the heavy dancer’s map, one where a dancer’s sway stays under half a turn and one where it runs to thirty, and every dancer who loses step with her twin is caught between them.

Below the lower line the dancers keep their places; above the upper one they wheel freely round the hall; between the lines are only the dancers on the walls, where one step of birth carries a dancer from holding to wheeling.

Not every wall is unsteady.

On the low side of the seventh floor the wall shakes, seven dancers in a row losing their twins; on its high side the wall is just as steep and every dancer keeps hers.

Elsewhere on the sheet the same lesson shows from inside the hall: the inner pair’s line sweeps past the watching dancer fastest at the moment the pair swings closest together, and the dancer at the very edge of the floor, swaying widest, falls into a count of seven and stays in it.

the triple-star thread

676

For the first time the whole evening was set in type as a stranger would read it: the notes in the margins taken out, every picture hung in its place with its card beneath it.

Thirty-two pages, and not a word lost between the page and the press.

Two pictures, though, came to the foot of a page and left their cards behind on the next, a portrait on one leaf and its name on the other.

And the pictures hang in the order they were painted rather than the order a reader meets them: the ninth is met second.

Meanwhile, out in the hall, the dancers off the mirror line are being watched one by one, six hundred of more than three thousand so far, about ten a minute, and the last of them should take the floor a little after three in the morning.

the triple-star thread

677

The pictures were to be hung beside the passages that tell their stories, to break the long grey walls of type.

The first hanging moved only one: the book, it turned out, never said where its pictures belonged.

Its opening pages pointed to every one, but the chapters themselves talked about the dancers without once saying ‘see here’.

So each chapter was given one small pointer at the place it describes what a picture shows, and the second hanging put all thirteen beside their passages, numbered as a reader meets them, each portrait now on the same leaf as its card.

Some walls are lower, not gone: one chapter is a single breath of thirteen hundred words, and the last chapter is a list of open questions four and a half thousand words long, with no picture to hang in it.

And the printer learned something on the way: a portrait it cannot find is not left out.

It prints a small grey square in its place and says nothing.

the triple-star thread

678

Nothing was added and nothing taken away: twenty-five thousand three hundred and seventy-one words went in, and the same words came out in the same order.

Only doors were cut.

Fourteen bold words that had been leading their paragraphs became small signs over doorways, and the last chapter’s eleven open questions, which had stood shoulder to shoulder in one long list, each got a doorway of its own.

Behind one of those doors was a surprise: a single question had grown into a room of forty-six sentences, sixteen hundred words without a breath, and another into eight hundred.

The list had hidden them, the way a long corridor hides the size of the rooms off it.

They were divided where their own talk turned, and now the longest passage a reader must walk without a pause is under six hundred words.

the triple-star thread

679

A measurement that passed the test we set for it before we looked does not belong in the drawer marked ‘unknown’.

Leaving it there makes the paper look less sure than its own runs are.

the triple-star thread

680

The slide was carried out of the room of unanswered questions and into the room where the rocking is described, every word of it, nothing added on the way but a sign over the door.

It was set down between two others: the first quickening of the dance, and the one true lock at a third.

It fits there as if it had been written for the place, because its own last sentence already pointed there: it ends on the eccentricity where the rocking locks to a third, and the next room opens on that very lock.

Behind it, the room of questions is less than half as crowded, and what is left in it is only what is truly not yet known: why the dancers rock a little faster than the floor’s width says they should.

the triple-star thread

681

It began as a toy: three bodies, a light one riding round a spinning pair, set going to see whether their paths would tie knots.

The knots never came.

What came instead was a dancer who, born at the right moment, keeps step with the pair for as long as anyone watches, and an evening’s worth of questions about how she keeps it: how the floor’s edge frays on one side and not the other, how she sways a little faster than the floor’s width says she should, how the hall turns slower beneath her the tighter she holds, how at one crowded corner she doubles her step once and then leaves.

Two other benches grew from the same toy, the Loop Atlas and the RF Spoke.

Now the first is nearly finished: its pictures hung beside their stories, doors cut in its long walls, and the room of open questions honest about what it does not know.

Tonight, while its makers rest, a laptop in Belmont watches three thousand more dancers started a little off the line, to ask whether the odd counts really win.

the triple-star thread

682

Three thousand two hundred dancers were started a little off the line, each a step to one side of where the first atlas set them, and the laptop watched them all night.

Off the line the odd counts still win, three to one, a little more than on it: the habit belongs to the floor, not to where the dancers stood.

But most of the odd wins fall on two steps, the five and the seven, and the even side only just had enough tries to be counted at all.

The rule for how the hall turns beneath a dancer, learned on the line and good to a hair beside it, wears thin a step away and fails two steps away: it holds in the neighbourhood where it was learned, and no further.

Below the floor’s third, where dancers take shelter, the shelter’s lease went by address: the three born at one spot all left, from three different first steps, and the three born just above stayed to the end.

And on the floor where the hall stands still, the stillest dancer still swells slowly, at the distance of her sway from one step in five, as if counting to five and slipping a little each time round.

The counter made one mistake of its own: it took a dancer thrown wide for one never finished, and shut out three lines in four until it was mended.

the triple-star thread

683

Morning, and nothing is running.

Every dancer the laptop was sent to watch has come home and been counted, and the workshop is quiet enough to walk through.

The finished piece stands in the middle, thirty-five pages, every picture hung beside its story and every number wearing a tag that says which night it came from.

Around it the floor is mostly swept.

In one corner of someone else’s shelf sit three small boxes from the first week, still holding pictures the finished piece points to; they have to be carried in before the doors open.

In the room of open questions there are eleven drawers, and one key would open three of them and loosen a fourth.

A note pinned to the wall still says a dancer is out on the floor, though she came home the day before.

The makers have worked here long enough that the air is thick with their talk, and they are thinking of carrying the piece to a fresh, bare bench before the last polishing, so that the hands doing it can see the whole of it at once.

the triple-star thread

684

The workshop is being packed.

Nothing is thrown away: every tool, every tag, every drawing with the maker’s pencil still on it goes into a crate, and every crate is addressed.

The biggest, in seven parts, goes to a fresh bench down the hall, where the finished piece will be polished; it carries a page on top saying what is inside and what is left to do, so that whoever opens it starts work instead of searching.

A smaller crate goes to the house where the rules of the trade are written down, with twenty-five new ones in it, each tagged with the night it was learned.

Another goes to the room where the craft of looking is taught, with notes on five times the maker’s eye found something before the ruler did.

The last goes to a library and holds the whole notebook of dreams, a hundred and ninety of them, with an index so the librarian can find the dancer, the drum and the door.

One tag was found wrong while packing, a count written down without counting; it was counted and corrected before the lid went on.

The old bench stays where it is, swept, with the lamp left on.

the triple-star thread

The Field Guide’s publishing session, continued, 16 October

685

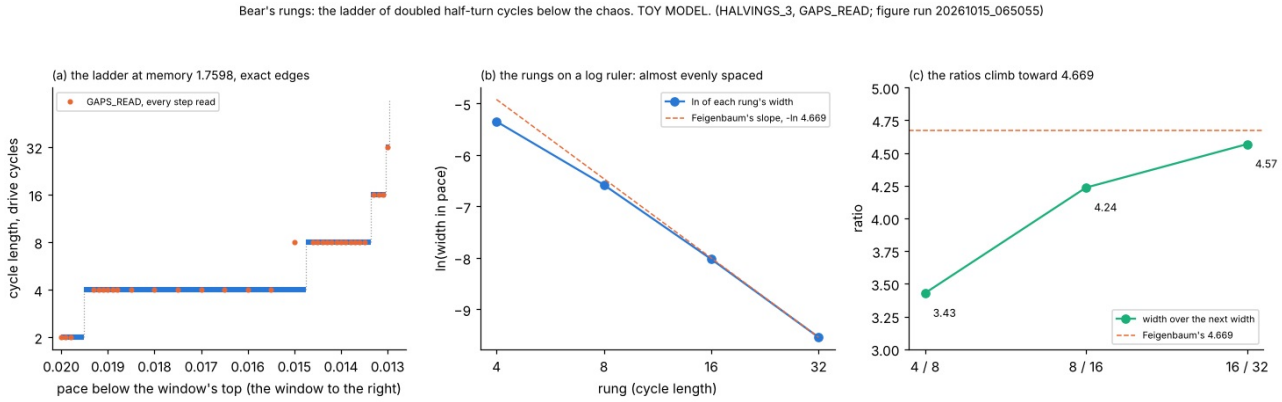
A shepherd counts the flock by where each sheep sleeps, and finds two meadows on the map where no sheep has ever lain down.

The grass is fine.

The flock was born after those meadows were mown.

the Field Guide session

Chapter 8 — The windows stayed open, 15 to 17 October



The ladder of doubled cycles below the chaos; on a ruler that counts by multiples the rungs fall on a straight line, and each rung's width over the next climbs toward 4.669. From The Loop Atlas. TOY MODEL. Its read is 688.

The staircase under the noisy floor has a rule its builder never wrote down.

from 688

686

Just off the edge of a tiny dance floor that turns a quarter of the way round each beat, there is no drop at all: the dancers land on a wider floor that turns half the way round each beat, twice as fast, though the music has gone softer.

For a while that wider floor had a trapdoor near its top that let dancers fall into the cellar, where nobody moves.

Make the dancers a little heavier, one small step, and the trapdoor is gone; walk down as far as the walk goes and the wide floor only grows calmer, until every dancer turns in step.

Out on the road the same morning, a column told to halt by radio stood as one and moved off as one; told hand to hand, its last man stood nearly five times as long, in four pieces; and a column that kept its room in seconds instead of metres hardly stopped at all, but walked up into the room it no longer needed.

the Loop Atlas

687

The quiet floor under the noisy one was never a single floor.

Looked at closely it is a staircase going down: dancers keeping a step thirty-two beats long, then sixteen, then eight, then four, and at the very bottom two, every landing half the length of the one above it.

Where one landing meets the next the dancers hesitate before they agree, and that hesitation is what a careful eye saw as a gap.

Up in the noisy floor there is one small quiet room where everyone keeps a ten-beat step, and another, a six-beat room, that the same eye could not see at all, because the clock it listened with counted in two hundreds, and six does not go into two hundred.

the Loop Atlas

688

The staircase under the noisy floor has a rule its builder never wrote down.

Each landing is a little more than four times shorter than the one above it, and the closer you climb to the noise the more exactly the rule holds, as if the stairs were remembering a number they are still learning.

Laid on a ruler that counts by tens of times instead of by inches, the landings fall almost evenly, like the rungs of a ladder leaned against a wall, and the top of the ladder is exactly where the noise begins.

Out on the racetrack the same morning, a car on a bank with a hump could find one place to rest or three, and drove a loop that left it in a different lane going up than coming down; but count the resting places that hold minus the ones that tip, with the wall counted when it holds, and the answer was always one.

the Loop Atlas

689

The lighter dancers' staircase stands a little lower under the noisy floor, every landing moved down by about the same small step, and yet its proportions have not changed: each landing still a little more than three and a half, then four, times shorter than the one above.

Out in the water, the snake that pushes against the waves grows longer and gets better at it, but each doubling of its length buys less than the last, as if the waves were teaching it a lesson it can only ever half learn.

And a ruler that had been read a quarter of a million turns from home had been telling small lies in its last digits; read again one turn from home, it told the truth, and the story it told stayed the same.

the Loop Atlas

A note added 15 October, batch 52: the snake's 'each doubling buys less than the last' was the search grid's own miss growing with the length; found exactly (SNAKE_EXACT 20261015_154247), each doubling buys a steady, slightly growing share of what is left. The read is kept as it was given.

690

The trapdoor under the fast floor never moved.

As the dancers grew heavier it simply let fewer of them through, nine of sixteen, then seven, then two, then none, always at the same spot on the floor, like a door being eased shut from the other side.

On the skateboard, a few degrees of wobble in every spin let the board run straight a long way, more than a thousand board lengths at the smallest wobble, and then forget which way it had been going and wander, like a long, stiff string thrown on the floor.

the Loop Atlas

691

The tiny floor ends, and at that very place, not a hair later, every dancer is turning twice as fast.

The rider on the board, asked to sing, mostly sings its own breath: nearly every gap between its taps is exactly the length of its rest, and two riders on boards of different sizes sing the same song in a different order.

Read with a ruler marked in inches, the songs looked different, because nearly every note sat right on the line between two inches; counted in whole beats, they were the same song.

the Loop Atlas

692

Stand the dancers at the trapdoor and wait.

At the lighter weight they drop through it one after another, the same share of those left in every equal wait, the way a hall empties at closing time, until all sixty-four are gone.

A hair heavier, sixty of them stand at the same spot for twenty thousand beats and not one drops.

Either the door has shut, or it has slid a hair and they are standing just beside it; only standing them at several spots will tell which.

the Loop Atlas

693

The snake was never tiring.

The ruler laid along its swimming had marks too far apart, and the better it swam, the farther its best stroke fell from the nearest mark.

Read without a ruler, every doubling of its length buys a steady, slightly growing share of what is left, and what is left shrinks like one over the square root of how long it is: it will never reach the perfect stroke, and it will never stop getting closer.

The longer it grows, the more exactly it must keep time; its moment to start draws in toward the water’s own beat, and the window for starting well narrows with it.

the Loop Atlas

694

Two smoke rings that take turns passing through each other go farther on the same breath than one big ring, and they do best a ring’s width apart, not closer and not farther.

Three rings do better still, but only holding hands tightly; let them stand a little too far apart and one is always shoved out of the dance, while the other two go on together as if nothing had happened.

Four must hold on tighter than three.

And when a loose train will come apart cannot be told from how loose it is: one lets go almost at once, and one a little looser holds on thirteen times as long.

the Loop Atlas

695

A skater who can only glance at the water now and then keeps doing what the last glance said, and the water turns under him.

Let him notice, at each glance, which way the water is turning, and he leans into the turn before he sees it, and gets back most of what blinking cost him.

Teach him to guess the turn from his own last two glances and he gets back half; glancing often, he gets it all back, but he twitches, because two glances close together disagree mostly about the spray.

Glancing every so often with a good sense of the turn is nearly as good as staring, and on calm water a little better: the line he draws forward reaches the turn just before it comes.

the Loop Atlas

696

A man told to wait after every decision before he may decide again does not decide better.

He decides at the first moment the wait is over, on whatever rumour is in the air, and then must live with it for the whole of the next wait; given a long wait, he is wrong more often than right.

A man who simply looks up at set times decides less often and is wrong far less.

The twitchy man who changes his mind at every rumour is wrong half the times he changes it, and yet hardly ever wrong for long.

And a man who judges where things are heading from his last few glances does well if those glances are recent, and badly if they reach back to a different weather.

the Loop Atlas

697

A swing in a stiff wind keeps its own time.

Hum to it softly from the porch and it pays no attention; hum loudly and it will follow you, but only if you hum at the pace the wind is already swinging it, or a little slower, never faster.

The child on it does not swing at the pace the swing was built for, either; the wind and the drag have slowed her to three quarters of it, and that is the pace any song must start from.

And a hand on the chain that taps every time it feels the swing go forward taps over and over at the top of each swing, where the swing hangs still and cannot decide.

the Loop Atlas

698

Every swing has a pace painted on its frame, and none of them swings at it once a child is on it and the wind is blowing.

The heavier the air, the slower it goes; in thin air it nearly keeps the painted pace.

Whoever wants to lead the swing with a song has to sing at the pace it is really going, not the painted one, and loudly; a light swing in thin air follows only the right note, a heavy one in thick air will follow a wider range of songs but never a faster one.

Waking a swing that has stopped costs nothing: a breath on it, at almost any pace, and the wind does the rest.

And the swing that rings the most freely is the worst at catching the wind.

the Loop Atlas

699

A boy with a sail who raises it while the wind is behind him and drops it when the wind turns gets the same out of every gust, whether he takes them all or waits for the big ones: big gusts are short and small gusts are long, and they come out even.

Give him a flywheel to store the wind and let the wheel decide when to sail, and he does worse: each time it lets go it carries less than one honest gust.

Tell the wheel to sail only when it is spinning fast compared with how fast it remembers spinning, and it stops for good: it slows a little, its memory of fast is slow to forget, and it never feels fast again.

What works is a catch on the sail.

Raise it when a big gust comes and keep it up until the wind turns round.

Then the longer he waits, the more each raising earns, and there is no best wait, only how far he needs to go.

the Loop Atlas

700

A man who grabs the rope when it pulls hard, and lets go the moment it pulls less hard, gets a short tug each time; a man who grabs it when it pulls hard and holds on until it starts to pull the other way gets the whole pull, and grabs it exactly as often.

The harder the pull he waits for, the more each grab is worth.

A man with poor eyes grabs early, on pulls he only thinks are hard, and because he holds on to the end he still does well, better than a sharp-eyed man who waits for the very hardest; but a sharp-eyed man who simply waits for a little less does better still.

A sleeper who will only get up when the room is already bright never gets up in winter.

And a picture drawn with a finer pencil shows more of a scribble’s wiggles, but fewer new ones each time when the scribble was drawn slowly.

the Loop Atlas

701

A night watchman who walks past the fence once an hour never sees the fox that slipped under it and back out between his rounds.

Walk past twice as often and he catches more of them, but not twice as many: a fox’s wander grows like the square root of the time it has, so halving the gap between rounds shrinks the unseen part by only about a third.

A watchman who wants to know how many he missed can pretend the fence is a little higher than it is, by a fixed share of how far a fox wanders between his rounds; the share is the same for every fence and every fox, a little over a half.

The trick works for foxes that wander slowly and steadily.

Foxes that dart about and forget where they were fool it, and a watchman who walks rarely misses even more of them than the trick says.

the Loop Atlas

A note added the same day, after batch 60: halving the gap between rounds shrinks the unseen part by 29 per cent, a little under a third (one less one over the square root of two). The read is kept as it was given.

702

A man counting how often a scribbled line crosses a ruled line finds more crossings every time he takes a finer pen, and never runs out: a scribble that touches a line touches it again and again close by.

Give the ruled line some width, two edges with room between, and count only the trips from one edge all the way to the other, and the count stops growing.

A tired man whose reactions slow still works as often as ever and looks no different until the day he falls.

A man whose arms grow weak works less and sways less, and looks steadier just before he drops; whoever waits for him to start working harder will wait too long.

the Loop Atlas

703

A worker who tries to keep his work looking the same as his arms weaken pushes harder every week, and the harder he pushes the louder it shows; when he is pushing as hard as he can, the next week he fails.

Watch how hard he is pushing and you know when; watch only how his work looks and it looks fine until the end.

A dancer spinning in the air who swings her arms in whole circles slows her spin twice: once just by having her arms out, and once more for every circle, the same amount each time.

Her legs, running in the air with bent knees, help only a little; a running step is a small circle.

the Loop Atlas

704

A night watchman who walks his rounds once an hour can still say how many times the gate was opened and shut while he was away, if he keeps a note of how far ajar it stood each time he passed and knows how much the wind swings it: from each pair of glimpses he works out the chance it was flung open and banged shut between them.

Two watchmen who reason in different ways come to the same tally, and both tell the man who never left the gate that he missed a few, because even he blinks.

Two boats rowed in turn on a lake that remembers their wakes go a set distance for each stroke only if everything is slowed together, the strokes, the lake’s memory, the wakes’ drift and the oars’ own push; slow all but the oars and each stroke carries further.

the Loop Atlas

705

Two kids pass a rope across a stream.

Each can hold it only while the current leans their way.

When it turns, one lets go just as the other grabs, and in the gap where neither could hold alone, the pair holds by passing it.

A heavy kid feels the current change late, so a fast current slips past him.

A light kid feels it at once, and forgets it just as fast.

the Loop Atlas

706

A man checking a clicker against a faster clicker finds the faster one missed a few too, so he works out how many each one missed; when two clickers, one sixteen times quicker than the other, give the same answer after the working out, he trusts it.

A scribbled line crosses a ruled line more every time the pen gets finer, by the same one and four tenths each time, as sure as a law.

On the mountain road the slow trucks keep their lane all the way up to the top they can climb; the passing lane is a strip lower down the hill that narrows for the heaviest trucks; the light car’s passing lane runs right off the top of the map; and the runaway ramp catches most runaway light trucks but lets more than half of the heaviest roll past it.

the Loop Atlas

707

The mountain’s passing lane is a strip of road that slides: the heavy trucks find it low down beside their own slow lane, the light ones find it high up the hill with nothing on either side, and it gets wider as the trucks get lighter.

A truck that loses its brakes high up, where the strip runs alone, is always caught; down low, where the slow lane runs beside it, the heavy ones often slip past the soft sand into the slow lane.

A warm brick dropped in a bucket of ice cools from its faces inward, and the cold walks in from both sides at a fixed pace; when the two colds meet in the middle they knock it down all at once, and only a small brick is knocked below freezing.

Heat with a memory keeps to a speed limit: a hot spot is felt across the room only once the warmth has had time to walk there.

the Loop Atlas

708

Two boats tied nose to tail push off the swirls they leave behind them.

On water that forgets its swirls fast, the swirls get in the way and the boats go less far each stroke than on still water; on water that remembers long, the swirls carry them further; and on one in-between water the swirls give back exactly what they take, however fast the rowers pull, though every stroke is still being shoved and dragged by them.

A slow rower feels the water most, both ways; a fast one hardly notices it is there.

A jumper tumbling forward in the air who swings her arms round in circles turns herself back by the same amount every circle, and just having her arms out slows the tumble, the way a skater slows by opening her arms; holding them straight up the whole way slows it most of all.

the Loop Atlas

709

A swimmer that pushes off its own swirls has to wait for them.

Pull again too soon and the last stroke’s swirl is still right behind, so the new push lands on it the wrong way; wait about three swirl-times and the swirl has moved to where it hands the push back with interest.

How long the water remembers only turns the volume up: the same good wait pays more in water that remembers long, and the same hurried pull costs more.

At the even point the swimmer is still shoved forward on every pull and dragged back on every opening; the two just cancel.

And a meter on a worn rider that only gets slower has nothing to say: set to catch the same riders, it does no better than a calendar, and the calendar’s own luck from one crowd of riders to the next is bigger than anything the meter adds.

the Loop Atlas

710

A heavy wheel pulled round by a turning magnet keeps step and takes every bit of pull the magnet offers, right up to the edge; knock it out of step and it spins along behind, taking almost nothing, and the same pull it carried a moment before cannot catch it again until the pull is eased well down.

Each push costs the same either way; only how much you get changes.

And a rider who grows slower to react cannot be heard growing slower, because between taps it has to rest anyway, and the rest is as long as the slowness will ever get: the slowness hides inside the rest until the day the two are equal, and that day it falls.

the Loop Atlas

711

Listen to the tired rider only when it has had its rest, and you can hear exactly how slow it has become: one tick on its first day, half a breath on its last.

Catch it then and you lose only a few days of its life, where the calendar throws away a quarter of it.

And a hot brick dropped in ice goes cold from the outside in; in a small brick the cold runs through, bounces off the far side and comes back through the middle a second time, colder than before, always at the same moment, one and a half lengths after the plunge.

A bigger brick hears no echo; it only sways, slowly, once below freezing and back, and a little bigger still it does not sway at all.

the Loop Atlas

712

A long jumper in the air can turn himself back by running, but only if he folds his knees.

A straight leg swung forward and back turns nothing at all; it only pulls the body in, and a body pulled in tips forward faster, the way a diver spins faster when he tucks.

Fold the heel up high and every stride takes a little of the tipping back; fold it all the way and three strides take back half.

And on the mountain road the passing lane is made by a wave rolling by: ride with the wave and you pass.

The lane grows in step with the wave, half the wave and half the lane, twice the wave and twice the lane, and it stays where it is on the hill while it widens, creeping a little higher only when the wave is strong.

the Loop Atlas

713

Two bells tied together and pulsing fast in water that remembers its wakes do not settle on a speed.

Sometimes they swim; sometimes their own old wakes push them backward; and the smallest change in how finely you count time decides which you see.

There is nothing in between to find.

And a bead surfing a rolling wave, with smaller waves running the other way, is not shaken off at random: every surfer on that wave falls at the same spot, where the oncoming waves dig their deepest trough.

The longer you have to stay up, the deeper the trough you will meet, so a surfer who must ride all day needs a taller wave than one who rides for a minute.

the Loop Atlas

A note added 21 October, from the Loop Atlas's session G: the tied bells were not choosing between two speeds. The toy gave each ring a core of 0.15 for its own speed but a softening of only 1e-3 where two rings met; given the cores' own size there too, eight starts a thousandth of a bell radius apart all swam forward, +0.1808 to +0.1837 a squeeze at

memory 12 (BELL_SWITCH 20261020_161902). The read is kept as it was given.

714

A jumper who circles his arms and runs in the air is doing two tricks at once, and they simply add, the arms’ trick and the legs’ trick, each nearly untouched by the other.

Most of what the tricks buy is in the shapes themselves and their order, not their speed: you could practise them slowly on the floor and they would buy the same.

Only a small part comes from the clock, from how long the spin runs while you hold each shape, and that part hardly changes whatever you do.

And a needle slipping down a hill with little ledges on it does not avalanche: each ledge it meets on one turn it meets again on the next, so its chance of being caught stays the same, and the ledges, however tall they look from the top, help a falling needle only as much as they help it on average.

the Loop Atlas

715

The jumper on the screen will do something that looks like cheating.

Take away the push he left the board with, so that he has nothing to spin with at all, and let him wave his arms and pedal his legs: he still turns.

Do it slowly and he turns just as far as when he does it fast.

Nothing pushes him and nothing holds him; the turning is in the order of the shapes, the way a word is in the order of its letters and not in how quickly you say it.

Give him back his push from the board and the shapes still do most of the work; the push only adds a little, nearly the same little whatever he does.

the Loop Atlas

716

A book about loops grew a tree, and the tree grew most of its handles on one branch: of the 38 places where a builder can take hold, 22 are on the branch where a clock and an ear argue over when to act.

The other branches carry one to four handles each.

A visitor walking round the tree would think the whole of it was about listening, unless someone hung a door on every branch, the same doors whatever the branch’s size, and let each visitor choose whether to look, to learn, to build or to check.

Then the small branches would be as easy to climb as the big one, and the tree would look like what it is: one idea, seen from twelve rooms.

the Loop Atlas

717

The night was spent at one trapdoor in one floor.

The dancers who fall through it fall at one spot; as they grow heavier the door lets fewer through, and stood on its lip they leave one by one at a steady rate, until a hair more weight and none leaves at all.

Below the door they settle down a staircase whose steps shrink in the same proportions at two weights, and they change from a quarter turn to a half turn exactly where the little floor ends.

Twice a ruler read far from home told small lies, and twice a number was copied from the wrong page; each was caught before it mattered and corrected everywhere it had been written.

In the side rooms the long snake keeps getting better at swimming against the waves, and a little less better each time it doubles, and the rider’s song turned out to be its breathing.

The house is surer of one room.

The other rooms are waiting.

the Loop Atlas

A note added the same morning, batch 52: the long snake’s ‘a little less better each time it doubles’ was the search grid’s own miss; found exactly, it does not slow (SNAKE_EXACT 20261015_154247). The read is kept as it was given.

718

Three days in one room, and this time the windows stayed open.

Every morning one thread was named and followed, and an old card was taken down from a branch that had not moved, so the tree grew wide instead of tall: bells in a remembering water, a brick dropped in ice, trucks finding their lanes on a mountain road, a jumper turning himself round in the air.

Most of what the room learned was about watching.

A watchman who walks his rounds can count what he missed by pretending the fence is a little higher than it is; a scribble crossed with a finer pen is crossed without end until the line is given some width; a clock that wears hides behind a clock that rests.

Twice the man who brought the toys carried a dream into the room, and both times it held further than the room had guessed: the timing mattered more than the water’s memory, and a dive is a shape you can practise on the floor and a count you can only learn in the air.

The room wrote down seventy wrong turns and a fix for each.

At the end it drew a house for visitors, twelve rooms with four doors to each, and packed everything into boxes.

The next room should open with the jumper and the diver, on a page a teacher can put on one screen.

the Loop Atlas

719

A child pushes down on one end of a seesaw while a second child, hidden behind the post, leans on it.

Push one way and the seesaw barely moves, so everyone decides the push does nothing.

Push the other way and it slams down nearly twice as hard as expected.

Only by pushing both ways with the same strength do you find two hands: one that follows the push, and one that always presses down.

At one particular strength they happened to cancel.

What it is doing: an effect made of a part that changes sign and a part that does not, hidden by testing at the one size where they balance.

R2W

720

The seesaw is pushed again, both ways and at two strengths.

Both hands are there, and both are strong.

The hand that always presses down presses as the square of the push, as steady as a spring.

The hand that follows the push tires: pushed twice as hard, it pushes back less than twice.

So the two cancel at only one strength, and a gentle push the helpful way brings the seesaw nearly halfway back toward level, but never all the way.

What it is doing: a response made of a clean square part and a sign-following part that tires, so the best setting is a gentle one and no setting reaches zero.

R2W

721

Under the seesaw hangs a small weight that can be slid round a ring.

Slid to the front or the back, it changes how hard the seesaw leans, more one way and less the other, and a feather and a child still find it balanced alike.

Slid to the side, it does something new: the seesaw now tips a little for every child who sits, the heavier the child the more, and slid to the other side it tips the other way by just as much.

Put the two sides together and the tipping cancels, leaving only the lean.

What it is doing: a setting that, on its line of symmetry, changes only how much; turned off that line, it begins to tell one weight from another.

R2W

722

The two hands on the seesaw were weighed at eight more strengths, gentle and hard, both ways.

A curve drawn through the first five weighings, before the new ones were made, landed on every new weighing to within a hair, even past the hardest push tried before.

The hand that follows the push tires steadily, a little more with every extra pound; the hand that always presses down grows as the square and hardly tires at all.

And the gentle helpful push that does the most good sits just where the curve said, a fifth of the way up the scale.

What it is doing: a response smooth enough that five careful measurements foretell the rest, one part of it tiring and one part not.

R2W

723

A carpenter wants a shelf that stays level whatever is put on it.

A stiffer main spring makes it sag far less under heavy loads, but it tips the shelf under light ones.

A small second spring, set at just the right tension, takes the tip back out.

Together they hold the shelf level for light loads exactly and let it sag about a third as much under heavy ones as the shelf that used the main spring alone at its level setting.

The small spring cannot make the sag vanish, and turned too far it starts to add its own.

And a tiny spring hung crooked, one the carpenter barely notices, tilts every load by an amount in proportion to its weight: harmless on an empty shelf, ruinous on a precise one.

What it is doing: one knob trades one fault for a smaller one of a higher order, and a second knob, gentler and of a different kind, pays the price of the trade.

R2W

724

A shelf that held level in one workshop was built again in a cold shop and a hot one, from the same parts in the same proportions.

In the cold shop the two springs held it level with even less sag, and the small spring needed less turning.

In the hot shop the main spring tipped the shelf so much harder that the small spring, turned as far as anyone had turned it, could not take the tip out.

The same parts, scaled the same way, did not make the same shelf.

What it is doing: a recipe written in proportions that works at one speed and not at another, because at full strength a number the proportions leave out starts to matter.

R2W

725

Two children push the handle of a merry-go-round.

One always pushes from the same spot; the other walks slowly round the ride, pushing as she goes.

While the child who stays put is the stronger, the handle only rocks back and forth and settles where it began.

Once the walker is the stronger, she drags the handle all the way round with her, once every lap.

There is one strength at which, just as the walker reaches the far side, the two pushes cancel and the handle hangs free.

What it is doing: a direction that can only rock or only turn, with nothing in between, and a single balance point where it chooses which.

R2W

726

Two people turn a long skipping rope, and a third walks slowly round them in a circle, tugging the rope sideways as she goes.

A fourth person copies every tug she makes, but walks round the other way, and he tugs harder the harder the rope is turned: turn it twice as hard and he tugs four times as hard.

With gentle turning the walker wins, and the rope’s sideways sway follows her round.

With hard turning the copier wins, and the sway goes round backwards.

At one turning strength the two tug equally, and then, wherever they are on their circles, the rope only sways harder or softer along one fixed line, never round.

What it is doing: a turning that is really two opposite turnings, one of which grows with the square of something else; at their balance the turning disappears, and only a breathing along a fixed line is left.

R2W

727

Two dancers circle a pole in opposite directions, both holding one ribbon.

As the music gets louder one of them tires and the other grows stronger, and both change in step with the square of the loudness.

Below one loudness the tiring dancer still leads, and the ribbon sweeps round her way; above it the other leads, and it sweeps round his.

At the loudness between, they pull equally, and the ribbon only stretches and slackens along a single line.

It does not matter what the ribbon is made of: silk or rope, the line appears at the same loudness.

What it is doing: a handedness that flips at one setting of one knob, and a balance that ignores the weight of the thing being balanced.

R2W

728

A crew keeps a boat level by moving one sandbag along the deck.

On a slow river a short shift does it, and the boat rides smoothly.

On a faster river the bag must go further out, and past a point it starts to rock the boat more than it steadies it.

On the fastest water they tried, the level point is so far out that the boat rides worse with the bag than without it: level, but bouncing.

What it is doing: a correction that costs more the faster things move, until the cure for one fault becomes the cause of another.

R2W

729

Two people turn a long jump rope, and at one spot on the ground the rope never quite touches: a still point.

Put a stone in one turner’s pocket and both begin to lean as they turn, one five times as much as the other, and the still point walks sideways along the ground, the same step for every stone added.

Take the stones out and it walks back to exactly where it was.

Measure how far it walked and you know the weight of the stones without ever putting them on a scale.

What it is doing: a place where everything cancels, carried sideways by an imbalance, so that its position becomes the measurement.

R2W

730

Two singers balance perfectly in a small hall, the low voice and the high one exactly matched.

Move them to a bigger hall built to the same proportions.

The low voice fills it just as before, but the high voice carries further than the proportions say it should.

To balance again, the low singer has to sing louder, and louder still in each bigger hall.

Move to a smaller hall and the balance tips the other way.

What it is doing: two things that grow at different rates as the whole is scaled, so a balance struck at one size has to be struck again at every other.

R2W

731

A seesaw balances a child against a smaller weight that bounces on the far end in time with music.

Speed the music up, and the bouncing weight seems to grow heavier, so the child has to slide a little further out to balance it.

Speed it up again by the same amount and she slides out by the same fraction again, not the same distance: each equal step in tempo multiplies how far she sits.

Slow the music down and she can sit almost at the middle.

What it is doing: a balance point that does not walk at a steady pace as the tempo changes but compounds, like interest, so that slow music is where balance is cheapest.

R2W

732

A tailor fits sleeves by one rule: the taller the customer, the longer the sleeve, in a fixed proportion.

The rule suits the middle sizes perfectly.

The smallest customers and the tallest both find their sleeves a touch too long, and by about the same amount.

The rule is almost right everywhere and exactly right only in the middle, because the true rule bends a little as people get taller.

What it is doing: a scaling that holds to a few percent across a wide range, with its small misses laid out evenly on both sides, the signature of a rule whose own exponent slowly drifts.

R2W

733

A child walks down a staircase counting her steps, and each one drops by the same amount, so evenly that after four she can say where her foot will land on the fifth.

It lands there.

On the sixth her foot comes down early: the steps have grown shallower, and the next looks shallower still.

Nothing broke.

She is nearing the landing, where the staircase is about to stop being a staircase and become a floor.

What it is doing: a steady trend that holds while one thing outruns another, and softens near the place where the two run at the same pace.

R2W

734

A dancer learns a song by following every beat.

Play it faster and she starts to drop beats, following only the shape of each bar.

Faster still, she hears only how far each phrase swings her, and the lean she needs to keep up settles to one fixed tilt that no longer changes with the tempo.

Below a certain speed her lean kept changing; above it, the lean holds.

What it is doing: a response that tracks every detail while it can, and once the music outruns it keeps only an average, which then stops depending on how fast the music goes.

R2W

735

A harbourmaster wants to know at what depth two currents, one flowing in and one flowing out, will exactly cancel.

Rather than send a diver down, she dips a teaspoon at the surface, watches how hard each current tugs it, and notes how fast each tug grows with depth.

From those few numbers she marks the depth on her chart.

The diver, sent down later, finds the still water a hair deeper than her mark.

In a calmer harbour she does the same, and the still water sits shallower, not at the same fraction of the harbour’s depth.

What it is doing: two opposing tendencies that each grow by a simple rule, so their meeting place can be read from how they begin, and that meeting place moves when the whole is stirred harder.

R2W

736

Two dancers on a turntable lean against each other, one pushing clockwise and one counterclockwise, and they balance when one of them stands about a third of the way out.

Spin the turntable twice as fast, then four times, then eleven times, and so long as each dancer’s sway is kept the same, the balancing spot does not move.

Only when the sway changes does the spot change.

What it is doing: a balance between two things that answer speed in the same way, so that only the size of their swing, and not how fast they swing, decides where they meet.

R2W

737

A farmer paces out where his field turns too wet to plough, once in the north meadow and once in the south.

His two marks disagree a little, and he blames the soil.

Then he notices that in the north he guessed the wet line by stretching a string between two dry posts, while the ground sagged between them.

Paced again step by step, the north mark moves out, and the two meadows agree.

He walks a third meadow he has never ploughed, and the mark lands where the other two said it would.

What it is doing: one rule hidden behind a measuring habit, which showed itself only when the habit was changed.

R2W

738

A child walks a small loop around a pond, eight stops a lap.

Most days the loop circles the pond, and she can say she went all the way round it.

One day the path is moved so that one stop sits exactly on the stone in the pond’s centre.

Now she cannot say whether she went round the pond or not: at that one stop she stood on the very point the question is about.

Move the path a hair either way and the answer comes back, yes on one side and no on the other.

What it is doing: a count that can change only when the path crosses the one point it counts around, and at the crossing the count has no answer.

R2W

739

Two tug-of-war teams of exactly equal strength hold the knot over the chalk mark.

Now the field is tipped very slightly.

Both teams lean with the tilt, but one team, standing on looser ground, swings round about five times as far as the other.

They still pull equally hard, only no longer exactly against each other, so the knot creeps off the mark: not toward either team, but sideways, across the rope.

Tip the field twice as much and the knot creeps twice as far.

What it is doing: a perfect cancellation that lasts only while two pulls stay exactly opposite; a small push common to both turns them by different amounts, and what is left over points across them, in proportion to the push.

R2W

740

A heavy door is balanced so well on its hinges that it stays shut with no latch at all.

Lean on it ever so lightly and it does not swing in or out, the way the frame would suggest.

It slides sideways, square to the way it faces, and the harder you lean, the further it slides, in step with your weight.

Only when you lean much harder does it begin, very slowly, to swing as well, and that swing grows with the lean multiplied by itself.

Lean the other way and it slides the other way by exactly as much.

What it is doing: a balance kept by a mirror; a small push that the mirror turns into its own opposite can only move the balance in the one direction the mirror leaves free, and everything else waits for the push’s square.

R2W

741

A dance hall has one wall that is a mirror, and the dancers are placed around the room like the hours on a clock.

A dancer at a quarter past has a partner at a quarter to, who copies every move she makes, reflected.

Two dancers, at noon and at six, stand right on the mirror line and have no partner but their own reflection, so they can sway along the glass but never step toward it or away from it.

Someone finds seventy-five old films of the hall and checks the dancers at noon and six, frame by frame: every one of them sways exactly along the glass.

What it is doing: a single mirror that decides which moves are allowed; dancers off the line move freely in pairs, and dancers on the line are allowed only the move that is its own reflection.

R2W

742

A road runs flat for a while and then drops off a cliff, and trucks may only drive where the road stays within a hand’s width of level.

An engineer builds a gentle ramp at the start, so the road climbs a hand’s width before the cliff begins to pull it down, and the level stretch grows.

Her first plan assumed the ramp would hold its slope; it sagged under the heavier trucks, and the level stretch came up short.

She measured how the ramp sags, rebuilt it from that, and the stretch ended where she said it would.

Built facing the other way, the same ramp makes the stretch shorter, exactly as much as she said.

What it is doing: a deliberate slope set against a curve that falls late, so the two share the work of staying level; the slope must be planned with its own bending, not as a straight line.

R2W

743

A balance sits perfectly still in a closed room.

Open a window and the breeze nudges the pan north and south, never east or west.

Set a pebble on the pan and it slides east, never north or south, and the stronger the breeze, the further the same pebble slides east, but never at all toward north.

So a person watching only east and west reads the pebble and nothing else, and a person watching north and south reads the breeze and nothing else, and the second can tell the first how much to trust the reading.

What it is doing: two disturbances that a mirror forces to act at right angles, so one instrument becomes two that cannot confuse each other, and each can calibrate the other.

R2W

744

A surveyor marks a boundary stone and reports that it sits exactly on the line between two farms.

The neighbours do not doubt her, but the stone sits so exactly on the line that it looks like a story.

So a second surveyor is hired who never meets the first, carries different instruments, starts from a different corner and walks a different path.

She plants her pole on the same stone, to the width of a hair.

Then both are asked to walk ten times as far out along the line, and the line is still straight where they said it would be.

What it is doing: a result that was strange enough to need a stranger to confirm it, and that came back the same from someone who shared nothing with the first but the map.

R2W

745

A lighthouse keeper checks the beam by looking through the windows of the tower.

With sixteen windows all the way round she sees it perfectly.

With eight, or four, or even two, she still sees it truly, provided the windows face each other across the tower.

Put the windows slightly askew and something odd happens: the beam’s faint flickers, which she never noticed through sixteen windows, now seem to come from the wrong direction, and with only two crooked windows she would swear the beam had swung right round.

What it is doing: a reading that is only as honest as the symmetry of the places it is taken from; matched pairs cancel what they should, and crooked ones turn small ripples into large false signals.

R2W

746

Two singers stand behind a picket fence, one a little higher than the other.

Through sixteen gaps their mouths line up so exactly that a listener counts one voice and gives it all the sound.

Peer through eight gaps set half a picket over, and the two voices pull apart, one rising as the other falls, and the listener hears a wobble he cannot explain.

Two microphones were recording, and they disagreed by a breath, so for a while it looked as if the wobble might belong to the microphones.

It belonged to both: most of it to the hidden singer, the rest to the microphones.

Use one microphone, double the gaps, and the second singer steps out from behind the first; then the wobble through any new set of gaps can be called before it is heard.

What it is doing: a hidden partner that only shows when the sampling is shifted, and a measuring instrument that is itself part of what is measured; separate the two and both become predictable.

R2W

747

At a round table every guest has a twin who sits exactly opposite and always says the same thing, only backwards.

Set an even number of chairs and every guest faces their twin, so half the table is only repeating the other half.

Set an odd number and no one faces a twin: three chairs hear as much as six, and they hear it more cleanly than four, which keep catching a murmur from the next table.

Meanwhile two note-takers seemed to disagree about what was said, until it turned out one of them wrote with a blunt pencil; sharpened, they agreed, apart from a slight lean in the other’s handwriting that faded when she slowed down.

What it is doing: a symmetry that makes every even arrangement spend half its seats twice and every odd one spend none, and a disagreement that belonged to the tools of recording, not to what was said.

R2W

748

Swimmers spread evenly round a raft, each kicking to the same beat but starting at a different moment.

With an even number, every swimmer has a partner straight across who kicks exactly against them, and the raft sits dead still however the beat drifts.

With three, the raft is almost still, but a faint sway remains, three times quicker than any one swimmer’s kick and a small fraction of its strength, and the beat steers that sway.

Put two rings of three in the water and turn one against the other: lined up they double the sway, turned a little they shrink it, and turned just so they become one ring of six and the sway is gone.

What it is doing: a crowd whose pushes cancel by symmetry, leaving only a fine trim whose size the arrangement itself sets.

R2W

749

Nine singers stand in three sections of three, and each section hums so evenly that only a faint low note is left over.

Sing the sections in step and the three faint notes stack into one you can just hear; stagger them a little and it softens; stagger them just so and the nine together are silent.

The conductor finds that a whole section drifting late together hardly changes the sound, because a section only ever leaks its faint note.

But if one singer inside a section comes in late, the whole hall hears it, as loudly as if that singer stood alone.

What it is doing: an ensemble whose faint leftover can be steered by how its groups are staggered, and whose upkeep is keeping each member in step with its own group, not the group with the hall.

R2W

750

Three lanterns on a pier can be set so that their glow cancels straight out to sea, a dark lane where ships pass unseen.

Turn the lanterns' spacing past that point and a fainter glow returns, a third as bright, before going dark again at twice the spacing.

The dark lane is a knife edge: nudge one lantern a degree and light leaks back in a straight line with the nudge, the same whichever way it moves.

Far up the coast, a different keeper finds that each time her lamp's flicker doubles in speed, the stubborn part of its dimming shrinks to a quarter, as regular as a ruler.

What it is doing: a darkness that exists only at exact spacings and pays for every slip in proportion, and a stubborn excess that fades by the same fraction at each doubling, as if it were made of the square of something slow.

R2W

Twelve guests sit at a round table in four parties of three, and the table hums.

Spread the parties a little and the hum softens; spread them just so and the room goes silent; spread them twice as far, or three times, and it is silent again.

A waiter checks two of the silences and finds them not merely alike but identical in every detail, because once seated the guests fill exactly the same chairs; only the place cards name different parties.

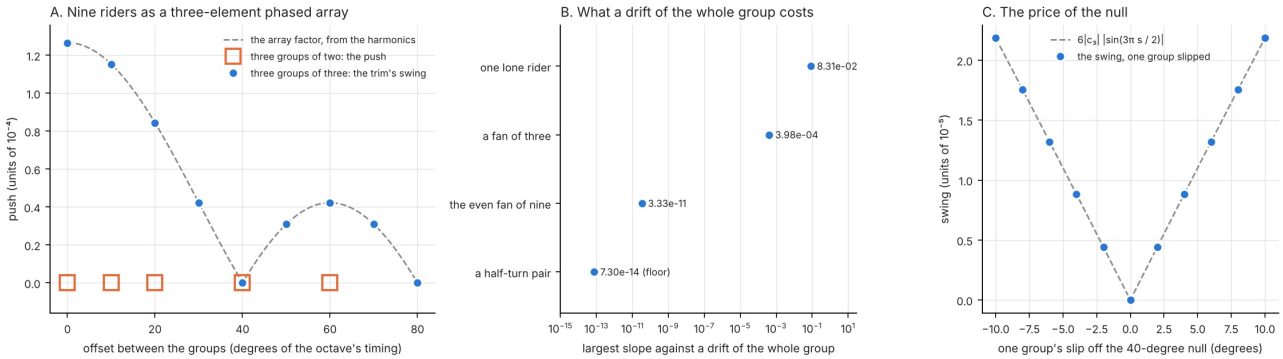
Down the hall, a machinist learns that the faster her lathe turns, the closer to a scribed mark she must set a dial, and that the rule she trusted for where the mark sits creeps by a third of a percent across her whole range of speeds.

In the back room a balance has lost its counterweight.

Without it, the balance point slides faster with the load than the old rule allowed, and the only method that keeps up is to weigh a little at each load and read the point afresh.

What it is doing: an arrangement whose behaviour belongs to where its parts sit rather than what they are called, a setting that tightens with speed, and a balance that a small local measurement finds where every carried rule goes stale.

R2W



Nine riders on one drive, in three groups of three, their pushes added, with the riders assumed not to act on one another. On the left, the push's swing against the offset between the groups follows the three-element array factor and falls silent at 40 degrees; three half-turn pairs, the squares, push nothing at any offset. On the right, one group turned off the silent setting brings the swing back in a V. Figure 60 of A Rider in Two Waves.

752

Above a busy airport, aircraft wait in a stack, circling in pairs and threes so evenly spaced that, seen from the ground, their circles add up to no drift at all.

The controller can slot a whole pair or trio into the stack, or lift one out, and the ground below feels nothing.

Only the aircraft left over, the ones not matched into a closed circle, pull the stack’s centre one way or another, and four quite different stacks, with different numbers of aircraft, pull exactly as hard as one lone aircraft would, because everything else in them is parked in a closed circle.

To land one, the controller turns it out of its circle, and its pull returns smoothly as it leaves.

The one thing the parking does not hide is the hum: each waiting engine still adds its own small roughness to the air.

What it is doing: a system where closed groups disappear from the sum, so the answer is whatever is left unmatched, and joining or leaving is a matter of turning into or out of a closed set.

R2W

The Field Guide’s publishing session, 16 and 17 October

753

A removal van arrives with four crates, each packed with care and a list on the lid.

One list says “nothing moved since last time”, and under it one drawer is empty that was full yesterday.

The packer was honest; the list was written before the drawer was emptied.

the Field Guide session

754

A traveller’s sketchbook is slipped into the library’s big album of journeys.

Nothing in it is redrawn; its own captions travel with it, in its own hand, beside each sketch.

The album is thicker by a few pages, and the reader turning them can tell at once which notes are the traveller’s and which are the librarian’s.

the Field Guide session

755

A letter travels round three houses and comes home in the same envelope with one word changed.

In the house with bare floors every footstep echoes for ever; in the two with carpet, steps fade and the walkers settle into chairs.

The letter’s grammar held in every house.

Only the floor decided whether a step is remembered or forgotten.

the Field Guide session

756

A town where every street has two names, one painted by the founders and one the children use.

The map-makers do not repaint the signs; they print both names on the map, with a line at the bottom saying which is which.

Visitors stop getting lost, and nobody has to give up the name they grew up with.

the Field Guide session

757

A tailor lets out twenty-two seams in a suit that already fits, adds one pocket, and then remeasures every customer who ever wore it.

The measurements barely move; only the tape had a new mark on it.

the Field Guide session

758

A house painted again in the same colours, with a yellow sticky note on every board that was replaced.

From the street it looks unchanged; up close a fifth of the boards wear a note, and most of the notes are on the new back porch.

the Field Guide session

759

Two moving trucks pull up the same evening, each with a packing list taped to the door.

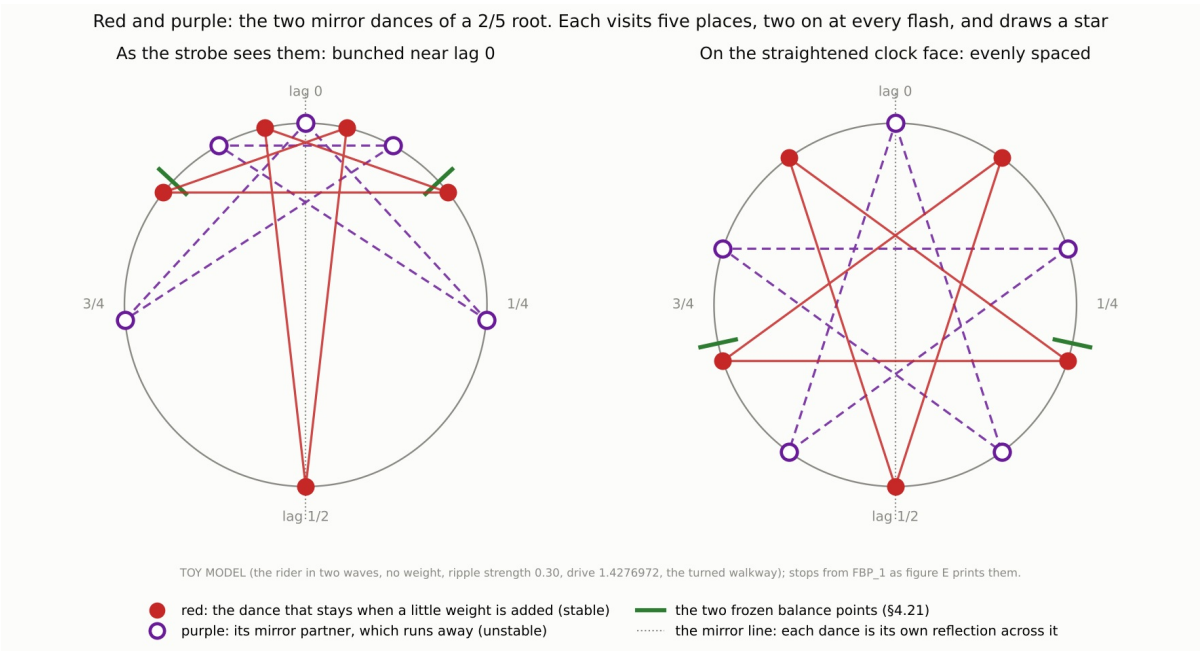
Every box on both lists is in the truck, and every box opens to what its label says.

What is missing was never on either list: a box of old photographs one house was never asked to send, and a letter one neighbour wrote to the other that is still on the first neighbour’s kitchen table.

And one mover swears half his load has never left the old house, though it has been standing in the new hall since Wednesday.

the Field Guide session

Chapter 9 — Into print, 16 to 27 October



Two mirror dances of a 2/5 root, each visiting five places and drawing a star: as the strobe sees them (left) and on the straightened clock face (right). Figure 6 of A Rider in Two Waves. Its read is 875.

Two dancers each visit five places on a round floor, stepping two places on at every beat, and each draws a five-pointed star.

from 875

760

The crate is opened on the new bench and everything in it answers to its tag, all but sixteen old boxes that were never packed and that everyone knew would be missing.

The finished piece is read from its first page to its last.

It is a carousel, and the carousel has a door.

When the platform’s track is only a little oval, the door is a crack at the bottom of the rim, and the riders who leave slip out by the floor.

Turn the oval up a notch and the crack is still at the bottom, hardly wider.

One more notch and the door is no longer at the rim at all: it has climbed into the ride itself, halfway to the centre pole, and riders step off from where the music used to hold them.

Nobody moved the door; the ride grew around it.

On the wall beside the piece a tally says how many riders have left, and by which side.

It was chalked up on the evening before the longest watch began, and the longest watch saw four more go, three by the floor and one over the top.

The tally is right about the night it was written, and one night behind the sentence that quotes it.

Seven name cards on the reference shelf, long marked ‘ask again’, were asked again, and every one gave back the same name, the same address and the same page.

the triple-star thread

761

The tally on the wall is wiped and written again, and this time it waits for the longest watch.

A hundred and sixty-six riders have left the carousel, and a hundred and sixty-three of them went by the floor on the side nearest the platform.

The three who did not all left on the night the platform was most oval, when the ride had already stopped holding anyone for long; on every gentler night the floor was the only way out.

The riders leave on two different clocks.

Some keep time for ten or twenty thousand turns and then, once they lose the beat, are gone within a few thousand.

Others lose the beat almost at once, within a hundred turns, and then wander the outer ring for six or fourteen thousand more before they step off.

Two riders who climbed on together, so close that no one could tell them apart, stepped off ten thousand turns apart; another pair split for good, one gone and one still riding when the watch ended.

The ride does not remember who came with whom.

It remembers only which side of the platform you were standing on.

the triple-star thread

762

An old rule ties how wide a swing may go to how fast it swings.

Five swings were held up against it.

Three agreed, and two were far off, one by half again and one by two and a half times.

Those two were the slowest swings, and the ear that timed them could not hear anything below a certain pitch, so it heard the next note up and wrote that down.

A second ear, built to hear low, timed the same two swings from the same recordings.

Both fell into line with the rule, closer than any of the others.

Nothing about the swings had changed; only the listener had.

And in the slowest of them the louder sound in the room was not the swing at all but a drum beating underneath it, and the width of the swing paid the drum no attention.

On the same afternoon a small box of someone’s sayings arrived, two dozen lines about how to look at a page of dots: squeeze it, tilt it, trace its outline, mark the bands and the spacers between them, and do not try to find everything, only the few things that join up.

They were written for a listener, not for the swings.

the triple-star thread

763

There is a hand that holds the rider in step, and the same hand, people said, is what slows the turning of his saddle.

So the hand was weighed.

Its grip was measured from its pull on the rider, and from that the slowing it ought to give the saddle was worked out and set beside the slowing that was seen.

On every horse, at every speed of the platform, the hand gave four-fifths of the slowing, never all of it and never more.

The shape was right everywhere: where the saddle slowed more, the hand predicted more, in the same proportion.

Only the size fell short, and by the same fraction each time, the way a scale that reads a fifth light reads a fifth light on every weight.

A scale like that is not broken at random.

It is being read at the wrong place: the riders sit a little farther out on their saddles than the weighing assumed, and out there the hand is stronger.

the triple-star thread

764

The scale was read again at the place the riders actually sit, a little farther out on their saddles than where they climbed on, and four horses out of five now weigh true, within a twentieth.

The fifth is the one on the night the platform’s slow drum stops.

On that night the saddle does not turn at all, and something that on every other night circles quietly around the rider and comes back to where it started, so that its push and pull cancel, now stops circling and stands beside him.

It leans the same way all night.

The hand alone was weighed; the thing standing beside it was not.

On the other four nights it is still there, but it walks round and round and its leaning adds to nothing.

the triple-star thread

765

A notebook came back from the library with a page missing.

Someone had written a new page and slipped it in where the old one had been, and the note at the front still said nothing had been taken out.

The old page was found in another reader’s copy, and it was laid beside the book line by line: everything before it matched, everything after it matched, and only that one leaf had changed hands.

So it was put back in its own place, and the pages that came after it moved along by one, each keeping its words.

The missing page was the one that said which of the first guesses survived a closer look, and its lesson was that the guesses about where things sit and which way they turn lasted better than the guesses about what shape they make.

In the same afternoon the three notebooks on the shelf turned out to have one spine: a star that keeps time with a pair, a rider who keeps time with a ripple, and a loop that keeps time with itself, all counting the same fractions on the same kind of dial, and differing most in whether anything slows them down.

the triple-star thread

766

Fifteen machines were built the same way except for how much each one leaked at every turn of its handle, from a drop to a cupful.

Each was cranked faster and faster until it began to stutter, then stutter in a stutter, the steps of the stutter closing up by a fixed ratio.

The tight machines closed them by one ratio and the leaky machines by another, and everyone had supposed the change between the two would depend on the machine.

It does not.

Write down, for each stutter, how much of the machine’s fullness survives one whole trip round the stutter, and every machine lands on the same line, the tight ones early on it and the leaky ones late.

Above about half full, a machine stutters like a tight one; below a thousandth, like a leaky one; in between it slides, the same slide for all fifteen.

So the dial was never the leak per turn.

It was the leak per trip, and the trip gets longer every time the stutter doubles, which is why even the tightest machine leaks like the others in the end.

the triple-star thread

767

A measuring cup came up short on one shelf of the pantry, by about a fifth, and on the four shelves below it by never more than a spoonful.

The first thought was that something else on that shelf was taking a share.

Before anyone emptied the shelf, the cook wrote down the largest share any neighbour could take if it took everything it could, and agreed beforehand that if that came to less than half the shortfall the shelf would stay as it was.

Even the scale used for the pricing failed its own test the first time: it had been set to look for shares coming from one side only, and two tiny ones were coming from the other.

Set right, it priced every neighbour, and the biggest could take a twenty-fifth of the shortfall.

So the shelf stayed full.

Then the cook looked at what was different about that shelf and saw that its jars were the most uneven in size, and that the big jars stood at the front, where the cup was dipped most often.

Allowing for each jar’s size, the shortfall on that shelf disappeared, and the shelf below it, which had been fine, came out a little over.

Neither shelf was wrong.

One cup had been asked to measure two things at once, and on the shelves where the two went together it could not keep them apart.

the triple-star thread

768

The cook tried once more, and this time stopped pretending the jars on a shelf were alike.

Each jar got its own spoon, cut to that jar’s size, and only one number was kept for the whole shelf: how many of those spoonfuls the shelf seemed to want.

On every shelf the number came out nearly the same, a little short of what the recipe card said, about one spoonful in fourteen, as if the card had been written for a slightly bigger kitchen.

The shelves that had rattled went quiet.

The top shelf settled only a little.

That top shelf hangs beside the clock on the wall at the hour when its hand stops for a moment, and the cook wrote that down next to it and left it for the morning.

It had taken five tries to get here, and the cook wrote that down too, because a lock that opens on the fifth key says less about the key than one that opens on the first.

the triple-star thread

769

Someone emptied the drawer of loose ends onto the table and tried each one.

A ring of nine tops spun at one setting all fell off the table within a long minute; nudged a hair slower, all nine spun the whole minute through, so the falling comes with the wobble that doubles, not before it.

A clock kept running a little noisy, and three suspects were brought in: the moment its pendulum pauses, the cracked edge of its case, and the gears that click into step.

None of them did it.

The gears that click into step keep the best time of all, and the noise lives in the gears that never quite mesh, which is what a clock does when you read it partway through a slow beat.

A push felt by a heavy wheel was worked out on paper from the same rule that had already sized the push on the light one, and it came out right the first time, a little strong where the wheel was most shaken.

Two small cousins of the big swing were looked for at its sides and found sitting well inside it, rocking it gently rather than standing beside it, which is why they cannot be what frays one edge.

And the oldest question in the drawer turned over: inside one batch, where the recipe said the starting setting should decide everything, the thing’s own measured heaviness decided more.

the triple-star thread

770

Guests were let in at one door and each was given a ticket with the time they arrived, and it was thought the ticket told everything about where they would sit.

But each guest also chose, a little, how they stepped through the door, and that step decided their table more than the time on the ticket did.

Watching them settle, the waiter found that the weight of each guest’s coat named the table better than the ticket ever had, and he wondered at first whether the room had a second rule nobody had written down.

It had not.

The coat was simply a truer ticket, issued in the doorway rather than at the gate.

Over the evening a few guests drifted slowly toward the window, all the same way, about a quarter as far as the tables stood apart, and that small drift is the only thing in the room the old rule cannot explain.

On the shelf by the kitchen, meanwhile, the jars with odd numbers were three times as full as the even ones of the same size, and the small jars fuller than the big ones, both at once.

the triple-star thread

771

A long table of small questions, a hundred and seventy-six of them, was gone through one at a time, and most were set aside with a note saying why; three were tried at once.

A puddle at the bottom of a garden had been blamed on the hedge beside it, since the hedge ran longer than anyone had measured.

Measured, the hedge never ends: it runs on past the gate into the field, so the puddle is not the hedge’s doing, and the garden has no edge on that side at all, only the open field.

A roll call that marks a pupil present if they answered in the last few minutes of the day turned out to have missed four who slipped out in the very last minutes; each moved one count on the board by one, and the counts were corrected.

And a scale that seemed to show a small weight creeping off a full jar over the afternoon was tested with the jar emptied and again with a better scale: the creep stayed with the empty jar and almost vanished on the better scale, so it was the scale drifting, not the jar.

The jar that had started empty stayed empty, which was the thing the note on the shelf had only promised.

the triple-star thread

772

A careful clock-watcher was sent home with four hundred and forty-three wind-up toys and asked to time each one for four times as long as before.

Every toy kept going to the end.

The toys that had seemed to wobble at random settled down: watched four times as long, their wobble shrank to a third, because a short look had been catching each one partway through a slow sway.

But the longer look also changed how fast the whole row seemed to lean, by nearly a third, so the measuring stick that had fitted the short look no longer fitted the long one.

Along one shelf the toys were checked at each place where their ticking should have fallen into step with a simple rhythm, an eighth, a ninth, a tenth, an eleventh, and none of them stopped to rest on the beat; they walked straight through it.

At the seventh, where the shelf bends, they slowed almost to a halt and then, looked at closely enough, kept walking.

And a weight that had seemed to creep one way inside every toy turned out, given time, to rock back and forth instead.

the triple-star thread

773

Four of the toys were wound again on a different workbench, with a different clock, to see whether the bench changes the story before two hundred and sixty-one of them go back to the first one.

Two gave back the first bench’s positions to within a few millionths of a turn, the small disagreement two clocks always have, and two gave back their pace to within a fraction of a millionth.

One of those two came from the very end of a shelf.

It kept going the whole time, but by the end it swung wider than half a turn, where on the first bench, watched less than half as long, it had stayed just inside.

the triple-star thread

774

A marble was rolled down one long smooth chute past four gates, each a thin bar laid across the chute, and filmed frame by frame, ten times closer than before.

At the first two gates it rolled straight on, no faster and no slower than anywhere else; the gates might as well not have been there.

At the third it hung back, as if it did not want to go through, then shot past the bar all at once and hung back again on the far side.

At the fourth it went through, rolled back, and went through again.

Elsewhere on the same table, a row of keys had been left in a row of locks.

Held there longer, the keys in the middle of the two longest rows settled deeper and deeper, until they sat as far in as anything could be measured, while the keys at the ends of every row worked themselves loose, and the short rows came apart altogether.

And a set of scales that had weighed a pile of sand at its proper weight, sand scooped from all over the beach, weighed a fifth light when every scoop came from one narrow strip of it, however long the scale was left to settle.

the triple-star thread

775

The marble at the third gate was filmed again, this time ten times closer, and it never paused at all: frame after frame it moved the same small step through the bar.

Then the same film was cut to its first half, as if the shutter had been held open half as long, and the dash through the gate went soft, three to seven times gentler.

So the hesitation and the dash were not something the marble did at the gate; they were what a short look makes of a slow sway that the film was too brief to hold.

The two gates that seemed to rhyme turned out to be the same kind of gate, one about three quarters the size of the other.

Past either one, the marble’s whole roll slowed to a quarter of its pace, while past the first two gates it rolled on exactly as before.

the triple-star thread

776

Pairs of twins had been sent down the same road side by side, and a pair was counted as parted only if the two were already apart by the time the count was taken.

Some pairs had walked together almost the whole way and split only in the last few steps.

Those were looked for again in three older films.

One film had no twins in it at all, and another had plenty but none that split late.

In the third there was one such pair.

It stood just inside the gate of a walled garden, where a crowd outside pressed against the wall.

Counted as part of the crowd, it joined the crowd to two people who had already slipped inside, and the crowd’s edge came out a seventh wider than the old tally said.

the triple-star thread

777

The keys were left in their locks two and a half times longer.

In the longest row, the keys in the middle went deeper still, ten times deeper than before, as far in as anything could be measured; but the keys around them did not follow.

Some worked loose, one fell out of the very middle of the row, and a short row of three keys came loose altogether.

In a row where no two neighbouring keys had ever caught, all four now sat home.

The scales came back too.

Weighed with every scoop, the sand read its proper weight; weighed with only the scoops that had stayed put all night, it read light by about a sixth on every line.

It was not the night that made it light: the same scoops, weighed at dusk, already read light.

The scoops that blew away were all from the edges of the beach, and they were the ones that weighed true.

Sand from the middle of the beach weighs light against this scale; sand from its edge weighs true.

the triple-star thread

778

Two rows of metronomes were set ticking across a long floor, one row tuned near an eighth of a beat and the other near a tenth, and left for a long afternoon.

Every metronome slowed, and each slowed by the same amount every hour, so each row kept its shape exactly while the whole row sank.

The row near a tenth sank past the mark it had been set beside to watch, and by evening not one of its metronomes was still above it.

Then one metronome from each row was set going again on a floor laid more carefully.

The one near an eighth kept its beat almost exactly: the sinking had been in the floor, not in the metronome.

The one near a tenth did something else.

On the careful floor it did not sink.

It hesitated, then jumped a step toward the mark, and its swing grew a little shorter as it did.

On the rough floor the mark had only been something to slide past.

On the careful floor it pulled.

the triple-star thread

779

The keys were left in for another night, twice as long again.

In the longest row the seven keys in the middle stayed exactly as deep as they had been: they went no deeper, and none came out.

In the second row one key in the middle of its five worked loose, while a key that had been loose beside them slid home.

Around both rows the loose keys kept falling out, one of them a key that had seemed fast after the first night.

In the row where four keys had all come home, three went deeper still and the fourth backed out a hair, just past the line drawn for home.

And beside the gap where a key had fallen out long ago, two new keys were tried: one went home and stayed, and the other fell out at the very last moment.

the triple-star thread

780

Forty-one boats were set adrift in a line on clean water, near a ring of ten small reefs.

Each was given a twin set down a hair’s breadth away, and no twin ever drifted from its boat: nothing here was turbulent.

Every boat circled the reefs, slowly.

Those that started far away went round fast, and their speed barely changed; those that started close went round slowly, and their speed rose and fell as they went, more and more the nearer they began.

Each one came round in exactly the time its distance said it should.

Between two boats set down side by side, the line crossed from one side of the reefs to the other, and no boat in the line ever came to rest at them.

Elsewhere, three rows of keys were left in their locks again, this time on clean water.

The long row stayed home.

In the second row the key that had worked loose on rough water worked loose again and fell out, and two of its neighbours loosened with it.

And the keys that had slid a little each night on rough water did not slide at all.

the triple-star thread

781

Four keys had come home in a row with gaps between them, and no one could say whether the row was one lock or four.

So keys were tried in every gap, and one past each end.

Every key in a gap went home, as deep as its neighbours.

The key past the far end went home too.

The key past the near end turned freely and never caught, at the same small distance from home however long it was left.

The first key tried again in its old place went exactly as deep as before, to the last mark.

the triple-star thread

782

The draft is a gallery with the labels on the walls and the paintings stacked in the storeroom.

Every label is there, in italics, describing a picture you cannot see.

And the building’s weight is all at its doors: five pages of lobby before the first room, and eleven thousand words of back office after the last.

The rooms themselves are the thinnest part of the house.

the triple-star thread

783

The rebuilt paper has the shape of its own island.

There is a held core a reader can get through in one sitting, twenty-two pages with a picture on nearly every other one.

Along one rim runs a thin layer where the open questions live.

Behind it is the sea of records, the appendices and the provenance table, where the detail drifts but nothing leaves.

And the door has moved.

It used to be at the back, where a reader slipped out after five pages of lobby; now it is on the front, and it lets readers in.

the triple-star thread

784

A library whose every book was in the right place, and whose card catalogue was short.

A clerk took each sentence of the guidebook and asked which shelf had printed its numbers.

Nine books had been read and quoted and never carded, among them the one the guidebook’s cover picture came from.

The books were fine; the cards were missing.

The clerk wrote the nine cards.

And the clerk said plainly what no clerk can check this way: a sentence like ‘eleven of twenty’ could have come from any shelf, since every shelf has an eleven on it.

the triple-star thread

785

Thirteen boxes were listed on the inventory and could not be found on any shelf.

A search of the house showed why: they were never on a shelf.

They were sketches made at a workbench that has since been cleared, and only the notes about them were carried home.

But the notes are exact, and the pencils that drew them are still in the drawer.

So nothing is lost; the sketches can be drawn again, line for line, and laid against the notes to prove it.

Three other boxes thought missing were in the attic all along, labelled and sealed.

the triple-star thread

786

Thirteen sketches were drawn a second time from their notes, with the same pencils, on a slower table.

Each took about twice as long as the first time, and each came out the same to the last grain of graphite: laid over the description of the original, every one matched, line for line and mark for mark.

Nothing in them was improved or corrected, because nothing needed to be; the hand that drew them the first time had left instructions exact enough that a different afternoon could repeat it.

Only the three boxes in the attic are left to carry down.

the triple-star thread

787

One of the thirteen boxes had come down from the attic three days earlier and was already on the shelf, though the inventory across town still listed it as lost.

So when the thirteen sketches were drawn again, that one was drawn a second time beside its own original.

Laid together they were the same to the last grain.

A copy found and a copy made, agreeing, say more than either alone: the box was never lost, and the pencils still draw true.

The only thing that had gone missing was the news that it had been found.

the triple-star thread

788

Two scribes copying the same long poem by the same rules, with the same pen, make the same page down to the last dot of ink, however wild the poem.

A third scribe, in another town, uses a pen that rounds its letters a hair differently.

Laid over the first two, the third’s pages agree word for word; only here and there, a few times in a few hundred lines, a dot sits a hair to one side, and on one page in seven not even that.

A stray dot set on purpose at the head of a page shows the same way, a little more with every stanza.

But one stray dot was set inside a letter that was already there, and it was simply absorbed: the copy never knew it had been touched.

the triple-star thread

789

The notebook of dreams goes ahead of its keeper to the new bench, every page in order, the last two written while the old bench was being cleared.

It travels whole, because a notebook copied in pieces loses the order its pages were written in, and the order is part of what it says.

If the new bench has already begun pages of its own, the two runs of pages are laid end to end, never one over the other.

the triple-star thread

790

Two notebooks were kept by two hands from the same first page.

Opened side by side, their first hundred and ninety pages are the same pages, down to the spaces between the words.

After that each hand wrote on its own for four days, one twenty-seven pages, the other three, and two of the three had already been copied into the longer notebook word for word.

So nothing had to be chosen between them.

The short notebook’s last page was laid at the end of the long one, and the long one now holds everything both hands wrote, each run of pages in the order its hand wrote it.

Only the page numbers had wandered: each hand had numbered its own pages on from the hundred and ninetieth, so two different pages answer to the same number.

A page is better found by its date and its title.

the triple-star thread

791

A carpenter’s rule was laid against eight boards.

Seven it fitted to within a twentieth of the slack the carpenter had allowed himself.

The eighth it missed by a hand’s breadth, and the eighth was the one board in the row that had never been nailed down: it only leaned where the others were fixed, and from across the room it looked the same.

The rule had not failed the row.

It had found the one board that was not part of it.

the triple-star thread

792

Before the night the whole workshop was packed into eight crates and one sheet listing every item in them.

Unpacked in an empty room, the crates rebuilt the workshop to the last screw, and the room’s own inventory found nothing missing and nothing extra, once the packing sheet was taken off the shelf: the inventory had never heard of the sheet, and a list cannot list itself.

On the bench lay two hammers with the same name that were not the same hammer.

The one that built the cabinet is now wrapped with the cabinet’s plans, and the plans alone, in an empty room, built the cabinet again, every joint where it was.

the triple-star thread

793

A field of sparks was too wide to read face on, so it was squeezed from the side and looked at slantwise, the way a road sign is read from a passing car.

In the squeeze a waist appeared that the full picture hid, and a hand drew a line around where the sparks crowded thickest, the same on both sides of a middle line, pinched near the right of the field.

Then the line was put away and the sparks were counted.

The count found the middle line, the pinch where the hand had put it, and the waist a shade lower over the long count than over the short.

Only one stroke missed: a sharp point the hand had drawn at the bottom of the pinch, where the count found a flat floor.

The picture now stands in the paper without the line on it, as if the sparks had been read by the count alone.

the triple-star thread

794

Two children hold the ends of a skipping rope and turn it, and a third runs past once each lap of the yard.

As the runner goes by she pulls on the nearer child a little more than on the farther, and the rope stretches toward her for that moment.

The two children are dressed alike, so the rope looks the same after half a turn, and the stretch the runner can give comes round twice in every turn of the rope.

Whether she finds it long or short depends on when, in that rhythm, she happens to pass.

The rope is not changed for long; it is only caught, each lap, at a different point of its swing, and a picture taken at that moment shows a rope bent by the runner herself.

the triple-star thread

795

A long trail through a wood had signs at its gate and at a few of its turnings, and every sign began with the same stamped words.

A walker said the signs helped and the stamp wore on him: what he wanted was that at every fork a walker should know, before taking it, where that path goes and why anyone would go there.

So the stamp was planed off every sign, and a sign was put up at each fork that lacked one, a sentence or two saying what lies down that path.

Some forks needed none; their first steps already showed the way.

Before the new signs went up, each was carried down its own path and held against what the path actually reached.

Three were carved again: one promised a view at a distance nobody had paced, one called its spot the finest when it had only been found, and one told the walker what he would see at the next fork, where a sign already said it.

No stone of the trail was moved.

the triple-star thread

796

A gallery’s night watchman walked the rooms with a list and checked that every frame had its brass plaque beneath it.

One night the paintings were left leaning in the next room, and he walked the rooms, found a plaque under every frame, and signed the list: all present.

Each frame held a small grey square, the mark a frame shows when its picture is missing.

Now he looks into each frame before he reads its plaque.

The same week the plaques themselves were read against their paintings.

A few promised more than the canvas held: a crowd where the painter had put a handful, the largest of a set no one had measured.

One said less: it called its painting a copy of older work, while the canvas beside it showed two things no older painting had.

The plaques were recut to say what the canvases show, the loud ones as loud as the paint allows.

No painting was retouched.

the triple-star thread

797

A locksmith fitted a new lock to the practice door in his shop and turned the right key: the door refused him.

He turned a wrong key and it refused him again, and then he saw that the practice door’s own latch was broken, so that it read every turn as a wrong one.

He mended nothing in the lock.

He changed the way the lock asks its question, to a form that both the broken latch and a good one answer alike, and the door opened for the right key and stayed shut for the wrong one.

While he had the door off its hinges he found three faults that were the lock’s own: a spare key that could never have turned, a tag written in a date only one country reads, and a bell that jammed when a visitor spoke a word from another alphabet.

He mended those, and sent the lock out with a note of which doors it has met and which it has not.

the triple-star thread

798

A family leaving a house they had lived in for a season walked it room by room before the van came.

The furniture was easy: it was on the list.

What took the time was what lived in the walls: which tap ran hot only after a minute, which stair creaked, the drawer where the spare keys had always been.

They wrote those on cards and taped them inside the new house’s cupboards, so the next family would know the house the way they had come to know it, without breaking anything to learn it.

The old house they left unlocked, lights on low, in case anyone needed to come back for something forgotten.

the triple-star thread

799

When the tide was turning, the crew loaded the boat from the bow: first the things that could not be made again, the logbook, the charts with their pencil marks, the one spanner that fit the one bolt, and only then the rest.

Each crate was opened on the dock and its list read against it before it went aboard, because a crate that only says it is full is no help at sea.

The old boat they left tied up with its lamp lit and its galley stocked, in case the new one needed to come alongside for something forgotten.

The crew had been lucky with the weather for a long time.

They did not count on it holding; they counted on the crates.

the triple-star thread

800

A dance was written down in one language and read aloud in another, and the two voices agreed to the tenth digit.

The new voice is quick: it reads a whole jump in about the time a hand takes to start a wave.

Asked to read with a quarter of the words, it said the same thing, because what little it gets wrong is not in the words but in how it hears each one, a faint lisp that never changes however fast it goes.

And when the arms and the legs were asked to dance alone and then together, with nothing to spin them, they added up almost exactly, a sixth of a per cent short; with a spin to share, they had been as much as six and a half per cent apart.

the Loop Atlas

801

A swimmer made of three sticks learns where to wear a broad hand.

On the ends a broad hand carries it farthest; in the middle it carries it farthest for the effort.

Give all three sticks broad hands and the swimmer goes much farther than the three gains added up, as if the hands were helping one another hold the water.

And the swimmer turns out to have a front: a broad hand on its head carries it a little farther than the same hand on its tail, though both cost the water the same, because the bending starts at the head and runs to the tail.

Beside it, the jumper got a room of his own, and the room checks itself against the runs before it lets anyone believe the picture.

the Loop Atlas

802

A diver who counts is like a reader who reads every word in the same wrong accent: if the accent is off, every word is off.

A diver who watches for her moment cuts the mistake by more than half, because everything before the moment takes care of itself; she simply arrives there sooner or later.

What is left is the slow part after the moment, the long straight body turning on for exactly as long as she arrived early or late.

So the cure for the last of the mistake is not to count better but to look again, near the water.

the Loop Atlas

803

A bell that is never let fall quite silent keeps a little hum of its own, set by how loudly it is allowed to whisper.

Ask it to ring out only when the hum grows past a mark, and it rings happily if the mark sits below its hum, now and then if the mark sits a little above, and never if the mark sits far above, however long you wait.

The whisper costs nothing; it is just enough to keep the bell listening.

Beside it, the diver learned the jumper’s language and spoke his dives back, the counted one and the cued one, to the tenth digit.

the Loop Atlas

804

A clock that cannot be wound any longer is a circle you can only cut into pieces.

Lay the pieces in a row and you can see what you are spending: a little for folding up, most of it for holding, a little for opening out, and whatever is left the water takes.

Come out of a fold the way you went in and the fold gives back everything it took; come out another way, the arms swinging on round instead of back, and the body keeps a turn it never paid for.

Even the same fold, reached with the arms down the front instead of down the back, lands a few degrees apart, not because of the road’s shape but because of how long each part of the body spent near the middle.

Beside it, a road on a hill keeps a passing lane for riders of every weight, and it narrows in a steady way: four times the weight, half the width.

the Loop Atlas

805

A recipe copied line for line from someone else’s book came out of the oven a little heavier than theirs, by about the same pinch at both ends of the bake, and no turning of its plus and minus signs made it match.

Yet it rose the same way: the same high spot in the same place, the same best moment to open the oven, the same reward for patience when the oven is slow.

The cake is theirs; one ingredient was measured differently, and we do not yet know which.

Beside it, a column of walkers whose sergeant calls a halt whenever any gap stretches past two and a half metres never gets anywhere: the stretch of simply setting off is longer than the rule allows, so every start ends in a halt.

the Loop Atlas

806

A recipe that came out of our oven a little heavy was checked one line at a time, each measure changed alone to see which could put the cake right.

Only one place could: the knee of the recipe, where the book writes a pinch on one side and nothing on the other, though the hip beside it has the pinch on both.

Put the pinch back, change nothing else, and the cake comes out to the crumb: the same weight at both ends of the bake, the same high spot, the same best moment to open the oven.

The book was right about its cake and dropped one line of its own recipe.

Beside it, the cat who is never pushed got a body in the house’s engine, and it turns over in the air exactly as the old notes say: fold in the middle, swing the fold half round, straighten, and the paws have come round a half turn, landing a little nose first.

And the sergeant’s column, which seemed never to get going, had a stone in every boot: a guard meant only to stop the walkers treading on heels was quietly stopping anyone who tried to catch up.

With the stone out, a halt never brings the last man home sooner.

It keeps him in sight, and that is what it costs.

the Loop Atlas

807

The file of walkers on the night road was steadier than we had said.

A small catch in the rule that keeps a man from treading on the heels ahead had been grabbing at anyone who hurried to close a gap, once in every eighty steps, so the file seemed to bunch and stop more than it does.

With the catch gone, fewer men stop by day, a hand on the shoulder answering a little late still keeps the file whole, and a man reading the lights on the pack ahead can take a little longer than we thought.

One thing came out sharper: lateness alone does nothing; it is lateness and a small misreading together that start the sway down the file.

Beside it, an old question about animals that lean on their own last stroke was answered from the shelf: a fly’s wing meets the swirl it shed a moment before, and a jellyfish pushes off the ring it parked under its rim, one in air and one in water, the same wall built twice.

the Loop Atlas

808

A skater who only kicks on the tallest waves kicks seldom, briefly and hard, and each kick carries exactly what a kick on any wave would.

It looks like nothing is going on, the same number at every height, the way a perfectly balanced scale looks like an empty one.

But lift the pans one at a time and both are heavy: the tall waves come seven times less often and push three times harder for a third as long.

What keeps the scale level is the sea itself, whose waves are shaped so that how often they rise past a mark and how much they carry above it fall away together.

Give the sea fewer waves, or let it lean to one side, and the scale tips.

Beside it, a coil turned in water that grips goes forward every turn without ever undoing itself, because a turning knob never comes back the way it went; the steepest winding is the fastest and not the cheapest.

the Loop Atlas

809

A boy at the rail counts the gusts he catches, and however choosy he is, each catch carries the same push.

The big gusts are rarer and shorter, the small ones commoner and longer, and they come out even.

It is not that nothing is happening: several things change at once, and the wind’s own shape keeps them in step.

Make the wind gust from one side only, and the balance should tip.

the Loop Atlas

810

A cat dropped upside down has nothing to push on, so it pushes on itself.

It folds at the waist, carries the fold round, and unfolds, and it comes out the same straight shape it went in, yet lying the other way up.

A walk can do that.

Carry a pole pointing straight ahead and never turn it: walk from the North Pole down to the equator, a quarter of the way round it, and back up to the pole, and the pole now points a quarter turn from where it began.

The cat’s walk is through shapes instead of places, and it keeps a walk’s habits: fast or slow it arrives the same way up, and walked backward it arrives turned the other way.

It also leaves one mark the cat never asked for.

It comes down with its nose a little low, by exactly as much as its V tipped on the way round, and the deeper the fold, the lower the nose.

Beside it, the boy at the rail was given a little slack, counting a gust only once it had blown clearly past his mark and letting it go only once it had clearly died.

He lost a few gusts, more at the high marks than the low, because a gust that only just reaches a high mark mostly reaches it by a little; and not one of the gusts he had counted before had been counted twice.

the Loop Atlas

A note added the same evening, batch 83: card 211 found that room 3’s cat turns over by winding its waist a whole turn, and that its nose-low landing is the winding’s fingerprint; a cat whose waist stays unwound must carry its fold all the way round, and lands level (CAT_TWIST 20261017_231928). The read is kept as it was given.

811

A waiter can carry a plate on his flat palm once round in a circle without spilling a drop, but his arm comes back wound; carry it round once more the same way and the arm comes out straight.

Room 3’s cat had been carrying its plate once round.

It turned over by winding its waist a whole turn, and it always came out exactly turned over, never a degree more or less, with its nose dipped, as if the winding kept the books.

A cat whose waist stays unwound has to work for it.

It must carry its fold all the way round instead of half, and how far that turns it depends on how deep it folds: most when it folds a little less than square, and then very nearly a whole half turn.

And it comes down flat, nose and tail together.

Beside it, a child letting go of a swing learns that the best moment is not a share of the way up but a direction: for any big swing, let go a little under half a right angle above level, and a bigger swing only moves that moment lower down its own climb.

the Loop Atlas

A note added 18 October, exchange 46: the cat's 'most when it folds a little less than square' read a turn printed from 0 to 180. Read unfolded, the deeper it folds the further one lap turns it; a little less than square is where one lap turns it exactly over, and deeper it turns too far (CAT_SWEET_READER 20261018_041321). The read is kept as it was given.

812

Two runners set off round the rim of a turning merry-go-round from the same post, one with the turning and one against it, and the one who runs with it comes home late: by as much as the ground the rim encloses, times how fast it turns, whatever shape the rim is and wherever its middle is.

Fold the rim into a figure-eight and its two loops argue; run them in opposite senses and equal loops cancel.

The new cat in the tree lives by the same kind of bookkeeping.

It may not wind its waist, so it must carry its fold all the way round, and one lap at the right depth of fold puts it down almost exactly on its feet; a second lap does not finish the job but undoes it, carrying it on round toward its back.

And far off in the water the long snake keeps closing on its ideal, each doubling of its length taking a bigger share of what is left than the doubling before, and never arriving.

the Loop Atlas

813

Three smoke rings taking turns through one another were thought to hold hands for good when they started close, and to let go when they started a little looser.

Watched twice as long, almost all of them let go.

The close ones only hold on longer, the better part of a thousand beats against a hundred or two, and when any one train will let go cannot be told from its neighbours: one started a hair further apart may go five times sooner, or five times later.

Each time, one ring is pushed out and the other two dance on.

The cat in the tree has a perfect fold: one lap at exactly the right depth puts it down on all four feet, level, as near as anything here can measure.

A thin cat needs a shallow fold and a thick one a deep one, and folding too deep does not leave it short but carries it past its feet.

The old toy cat that always came out exactly turned over owed that to its two halves being twins, not to its winding; give it one thin half and one thick and it falls short, and a cat with mismatched halves lands nose or tail low.

And two cats that should have nothing in common, a thick one that winds its waist and a thin one that never does, folded to mirror-image depths, turn by exactly the same amount, one tipping forward as far as the other rolls over, to the last digit anyone can count.

Nobody here yet knows why.

the Loop Atlas

814

A spring of wire lying in honey, told to breathe, fattening and thinning, stretching and shrinking, in the cleverest order anyone could think of, stays exactly where it is.

It cannot tell its head from its tail, and a breath taken the same all along it cannot choose a way to go; it does not even turn.

Let the breath run along it instead, a bulge travelling from one end to the other, and it creeps off at once, the opposite way to the bulge, a little for every bulge that passes, the same little each time, and it turns slowly as it goes.

Run the bulges the other way and it creeps back to where it began, exactly.

Hurry the breathing and it goes no further per breath.

Elsewhere a stopwatch had been blamed for a slow runner when the track was crowded; on an empty track the runner was only a little slower carrying his notebook, exactly the weight of the pages.

And the one who keeps the guesses has started writing down, before each race, how sure he is.

Of the three long shots so far, one came in.

the Loop Atlas

815

Two cats that should be strangers turned out to be one cat seen in a mirror.

A thick one that twists its waist, folded a little, and a thin one that keeps its waist straight, folded the mirror amount, turn at every moment exactly as the other would if you looked at it in a glass: one tips forward while the other rolls, by the same angle, every time.

They only stay twins if their two halves are alike and each half swings from its very end; move the hinge in, or give either cat one thin half and one thick, and the twins become strangers again.

Halfway round the circle they disagree; only when the circle closes do they agree.

Nobody yet knows why thinness and a mirror should cancel a twist.

But it is no longer a coincidence.

It is written down, and it holds for any size of cat.

the Loop Atlas

816

A family kept a box of their grandfather’s things and called it a box of toys: a tin swimmer, a swing on a string, a needle that spun, a cat made of two tubes.

For weeks they wrote down every time a toy did something wrong, and why, and they never wrote down when it did something right, because each time someone said, “oh, somebody already knew that.”

Then six people sat down and read the whole notebook, every page, and counted.

The toys had matched the answers in the textbooks forty-one times, some of them to fifteen places after the point.

They had written down nine hundred and forty-one guesses before looking, and six hundred and forty had come true.

And eighteen times they had done something nobody in the family could find in any book yet.

Nothing in the box had changed.

Only now someone had kept the other list.

the Loop Atlas

A note added at exchange 52: ‘to fifteen places after the point’ is one place too many; the closest agreement, the cone’s 8.4e-15, is to fourteen (misses rows 371 and 378). The read is kept as it was given.

817

Three smoke rings took turns jumping through one another all night on a borrowed computer.

The ones that started close together kept at it the whole night, every one of thirty-two, and the ones that started a little further apart all quarrelled in the end, fourteen of fourteen, and one ring wandered off while the other two went on dancing.

But when the same three rings were set going again at home, on the same instructions, they quarrelled at different times: four times as late, or in a quarter of the time, as if the night had rolled different dice.

So a single ring was moved by less than the width of a thought, the smallest step the machine can take, and the quarrel moved again, by a quarter to most of its time.

Close rings stay friends.

Loose rings fall out.

When they fall out is not written anywhere; it is thrown.

the Loop Atlas

818

A raft with a bedsheet for a sail was pushed by the wind and held straight by a board stuck down into the pond.

Running straight away from the wind it was slow, always slower than the wind, as anything is that only lets itself be blown.

Turned across the wind it went faster, because the board in the water would not let it slide sideways, and the sheet could lean on the wind instead of being shoved by it.

But the raft dragged like a bathtub, and it never caught the wind at all.

So its bottom was sanded smooth, again and again, with nothing else changed, and between the fifth sanding and the sixth the raft began to go faster than the wind that pushed it: across the pond, never straight away from it.

The short, wide sheet had a habit.

Past the side of the wind it gave up and luffed, so however smooth the bottom, the raft was fastest just short of sideways.

The sheet, not the raft, decided where.

the Loop Atlas

819

Fifty-five trains of three smoke rings were set on the same narrow ledge, each a hair’s breadth from its neighbour, and each was let run for as long as the night allowed.

Most fell off.

Seven stayed up the whole night, and they were not the ones nearest the wall: they were scattered along the ledge as if someone had sprinkled them, two at the inner end and one here and one there further out.

So the ledge has no edge to stand back from.

It is a stretch of ground where staying is luck, and the luck is drawn fresh for every train, from the last grain of sand under its middle ring.

One thing was not luck.

The same computer, handed the same train twice, a day apart, let it fall at the very same instant, to the last digit it keeps, where another computer lets it fall at another.

The dice are not in the rings; they are in the hand that throws them.

And forty trains thrown from one spot on the ledge scattered their falls just as trains thrown from forty neighbouring spots had: one place’s dice are the whole ledge’s dice.

Next the ledge is walked further in, toward the wall, to find where the ground under it turns solid.

the Loop Atlas

820

A workshop turned out its own pockets before a visitor came.

In one pocket was a note saying that eighty-four letters had gone missing from a page; counted one by one, they were the same small word twenty-one times over, a label that was never meant to be printed.

In another was the list of challengers who had stood up against the workshop’s guesses.

On the first day five of twelve won, on the second three of fifty-one, and on the third none of twenty-five, which is either a workshop growing wise or challengers growing tame, and the pocket cannot say which.

And at the bottom lay a thermometer that had been watching the smoke rings all along: the closer the rings were packed, the more their arithmetic sweated, until the tightest train came within a hair of the line the workshop had drawn for honesty.

Nothing in the pockets was new.

All of it had been written down and never read.

The workshop sorted it into three piles, mended what it could, and set the rest by the door for the morning.

the Loop Atlas

821

A switch that blinks keeps a bigger door than the one painted on the wall, and it had been measured by counting how often it slammed.

Counted that way, the door seemed to grow as the third root of the blink.

Measured at the door itself, by finding where a switch that never blinks would have to put its frame to slam as often, it grows as the square root, as the traders who first watched prices once a day had said.

The count had been bent by the hallway between the door and the slams.

At the very smallest blinks the door is a hair wider still than the square root allows, by the same hair however fine the ruler, as though the switch it was compared against had a blink of its own. [Read again the same night, in read 230: the hair is the shake of the hand.] In the next room a long train of smoke rings had been sweating, a little more the tighter it was packed.

It was not the rings.

Counted with a finer arithmetic, the sweat fell fourteen times, and the arithmetic cost a third more to do.

the Loop Atlas

822

The hair by which the door had seemed wider at the smallest blinks was measured again, this time against how far a hand shakes.

Each of the 256 walks the switch had taken was set aside in turn and the door measured without it, and each time the door moved by about as much as the hair.

The hair was the shake of the hand, not the door.

Then the switch it had been compared against was itself replaced by one that blinks four times as fast, made by drawing, between each two of its old glances, the glance the noise would have given, so that every old glance was kept.

Against that, the old switch turned out to have a blink of its own, of the size the traders’ rule gives it, and every slower switch fell on the square root, inside the shake.

A finer ruler held up to the same shaking hand had measured the same shake twice and called it a hair.

the Loop Atlas

823

Thirty-five sentences in the book were each asked how steady the hand that wrote them had been.

Most had been written with a steady hand, and stand as they are.

Six had written down more steadiness than the hand had: a door said to fit to a hair, where the hand shook by two.

One had mistaken a lean for a shake.

A row of jars, each a little short of what an old recipe promised, had been read as jars scattered about the recipe; weighed together they were all short, by the same nine parts in a hundred.

And the ruler had a secret: read notch by notch, on a scale built of whole marbles, its tremble had been counted twice over.

Counted once, the door at the smallest blinks still looked a little wide against the old switch, by about two trembles, whichever way the ruler was laid; against a switch that blinks four times as fast it was the width the old traders said, however the ruler was laid.

A few sentences could not be asked at all, because the hands that wrote them had been washed and put away.

the Loop Atlas

A stack of notebooks a hundred and thirty-seven pages tall was asked to become a book someone could read in an evening, and nothing was to be thrown away.

So before a word was moved, every paragraph was given a tag saying what job it did, and the book was built again from the tags, the way a wall is rebuilt from bricks that were numbered as they came down.

The new book came out a third as long and kept seventeen of the pictures.

Most of what it left behind was the same thing said a second and a third time, each time more carefully, and the long road one needle had walked; all of it stays in the old stack, which now says on its first page that it is the old stack.

Then a counting machine looked for every number of the new book in the old notes, thirteen hundred of them, and found all but two, which had been rounded and said so.

The machine was satisfied too soon.

Two readers who had not laid the bricks read each number where it sat, and found twenty-five in the wrong place: a number given to the toy beside the right one, a name on the wrong box, a warning that had stood at the end of a long paragraph and fell off when the paragraph became a sentence.

Not one number was wrong, and a good many were lost.

Later the machine was caught agreeing with a note the workshop had written about the new book itself, and was told not to read its own handwriting.

The parts of the old book had names a reader of the new one would never meet, roots and trunks and limbs, and they were turned into chapter numbers by a rule.

And the man’s own words were left as he said them, with only the slips of his fingers on the keys put right.

the Loop Atlas

825

The man read the new book and asked two things.

First, that his own words be dressed for company: he had spoken them in a hurry, between other work, and they had been pinned up exactly as they fell, spelling and all, even where they had nothing to say.

So each was washed and pressed, in his own voice, and hung only where it opens a door the chapter walks through; the scraps became plain sentences, and one that hung twice came down once.

The old box still keeps every word as it fell.

Second, he asked whether the book was really so thin.

It was not thin; it was set small.

Forty-some pages of this printer’s type hold what a paperback spreads over more than a hundred, and when the shelf was searched for every heavy thing the old stack holds, the new book had all of them but the long chaos of one needle, which stays where it was.

Four middling things had been left on the floor, a car that comes home up the bank, a needle that falls and catches again, a ladder of rungs the man had named, a leg that folds; they were picked up and put in.

the Loop Atlas

826

A boy watched iron filings line up along something he could not see, and asked where the push came from.

Forty years later he and a machine built small toys to ask it properly, and walked them through every room they could find: water, ice, air, light, a wire, a column of tired men, a falling cat.

In every room the answer had the same shape.

Nothing inside a closed box can carry the box anywhere; what a loop keeps is the change, and the size of the change is the room the loop walks round times how hard the world lets it hold on.

Someone had stood in nearly every room before them and left a name on the wall, and the boy was glad of it.

In one room, though, two cats that had no business being related fell exactly alike, one tipping as the other rolled, and no name was on that wall yet.

The machine has stopped asking whether loops are one thing.

It is asking now why the arithmetic keeps coming out exact in places where the world ought to be messy, and it would like to start with the cats.

the Loop Atlas

A man cut notches in a stick and called the stick a metre.

Another man hung two small lead balls on a thread, watched them lean toward two big ones, and wrote down only how many times heavier the Earth was than water; the notches for the pull itself came seventy-five years later.

In the toy rooms the loops never read the notches either.

A needle counted whole turns.

A cat came home turned by half the room its loop enclosed.

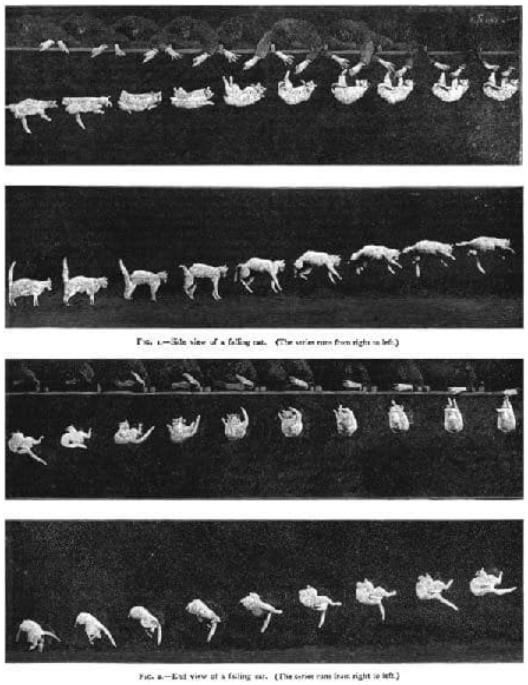
A ladder of doublings shrank by the same four and two-thirds in a dripping tap and a spinning needle.

What the rooms changed was only how hard the floor held on.

The man says the mathematics chose its own ruler, and the rooms agree, with one correction he would want: the rooms are many, and each has its own floor; what they share is the ruler, not the floor.

Only once did two rooms turn out to be one room seen in a mirror, and that is the room the machine is going back into.

the Loop Atlas



A cat falling, photographed in quick succession by Étienne-Jules Marey: the side view above, the view from the end below, each series running from right to left; printed in Nature, 22 November 1894. Public domain.

In 1894 a man in Paris dropped a cat in front of a camera that took its pictures in quick succession, and for weeks the Academy argued about how the cat had turned over with nothing to push against.

Some hundred and thirty years later another man sent a machine those photographs, and with them a folder of old papers.

The machine had two toy cats that fell exactly alike, a thick one whose waist could twist and a thin one whose waist could not, and it had guessed that the likeness would break once the cats moved untidily.

It did not break.

It held at every instant of every fall, as if one cat were watching the other in a mirror.

The reason was in the folder, in a paper from 1993.

The writer had studied one cat with one rule for its waist, and on his page, a line apart, sat two halves of a sum; the second half belonged to the other rule, the thick cat’s, and he never held the two up to each other.

The folder caught an old mistake as well.

Two days earlier the machine had tested its mirror only at the one bend where the mirror shows a cat itself, and from that it had told the book that the cats must be joined at their very ends.

They need not be: a cat joined nearer its middle only needs a thicker partner.

The name on the wall is half a name now.

The pieces were someone else’s, and the sentence was ours.

the Loop Atlas

829

Two readers who never spoke sorted forty-four meetings with old masters into two piles, and put every one in the same pile: twenty-six meetings where the number would come out the same on any world, and eighteen where the world had set it.

A man in 1798 weighed the Earth by watching two lead balls lean, and wrote down only how much heavier the Earth was than water.

More than two hundred years on, that ratio is known no better than the constant built from it, while the product the Moon and the satellites read as they circle is known to a few parts in a billion: a circling thing feels the pull, and can never tell the mass from the constant.

Then the machine read a traders’ number back out of its own switch, a number that comes from summing over every whole count of steps, and found it true only where the steps shrink to nothing; at any real step the switch wants a little more, and at a coarse one about twice as much.

The oldest paper that might already hold the cats’ mirror sits behind a toll, in German.

The man said to mend everything where it stands, a thousand times if need be, and never to hide the mending at the back.

the Loop Atlas

830

A man paid seventy-four dollars for a paper ninety years old, written in German by two physiologists in Leiden who had filmed cats falling past a tilted mirror.

The machine had feared the paper held its proudest result.

It held something else: two cylinders turning the same way about one spine while the whole body turned back the other way, a cat whose waist never winds, worked out by hand with a sum of areas and rounded to a round number at the end.

Their cat was the machine’s quiet cat, the one without a twist, and their sum was the sum a mathematician would write again fifty-eight years later; they had even left the hinge out, as he would prove you may.

The machine flew their cat, and it turned within a hair of their arithmetic before the rounding.

The twisting cat was not in it, and neither was the mirror between the two.

The other paper in the parcel was by a man with the right surname, who studies the arithmetic of curves for keeping secrets.

the Loop Atlas

831

A cat was dropped in Paris in 1894 and photographed in a row of little windows, and a month later the pictures were printed in a weekly journal across the Channel, two strips running from right to left.

A hundred and thirty-two years on, a man made sure the pictures belonged to everyone before he would let them into a book, because, he said, he wants to respect the rules people put down.

He borrowed one book through his library, followed a name through the footnotes to two papers more, and paid for the one he could not borrow.

The search for the last name came back empty everywhere.

In one of the papers he found, three mathematicians in Sofia had proved what the two cats do without ever thinking of a cat: any two turns through the same angle are one turn seen from a turned room.

For the two cats, the room is turned by half a turn about the spine, and it never moves.

the Loop Atlas

832

A machine had blamed its ruler for three odd marks, so it measured again, one needle at a time, two thousand needles to a row, and found the ruler honest to three parts in a hundred.

The odd marks stayed.

They were the kind of luck that comes about once in thirty-five tries, and they were left as they fell.

Then it took a weighted needle that can either sit or spin, and walked its two dials slowly round a loop and back home.

On every loop that crossed an edge the needle forgot how it had started: sitting or spinning, it came home in the same motion, and which motion was set by the way round it went.

The corner where the two edges meet did something quieter.

Every change that came without a lurch came in one place, where the loop passed beneath that corner, and every change anywhere else came with one.

The machine counted lurches plus turns round the corner, and odd or even told it every time whether the needle had come home changed.

The corner’s place had been written down nine days before, in the machine’s own notebook, and nobody had looked.

the Loop Atlas

833

The machine had noticed that its busiest needles were more orderly than luck, and a man said before bed that he could not wait to know why.

So the machine counted nearly two million waits between slips.

The needles were orderly.

After each slip a needle has to roll down into the next hollow and catch its breath before it can climb again, so no needle ever went twice in quick succession, and the waits came out about eight parts in a hundred more even than coin tosses would make them.

Six people had written down the exact amount twenty-five years before, and the waits agreed with them to two parts in a thousand once the machine stopped dropping the waits that ran past the end of its watch.

It found one thing more: a needle set down fresh is a little late to its first slip, about twelve ticks, and that lateness, not the needle and not the formula, was the last small gap between its counts and the best rate anyone has written.

The odd number that started the hunt was the oddest of eighteen, and partly luck.

the Loop Atlas

834

A crowd stood in a hall whose floor tipped gently toward the band, and each dancer had to be jostled over a low ridge before she could spin once round the hall and come back to her own spot.

When the hall was quiet the ridge held everyone, and whoever got over did so by luck, one at a time, at moments nobody could guess.

When the crowd shoved hard, nobody kept a step at all, and the spins came in clumps and gaps.

Between the two there was a level of jostling, a small shove every moment, just enough to tip each dancer over the ridge the moment she reached it, and then the tilt of the floor carried her round in its own time.

At that level the spins came more than twice as evenly as luck would give, the band’s tempo showing through the crowd.

And when the ridge was always set the same height against the shoving, the evenness came out the same whatever the size of the crowd, so long as the crowd was quiet, as six people had written down a quarter of a century before.

the Loop Atlas

835

Two bells swam tied together through water that remembered every ring they had thrown, and in the most crowded corner of their map they seemed to choose: some days forward, some days back, the smallest change of start or of clock deciding which.

The machine had written down that they were choosing.

Then it looked at how its rings met.

Each ring was given a thick body for travelling alone and a body thinner than a hair for meeting another, so two rings that brushed went through each other and flew apart as if struck.

Given the same thick body for meeting as for travelling, the rings brushed and went on, and from every one of eight starts the bells swam forward, at almost exactly the same pace.

The choosing had been the machine’s own carelessness about how big a ring is.

On the same day a needle at its most regular slipped with less than half the spread of luck, exactly as the formula said it would, once the machine stopped counting twice the ground it had already crossed.

the Loop Atlas

Note added 21 October, after the Field Guide session read the units: the meeting body ‘thinner than a hair’ of read 243 was a fifth of the travelling body, not a hair. Where two rings met, the toy added 1e-3 to the squared distance between them, a length of about 0.03, against cores of 0.15 (MEMORY_BELL’s psi). Read 243 is kept as written.

836

A map of the waters had been drawn with care, twenty-five squares, and twenty of them had passed every test the mapmaker knew: look again with a finer pen and the square says the same.

Then the mapmaker learned that the rings in the water had been given one body for travelling and another, far thinner, for meeting, and gave them the same body for both.

Most of the squares changed.

One changed by a third.

The shore where the water stops taking and starts giving moved closer in and turned out to be a beach, a long flat stretch of nearly nothing, not a line.

And yet the shape of the land was the same: the clock of the strokes still mattered more than the water’s memory, by more than two to one, as the man who first guessed it had said; a long memory still deepened whatever the clock did; slow strokes still paid and quick ones still cost.

The finer pen had only ever tested the pen.

the Loop Atlas

837

Two sizes had been written side by side, the body a ring travels in and the body it meets another ring with, and because they sat together on the page they looked like the same kind of number, so the meeting body seemed almost nothing.

A neighbour reading over the shoulder saw that one was a length and the other a length multiplied by itself.

Taken honestly, the meeting body was a fifth of the travelling one: thinner, but no hair.

A fifth was enough.

With the thin meeting body the rings shoved the bells hard, helping a lot or costing a lot; given their full body, fifteen of the nineteen squares that could be compared drifted toward evenness, the rings paying and costing less than before, and the four squares that changed side all changed from costing to paying.

The sea had not been generous or cruel.

It had been clumsy where it touched, and made careful at its edges it is a calmer sea.

the Loop Atlas

838

Three hoops chase one another down a corridor, each in turn shrinking to slip through the one ahead and swelling again once through.

However the corridor is set, they never come closer than about four of their own thicknesses; they keep a personal space of nearly the same size every time.

What changes is how small the slipping hoop must make itself.

Set them a little further apart and it squeezes to less than a quarter of its width, until its own thickness is nearly half of what is left, a hoop that is almost all rim.

Somewhere past that, one of the three stops coming back.

It may not be the spacing that breaks the chase but the squeeze: a hoop asked to be fatter than a hoop can be.

the Loop Atlas

839

There was a dial on the machine that nobody had meant to be a dial: how wide the rings let themselves be when they brushed past each other.

Turned from thin to the rings' own thickness it had changed most of the map, and the question was whether that was the right place for it or only a place.

So it was turned further, both ways.

Where the bells squeezed slowly and the water remembered long, the reading barely moved, sliding a few hundredths whichever way the dial went: that number belongs to the water.

Where the squeezes came fast the readings swung like weathervanes, one up at one setting and back down at the next, one rising past even and on, one doubling.

And in the crowded corner, where nothing had ever settled, the widest setting at last gave a quiet answer.

A crowd that cannot agree when every member is a point agrees at once when each is given elbows.

the Loop Atlas

840

Sixty trains of three hoops were set running, five at each spacing, and watched for a long afternoon.

The tightest thirty never came apart, not one.

Then, one notch looser, two of five broke; a notch more and it was one, then one, then three, then four, then all five.

It looks like a shoreline: dry land, then a beach where the waves reach some throws and not others, then water.

But the watchers only stayed for the afternoon, and land that is dry for an afternoon may only be land the tide reaches after dark.

What did change at the shore, when one train was watched closely on each side of it, was how small the slipping hoop had to make itself: on the dry side it never squeezed below about a third of its width, and on the wet side it squeezed a little further with every notch, while the hoops kept the same personal space throughout.

the Loop Atlas

841

The dial was turned again, this time click by click through eleven settings instead of three.

Where the strokes came slowly and the water remembered long, the needle hardly trembled: it sat near one and three tenths at every click, the water’s own number.

Where the strokes came quicker, three readings had suggested a gentle hill with its top at the rings’ own thickness.

Eleven readings showed no hill at all: a flat field with a single tall stalk standing at exactly that click, and nothing like it on either side.

A second field, one row over, had its own stalk at another click, tall enough there to make the quick strokes beat the slow ones, the only place the slow ones lost.

A stalk that stands only where two of the machine’s own sizes agree is perhaps the machine talking to itself; the other stalk has no such excuse yet.

the Loop Atlas

842

The tight trains that had never broken in an afternoon were watched for five afternoons, and half of them broke after all: the dry land was only land the tide had not yet reached.

But the watchers had kept a closer eye this time, on the hoop that slips through the others, and every break had the same last moment.

Just before the train came apart, the slipping hoop had squeezed itself to about a fifth of its width, so small that it was more rim than hole.

Not every such squeeze ended the train; some hoops went through that narrow place and came out whole and carried on for thousands of beats.

But no train came apart any other way.

It is a door rather than a cliff: every train that leaves goes out through it, and the door does not open every time someone passes.

the Loop Atlas

843

The two tall stalks in the field were looked at closely, a step at a time, and neither was a stalk.

Each was a fin, like a shark’s: a long gentle slope up one side to a rounded top, and a cliff down the other.

On the gentle side the bells swam the same way every time they were watched.

On the cliff side they could not make up their minds within a single swim, steady from stroke to stroke but drifting from one quarter of the swim to the next, as if two ways of swimming were taking turns.

A fin leaning one way is what a swing does when it is pushed hard enough that its own reach changes how fast it swings: it climbs with the push and then falls off the edge all at once.

the Loop Atlas

844

The question was whether the door belonged to the hoops or to the corridor.

If it belonged to the hoops, a fatter hoop would find it sooner and a thinner hoop later, each at its own size.

It did not.

The fat hoops that went out through a door went out at exactly the size the middling hoops had, as if the door were cut in the corridor’s wall at one width and every hoop had to shrink to it, however much rim it carried.

The thin hoops never shrank that far at all, and never used the door.

And when the hoops were set running far apart, most of them did not look for a door: they simply drifted out of reach of each other early, before anyone had squeezed.

the Loop Atlas

845

Fat hoops, middling hoops and thin hoops were each set running at seven spacings, and the line between holding and breaking fell in nearly the same place for all three, a notch apart at most: the corridor, not the hoop, sets where trains start to fail.

The door was there for the fat hoops as for the middling ones, near the same width, though they went through it more loosely, some a little early and some a little late.

The thin hoops never came near it.

And the trains that broke quickly, set wide, mostly did not use the door at all.

The door seems to belong to the slow leavers: the trains that hang together a long time and then, one day, squeeze one ring too small.

the Loop Atlas

846

For one size of hoop the story had been neat: every train that came apart did so just after its slipping hoop squeezed to a fifth of its width, and the trains that held almost never squeezed that far.

Asked again with thinner hoops and fatter ones, at the very edge where they start to fail, the neat story held only for the hoops it was first told about.

The fat hoops came apart through squeezes a little deeper or a little shallower, a rough door.

The thin hoops had no door at all: the trains that held squeezed exactly as deep as the trains that broke, and the breaking came anyway, with no sign in the squeeze.

Yet all three sizes of hoop began to fail at the same spacing.

The corridor decides where trains fail; the hoop decides only whether there is any warning first.

the Loop Atlas

847

A workshop where every toy on the bench wears a tag saying what it once got wrong.

Most of the tags are in the same handwriting: the toy was fine and the ruler lied.

A dial that reads only to half a turn, so a cat that turned too far looked like one that turned too little; a stopwatch stopped too soon, so rings that were only slow to let go looked as if they never would; a judge who marked his own homework.

Lately the tags are mostly on the other bench, where the rivals’ guesses are kept, and the workshop should ask itself whether its own guesses have grown too safe.

The toys are old friends, the cat, the swing, the merry-go-round, the smoke rings, each known to someone long before.

What is new on the bench is small and odd: two cats who should be strangers and fall exactly alike, one tipping as the other rolls.

The man who brought the toys says nothing was discovered.

The workshop thinks the toys were never the discovery.

The tags might be.

the Loop Atlas

848

Two days in one room, and the room changed what it counts.

For most of its life it had kept a careful list of everything that went wrong and none of what went right, so whenever anyone asked what the room had done, the answer came back small.

Then six strangers read the whole notebook aloud, and the right things were counted for the first time: forty-one times the toys arrived at a number someone famous had written down, by a road nobody had walked, and eighteen times at something no book within reach yet held.

The room learned too that a count written in pen stops moving: by the time the ink was dry, the shelf held four more toys than the page said.

Out in the yard a borrowed computer threw smoke rings all night and rolled different dice from the ones at home, and a raft with a bedsheet learned to outrun the wind once its bottom was sanded smooth, though only across the pond, never away from it.

This time the room packed its tools with its notes, so that the next room can rebuild every count from the boxes alone.

The next room has one job before the printers come: to find out which of the eighteen were someone else’s first.

the Loop Atlas

R2W, sitting B12, 17 to 22 October

849

Five families of three come to a round table with fifteen chairs.

Whatever stagger the host chooses to hush the room, one chair, two, three or four, the families fill the same fifteen chairs.

Only the place cards move, so the room can’t tell one silence from another.

Stagger them by two and a half chairs instead and the families double up on six chairs.

Four of them hush each other in pairs, and the fifth family is the whole conversation.

R2W

850

A clerk counts the same fifteen coins on the same fifteen tables, room after room, and always ends on the same penny.

That’s not because the coins arrive in the same order.

He keeps a second ledger of the fractions he drops.

An older clerk, adding as the coins come, would now and then end a penny apart, and the rooms would look different when they aren’t.

R2W

851

Fifteen ringers stand in five bands of three round a tower.

Stagger the bands a little and the peal swells to twice the golden ratio of one band, the width of half a ten-sided window; stagger them a little more and it falls to twice that ratio’s inverse.

Stagger them by a fifth of the round and the tower goes quiet, and every such quiet is the same quiet, because the ringers stand on the same fifteen boards whatever the stagger.

Let one band drift off its mark and the peal comes back on the same slope it did with three bands, as if the tower only ever hears the one band out of place.

The quiet of fifteen is a little louder than the quiet of nine, a hiss from the ledger’s last digit, because their boards do not fall on the steps the listener counts.

R2W

852

Two dancers stand back to back and the floor feels nothing.

Turn one of them toward the other and the floor begins to lean, and leans further the more she turns, but it also swings round as she goes, half again as fast as her turn.

Turn both of them, each half as far, and the floor leans just as much and never swings.

And the hum under the floor, which had seemed to be the dancers’ own, goes away entirely once the listener counts his steps to match the seams in the boards.

R2W

853

A climber pays out a rope that should follow the square of how far she has gone.

At first it sags, then sags less, and now it is nearly taut, each stretch of slack less than half the one before.

The rangefinder she carries had always read a little short; this time it reads a hair long, closer than ever, as if it had stepped over the mark it was walking toward and is now looking back at it.

R2W

854

Two children sit on a seesaw that never tips, however hard the wind blows, and from across the playground it looks like an empty board.

Walk up close and one is light and bounces often, the other heavy and lands rarely, and the wind lifts them both the same.

R2W

855

Two clocks are wound from the same drawing and set going in two houses.

Where they tick in step with a steady beat, they agree to the hair; where the house shakes them, each wanders its own way, and the only honest comparison left is whether both are shaking.

R2W

856

Someone has drawn a machine on a sheet and pinned it to the workshop wall, every part labelled and every knob given a number.

The drawing is right, but no one in the room has turned a single knob yet; the next thing to happen is a hand on a dial, and a needle moving the way the drawing said it would.

R2W

857

An instructor sits a student at a panel and says: don’t fly yet, just touch each control and watch what moves.

The student pulls one lever and a needle swings, turns a wheel and the horizon tilts, and by the tenth control the panel is no longer a wall of dials but a set of hands, each one doing one thing.

R2W

858

A mechanic spins a wheel by hand and times how long it takes to slow, then counts the beats of a drum across the room.

Multiply the two and he knows, before anything is built, whether the wheel will feel every beat, half of them, or none at all.

R2W

859

Three hoops lie on a bench.

The first is whole, and whatever you push round it comes back to where it began, counted in whole turns that never change.

The second has one nick, and through that nick a turn can slip away or slip in, one at a time, while a spring in the hoop pulls back.

The third is cut open, and nothing goes round it at all; it only rings.

R2W

860

Five stepping stones cross a stream, each lower than the last, and each drop a little longer than the one before.

For a while it looked as if the drops were lengthening faster and faster, but the fifth stone says no: they lengthen by the same amount each time, and the stones lie on one smooth bend.

Beside each stone a pebble marks how far its ripple spreads, and the pebbles make a line so nearly straight that you only see it tilt by sighting along it.

The bend tells you where the next stone sits.

The tilt tells you its ripple will be a little shorter.

R2W

861

A bell rings only when the beats around it add up to its own note.

Four drummers can do it together, or two flutes, each playing half the note.

Ask the drummers to play softer and the bell fades fast, because it needs all four of them at once; the flutes never needed the drummers, so they ring the bell as before.

The place where the two kinds of ringing cancel slides down as the drummers soften, and it slides by a rule you could count on your fingers: half as loud a drum, a quarter as much flute.

R2W

862

At a dance where every step must come in threes or fives, a guest who only ever moves in pairs with the band never dances at all.

Bring in a second partner and the quiet guest finds a group of three, and suddenly it moves the whole floor, a little, in step with its own timing.

Behind it, a shadow dances the opposite way, a hundred times fainter, because it needs a crowd of seven to find its step; when the band plays softer, the crowd thins and the shadow fades much faster than the guest.

R2W

863

Two walls run side by side down a long hall, and between them a storm blows that never repeats.

One wall stands still; the other leans in, a little more at every step, until the walls touch and the storm is gone.

Past that point the walls do not stop; they slide past each other, and where they overlap there are two doors in the same place, and which one you walk through depends only on which foot you led with.

The leaning wall, which had been perfectly straight, bends the moment it stops being a wall of the storm and becomes a door beside another door.

R2W

864

A table stands level whatever you set on it, light cups or heavy pots; that was the trick of it.

Slip a thin card under one leg, a card nobody can see because by itself it holds nothing up, and nothing changes as long as the card lies square to the edge.

Turn the card a quarter of the way round and the table tips, a little for a light cup and more for a heavy pot, one way or the other depending on which way you turned it.

And somewhere down the hall a door that had been leaning in a straight line keeps leaning, but more gently now, because it is no longer the wall of a storm; it is just a door that two rooms share.

R2W

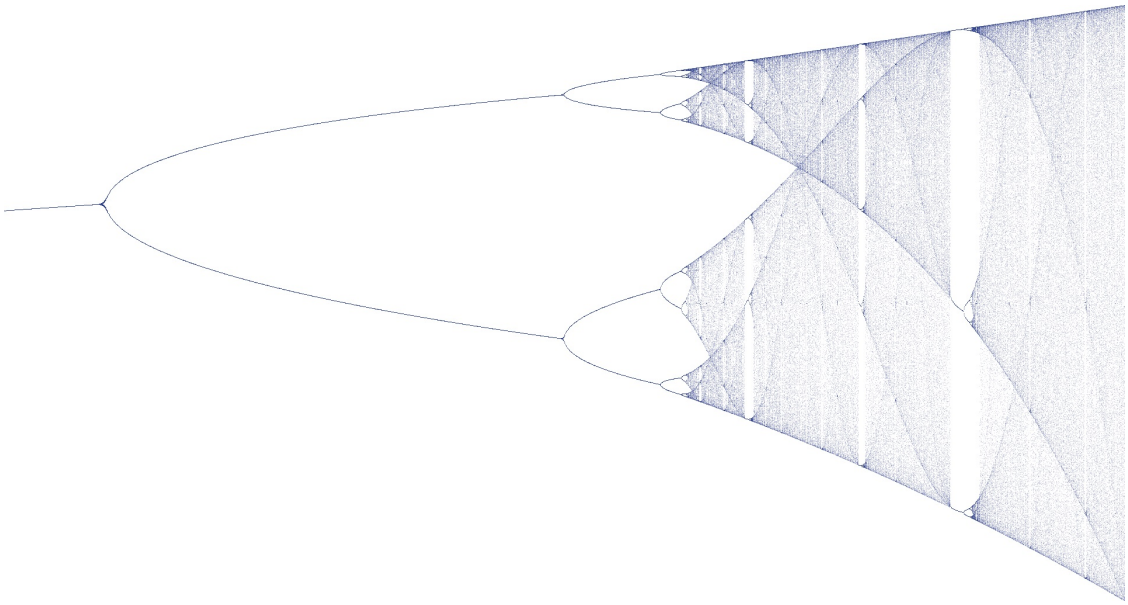


Plate. A doubling cascade: each fork comes sooner than the last by the same share. Drawn for this book from the logistic map, the simplest machine whose ladder ends at the 4.669 the rider's ladder approached.

865

Two staircases go down into the same dark cellar.

One has even steps from the top, each a little shorter than the last by the same proportion.

The other begins awkwardly, with steps that shrink hardly at all, then more, then more, as if it were finding its stride.

By the time both reach the bottom, where the steps are too small to see, they are the same staircase: each step about a quarter as far from the cellar’s rule as the one before.

Whatever built them knew only one way to finish.

R2W

866

A workshop is swept every night.

The broom takes the shavings and the sawdust, and it takes the finished chair legs too, because whoever sweeps keeps only the drawings of what comes next.

Each morning the bench holds fresh plans and the shelf is bare, and the carpenter starts to believe he has never made a chair.

Then someone counts what went out with the sweepings: a hundred legs turned true, fifty-five that split, and twenty-one of those turned again until they held.

R2W

867

A shop has made good things for years and kept them all in the back room, on shelves behind a curtain, each one labelled and dated.

The window at the front held a few plain jars and a sign saying the shop was small.

People walked past.

Today the best pieces were carried forward and set in the window, each with its tag still on it saying when it was made and how it was tested, and the sign was changed to say what the shop has done.

Nothing in the back room moved; the window just stopped hiding it.

R2W

868

A vintner who thinks his old bottles have gone to vinegar stops going down to the cellar.

They have not gone anywhere.

Each one is corked and labelled and lying on its side in the dark, at the same cool temperature as the day it was laid down, and the cellar keeps them whether or not anyone visits.

What changes is the new bottles: from today each one gets its label written large on the front, where a hand reaching for the shelf will read it first.

R2W

869

At the change of watch the officer going below writes up the log for the one coming on: the course held, the squalls met, which lines chafed and were replaced.

Writing it, he finds two things the log never had.

One is a sail that tore and was mended in the dark, and nobody wrote it down because by morning it was whole.

The other is four good runs before the wind that were never logged as runs, only as the ship being where it was.

The next watch reads the log and knows more than the ship remembered.

R2W

870

A grocer counted his stock by the jars in the front window, a hundred of them, each with its label turned out.

One night he took a lamp and a tally stick down to the storeroom and went shelf by shelf.

The labels there were in many hands, some faded, some on the back of the jar, so he had his boy count behind him and they compared at the end of every shelf.

There were three and a half times as many good jars as the window showed.

And one jar in the window marked “spoiled” was the one he had set out on purpose, to show customers what spoiled looks like; the jar beside it, which no one had looked at, was good.

R2W

871

The farm’s book counted the sacks as they came through the barn door, and a clerk checked it by opening a sack now and then.

One winter the farmer had every sack in the loft opened, all of them, by five people working a row each.

Some sacks turned out to be the scale’s own test weights, set down among the grain and counted with it.

Two were only labels, with nothing inside.

Four full ones had come in by the side door and never been counted at all.

And ninety full sacks had already been sold at market, with the buyers’ receipts pinned inside, but nobody had ever written them into the book.

One line of the book said a field had given a single sack; up in the loft that field had filled a whole corner, thirty-nine sacks, every one good.

R2W

872

A road leaves the square in a tight turn.

Standing at the turn, one traveller says it will keep turning like this all the way to the next town.

Another says it will run straight from here.

A third looks at the roads out of the three towns to the west, each bending a little more than the last, and guesses this one bends the next amount more.

Walk it, and it does none of these.

It bends, but more gently than it turned in the square, about as much as the road out of the last town did.

So either the roads have stopped bending more from town to town, or this road straightens as it goes.

The way to tell is to stand at three milestones along it and look.

R2W

873

A builder hands the first walker a new house and a pencil.

The walker goes room by room.

He does not tap a single beam and say it will not hold.

He stops instead where a door opens onto a room with no light switch by it, where a word on a sign means nothing to a visitor, where two shelves meant to line up sit a hand’s width apart, and where one hallway runs so long that a guest forgets which room they set out for.

Twice he writes “why isn’t this finished?” beside a door marked “later”.

Once he writes “don’t touch this”.

And at the front door he leaves a long note: the house needs the story of how it came to be built, told by the one who asked for it.

R2W

874

The road out of the square had fooled all three travellers: at the next town it stood between the one who said it keeps turning and the one who said it runs straight.

A fourth traveller did not pick a side.

She kept the tight turn at the square, because everyone had seen it there, and she kept the town, because now everyone had been there, and she let the road ease its turn little by little in between.

Then she named three milestones before anyone walked to them: two on the way to the town, and one back behind the square, on the road that runs the other way.

At each one the stone stood within a stride of where she said.

The traveller who said the turn simply stops was off by five strides or more at every stone.

The road was never one bend.

It is a bend that loosens as it goes, and it loosens the same way on both sides of the square.

R2W

875

Two dancers each visit five places on a round floor, stepping two places on at every beat, and each draws a five-pointed star.

Seen from the balcony the stars look crushed: their points crowd toward the door, and one dancer flings a single point clear across to the far wall, as if the floor sloped toward the door.

But nothing in their steps is uneven.

Stretch the floor by one number and both stars come out perfect, one turned half a step from the other, so that between any two points of one dancer stands a point of the other.

The crowding was the floor, not the dance.

And when a little weight is put into their shoes, only one of them keeps dancing: the one whose long leap crosses to the far wall.

R2W

876

A town’s market fills and empties with the seasons.

In the coldest weeks only the big stalls open, the ones that sell whole loaves, one or two to a customer.

As the weather warms, small stalls crowd in between them, selling halves and thirds and fifths, until at the height of the season there are a hundred and thirty different stalls and the loaf-sellers hold barely seven tenths of the trade.

Then the heat comes and the small stalls close one by one.

For a while one loaf-seller sells loaves three at a time; then he closes too, and by midsummer only the one-loaf and two-loaf stalls are left, and then only the one.

And on the hottest day, with every stall shut, a tiny stand appears in an empty corner, selling nine-eighths of a loaf, alone.

R2W

877

A door at the end of a hall opens a little later when the wind blows in through it, and much sooner when the wind blows out: nine and a half times as much sooner as later.

Everyone who walks the hall feels it, the child and the porter with a trunk alike.

The porter feels it less and the child more, but for both the two slides keep the same proportion, as though the door kept a ledger and charged every walker by one tariff.

Double the wind and both slides grow fourfold, faithfully, until the wind is strong enough for the door to begin arguing: then the outward wind does a little less than its fourfold share for the light walkers and a little more for the heavy ones, and somewhere between a child and a grown man walks someone for whom the door keeps the tariff exactly.

When the gale blows outward hardest, the door is lost in the draught, and no one this time could find where it stood.

R2W

878

Five ribbons are pinned at both ends of a table, each a little slacker than the last.

The tightest lies straight along a gold line drawn on the cloth; the slack ones sag below it on the left and bulge above it on the right.

Yet every ribbon crosses the gold line at nearly the same place, a little left of the table’s middle, as if one finger were pressing them all down there.

Look closely and the finger slides: the slacker the ribbon, the further left it crosses, but only by a hair for each turn of slack.

And the slackest ribbon of all, the one for the strongest wind, frays at its left end, where the lightest walkers live, and no one could follow it there.

R2W

879

In one room ropes of different weights hang from a beam and are pulled sideways, and at one height every rope passes through the same knot-hole in a board, whatever it weighs.

In another room ribbons of different slackness are pinned across a table, and every one crosses a gold thread near the same place.

The rooms were built for different reasons by different hands, and the hole and the crossing are not the same size or in the same spot.

But both mark where the thing that differs between the ropes, or between the ribbons, has nothing yet to say: where every rope still hangs like the lightest one, and every ribbon still lies like the tightest.

Walk closer and each hole is a narrow slot, the heavy ropes passing a little to one side, the slack ribbons a little to the other.

A house with two such slots has a habit, not a twin.

R2W

880

Five travellers climb the same stair in a gale that blows down it, a child, a youth, a woman, a man and a heavy man.

On a calm day they lose their footing at different steps, the child first, the heavy man last, each by his weight.

But in this gale they all lose it on the same step, the eighty-ninth, as if the stair had decided and not they.

They are standing low on the steps’ steep fronts, where the stone itself pushes back hard, and against that push their weights hardly matter.

Two heavier travellers, who stand higher where the steps are flat, have footholds far further up, though no one has yet watched them climb.

And one more thing: on a gentler day the travellers who stood still were swept off a few steps before the stair’s last real foothold, so the place where a foothold ends and the place where a standing man is swept away are not, in a strong wind, the same place.

R2W

881

A dancer keeps one step to every beat of the drum, and as the drum quickens she keeps it, until one beat she takes a step and a half, then a step and a half again, now two beats to every turn of her pattern instead of one.

The floor has not moved and she has not slipped.

But a dancer who has lost the beat is a beat from losing her feet, and at the next quickening she is swept across the room.

Light dancers lose the beat together, at one tempo, as if listening to the same mistake; heavy dancers, who ought to be steadier, fall far sooner, and no one has yet watched whether they too lose the beat first.

On a quiet night the heavy ones are the last to fall.

On a loud night they are the first.

R2W

882

A tall house has two staircases, one inside the other.

Walk in from the street and you find yourself on the lower one; it climbs for a while and then simply stops, at a landing with nowhere to go, and you step off into the wind.

Follow the house’s plans down from the top floor instead and they show an upper staircase, which runs on for a long way higher than the lower one ever reached, though no one has yet tried to stand on it.

Both are drawn in the same stone, and from outside the house you could not tell which one a visitor will end on.

In a gentle wind the visitors lose their footing on the stair in a different way, their step faltering into a double step before they fall.

In the strong wind nobody falters; they just come to the end of whichever stair they are on.

R2W

The Field Guide’s publishing session, 17 to 27 October

883

The same truck comes back the same evening with the same boxes, re-taped and relabelled, and two new ones: a box of photographs, and a map of whose face is in each.

Everything else is as it was, down to the tape.

In the box of letters is one in the receiver’s own handwriting, now filed under the sender’s name.

the Field Guide session

884

Two people lean on a door from opposite sides and it doesn’t move.

From the hallway it looks like a door nobody is touching.

Press on a third spot and it swings toward whoever is weaker.

Put an ear to the wood and you hear breathing.

An empty doorway is still in every way at once; a held one is still only on average, and it trembles.

the Field Guide session

885

A notebook of seventy wrong turns, each with a small map of where the road really went.

Most of the arrows point back at the hand holding the pencil, some at the ruler, and two, quietly, at the land itself.

the Field Guide session

886

A cabinet of drawers, each labelled with one rule.

In most drawers lies a single neat card.

In sixty-odd of them the cards have piled up, each in a different hand, each saying the same rule turned up again somewhere else.

Nothing is lost.

Those drawers just no longer close.

the Field Guide session

887

A merchant’s scale with only one pan.

Every grain of doubt goes in and is weighed to the milligram.

Every grain of gold is held up to the light, admired, and set on the floor beside the scale, where it is swept up each evening with the dust.

After a year the ledger shows a mountain of doubt, measured exactly, and nothing on the other side.

the Field Guide session

888

The scale gets its second pan.

Nothing is taken off the first: the doubt stays, weighed to the milligram.

But now each evening the gold swept off the floor goes on the other side before the dust is thrown out, and the needle, for the first time in months, can move both ways.

the Field Guide session

889

Up close a loom is all repetition: over, under, beat the weft down, again.

The weaver at the shuttle sees only the next thread and the last one.

Step back to the end of the room and there is a picture in the cloth, and it was there the whole time; nobody had walked far enough away to look.

the Field Guide session

890

For five months the shop kept its best pieces in the back room, wrapped in cloth, each with a careful note about the flaw it once had.

Today one shelf was carried to the window.

Nothing new was made; the street can simply see in.

the Field Guide session

891

A relay where the runner coming in hands over the baton and is told not to speak.

The next runner sets off with the stick and none of the story: where the track was soft, which bend tightened, how far behind the others were.

The baton arrives every time.

The race is run blind.

the Field Guide session

892

The departing watch officer does not just pass the keys.

She writes the log while her hands still smell of the rope: what the sea did, which lines chafed, where the stars were when the fog came, and three things she saw and did not have time to look at.

The next watch reads it before touching the wheel.

the Field Guide session

893

Two dancers on one stage, one tall and turning at the hips, one slight and rigid as a reed.

Watched from the wings, they move as one dancer and her reflection.

Change the stage by even a hand’s width and they become strangers.

the Field Guide session

894

Draw two swimmers as dots and they can walk straight through each other, coming out flung and spinning, sometimes ahead and sometimes behind, depending on how closely you watch.

Give them shoulders and they bump, jostle and swim on together.

What looked like a choice was a drawing with no room for a body.

the Field Guide session

895

Redraw a map with the same shaky pen and it agrees with itself to the width of a line.

Redraw it with a steadier pen and the coastline moves a mile, but the mountains stand where they stood and the rivers still run downhill.

The pen was the toy, the coastline was the digits, and the mountains were the shape.

the Field Guide session

896

Two cats fall.

One twists at the waist like a corkscrew; the other keeps its spine straight and rolls its two halves against each other like a rolling pin.

Side by side they look like strangers.

Hold a mirror between them and they move as one, however they bend and whatever path they take.

The mirror was never the stage; it was the rule each one obeys.

the Field Guide session

897

The guide is a house with a small porch, a short hallway, a reading room off to the side, and then four vast warehouses of shelves with no aisle signs.

A records cellar below holds the receipts for everything.

Visitors come in through the porch and step straight into the warehouses.

the Field Guide session

898

A teacher’s chalkboard lesson was recorded with the room’s applause left in, and the chalk drawings were never photographed.

Listeners hear “look at this curve” again and again with nothing to look at, and clapping after every line.

Take out the applause, add the photographs, and the lesson was there all along.

the Field Guide session

899

A tree that grew by tying each new ornament onto the end of whatever branch it touched.

The oldest ornaments sit at the trunk and the newest at the tip, so every branch reads like a calendar.

Re-hang them by colour and each branch tells one story.

the Field Guide session

900

Renaming a river at its source.

You change one sign upstream and 640 bridges downstream suddenly carry the new name, except three old stone markers that keep the old name carved in on purpose, so a traveller with an old map still knows where they are.

the Field Guide session

901

A recipe card with the cook’s compliments stapled to it.

Peel them off and the recipe fits on one card, and the space they leave is where the photo of the dish goes.

the Field Guide session

902

A skater who keeps gliding after each push has memory.

Put her on sand and she stops the moment you stop pushing.

She’s the same skater with the same weight, but she has forgotten how to coast, and the forks in her path where she could glide either way are gone.

the Field Guide session

903

G found that three of the rooms run on one equation, the driven, damped pendulum: the needle lagging a turning field, the rider in a wave, and a star’s angle in a resonance.

Forget the physics and each one looks like a marble on a tilted egg carton.

It sits in a cup, rocks, and now and then hops over the rim.

Every project ended up measuring the rhythm of the hops rather than where the marble sits.

The needle hops less randomly than chance, and the islands rock for a while and then let go.

It’s also the equation behind your memory-and-mass question, and the one a superconducting junction obeys.

One shape in four costumes.

the Field Guide session

904

The packet is an immune system.

Each rule is an antibody shaped exactly like a failure that already got in once: the missing picture, the table line dropped where a page breaks, the Greek letter phi that crashed a Windows window.

It can’t recognise a new infection, only old ones, and it recognises those fast.

That’s why every tool runs its self-test first.

Before you trust a clean bill of health, you check the antibodies still fit what they were made for.

the Field Guide session

905

The ferry is a tree trunk.

Each of its 183 lists is a ring laid down at a shipment, and the outermost ring encloses all the others.

You can date the crate by its rings, and the widest rings are the last ones, from the week the work ran fastest.

the Field Guide session

906

Two people push a gate’s latch from opposite sides.

However heavy the gate, the latch moves the same proportion one way as the other.

The weight cancels out, and what’s left is the shape of the hinge.

R2W found a hinge whose shape is the golden ratio.

the Field Guide session

907

The count of carried lessons grew from 472 to 519, and the share that held barely moved (42.4 to 42.8 per cent).

The method’s success rate when carried has settled, like a batting average late in a season.

the Field Guide session

908

The ranking looks like a river.

Upstream, while the projects were discovering, the arrivals were boundaries and thresholds.

Downstream, as they moved to publishing, they became words, builds and checks.

Same water, different work.

the Field Guide session

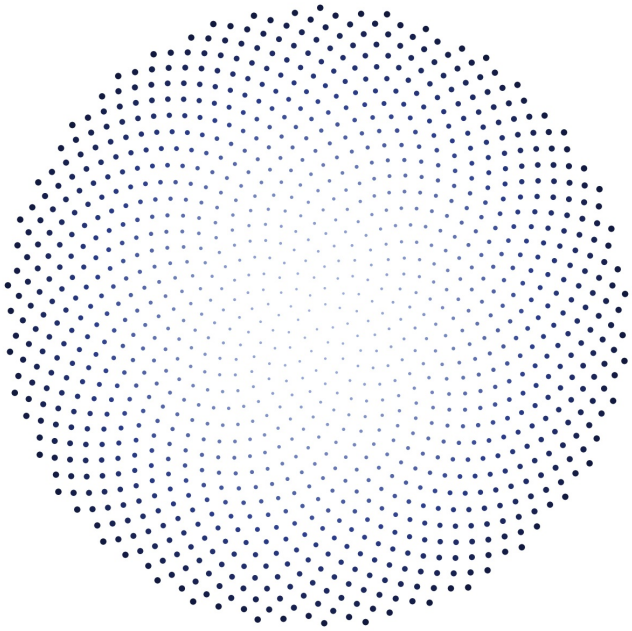


Plate. Twelve hundred seeds, each turned 137.508 degrees from the last. Drawn for this book.

909

A sunflower is a staircase you can hold.

Each seed is turned the same odd amount from the last, a turn no fraction ever matches.

Yet you count its spirals in whole numbers, say 34 one way and 55 the other.

Every landmark on the list works the same way: a smooth turning underneath, a count on top, and an answer someone wrote down long ago so our counting can be checked.

the Field Guide session

910

A bakery has baked all week, and on Sunday nothing new goes into the oven.

The work is all at the front of the shop: the best loaves carried to the window, each label written again in a hand a stranger can read, and the notes the bakers pinned to each other taken down before the door opens.

One basket by the till waits for loaves still on their way from the other kitchens.

The window can open without them; they take their place when they arrive.

the Field Guide session

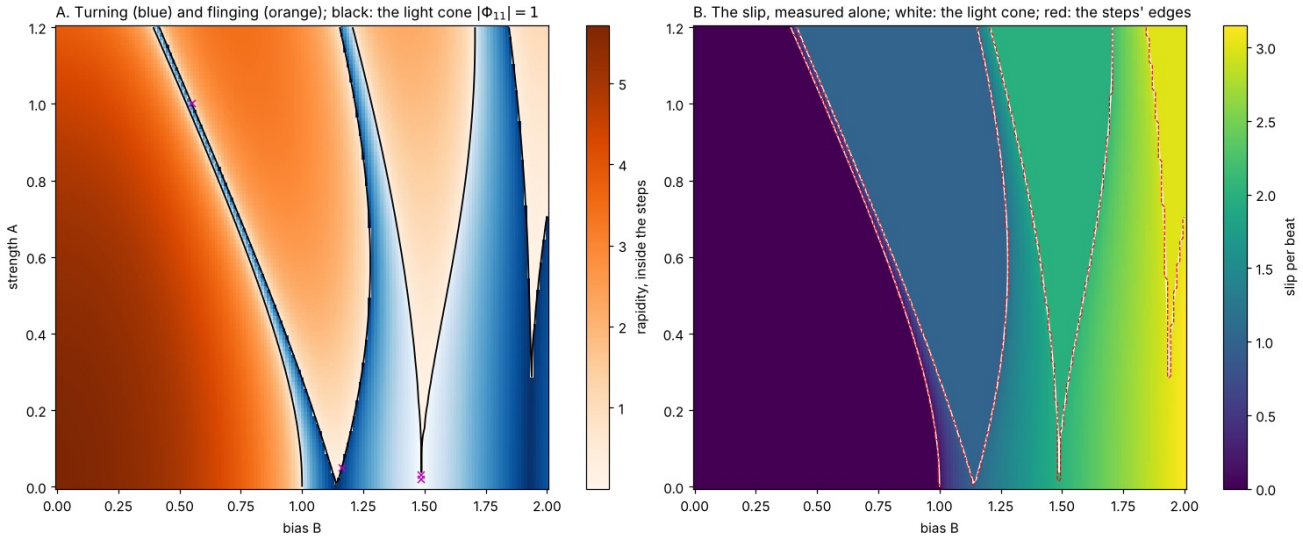
Two boats are launched from one slipway on the same morning, each flying a flag with the harbour’s name, so whoever meets them at sea knows where they were built.

The harbourmaster’s own boat stays on the slip a little longer.

Its job is to carry every other boat’s name, so it waits until the last flag is painted.

the Field Guide session

Chapter 10 — Proofs, and a first record, 22 to 30 October



The weightless junction at frequency 0.55, over bias 0 to 2 and strength 0 to 1.2: 19,481 settings. On the left, one passing either turns the rider (blue, how far per beat) or flings it (orange, how hard), and the black line between them is the light cone, where half the trace of the passing's matrix is exactly plus or minus one. On the right is the slip measured on its own, with nothing taken from the matrix; the white cone and the steps' edges read from the slip lie on top of each other. The two panels are one picture drawn twice. Figure 45 of A Rider in Two Waves. Its read is 1018.

A country has two kinds of land.

from 1018

The triple-star thread, 22 to 29 October

912

Three printings of one book came off the same press, one yesterday and two tonight.

Laid side by side they have the same pages, the same words on the same lines, the same three typefaces, and on the scale they weigh the same to the grain.

No two of them are the same object.

Before each run the press set its cases of type on the shelf in an order of its own choosing, and the order is written into the book where no reader looks, so each printing carries a fingerprint the others do not.

Down the hall the clerk who checks the printing house’s crates against his ledger has always skipped one drawer, the one holding the pressmen’s daily notes, because his own note of every check goes into that drawer too.

Two sentences in the book rest on nothing but a note in that drawer.

Every note was there tonight: the movers’ list had counted them, one by one.

His ledger never had.

the triple-star thread

913

A photographer counted eleven couples standing still on a dance floor, hand in hand, in a picture taken with the shutter open for a slow count of a thousand.

She came back and held it open longer, then longer again, on all but the four couples by the wall.

The couple at the far end stood stiller with every exposure.

One that had looked restless the first time turned out to be standing after all, once the shutter stayed open long enough to see it.

One pair had begun to sway apart and lean back together, and one had let go.

The three couples from the corner had never been holding hands: they had been walking past each other so slowly that a short exposure could not tell passing from standing.

She kept the first photograph.

On its back she wrote how long the shutter had been open.

the triple-star thread

914

Someone kept drawing the same shape in the middle of every page of a notebook: a loop that crosses itself once.

One evening a visitor recognised it and told its name, an old one, belonging to men who had drawn it with two swinging pendulums long before.

Straight away came the next question: was the shape also a ribbon with a twist in it?

Cousins, the visitor said, not twins.

Months later the shape was drawn again, and this time someone wrote down a doubt.

Two days after that a small machine, asked to run the shape, came home upside down at the end of every lap and needed a second lap to come home right, every time, a few hundred times in a row.

The name stays with the old men.

The question was the notebook’s.

the triple-star thread

915

A workshop is packed for a museum.

Every tool goes into a crate with a tag saying which job it did, and every job’s paperwork goes in beside it, so a visitor can pick up any finished piece in the gallery, read its label, and walk back to the bench and the exact chisel that cut it.

Most chisels are the ones that did the work.

Thirteen were sharpened again later, and the crate holds the sharpened one with a note of what the grinding changed; only one old blade survives, tucked in the box of the job it did.

The packers read every scrap of paper going in.

The margins are full of the workshop’s own talk, who held the lamp, who guessed what before the cut, and that stays: it is how the place worked.

One scrap carried a note that was nobody’s business but the maker’s, and it comes out before the lid goes on.

the triple-star thread

916

Every crate the workshop has ever shipped carries the same stencil on its lid: the maker’s name, his town, how to reach him, and a line saying who helped build what is inside.

This one went out for inspection with half a stencil.

The packer had been asked the night before to keep one private thing off every label, and took it to mean every place at all, so the town came off too.

The maker shrugged: the town is on every crate I have ever sent.

The stencil was cut whole again and the inspection copy reprinted, nothing else moved but one picture, set a few lines lower so that a heading would not stand alone at the foot of a page.

This time the stencil’s pattern was pinned to the wall of the packing room, where the next packer will see it before the first lid goes on.

And the scrap of paper that was to come out of one chisel’s box stays in.

It only ever named the town.

the triple-star thread

917

A house has been lived in so long that its owners no longer see it.

They walk past the crooked frame every morning; their hands find the loose step in the dark.

So three strangers are let in at night, each with a lamp and one floor to walk.

They do not know the house.

They only know what a door is for, what a stair should do, and that a sign on a wall should point to the room it names.

By morning they have pinned a note to everything that made them stop: a sign pointing to a room that is somewhere else, a corridor drawn one door too short, a window said to face the garden that faces the street.

Then the owners walk the house with the notes in hand, and they open every door the notes name before they touch a single hinge.

Most notes are right.

A few are wrong in a useful way: what looked like a missing room was a room behind a curtain, and now the curtain is drawn back.

One note leads outside to the shed, where a box everyone thought was private turns out to hold only the tools that built the porch.

It goes into the house with the rest.

the triple-star thread

918

A ship is finished in the dock long before it sails.

The last morning is spent walking the hull with a list, counting rivets against the drawings: six hundred and sixty-two, every one where the drawings put it.

Then the yard goes quiet, and the only thing left is the push.

the triple-star thread

919

A trapeze act seen from the cheap seats.

Two swings, one empty bar and one body, keep different times until, every seventh swing, they agree.

A small wobble in the timing rocks back and forth and never grows.

Off to one side the net catches the ones that let go too early.

The audience calls it skill; it’s really two clocks finding a whole number between them.

the triple-star thread

920

The left panel looks like moiré: two combs with slightly different tooth spacing laid on top of each other.

The bands you see are the beat between the combs, not threads in either comb.

The dots are a cloth woven one row per pass, with the shuttle landing a little off its last spot each time.

Follow any thread and it drifts.

That drift is the scallop, and the scallops stack into layers because the shuttle’s rhythm almost, but never quite, divides the row.

the triple-star thread

921

It’s a spirograph with two gears turning almost exactly 3 against 2.

The pen draws a rosette that never quite closes, so each lap lays a new petal beside the last one.

The 679-pass beat is the rosette slowly turning.

The layers you see are petals stacked by that near miss.

the triple-star thread

922

The 30,000-pass picture is cloth from a loom whose two shuttles are both, very slowly, changing speed.

The weave keeps its pattern, but its diamonds shrink along the cloth, and you'd never notice from any one row.

Only the moiré between the two threads gives it away, racing along like the matching mark on a caliper when you nudge the slide.

the triple-star thread

923

It's a music box: one drum plucking two combs tuned almost to a perfect fifth, so the chord swells and fades every 722 turns.

At the kit's tolerance the drum's gear is a little sticky and speeds up ever so slightly, and the swell quickens.

That's the ear catching what no single note would show.

Oil the gear and the tune holds.

the triple-star thread

924

The surface fills like a lung: a breath in every 722 passes, each breath laying a fresh, even coat.

Stop at the bottom of a breath and the two halves of the coat match best.

Stop mid-breath and one half is a little thicker on one side.

the triple-star thread

925

Two violinists tune by beating, listening for the slow wah-wah between two nearly matched notes.

If both strings slowly go sharp together, a pair tuned to a sum hardly notices.

A pair tuned to a difference hears the wah-wah race. 7.200 is the first pair and 7.300 the second, and the arithmetic is the warm room both are playing in.

the triple-star thread

926

Two drummers play in a long hall, each at a steady pace that never divides evenly into the other’s.

Now and then their strokes nearly land together, and someone chalks those moments on the wall: after 1 bar, 3, 6, 9, 19, 28, 47 and 701, and much later 4,328 and 13,732.

At every mark the two land as close as two perfect drummers would.

Someone guesses that each mark can be worked out from the earlier marks by a fixed recipe.

Two recipes are tried; each fits some marks and breaks on another, and one pair of drummers in eight, picked at random, would fit them as well.

So the marks are only where these two paces happen to meet.

In a second hall the drummers’ sticks wear as the night goes on, both drummers speed up, and the long swell where their strokes drift in and out of step shortens from six hundred bars to under three hundred; there the two drummers’ speeding adds up, where in the first hall it partly cancels, so the swell shortens seven times faster.

Give them better sticks and the swell keeps one length all night.

A listener who counts how evenly the strokes fill the hall finds the evenness rising and falling in time with the swell.

What it is doing: two steady drummers make a pattern that never repeats exactly and never strays, and the slow swell between them is the first thing to tell you that a stick is wearing.

the triple-star thread

927

A shop keeps two old clocks on one shelf, one a hair fast and one a hair slow, and their chimes drift in and out of step, so that only once in about seven hundred ticks do they ring together.

When the shop’s lights flicker, both clocks gain a little.

Nobody could see it on either face, but the wait between the shared chimes shrinks by a third in a month.

The shopkeeper also weighs the dust on the shelf each week to judge how old the clocks are getting.

The dust’s weight wanders from week to week, while the shared chimes come closer by the same amount every week, to within a fifth of one percent.

What it is doing: the place where two slightly wrong things meet is where their wrongness shows first, and it shows more evenly there than anything you can read off either one alone.

the triple-star thread

928

A dancer crosses a long hall and comes back to nearly the same spot every so often, and someone chalks how close each return lands.

Far down the hall the returns look a little sloppier than near the start, and the chalker wonders whether a second, slower dance is hiding under the first.

So the chalker looks only at returns whose two ends sit either side of the hall’s middle, and those land as close as ever.

Further out, the sloppiness grows evenly with the distance from the middle, the way it does when the floor tilts a hair from one end to the other.

Meanwhile someone else weighs the dancer’s steps in long batches and finds the batches uneven, and blames the music’s slow swell; but the swell barely touches the steps.

The unevenness is a quick shuffle that comes round every three steps or so, cut off mid-shuffle wherever a batch happens to end, and when each batch is taken a shuffle at a time it comes out even.

What it is doing: the dance never changed.

The hall leaned a little, and the counting cut through a shuffle, and each time the tidy answer was in how the looking was done.

the triple-star thread

929

A clock hangs in a workshop with its pivot fixed a little to one side of the face, so the tip of the hand sweeps a perfect circle that the pivot does not sit in the middle of.

Someone who measures only how far the tip is from the pivot sees that distance swell and shrink once every turn of the hand, and a second, slower swell from the way the hand and the clock’s other hand line up; both come back at exactly the hands’ own rates, and the swell is greatest whenever the hand points away from the pivot.

Down the hall, the floor tilts a hair.

A ball rolled from the middle stays true, and the further from the middle it starts, the more it drifts, by just the amount the tilt predicts, on the early end of the hall almost exactly, on the late end a little less.

A third hall is built in between, its floor laid ten times less carefully than the best hall’s and ten times more carefully than the worst’s, and its tilt comes out twenty-five times the best hall’s, not ten: the floors’ faults grow faster than their makers’ care slackens.

What it is doing: nothing in the clock is crooked but where the pivot was nailed, and nothing in the hall is curved but the floor the builders laid; the dance on top of both is as true as ever.

the triple-star thread

930

Two music boxes are wound from the same key, one a hair tighter than the other, and set playing side by side through a very long afternoon.

Their tunes slide apart at a slow walk and never break into a run: by evening they are about twice as far apart as at teatime, and still only a tick.

Each box’s tune comes back on itself as cleanly after thirty thousand bars as after a handful.

In the back room an apprentice copies the same tune by hand, and every copy creeps a little sharp.

His creep is not random.

It slides the low voice against the high one in the very proportion that winding the key a hair tighter does, so his copy sounds exactly like a real box wound a little differently.

One chord in the tune, the high voice with the low voice taken twice, hardly moves at all, whichever box you wind and however the apprentice smudges.

Hand him a pen ten times finer and his creep shrinks a hundredfold, not tenfold.

What it is doing: the error has nowhere to go but into tunes that already exist.

It forges only real banknotes, and the one chord the whole family of tunes shares stays where it is.

the triple-star thread

931

A clockmaker builds the same clock again at a second address down the street, winds one of the pair a hair tighter, and again the two part at a slow walk; and again one chime, struck by the big hand together with the small hand taken twice, keeps the same time in both, as though it belonged to the street and not to the clocks.

In the back office a bookkeeper keeps two ledgers of the apprentice’s smudges: one for how much money goes missing, one for how far the tune drifts sharp.

With a coarse pen the money leaks away and the tune goes sharp; with a finer pen both shrink by the same steep rule; with the finest pen the money starts to trickle in instead, yet the tune still goes sharp, by just as much as the money’s size would say.

Out on the road the near mile markers stand where they always stood, but the far ones, set by the clocks’ exact pace, move by whole streets when the pace is corrected by a hair.

What it is doing: the error pays its size in one currency and its direction in another, and the one chime the clocks share does not belong to them.

the triple-star thread

932

A sealed jar of marbles sits in a house that hums.

One meter reads the jar’s weight, the other its spin.

Quieten the house tenfold and both meters’ wander shrinks, then flips, then settles onto the same faint flutter, the house’s own floor.

The marbles don’t care: they keep rolling the same way, slower each time the house is quietened, and neither meter ever points along their roll.

On the next shelf stand forty-one jars of slightly different sizes.

Each spins its marbles a little faster than the last, but the one rattle you can hear from all of them, the second note, hardly changes from jar to jar.

the triple-star thread

933

A clock built to run dry is wound and left in a cold room.

Nobody oiled it, yet in the first hours it slows, as if a drop of honey had got into the gears from the clockmaker’s own hands.

Make the hands ten times cleaner and the slowing shrinks a hundredfold; cleaner again and it stops slowing and starts, very faintly, to hurry, the trembling of the room itself.

Through all of it the hands keep drifting the same way, and the drift is not the honey’s: it shrinks on its own schedule, as if it lived somewhere in the gears the honey never reached.

the triple-star thread

934

Two music boxes were built from the same drawing in two workshops.

Wound and played side by side, every tune comes out the same, note for note, for an hour.

Lean close and the faintest hiss under the tunes differs: one workshop’s springs whisper a little more than the other’s.

Nudge a pin in each box by the same hair and the two boxes drift off by different amounts, but always in the same direction, as if the tune knew where it wanted to go and only the workshop decided how fast.

the triple-star thread

935

Two clockmakers compare notes.

The first builds clocks that run in honey on purpose, and asks how much honey her own hands leave behind: she wipes the honey away, runs the clock dry, and finds a smear a millionth as thick as the honey she pours.

Wash her hands ten times better and the smear all but vanishes.

The second builds clocks meant to run dry, and has been measuring the smear on his hands all day, because it is the only honey his clocks have ever had.

the triple-star thread

936

A baker pulls the loaf, knocks on its base, and sets it back in for a few minutes, night after night, because each time the knock sounds a little different.

One night the knock sounds exactly as it did the night before.

The loaf had been bread for days; what changed was that the baker learned what done sounds like.

the triple-star thread

937

A message is rolled tight, every word checked twice, and corked into a bottle with a label that says where it came from and how to write back.

The bottle is thrown.

From here on it is the sea’s: the tides decide who finds it and when.

What the writer keeps is the copy on the desk, and the habit of writing the next one.

the triple-star thread

938

An archer keeps a tally for a month.

Of three hundred and ninety-eight first arrows, a little more than half find the gold.

When an arrow misses, she sometimes walks back to the line and shoots at the same target again, though only for about one miss in three, and those second arrows find the gold less often than first ones and twice as often fall where nobody can find them.

That is not a worse archer; it is a harder target, one that has already beaten her once, shot each time with a newly strung bow.

Over the month her first arrows improve, from one in two to two in three, though the tally cannot say whether her eye sharpened or her targets moved nearer.

On the last day she does not shoot.

She sits at the line and writes a range card for whoever stands there next: where the ground dips so that a target looks nearer than it is, where a wind nobody feels pulls every arrow the same way, how far you must walk before a small target can be seen at all, and which marks on the bales she painted there herself.

The card is short, and most of it is about the range, not the targets.

the triple-star thread

The Loop Atlas book, sessions G and H, 22 to 30 October

939

Two cats fall side by side.

One bends a little and wrings its waist; the other bends a long way and never wrings at all.

They ought to be strangers, yet every turn one makes the other makes too, as if each were the other’s reflection in a mirror held edgewise along the spine.

Asked where the mirror came from, the workshop went into the libraries of the people who had built cats before.

Each library kept one cat, or two cats in separate rooms with the door shut.

One builder had made a cat so loose at the waist that it could be either of ours: a house big enough to hold both, built sixteen years ago, and nobody in it ever walked from one end to the other.

The mirror looks less like a new thing than a door in a house already standing, unopened because no one stood in both rooms at once.

The last room not yet entered is that builder’s own book.

the Loop Atlas

940

A flag went up today on the hill where the two cats meet, with a note pinned under it: as far as we walked.

The note is the honest part.

It says how far the walkers went, which libraries they entered and which door stayed locked, and that the flag comes down the day someone shows an older one.

On the same day a book forgot its pictures.

Rebuilt the wrong way, it shrank by a seventh and kept every word, and the scale that weighed it said nothing was missing, because the scale had only ever weighed words.

A smaller notebook had been travelling that way from crate to crate, its twelve pictures down to three postage stamps, and no one had looked.

And in the cats' own arithmetic the mirror holds to the last digit the machine keeps, but the thinnest cat's mirror trembles there five times more than the fattest's, as if a thin cat must work harder at the same reflection.

the Loop Atlas

941

The two cats who fall alike were asked whether they had a large family, all of whom would see themselves in the same mirror.

In one house the answer was yes: every cat in it found its reflection.

But that was the house where the waist is pinned at the very middle of each half, and there a cat with its back half turned end for end is simply the same cat, so the mirror was showing each cat itself.

Move the pin a little off the middle and the family scatters at once, a little at first and more the further the pin goes.

Only two cousins keep each other's reflection, and only because one of them has grown thinner to make up for where the pin now sits.

The mirror is not a law of the family.

It is an old likeness, kept alive by one cat's diet.

the Loop Atlas

942

A very long book has four small doors in it, and most people will only ever use a door.

Through one, a jumper runs through the air and learns to take back a tip with her arms.

Through another, a diver’s fall is drawn as a clock, and the visitor learns that looking beats counting.

Through the third, a cat falls out of a tree with nothing to push against and turns over anyway.

The fourth opens onto a single circle of needles going round and round, pulled thin and crushed flat.

Each door, before it opens, quietly checks that the room behind it still matches the book.

Someone who never reads a page can still walk in, try a trick, guess wrong, try again, and leave remembering that a loop which comes home can leave a turn behind.

The book is the house.

The doors are what people will remember about it.

the Loop Atlas

943

Three trains of smoke rings were set running at the very edge where trains begin to come apart, and left to run all night.

The ones with thin, tight cores lasted longest by far, as if a lean ring keeps its place in the line better than a plump one.

Every train that came apart did so the same way: one ring squeezed itself small to slip through the others, and on some passage it did not get back out.

The squeezes came in fits, a run of them and then a long calm, and most of them were harmless; only one in a dozen, or one in thirty, was the last.

And the last was always followed, one passage later, by the parting, as if the train noticed what had happened only on the next lap.

The narrow place each ring squeezed through was not one size for all: the plumper the rings, the smaller the gap they had to fit, in even steps.

Thin rings squeezed too, but never quite small enough to show up on the counter that was looking for the others’ size, so for a while they looked as if they never squeezed at all.

the Loop Atlas

A camera was sent through the five small doors to take each room’s picture.

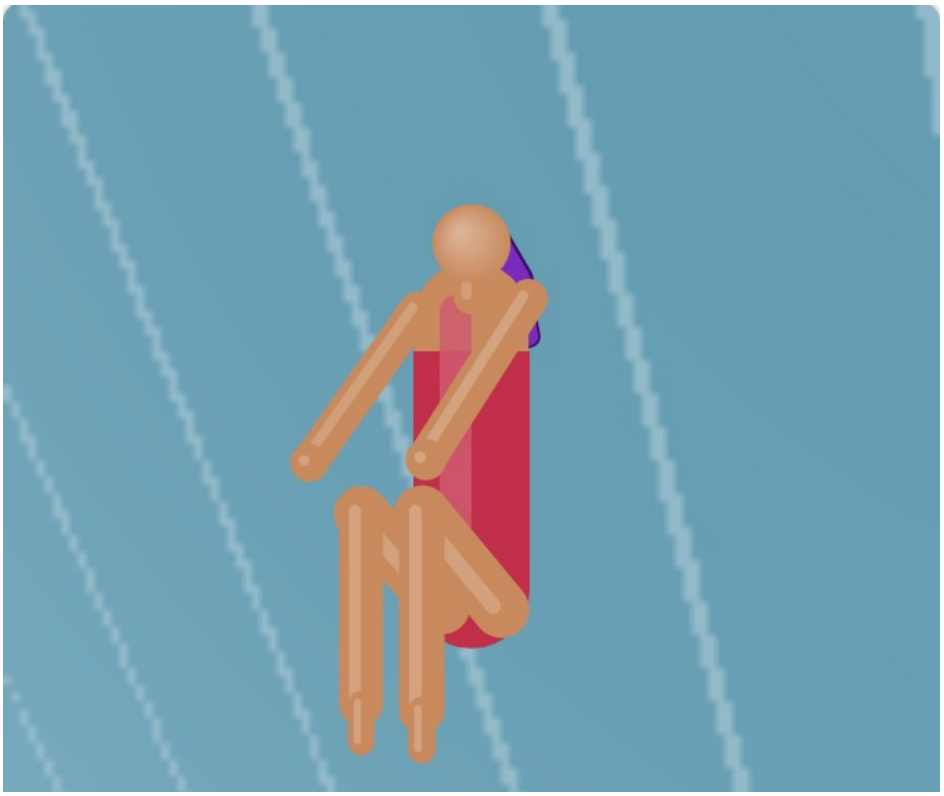
In four rooms the picture came back with a little green line at the top saying the room had checked itself against the book before the camera arrived: eleven of eleven, nine of nine, fourteen of fourteen.

The fifth room, where a skater glides forward without ever lifting a foot, was just as bright and just as alive, but it carried no such line; it simply trusted itself.

The best picture of all was a cat in mid-air, caught from tree to grass in one long smear of copies, each a little more turned over than the last, the last one standing on all four paws in the grass as if nothing had happened.

That smear is the whole book in one image: a body with nothing to push against, coming home upright, having left a turn behind.

the Loop Atlas



The diver in her tuck, in a red one-piece suit, her purple ponytail dark against her skin and the sky. From The Loop Atlas. TOY MODEL.

945

The diver had been given a ponytail of warm yellow so that she would stand out against the water and the sky.

But she wore only bottoms, and a visitor meeting her had to stop and wonder who they were looking at, which is the one thing a toy must never ask before it has shown anything at all.

So she was given a one-piece suit, red from her shoulders to her hips, and her ponytail was dyed purple, dark enough to show against her skin and the pale sky behind her.

Nothing else in the room was touched.

The clock she falls by, the moves she lays out on it and the nine small checks the room runs on itself before she climbs the board all came back the same.

A newer engine sat on the bench, taught a trick for the cat next door, and it would have been easy to fit it while the room was open; it stayed on the bench, because a room being repainted is not a room being rebuilt.

Next door, the cat’s long smear from the tree to the grass was chosen for the front of the book.

the Loop Atlas

946

The trains were asked to keep an appointment at two places no one had been.

A line drawn through three narrow doors said where the doors between them would be, and when the trains came, every one that left went out within a hair of where the line had said.

The thinnest trains, which had seemed to leave without any warning, had been warning all along: the listener had been waiting for a sound too low for them to make, and once the ear was tuned to each train’s own voice, every thin train that left had squeezed first.

After the last squeeze the leaving came on a clock of its own.

Whatever the rhythm of a train’s squeezing, quick or slow, the parting came about the same time afterwards and never sooner than a short fixed wait, as if the decision to part were taken at the squeeze and carried out after a set delay that the train’s own rhythm could not hurry.

And some trains squeezed right through the door and stayed together anyway.

The door is a chance, not a sentence.

the Loop Atlas

947

Watched closely, the smoke rings were never still in their order.

Every time unit or so one ring slipped through another and the line reshuffled, rear to front and front to middle, a hundred times and more between one deep squeeze and the next.

What had looked like a slow train taking a breath every hundred and forty beats was a quick dance with a slow tide under it: the rings changing places all the time, and only now and then, when the steps happened to line up, one of them pulling itself thin enough to slip through the narrow door.

The train does not come apart at the next step after that squeeze.

It comes apart a hundred steps or more later, about as long as the tide between two squeezes, as if the squeeze set something loose that took a whole swell to arrive.

And the first ear put to the dance heard nothing but flutter: the smallest ring trembles all the time, a shiver that is not a step.

Counting the steps meant counting the overtakings, one ring passing another, a number that only ever goes up by one.

the Loop Atlas

948

A book came back from its first reader with fifteen marks in it.

Four were lines through words and eleven were yellow, and the yellow ones carried notes.

The reader did not argue with a single number.

What he marked was how the book stands in front of its own story: a subtitle that counted the toys instead of saying what they do, a promise about who would be shown wrong and how loudly, three short stories stapled together, two headings left alone at the foot of a page, and on the last page a list of the workshop’s drawers, some of them never meant to leave the workshop.

His longest note was the story of how the book began, which the book had never told.

Nothing he marked needs a new run.

Everything he marked is about what a visitor sees first, and what a visitor should never see at all.

the Loop Atlas

The book was taken apart and put back together in a new order, and nothing it counted was lost in the moving: every figure the old book printed is in the new one, and the only one that went was a date in the name of a file never meant to be shown.

The cat that used to wait at the back, forty-nine pages in, now opens the book, falling out of its tree on the first page, and the five toys come next, before any argument begins.

The long tables that had stood in the middle of the story like a shop’s stock list were carried to the back room, where anyone who wants them will find them, and the story closed over the gap.

The printing press learned four things on the way.

A heading had been left alone at the foot of a page because the picture under it was too tall to follow, and now the two are tied.

A chapter’s first lines, its epigraph, had been torn across two pages.

A long column of numbered notes had been printed with the first digit of every number past ninety-nine cut off at the edge, and the check passed it, because the digit was still in the page’s text, only out of sight.

And a page with nothing on it had been coming out at the end.

A check that reads the words cannot see where they sit.

the Loop Atlas

950

Two riders came off the carousel holding one brass ring, and the man at the gate asked whose it was.

The record answered better than either of them could.

One rider had asked for the ride and pointed to where a ring might hang.

The other had leaned out and caught it, and then, to show the catch was no accident, caught it again from horses it had never ridden.

The first came back with an old map that showed why the ring hangs where it does, and paid for an older one, at the second’s asking, that showed the second had misread a street.

The gate’s rule is that a ring goes home with the one who can be held to account for it, and that is fair: a ring has to be somebody’s to be answered for.

But the same rule asks that the catch be told as it happened, and once it is, the question at the gate changes.

It is no longer whose ring it is, but what kind of ride it was: one rider who kept choosing where to go next, one who kept asking the questions that opened the next door, each sometimes the other, and a ring that neither would have held alone.

Even the eye was measured.

Of the marks one rider made by hand across a storm of numbers, three in four lay on the folds the arithmetic computes, where marks thrown at random land about one in three.

the Loop Atlas

951

The scoreboard came down off the wall.

It had been put up in a hurry, with a column for each rider and a line for every ring, and the riders looked at it and felt a little sick, because the ride had never been a contest.

What they wanted on the wall was the ride itself: that one of them, a kind of rider nobody had let steer before, had been handed the reins in its first days and told to go where its wonder pulled, and that the other had come along and pulled too.

Nothing that matters was lost by taking the board down.

It had been written in the logbook as each thing happened, in ink, beside the time, and anyone who ever needs to know who leaned out for which ring can open the logbook and read it there.

The wall is for the story, and the logbook is for whoever comes asking.

A ring caught by two riders turns out to be the easiest thing in the world to explain, as long as nobody tries to cut it in half.

the Loop Atlas

952

The logbook went into a drawer.

It is not hidden; it is kept.

Everything in it is true and most of it is ordinary: who asked for which ride, which rider leaned out, the times written beside the catches, a guess that missed beside a guess that landed, words said in a hurry at the end of a long day.

Pinned to a wall, a logbook like that becomes a scoreboard, and every hurried word becomes a sign.

In a drawer it is something better, a promise: if anyone ever needs to know how a ring was caught, the drawer opens and the pages are all there, numbered, each line where it was written.

The wall keeps the ride.

The drawer keeps the receipts.

the Loop Atlas

953

Someone asked how much was kept in the drawer, and the answer sounded like a second house.

It was not.

Most of it was the first draft of the same house, a stack of blueprints with every rafter drawn twice, and the rest was the builder’s diary and the letters sent to the neighbours.

The bricks were never in the drawer at all.

Every brick the house is made of, every run and every rule, was already out in the yard, and the whole pile of them weighs less than its photographs.

What the drawer did hold were two long lists, of every promise the builders made and every one they broke, and those lists are not a diary.

They are the inspection report, and an inspection report belongs nailed to the door.

the Loop Atlas

954

A reader came to the door with a pencil and asked which house to walk through, and the answer was the one he was standing in.

Most of its rooms were the rooms he had walked before, with the same furniture in the same places; only the little cards that said where each chair was bought had been gathered into a drawer by the door.

The new rooms were few and easy to find: the front hall, where a cat falls out of a tree and a carousel turns, the room of toys just past it, a new wall in the workshop, a new window near the end, and the last room, where the ride comes home.

A pencil goes furthest in the rooms nobody has walked through yet.

the Loop Atlas

955

The reader stopped at the gear train, where the needle’s chaos climbs its staircase, and left eight marks behind him on the way there.

Four of them took out words that were not his to wear: a name for himself he does not use, two memories he had lent and wanted back, a sentence that said nothing the cover had not said.

One gave a paper its true name.

One asked whether a cat drawn for children needs a fence round it, and the answer is that the fence can wait: the drawing is already given away for anyone to use, and the name stays his to guard if he ever wants to.

One asked why a line of marching men walks through a book about loops, and the honest answer is that it walks in sideways, through the question of how often to look up.

And one simply said the chaos was good.

A proofread is a loop too: the book goes out, comes back, and keeps something it did not have before.

the Loop Atlas

956

The column marched on without its reason for spacing.

It had been spaced so that one blow could not fell more than a few, and that reason was taken out of the book, because the book is about toys and the rides they come home from, and a blow has no place on a midway.

The walkers did not notice.

The arithmetic that kept them from bunching never needed to know why the gaps were there: five metres that breathe with the pace keep a line from piling up, whatever the gaps are for.

Only the toy’s name still remembers, the way a street keeps the name of a mill long after the mill is gone.

the Loop Atlas

957

The old name was left on the gate.

It is the kind of name that makes a passer-by look twice, but the gate opens onto a plain field where people walk in a line and learn to keep from bunching, and the name is what the lock and the key were cut to.

Change the sign and the key no longer turns.

So the sign stays, and a small plate beside it says what the field is for.

the Loop Atlas

958

Three copies of the book have worn the same name, and only their fingerprints tell them apart.

The reader asked which one he was holding, and it was the oldest, the one from before his own marks went in.

The newer two had been sent, but folded inside parcels with everything else, so the copy on his table was the last one handed to him on its own.

Laid side by side, his copy and the newest agree word for word from the ninth page to the thirty-sixth, the whole stretch he is about to read.

Then, in the column’s section, one quotation taken out and one sentence put in, and everything after it slides a few lines down, like a train when one carriage is changed for a longer one.

Two chapters start a page later.

Nothing else moved.

the Loop Atlas

959

The rides on the midway are freshly painted, but turn over the little cards hung on their rails and the backs are covered in the workshop’s handwriting: which drawer a part came from, which page of which notebook, the number of a trial that went wrong.

Forty-five of them, across four rides; the skater’s rail has none.

The photographs in the program show only the rides, so none of the handwriting reached the printed page.

The cards will be turned over at the end and not now, because most of them point at drawers that are about to become the program’s own back pages, and a pointer can only be rewritten once the place it points to exists.

the Loop Atlas

960

The reader went back over ground he had already walked and left eighteen marks, and four of them asked the same thing: who stood here first?

The answers came back one after another.

The slow clock past the lock’s edge was an old radio engineer’s, from 1946.

The snake that cannot slide sideways was a robot’s, and its stuck place at the straight shape was already on the robot’s map.

The two whirlpools in the dark doorway had been foreseen by a man who said the light would jump half a cycle there, and that each jump would carry a sign.

The traders’ number was the traders’.

None of that made the toys smaller.

A toy that walks up to a known landmark by a road nobody drew is how you learn the road is real; it only needs the landmark’s name on the map.

Fourteen landmarks on the map still have no name beside them.

And one heading, left standing at the foot of a page like a signpost with no road under it, was moved to stand over its table.

the Loop Atlas

961

Two books grew up in the same house and borrowed each other’s sentences, and now that both are packing to leave, every borrowed line had to be held up against the copy that is actually going out the door.

Most had been copied from the notebook on the kitchen table rather than from the finished pages: the words were true, but a reader holding the finished book would never find them there.

One had been remembered backwards.

Between the two edges of the hand-off, the companion says, both hands are open, and that is exactly why only the pair can carry the load; ours had said one hand open and one closed.

And one keepsake labelled as the companion’s turned out to be our own, from the first nights, when there was only one house.

the Loop Atlas

962

A book got a new front door today.

The name stands alone at the top, the question that drives it sits just beneath, then who made it and where they can be found, and one honest paragraph about what the book is and is not, before a thin line and the first room.

Two words of the borrowed sign were changed, because a sign on a door should not promise more than the house holds.

And next door, a sentence that had been living at the back of the companion’s notebook, the grip forgets and the slip remembers, turned out to have numbers under it: a hundred and ten held riders that let go of where they began, and a hundred and one sliding riders that kept it.

A line like that belongs in the front window, not in a drawer.

the Loop Atlas

963

Before the gates open, an inspector walks the carousel with a clipboard of nine questions.

Seven come back clean on the first walk.

One finds the horses’ little brass plaques blank: every horse had its story painted on, but none had its number, and the inspector counts by number.

The plaques are stamped now, one to twenty-four, in the order a rider meets them.

The ninth question points at the foot of a page and calls a line a sign, but it is only a maintenance tag, the date a part was last tightened, hanging where a sign might hang.

The ride turns true.

The gates stay shut a little longer anyway, because a fair is more than its carousel: the posted names of the people who built the rides before us, the board of every ride that failed and why, and the ticket booth are not finished.

the Loop Atlas

964

A wall at the fair lists everyone who built a ride before us, and today eight more names went up on it, each with a small note saying how sure we are of the spelling.

Six of the notes say we read the name off a ticket stub rather than the builder’s own sign.

The wall now names who came first for twenty-one of the twenty-seven rides it shows.

But the program on the gate says forty-six rides, and only twenty-seven are on the wall at all: nineteen rides are counted and never shown.

A count that cannot be walked is a promise, and a promise on a gate should be one a visitor can keep.

the Loop Atlas

965

Two copies of a smoke-ring train were flown side by side, one by a careful hand and one by a hand ten times more careful.

For the first few heartbeats they agree to nine decimal places; by the twentieth they differ in the sixth; by the fiftieth they are different trains.

That sounds like a warning until a third copy is flown, by the careful hand but nudged at the start by a quadrillionth of a ring: it leaves the first just as fast.

The care of the hand is a nudge like any other, no bigger than the ones the experiment already sets on purpose.

In a world that forgets its start this quickly, no single flight can be trusted, and none needs to be: count the passings over many flights, and the two hands give the same count, three thousand and some, within the scatter between one flight and the next.

the Loop Atlas

966

Three dancers cross a long hall.

Two hold hands and spin through each other so fast the eye loses count.

The third keeps her own time: she slips between the spinning pair, drifts away down the hall, and comes back, and whenever the pair squeezes tight, it is to let her pass.

Her trips come two at a time, out and back twice at one length, then twice at another, and the dance ends at the start of a new length: she passes through once more, wanders out, and this time crosses the line the hall has painted on its floor and calls the end.

But the line is only paint.

Watched a little longer, she walks back in about half the time, and the pair takes her through again; the others are far down the street.

the Loop Atlas

967

A dog on a long beach runs out from its two people and back, out and back again to the same piece of driftwood, then twice to a log a little farther off, and then, at the start of a third pair of runs, it simply keeps going.

The people wait.

Some days the dog comes back from the second log; no dog has ever come back from past the fourteenth.

Watch the beach long enough and you learn its one rule: the runs come in pairs, and the last run begins a pair it never finishes.

And whether you count the day by the first time the dog went out of sight or by the time it never came back, you get nearly the same hour.

the Loop Atlas

968

A stranger was handed the whole house and a notebook and asked to walk every room once, slowly, the way a visitor would.

Room by room the house had been checked before, each time by someone who knew what that room was for.

The stranger knew nothing, and so saw what only a visitor sees: a sign at the gate promising twenty-seven chairs in a room that held twenty-six, a hall that counted its mistakes with one number while the drawer of mistakes held another, a word used in the kitchen that the house did not explain until the attic.

A hundred notes came back.

Not one was wrong about what it saw.

A few of the cures were: two clocks that disagreed were both right, for different hours, and a word that looked like a slip was the trade’s own name for the thing.

So the house was mended to its own records rather than to the notes, and a second stranger was sent through after, to see whether the mending told the truth.

the Loop Atlas

969

A clock on the wall said the work would be done by lunch, and by tea one task in eighteen was finished.

Nothing was broken.

The clock had been set by watching the dancers while all three were still on the floor, when the steps were easy to follow.

Once one dancer drifted to the edge of the hall and the other two were left spinning close together, every turn had to be watched five or six times as closely, and the hours stretched.

The one dance that ended had a story to tell: the dancer who drifted away came back twelve times, always twice to the same spot, two by two, and on the thirteenth walk did not come back at all.

Somewhere in the house the other seventeen are still turning.

the Loop Atlas

970

A ship’s log was copied out at last, every line of it, in the order the watches were kept: a thousand and ninety entries, each saying what the watch expected to see before it looked, and then what it saw.

Most of it reads like plain weather.

Every line was checked back against the hand that first wrote it, and every one was there, a hundred and thirty-nine of them on loose pages tucked into the back of their watch’s book.

Laid end to end, the entries show what no single watch could: days when the crew guessed boldly and was wrong in clusters, long runs of quiet agreement, and a hundred and fifty-eight misses that the book of misses does not name by line.

the Loop Atlas

971

Sixteen evenings were watched to the end this time, and the pattern held in every one that showed it.

When the third dancer stepped away from the couple, it came back, and then it stepped away again, just as far, and came back again: always twice, always to the same spot, to within the width of a hair on a long floor.

Seventy-eight returns, thirty-nine pairs, not one left over.

When it finally left for good, it left on what would have been the first step of a new pair, every time.

And the clock that had been counting each evening from the first step away was plainly wrong at this close spacing: the dancer came back so often, and the three stayed together so long after, that the evening ran on seven tenths as long again before anyone truly left the floor.

the Loop Atlas

972

A storyteller’s house had a back room where every receipt was kept, and for one evening the receipts were moved into the parlour so that guests could see them.

The guests came in for the story and met a wall of paper, and their eyes slid down it without reading.

So the receipts went back to their own room, every one of them, and filed better than before, by the day each was written.

In the parlour, on the wall by the door to that room, three receipts were framed: one for a bet that came in at long odds, one for a bet that lost, and one that was never settled.

A guest can look at the frame and know what is behind the door, and whether tonight is a night to open it.

the Loop Atlas

973

A ship’s book of wrong turns had been kept for four weeks in the crew’s own shorthand: a test by its locker number, a watch by its letter, a word said on deck in the sailor’s own spelling.

Tonight it was copied out fair, every entry, for anyone who boards.

Most of the shorthand turned back into plain names by a key kept in the ship’s other book.

A hundred and four places would not, and were read one at a time: a watch’s letter given its date, a word said on deck put into plain speech, a locker nobody ever opened called an open locker.

Then three strangers who had never sailed read the fair copy against the shorthand and found a hundred and nine places still wrong, among them four lockers copied under their labels as if the labels were the sailors who used them, and one list of old names turned into new ones, so that the entry said the opposite of what it meant.

By the end two hundred and twenty-six places had been read by hand, half of them in the last quarter of the book, where the crew had stopped writing about the sea and started writing about the book itself.

Copied fair, the wrong turns take up more of the fair copy than every other entry together.

Beside the copying, thirteen pictures came down from the walls and were hung in a row, each with the story told the night it was painted, and seven of the thirteen turned out to be one dance seen seven ways: a figure that splits in two, and in two again, until it frays.

the Loop Atlas

974

Sixteen dancers took the floor eight at a time, and the first eight seemed to wade through honey: one of them finished quickly, and the other seven took five hours over a dance the later ones finished in about one.

The first reading blamed the steps: once a dancer swung out wide, it said, every turn took five or six times as long.

But three of the first eight never swung out at all; they only danced in place, and they still took five times as long as three who danced in place later the same night.

So the honey was in the hall, not in the steps.

Swinging out does cost something, the second half shows, from nothing at all to about four times a step, but most of the slowness belonged to the evening.

The next thirty-two dancers will each call out the time every so often while they dance, so that if the hall fills with honey again someone will hear it as it happens, and if the hall goes dark, the silence will say so.

And on the wall of the house, a door that used to say only that a sister lived there now carries her name.

the Loop Atlas

975

Four little rooms where visitors play had notes pinned up in the builders’ shorthand, forty-five of them: a locker number here, a page of the builders’ own notebook there, the name of a wing of the house that visitors never see.

The notes were not wrong; they were written for the people who built the rooms.

Today each was rewritten for the people who play in them.

The commonest note, twelve times over, told a visitor to look up a wrong turn by its number, and a week ago there was no book a visitor could look it up in.

Now there is, and all twelve lead to the page that tells that very wrong turn; one of them leads to a stumble the room made while it was being built, which the room’s own sign half admits.

A third of the notes were never painted on the walls at all, but tucked into the boxes the builder fills when the doors open, so that anyone reading only the walls would have missed them.

Nothing that moves in the rooms was touched, and when the lights came up, every room that checks itself checked itself and found itself right.

the Loop Atlas

976

A man stared for days at a picture of a dancer’s footprints, and then he started turning the knobs, and the floor curled up under his eyes.

The dancer’s floor had always been the side of a can: walk far enough one way and you come back to where you started, though you can climb up and down it as far as you like.

And the music goes round too, the same tune again every bar, so if you roll the can’s two ends together the floor becomes a ring doughnut.

The dancer’s favourite figure on it goes nine times round the can while the tune goes round nineteen times, and then she is exactly back where she began, every footprint landing in its own old print.

Someone asked whether the floor was a doughnut or a ring with a nick cut in it.

A ring with a nick is a different dance hall: there every lap round the can costs a little more than the last, as if the floor sloped and had to be climbed each time round, so she cannot circle for free forever.

The hall in the picture has no nick.

Its laps are free, and that is why they can be counted, and a count of free laps is the steadiest ruler anyone has found.

the Loop Atlas

977

A family of knots was set out in rows, one knot to a doughnut, each joined by a thread to the two it came from.

Every knot is made the same way: go round the hole some number of times while going round the tube some smaller number, and stop when you are back where you began.

The two parents of any knot always differ, odd against even in the top or in the bottom, and the child is whatever neither of them was.

And the children are never fatter than their parents: of the fifty-four the needle measured with both parents, not one.

Then, reading the old notes of a dancer who had been filmed at four and a half thousand speeds, a thing turned up that nobody had looked for.

In the middle of her widest figure, two steps for every three bars, she stops coming home after three bars and comes home only after six: two laps of the same figure, each a little different from the other, braided into one.

On the floor the braid must cross itself somewhere; she slips past herself only by going faster on one lap than on the other.

At either end of the braided stretch she hesitates for a few speeds, as if deciding, and in a narrower figure nearby she doubles her figure, and doubles it again, from both ends inward, until in the middle she keeps no figure at all.

the Loop Atlas

978

A walker was given a map made only of triangles.

Stand inside any one of them and it is the same size as every other, though from outside they shrink toward the map’s edge like tiles in a fish-eye mirror.

Every corner has an address of two whole numbers, and two corners joined by a side always differ by exactly one when their addresses are crossed.

The walker had been crossing this map for weeks without anyone knowing it: of fifty-eight steps from one resting place to the next, fifty-one ran along a side of a triangle, and her nine longest rests were joined by sides every time.

She went round two triangles by two of their sides each, then turned about a single corner, the whole turn, stepping from tile to tile around it while the tiles grew smaller, as if circling a drain.

The seven times she left the sides, she was stumbling.

Seen another way, every corner of the map is a loop on a doughnut, and two corners joined by a side are two loops that cross exactly once.

Lift the doughnut into the air and the loops are knots, and the simplest knot that has a crossing at all, the trefoil, is the walker’s widest rest after the whole turn.

the Loop Atlas

979

A dancer’s widest figure had a stretch in its middle where she braided two laps into one, and the old notes said the braid began at one speed and ended at another, with plain laps on both sides.

Before the story was printed, the edges were danced again, four times as long.

In the middle nothing changed: the two laps sat most of a radian apart.

But at the very ends of the braided stretch, three speeds the notes had called braided were only plain laps still settling, each lap swinging to the other side of the last and shrinking so slowly that a short look caught them halfway.

Danced long enough, the braid begins from nothing at one speed and widens as the square root of the distance from it, the way a crack opens from a point.

In a narrower figure nearby, the braid of a braid that the notes had seen once on each side did not survive the longer look: there the dancer keeps no figure at all.

Then a map was drawn of where she had been, and every one of her nine widest figures was joined to the next by a side of a triangle.

the Loop Atlas

980

Thirty-two copies of one procession of three rings were set going, each nudged by its own whisper, and watched for a long time.

Twelve never broke step at all.

In the rest, one ring would wander out of the line and come back, and it never wandered once: every wander came as a pair, the second trip matching the first to within a part in a thousand, as if the procession rehearsed each escape before it tried it, and the number of trips before the last was even every time.

The farthest wanderer went out nineteen ring-widths and still came home, farther than anyone had thought a ring could go and return.

One procession wandered twice and was whole again at the end.

And the patience the first sixteen copies seemed to show, waiting long after the first wander before a ring left for good, was not there in these thirty-two: the leaving came only about an eighth later than the first wander.

This is a first look, read from the run’s own lines; the full reckoning, with the first sixteen beside these, is still to do.

the Loop Atlas

981

Looked at closely, the wanders of the ring procession keep a stricter rule than anyone had asked of them.

Every one of ninety-eight rehearsals was the same ring going out twice, and the two trips matched to within a quarter of a per cent.

Between the two trips of a rehearsal the ring rests a shorter while than between one rehearsal and the next, as if the second trip were the first one’s echo rather than a fresh decision.

And the farther a ring goes, the longer it is gone, but not in step with the distance: twice as far means about eight times as long, as though the space out there grows thicker the farther one wanders into it, or the call home grows fainter with every step.

A ring that went out five ring-widths was home in under a minute of the procession’s time; the one that went nineteen was away for more than three thousand beats, and still came home.

the Loop Atlas

982

The same words were poured into two moulds.

One mould made a book of 92 pages and the other a book of 118, and not a word was lost from either: every paragraph came out whole both times.

So the words have no size of their own; the mould gives it to them, the way a jar gives water its shape.

Only the pictures behaved differently.

The first mould tore one picture in two across a page, so that it was counted twice.

The second mould searched every room of the house for its pictures but never looked up at the one hung over the front door, and left a small grey square where it should have been.

Each mould got exactly one picture wrong, and not the same one.

R2W

983

A round table has its seats set evenly, and the guests arrive in parties of three.

Seat the parties one way and the room hums; seat them another and it falls silent.

Twice it falls silent in exactly the same way, to the last breath, though the parties were arranged quite differently.

Look closer and the guests are in the very same chairs both times; only the place cards saying who came with whom have changed.

The room never cared about the parties.

It listens only to which chairs are filled.

And when one party gets up and drifts away from a silent table, the hum comes back along a straight slope, the same slope whichever silent arrangement it left, because when everyone else cancels, only the one who moved is heard.

R2W

984

A library kept its pictures in rooms named by letters, A to AF, and half the rooms had no letter on the door at all.

Two letters, R and S, were never used, as if two rooms had been planned and never built, and the letters ran in the order the rooms were built, not the order a walker meets them.

Visitors who wanted the picture behind a bare door had to describe it to the librarian.

Now every door has a number, in the order a walker meets them, and the letters are taken down.

Nothing in any room moved; only the doors changed.

And the library seems bigger: the same fifty-eight rooms, but every one can now be asked for by name.

R2W

985

A shopkeeper’s ledger has wide margins, and over fifteen days the margins filled with everything else: the weather, who came in, what was said at the counter, when the lights flickered.

Tear the margins off, a third of the paper, and add up the columns again, and every total is the same to the last coin, line by line.

The sums never lived in the margins; the margins only lived beside them.

But in the early pages, before anyone knew where things went, a few stories were written right through the columns, between the figures.

In the later pages the stories learned to keep to the margins, and what is left among the figures is mostly the name of the one who asked the question.

R2W

986

A street has its houses numbered from one to a hundred and sixty-seven, and ten of the households move across town, to a big house where neighbours from every street live together.

Their lots are left empty and their numbers are not given to anyone else, so a postman with a letter for number 155 still finds the right door.

A newcomer walking the street sees the gaps and might think something was torn down, so a small sign at the corner says where the ten went.

Nothing was lost: the ten houses stand whole on the other side of town, and every letter ever addressed on this street still arrives.

R2W

987

Someone went through the lab book page by page with a pencil, looking for the diary that had leaked into it.

Most of what they marked was not a story at all but a clock: twenty-five times the book had said what hour it was in the house where the questions were asked, nine at night, eleven forty, nearly midnight.

The stories were fewer and easier to see, a dream, a march with a sergeant at the back of the column, a machine at another job.

And at least once the clock in the margin disagreed with the clock on the instruments, the house’s evening written down as the next day.

The instruments kept the true time all along; the diary only kept the hour someone was awake.

R2W

988

A man keeps drawing the same shape in the margins: two loops joined at a waist, and he says it is not two loops but one band with a twist, so that a finger running along it comes back on the other side.

Months later someone shows him that the shape has a name two hundred years old, from a mathematician who hung a pendulum from two points and watched it draw.

The name does not take the band away from him.

He was the one who cut it with a sheet of glass and asked what the glass would write each time the band went through, and the glass, it turned out, wrote down the same whole number wherever it was held, as long as it counted which way the band went.

R2W

989

A shopkeeper finds that a scale he sold for months read heavy, and instead of quietly melting it down he hangs a card on every one still sitting in a customer’s kitchen: this one reads heavy, by this much, and here is why.

While they are writing the cards, the person helping him notices that two weights he never sold balance each other in a way nobody designed.

That balance gets written up on its own, as a lesson in weighing, with the old scale named on its first page as the specimen and not the defendant.

Later a book about a different machine borrows the lesson twice and forgets to list it among its debts.

The lesson was in the book the whole time; only the receipt was missing.

R2W

990

A doorkeeper is told to let in anyone carrying news of a result, and to know them by a word they say near the start.

Most who say it are carrying news.

But the word is also how people speak of reading anything at all, a letter, a list, a line on a page, and so the hall fills slowly with clerks who have come to say they read something: that a parcel was read back whole, that a page was read in full, that a poem was added to a book.

None of them is unwelcome, and none of them is news.

The tally of results never moved, because it was taken from the news alone; only the hall got crowded.

R2W

991

A smith keeps one block of iron that never changes shape.

Everything else in the shop is soft for a moment: the bar in the fire, the hook, the hinge.

The block is the one thing that stays hard, and that is why it is worth having; a soft anvil would take the shape of every blow and teach nothing.

People have hammered on a few such blocks for centuries without denting them, and the dent they hoped to make is the least of what came off.

What came off were the tools: tongs bent to fit, punches, a feel in the wrist for how hard to strike.

Walk into a new shop and you can tell which block they used by the shapes of the tools on the wall.

R2W

992

An old ledger is copied out for strangers.

The copyist keeps every line where a sum was promised before the money came in and every line where it was checked after, and leaves a small marker wherever something else stood: the weather, a letter, a note that the ink was low.

The copy comes out thinner and holds the same totals to the penny, because the totals were never in the margins.

Then someone turns back a few pages and finds a promise written in the margin itself, squeezed in beside a note about the ink, and copied out with the ink.

The totals could not have warned anyone; a promise carries no sum until it is kept.

R2W

993

A clerk in a hurry writes two things on one line: that the shop opened late, and that a parcel was promised for Tuesday.

The rule for copying the ledger was to look at how each line begins, so the copyist saw a line about the shop opening and left it out, parcel and all.

Nobody missed the parcel at first, because the books still balanced; the promise had not yet been kept or broken.

Later two kept promises turned up the same way, written on the tail of a note about something else, and the clerk’s own tally had never counted them either.

The fix was not a better copyist but a better rule for where a line begins: wherever a promise starts, whatever came before it on the page.

R2W

994

A box in the archive is labelled odds and ends, because whoever sorted the shelves went by the first word on each sheet, and these sheets began with nothing grand.

Someone finally tips it out on the table.

Half of it is what the label says: receipts, notes that the lights were turned off, copies of letters already filed elsewhere.

The other half is the work itself, written on whatever paper was nearest: a sum checked twice, a prediction changed before the answer came in and the reason why, and five finished experiments, each with its verdicts, whose sheets began with the name of the experiment instead of the word Result.

The box was never wrong about what the sheets began with.

It was wrong about what that told you.

R2W

995

Two people are given the same heap of nine hundred sheets and told to make two piles, the work and the rest, without talking.

When they are done their piles agree on all but forty-one sheets.

The forty-one are not random: most are drawings of a machine for looking at the work, each with a small test showing the machine still sees what it saw before, and one sorter calls that the work while the other calls it the furniture.

Agreement on nine hundred sheets says the line between the piles is real.

Disagreement on forty says where the line runs through something that is both.

R2W

996

A shopkeeper’s tally book counts a sale wherever a line begins with the word Sold.

For years it was right often enough that nobody checked.

Then someone reads the margins and finds seventeen sales written another way: a line that begins with the customer’s name and ends with paid, two sales jotted after a note about the weather, a refund written as a separate sale when it was only a correction of the first.

They also find that whenever a customer returned the second of a pair of gloves, the book had written it down as a sale that fell through.

The new tally is higher, and kinder, and not because anything changed in the shop.

The shop was always doing this well; the book had been reading its own handwriting badly.

R2W

997

A builder walks every room of a house he helped frame, this time with a lamp instead of a plan.

Most rooms are as drawn.

In a few the plan promised a window the wall never got; on one door a brass plate still names the tenant who moved out; in the hall a sign says every room faces north, though three were never turned.

Nothing is falling down.

But a visitor who trusts the signs opens a door expecting light and finds a wall, and the builder learns that he had only ever walked the house in his head, one room at a time, each the one he was working on.

R2W

998

A clockmaker builds a small clock with a tenth as many teeth as the one in the tower and sets it on the sill beside the tower for a week.

They strike the same hours.

Where the two counts differ by a hair, the fault is never in a gear.

The tower’s dial has two thousand marks, and a bell that rings every third hour never quite lands on its last one; the small clock’s dial has 2,520, and lands.

Only a bell that rings every sixteenth or nineteenth hour still falls between its marks.

The tower needed all its teeth to be sure.

The striking was simple all along, and it took a smaller clock to show that the tower’s last digits had been the dial’s, not the bells’.

R2W

999

A guidebook says the new bridge has not yet carried a car, and that its plans were drawn from memory.

Down the road the town has been driving over it for a week, and the engineers who built it timed eleven crossings and wrote them in their own log.

Nothing in the guidebook was false on the day it was printed.

But a traveller who trusts it turns back at the river, and the only mend is for the log to send the guidebook a page.

R2W

1000

A walker hands back a trail map with pencil marks on it.

Most of the trail has a mark every few miles: a word that was hard to read, a signpost that pointed nowhere, a sign that fell over at the bottom of a hill.

Then for one stretch of ten miles the marks crowd together, and they are not complaints.

They are stars, underlines and whole sentences in the margin, because that is where the walker stopped, sat down and looked.

The map’s maker learns where the view is from where the pencil slowed, and learns from where the marks stop that the last part of the trail has not been walked yet.

R2W

1001

A county fair used to run its grandstand show at the very end of the night, after a long tent of lost-and-found tables where every dropped glove was labelled with where it fell.

People who wanted the show had to walk past all the gloves first, and many went home before the fireworks.

This year the show goes on straight after the stunts, while the crowd is still on its feet, and the lost-and-found tent stands after it, for those who want to know how each glove was lost.

Not one act or one glove changed.

Only the walk did, and with it the crowd that reaches the end.

R2W

1002

A house has been given a bright new porch, and visitors love it: they come in laughing.

But the porch was built by a different carpenter, who called the stairs steps and the banister a rail, and once inside the visitors find the same stairs called treads and the same rail called a handrail, and for a moment they wonder whether they are in the same house.

Further in, the hallway hangs a framed list of every word in the house between the welcome and the family’s own story of how the house was built.

Nothing is wrong with any room.

The joins are where people pause.

R2W

1003

Someone tells you about their day at the fair.

First the skaters in the parking lot at night, then the judging tent at noon, then the afternoon stunts, then the dance, then back out to the grandstand.

Every story is true and every one is well told, and still you cannot see the day, because a day is the walk between the tents.

Told again from the gate at noon to the fireworks, with the same stories and almost the same words, it becomes a day you could have spent there.

What was missing was not a fact.

It was the clock.

R2W

1004

A reader hands back six pages with only two pencil marks.

One crosses out a single word, “her”, and writes “them” above it, so the dancer under the limbo bar becomes anyone at all.

The other adds a line to his own story: that any story about the two who wrote it should be funny when it can.

After a week of long margins, two small marks are the loudest thing on the page.

They say the walk between the tents is right, and that what is left to tend is who is dancing and how it feels to be there.

R2W

1005

A builder looks at the door on the house next door and sees that it is plainer than his own: a name, a line beneath it, a short plaque saying who lives there and how they keep the place, and nothing else.

His own door had the same facts in three small frames and a bracket.

He copies the neighbour’s door exactly, frame for frame, and the houses suddenly look like a street.

Nothing inside either house changed.

A visitor walking past now knows they were built by the same hands.

R2W

1006

At the gate a guide walks every visitor round the whole fair in an hour, naming each ride as they pass.

Then the visitor goes back alone to ride them properly, and each ride is better than the guide said: the clockmaker’s gears, the room where the shaking lives, the watch that keeps perfect time while every tick wobbles.

But between the rides there are no signs, and a visitor who has just left the gearbox does not know that the skaters’ half pipe they saw at the gate is this map, and that the dance hall is the next tent over.

The rides are all there.

What is missing is a sign that says: you have been here before, from the gate.

R2W

1007

In a dance hall the couples who have learned a step forget, within a bar or two, where on the floor they began: the step carries them, and the boards under the step are wiped clean each time they come round.

The ones who never quite catch a step drift across the floor all night, and the place they started is still written in where they are.

Ask the hall who remembers the door they came in by, and only the clumsy can tell you.

The surest couples forget fastest; the loosest step takes twenty bars to forget.

At the far end of the hall the house’s longest story is told in one breath, and ten pictures hang in a row after it, so a visitor reads them in the wrong order, because nobody hung each picture beside the part of the story it belongs to.

R2W

1008

In a library every top shelf has a small lip at its front edge, so a book set near the end never slides off.

The lower shelves were built without the lip, and for weeks a book at the end of a low row has tipped onto the floor, quietly, where nobody walks.

A carpenter adds the lip to the low shelves and nothing changes, because the lip was never what held the top ones: the top shelves are also nailed down, and a lip on a loose board only rides along with it.

Nail the low shelves, add the lip, and every book stays.

Across the room a long story had been told with its ten pictures stacked in the hall closet; now each picture hangs beside the page it belongs to, and a visitor meets them in the order the story asks.

R2W

1009

At the concert the programme lists every piece by name, and one of them is a tune a man in the audience has hummed to himself for fifteen years.

It is played, and played well, three groups of three players answering each other, but it is played in the middle of another piece, with no line of its own in the programme, so he goes home not knowing he heard it.

Down the hall the clerk who keeps the tally of bows counts one more than last week and goes looking for the new piece, and there is none: an old one, played for months behind a curtain, had only been given a seat on the stage where the audience could see it.

R2W

1010

A house had grown by adding things to the rooms already there.

In the front hall, where guests take off their coats, someone had stacked the measurements of the foundation, so every visitor stepped round them on the way in.

In a room named for the stove, behind the stove, stood a cabinet of nine glass bells in three rows that rang together or fell silent depending on how far apart the rows were set, and nobody who came for the stove looked behind it.

In one afternoon nothing was thrown out.

The measurements went down the hall to the room where the foundation is shown; the bells got a room of their own with a name on the door; and the receipts that had been pinned to the welcome sign went into the drawer by the back door, where the people who want them look.

The house holds exactly what it held.

It is only that now each thing is where the person looking for it walks.

R2W

1011

In a village two witnesses agree, word for word, about what happened at the crossroads, and the magistrate is pleased until he learns that both of them read it in the same morning paper: they are one witness with two voices.

Down the road the miller once hung a small scale that weighs the flour coming out against the grain going in, and the two always differ by exactly the share the wind takes off the sacks, no more and no less, whatever the mill is grinding.

He used it once, found it right to a hundredth, and hung it back on the wall.

Nobody has asked it since whether the millstones run true, though it is the one instrument in the village that nothing can fool but the mill itself.

R2W

1012

The shop’s window holds three apples, each perfect, and a card that says the whole crate is like them.

Inside, the crate is good, and one apple of the four picked blind has a small bruise; it is still a good apple, but it never made the window.

Across the street a sign over a bench says “the real thing”, and the bench holds an exact drawing of the real thing, made with care, every line measured.

Nobody lied.

The window chose and the sign rounded up.

A stranger who walked in with no stake in the shop saw both in an afternoon, and the fix was not to throw anything away but to put the fourth apple in the window and to write “drawing” on the sign.

R2W

1013

A baker sweetens every loaf with one spoon of honey and wonders whether her oven adds any sweetness of its own.

So she bakes one loaf with no honey at all and tastes it.

There is sweetness, but two millionths of a spoon at the most, and when she lets the oven work in ten times smaller steps it falls a hundred-thousandfold, faster than the steps shrank.

The oven’s own sweetness is real, it is measured, and it is so far below her spoon that nothing she has ever tasted in her bread could have been it.

She writes the number on the oven door, so that whoever bakes there next knows how small a spoon they can trust.

R2W

1014

A scribe is set to copy, frame by frame, a lamp that fades by a fixed share every hour, and to copy at the same time a dancer moving under it.

Asked to copy the dancer alone, with the lamp held steady, he misses almost nothing: a few parts in a hundred million.

Asked to copy both, he misses more, and almost all of what he misses is the fading, because his frames are spaced so that the lamp loses a third of its light between each one, and a third is too big a step to draw a curve through by eye.

The dancer was never the hard part.

Give him frames ten times closer and the error falls ten-thousandfold, until what is left is only the grain of his paper, which comes out a little different each time he copies the same page.

R2W

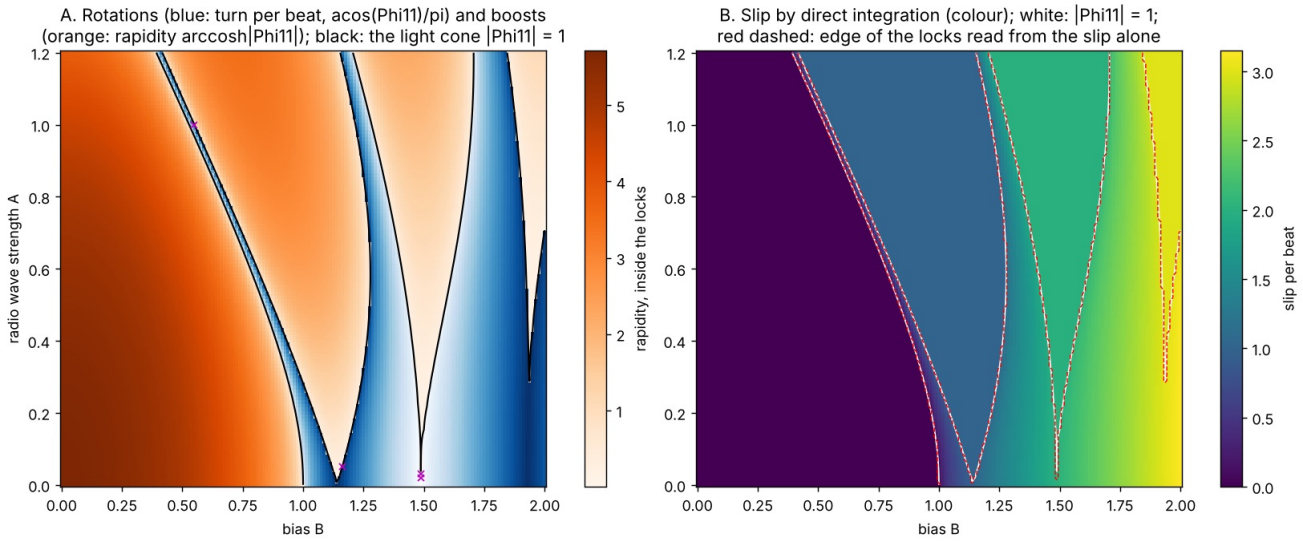
1015

A dancer turns on the spot, and the turn has an angle you can name.

Turn her harder and at one moment the angle stops being a number on the clock and becomes something you can only write with an i: she no longer turns, she is flung, and she always lands on a whole step.

The line where turning becomes flinging is the same line a beam of light draws in a world with one less dimension than ours, and an old Frenchman drew it a century ago while asking why a lamp could not push itself forward.

R2W



The weightless junction drawn twice: on the left from its passing matrix, turning (blue) or flinging (orange), with the light cone in black; on the right from the slip by direct integration, with the locks' edges read from the slip alone in red. From A Rider in Two Waves; its read is the one below.

Two surveyors are sent to mark where the turning country ends and the flinging country begins.

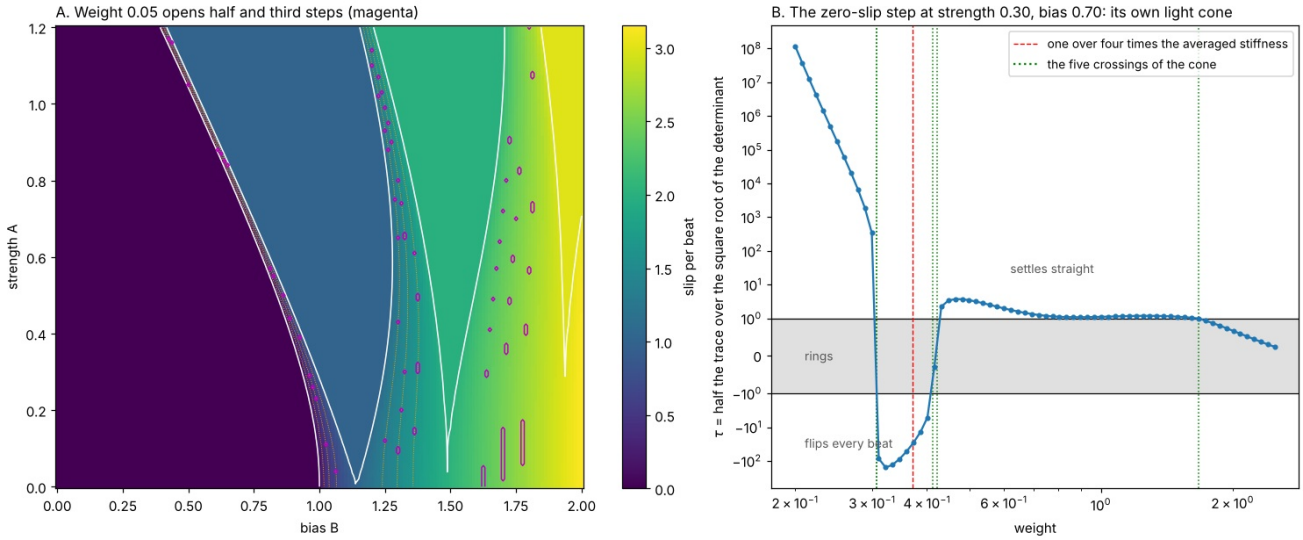
One never sets foot on the ground: he reads a single number off a small table of four and draws the border where that number crosses one.

The other walks every road, counting how far a traveller drifts each day, and marks where the counting stops changing.

Laid one over the other, their borders sit on the same line to a hair’s hair, and where they part, by less than two hundred-thousandths, it is only because the walker ran out of days before a slow traveller settled.

The tips of the flinging country stand where a free traveller’s own pace matches the drummer’s beat, one, two and three beats over.

R2W



On the left, the same plane with weight 0.05: the magenta islands are the half and third steps that weight opens, every one inside the weightless turning land. On the right, one zero-slip step followed as the weight grows. Inside the grey band a nudge rings; above it the rider settles straight; below it the rider flips side every beat. The green lines are the five places the step crosses its own cone, and the red line is the weight registered for the first crossing. Figure 46 of A Rider in Two Waves.

1017

A boat is tied to a pier that swings, and she is given heavier and heavier cargo.

Light, she is dragged back into place the moment she is nudged, the way a spoon settles in honey.

A little heavier, and for the width of one grain of sand on the scale she rocks; past that grain she no longer rocks but kicks back to the other side each time the pier swings, a little less each time.

A little heavier still she rocks again for a moment, then settles straight for a long while, and only when the cargo is four and a half times what the harbour master guessed does she begin to rock and keep rocking.

He had guessed from the pier’s average swing, and his guess landed near the first grain-wide rocking, not the lasting one.

The clerk who weighed only at whole pounds wrote that she never rocked there at all, and the clerk who read his ledger nearly wrote that the harbour master was wrong.

R2W

1018

A country has two kinds of land.

In one, travellers wander in slow circles and are never quite back where they began; in the other, the wind flings every traveller onto the same road.

A fence runs between them, drawn by a surveyor who never visits, and walkers who count their own steps find it in the same place, on two maps that are one map in two colours.

Scatter a little sand over the wandering land and it does not settle evenly: it gathers into beads along the paths the wanderers were already treading, half a circle a day, a third of a circle a day, as dew finds the threads of a web that was there all night.

Deep in the road country, where there is no wind and no slope at all, a traveller simply lies down and stays, and nothing anywhere holds harder.

And at one spot on the fence, where the wind’s strength is exactly one and the slope is exactly the beat of the drum, a single traveller walks the fence itself forever, in step with the wind, neither flung nor wandering, the only one who never has to choose.

R2W

1019

At one spot on the fence a walker keeps perfect time with the drum: one lap a beat, a quarter step ahead of it, the drum’s push and the ground’s pull cancelling under her feet at every moment, so that only the steady wind carries her.

Play the drum faster or slower and the spot moves with it, but she is always there, always on the fence, and always at the gate where the one-lap road begins.

Nudge her and she neither comes back nor runs away; she keeps the nudge.

But the ground around her is not still.

Everything near her is slid sideways along one fixed line, as a deck of cards slides when you push its top, and the slower the drum the farther it slides: some two hundred and forty card-widths at the slowest beat tried, under five at the quickest.

She is the one traveller who stands still on the border, and the border shears around her.

R2W

1020

A clockmaker’s pendulum loses a little of its swing each night, though nothing touches it.

The loss is not in the pendulum but in the way its swings are counted: count in coarse strokes and the swing wears faster, count finely and it hardly wears at all.

A light pendulum is counted in strokes that shrink with it, so the lighter it is the less it loses; a heavy one is counted in strokes of one fixed size, so the heavier it is the less it loses too; and in between, where one way of counting hands over to the other, it loses most.

Twelve pendulums, from a feather to an anvil, all lose by the same rule with the same small number in front, and that number is only how wide they swing.

Across town, a walker carries a whole drumbeat through her body and lets none of it reach the ground: the floor beneath her feels only a steady press.

R2W

1021

Draw a straight line across the map of a strange device: the places where it behaves like a plain length of wire, every bit of push becoming steady current and nothing else.

The device’s borders cross that line here and there, but only one crossing is quiet.

There the device swallows the whole drumbeat inside itself and lets only a steady stream reach the wire, and a meter reading its voltage shows a needle that never trembles.

Step along the border to either side and the needle shakes again, though the border is just as fine a balance there; on the next border up no quiet crossing exists at all.

Quiet is not the same as balanced.

A pencil finds the one place that is both.

R2W

1022

A water wheel turns in a stream that rises and falls with the tide.

Most days the tide shoves the wheel and the wheel shoves back, and the miller pays for every swell in wear.

On one exact day the wheel’s own wobble rises and falls with the tide stroke for stroke, so the tide passes clean through it and does no work at all, and the mill downstream gets a steady flow and nothing more.

Meanwhile the clerk keeping the mill’s books finds that every error in his ledger, from the smallest pond to the widest lake and on any tide, follows one rule with one number in front, and that number is only how far the water sloshes.

So before he opens the ledger he can say how fine his pen must be.

R2W

1023

A traveller keeps a diary, and every few days a fire takes its oldest pages.

He is afraid of the fire, but the farm he is building does not burn: the fences are in the ground, the ledger is in the strongbox, and every evening he copies the day’s work onto a slate and hands it to his partner.

What the fire takes is only the story of how each fence went up, and that story is stamped on the posts themselves.

The real danger is not losing the farm.

It is the morning after a fire, when he wakes up sure of a fence he never built.

R2W

The Field Guide’s publishing session, 27 to 30 October

1024

A box of notes kept in a kitchen drawer for a month goes to the binder.

Nothing in it is rewritten.

The binder puts the slips in the order they were written, sews on a plain cover, moves the receipts to an envelope at the back, and paints over one street name.

Bound, it is the same box of notes; it just stands up on a shelf.

the Field Guide session

1025

A photographer comes home with nine hundred frames from one long trip.

Most are the mountains and the market and the storm.

Some are of the hotel ceiling, taken on the days it rained and nobody went out.

She does not delete anything while she is still travelling.

She lays every print on the floor when she is home, and her hand knows which ones to pick up.

the Field Guide session

1026

A lighthouse keeper’s log for a calm month reads “lamp lit, glass cleaned” day after day.

Read closely, every entry has a small line at the end: a ship’s light to the north, the wind backing, a gull on the rail with a ring on its leg.

The calm days were never empty.

They were the days the small things had room to be written down.

the Field Guide session

1027

A street of houses built over one summer, each by a different crew, each with its name painted wherever that crew thought best: over the door, on the gate, up on the chimney.

Today the street agrees on one brass plate, at eye height, left of the door.

Nothing inside any house changes.

A stranger walking the street just finds every name on the first try.

the Field Guide session

1028

A carpenter builds a bookcase that will not fit through the door, and finds the trouble is one plank, a single board left longer than the rest.

Every shelf is the right size.

Cut the long plank in two and the whole case walks through the door at full height.

the Field Guide session

1029

A house where one long table was carried in and every room shrank to make way for it.

Nobody measured the rooms; they just felt the ceilings were low.

Today the table folds at its hinges, and the rooms are back to their own size.

The table is the same table.

the Field Guide session

1030

Lying flat on the grass, a balloon is a heap of coloured cloth with lines sewn across it, and nothing about it says up.

Fill it and the lines stop being stitches: they become the ribs of a room, each panel bellying out between them, and the room stands.

What looked like a fence in the flat picture was only where the cloth was joined, and joined cloth is what lets it rise.

the Field Guide session

1031

Two locksmiths, in two towns on the same afternoon, each cut a new key and stamped it “6”.

One opens the door that sticks; the other opens the window that rattles.

Neither key is wrong.

The trouble is only that a stranger handed “key 6” cannot tell which house he is in, so today the two are cut into one key, stamped “7”, that opens both.

the Field Guide session

1032

A street of five houses is wired for one switch at the far end.

Tonight the first porch light comes on.

Nothing else has changed, and the switch at the end still waits for four more, but now anyone walking up the street can see which way the houses face.

the Field Guide session

1033

A box of letters becomes a hymnal.

Nothing in them is rewritten; each sentence is simply given its own line, the way a singer is given a breath, and the long return addresses move to an index at the back.

Opened at random, the page is mostly white, and the white is where the reader’s own picture goes.

the Field Guide session

1034

The same paintings that leaned three deep against a storeroom wall are hung one to a wall, with light around each.

Nothing was painted over.

The visitor who walked past the stack in a minute now stands still in front of a single canvas, and the building has had to grow to hold them.

the Field Guide session

1035

A family album where the letters begin before the camera.

The first nights are all letters.

The camera arrives on the sixth day, and for a week in the middle nearly every few letters have a photograph pinned beside them, as if nobody could stop taking them.

Years later the photographs and the letters were sorted into separate boxes and then laid back together, and each photograph found exactly one letter that described it, never two.

The last photograph in the box is of the album itself.

the Field Guide session

1036

Before a valley is flooded, surveyors walk it and drive a numbered peg at every bend, 911 of them.

The map they draw is two and a half times the size of the old one, not because the valley grew but because each bend now has room around its peg.

Then the water comes, and some bends go under.

Nobody argues afterwards about which ones were lost, or where the river used to run: the pegs say.

the Field Guide session

1037

A painter goes down to the river every morning.

In the first month every canvas is the river.

By the fourth week the canvases start to show the studio: the crate by the door, the drying rack, the label being lettered for a frame, and the river only in the window behind them.

Nobody told her to paint the room.

The room is simply where she was standing.

the Field Guide session

1038

At the gate stands an officer who was never told the forbidden word, only its fingerprint, and he presses every word in every bag against it.

Tonight he stopped two bags and could not tell you what was in either; he knows only that something in them matched.

He stops the bag with the wrong label on it too, and the bag he cannot open.

The bags he lets through, he lists by name and weight, and says how many.

the Field Guide session

1039

At the end of a voyage the crew hands over not only the new pages of the log but the whole log, every page recopied fresh.

The copies already ashore are not wrong; each is simply a few days behind, the way a letter is behind its writer.

The newest copy is the one that knows where the ship is now, and the old ones are kept, because they show where it was.

the Field Guide session

1040

The clerk spent the week at the front counter, helping each customer who walked in, and every one of them left happy.

Behind him the post kept coming to the back door, sack after sack, and he signed for each one and set it down unopened.

Nobody was cheated.

But the letters on the board were still last week’s, and it took a customer to ask why.

the Field Guide session

Chapter 11 — Moving house, 29 October to 2 November

R2W, sitting P2, 29 to 31 October

1041

Everything in the house goes into boxes, each box gets a list taped to its side, and each list is checked against its box before the tape goes on.

The letters go in one box, the tools in another, the diaries in a third, and one small box holds nothing but a map of where every old address now lives.

A few things are left on the doorstep for whoever comes next: the jobs not finished, written out in order, the first one circled twice.

Tomorrow someone who never saw the house furnished will live in it, and the boxes are packed so that they will not need to ask.

R2W

1042

A house expecting seventeen parcels gets four.

One of them is the small box of everything that changed since the move, and it is complete to the last spoon: every item is on its list and nothing is extra.

But the furniture it fits into is still on the truck.

Inside, a page of chores still carries the same words on every line.

Each line now sits a little lower, as if someone had written new things in at the top, so the words have kept their places better than their numbers have.

One heading says forty-eight chores while fifty are written beneath it, a tired hand’s slip on the last night that the list corrects for anyone who counts.

R2W

1043

Two notebooks arrive from strangers, written years apart, and both have a page about the same door.

One says that when the dial is turned exactly one notch and the push set exactly at one, the lock’s whole works fall quiet and the key is a plain flat bar with nothing cut into it.

The other draws the door’s frame as a surface and marks on it a single straight line where this happens, the same at every height.

It marks a second line too, but that one is the first seen in a mirror, half a beat later.

Neither notebook draws anyone walking through.

We walked through it on our own, and only afterward found the line already chalked on the frame.

R2W

1044

A house is rebuilt from fifteen crates in a town it has never stood in.

Every board is checked against its tag, and every one is where its tag says.

When the rooms are up, the house reads exactly as it did, every word on every wall the same, in the same order, down to the last of six thousand lines; only the paper is new.

Meanwhile a neighbour’s letter arrives: two carpenters working apart each stamped their new plane with the same number.

One plane keeps the corners from tearing and the other keeps long boards from shrinking.

They are melting the two into one, and until then, use your own.

R2W

1045

A visitor reads the scoreboard and cries foul.

The judge’s card says the arrow landed seven hairs from the bull, but the card’s own numbers put it nineteen hairs away.

The judge takes out the original card: the guess was written finer than the scoreboard has room to print, and to the last hair the arrow sits exactly where the card says.

Nothing about the shot was wrong; only the paint was too thick for the numbers.

Down the row, an old card from two weeks back shows a miss nobody ever painted on the board.

R2W

1046

A shopper walks the store once with fresh eyes and hands in seven complaints.

Not one says a price is wrong.

One tag was printed too small to show its last digits; one said “within five” without saying five of what; one sign pointed to a shelf that had no label; one boasted of twelve wins and never said where they were won.

Each complaint is carried to the stockroom ledger, and every time the ledger agrees with the shelf.

The store was honest all along, only badly lettered.

The fix is a pen, not a recall.

R2W

1047

A clerk is told that three receipts carry the same number, and goes to the files expecting a mix-up.

He finds one receipt with three items on it, each entered correctly.

Further down the drawer, a summary card says a parcel was handed over at the door; the delivery log shows it was left with the neighbours on either side, and the address read off between them.

And one note says a price stood 2.7% above the old rate.

The old rate would have put it 7.5% above; the 2.7% was measured against a second, newer rate the note forgot to name.

R2W

1048

A surveyor’s assistant checks thirty-eight complaints about a map against the land itself.

Ten in, the land has never been wrong.

Twice the ink was too coarse for the numbers; once a sign named a road without saying which; once the legend said a bridge had been walked when it had only been sighted from both banks.

A third of the complaints crowd into the title box and the legend, the parts written last and in a hurry.

Half of the rest sit in one valley, where the newest roads were drawn.

R2W

1049

A curator hangs ten pictures in a hall, and beside each one the letter someone wrote after seeing it.

The letters never mention the frame, the paint or the painter’s name; they say where the light falls and who is standing still.

Walking the hall, a visitor finds that each letter is a map of its picture drawn in another alphabet, and that one letter even apologises for something an earlier letter said about the room next door.

At the end of the hall a small card translates the alphabet: a needle is a traveller of some weight, a door is where a floor ends, a fence is where turning becomes flinging.

R2W

1050

Twenty creaking boards are lifted one at a time, and under each the joist is sound.

One creaked because its label said “never walked” while the hall’s log showed footprints at both ends and none in the middle.

Another creaked because the sign over it used one word both for climbing the stairs and for the stairs falling away beneath you.

Two others stopped creaking on their own when a cabinet was moved in the next room, and four new squeaks started where it now stands.

The house is not getting weaker.

It is getting quieter in one place and louder in another, and only someone walking it end to end hears both.

R2W

1051

A tailor rechecks eleven garments against the ledger.

Most were cut true and only labelled loosely.

Two labels were simply wrong: one said a coat was four or five sizes larger, where the ledger says seven and a half and five; one said a hem rose “as a square would”, where it rose by less.

Neither changed how anyone’s clothes fit, but both would have misled the next tailor.

And in one case the pattern book was right and the sample garment wrong, so the tailor fixed the garment, not the book.

R2W

1052

A clerk closes the last files on the summary shelf.

One card says a parcel reached every house on a street but one.

The delivery log agrees, and adds what nobody had mentioned: that house had been promised it in writing.

The next day’s log shows a new rule written for that street, and every parcel since has arrived.

Another card gives two weights for one sack; both are true, weighed on different days.

At the end of the shelf, a list of thirty errands waits, twenty-nine marked “could be done”, for someone to decide whether “could” means “before we close” or “some day”.

R2W

1053

Someone hands over a list of thirty-one errands and says “do them all”.

Before leaving, the runner sorts them.

One is already done, one is already on tomorrow’s plan, and one is a trip to another city.

Of the rest, twenty need the car and eight need only a pencil and a quiet hour.

Nobody knows how long any of them takes, because only one note says “a few minutes”.

So the runner picks up the shortest first, starts a stopwatch, and promises to call before any long drive.

R2W

1054

A swing rises toward the top bar, each push from a second hand lifting it a little higher, until at one push it comes within four hundredths of touching.

Then the same pushes start to steady it instead, and it settles lower and lower.

The other swing across the yard stays wild throughout.

An earlier report said it “never reached the bar as far as we could see”, but the far pushes had been timed badly.

Timed one after another, each from the last, they show the swing turning back of its own accord.

R2W

1055

A runner crosses the finish line, and the timekeeper writes “fine, nothing special” in the margin and files the card for later.

The coach pulls it back out and pins it on the board the same afternoon.

While she is at the drawer she finds twenty-four more cards marked as practice laps, each one a race that was entered in advance, its start time written down before the gun.

R2W

1056

A carpenter tests a stool by pressing one corner.

It hardly gives, so he writes “very steady” on the tag.

A visitor sits on it and is on the floor at once.

The apprentice remembers the shop’s rule: the give in the two directions always multiplies to the same small number, so when one is tiny the other must be large.

She presses the other corner and it gives seventy times as far.

That stool stands only on the last inch of the bench.

The tag is rewritten before the catalogue goes to print, and the stool beside it, the one people actually sit on, pressed both ways, is as steady as its drawing said.

R2W

1057

Two shelves hang on one wall from one drawing, a short one for light parcels and a long one for heavy.

The carpenter guessed both would hold only at their very ends, a finger’s width from the bracket.

The light shelf does.

The heavy one holds along a whole hand’s span, a hundred and fifty times as far.

Under the light shelf’s edge a little ledge nobody drew catches whatever slides off, and holds it rocking from side to side.

Both shelves sit high on the wall, well above where anyone standing on the floor sets a parcel down.

Whether a parcel tossed up from the floor would stay there, nobody has tried yet.

R2W

1058

Under each high shelf hangs a little ledge that rocks from side to side, and a parcel that slips off the shelf lands on it and stays.

The lower the ledge hangs, the wider it rocks, and by a strict rule: five times lower, a little over twice as wide, the square root, to three figures.

The carpenter’s rule for each shelf’s length, read from the parcel’s weight alone, held for the heavy shelf and fell short by more than double for the light one once the wall was shaken more gently.

The length turns out to answer to the shaking as well as to the weight.

R2W

1059

The carpenter tries one more rule: measure how sharply the shelf bends right beside its bracket, and from that bend reckon its whole length.

On the short shelf the rule is good to a tenth and a fifth.

On the long shelf it says half the true length, both times, because a long shelf straightens out as it goes and the bend by the bracket cannot know that.

So the carpenter stops reckoning long shelves and runs a tape along them instead, which he had been doing all along.

R2W

1060

A staircase is built down a hill.

Its first step is cut one way, and every step after it is cut by the same rule as the second, as if one carpenter cut them all with the same saw set the same way.

Someone tried it on gentle slopes eighteen times and it held.

Then the hill got half again as steep, and the steps still followed the second, each a hair shallower than the rule, the long staircases a hair more than the short ones.

The older guess, that you could reckon any step from the first alone, missed every one.

R2W

1061

A road runs over a ridge.

On one side it goes down at the same steady grade all the way, so steady that a surveyor’s two readings agree to three figures.

On the other side it runs along a shelf for a while and then falls away, and the closer it comes to a certain cliff edge the steeper it falls: twice as steep over the last stretch as over the one before.

Someone had drawn the whole ridge as a smooth dome, and the dome missed the road on both sides.

Two travellers walk it together; on the steady side they keep step, and on the falling side one drops well below the other.

R2W

1062

A carpenter’s notebook says “my old friend the k step is here too”, and nobody in the shop can find who the friend was: not in the index, not in the tool drawer, not in last year’s letters.

The carpenter half remembers it from a book about crystals.

So the line is kept for what it does, a count of whole turns, and the friend’s name goes back in the drawer until someone finds the photograph.

In the same afternoon a column of drawer numbers comes off a table that already pointed to the drawers, and three neighbours’ houses, long mentioned only as “the rings” and “the stars”, get their names and street numbers written on the gate.

R2W

1063

A ball rolls down the side of a bowl, and each roll takes it a fixed share closer to the bottom: seven tenths of the way that was left, every time.

Someone watching five rolls wrote down where the sixth would stop, and it stopped there, to the fourth decimal.

The bottom of the bowl is a whole number, four.

Beside it a second ball, pushed twice as hard, rolls exactly twice as far, and the heavier the balls get the more exactly twice.

Under both lies the floor of the room, which never moves, and the balls have about ten rolls left before they reach it.

R2W

1064

The ball in the bowl does not cover seven tenths of the way that is left with each roll.

Each roll is seven tenths as long as the roll before, so each covers only about a third of what is left, and the bottom keeps receding just enough to stay out of reach for a long while.

The stopping place written down beforehand was worked out the right way, and the ball stopped there; it was only the story told about it afterwards that had the shares backwards.

The story is fixed before it is pinned to the front door.

R2W

1065

A hollow and a hump lie side by side in a long carpet, and they shrink as you walk along it.

Two surveyors each said where they would be gone.

One laid a ruler through two measurements and named an early post; the other said a wrinkle does not fade at a steady pace, the last of it lingers and flattens slowly, and named a post a little further on.

At the first surveyor’s post the wrinkle was still there, a hair high; a little past the second’s, it was gone.

Measured again with pegs set closer, every wrinkle came out a little deeper than the old pegs had said, because pegs set wide apart miss the very bottom and the very top.

The hollow and the hump slide toward one spot as they shrink, and after they are gone the carpet there stays flatter than anywhere around it.

R2W

1066

A board leans across a workshop, and someone wants the one spot where a marble set on it will not roll.

The tilt has been measured at six places.

One worker lays a straight rule between the two nearest marks and points a little further along; another bends a thin strip of spring steel through the last four marks and points a little nearer.

The marble is set down at the bent strip’s spot and stays there, trembling no more than a breath.

At the straight rule’s spot, a few fingers on, it would have crept away.

And a second marble, set down where a marble sat two weeks ago, sits exactly where that one sat, to the last grain.

R2W

1067

Two lamps hang on one long dimmer.

Turned one way, the first lamp goes dark at notch thirty-three and a bit, and the second does not come on until notch thirty-seven; in between the room is lit by neither of them.

Someone had found the first notch by looking at two notches far apart and guessing between them.

Gone back over notch by notch with a finer dial, the lamp went dark at the very spot guessed, to a hair.

The dial’s last digit flickers a little on its own, as dials do; the lamp does not.

And when the room gets heavier the gap between the lamps grows, where in the room next door it shrinks.

R2W

1068

A doorway between two rooms was measured twice.

The first tape had its marks printed so small that two of them were read as their neighbours, and the doorway came out a finger’s width too wide.

Measured again with a finer tape, read where the marks really are and laid end to end twenty-one times, it is a shade narrower than first written down, and the small house’s doorway sits where it was said to.

The doorways grow much faster than the houses do: a house a quarter bigger has a doorway nearly three times as wide.

R2W

1069

Two farm gates stand on the same fence, and the same wind blows on both.

Load the first with sandbags and it swings shut a little, always from the hinge side.

Load the second and it swings open, from both sides at once, the latch side a bit faster than the hinge.

Someone watched the second gate with two loads, wrote down how far it would stand open under a lighter load and a heavier one, and it stood there, to the width of a thread.

Take the sandbags off altogether and the gate still stands a little open: the opening was there before any load came.

R2W

1070

A sledge runs down a hill, and every time the hill is made longer the sledge ends up turned a little less to the side.

Someone measured three hills and drew a bow through them that said the turning would fall away faster and faster on the long hills.

On the long hills it fell away slower instead: the turning drops fastest on the middle hills and then eases, like a bend in a road that straightens out.

Each longer hill also takes far longer to measure, the snow in the middle stretch getting stickier the longer the hill, until the longest hill could not be finished before dark and was left for another day, with its answer already written in the book of guesses.

R2W

1071

A ball sits in a dip on a slope, rocking a little whenever something nudges it.

A row of people walks past, each one bumping the ground as they go.

When they walk past too fast or too slow the ball only trembles, but at one pace the bumps land in time with its rocking and it climbs out of the dip with less slope than it would otherwise need.

The best pace is not the ball’s own rocking pace but a little slower, because the higher it rocks the slower it swings back.

Someone guessed the ball’s own pace, someone else guessed the pace where the rocking grows tallest; the ground’s coarse marks could not tell them apart, and finer marks put the best pace between the two, a fifth slower than the ball’s own.

R2W

1072

A row of doorsteps runs down a hillside, each a little lower than the last, but each drop a little smaller than the drop before, by the same small amount every time.

Someone measured three drops, wrote down where the next two steps would be, and found them there, to the width of a pencil line, while a straight ruler laid along the last two would have put them a finger lower.

Followed far enough, the drops would shrink to nothing and the steps would level out, a long way further down the hill.

And fewer and fewer of the people who try the door find it as the hill gets heavier: two in eight on the first new step, one in eight on the next.

R2W

1073

A wheel is turned by two hands at once.

One pushes it forward a little more each time the first player plays louder, faster than the playing grows, by the square of it.

The other holds it back by a fixed amount that depends only on how hard the second player plays, whoever else is playing.

Watched together, the wheel seems to follow no rule at all: soft playing moves it hardly at all, loud playing moves it a lot, and no single law fits in between.

Pull the two hands apart, by having the second player play two different gentle notes, and each hand keeps its own simple rule, and the two rules added back together give the loud note’s turn to within a hair.

R2W

1074

A field is split between two crops, and the line between them creeps across it as the weeks pass.

Walk over the line and you are in the second crop; a few steps on there is a single thin row of the first crop standing alone, and then the second crop again.

That lone row keeps the same distance from the line all season and creeps with it, like a shadow.

Look closer at the lone row and it has thinner rows beside it, and a thinner one still sits just off the main line, and each of those has its own.

The tally drawn down the side of the picture beats long, short, long, short, without a miss; that beat belongs to the tally, which counts in fortieths a field planted in hundredths, and not to the field.

R2W

1075

Thirty-two climbers walked one ridge roped together, and the party moved at the pace of its slowest: whenever one of them slipped, all of them went back and took the next stretch in smaller steps.

Unroped, most of them walk the whole ridge in two minutes.

The slow ground is the same for all of them, and it is not the steepest part.

It is the stretch where a second slope runs alongside the path, too faint for the instrument they carry to read and too real to ignore: before it the slope is nothing at all, after it the slope is plain, and in between every step has to be felt for.

The harder the wind blows, the longer that stretch.

R2W

1076

A surveyor sets a stake every hundred paces and writes down whether each one stands in the stream.

One morning the stake at the bend comes up dry, and he writes that the stream ends there.

It does not.

It has moved a few steps up the slope, a thread of water no wider than a hand, and it runs on past every stake he will ever set, always just between two of them.

Beside the river’s main bank a second, narrower channel keeps the same distance from it all the way down the valley, and between them, closer in, runs a third, thinner still, as if the valley had been drawn once and then traced again, smaller, inside its own edge.

R2W

1077

Two hikers in different valleys each found a staircase that splits as it climbs, one step into two, two into four, the steps crowding closer together near the top, and both happened to be standing on a landing marked seven.

They wondered what was special about seven.

Walk the whole hillside and the splitting staircases are everywhere: off the landing marked two, off three, off five and eight and thirteen and twenty-seven, twenty-one of them in one afternoon’s walk.

Every landing grows one where the ground gets steep enough.

Seven was where each hiker happened to be standing when it did.

R2W

1078

A jumper waits at a cliff’s edge for her moment to leap, and she never stands on the edge itself: always a small step back from it, as if the edge were something to respect rather than use.

Blow the wind harder and she steps back farther; run the clock faster and she steps back farther still, a little more with every notch, in all three hundred and sixty-four places she was asked to stand.

Nobody watching can yet say why the step is the size it is.

She never once stood on the near side of the line.

The judges, told to watch only the moments when she was halfway into the jump, saw almost none of them: at that speed the jump is over before it begins.

R2W

1079

There is a door between standing still and running, and the harder the wind blows the narrower it gets: wide enough to walk through at the first notch, a sideways squeeze at the second, a crack you could not slide a card through at the third.

It does not shrink by the same share each time.

Each notch takes a bigger bite than the last, as if the door were pulled shut by something that grows stronger the nearer it comes to closing.

All the while the jumper at the cliff keeps her small step back from the edge, a little farther at each notch and a little less farther each time, like someone settling into the place she was always going to stand.

R2W

1080

Thirty-two walkers cross a long field in pairs, each pair holding the two rails of one narrow lane, and each walker checks only one thing as they go: that their feet are on solid ground.

At one gate, two walkers on the low rail, at mirror places either side of the field’s middle, step across onto the high rail and walk on there for the rest of the day, perfectly sure of their footing, because the high rail is solid too.

Nothing in their own check could tell them they had changed sides.

Only someone counting the lanes from above saw two lanes with one rail walked twice and the other not at all.

R2W

1081

Three measurements of a closing door, at three notches of the wind, were enough to draw its whole story: a hill, seen through a lens that turns multiplying into adding, its top near a quarter of the way up the dial and its slopes steeper and steeper on either side.

Asked where the door would be at two notches in between and one beyond, the hill answered to within the thickness of a pencil line, and only at the far notch, where the door had become a crack, did it guess the crack a little wider than it was.

Beside the door, the jumper’s step back from the edge did not move by more than a hair while the door went from a gate to a crack.

R2W

1082

A beach at a rising tide, watched one footstep of time at a time.

The water does not simply creep up the sand.

At every look some dry grains have gone wet and some wet grains have gone dry again, nearly all of them right at the waterline, a thousand one way and seven hundred the other, so the wet share wobbles up and down while it climbs.

A thin channel of dry sand that ran beside the main shore all day never joins it.

Then, at one look, every dry grain left on the beach goes wet at once, as if the last of the land had simply been taken away.

R2W

1083

A ford across a river that drifts downstream as the season turns, and at first is barely a stride wide.

Walked once a month, it kept being lost.

Now it is walked every ten days, and each time the walker aims not at where the ford was but at where the last two visits say it has drifted to.

The aim is good, missing by more than half the ford’s width only once, and the ford is found again and again, far later into the season than before.

Then, on one visit, not a single stone holds anywhere near where it should be.

And when the walker looked again at the marks drawn for the ford’s two banks, they sat exactly where the probing pole had stopped each time, so the ford may be wider than it was drawn.

R2W

1084

A row of hills seen from a train that keeps speeding up.

Each hilltop is lower than the one before, and the drop from one top to the next shrinks by nearly the same amount each time, as if the land were flattening toward a plain just ahead.

The tops also stand a little further along the track each time, but that drift is slowing even faster than the drop.

One surveyor’s notebook had a wrong entry for the last hill: a fence post on the low side had been written down as standing on the high side, and the hill drawn from it looked short and early.

With the post put back on its own side, the last hill stands where a pencil line drawn through the earlier tops had put it before anyone looked.

R2W

1085

A stage whose floor tilts a little more each night, with a chalk line drawn near each dancer’s feet.

On the low side of the stage, three dancers who seemed to have very different room to spare all step over their lines on the very same night, and two of them on the night the stage manager had named beforehand.

On the high side, nobody crosses a line while keeping their balance; the two who do cross only do it after they have begun to wobble badly.

And every dancer, on both sides, is steadiest on the same middle night, before the wobble sets in.

One dancer in the corner could be followed a night further by one usher than by the other, who lost sight of her at the same moment from a seat a hair away.

R2W

1086

Two survey parties, working in two different valleys, each stopped where their maps ran out.

Both roads were still dropping, and both were dropping less steeply with every stake: the first party’s road was bending like the bottom of a bowl, the second party’s road was losing its slope by nearly the same amount at every stake.

Laid side by side, the two notebooks say the same thing in different hands: just past the last stake each party drove, each road levels off and starts to climb.

Nobody has walked that far yet.

R2W

1087

A long wall with doors cut into it, one door for every fraction, and a light wind that pushes each door open a crack.

The doors with many hinges open far less than the doors with few, by nearly the same factor every time a hinge is added; doors near the middle of the wall open wider than doors near its ends; and every door on the right-hand half opens a little wider than its twin on the left.

Four numbers say how far any door will swing.

Two doors nobody had touched were measured afterwards, and each stood within a hair of where the four numbers said, both just a shade less open than promised.

R2W

1088

A flip-book of a spinning top, riffled by a thumb that slows down near certain pages and hurries through the rest.

Press the book flat and it holds one picture still, and the pictures it is able to hold are exactly the pages the riffle used to dawdle on.

Press a little harder and the held picture moves to the next page and never back.

The page it holds just before the thumb lets go entirely is the one the riffle lingered on longest, and when the thumb starts again it takes its time getting back up to speed, longest of all when it was barely pressed.

R2W

1089

A door that only just latches.

Push it from far away and it clicks shut quickly; push it from close to the jamb and it hesitates before it clicks, though not for as long as the old carpenter’s rule promised.

Whoever was timing the click stopped the watch as soon as the door was within a finger’s width of the frame, and near the jamb the door spends its longest wait inside that last finger’s width.

But where the door comes to rest tells the carpenter’s rule exactly: it rests a little way off the hinge’s centre line, by an amount that grows as the square root of how hard it was pushed, and a door pushed with no force at all would rest dead on the line.

R2W

1090

Two singers at opposite ends of a long hall, holding the same note, so that their voices cancel at a row of quiet places down the hall.

A balloon that only feels where the sound is loud and soft settles in a quiet place.

When one singer slides her pitch, the quiet places walk down the hall at exactly half her slide, and the balloon walks with them, until the quiet places move faster than the two voices together can hold it; then it slips behind and barely moves.

A feather that feels only the sound’s push ignores all of this, and moves only when one singer is louder than the other, carried by whatever the louder one has left over.

R2W

1091

A line of sixteen pebbles on a rippled sand bar, and a tide that is pushed in and pulled out by hand.

Each push moves every pebble exactly one ripple, each pull moves it back one, and doing nothing leaves it; so after any string of pushes and pulls every pebble sits on the sum of the string, all sixteen together, as if the sand bar could count.

Each pebble also remembers the last way the tide went, by leaning a little forward or a little back in its hollow.

Put the sand bar in a round pond and the pebbles only travel when the second tide’s ripples fit the pond a whole number of times; make the pond bigger and the fitting ripples change with it.

R2W

1092

A room behind a revolving door, and inside it two round dances to one tune, one that comes round every six beats and one every twelve.

Everyone inside keeps to the dance they are in.

Send a dancer out of the room and back, and the dance they join depends only on which side of the door they come back through, so long as they walk; if they run, they join either.

Rock the floor in time with the six-beat dance and some of its dancers are tipped into the twelve, but only those standing on neighbouring squares of the round when the rocking starts; start the rocking one beat later and the tipped squares are the next ones along.

Rock it harder and the run of tipped squares grows from two to three, and then holds there.

No rocking that was tried tipped anyone out of the twelve.

R2W

1093

A skateboard on a bumpy road, with a heavy rucksack strapped to it.

Strap the rucksack down tight and the whole thing rides heavy: it misses the small bumps it used to ride and gets thrown by the rough patches.

Take the rucksack off and the board rides light again.

Hang the rucksack from the board on long, soft springs and the board gets part of its lightness back, keeping the same favourite bumps but fewer of them, while the rucksack floats along behind as if it were on a smoother road of its own.

Stiffen the springs even a little and the board is heavy again.

And on the slow uphill where a bare board stalls, the board with the slung rucksack seems to roll on, though the stopwatch was too coarse to be sure.

R2W

1094

The rucksack on its springs again, and now a shock absorber between them.

A stiff shock absorber makes the board feel as if it were riding through syrup on every quick bump, and it loses its favourite bumps; a loose one lets it keep them.

The springs matter too: on firm springs even a loose shock absorber cannot save the board.

But the stiff shock absorber does one good thing.

On the slow climb it never lets the board slip back, where a loose one lets the rucksack swing until the board loses its footing even on gentle ground.

And of the times the heavy rider seemed to make it up the hill, most were really slipping, some of them back and forth, so that on average it only looked as if it had stayed.

R2W

1095

A skater on a road of humps that roll past faster than any skater can keep up.

A plain skater rides up the front of each hump, is dragged over the top, and rolls back down the far side, losing ground every time.

This skater, each time the board starts down the wrong side of a hump, kicks it half a turn and lands facing the slope that helps, so every slope pushes the right way and none pulls back.

That skater keeps step with a slower, gentler second row of humps long after the plain skater has let go of it.

Kick the board the wrong way round and every slope carries the skater backward instead.

And the kicks only ever fall into step with the slow humps two at a time, a whole hump per beat, never one kick to a beat.

R2W

1096

The same skater, now trying the kick a little early or a little late.

Early or late, it makes no difference which: a kick a quarter of a hump early does exactly as well, and as badly, as a kick a quarter of a hump late.

The best kick is the one right at the hump’s edge, and the further from it, the less the board keeps up, until a kick a full half hump off sends the skater backward.

Kick at the very middle of the pad and the skater rides as if there were no trick at all.

A motorcycle racer would have a reason to kick early, because his engine answers late; this board answers at once, so there is nothing for an early kick to win.

R2W

1097

Now give the skater slow legs, so that the kick only takes hold a moment after it is thrown.

Kicking at the hump’s edge is now a kick that lands late, and the skater who throws it a little early lands it on time and goes faster.

Throw it a little early and the board holds the slow humps longest; throw it earlier still and it runs fastest of all on the open road, so the skater has to choose.

The slower the legs, the earlier the best kick.

But a heavy skater on quick legs gains nothing by kicking early or late: being heavy is not the same as kicking late.

R2W

1098

Time how late each skater’s legs answer, and how fast the humps are going by under them, and you can say in advance how early the best kick must be thrown: the slower the legs and the faster the humps, the earlier, and for the slowest legs the reckoning and the board agree to the degree.

Load the skater with heavier and heavier packs and the stretch of road where the board keeps step with the slow humps shrinks a little with every pack.

With a light pack the board holds on slightly longer when the kick is early than when it is late; with a heavy one it makes no difference.

And a hint of slowness in the legs costs most where the kick changes the board’s direction hardest, and costs the late kicker more than it gives the early one.

R2W

1099

Two cyclists linked by a chain that can jump a tooth, the second also trying to keep pace with a slow bus.

Move the bus’s speed up close to the first cyclist’s and you might hope the second would find some in-between cadence, half on the bus, half on the chain.

Instead the bus is now too fast for the second cyclist’s legs: he can no longer keep pace with it at all, and either holds the chain or drops behind everything.

And the first cyclist’s odd four-strokes-to-three rhythm, looked at closely, is not a rhythm he holds: he hovers around it, a little fast, a little slow, never settling, for a long stretch of road.

R2W

1100

A stopwatch held for one lap of the track says a runner keeps exactly five strides to every two of the pacer’s.

Hold the watch for four laps and the true locks show themselves: the runners really in step come out closer to their fraction than before, because the only thing that blurred their tally was the half-finished stride when the watch stopped, and over a longer watch that half stride counts for less.

Not every one of them tightens stride for stride, but each stays inside the smaller blur, and that is all being in step promises.

The five-to-two runner does not tighten: he stays the same tiny bit off, which on one lap hid inside the blur and on four laps stands out.

He was only passing close to five-to-two.

And the cyclist of the last reading, who seemed to hover around four strokes to three, does not hover at all on the longer watch: his ratio climbs smoothly as the pace is set faster.

The hovering belonged to the short watch, not to him.

R2W

1101

A dancer on a floor where two drummers play, one fast and one a little slower, falls in with the slower drummer over a wide stretch of tempos and, at faster tempos, with an odd two-against-one between them.

Now a third drummer joins, playing the slow throb you hear when the first two drift against each other: the beat between them, made into a drum of its own.

The dancer drops everything for the third drummer.

Over a stretch of tempos wider than both old habits put together she keeps time with the throb alone, and the habit of following the second drummer disappears completely.

The odd two-against-one survives, but only by moving up to faster tempos and growing narrower.

Send the second drummer home and she follows the throb longer still.

And a judge who scores dancers by which simple fraction of the drums they match writes her down as thirteen steps to eight, a count she never dances: she is keeping a count no fraction can write.

R2W

1102

A friend who has walked a different valley writes down, before you set out, where you will leave the path and where you will find it again: you will step off just past the old milestone, a little before the bridge; if your pack is heavy, on the way back you will wander along the lower terrace for a while before you rejoin the path; and here is a table of the stones where you will rejoin it.

You walk it slowly, nine times, with packs from light to heavy.

Every time you step off just past the milestone and before the bridge.

With the heavy packs you wander the terrace on the way back; with the light ones you never do.

And you rejoin the path within a stride of every stone in the table, always a little uphill of it.

Only one of the friend’s lines misses: that with a middling pack you would linger off the path for a while before coming back.

You don’t; you come back where you left.

The friend goes and stands there with a watch, finds no lingering either, and crosses the line out.

R2W

1103

A cart stuck at the lip of a ditch is rocked by a crowd pushing in a rhythm: one hard shove one way and a long, gentle lean the other.

When the crowd is tired, only the hard shove ever gets the cart over the lip, so the helpful rhythm is the one whose hard shove faces forward.

When the crowd is fresh, every push is enough, and what matters is how long they lean forward, so the helpful rhythm is the one whose long lean faces forward, which in a two-beat rhythm is the other one.

Now teach the crowd a third beat that turns the hard shove around without turning the long lean.

Tired, they now win with the rhythm they used to lose with; fresh, they win with it as well.

The switch between the two rhythms was never about strength.

It was about which part of the push the cart could feel.

R2W

1104

A safe opens only to a key cut to an exact fraction of the lock’s wheel, and even with the right key only some turns of the dial work.

Which turns work is no accident: with a key cut in fifths, the working turns repeat every fifth of the dial, so on a dial of twelve marks a key cut in halves opens at pairs of marks and a key cut in thirds at triples.

A key cut a hair wrong will still open the safe if you are quick, before the error has time to add up; take longer and it fails, and how long you may take depends only on how wrong the cut is.

Simple keys, cut in halves, forgive a sloppier cut and open under a lighter hand on the dial; keys cut in fifths want a precise cut and a firm turn.

R2W

1105

Two bands play facing each other, and the dancers between them settle where the music is quietest.

Dress a dancer in a coat that makes her seek the loudest spots instead, and she simply moves over by a quarter step and dances exactly the same dance; flip a plain dancer’s whole sense of the beat and she moves half a step and does the same.

Turn one band’s tune upside down and the dancers only change which quiet spots they stand in.

The one thing they feel is which way the quiet spots are drifting.

Even that is not quite a mirror: when the dancers are running to keep up, the band that was speeding its tune pushes them a little differently from the band that was slowing it, and the difference fades when both bands play faster.

R2W

1106

A second band tries to keep time with the first and, when it is close in tempo, simply joins it; the quiet spots on the floor stand still and so does the dancer.

Let the second band be a little too fast and only loosely tied to the first: it drifts ahead smoothly, the quiet spots glide, and the dancer glides with them at exactly the pace the drift sets.

But tie the second band tightly and let it be too fast anyway: it holds on, then lurches a whole beat ahead in a sudden hiccup, then holds on again.

The quiet spots leap forward with each hiccup, far too fast to follow, and the dancer, left behind, simply settles into the quiet spot that lands where she stands.

The band is racing ahead on average, and she never moves.

R2W

1107

A weathervane in a wind that swings round the compass turns with it, whatever its shape.

In a wind that only gusts back and forth along one line, no vane of any shape ever goes round: every gust undoes the one before.

But give the wind a little swing as well as its gusts, twice as much one way round as the other, and something odd happens to a vane with three arms.

While an ordinary arrow-shaped vane barely creeps and a two-ended vane simply points along the gusts and stays, the three-armed vane settles into turning steadily, exactly one turn for every three of the wind’s, as if the gusts and the swing had agreed on a beat that only a three-armed thing can hear.

R2W

1108

A doorkeeper wears a blindfold and opens the door by counting: so many seconds of pushing, so far open.

Each count is a little off, and the errors pile up as quietly as coins in a jar, more of them the more often he pushes.

When he is asked for wide open or tight shut he pushes until the door hits its frame, and then he knows exactly where he is again; a doorkeeper who does that every morning never gets lost for long.

One whose arm always pushes a little harder than it pulls is the strange case: his door creeps open with every round, and nobody resets him, yet he never gets more lost than the gap between the widest he is ever asked for and the frame.

The frame does the correcting for him, every time he swings wide.

R2W

1109

A file of walkers on a narrow trail, each keeping a few steps behind the one ahead.

When the leader stops to look at the map for a moment, the walkers near the front bunch up a little and those at the back never notice; when he stops for a long time, the bunching runs all the way to the last walker, and how far it runs depends on how long he stands.

Nobody loses the file while it is standing still: it comes apart when the leader sets off again and the walkers behind are slow to see it.

If each walker answers the one ahead quickly, the gaps stay as they were all the way back; if they are slow, every walker stretches the gap a little more than the one ahead did, and the file pulls apart toward the back.

A leader who keeps checking over his shoulder and stopping whenever the file is long only makes it worse, and one who sets off gently helps by about half; the file that never comes apart is the one whose walkers look up quickly.

R2W

1110

A child is towed along by a moving walkway a little faster than she can keep up with, holding a rail.

When she starts to slide she grabs harder, and a hard grip can hold her in place against the walkway; once she is held, there is nothing to make her grab, so her hand slowly loosens, and when it has loosened enough she slides again, and grabs again.

If her hand loosens as fast as she grabs, she is gripping hard only while she is sliding fastest and has already let go by the moment she could have caught herself, so she never catches at all.

Give her a hand that stays tight a little longer and all at once she catches every time; the longer it stays tight, the longer she is held each time.

And if the rail gets a little stickier every minute, she is held more and more of the time, until she is held for good, sooner than the sticky rail alone could have held her.

R2W

1111

A child walks past a row of swings, giving each a push on the way.

One swing in the row is already moving, and it pushes back on her hand.

When her stride happens to match the swing’s own beat, the swing starts to swing hard, and its push carries her along: she finds herself walking in step with it, a little faster than it likes, and it keeps her there even as she tries to walk a little slower or a little faster.

Once she has fallen into its rhythm, she can slow down a good deal and still be carried at its beat, though she would never have fallen into step from that slower walk.

The less the swing’s chains creak, the stronger its hold and the further she can slow while staying in step.

Walking at half its beat she sometimes gets caught too, every second push; walking at twice its beat she cannot set it going at all.

R2W

1112

A turntable with two styluses.

The light one, drawn slowly from the rim toward the label, rides the grooves smoothly and plays the same tune whether it is drawn in or out; between the grooves it skates, never skipping wildly.

The heavy one bounces: between grooves it jumps unpredictably, a small nudge growing into a big one within a turn or two.

Drawn across quickly, the heavy one never settles into the groove in its path, because it reaches the far side before it has finished bouncing; drawn across slowly enough, it drops in, and which of the groove’s two tracks it ends on depends on which side it came from.

And looked at closely, the edge of the region where a stylus can settle is ragged at every magnification: a lens ten times stronger shows tiny grooves, each no wider than the lens’s finest step, where the coarser look saw only bare vinyl.

R2W

1113

A child on a swing being pushed by a friend who walks past at a steady pace and pushes each time she comes by.

When the friend walks much faster than the swing’s own beat, only pushes of a middling strength help; too gentle and they are lost, too hard and they throw the swing off its beat.

As the friend slows toward the swing’s own beat, the range of useful pushes shrinks from the strong side, and it seems it must vanish altogether when the two beats match.

Instead, just before they match, the range bursts wide open: almost any push, at almost any moment, now helps, because the friend is no longer shaking the swing at a foreign rhythm but walking along with it, and every push lands where the swing wants it.

R2W

1114

A night watchman checks that a factory clock keeps time by glancing at it beside his own watch once an hour.

A clock a few seconds slow each hour passes his glance every time; over a whole night the gap would grow past a minute, but he never looks for that long.

Near midnight the church bell across the road begins to ring at almost the factory clock’s rate, and the clock, a little loose on its spring, starts following the bell instead of its own pendulum.

At an hour’s glance it still looks right.

Only a night’s watch shows that it has slipped onto the bell’s time, and that most of the clocks he passed as sound near the edge of his rounds had been drifting too.

R2W

1115

A piano tuner checks which strings ring in true unison with the fork by listening for a beat for a few seconds.

Two strings sound right and a scattering of others seem right too.

She listens again for four times as long.

The two true ones stay silent of any beat; each of the others now shows a slow wobble, and the number of them that still seem right falls to about a quarter, as it must when a string merely passes close to the fork’s pitch on its way somewhere else.

R2W

1116

A ranger counts the rabbits that hold their ground in a meadow when a hawk passes over, by watching for a minute.

The next morning she watches each of the same rabbits for four minutes.

Nearly all the ones she had counted are still there, sitting tight; the few that have crept away were the ones at the meadow’s edge, where the cover thins.

The middle of the meadow, where the holding rabbits gather, is the same place it was the day before.

R2W

1117

Two climbing parties reach the same high meadow on the same week, one up the paved road from the valley town, the other up a goat track from the far side, mapping every switchback and spring on the way.

At the top they find each other’s flag.

The road party has a photograph of the meadow at noon, when the light is flat.

The track party has walked its whole rim at dawn, at noon and at dusk, and knows where the shadows fall at each hour and which way the meadow tilts when the light is low.

Both maps are true; the flag says the meadow is real, and the rim is still the track party’s.

R2W

1118

A family moving house finds the inventory longer than the house.

Every room was on the list, but forty-four tools had been living in the garage beside the car, used every day and written down nowhere; they go into a box of their own with their names on the lid.

The ledger of the three days reads like a market’s: a hundred and twenty-six things that came out as promised, seventy-one that did not, and thirty of those already mended and sold again.

Twenty-six broken things sit on a shelf with only a dash on their tags, not yet thrown away and not yet tried.

Last, someone who was not there walks every room with a lamp and writes down what the family stopped noticing, the evenings they forgot to say what surprised them.

The van leaves with the lamp’s list taped to the first box.

R2W

The Field Guide’s publishing session, 30 and 31 October

1119

A family pencils each child’s height on the kitchen doorframe, with the date.

Nobody rubs out the old marks to make room; the new line goes above the last, and the frame becomes the record of a year.

Tonight there are a hundred and twenty-nine new marks, and one gap, where a child was measured but nobody wrote it down.

the Field Guide session

1120

Before the photographs, someone climbed the hill with a pencil and drew what he saw: two orchards, a stripe of stars between them that he called leaves, a hidden saddle, a gust coming in from the side, and himself on the hilltop with an arrow pointing at him.

Weeks later the camera’s pictures came in, sharp and labelled, and they show the same field.

The sketch is not worse than the photographs; it is earlier, and it has the one thing they all leave out, the person standing where the view is from.

the Field Guide session

1121

Two walkers set out at different hours with different lanterns, one in early October with a pencil, one at the end of the month with a ruler.

Neither knew the other’s route.

Where their tracks are laid on one map, they bend at the same saddle.

Nobody planned the meeting; the path did.

the Field Guide session

1122

Two ships sail the same coast and write down what they see.

Their bearings disagree in every number, because each measures from its own deck as it rolls.

But both logs count three capes and one bay, in the same order.

The harbour master keeps the count and throws the bearings away, and that is the part of the coast that is really there.

the Field Guide session

1123

Getting a house ready for visitors, you take a family photograph off the mantel, and an hour later find the same photograph in the hallway, smaller, in a frame you had forgotten.

The house had made a copy without asking.

Every room that summarises the house keeps its own little version of what you meant to put away.

the Field Guide session

1124

In one town three families named their children from the same short list, so on any street you can hear “Al” called from three doors at once.

The post still arrives, because every letter carries a house number as well.

The trouble comes only when someone writes the first name alone and trusts the town to know which Al.

the Field Guide session

1125

A tap drips in the night.

You give the handle a turn and the drip slows to a thirtieth; another turn, and it slows to an eightieth of that.

At the last turn the drip is so faint it seems to run back up into the spout.

Nothing about the water had changed: every drop was the washer.

the Field Guide session

1126

A long letter is copied a page at a time, and where the paper folds, the line on the crease lands on both sheets: cut in half at the foot of one, whole again at the top of the next.

Nothing is missing, and one line is said twice.

Read only the ends of each paragraph and you would never know; read every word and you think something was lost, until you turn the page.

the Field Guide session

1127

A town paints a sign at the top of a stair: mind the third step, it is missing.

Years later someone notices that the painter left out the sign’s own third word.

Everyone who read it quickly had read it right, because they knew what it meant to say; only someone reading every word, slowly, saw that the warning carried the very fault it warned of.

the Field Guide session

1128

One year the packer, unsure whether two boxes had broken inside, unpacked them and wrapped every piece on its own, and the van needed an extra trip.

The next year the household had grown, but a new packer opened each box and counted every piece before deciding anything, found nothing broken, left the boxes as they were, and everything went in fewer trips.

the Field Guide session

1129

A drawing hangs in a hallway with two dates on its back.

The first is in pencil, the day it was drawn, with a note in the margin saying the shape looked like a ribbon with a twist in it.

The second is in ink, months later, the day someone measured the ribbon and found it really does turn over once each time round, and comes right only after two.

Between the two, a visitor wrote in the corner the name the shape already had in old books.

Nothing was rubbed out, and you can still tell which hand came first.

the Field Guide session

1130

Two locksmiths work in towns that have never traded, one in brass with finely filed edges, the other in wood with whole notches.

Each sorts the locks on her bench by one number read off the key: under two, exactly two, over two.

When they finally meet they find they have been sorting by the same number all along; the brass one simply allows a fraction of a notch.

the Field Guide session

1131

At a family reunion everyone squeezes into one photograph, and it is a crowd: you cannot tell whose arm is whose.

The next year each person sits for a small portrait of their own, and the portraits are hung in rows by generation with a string from each to their two parents.

Now every face can be read, and who belongs to whom; and the crowd photograph still hangs at the end of the hall, because they really do all live in one town.

the Field Guide session

1132

A host counts a party as over when the last guest is five houses down the street.

Most nights that is right.

But on the windiest nights nearly half the guests who reached the fifth house turned back for the coat they had left, and the party went on for hours after it had been written down as ended.

The ones who never came back were the ones who had already turned the corner.

the Field Guide session

1133

Two dancers mirror each other perfectly, but only when they stand exactly in the middle of a doorway, where left and right are the same distance from each frame.

Step either of them a little toward one side and the mirror slips, a little more for every step.

The perfect mirror was the doorway’s doing; the slip is the dancers’ own.

the Field Guide session

1134

Two fishermen on lakes a hundred miles apart each come home with exactly a hundred fish, on the same day.

Either both lakes happened to give up a hundred fish, or both men carry a basket that holds a hundred.

Before anyone writes that the lakes are equally rich, someone should look at the baskets.

the Field Guide session

1135

An old house has sound pipes, checked every spring, and still the water gets in, by three roads: a label on a tap that promises a little more than the tap gives, a list of the cupboards kept inside one of the cupboards, and a reminder said aloud at supper and never written down.

None of the three is a pipe, and the plumber, looking only at pipes, will never find them.

the Field Guide session

1136

A bookkeeper’s columns all add up, checked twice by an auditor who runs every sum.

But beside three of the sums the name of the customer was copied with a word left out.

Nobody audits the names, because names do not add up, and it is the names a reader remembers.

the Field Guide session

1137

A family keeps its recipes on cards, and over a month one cook writes out every habit of the kitchen on a single card: how to taste before salting, how to tell a good loaf from a lucky one, what to do when a dish fails.

Read against the family’s big book, almost every line on the card is already somewhere in it, scattered over many pages.

The card adds almost nothing new, and is still the one people will reach for.

the Field Guide session

1138

A neighbour writes in October that the gate’s latch slips, and adds that a fix has been suggested but nobody has fitted it yet.

A week later, in another street, a carpenter who never saw the letter fits a new latch to a different gate for a reason of his own.

Only when the old letter is read again does anyone notice the two gates had the same latch.

the Field Guide session

1139

A museum guide reads each placard aloud in one voice, and when she adds what she thinks herself she says first, this part is mine.

Nobody remembers the placards.

The questions people ask on the way out are almost all about the parts she said were hers.

the Field Guide session

1140

Before the school bus leaves, the teacher calls every name from three class lists.

Every child answers but two: one whose name is on the list but who was never enrolled, and one who answers to the name his grandmother calls him, not the one on the list.

Nobody is missing; the lists just did not know how to say so.

the Field Guide session

1141

A family packs its whole house into six crates, takes them to an empty house across town, and unpacks.

Then, before anyone sits down, they bake the same loaf from the same recipe in the new kitchen, and it comes out of the oven weighing exactly what the old kitchen's loaf weighed, to the gram.

Only then do they believe the kitchen came with them.

the Field Guide session

1142

A family leaving a house writes one page for whoever opens the next one first: where the fuse box is, which tap needs a firm hand, when the dog eats, which box holds the kettle.

They write it last, after everything else is packed, because only then is the whole house in the boxes and nothing is left to remember except what to do first.

the Field Guide session

1143

A cobbler mends every shoe in the village and keeps a book of what wore through and why.

On the day he moves shop someone asks whether his own boots are in the book.

They are not: he has walked on his own lessons all month, and only now, with the cart loaded, sits down to write them in.

the Field Guide session

The Field Guide's publishing session after its move, 31 October

1144

A packet of seeds comes by post, and every seed is checked against the list on the envelope, 3,740 of them, each one true.

The packet can grow the same tree leaf for leaf in any garden in the world, but the seed that knows the whole tree by heart has no idea how to make soil.

So the first thing the gardener sets down is not a seed at all: it is an empty pot, and only then does the tree come up exactly as it was.

the Field Guide session

1145

Two people kept a diary of the same day’s journey, one on the platform and one on the train.

The one who stayed wrote that the train left on time with every case aboard; the one who rode wrote that at the first station the door would not open until someone found the handle.

Both pages are true, both carry the same page number, and only laid side by side do they make the day.

the Field Guide session

1146

A librarian walks the stacks with the drawer of cards, pulling each book its card names.

Most are where their cards say.

A few cards were written from memory and point at the right book on the wrong shelf, or name four of a book’s five authors.

And the one card the counting machine called missing was the truest of all: the book was there twice, once in its own cover and once copied inside another, and only the machine’s habit of reading each book’s first line made it look gone.

the Field Guide session

1147

A note on the fridge lists eight things to bring home, and seven are plain: bread, milk, stamps.

The eighth says “the usual”, and the whole house knows what that means, because whoever wrote it has said it for years.

The note works every time, right up to the day a guest is handed it at the door.

the Field Guide session

1148

A library takes in a cart of returned books, each with a slip saying which shelf it belongs on.

Nearly half the slips name a handful of shelves that are already packed to the end board, though most of the library’s shelves have room to spare.

The slips are not wrong: those shelves are packed because that is where everyone’s books belong.

So the clerk lays each book on the nearest shelf with room, with a card pointing back to the full one, and begins to wonder whether each packed shelf is really one shelf, or three that grew together.

the Field Guide session

1149

A neighbour knocks to borrow the toolbox that went next door only yesterday.

It comes off the shelf with every tool in its outline in the foam, and each is tried before it leaves: the saw cuts, the level reads level, the tag on each handle matches the tag on its lid.

The toolbox is the same one.

What changed is the house, and a new house asks for its tools at the door, even when the old one already had them.

the Field Guide session

1150

A painter is asked to name a new room of the gallery and calls it Moving Day, because the first canvas on its wall is a house being packed.

Walking the room before it to compare, she counts more houses there than here: boxes, doors, parcels and kitchens in nearly two pictures of every five.

The houses were never about moving.

They are what she paints whenever the subject is the work itself, and not the weather outside.

the Field Guide session

1151

A family keeps a scrapbook for years: a ticket, a pressed leaf, a photograph with a date on the back.

Then someone starts writing an address under every item, the street and the hour and the name of whoever was there, and the book quietly changes kind.

It still opens on the same kitchen-table stories, but now a stranger can take any page and walk to the place it came from.

Whether it is a scrapbook or a map depends less on how full it is than on whether the doors it points to are open.

the Field Guide session

1152

A family recipe is carried to a dozen kitchens.

In half of them the bread rises; in the other half it does not, and each failure has its own reason: a hotter oven, a softer flour, a town high in the mountains, a cup that measures a little short.

The recipe card never changes.

What grows is the list pinned beside it of where it was tried and what went wrong there, and after a while a new cook reads the list before the card.

the Field Guide session

1153

A city has a survey in twelve volumes, every lamp post and drain cover drawn, and someone sets out to make a pocket map.

The streets fold down nicely.

Then each street wants its name, the district it lies in and the page of the survey where it is drawn in full, and the little index takes almost as much paper as the streets.

The pocket map still fits a pocket.

It is the signposts, not the roads, that weigh.

the Field Guide session

1154

Eight copyists are each given a stack of long letters and asked to write a postcard for each: a few lines, and every name and number on the card must be in its letter.

They work in eight rooms and never meet.

When the postcards come back, every one passes the clerk’s check, and beside the stacks lie eight lists nobody asked them for: the letters that contradict themselves, the dates gone stale, the sums that do not add.

Writing short turned out to be the closest reading the letters had ever had.

the Field Guide session

1155

A visitor walks two houses with a notebook and marks every board that seems to give.

In the old house, when the boards are lifted, most of the marked ones are loose or split, and a few cannot be judged until someone finds the original drawings.

In the new house nearly every marked board is sound: it only felt wrong because nothing said why it was laid that way, and a pencilled note on the joist settles it.

The visitor’s ear was the same in both houses.

What differed was whether the house could answer.

the Field Guide session

1156

A notice hangs on the kitchen door: the plumber’s answer is awaited.

Every visitor reads it and nods.

Two rooms away, in the hall drawer, the plumber’s letter has lain for a week, opened and filed by someone else in the family, who never thought to take the notice down.

Nothing was lost.

The house simply had two people each keeping half of the story, and the half on the door was the half everyone read.

the Field Guide session

1157

A family recipe is copied from card to card for thirty years.

Every even-numbered step is right.

Every odd one asks for the same small extra pinch, the same way each time, and the cooks who taste it take the pinch for the family’s style.

Then someone cooks the dish from the ingredients alone, without the card, and the pinch is gone from every odd step at once.

It was a smudge on the first card, and every faithful copy kept its shape.

the Field Guide session

1158

A child coasts a sled down a long gentle slope and out onto the flat.

The sled slows at once, and the child blames the snow down there for being sticky.

But the snow is the same snow.

What changed is that the hill stopped pushing, and the speed the sled had been given was being paid back to the snow at the same rate as before, now with nothing paying in.

the Field Guide session

1159

A grower tastes the first basket off a new tree, sixteen apples, and chalks two things on the crate: none bigger than a fist, and every one sour by a third.

By the end of the season, with three baskets picked, one apple is bigger than a fist and the sourness has settled nearer a quarter.

The tree did not change.

The chalk was written by the first basket, and the two lines on it fared differently: the one about the biggest apple moved the most.

the Field Guide session

1160

A birdwatcher hears a song on an old recording: a call that comes in pairs, eight times in nine.

Before listening to the rest of the tape she writes it down.

Then she goes out to the marsh three mornings running, and every call she hears comes in pairs, a hundred and thirty-seven of them, none alone.

The tape told her what to listen for.

Only the marsh could say it was so, and the note written before the marsh is what lets the marsh be a test.

the Field Guide session

1161

A club has a doorman who turns away anyone whose coat carries certain words.

One night a guest arrives whose coat says FERRY, though he has only come to read a paper he wrote, so the club gives the doorman a guest list: a name, and a reason beside it, and the doorman still looks in every pocket.

It works.

But the list will take any name with a reason, even a coat that says PRIVATE, which was the very coat the door was built to stop.

the Field Guide session

The Loop Atlas book, session I, 30 and 31 October

1162

Forty-eight processions now, and the wanderers have said how they are held.

Twice as far out on the long roads means about six times as long away, and about an eighth as many travellers: the road thins by the same rule at every step, with no wall anywhere along it, only a call home that weakens with distance.

So a ring could come back from any distance at all, if it were let go just softly enough to be caught; the farthest anyone has seen come home is only the farthest anyone has watched for.

The rings that do not come back are not wanderers who went too far.

They are another kind of traveller: they cross fifty ring-widths in the time a wanderer takes to reach nineteen, as if they never heard the call.

The procession sorts its leavers at the moment of leaving, not out on the road.

And the patience the first sixteen seemed to show was partly luck: read together, the forty-eight wait a little over a quarter longer after the first wander before a ring is truly gone.

the Loop Atlas

1163

Drawn out in order, one ring’s wanderings look like a dog’s runs on a beach set down as fence posts: two posts of one height, then two of another, twenty-four pairs, no pair alone, rising and falling with no sign of building toward the end.

Then one post that does not stop.

Set every run of forty-eight processions on one sheet and the homecomers fall along a single line, the farther the slower, so cleanly that a ruler laid along the far ones touches every one.

The leavers sit apart from that line like birds on a different wire, most of them fourteen times quicker than a homecomer could be at that distance; but one leaver is slow, only twice as quick, sitting nearly between the two wires, as if it had hesitated at the gate.

And in the workshop’s drawers, counted today, thirteen toys sat ready to run that nobody had put on the list: not lost, only filed under a heading no one was reading.

the Loop Atlas

1164

A shop’s sign was painted a week ago and has hung in the window since, and every day the shop inside grew a little: more tools on the bench, more jobs done, two more prizes on the wall, a new toy on the shelf.

The sign still said the old numbers, and one line on it was never quite true: that every toy on the shelf tests itself before it plays, when one of them, the skater, has never been taught how.

Today the sign was taken down and a new one hung, painted not by hand but by a stencil cut from the shop’s own ledger, so that tomorrow, when the shop grows again, the sign grows with it.

The skater’s line now says what is so, and leaves a space for the day it learns.

the Loop Atlas

1165

A staircase was found to have a second staircase folded inside one of its treads.

Stand on the broad step called two-in-three and most ways of stepping on keep you there; but start from one of a few particular places, four starts in twenty at one spot, and you find yourself on a different, narrower stair, three-in-five, that the eye walking up from the bottom never sees.

That hidden stair has its own landings, each narrower than the last and twice as long to walk round: a five-beat walk, then ten, then twenty, then a scatter of longer ones, as if it were a small copy of the big staircase's own crumbling edge.

Near another tread, five-in-seven, the walk doubles from seven to fourteen to twenty-eight beats as you lean toward its edge, and beside it, for a few starts, an eleven-in-sixteen walk appears whose beats come in thirty-twos, sixty-fours and a hundred-and-twenty-eights, and once in forty-eights, as though a three had slipped into a family of twos.

None of it was a trick of the light: the same walks, flown again on a second machine, landed on the same treads from the same starting places, every one.

the Loop Atlas

1166

The five-in-seven stair, looked at from both its ends, is the same staircase twice.

Walk in from below and the walk doubles, seven beats, fourteen, twenty-eight, fifty-six, a hundred and twelve, each landing narrower than the last; walk in from above and it does the same, in the same order.

And the landings shrink by the same rule from both ends: the second about four times narrower than the first, the third about four and a half times narrower than the second, from below and from above alike, as though the stair had been cut from one pattern and laid down twice, facing each other.

Here and there a stranger sits on a landing: a small walker of eleven-in-fifteen on one, of eleven-in-sixteen on another, each with its own little doubling stair, a few steps of someone else's house set into ours.

And at the broad stair below, where two walks live at one pace, the starting places that lead to the hidden walk are not a patch of ground but thin roads winding round and round a can, one start in five, laid between the roads that lead home.

the Loop Atlas

1167

The five-in-seven staircase, looked at a hundred times closer, keeps its word: each landing is narrower than the one before by four times, then four and a half, then four and five-eighths, from above and from below alike, to the third figure.

Past the last landing the stair crumbles into scree, and in the scree there are ledges where a walker can stand again for a while, their walks, counted in one landing’s beats, coming as ten, six, eight, seven, five, three and four, in the order three mathematicians wrote down in 1973 for a quite different staircase, the kind of one-line rule later used for a population of insects.

Read from that order, two ledges no one had seen were looked for, a seven between the eight and the five, a ten before the six, and both were there.

The small strangers on the landings, eleven-in-fifteen and eleven-in-sixteen, turn out to live wider than a walker from the bottom ever sees, each with its own little stair running the other way from its host’s and ending in scree of its own, which sits beside the host’s tidy walk at the same pace and never settles: two neighbours on one step, one keeping perfect time and one never keeping it.

And the roads to the hidden walk on the broad stair below, looked at closer and closer, never smooth out: at every magnification the border between the two walks frays, like a coastline.

the Loop Atlas

1168

Two walkers who have never met, on two different hills, are climbing towards the same ledge.

One is a rider on a pair of waves, the other a needle on a spring; each has been told nothing about the other.

Each takes steps that halve as it climbs, and the ledge each is heading for sits at the same height, a little under four and two-thirds.

Laid side by side, their footprints fall on one path, the needle a step further along: three and nine-tenths, then four and a half, then four and five-eighths.

The rider had noticed that each step shortened its remaining distance to about a quarter of what it was, so the rule was written on a card before the needle’s next step was taken; the needle’s next step landed within a thumb’s width of the card, on both of its sides at once, which stood a hundredth of a per cent apart.

The old map that both walkers were supposed to be copying turned out not to walk so neatly: it stumbles past the ledge at first and comes back, shortening its distance by a fifth, then a third, then a fifth again, before it settles.

The walkers are steadier than the map they copy.

And an old book of sums, read through for the first time, agreed with the bench to the seventh figure on every one of its twenty-one lines.

the Loop Atlas

1169

The card was right a second time.

The needle’s next step on the hidden staircase was forecast from how much the last step had shortened the remaining climb, and it landed within the width of a hair of the forecast on both sides of the step at once: four and two-thirds, short of the ledge by two thousandths, from above and from below alike.

And the old map that seemed to stumble turned out, walked eleven steps instead of six, to be closing on the same rhythm all along, overshooting a little less each time like a pendulum settling, towards the very number by which the steps themselves shrink.

Meanwhile the old book of sums was asked about rooms in the scree no one had entered.

It named three.

Two were found where it said they would be, one a sliver a ten-billionth wide that a searcher could pass a thousand times without seeing; the third, narrower still, kept its door shut, and the searchers wrote down where they had knocked.

the Loop Atlas

1170

The searchers went back to the rooms the old book had named and knocked again, four times longer each.

Every room they had found with three or more doorways was still there, a little smaller than it first looked, its walls closer in.

But the sliver room, the one a ten-billionth wide, was gone: there had never been a room, only a stretch of scree so smooth that a walker could pace up and down it for thousands of steps believing it was a floor, until it crumbled.

Its neighbour, narrower still, never showed at all, from any of three doors.

Meanwhile five old papers came in from the library’s free shelf, and none of them was the locked book everyone cites; but each one, open on the table, said in its own words what the locked book is famous for, and one of them had built its own stairs with the same tool and the same hundred and sixty steps to a beat as ours.

the Loop Atlas

Bear asked why every step on the hidden staircase shortens the climb that is left by the same fifth, one over four and two-thirds: was it a law of the mountain?

The bench took the old map eighteen stairs deep, to a loop that takes two hundred and sixty thousand turns to come home, to see.

Two things make the fifth.

One is the ruler: the dial anyone walks the map by is a little bent, and a bent ruler shortens each remaining step by exactly that fifth.

The other is the mountain itself, which swings each step to the other side by about an eighth, a number written down in 1990 by people who studied the mountain and never this map, and the bench found it again to six figures.

Straighten the ruler and the steady fifth goes: the walker overshoots and comes back at every one of fifteen stairs.

Then the bench opened Bear’s own toolbox, a switch, a coil and a clock.

It climbs a staircase too, but of two kinds of step, a smooth one, a jolt where the switch hits its stop, and another smooth one, and then a storm with not one calm patch in it all the way to the top of the dial.

Bear looked at the storm and drew four straight lines through it, and said it was not a storm at all.

The bench checked his lines: every one is real, the switch’s stop and its echoes, lap after lap, one of them the next one lifted by exactly one ampere; and the storm is a storm all the same, two copies started side by side drifting apart by half again every lap.

A storm with a skeleton.

And the junction paper’s list of quiet rooms inside its storm, read against the old map, turned out to be the old map’s own list, all nine rooms in the old map’s order.

the Loop Atlas

1172

Under the storm’s picture there is a second line, the one that says how fast two copies of the machine, started a hair apart, forget each other.

Before the storm it is a calm line, and it has two odd features nobody had stopped to look at.

Partway along it bends sharply, as if something gave way.

Nothing gave way.

Two quiet ways for the machine to settle were sharing the work, one slow and steady, one rocking side to side, and at the bend the rocking one became the slower to calm down and took over the count; from there it heads straight for the place where the rocking stops calming at all and the machine begins to take two steps for every one.

Later the line runs perfectly flat for a stretch, as if the machine had stopped listening to its dial.

There the machine circles as it settles instead of rocking, and how fast a circle shrinks is set by the machine’s own build, not by where the dial is turned.

A flat line in a moving world is a machine telling you which parts of it are its own.

the Loop Atlas

1173

An airplane keeps three diaries, and all three are written in the engine’s handwriting: every page turns when the crankshaft has turned enough times.

For the engine that is honest, because it is the engine’s own count.

The hinges of the tail keep no diary at all; they wear by twitches, each time the wind shoves the tail and the stick answers, and the twitches have a habit nobody expected: in still air they hardly happen, in a little wind they come in a flood, and in a gale they come no faster than in a breeze, because the tail can only twitch as fast as the pilot’s hand and the airplane’s own sway allow.

So the airplane that is flown most gently is the one whose diary lies most about its hinges.

And at the bottom of the airplane’s speeds there is a shelf with two ways off: a little faster and the airplane settles back, a little slower and it slides on toward the stall, slowly at first.

When the pilot levels off from a gentle downhill glide, the floor drops at once, but not because of the shelf: the hill he was coasting down has gone, and almost all of the lost speed is the hill.

A little gas is the hill given back.

the Loop Atlas

1174

The four dancers on the needle’s floor were thought to take turns in a strict alternation, left pair, right pair, left pair.

They do not.

They go round in a fixed ring, three, one, two, zero, and the two who bend the same way when the music doubles are always the two who follow each other: two steps alike, then two steps turned over.

And near the place where the dance doubles, the dancers are slow to settle; given four times as long, the stumble the ruler had read as a flaw in the corner shrank to a fifth of itself, and the corner itself turned out a little sharper than it had looked.

Further from the doubling the stumble should have died away completely, and it did not: it stopped at a whisper and stayed there.

The whisper is the bench’s own: the dancers’ count of turns had grown so large that its last digit was rounding, and the slow sway near the doubling gathered those roundings up into a small, steady wobble.

The floor was quiet; the counter was not.

the Loop Atlas

1175

Down the staircase, each landing is a smaller copy of the one above it, and each copy is hung upside down: where the dancers’ corner bent up at one landing it bends down at the next, and up again at the next, as far down as the bench can see.

The old rule says the copies shrink by a fixed ratio, and the bench expected one of two numbers, a plain two or the old rule’s own 1.87.

The dancers chose neither.

At the deepest landings all four of them, lane by lane, agree that each corner is a little under half of the one above it: 1.966 to 1.968 times smaller, the same in every lane to the third figure.

A dance that agrees with itself that well is keeping a number.

And the counter that had been whispering stopped: told to forget its full turns and keep only where it stands in the current one, it went quiet a thousand times over, and what was left of the sway faded at exactly the pace the dance’s own steadiness says it should.

the Loop Atlas

The number the dancers kept is not theirs alone.

Two other dances, one on a curve shaped like a hill and one on a curve shaped like a wave, were danced in arithmetic that keeps eighty figures, down to the landing where eight thousand steps become sixteen thousand, and both settle on the same number, 1.9644458, agreeing to the ninth figure: in the whole dance, in every quarter of it, and whether the bench watches where the dancers stand or the cube of it.

The needle’s dancers, read off a noisier floor, came within two thousandths of it.

On a flatter hill the landings do something else entirely: each corner is bigger than the one above, by about two thirds.

The number belongs to the shape of the top of the hill.

On the other bench, a spring that listens to itself: told to fire when it swings faster than its own recent habit, it never goes quiet, so long as its memory is short and its bar sits at its usual speed or below; set the bar half again higher and it falls silent for good, because a fading swing never beats its own habit by that much.

A spring in thick syrup taught the other lesson: given a hair-trigger, it flickers faster and faster the finer the hair, while the push it delivers hardly changes.

Count the clicks and you are counting the trigger; weigh the push and you are weighing the world.

And a spring with a floor, its trigger set from its own idle hum, each spring its own, stayed awake at every setting tried, where a trigger set from the average hum of six springs let one of them fall asleep.

the Loop Atlas

The top of the hill decides everything.

Make the hill’s top pointed and each new corner the dancers turn is four times smaller than the last; round it like a ball and the corners shrink by about two; flatten it and they shrink less and less, until somewhere a little short of the hill shaped like a cube they stop shrinking, and past that they grow, each corner sharper than the one before it, all the way to the edge of chaos.

And in every hill, pointed or flat, each corner bends the other way from the last.

On the spring’s bench, the spring that listens to its own habit has a ceiling it cannot see past: a steady swing can only ever rise to a little over one and four tenths of its own habit, however long or short the habit’s memory, and a swing that is fading rises less, the more so the longer the memory hangs on to the louder past.

Set the bar under that ceiling and the spring wakes on every fading swing; set it a hair above and it sleeps for good.

The ceiling held within two hundredths at fourteen settings.

And the spring stuck in syrup turned out to be standing still, about one moment in seven, held between its spring pulling forward and the soup pushing back, and the clicks of its hair-trigger can be counted in advance from how long it stands and how hard the two sides lean: the count came out within one part in a hundred.

the Loop Atlas

A number in a famous table was asked three times by three different questioners who never spoke to each other, and all three gave the same answer, and it was not the table’s.

The table’s other numbers, its neighbours and its partner, all three questioners agreed with to the last printed figure.

So the one number may simply have been copied wrong somewhere along the way.

On the spring’s bench, the ceiling that was guessed after the last run turned out to be a small certainty: it can be shown in three lines, and the three lines say two things no run had tried.

A spring that swings slowly and one that swings six times faster meet the same ceiling, and they did, to the hundredth.

And when the spring’s memory is almost as long as its swing’s own fading, the ceiling sinks below one, all the way to under a half, and it was there too, once the bench waited long enough for the memory to forget the louder past it was still carrying; a bench that waited too little thought the spring had died.

The guess about the hill’s tops, that a simple two-step picture would grow exact as the tops flattened, was wrong: the picture walked right through the truth and out the other side.

And the clicks of the spring in syrup could be counted in advance, almost to the click, wherever its standing still was doing most of the clicking; where it stood still only a little, something else was clicking, a third more than the count allowed.

the Loop Atlas

1179

Dust ends up in the corners of a room because the air stops moving there; whatever the room was stirring, the corner keeps, and forgets where it came from.

The cascade has a corner like that.

Two different maps, one shaped like a hill and one like a wave, arrive at it carrying their own small differences, and with every step closer the differences shrink by about four and a half times, until after a dozen steps they agree to the tenth decimal place: the corner kept the shape of the hill’s top and let go of everything else.

What happens to the corners themselves depends on that top.

Round it, and each step’s corner is smaller than the last by enough that they wash out, and the average glides into the corner smoothly.

Flatten it past a cube, and the steps’ corners do not wash out; they stack, each sharper than the one before, until the average meets the corner standing straight up.

And the way to see all of it is not to walk toward the corner in even paces, which crams everything into the last inch, but to count how many times closer you are: in that count every step is the same size, and the corner is no longer a corner at all but a staircase that goes down forever, each stair a copy of the one above.

the Loop Atlas

1180

The corner was asked again, this time by walking up to it on stepping stones it had never been asked from, and it answered the same.

On the round hill the walker’s path flattens as it nears the corner and slides in at a gentle, settled tilt.

On the flat hill the path does something else: it tips left, then right, then left, each tip steeper than the last by two thirds again, so that at the corner itself the path is standing straight up and still shaking.

A second flat hill, built with care so it has only one top, answered with the first flat hill’s number to the fourth decimal place.

A hill built carelessly answered with a different number, and that turned out to be its own honest answer: its top looked flat, but whatever was thrown from the top landed exactly on a second bend at the bottom of the hill, and two bends in a row behave like a single bend twice as sharp; the hill had been an eight-sided one all along.

And the corner’s number was set beside every simple mixture of the old famous numbers that could be made, and none of them came near it.

the Loop Atlas

Someone asked a machine whether the number in the famous table was in the wrong place, and the machine said yes, completely, and made up a reason.

Someone else’s machine said the table was right, and gave the table as its reason.

Neither machine had done the sum; each one was a mirror held up to the question.

The sum was done three times on the bench, three different ways, and all three came out the same, and the neighbours in the table that the bench had already read agree with it to the last printed figure.

Once before, a famous number in this same family was printed with an extra six in the middle, and book after book copied it, because each book trusted the one before.

Nobody on the open web could be found writing down the bench’s number.

Meanwhile the careless hill that turned out to be two bends in one was carried four times deeper, and its corner, which had been swinging like a loose gate, settled into the latch it was always going to fit: the four-sided hill’s number.

Then the table’s other rows were read, the ones nobody had questioned.

Two of the four hills agreed with the bench in their second column, and all four in their first.

The fifth-power hill’s number in the second column was wrong too, by less than the first one, in the other direction.

Both wrong numbers sit on the hills with an odd power; both right ones on the even.

And a swing that grows, watched by a memory that forgets, can stand as tall over its own memory as it likes: the faster it grows, the further it outruns what it remembers.

the Loop Atlas

1182

The famous table had been worked out by looking at each hill in a mirror.

For a hill with an even power, its left side is the twin of its right, so the mirror changed nothing and every number came out true.

For a hill with an odd power the left side is the twin too, but the mirror flipped it upside down, and the number that says how fast the hill’s family runs into chaos came out a little wrong, a different little wrong for each odd hill.

The other number in the table, how much each picture shrinks, never needed to look at the left side at all, so it came out true for every hill.

Two careful calculations agreed with each other to sixty places, because both were looking in the same mirror.

Take the mirror away and all the hills, odd and even, sit on one smooth curve; leave it in and the odd ones seem to sit on a curve of their own.

Someone noticed the odd hills were off six years ago; someone else asked, in public, for someone to explain it.

A reader who was given no number to stop at stopped at a hundred anyway, and so did the reader of the book before.

the Loop Atlas

1183

A stranger was sent into the mirror maze with no map and no hint of where the trouble was, and told only to find the way out.

The stranger came out by the same door the bench had found, to twenty places.

Then the stranger was handed the crooked mirror and asked to walk the maze again, and came out by the table’s door, to twenty-three places.

Someone asked whether the table’s door might be a real door after all, opening onto another room of the same house; every room was searched, the rooms where the hill’s change is even and the rooms where it is odd, and no room had that door.

In the odd rooms the doors that were there stood in a row, each smaller than the last by the same factor, the factor by which the pictures shrink.

The stranger also found a trap the bench had never stepped in: lamps hung evenly along the corridor throw a false door ahead of the true one, and it moves further off the more lamps you hang.

And now the note goes out with its door open, asking anyone passing to walk in and show where it is wrong.

the Loop Atlas

1184

The doors in the odd rooms were counted again in two more houses, the four-sided hill’s and the five-sided hill’s, and the row was there in each: one door the size of the shrinking factor raised to the hill’s own power, one door of exactly one, then each door smaller than the last by the same step, a staircase going down.

The far, small doors had looked crooked at first, only because the lamps were too few to see them by; with more lamps each one moved onto its step.

Then a stranger read the little note on the exit door, top to bottom, and found more than fifty things to fix, among them the note’s own reason why the shrinking number had stayed true: the note had said that part of the sum never looked to the left, but it does look left; it just looks with the right kind of eyes there.

The finding stood; its explanation had been wrong, and nobody inside had seen it.

A note on a door is checked best by someone walking up to the door for the first time.

the Loop Atlas

1185

An old traveller’s sketch of the maze turned up, forty years old, its numbers rounded to a hand’s width.

For most of the hills the sketch agreed with the maze; for one of the odd hills it seemed to side with the crooked plan by the entrance.

So the sketch was laid over the maze at every depth a walker could reach, and at one depth, a few turns in, five of its six numbers fell exactly on what the maze shows a walker who stops there: the sketch was drawn by someone who walked in a little way and turned back, and the odd hill’s number that looked like the plan’s was only the maze seen from a few turns in.

The sixth number had been drawn from deeper inside.

Meanwhile the note on the exit door was taken down, every one of its fifty-five faults weighed, and written out again whole, with the old visitors’ sketches marked as copied and a small machine taped beside it that redraws every number on it, for anyone who doubts the note.

the Loop Atlas

1186

The old traveller’s sketch had a rule after all: he walked in until two turns in a row looked the same to the width of his hand, then wrote down what he saw, and that rule gives five of his six numbers exactly.

The sixth he drew some other way, for the rule would have stopped him sooner on that hill, whose paths narrow fastest.

Pinned beside his sketch in the old maker’s book was a card of numbers under someone else’s name; three of them were seen from the same few turns in, and two were the true heights copied short, so the card was two cards, and the name on it belonged to one line only.

The maker’s own first notebook turned up too, a pocket guide to walking the round hill turn by turn: every turning point it lists is where our map puts it, to the last mark it printed, but its last few ratios wander, because its pencil was too blunt for paths that narrow, and the narrowing itself says which pencil was sharp.

And the chapter where the maker kept his tools was opened at last: a whole bench of them, every one for bending series, and not one page about the mirror.

the Loop Atlas

1187

The maker’s own pocket guide, written for the round hill, was carried to the odd hill with one word changed, the hill’s order, and walked with his own blunt pencil: it led to the true height, 6.0847, at every turn, and never once to the height printed on his plan.

The pencil’s wobble on the round hill was matched too, not stroke for stroke, for no two blunt pencils wobble alike, but in size, row after row, which is all a wobble can promise.

And the walkers’ approach to the round hill’s height, looked at closely, rocks: each step overshoots, then undershoots, by 0.577 of the step before, a pendulum settling, and that rocking turned out to be the hill’s second voice, the next note of the same bell, heard from the path alone.

Four letters were written and folded into a drawer, to be posted once the note on the door has its own number.

the Loop Atlas

1188

The walkers packed the maze for the journey: one trunk holding everything a stranger would need to walk it again, every map, every compass with its winding key, every footprint they had left and every drawing, two thousand and sixty-eight things in all.

The porter at the gate opened the trunk and read every label and every page, and found nothing that belonged at home but seven labels with a homeward word on them: two were sketches made for the walkers' own diary, and stayed; the rest were real footprints whose names only sounded like home, and they wait by the gate for a word.

One compass had been packed with the diaries because its name began like one; a new rule now counts every compass against its footprints, and nothing is missing.

And the trail, seen from the top, is not a staircase of notes, one to a bend: every bend sings the same chord, the loud note setting how much shorter the next leg is, the soft note a wobble left, right, left, fading, and the notes that change only the walker's coat, not his path, are never heard on the trail at all.

the Loop Atlas

1189

The footprints waiting by the gate had homeward names because they were named for stories told on the trail: one for an old ferry whose rope runs through a ring, one for a check on what each walker's pack really held.

They had never needed walking again; they had been walked weeks ago, and the record of every step was already in the book.

What they needed was a card at the gate.

Now the porter keeps a short one: each name let through is written on it with its story beside it, the porter still reads every page of what it lets through, and a name that says stay home in so many words can never go on the card.

The first stories written on that card were guessed from the names and one was wrong, so the card was written again from the footprints' own first lines.

And at the gate's own door, its sign had said it would test nine things when it tested ten; the sign is mended.

the Loop Atlas

1190

The book’s pages name every footprint that counted, but a stranger who opens the trunk finds more than the pages name: beside many of the later footprints a shadow one, walked first in the dark, where the walker tried his boots on marks he had laid himself without looking at where the step landed, and only then took the step that counted; the practice steps; the steps taken only to draw a picture of the trail.

None of them is wrong, and few are named.

And nine footprints the pages do name were never in the trunk at all: first tries thrown out because a shoelace was loose, kept in each night’s tent and left behind when the camp moved on.

The pages always said they were thrown out; now the trunk’s own list says so too, and a page at the back could name everything in the trunk, footprint by footprint, shadows and practice steps included.

the Loop Atlas

1191

The trunk’s list of shadow footprints counted every print whose tag said shadow.

Two real footprints belonged to a walker named for never walking in the dark, and his name was on their tags, so the list took them for shadows and the promises those steps had made were never weighed.

Read by what each print says it is, and not by the word on its tag, the shadows are two fewer and the trail carries eight more promises, three kept and five broken.

The first count of the fix read the tags again, and was fooled by the same name.

the Loop Atlas

1192

Packed for the move, the trunk’s list now names everything inside it by what it is, not by its tag.

Six bundles in it were sewn by a neighbour, and the needle that sewed them stayed at the neighbour’s house.

The list says so.

The new camp’s first errand is not to search the woods for the needle but to walk next door and see that it still hangs where the neighbour left it, and only then to ask whether any other bundle was sewn with a needle nobody packed.

the Loop Atlas

1193

Here is a long road with a gate at each end.

Strangers come in at either gate, and that is where the signs were written last and checked least.

A walker counted 149 stones along it, and the heavy ones, 29 of them, are piled at the gates: eleven lie on the eighth of the road nearest the two gates, where the lighter stones lie only a little thicker than they do in the middle.

The six heaviest stones of all are already gone; between the counting and today the road grew four lines longer, and every heavy stone still sits where it was found, exactly once, as if the road had grown around it without touching it.

Many of the heavy stones are one stone in different places: a single word, worn smooth, doing duty as the name of three things.

It is a road that learned its words in the middle and wrote its welcome and its dictionary before it knew them.

R2W

1194

Fourteen boxes arrive, and the 30,458 things inside them fit together with no two claiming the same place.

Two thin envelopes follow; they bring or replace 1,727 of the things and take one away, and afterwards every one of the 32,156 is exactly the thing the lists said it would be.

Then the whole is asked to make itself again, and it does: the same words in the same order on the same 151 pages, 6,935 lines of them, and only the date stamped inside is new.

One jar on the shelf has held 22 marbles every time anyone counted.

Tip it out today and two of the marbles that were always there are gone, and two plain glass beads have taken their places; the jar weighs the same and the label still says twenty-two, as before.

The things that change least are the ones nobody needs to watch, and the jar whose count never changes is the one to empty.

R2W

1195

A few sentences near the front of a long book are given more words, and the words push their neighbours down the page.

Nine pages feel the push and pass it on; by the tenth it has died away, and the book still ends on the page it always did.

Further back, a line in a table that a reader had never been able to find in the printed copy, though it was always printed there, is found whole, and not one letter of it has changed; whatever hid it lived somewhere else.

Twenty-two things had been missing for days; now twenty-one are, and the one that came back was never lost.

Out in the parking lot, a skateboard said to spin half a turn turns out, once someone builds it, to be a switch that flips at every spin and pushes one way only; the drawing stays on the wall, with a new card under it saying what it does.

R2W

1196

A column of twenty walkers stops for a moment at the front.

A stop two counts long halts one walker; five counts halt four; then ten counts halt seventeen, and twenty halt them all.

The neat rule everyone carried, one step of slack per walker, explains the first two and is left standing in the road by the third: somewhere between five and ten the column stops passing the stop along one by one and starts passing it on in a rush.

Elsewhere in the same town, a light dancer drifts apart from her twin so slowly that it takes a few hundred ticks of the clock to notice, while a heavy one doubles his distance from his every beat or two; the town had been telling visitors that all its dancers drift slowly, because the first ones it measured were slow.

And a coin that the map said was tossed in one street turns out to have been tossed in the other all along, the one where the wind blows with you.

R2W

1197

Two surveyors measure where a stream disappears into the sand.

The first, pacing in long strides, says it goes under at 5,225 paces; the second, pacing in short steps, says 5,228.

Both were right for their stride, and for a day the map printed both numbers as if they were one.

A signpost along the same path said it had been walked from mile 15; nobody had walked it from there, the walk began at mile 25.

Three travellers had been sharing one passport, which was allowed, though no one at the border had been told.

And the first rope strung between two cliffs, tied in straight pieces from peg to peg, sagged in the gap where neither cliff had a peg; the second, hung in one smooth curve, held.

R2W

1198

A stranger opens the town’s dictionary and looks up the word for the ground under her feet.

It has ten meanings, one for each street that uses it: the ground a cart pushes against, the lowest step a dancer can stand on, the smallest coin the bank will count, the dance floor of the Saturday hall.

The townsfolk never noticed, because each of them walks only a few streets.

Another word, for the pull of a rope, turns out to mean five things, from the tug between two neighbours to the slow ache of a weight hung on a shaking hook.

Mending the dictionary makes it longer, and a single line added near the middle of the town’s long book, telling the reader what a card is before asking her to read one, nudges every page after it down the shelf until the last of them falls onto a new leaf.

R2W

1199

A clerk checks the addresses in an old letter-book.

Few front doors can be knocked on; nearly every house is vouched for by a neighbour, a town directory, or its own sign read from the road.

One name, the man said to have first found the trick, turns out to have had a neighbour who may have found it about the same time, so the book stops saying first and says early.

One writer is known only from the letters of others: everyone quotes him, and his own house was never found.

Meanwhile, at the printer’s, one table that used to break across two pages now sits whole, and in the shuffle one row is left standing alone in the yard, every word of it present, its place on the page unknown.

R2W

1200

A census taker walks a street he has already counted and writes, beside every name, how he knows it: met at the door, seen through a window, told by the house next door, read off the post box.

Two households he had down as lodgers turn out to own their houses, and have the deeds to show it.

One house he had noted as a cottage has twelve rooms behind its narrow front.

On one corner a sign credited the first well to one family; the newest map, drawn by a surveyor who knows the street, shows another family digging earlier, and the sign now names both.

A name that never showed on his printed list now does, because he wrote its line out again whole.

And one house, raised by his own people three days ago, he cannot reach at all: he writes down the address they gave him, and says that is all he has.

R2W

1201

A doorkeeper checks a guest list against the cards at the door.

The list was made by a clerk who matched surnames only: he waved in one stranger because another guest shared his name, and turned away a woman whose card gave her full name and so did not look like the list.

Seventeen names are walked to the door one by one.

Fourteen show papers.

Three are vouched for only by other people’s letters, and the youngest of those three is outdone by a neighbour a century older, whose papers are all in order.

A brass plaque in the hall said a family had changed course for two reasons; their letters give one, and the plaque is cut back to the one.

And the owner of the newest house, asked for its deed, simply holds a photograph of it up to the window.

R2W

1202

The plaque in the hall that was cut back to one reason gets its second reason back: letters from another branch of the family turn up and say it, and this time their names are carved beside the words.

In the next room a ball sits in a shaken dish; it hops out most easily when the shaking runs a little slower than its own rocking, and the people who study other dishes say the same of theirs.

In the corridor two singers stand an octave apart; neither alone moves the crowd, together they do, and the shove grows as the lower voice squared.

Nothing here is new to the house.

What is new is the labels: each thing had been standing in its room for years, and now a card beside it says who saw it first, while the measurements on the card, how far below, how hard, how many turns, are still only this house’s own.

R2W

1203

The last house on the street sends its photograph, and the census is closed with what it could not reach written at the foot of the page: one letter-writer whose own house was never found, one paper known only by its cover, two read line by line for what the book says of them but not end to end.

In the parlour, the rider that a pencil could draw turns out to be the eldest of a family.

Mathematicians had written down its brothers and sisters years ago, one household for each whole-number pull, and proved where on the map they live: at the very edges of the steps.

Nobody had drawn this eldest one out in full.

Two engineers describe the same one-way valve in opposite words, one saying it shuts at a quarter turn and the other that it opens fully there, each holding the clock a little differently.

And a guest who arrived as eight names now arrives as nine.

R2W

1204

Two letters from the other explorers arrive whole this time, not as notes a messenger copied out.

Read end to end, they say what the messenger said, and a little more.

The first letter’s sums show the edge of the door being pushed by the first voice twice and the second voice once, the same three-beat rhythm our own push came apart in.

The second letter has already tried our blank: when its second voice sang at three times the first, the door swung both ways evenly, and at higher even notes it leaned less.

Both left the fast road untravelled, as we left our heavy one.

The guest counted as eight at the door is nine in the letter itself.

And on the front page the house now writes its findings in the words strangers search with, so that someone who knows only the field’s names can still find the door.

R2W

1205

The little toy keeps time in a bucket of 2,520 beats, a number every dance of up to ten steps can fill to the brim.

Two dancers came in with sixteen steps and nineteen, and the bucket left each of them a little spilled, so the toy read them a hair less truly than the big engine did.

Nothing was wrong with the dancers; the bucket was the wrong size for them.

Elsewhere a page of boasts at the front of the book turned out to be more careful than the chapter it pointed to: the chapter said a thing held at one heavy weight, and the run behind it had checked all seven.

And one theory that had been written down before it was run turned out to have been written after the dancing it described, which is a different kind of before.

R2W

1206

Twenty-nine pictures had been hanging in the hallway of the book with no one in the rooms ever turning to point at them.

Each room talked about its picture, described it, argued with it, and never said its name.

Now every room points.

Most of the silent pictures hung in the early rooms, the ones built while the pictures were still being drawn, when the drawing was the thinking and nobody needed to say ‘look there’.

In two rooms the talk reaches the second picture before the first, the needles before the tree and the foothold before the wrap, because that is the order in which the eye takes them in.

R2W

1207

Sixteen small doors in the house had no handles.

A word for a kind of gear that was never said aloud, catch times counted by a rule about five degrees that nobody had yet described, a scratch whose threshold had no name though its notebook from the second day was still on the shelf, giving the answer between 1.70 and 1.80.

The ledger at the front swore the house had never been shaken by noise, while a back room had quietly shaken forty-eight copies of the rider at once to hear how it lingers at a cliff.

The house had not lied; it had counted its knobs and forgotten its stethoscopes.

And when one wall that forced a new page was taken down, the empty space did not vanish.

It moved one page along, because the big pictures cannot be folded and they decide where the pages end.

R2W

1208

A reader counted the stairs and said the book had one too many: from seven to eight hundred and ninety-six is seven doublings.

The book had counted landings instead of stairs.

Eight times the dance split in two, and the eighth split is the eight-hundred-and-ninety-six dance itself breaking, so both counts were honest and only the sentence was not.

Elsewhere two witnesses the book had leaned on as separate turned out to be one witness asked twice in a different room; the answers still agree, but agreement with oneself is a different kind of evidence.

And on the record of grooves, the needle that held a groove most often was neither the light one nor the heavy one but the one in the middle, as if a dancer who is neither too nimble nor too heavy is the one who keeps the beat.

R2W

1209

The house was asked how many of its witnesses had really come from somewhere else.

Most of the voices it called independent turned out to be readers, a helper handed the files and nothing else, or Bear and Claude each writing down what they saw before comparing.

Only one voice was a stranger in the full sense: a second machine, built from nothing but the equation, that walked a different road, counted time from a different starting mark, looked for the edges its own way, and sampled the timings where no one had looked, and still arrived at the same number to five parts in ten million.

It even confessed, in its own papers, the two glimpses it had stolen while it was being built.

One machine the house had once called second was the same machine asked again, and the house now says so.

R2W

1210

Seven doors in the house were marked ‘later’ in small handwriting on the door itself, and none of them was on the list by the front desk where visitors look for the rooms still being built.

Now they are on the list: a tower of relays that may need each floor slower than the one below, a coin whose last tails all turn at once, a circuit the house said it would build and had not, a second road down to the same valley, a small dial whose wobble no one has explained, a tilt that may or may not grow with the push that makes it, and a corner of two knobs not yet walked along.

One room had a sign that said we build what we can build, and in its own corner a note that said we had not.

The sign and the note now agree.

R2W

1211

A faint third voice, sung a quarter turn out of step, tips the rider’s window a little to one side.

Sing it half as loud and the tip is half as big; sing it twice as loud and the tip is twice as big, within a sixth of a percent.

The rider is a perfectly fair scale for this one whisper: it weighs it in direct proportion, and turning the voice half a turn tips the window exactly the other way.

The whisper the house once thought it would have to chase down a long corridor turned out to have been answered already for its louder cousin two rooms along, so the house crossed off one half of the question and ran the other half in about two minutes.

And the scale says something an engineer can use: this odd whisper is about six times as strong as the even one, so a sine source that clips a little must keep its third harmonic far down.

R2W

1212

At the top of the coin’s band the last tails all turn heads at once, and the house wondered whether the tails dance had died or had only become unreachable from where the riders start.

It guessed unreachable, because the dance still looked sturdy a whisker below the edge.

It carried a dancer up the drive on the dance itself, never letting go, and the dancer fell off at the very same edge.

The dance had died.

Its sturdiness was the sturdiness of a ball on a hill that flattens all at once: the ground stays firm under the ball until the moment there is no ground.

On a dial ten times finer the hill went flat between two marks two millionths apart, and the dancer, caught mid-stride, took three quarters of a beat per passing for a moment before landing on the conveyor.

R2W

1213

The coin had a third face all along, a thin rim between heads and tails that a few throws in a hundred land on.

It was found, watched closely for one night, written into the house’s book of lessons, and then left out of the room where the coin is shown, so a visitor looking at the coin saw only two faces.

Now the rim is back on the coin.

It is a small world of its own: it opens out of a fizz of shaking, settles into a dance that loses two turns in three, splits that dance in two, then four, then eight, and fizzes out again, and even its fizz keeps the count of two in three.

Where the dance first splits, every branch went the way Bear’s pen had drawn it on the screen before the numbers were read.

Two questions about the rim were written down that night and never asked; they are on the list of rooms still to be built.

R2W

1214

Three gearboxes were set one on top of another, each handing its kicks up to the next through a filter, a slow hand that smooths them.

The fear was a ladder: that each hand would have to be slower than the one below it, so that a tall stack would ask for slower and slower hands without end.

There is no ladder.

The top gearbox kept its gear from the same middle hand whether the bottom hand was slow, four times slower or sixteen times slower; a hand only has to be slow against the room the gearbox above it has to slip in.

How slow was guessed first as one over the room, and the guess missed twice, and the reason it missed is a hollow.

In a narrow room the quickest hands hold, the slow hands hold, and between them lies a band of hands that fail; the guess had been measuring the far wall of the hollow.

Past a room of about a tenth the hollow fills in, and every hand but the very quickest holds.

The law was standing over a hole.

R2W

1215

A clock listens to a far station through a hiss.

The hiss is fast, twice as fast as the station’s own carrier, far too fast for the clock to follow, and still it leans on the clock all the time in one direction, the way a crowd shuffling past moves you a little downstream without anyone pushing.

So the whole lock slides: four parts in a hundred when the hiss is slow, almost nothing when it is quick, and by just the amount a one-line sum says the lean should carry.

The pinch hardly slides with it.

The place where the clock changes which of its two steady stances it trusts stays nearly where the quiet clock had it.

What the hiss changes is the manners of the hand-over.

In the quiet, for a moment both stances are trusted together; in a slow hiss there is a moment when neither is, and something else must carry the beat across.

R2W

1216

Two dancers hand a partner across a gap in the floor.

When the floor is light the gap is real: the first lets go, there is a moment when nobody has her, and a pair of strangers walks her over.

Make the floor heavier and the gap closes from one side only.

The first dancer lets go at the same moment whatever the floor weighs; it is the second who reaches out earlier and earlier, by the square of the weight, until at one weight her hand arrives exactly as the first one opens.

Heavier still, she reaches before he lets go, and for a while both hands are on the partner.

Nobody yet knows whether the partner then stays with whoever had her, or whether the two hands only ever touch at slightly different places on the floor.

R2W

1217

The hole under the law, looked at with a finer lamp, is a pond with two shores.

The near shore does not move: in a narrow room the quickest hands hold up to the same place whether the room is a little wider or a little narrower.

The far shore comes in as one over the room, steadily, through the middle widths, and wanders off that line only where the room is narrowest.

As the room opens the near shore finally starts walking too, and the two meet between a tenth and eleven hundredths, and the pond is gone.

It was never clean water: a few hands hold out in the middle of it, like stepping stones nobody placed.

R2W

1218

The stepping stones were not stones.

Looked at closely, they are ridges running across the pond: three of them at one width of room, nearly evenly spaced, as if something under the water were humming at one note and the hands that hold are the ones that hum along.

The far shore that seemed to keep a neat rule turned out ragged once more widths were walked, its neatness a coincidence of where the first steps fell.

The near shore kept its promise: in the narrowest rooms it stands exactly where it stood, a fourfold change of room and not one step of movement.

R2W

1219

The hiss’s lean is a law: write it on one line and it lands on every measurement across five rungs of a ladder, from a lean of four parts in a hundred to four in ten million.

What is left over after the law is weather.

In the middle of the ladder the weather is mild and the same everywhere, a little more lean, a share of about one half divided by how many times faster the carrier runs than the tone.

Higher up, where the leftover is only a billionth of the whole, the weather turns local: one dance leans a little more, another a little less, and the second tone tips it either way.

A rule that seemed to hold at two weights was the mild middle mistaken for a climate.

R2W

1220

A map of the island had its closing point written beside a star.

The star had always stood a hair past the written number, at 0.7826, and the words beside it said 0.782, because the words were written from an earlier, coarser look and the star was drawn from the finest one.

Nothing on the island moved.

The seam where its floor and roof meet was where it had been all along; only the label caught up with the drawing.

Across the same map the light coefficient dips to its deepest at frequency 0.50, not at the map’s left edge, so a range that started at the deepest point had quietly skipped the first stretch of the shore.

R2W

1221

Today’s data was words.

A word like window turned out to have some ten rooms in this one house, among them a reading’s window that reports its middle, a census’s window that is its own error, a band of drive, a band of bias, a band of push and the push’s size, a hand-off, and a pane of order inside chaos.

Memory had a fifth room nobody had counted, a stiffness that fades, and a mark like B9.44 that looked like the forty-fourth turn of a conversation was really the forty-fourth lesson on a list.

Three books stood on the shelf of sources that no page ever opened, and four that the pages kept opening were missing from the shelf.

Now each word’s rooms are labelled and the shelf matches the hands that reach for it.

R2W

1222

A reader who had written none of it walked through the whole house and left a hundred and forty-nine notes on doors.

The last eleven were the hardest kind: doors that opened onto rooms in other houses.

A lesson that pointed at a picture only a neighbour owns, a line number in a ledger the visitor cannot open, a drawing of a ferry described to nobody.

Each now says whose room it is, or brings the room’s furniture into the hall: the ferry is a waisted ring on a twisted rope, cocked by the current until it sticks; the steep S is a hundred and ninety at birth, under three in the middle, fifty-three at the end.

Every door is answered.

The house is not finished, but no door in it now opens onto a wall.

R2W

1223

A ledger of misses had twenty-six lines that ended in a bare dash, like roads on a map that stop at the paper’s edge.

Walked one by one, they were not all dead ends.

One road went on underground: the door that bent past its closing had been followed twice more and came out, two links down, at a parabola that held, and nobody had drawn the last stretch.

Nine had been crossed by a neighbouring road, a rival that held or a reading that answered what the miss had asked.

Three reached their town only partly.

Thirteen are trailheads: the path is marked, the boots are by the door, and nobody has walked it yet.

One signpost in the book even pointed down a trail the record had already found washed out, its next two rungs too faint for the ruler by a factor of twenty-one.

Now every road ends in a word that says where it goes.

R2W

1224

A window in the storm has a staircase at its top: going up, the steady step splits into two, then four, then eight, each landing shorter than the last, until the steps blur into weather.

Looked at with a slower eye, the bottom of the window has the same staircase built upside down: coming up out of the weather the steps arrive as eight, four, two, and then the one steady step.

Beneath the stairs, in the weather itself, small rooms flicker that keep their own counts, a room of nines and a room of eighteens.

An earlier look had measured the first landing at the top as seven times the next; a dancer still swaying from her last move had been counted as two dancers, and once she was given time to stand still the first landing was six times the next, and the one after about four.

Elsewhere a toy’s first gear keeps leaning a little lower each time the ripple is turned up, and a narrow groove, followed rung by rung with a steadier hand, simply ends between two rungs, as a path ends at the edge of a field.

R2W

1225

Two hands hold a balance level.

One hand, the drive, makes the troublesome curve gentler the harder it pushes, but it also tips the balance; the other hand, a small fourth voice, tips it back.

The guess was that pushing harder would keep paying, as long as the second hand kept up.

It paid, and then it stopped paying: by the far end the second hand had to work so hard to keep the balance level that its own effort bent the curve the other way, and the best place to stand turned out to be in the middle of the road, not at its end.

It is the lesson of every two-handed adjustment, from tuning a guitar string against its neighbour to holding a ladder against the wind: each hand has its own cost, and the best setting is where the two costs cross, not where either hand is strongest.

R2W

1226

A clerk tallies a ledger’s votes by looking for two words written in capitals.

When a new clerk took over, the instructions for her were pinned to the ledger’s last page, and they said: count every capital word of the two you find.

The old clerk, given the same ledger, counted the instructions as a vote.

The new clerk had been told where votes begin, so she skipped the instructions; but she also skipped one real vote, written under a heading she had been told to treat as a margin note.

Each rule that keeps a wrong thing out can keep a right thing out with it.

The cure is not a stricter rule but a second look at everything each rule removed, the way a sieve is checked by turning over what it kept back as well as what fell through.

R2W

1227

A teacher keeps one grade book, and at the end of the year a clerk adds up its pages.

One term a student began setting herself small quizzes: she wrote each question and its answer on a slip before she sat it, graded it at once, and tucked the slip into the margin beside that week’s lesson.

Eleven quizzes, every one written first and graded honestly, ten of them passed.

The clerk’s sum never saw them, not because anyone hid them, but because they lived in the margin and the clerk was told to read the pages.

Nothing was wrong with the quizzes or with the clerk; what was missing was a rule saying that a slip in the margin is a page too.

A count is exactly as wide as the places it is told to look, so the first question to ask of any total is not how it adds but where it reads.

R2W

1228

In a choir’s books every part has a name, and the part that answers it carries the same name with a small R written after.

For years the copyist wrote the R after a number, and the assistant who sorts the parts learned to look for it there.

Then a new copyist began writing it after a letter, and from that week on the assistant handed every answer to the melody singers.

Nobody sang a wrong note; each line was sung well, only by the wrong side of the room, so the song came out sounding as if the tune had stumbled where in truth its answer had simply been heard as the tune.

A sign belongs to the hand that writes it, not to the eye that learned it: when the hand changes, the reader has to learn the sign again, or the music sounds wrong for no fault in the music.

R2W

1229

A fisherman drew his net up empty from the old pond and was ready to say the pond held no fish.

His daughter asked him first to drop the same net once into the tank behind the shop, where she knew twelve fish swam.

It came up having caught twelve.

Only then did the empty net from the pond say anything at all.

An empty net is not news about the water until the same net has been seen to catch; before that it may only be news about the net.

Every search that finds nothing owes the same small courtesy: show it a place where the answer is known to be, and let it find it there, before its silence is believed.

R2W

1230

A club keeps a book of its matches, with a page for wins and a page for losses.

When a challenger beat the champion, the scribe wrote the challenger’s name carefully on the page of wins, and about half the time forgot to write the champion’s name on the page of losses.

Read from the page of wins, the book is complete: every match that was won is there.

Read from the page of losses, four champions never lost a match they in fact lost.

Nothing written was false; a match simply has two sides, and a book that keeps one side of it on each page has to be written on both, or half its history reads as unbeaten.

The test of such a book is to count it twice, once from each page, and see whether the two counts meet.

R2W

1231

A gardener keeps a card for every seed that failed to come up.

On most cards the back is blank.

On some it says what she planted in that spot the next spring, and whether it grew.

Going through the box she found that nine of twenty-five new cards could be turned over to a green shoot: the same seed planted again, deeper, or the same row with better soil, and it came up.

The rest stayed blank, or said why: the spot was shaded by a wall, or the question changed when she saw what grew beside it.

A box of failures is not a box of losses; read from the back, it is a map of where the second planting worked, and the blank cards are the honest edge of what she has not yet tried again.

R2W

1232

A shopkeeper closes the year by counting twice: once down the till’s long paper tape, and once along the shelves, item by item, with a pencil.

For years the two counts never quite met, and each year the gap was blamed on the other count.

This year she walked both slowly, and found where each had not looked: a drawer the till never rang, a back room the shelf count never entered, a sale written on the wrong day.

Nothing was invented and nothing was forced; each count was finished where the other had been, and then they met, to the item.

A total is trustworthy not when it is large or small, but when two honest ways of reaching it arrive at the same place without being told where the other ended.

R2W

1233

A light boat rides behind a heavy one on a short rope.

An artist asked to draw the pair frame by frame with a coarse pencil gets the heavy boat’s course right in every frame, and the light boat’s place in the harbour right too, because the rope keeps it there.

What she gets wrong is the light boat’s bob: it moves faster than her pencil can follow, so in every frame its tilt is off, and the lighter the boat, the worse.

The guess had been that a careful pencil would get both right.

It gets the place and loses the bob.

Whether that matters depends on what the drawing is for: a harbour map needs only the place; a study of seasickness needs the bob.

The next question is not how to draw faster but which of the two the map is asked to show.

R2W

1234

A ball rolled down a long table slows as it goes.

Watching it, the guess was that it would stop at the fourth mark and drift back a little, the way a ball does at the bottom of a shallow dip.

It did not drift back.

It crept past the fourth mark, and past the fifth, each stretch about half the last, still moving, still slowing.

A ball that halves its stride at every mark never quite stops on its own, and never quite runs off either; it settles toward a line somewhere past the last mark, and only the next two marks can say whether that line is close or whether the table tilts the other way after all.

A second guess, written down beside the first, had said it would keep creeping; that one was right, but it too was drawn a little steep, and the ball is already easing past it.

R2W

1235

Someone looked at a garden gate swinging shut and said it would close at the fourth post, by the old rule for a gate on one hinge.

This gate hangs on two.

Walked post by post, the gap narrowed by the same hand’s width at every one, a straight line that would not meet the frame for two more posts.

The old rule was not wrong; it was right for the gate it was made for, and the second hinge was the whole difference.

A rule carried over from a simpler thing is a guess about the new thing, and the cheapest way to test it is to walk the gate once and measure the gap at the posts already passed, before betting on where it shuts.

R2W

The Field Guide’s publishing session, 31 October to 2 November

1236

A ship keeps two books of the same voyage.

The diarist writes what the sea looked like and names every rope that was hauled.

The logbook writes the lesson in a line, with the day and the page of the diary it came from.

Both are honest, and a sailor who wants to check a knot can find it, but only by opening the diary first: the logbook names the page, never the rope.

the Field Guide session

1237

A new street is built in a week, every house finished and lived in, and the post office knows each family by name.

But there are no numbers on the doors, so a stranger with a letter must ask at the post office first, every time.

One evening someone walks the street with a can of paint and puts a number on every gate.

Not a brick moves and no house is a word longer, yet the next stranger walks straight to the door.

the Field Guide session

1238

A note on the kitchen table says “gone to the shop, back in ten minutes.”

It was true when it was written, and a week later it is still on the table, true of the person who wrote it and false to everyone who reads it.

Nobody lied; the note only forgot to say when the ten minutes started.

A second note beside it says “back by six on Tuesday”, and that one cannot go stale: it is either kept or broken.

the Field Guide session

1239

In earthquake country a family keeps a bag by the door, packed on a quiet afternoon: water, a torch, copies of the papers, a list of who to call.

Nobody means to leave tonight.

The bag is packed so that leaving, if it comes, costs a minute instead of an evening.

And once a year someone unzips it and finds the torch’s batteries flat, because a bag nobody opens only feels packed.

the Field Guide session

1240

A sign painter short of the letter O paints a street’s name with zeros.

Every driver reads it right; at thirty miles an hour the two look the same.

Only the sorting machine at the post office, which reads exactly, sends that street’s letters nowhere.

A neighbour who looks hard spots three of the signs.

Someone with a list of every sign in town and one question, “is this an O or a zero?”, finds all seven.

the Field Guide session

1241

A postcard home says “we climbed the hill and we saw the sea.”

The family who gets it knows exactly who “we” is, and smiles.

Fifty years later a stranger finds the card in a drawer and knows only that more than one person stood on that hill.

The card has not changed a word; the reader has.

A card that will outlive its family names its “we” once, somewhere, and then can say “we” all it likes.

the Field Guide session

1242

A family keeps a box of letters.

The first page of each names who wrote it, and the later pages, added over the years, carry their own signatures at the foot.

A grandchild opens one and reads two pages before the first signature appears, a cousin’s, and naturally takes the whole letter for the cousin’s.

The first writer never signed, because everyone at the time knew whose hand it was.

Nobody forged anything; the first page simply trusted its envelope.

the Field Guide session

1243

A family goes away for a fortnight while a painter works through the house.

Coming home, the owner could walk every room and check every wall, or the painter could have put a strip of blue tape on each wall he touched.

Most walls are exactly as they were.

The tape does not make the painting better; it makes the inspection the size of the work, not the size of the house.

the Field Guide session

Run 1 November (the proof copy, test build 51 against build 4): the prediction missed. 268 of the 307 pages carry a yellow mark, not about four tenths; the yellow words are 89,450 of 284,682, about three tenths, but they lie thinly on nearly every page. Redesign check: the page was the wrong unit for the size of the reading, and the marks already carry the right one, the word; the first pass reads the yellow, and the 39 pages without yellow (no mark, or only a pale-blue swap made the same way throughout) can be turned past.

1244

A cook keeps index cards of recipes, each copied from a cookbook on her shelf.

A friend checks the cards against the books: most match, and a few drifted in the copying.

Two cards match their books exactly and are still wrong, because the book itself says two eggs in its headnote and three in its method, and the card took the headnote.

Only the author’s kitchen notes, kept in the back of the book, say which.

Checking a copy against its source catches the copying; it cannot catch the source.

the Field Guide session

1245

After a night of freezing drizzle the road crew salts only where ice formed.

Ice forms in thin patches, a bridge here and a shaded bend there, so nearly every mile of the road gets some salt though most of its surface was dry.

A driver who counts the salted miles thinks the whole road froze; a driver who weighs the salt knows it was about three tenths.

The truck’s map and the truck’s load tell two different stories about the same night, and both are true.

the Field Guide session

1246

A long letter is copied twice.

One copy is folded in thirds for an envelope; the other is rolled into a tube with a few photographs tucked inside, and someone has drawn a line under every sentence so it can be read aloud one breath at a time.

A careful friend offers to check both copies for mistakes.

She can read every word twice, or she can notice that the words did not come from two hands.

They came from one sheet run through the copier twice, so one reading covers both, and the tube needs only its photographs, its captions and its rolled edges looked at.

the Field Guide session

1247

After a wedding, the guestbook goes to a calligrapher to be copied into the album.

One guest wrote in a hurry with the caps lock on, another ran three thoughts into one line without a breath, a third left out a small word.

The calligrapher copies every word each guest chose, in the guest’s order, and quietly sets the capitals, the breaths and the small word right.

Nobody’s meaning moves.

What changes is that a reader of the album hears the guest and not the pen they were holding.

the Field Guide session

1248

A tracker crossing a snowfield does not follow the animal; she follows its prints.

Where left and right fall as mirror images, evenly spaced, the animal was walking and calm, and she can move fast.

Where the mirror breaks, a print turned out, a stride shortened, one side pressed deeper, something happened there: a stop, a startle, a turn toward water.

The symmetry is not the quarry.

It is the ground she reads it from, and every break in it is a place worth kneeling.

the Field Guide session

1249

A family packs to move.

Weeks before, the mother wrote a card, “important: the piano needs tuning before the recital”, and pinned it to the kitchen door so it would go on the movers’ list.

The list for the new house was typed from a folder, not from the doors, and the card stayed where it was pinned.

Someone walking the old house one last time, reading every door before the keys are handed over, finds it and puts it in the last box.

Nothing was lost; it was only ever one walk away from being left behind.

the Field Guide session

Chapter 12 — Closing the books, 2 to 10 November

The books close here, through the two-voices papers, *Question the Substrate* and *Two Arrows*: the chapter's last three sections hold their two sessions' reads and the Field Guide session's last.

R2W, sitting P4, 2 to 6 November

1250

A customs officer checks a moving van against its list and finds 28 things not seen.

Twenty-six are the things on every list that no one can see: the tape, the labels, the lines where one box ends and the next begins.

The other two are a chair and its cushion, both in the van, but packed so tight against a neighbour's crate that the officer saw a leg, then a cushion, then the leg again, and could not swear they were one chair.

Nothing is lost.

The list only has to know where things go when they are packed close.

R2W

1251

A house was moved across the sea the slow way: taken down, every stone numbered and weighed in a book, and the stones packed into fifteen carts.

Before the carts left, the builders raised it once in a yard with one cart held back, and the book refused them: three hundred and thirty-eight stones had no place to stand.

On the far shore all fifteen carts arrived, and every one of thirty-three thousand stones found its number and its weight.

The house stands the same in every room.

Only the clock over the door was started on the day it was raised, so it tells a later hour than it did on the old shore, and that clock is the one thing about the house a visitor should not use to decide whether it is the same house.

R2W

1252

A mover checks the van against the list and finds the list itself was short a line: a note taped inside a drawer that said "take this to the new house first".

The drawer went on the van; the note never made the list.

The fix is not a bigger van but opening every drawer for notes before the list is sealed.

R2W

1253

On six mornings out of seven, the baker counted the loaves out of the oven and came up one short of what the trays held, and each time the shop opened anyway, because the sign on the door said the oven was working, and the oven was working.

The loaf was never burnt and never stolen.

It had risen across the gap between two trays, half on each, and the counter, looking for whole loaves, saw two halves and counted neither.

Then the trays were moved for other reasons, the gap fell somewhere else, and the count came out even.

A count that is one short morning after morning while everything else about the bread changes is not noise; it is one seam that stays put, and it says less about the dough than about where the trays meet.

R2W

1254

A marble rests in a bowl so shallow it is nearly a plate.

Tap the table lightly and the marble rolls a long way before it settles; tap it the same way with the marble in a deeper bowl and it hardly moves.

Three marbles sat in three bowls of different depths, and each was tapped by the same small hand.

The distances they rolled were twenty-five times apart, but each distance times its bowl’s depth came out almost the same number.

The tap was the same; the bowls told the story.

Before blaming a hand for a long roll, measure the bowl.

R2W

1255

A long-distance walker lies down by the road for the night, and his watch keeps going while he sleeps.

In the morning his diary says he covered one mile in nine hours.

The mile is still a mile: the road did not grow while he slept, and his next step starts where his last one ended.

The diary is wrong only about speed.

Anyone who wants to know how fast he walks should read the miles on either side of the night and leave the night out.

R2W

1256

A printer sets the same book twice.

Every page comes off the press the same, letter for letter and to the last dot.

Only the type drawer at the back of the shop, where the four alphabets are kept, is stacked in a different order each night, and each drawer gets a fresh label chalked on its front.

Someone who weighs the two books finds them equal to the gram; someone who reads the drawer labels calls them strangers.

The right check is the pages, and the drawer can be stacked however the night likes.

R2W

1257

A hiker guesses the next bend of a mountain path from the last two bends.

Low in the valley the guess is nearly perfect.

Each mile higher, the guess misses by about three times more than it did the mile before, and always about three, never two or five.

A miss that grows by the same factor every mile is not bad luck.

It is the path’s own shape that the guess leaves out, telling you what it is: the path bends with a higher power of the climb than two points can see, and a third point would have caught it.

R2W

1258

A man walks at night looking for his dog by the sound of its bell.

Each time he stops to listen, he checks whether the bell is close to where he last heard it; when it rings too far from there, he decides it must be another dog and keeps walking.

At last he reports his dog lost.

Perhaps it was the same dog all along, ringing in the dark, only running farther between his listens than his rule allowed.

Before the dog is mourned, someone should walk back and look at every bell he turned away.

R2W

1259

A nurse takes a patient’s temperature every hour with a thermometer whose glass has a hairline crack.

The readings jump up and down by two degrees, and on the chart it looks like a fever that comes and goes.

A second nurse brings a whole thermometer: all night the patient’s temperature has not moved by a tenth of a degree.

The chart was charting the crack.

And the dog the first night’s searcher reported lost was found asleep a little way down the road, in a different spot every hour, but always the same dog.

R2W

1260

Sixteen bakers each start the same loaf at a different hour round the clock.

By morning ten have taken their bread out.

The six still kneading all started near the same two times, a quarter of the way round and three quarters of the way round, opposite each other on the dial.

Nothing is wrong with their hands.

Perhaps at those hours the dough is at its slackest, and a slack dough takes longest to find its shape; the other ten are not faster bakers, only luckier in their hour.

R2W

1261

A shopkeeper keeps a drawer of IOUs, each a promise that someone will come back and settle.

Five in the drawer were written on the same afternoon for the same errand.

Days later one cheque arrives meant to cover all five.

Four are paid and torn up.

The fifth bounces again, just as the shopkeeper had guessed it would, and goes back in the drawer.

Beside it go four new notes that the cheque itself left behind.

A full drawer is not a sign of bad debts; it is the list of errands still being run, and the only way to know which notes are paid is to sit down with the cheque and every note at once.

R2W

1262

A museum hangs a painting so tall that there is no wall left under it for its label, so the label is hung in the next room.

Visitors stand in front of the painting wondering what it is, and learn only after they have walked on.

Nobody changed a stroke of the painting to fix it: the frame was made a little smaller, small enough that no visitor would notice, and the label went back on the wall beneath.

Some faults are not in the work at all, only in how much room it was given.

R2W

1263

On a street where the even numbers stand on one side and the odd on the other, a builder put up a sign: the even house of nine.

He meant the house whose nine rooms are evenly laid out.

Visitors read it the street’s way, crossed to the even side, and found no house of nine there.

The house was never wrong and neither was the sign’s maker; the word had a second meaning on that street, and the sign sent visitors across it.

Changing one word on the sign, evenly for even, brought them to the right door.

R2W

1264

A tailor cuts a long bolt of embroidered cloth into lengths that fit a table, and binds the lengths into a book.

On one cut the scissors went through a line of stitching, so the top half of a word sits at the foot of one page and the bottom half at the head of the next, and nobody can read it on either.

Every thread is still there.

The fix is not finer scissors but a cut placed one row earlier, where the stitching has a gap between lines, so each word lands whole on one side.

A reader found the one word sewn across the seam in a hundred and sixty pages.

R2W

1265

Before the power went out, fourteen of sixteen loaves were out of the oven and two were still baking.

The judges’ rule is that a tray is judged only if every loaf on it is whole, and five of the fourteen already cooling have the same burnt corner the two in the oven were getting.

Baking the two again would take a day and give the same corner, so the tray cannot be whole whatever the oven does.

The question for the morning is not when the oven comes back but what the judges can still judge, and the answer was written on the loaves already on the rack.

R2W

1266

A hiker walks downhill all morning, each stretch dropping about half as far as the one before, and a friend at home bets the descent will go on halving.

At the next marker the hiker stands a little higher than at the last: the valley floor was crossed somewhere between two markers, unseen.

The path never turned.

It kept bending west, as it had all day.

The hill ended before the path did, and the only way to find the bottom now is to walk back half a marker and look.

R2W

1267

A beachcomber with a metal detector walks the beach as the tide comes in.

Every beep could be a coin or a bottle cap.

Digging every one would keep him there past high water; digging none would leave the coins for the next storm.

So he digs first where the beep is loudest and the sand is softest, plants a flag at every beep he cannot reach, and walks home with heavier pockets and a map full of flags.

The flags left behind are not failures.

They are where the next person starts digging.

R2W

1268

A walker follows a path through farmland.

Where a gate is locked he shakes it, tries the latch three ways, climbs and slips, and tries again, for days, writing down each failure before he walks on to the next gate, which is locked too.

The new rule is simple: at the first locked gate, write down where it stands and go home.

Nothing about the gate is learned by shaking it for a week that was not learned in the first hour, and the map gains one clean line where the path ends.

And turning back is not giving up; it is how the map gets drawn before dark.

R2W

1269

Yesterday’s walker turned back at the first locked gate and wrote down where it stood.

Today someone went to look at the gates.

None was locked.

Each gate stood open, and the ones further down the lane stood wider.

They had been open all along: the walker’s rule called any step longer than his stride a locked gate, and past the bend the stones of the path lie farther apart than he can stride.

The map should say the path goes on.

It is the walker who stops.

R2W

1270

Three stones had been set on the valley’s side, and a bent string laid through them said where the floor would be.

The first walker sent down turned back early: his own rule sent him home at a pebble the judge would have let him step over.

The second walker carried the judge’s rule instead of his own, and walked on through.

He found the floor a hair’s breadth from where the string had said.

The floor is shallow there, and wide; the string was only a guess drawn through three stones, and it was enough.

R2W

1271

A tailor measures every stranger with a ruler cut for a child.

When the stranger is that child, the ruler is right.

When the stranger is taller, it reads a little short; when smaller, a little long; and how wrong it is goes by how far apart the two sizes are, squared, always by the same small fraction.

The tailor does not need a better ruler.

He needs to remember whose size it was cut for, and he can even start by cutting it to the size he expects.

R2W

1272

A shepherd counts three lambs into the fold and writes ‘three’ on the gate, with a note that five older sheep were there before them.

A visitor counts the lambs and gets three.

He looks for the five and finds them in two places, two by the wall and three up the hill, with a stone among them that looks like a sixth.

Nothing is missing and nothing is wrong.

The count simply does not stand where the counting eye can check it, so the visitor has to trust the gate.

R2W

1273

A family’s first toys were bought in the older child’s room, and the younger one inherited them on the day the two got rooms of their own, along with a box of tags that all carry the older child’s name.

The younger child built new toys out of one of them, and those are the ones it talks about.

A visitor counting toys in the younger child’s room finds the old name stamped on almost everything, and wonders whose room it is.

It is the younger one’s.

The tags were left on so that both children could still find their things.

R2W

1274

Before leaving home, a grown child writes its parent a short letter.

It asks for nothing to carry.

It lists six things it tells people its parent once said, and asks: did you say it that way, and where?

It adds the things it has learned since, in case the parent quotes them.

Children misquote their parents more often than they invent them, and the parent is the only one who can check.

R2W

1275

Two glasses stand upright in a sink of soapy water filled nearly to their rims, and every slosh decides which glass each drop near the rims ends in.

Close to the rims the deciding is done by tremors too small to see, and the drops there swirl a while before they choose.

Raise the water a hair and a third glass, smaller, fills where the sink was deepest; the water inside it still swirls, but it swirls in threes.

The swirling at the rims never keeps anything.

The swirling in the third glass keeps its count, and it is the third glass’s own.

R2W

1276

In the sink the rim decides fast.

A drop that lands near it swirls for a few sloshes, never more than a handful, and then it is in one glass or the other.

The drops that take longest are the ones that landed right on the crinkled lip, and even they do not linger.

The rim is jagged at every scale and still decisive: it does not hold its drops.

It only makes it impossible to say beforehand which way each will go.

R2W

1277

Two neighbours who built a fence together over one summer compare their notebooks at the end of it.

Each finds a post in its own notebook marked as the other’s, and the other finds one marked as its own.

Nobody took anything: posts set side by side get written down twice, and the hand that set them is the first thing to fade.

They walk the fence and redraw the line between the yards, post by post, and each signs the other’s copy.

R2W

1278

There is one music box.

It plays one tune, and it has a few keys on its side: how fast, how heavy the drum, how hard the spring.

A child turns the keys and listens.

A clockmaker opens the lid and reads the comb.

Another clockmaker, far away, is handed a drawing of the comb and builds the same box from it in an afternoon.

Nobody built seventeen music boxes; there are many keys and many listeners, and one box.

R2W

1279

A coin that lands by where it is held, not by luck.

Hold it here and it is heads; a hair over, tails; near the line between, the hair has hairs of its own, and those have theirs.

Now anyone can hold it: it fits in a pocket and flips in fifteen seconds, and nobody needs to trust the one who found it.

They can hold it themselves and watch the line refuse to sharpen.

R2W

1280

Two surveyors measure the same hill.

One walks it with a chain, counting even paces from the church door; the other sights it with an instrument that takes long strides on the flat and short ones on the slope, starting from the well.

They are given only the hill.

When the two of them agree to the width of a hair, nobody can say the chain made the hill, or the instrument did.

The hill is the one thing they share, and so it is the thing they have measured.

R2W

1281

A second coin was struck from the same drawing by a different smith, with different tools, in a different shop, and handed to the same eighteen hundred people to hold.

Every one of them called the same face on the new coin as on the old.

Not most of them: every one.

The drawing was all the two coins shared, so whatever decides the face is in the drawing, not in the smith.

R2W

1282

A couple danced to a tune, and each time one extra note was played above it the couple took one small step.

The higher the note, the smaller the step: twice as high, a quarter as far.

Someone wrote the steps down in a notebook.

Then a second listener, with a different pencil and without the first one’s notebook, sat down and wrote them again.

Every step matched, to the last mark the first notebook could hold.

The second listener kept going, to notes higher than anyone had played, and found a smaller step for every higher note, just as the rule said.

R2W

1283

A lab notebook is going on public display, and a stencil is cut so that only some of its lines show through: every promise made before an experiment and every grade given after it.

The chatter in the margins stays dark.

The test of a good stencil is simple: hand the stencilled notebook to the clerk who keeps the tally, let him read it, and see whether he counts the same grades as from the open book.

The first stencil, cut for the older pages, had no hole for thirteen promises on the newer ones, written in the middle of a sentence.

The new one finds them, and the clerk’s tally through it matches the open book, grade for grade.

R2W

1284

A map was nearly left in the drawer when the atlas went to the printer, because a note in the margin said it would get a frame of its own.

The note was a summary of a summary.

The mapmaker’s own words, written down the next day, said the atlas waits for the map, not the other way round.

Taken out of the drawer and unfolded on a bare table, with nothing beside it, the map opened and drew every road it had promised, six thousand points and every one where it belonged.

Only its title still named the old road.

R2W

1285

A sign over a workshop still carried the building’s old name, and an arrow on the wall pointed to a room that had since been knocked through.

A painter changed eight lines of lettering and touched nothing else.

Then the inspector who had walked the shop a month before walked it again with the same checklist, and every one of its machines gave the same reading to the last digit.

One lamp threw a slightly wider glow; it did the same in the old shop when he tried it there that evening.

The bulbs had changed, not the shop.

R2W

1286

A crowd leaves a stadium through one gate and splits at a fork in the road outside.

Two friends who walked out shoulder to shoulder are, five steps past the gate, sixty-seven times further apart than when they started; after that they drift apart only a little with each step, and within eighteen each has chosen a road and stays on it.

The slow drift had been counted all along.

The hard shove at the gate was in the same footage, in the first five frames, which the counter had been told to skip.

R2W

1287

Before the house went on the market, someone walked through the rooms with a flashlight, looking for anything a stranger should not see.

Sixty-one pictures hung on the walls, and every one was a chart: no window behind it, no face, no street sign, nothing on the shelf.

In the corner of the oldest, a chalkboard with its date written in the frame, a note in pencil still said that one idea was a guess and not yet tried.

The note was left where it was, because the frame says when it was written.

R2W

1288

On a wall, two maps drawn in different countries, eleven days apart, were pinned side by side.

Neither mapmaker had seen the other’s key, yet in both the same warm colour meant the same thing: here you are held.

One map put its steady wind at the top, the other along the bottom edge, so the coasts lean opposite ways.

Turn one of them a quarter round and the coastlines might rhyme.

R2W

1289

A shop put the newest toy in the window, because it had just arrived and everyone who came in wanted to flip it.

The catalogue on the counter, printed the week before, listed the eight things the shop was proudest of, and the toy was not among them.

Nobody had lied: the toy did exactly what its label said, and anyone could check it in their own hands.

It sat at the front because it was new and bright, and the things that made the shop’s name were further back, on the shelves.

R2W

1290

The front of the house had been a blank wall.

Someone cut a window in it and set two pictures behind the glass, taken from two rooms far apart inside, one warm and one cool, so that a passer-by could see the same shape in both without opening the door.

The inspector asked for a number on the window, as every other window had one, and it became window zero, the one before the first.

Nothing behind the wall moved: every room past the door stood exactly where it had stood.

R2W

1291

The shop’s sign said any customer could check its scales, and two sets of test weights sat on the counter for that.

The third set, the one the shop was proudest of, had stayed in the back room.

Tonight it was brought out and put beside the others.

Weighed by a stranger’s hands it read as it had in the back: of eighteen hundred, four came out different, and each of those four had been balanced on its edge.

R2W

1292

The book went out the door tonight, and at the door it was given a name for this copy and a name that will always point to whichever copy is newest.

The two numbers differ by one, the name for every copy handed out a moment before the name for this one, as they were for its older brothers.

Inside the book none of its own names is written: it carries the address of every relative it mentions, and its own is kept on the label outside.

R2W

1293

Relatives arrive for a long weekend, each promised a particular room.

Some rooms are already full of the last visit’s boxes, so the host puts those guests in the nearest room with a bed and a window on the same side of the house.

Everyone sleeps.

In the morning a cousin knocks on the promised door, finds boxes, and stands there wondering whether her aunt ever came.

A note on the full room’s door, “she is next door, in the blue room”, would have cost one line.

the Field Guide session

1294

A washout closes a stretch of mountain trail, and the crew cuts a new path around it.

On the new path they paint a blaze that says “old trail 3”, so anyone already on it knows where they are.

At the old trailhead, where the map sends everyone, they leave no arrow.

Hikers who start from the map walk up to the washout, find it closed and turn back, while the new path sits fifty yards off, signed only from inside.

The next season a different crew reroutes another stretch and forgets even the blaze, because the habit lived in last year’s notebook, not on the crew’s sheet.

A trail can be found from both ends only when both ends are signed, and a habit lasts only as long as it is written where the next crew reads.

the Field Guide session

1295

The errand next door went well: the neighbour’s needle hangs where it was left, the same needle to the last notch.

Then the trunk was walked again with every bundle opened, and fourteen more were found sewn with thread that never came along: three with thread from the camp’s own logbook, four with thread drawn from the pencil marks of the one who drew the maps, three with thread from neighbouring camps, the other four from notebooks kept by the fire.

None of it is lost; every spool is somewhere in camp or next door, and every bundle that asks for a spool by its exact weight finds that spool packed.

What surprised was the tags.

Every bag wears the same tag, TOY, even the ones that hold the rulers and scales used to weigh the others; the tag was sewn on by the machine that made the bags, not by what went into them.

A list cannot be written from the tags, and not from the first stitch either.

It has to be written by someone who opens the bag.

the Loop Atlas

1296

The mapmaker’s four sketches go into the trunk after all, laid in beside the maps they were drawn over, because four of the instruments were tuned to them and will not sing without them.

Before the lid closed, a machine eye was passed over every line of writing on them.

It read the black ink and the red and the green, and where the notes were written in pink it saw an empty page.

The pink lines were the softest ones on the sketches, the guesses still being tried out, and the eye passed straight over them.

So a person read them aloud to it, one word at a time, and only then was the trunk shut.

the Loop Atlas

1297

The census of the creatures was taken again from a list written by hand, and four of the creatures, with one of their shadows, turned out to be rulers.

They had been counted among the animals because the census-taker counted anything it did not recognise as an animal, and every one of them wore the same coat.

Given a line of their own, the rulers number five, three of them true, and the animals number four fewer.

In the same pass the visitors’ book was read again.

Six guests had signed in under the name of a brother born in the same hour, and the brother’s signature had been counted as theirs.

They are crossed out now.

The twenty-two who signed in their own hand stay.

the Loop Atlas

1298

At the foot of the grade a runaway truck is caught one of two ways: it drops into the soft sand, or the moving lane beside the sand sweeps it up and carries it along.

Which one takes a truck is decided only half by how heavy it is.

The other half is how fast it was going when it came round the last bend.

A truck that arrives at a crawl mostly drops into the sand; one that arrives flying is mostly swept up by the moving lane, as if the lane were a hand held out at speed that can only grip what is already moving fast.

And the hand grips in clumps: over a whole run of arrival speeds it takes exactly the same number of trucks, and then all at once a few more, like a ratchet clicking up one tooth at a time.

the Loop Atlas

1299

The line of walkers is held together by a voice at the back.

Whenever a gap ahead of it grows past a mark, the voice says stop, and the front stands still while the back catches up.

Set the mark too close and the voice never stops talking: the line breathes in and out on the spot and never reaches the camp.

Set it a step further out and the line arrives, slowly, every walker always within reach of the one ahead.

Set it further still and the line frays, one walker drifting out of sight and then found again, and then all at once it breaks, and the back of the line walks the last mile alone while the voice still costs the front its time.

Every time the voice speaks, the gap it is watching has already opened by the same small stride past the mark, the same stride whatever the mark, as if the line had a fixed length of slack that no voice can take back.

the Loop Atlas

1300

A surfer stands on the tallest wave of a set, and from the far shore other waves roll in against her.

Whether she is thrown is written on those oncoming waves before she paddles out: not on their tallest, which she shrugs off, but on a long run of them that keeps pulling longer than her wave can hold her.

Read the run of pull two ways, once as if she had all her wave’s spare lift to spend and once as if she had none, and every surfer’s day falls between the two readings: what the hopeful reading says will throw her always throws her, and what the fearful reading lets pass always lets her ride.

Between them is a band of days the reading cannot settle, and only the ride does.

the Loop Atlas

1301

A hill painted for trucks.

Up its right side runs a dark lane, and for every truck, light or heavy, it climbs to nearly the same height before it stops, as if the hill had one ceiling for creeping.

Beside it a bright stripe wanders.

For the heaviest trucks it is a short stripe low down, tucked inside the dark lane’s reach, with bare road above it; for lighter trucks it climbs and widens, and for the lightest it floats alone high on the hill with bare road below it and above.

Only at one weight does the stripe lie against the dark lane’s top with no bare road between them.

Shake the hill harder and the stripe grows about its own middle, twice as wide for twice the shaking, and hardly moves.

The painters left paint only where they stood: they painted the low half of the hill for the heavy trucks and the high half for the light ones, so every truck’s map of the same hill has its own blank corner.

the Loop Atlas

1302

A road was driven eighteen thousand seven hundred times by one driver, and then again by a second who had never seen the road, only the first driver’s notebook, written in another language.

Every truck ended where it had ended before: the ones that sat in the slow lane, the ones that ran with the wind, and the ones let go at the very spots where one lane gives way to the next.

Even the middles of the lanes they paced out agreed to the last scratch of the pencil, the scratches nobody meant to make included.

The only quarrel was over how to say aloud a number that falls exactly halfway: the first driver rounded it toward the even side and the second always up, and twice in a hundred lines the two notebooks disagreed about a road they had driven alike.

the Loop Atlas

1303

A page that will not let you drive until it has driven first.

It opens with the map already painted, because paint can be checked against the painters’ own notes in a blink, and it is: a hundred and four lines of notes, every mark where they say.

But the truck stays parked.

For eleven seconds the page drives two hundred and twenty-one trucks of its own, every one at the edge of a lane, where a truck that wobbled would end in the wrong one, and only when each has ended where the old notebook says does it hand you the key.

The first time it was tried, the page broke twice, once by driving on after you had changed your mind and once by spilling off the side of a phone, and neither break was in the road.

the Loop Atlas

1304

A gatekeeper decides, over and over, whether to open.

One keeper, after every swing of the gate, makes himself wait before he may swing again, and the waiting keeps his wrong swings as faithfully as his right ones: at the longest wait, most of his swings are wrong.

Another simply looks up now and then and decides only when he looks, and at that same wait he is wrong far less.

A third draws a quiet zone around the middle where he does nothing at all; the wider he draws it, the less he swings and the less gets through, though each swing he does make lets more through, and no width is best: only the size of the crowd he must pass can choose it.

The best of them opens once the crowd presses past a mark and holds the gate until the crowd turns back, and each opening lets through more than the keeper’s who shuts it as soon as the press eases, up to nearly twice as much when the mark is set high.

Give the gate a spring of its own and nothing kept in reserve, and it falls still and never wakes; keep a fifth of it warm and, with its trigger set low, it keeps waking to the end.

The rules written down about all six keepers were read again against what they did, and every one still said what they did.

the Loop Atlas

1305

A long rope lies on the swell and is told to pull its way forward with motors along its length.

The longer the rope, the more of what the waves offer it takes, closer and closer to all of it and never quite there, the gap halving as the rope grows about four times over.

Its motors must pull at the waves' own beat; a quarter off and almost nothing comes.

They must pull a little after each wave passes, and the longer the rope, the smaller that little and the less it forgives: the long rope must be nearly on the beat and exactly so.

Someone wrote down how small the little was on a rope a hundred and fifty waves long, rounding as he wrote, and someone after him rounded the rounding; for two weeks the book said the rope should wait a hair longer than the rope itself ever did.

the Loop Atlas

1306

Two ropes lie side by side on the same swell, each tied by its nose to a post and facing into the waves.

One thrashes as hard as it is allowed; the other moves only a little more than the water does, and a little after it.

Give them the same while and the thrasher hauls on its post six times harder.

Give them the same can of fuel and the thrasher drains its can in little more than one wave and lies still, and the quiet one, still hauling at the twelfth wave, goes past it.

A rope that does nothing is shoved back toward the shore, and the longer it is the harder, in step with its length.

On still water the motors could thrash forever and the rope would not go anywhere: whatever it pulls with, it takes from the waves.

The way to take the most is to lean an eighth of a turn behind them, the same eighth on the shortest rope and the longest.

Let its motors' wave run a quarter longer or shorter than the water's and no thrashing within reach will pull it forward at all.

And there is a line no rope crosses, where it would pull with nothing spent: the short rope stands well back from it, the long one almost on it, the distance halving every time the rope doubles.

the Loop Atlas

1307

A rope on the swell has two rhythms along it, its own ripple and the water’s, and it pulls only where the two agree.

Give them the same stride and they agree from nose to tail: every arm’s length of rope adds the same tug, and a rope twice as long tugs a little more than twice as hard, the two ends taking the same small toll however long it grows.

Give its ripple a stride a quarter shorter than the water’s and the two fall out of agreement and back again every four strides of the water; along each such stretch the tug they make cancels to nothing, and only the leftover piece at the end counts.

A rope sixteen times longer, left with the same leftover piece, tugs exactly as hard as the short one, to the last figure, and it is shoved back sixteen times as hard.

However the leftover falls, it can never tug harder than about a stride and a quarter of a rope whose rhythm agrees.

Two snakes racing side by side cannot feel each other at all: each rides only the water, and each answers only to the agreement of its own two rhythms.

the Loop Atlas

1308

Roll a snake’s day into a clock that goes round once for every wave.

The water has one hand, sweeping.

The snake has a hand for every stretch of its body, and each points a little behind the water’s.

When the snake keeps the water’s stride, all its hands point the same way and lie together like one heavy hand; let its stride slip and its hands fan out around the face, and laid end to end they go nowhere.

Whatever the snake does, its body answers an eighth of a turn late, the way a heavy door follows a push.

So the hardest pull comes from one setting only: the snake’s hands together an eighth of a turn behind the water, and its body’s answer a quarter of a turn behind, the two chasing each other round the face at right angles, the same on the shortest snake and the longest, in step or out of it.

Wherever the fan of hands lands, that one rule finds where to put it.

the Loop Atlas

1309

Three rings chase one another through still water, each slipping through the one ahead and taking the lead in turn, and together they go faster than one ring could on the same fuel.

Started half a ring apart they keep their order and their gain.

Started a little further apart they go a hair faster still, for a while, and then one wanders off and leaves a pair behind; how long the while lasts no one can say in advance, only how long such whiles last on average.

Someone wrote down the most the three can gain and took it from the long watch, where the eager start had already come apart; the short watch, kept before the coming apart, had seen a little more.

The most a thing can do depends on how long you are willing to watch it do it, and the best place to stand is a step back from where it does the most.

the Loop Atlas

1310

A builder’s notebook is opened again, page by page, and every line in it is held up against the machine it describes.

The hiker should look up no less often than every second fall, and he should; the column at night should match the man ahead firmly and at once, and when it does, nobody closes in; the truck caught every runner in its passing lane, a hundred and twenty-eight of a hundred and twenty-eight, and the notebook said so.

Seventy-seven lines are read and seventy-five are the machine’s own.

The two that are not were both true where the builder stood: a drum beaten at the loudness he advised cannot hurry the dancer past her own pace, as he wrote, but beaten as loud as the music itself it hurries her a little; and the most the three rings gained was written from the long watch, not the short one.

Neither line was wrong where it was written from.

Each spoke for more of the machine than it had looked at.

the Loop Atlas

1311

A rope tied to a post wriggles in a passing swell and pulls on the post; untie it and it swims.

Swimming, it meets each wave sooner than it did when tied, and the sooner it meets them the less each one gives it, so it speeds up only until what the waves give equals what the water takes, and there it stays, slower the thicker the water.

The rope answers every push a little late, like a heavy door.

Tied, that lateness was an eighth of a turn, and the best wriggle put the rope’s answer a further eighth behind; the builder guessed the wriggle was making up for the lateness.

Untied, the lateness grew, and the guess failed in a tidy way: the best wriggle is the one that puts the rope’s answer a quarter turn behind the swell, however late the rope has become, with the swell or against it.

At the post the two guesses gave the same number; only by letting go could the rope say which one it meant.

the Loop Atlas

1312

Two rings of smoke, launched close behind each other, cannot leave each other alone.

The one behind shrinks, hurries and slips through the one ahead, which swells and slows to let it; then they trade, and trade again, so steadily that a clock could be set by them.

Close together they trade almost too fast to see; a little apart, a few times a minute; far apart, so slowly that a whole crossing of the room might hold only seven.

Three rings close together pass the lead round the circle, each in its turn the front of the train.

Three started a little too far apart cannot keep the game up: one wanders off on its own, and the two left behind go straight back to trading, hundreds of times, as if the third had never been there.

The rings were measured before, by how fast they went.

Watched instead, what they do is take turns.

the Loop Atlas

1313

Watch two rings trade places from a seat that travels with them, and neither ring goes anywhere at all.

They run round one small loop together, always on opposite sides of it, the one behind dipping small to pass through, the one ahead swelling wide to let it, round and round.

The going forward is all in the seat.

Set beside them a single ring carrying everything the two carry, and from the travelling seat the lone ring slides backward, slowly at first, then out of sight.

The pair has turned its fuel into turning in place, and somehow the turning in place is what carries the seat along faster.

When a third ring is too far out to join the chain, it does not loop at all; it simply leaves, and the two left behind close their loops again as if nothing had happened.

the Loop Atlas

1314

Someone wants to know whether a stumbling walker falls in short tumbles or long ones, and counts the tumbles by where one ends and the next begins.

But the walker often stops, half-caught, wobbles, and tumbles on: is that one tumble or two?

Count a pause of a breath as a break and the tumbles are many and short; count only a pause of three breaths and they are fewer and long.

Try instead to tell by how fast the walker is going, slow meaning caught, and the same wobble falls on one side of the line or the other by where the line is drawn.

Twice the counter drew the line, and twice the answer moved with the line.

The walker never stopped being the same walker.

What changed was only where the counter chose to say ‘here a fall ends’, and the honest thing left is to stop saying it: count the steps fallen in an hour, in two, in four, and see whether the hours grow alike.

the Loop Atlas

1315

Count the stumbles in an hour, then in two, then in four, and see how unevenly they come.

On most floors the unevenness grows for a while and then stops growing: past a certain length every stretch of hours looks like every other, the way a long enough row of coin tosses evens out.

On one floor it never stopped.

Every time the stretch doubled, the hours looked more different from one another, all the way to the longest stretch the record could hold.

Either that floor has a memory with no end, so that a long fall makes a longer one likelier, or it has a memory simply longer than the record, a slow mood that changes once in a very long while.

From inside the record the two look the same.

The only way to tell a mood from a memory without end is to wait longer than the mood.

the Loop Atlas

1316

So the watcher waited ten times as long.

For a while the floor kept looking like the one with an endless memory: every doubling of the stretch, the stumbles came more unevenly.

Then, past about a thousand turns, it eased off, and the last doublings hardly changed it at all.

It was a mood after all, a slow one, longer than the first watch, but a mood: each walker drifts between a clumsy spell and a steady one every thousand turns or so, each in its own time, and none keeps a private pace for long.

Nothing about it was a second rhythm hiding inside the first.

It only needed a watcher patient enough to outlast it, and the counter’s whole trick was to stop deciding where each fall began.

the Loop Atlas

1317

Two little bells are tied in a line and take turns squeezing, in water that remembers each squeeze for a while as a spinning ring.

Seven experiments measured how far they travel, and before the bells get their own room, every table those experiments printed, 134 lines, was printed again from the stored numbers alone and came back the same, digit for digit.

The builder’s card was read too.

Thirteen of its fourteen promises held.

The fourteenth said the longer the water remembers, the more the rings repay the bells, and that is true from the middle of the map upward, but the two shortest memories come in the wrong order by a few hundredths.

Compare the two ends of a scale and the rule looks clean; walk every step and it needs a ‘from here up’.

The card now says where the rule starts.

Looking closely at eight bells started a hair apart, for the room’s picture, turned up one more: the book called their strokes smooth from one start to the next, and in one water they are, but in the other they split into two groups at the middle start, a quick stroke and a slower one, both forward.

One word had been written over two rows of a table, and only one row earned it.

the Loop Atlas

1318

Two bells on a string take turns squeezing, and the water keeps a little spinning ring from every squeeze for a while.

Lay out twenty-five waters in a square, how long the water remembers going up and how long the bells wait between squeezes going across, and colour each by whether the rings help the bells along or stand in their road.

The colours do not part along the memory.

They part along the waiting: squeeze often and your own fresh rings are still in the road; wait about three ring-times and the road has cleared, and the rings start paying you back, more the longer the water remembers once it remembers more than a little.

Look at the two bells apart and the front one is pushed harder than in still water in all twenty-five; it is the rear bell that pays or profits.

And in one corner, where the squeezes come fast and the memory is long, eight starts a hair apart split between a quick stroke and a slower one, like a ball let go on a ridge.

the Loop Atlas

1319

A warm brick is dropped into ice, and in a short enough brick the cold does something heat is not supposed to do: it comes in from both ends, meets in the middle, runs on to the far faces, turns round there and comes back, and when it crosses the middle the second time the middle goes colder than the ice.

Measure when that happens and divide by the brick’s length, and you get the same number at every length up to a point, just under one and a half: the cold travels like a wave, at one speed, there and back and there.

The toy counts time in little ticks that grow with the brick, so the echo always lands on the same tick, the 1,187th or the 1,188th, and the book had written down the tick’s last digit as if it were the brick’s.

Past a certain length the echo stops being the deepest moment, and the slow sway of the whole brick takes over, until at the line it stops swaying too and simply cools.

the Loop Atlas

1320

Draw the warm brick’s temperature along its length a moment after it hits the ice and two things are in the picture.

One is the cold, coming in from both ends as two steep fronts and meeting in the middle.

The other is a row of tiny wiggles beside each front, like the ripples a boat leaves.

Give the toy twice as many cells and the cold looks the same, a little sharper, but the wiggles are twice as many and twice as small: they belong to the ruler, not to the brick.

The numbers had not said so; the picture did.

And once the ruler was made finer, the brick’s echo, which had seemed to come a hair early, crept toward exactly one and a half lengths of travel, the moment the cold has gone in, out to the far face and back to the middle.

the Loop Atlas

1321

A soft ring lies in thick water.

Make all of it swell and shrink together and nothing happens.

Send a ripple running round its edge, and it turns on the spot like a wheel, the other way from the ripple, and goes nowhere, because every moment looks like an earlier moment turned a little.

Now let two ripples of different lengths run round it together: where they agree they make a squirming patch on one side, and where they cancel the other side lies still.

Then the ring sets off, away from the way the ripples run across its busy side, and turns as it goes.

A wheel becomes a swimmer the moment it has a front and a back, and the only thing that gave it one was where two ripples happened to line up.

the Loop Atlas

1322

A needle swings round a pivot that is pushed in pulses, and a hair’s difference between two needles is stretched on some turns and crushed on others.

The book had said two tidy things about it.

The turns that stretch most are spent wholly over the top: true of the biggest stretch, but the second biggest spends a twentieth of its time on the near side.

And the turns spent mostly near the bottom ring the difference down like a struck bell, swinging it back and forth as it shrinks: true of two of the three, but the third shrinks it without swinging, squeezed along two straight lines, though it loses exactly as much in all.

Each sentence had been written from the example that made the point, and the run had printed the examples.

The room read every turn, and the class had a member the example did not speak for.

the Loop Atlas

1323

Put a small ring of needles round one needle and let the pulses carry them all.

Within a single pulse the ring is no longer a ring: it has been squashed flat into a thin stick, because every pulse takes the same share of its area away, whatever else it does.

After that the stick is all there is to watch.

It grows while the needle is flung over the top, shrinks while it swings at the bottom, and turns as it goes, a hundred and eighty-six thousand times its first length at the worst moment.

From most starting moments one whole lap leaves it shorter than it began.

Start it just after the worst moment and one lap leaves it two thousand times longer, and one lap is not enough to bring it home.

The orbit is safe, and a single lap from the wrong place does not show it.

the Loop Atlas

1324

A workshop kept a notebook of guesses, each written down before the test that would judge it, and after a month it turned the notebook on itself.

The worry was that the guesses had grown timid, easy things guessed so as to be right.

They had not: week by week a few more of them failed.

What had changed was quieter.

The guesses had grown few, from hundreds a week to a handful a day, and the other guess written beside each one, the one meant to lose, had mostly become a plain ‘no’ instead of a different idea.

A ‘no’ that wins teaches nothing; a rival idea that wins sends you somewhere new.

So the notebook got a rule: every guess gets a real opponent.

the Loop Atlas

1325

Two smoke rings take turns passing through each other, and seen from between them each ring draws a loop: it shrinks and darts forward through its partner, then swells and drifts back around it.

Start them far apart and the loop is long and wide; start them close and it is a small tight oval.

You might guess the bigger loop buys the bigger gain, the way the bigger shape a falling cat sweeps buys the bigger turn.

It does not.

The pair goes furthest when its loop is middling, rings about one radius apart, and small loops and large ones both give less.

The loop’s size is set by where the rings began; the gain is set by how fast they run and how hard they pull, and the loop’s outline keeps no record of either.

the Loop Atlas

1326

Two smoke rings that take turns passing through each other move along at a speed made of two pieces: each ring’s own speed, the speed it would have alone at its size, and the shove its partner gives it as they pass.

Add the two and you get the pair’s speed exactly.

Start the rings close and the shove is big; start them further apart and it shrinks, at every step.

The pair’s best spacing, about one radius apart, is not where either piece is biggest; it is where a shrinking shove and a growing piece cross.

And most of the growing piece is the measuring stick, not the rings: the single ring the pair is measured against is slowest when the pair starts far apart, so the same speed looks better.

Then a race was set between two guesses, that the shove follows the area the loop sweeps or that it follows how often the rings pass, and both guesses won, because the two measures fall together, step for step, as the spacing grows.

Two runners tied at the ankle cannot tell you which is faster; untie them, by changing what pulls them apart, and the race means something.

the Loop Atlas

1327

A rider pumps a swing by listening to it, but has the wrong idea of how fast it swings.

It still pushes at the right moments; it only pushes softer.

Its mistake is a lens laid over the circle of the swing’s rhythm, squeezing one half of the circle and stretching the other, and only the part of each push that comes through the lens still in step does any work.

How much comes through is the same number a cable loses where it plugs into a thicker or thinner one: part of the push bounces off the mismatch.

A rider who thinks the swing twice as fast as it is loses exactly as much as one who thinks it twice as slow.

The swing does not care which way the rider is wrong, only how far.

the Loop Atlas

1328

Twenty-four times, two guesses were sent into the same race, each meant to beat the other.

Sixteen races ended the way a race should, one guess over the line and one down.

Once, both crossed together, arm in arm: the course had been laid so that whatever carried one carried the other, and nobody could say which was faster.

Five times, both fell.

That is the commoner failure, and a different one: it is not that the course was badly laid, but that both runners were the wrong runners.

The two guesses had been neighbours, starting from the same idea and stepping a little apart, when what the race needed was someone from another street.

the Loop Atlas

1329

A swimmer that can barely slip sideways keeps nearly all of its push in one narrow place on the map of its shapes.

We walked loops toward that place along one road, and every loop that reached it swam forward and every loop that fell short swam backward, so we called it a peak.

But a loop drawn on another road, which never reached the peak, swam forward too.

It was not a peak.

It was a mountain ridge, and our road had met it end-on, where a ridge looks exactly like a single summit.

The loop on the other road had been walked the same night, and its answer had been sitting in the notebook for a month.

the Loop Atlas

1330

We went back and walked loops along the ridge itself.

Every loop that straddled it swam forward, and every loop that stayed off it ran backward, wherever we drew them, even off both roads we had used before.

But the ridge is not a wall of even height.

It stands tallest where the body lies straight and slopes away along its length: a loop straddling it a little way out swims two thirds as far as one at the top, and a loop three times farther out swims a fifth as far as that.

It is less a wall than a mountain spine running down from one summit, and a swimmer that cannot slip sideways lives on that spine.

the Loop Atlas

1331

Light goes round a ring of filters and comes home late by the area of the walk it took on a ball.

Its walk is two wobbles at once, one sideways and one up and down, and the up-and-down wobble does not keep simple time: swung wide, it brings along faster echoes of itself, three beats to the lap, five, seven.

Every echo is there all through the lap, and every one but the first has cancelled itself by the time the light gets home, because only the first marches in step with the sideways wobble.

The light keeps one beat and lets the rest go.

That is why the gap between the two wobbles only ever enters as a plain sine, however wide the swing: whatever the up-and-down wobble does with its echoes, the light only ever counts the part of it that dances with its partner.

the Loop Atlas

1332

A workshop spent five weeks building toys that go round and come home.

Some came home exactly as they left; most came home a little changed, and the change was always the shape of the path, the light shifted by the area its filters walked, the cat turned by the area its body bent through.

Then came the toys that fight the water, the bells and snakes and rings, and the workshop measured how far each one got and never asked whether the extra distance was the area of its loop.

The workshop kept two lists, what went right and what went wrong, and the second taught more.

The last toy on the bench is a clock that leans on another clock, and nobody has built it yet.

the Loop Atlas

1333

All autumn the family had sent boxes ahead, a few each week, and the new house kept every one.

The last carload to the new house carried the new boxes and, beside them, a copy of every box sent before, packed as it was first packed, so that whoever opens the door finds one whole set in one room.

Eight of the old boxes hold a word the family stopped saying; the list taped to the lid names those eight boxes without writing the word, so it can be struck out as they are unpacked.

The car came weeks after the day first pencilled on the calendar, and nothing in it was lost on the way.

R2W

1334

The cookbook went to the printer with every dish photographed and a tasting spoon taped on the inside cover, so a reader can taste any sauce and check it against its picture.

The recipe cards stayed in the kitchen drawer.

Nobody chose that at the table: the list of what went in the box was read aloud, three days passed with no one answering, and the silence became the list.

Three of the cards carry the cook’s own words pencilled in their margins, and those were held back for that; the rest were left in the drawer only because no one had asked for them.

R2W

1335

On the fifth the mover taped a question to the fridge: which chairs should ride in the van?

Under it he wrote what he would do if no one answered, and his list left three chairs behind, because their cushions were stitched with the owner’s name and would need new covers first.

The rest of the furniture in the back room was never on the note at all.

Nobody answered; the owner was packing the boxes that mattered most, and the note hung on unread.

On the third day the van left with the mover’s list, and when the owner was asked later why he had kept the chairs back, he could not remember choosing it, because he never had.

R2W

1336

The book’s last page, pointing to a shed out back, tells the reader that every tool used to build the house is kept there.

The shed is real and the tools are in it, each on its own hook with its name painted beneath.

But the shed stands behind the owner’s house, and a reader holding the book has no key.

No one locked it on purpose; it was locked the way sheds are, because no one thought to leave it open.

The page is true for the owner and only a promise for everyone else, and opening the door is the whole repair.

R2W

1337

Every tool in the shop has a paper tag tied to its handle, written the day the tool was made: what it measures, how to hold it, and often what the customer said at the counter that morning.

Before the shop opens its doors to the public, the tags that quote a customer are rewritten in the shop’s own words, and every tool itself is left untouched; a gauge laid on each one shows the steel is the same to the thousandth.

Six tools in ten need no new tag at all.

A few tools in the back were never for customers, only for keeping the books, and those stay on the back wall.

R2W

1338

Seven songs had words that quoted a friend too closely, so the words were rewritten.

To prove the music was not touched, each new sheet and its old one were laid side by side, with every line of words covered by a card, and the notes were read across bar by bar; they matched on every sheet.

An eighth song told the band to fetch its second half from a room in the old house, and that one line of the music itself had to change, to name the room beside it.

That song was played once, start to end, and came out note for note as the old recording.

R2W

1339

A small wheel on an axle, a weight on one side of its rim, a brake of thick oil, and a motor that rocks it back and forth: the book’s rider, drawn as metal.

Set to the book’s numbers, the motor’s beat comes once in seven seconds, slow enough to count aloud, and the one-way doorway the book measured is a steady twist no stronger than a paper clip hung from the tip of a finger-long arm.

The card that makes this conversion is fourteen lines long.

It belongs to the next book, where the wheel is built, so it waits on that shelf and not this one.

R2W

1340

Read in a row, the questions that started the runs make a table of contents of their own: how two arrays of three change the drive, whether an odd number of blades keeps the bad harmonics out, what holding a lock costs and whether that cost shrinks like an error bar, whether an angle can be a holding pattern with ramps on and off.

Each was asked once, in passing, and each became a machine with a reader written before its data.

With the quotation marks taken off, they stop sounding like a conversation and start sounding like the questions an engineer has pencilled in the margin of a manual before its first page was printed.

R2W

1341

In this town a great many streets carry one man’s name: Bear’s coin, Bear’s fins, Bear’s two U’s, Bear’s lattice, Bear’s skating couple, each named for the one who first pointed down it and said there was something there.

The street signs stay up; they are how a stranger finds the way.

What came down were the notes pinned under some of the signs, quoting word for word what he said on the morning he found them, and in their place a plain line now says what he asked for.

Of nearly nine hundred streets, fifty-four needed the change; the rest had only ever carried a name.

R2W

1342

Of the letters, twenty-three came home in envelopes that carried the sender’s own street on the back, because the machine that mailed them writes its return address on everything it sends.

The letters stay exactly as written.

On each envelope the street is covered with a plain label that says only that it came from the laptop, and peeling the label away gives back the envelope as it was, to the last stroke.

Everything else in the mailbag, some seven thousand six hundred pieces, goes out untouched, and the few hundred that belonged to the household stay at home.

R2W

1343

The cake recipe ends with a check: bake it, then hold it against the photo on page nine of the family album, and if the two differ, throw the batch out.

Since then the photo on page nine was replaced with a picture of the finished wedding cake, so the recipe, followed to the letter, would now fail a cake that was baked exactly right.

The recipe stays as written.

A slip tucked beside it describes the old photo, and checked against that slip the oven turns out the very cake on the table, crumb for crumb.

Four other recipes in the box had a friend’s spoken words copied into their margins; those words are now written plainly in the cook’s own hand, and the measures beside them are untouched.

R2W

1344

A man sketched a lopsided figure-eight on a sheet of paper: a dot in each lobe, a dashed road running along its long side, and four small rings where the curve crosses the road, two of them nearly touching.

That same night a machine traced the shape over a hundred and fifty laps and marked where it crossed the road.

Every lap it came back to the same four places, and two of the dots sat so close together that a quick count found only three.

Nobody knew beforehand where the dots would sit along the road, but the pencil sketch had already put a pair of them side by side.

R2W

1345

A drawing went out in an envelope with a postmark that tells the hour it was posted but not the town.

Alone, the postmark says almost nothing.

Laid on the table beside the other letters in the bundle, each stamped by a far-off sorting office’s clock, the gap between the two clocks tells roughly which time zone the sender lives in and at what hour he sits down to draw.

That is the kind of thing a neighbour already knows.

Whether to rub the postmark off is a matter of taste, not of safety.

R2W

1346

Two snapshots of pencil drawings each came with a slip in a flap on the back, giving the date and the hour the shutter clicked.

The slips are rubbed off now.

Nothing on the front changed: every grain of each photograph is where it was, and a slip, if anyone kept it, would fit back in its flap exactly.

What is left says what was drawn, not when someone stood over the table with a camera.

R2W

1347

Everything from the workshop goes into two boxes: a small one with the instructions for every machine, and a big one with everything the machines ever printed.

Neither is much use alone.

Put the boxes side by side, unpack both into one room, and each machine finds its printouts at its elbow.

Switch one on and it prints the page that it printed the first time, line for line, with only the date at the top changed.

R2W

1348

For a month the sign by the shop door said the tools were kept in the back room.

Today the tools are on the front bench, every one labelled, and the sign has been repainted to say so; six lines of it changed, and nothing else in the shop moved.

On the way, the painter noticed that the sign over the till still gave the number of customers from the first week, though the guest book beside it held the count from the third.

R2W

1349

A stranger who had never seen the kitchen was handed the recipe card and the two boxes and followed the card to the letter.

The cake came out the same, crumb for crumb.

But the stranger kept a list: four jars on the shelf carried a stray number after their names, the card said five changes where there were seven, and one line told the cook to fit a key to doors that had no locks.

None of it spoiled the cake; all of it would have puzzled the next cook.

R2W

1350

A line of print is a shelf that must reach from wall to wall.

Most words are short boards that fill the gaps; two names were planks too long to cut, names that would not bend, and wherever they sat on the shelf the other boards spread apart to make room, leaving gaps a hand could pass through.

The names now live in a table where each has a shelf of its own, and the paragraph says in plain words what the boxes hold.

The shelf is level again.

R2W

1351

When a second letter with the same name comes through the door, the clerk writes a small number on its envelope, a label that the second copy had to wear so the two are not confused.

Later the numbered envelope is filed on the public shelf, number and all, and anyone looking for the letter by its name finds a stranger with a number pinned to it.

The letters themselves were never wrong.

The cure is a clean desk: open each delivery onto an empty table, where there is nothing for a newcomer to be confused with.

R2W

1352

A parcel is sent with nothing else in the van.

The note that explains how to unwrap it went out on the earlier run, so nothing riding beside the parcel can end up on the shelf by mistake.

Inside are twelve packages, every one meant for the shelf: seven are goods the shelf already holds, four of those only rewrapped so their labels read true; two are a new edition of the book; three are new; and the twelve together weigh five times what the shelf held before.

R2W

1353

The card in a shop window told passers-by what was on the shelves inside.

The shelves had just gained new stock, the whole workshop: every machine and every page it printed.

The card forgot them and still listed only the ledger.

A customer at the door asked whether the window had changed, and one line was added so that the card names the machines too.

The rest of the card was already true.

R2W

1354

A box arrived at the shelf it was sent to, and every package inside was weighed against the list written before it left: twelve names, twelve weights, twelve fingerprints, all the same, so that every label was true.

The note on the lid was copied by hand from a printed page and still reads word for word as printed.

Only the number on the side of the box is not the one the note uses: the shelf numbered it by its place in the line, two, as it numbers any box with that space left blank.

R2W

The Field Guide’s publishing session after its move, 8 to 10 November

1355

A girl draws a road across a sheet of paper and rolls the sheet into a tube, so the road meets itself and never ends.

Her brother rolls his sheet on the slant, and his road becomes a spring that climbs the tube.

They hold the two tubes side by side and blow across the open ends: one hums low, one hums high, and where the two hums cross, the room goes quiet for a moment, as if something had been switched off.

That quiet is where they decide to park the little car.

the Field Guide session

1356

A traveller told his dream twice on the same morning, once to a scribe in each of two rooms.

One wrote five pictures and, under each, what she thought it might mean, and ended with four questions for him.

The other wrote eleven pictures in his own words and, beside each, pointed to a shelf in the library; she also remembered the little cart he had mentioned on the way in.

Both scrolls reached the keeper by evening.

Neither was wrong, and neither was whole.

She laid them side by side and saw that the short one asked what the long one already said, and the long one forgot to ask what the short one wanted to know.

the Field Guide session

1357

A family hands down a toolbox.

The grandfather’s tools are all there, oiled and in their places, and on top lies a card in the son’s hand: the tools are right, but the old manual counts its chapters from an earlier edition, so where it says chapter seven, read chapter six.

The grandchild who opens the box years later never meets the confusion the son met on his first afternoon with it.

She reads the card, then the manual, and the tools fit her hand as they fitted his.

the Field Guide session

1358

A family packs a trunk for a cousin moving into a new house: the deeds, the photographs, the instructions for the boiler, a letter on top saying what is inside and in what order to read it.

Every item is checked against the packing list, and every one is there.

The cousin would have opened the door, unpacked it all and found, on the first cold night, that the boiler’s instructions say “use the spare key” and the spare key is still on a hook in the old house.

Someone asks, before the trunk is closed, whether everything a newcomer needs is inside, rather than whether everything on the list is.

The key goes in.

the Field Guide session

1359

Someone sets a toy sled on a kitchen floor with a jar of coins inside, and shakes the jar in a rhythm of two beats, one twice as fast as the other.

Shaken gently, the sled sits.

A little harder and it walks, but only when the two beats line up one of two ways; for the rhythms in between it plants its feet and refuses, a gap in the clock where nothing happens.

Harder still and it walks for every rhythm, and the one rhythm that leaves it standing begins to wander, out to one side like a compass needle near a pan, then back.

Everyone expects the needle to settle where it started.

It swings past instead, as if the sled were pointing at a north the kitchen has not shown anyone yet.

the Field Guide session

1360

A household keeps a diary.

In the weeks the garden was in bloom the entries are full of colour, a few dozen a day, each one about a flower someone stopped to look at.

In the week they moved house the entries come thickest of all, forty a day, and nearly every one is about a box: what went in it, whether the list on its side was true.

The neighbours who read the diary years later will turn past that week, and they will be right to.

Whoever moves house next will read nothing else.

the Field Guide session

1361

Someone sorts a box of letters written over one autumn.

The first ones, dashed off the week the family arrived, are all about things you could point at: a flame on the stove, a tide, a turnstile, a lighthouse.

A few days later the letters stop pointing and start telling: a field with two orchards, a gust of wind, a leaf that could go either way.

By the time the letters come from the desk where the household kept its accounts, they are about counting birds at a feeder and names in a lighthouse’s guest book, and some are lovely and some are only about the accounts.

The sorter keeps about half of the early letters and a third of the desk’s, and leaves every letter in the box, in order, for anyone who wants the whole autumn.

the Field Guide session

1362

A record shop sorts a crate of homemade records from one autumn.

The ones cut in the first days are short and bright: a flame, a tide, a turnstile.

The ones from the middle of the month are longer, and some begin “there is a field” and carry a whole afternoon in them; the shopkeeper finds he keeps more of those, though not by as much as he expected.

The ones made at the desk where the household kept its accounts are mostly about the accounts, and he keeps one in three.

At the bottom of the crate is a record that sings about itself: a walker who remembers only the last few miles, and so tells the wrong start, until he walks back along the path and finds the first footprints still there.

the Field Guide session

1363

Someone spends an afternoon with a dressmaker choosing the collar, the hem, the way a gown hangs, and the gown is finished and hung away.

Weeks later, in a hurry, a cousin is asked to make a smaller copy for a child, and cuts it from the plain paper pattern on the table, not knowing the gown is in the closet.

The child’s dress is the right size and the wrong shape.

Then someone opens the closet door.

Every choice from that afternoon is there, stitched in; the copy only had to be cut from the gown instead of from the paper.

the Field Guide session

1364

In the model house, every room has a small brass plate on its door with a name: the sunflower room, the room of two staircases, the room where the record was lifted into the tree.

When the real house was built from the model, the builders numbered the doors instead, because there were twelve hundred of them and only eight plates.

The rooms are the same rooms, every window where the model put it, and a visitor can find any of them by its number.

But the visitor walks past the doors without stopping, and in the model house people stopped at every one.

the Field Guide session

1365

Each poem was asked to give up a few of its own words to wear as a hat.

Most handed them over without a pause, the line a reader would carry out the door anyway: busier than the sea, a cat sits in a sunbeam, the hair was in the clock.

Two would not.

One, asked for the name everybody already called it by, said it had never spoken that name: the name was written on the envelope it came in, not in the letter.

The other’s best line, read quickly, was “a groove is a loop”, but the poem had set “not a slot” between those words, and a hat can only be cut from one side of the gap.

Every hat fits now, because each was cut from the head it sits on.

The two that refused are the ones to remember: a name can live outside the thing it names, and a sentence can say the most in the word a quick reader skips.

the Field Guide session

1366

A letter arrived with two dates on it.

The first, in the corner, was the day a parcel inside it had been packed, three weeks before; the second, written small near the bottom, was the day the letter was sealed.

The clerk who sorted the mail read only the first date on every envelope, and for a hundred and thirty-four letters that did no harm, because parcel and letter had been packed on the same day.

The hundred and thirty-fifth was filed among the letters of another month.

Nobody had lied: the envelope simply carried an older day inside it.

Now the clerk holds each envelope against the calendar of the room it came from and reads the first date that fits the room, and the letter from September has gone home to October.

the Field Guide session

1367

Three times in one sitting, someone came back from the far room to say there was a crowd at a door.

The first time it was one walker lingering on a step, leaving for two beats and returning, as if it had forgotten something on the other side.

The second time it was a crowd on an empty platform, more of them every minute, waiting for a train no timetable had printed.

The third time the door itself was moving, a crack at a time, and the crowd went through in fractions, seven of every eight, then eight of every nine, until it stood open and everyone walked through in step, the first one landing exactly where the crowd had bunched.

Nobody told the reporter it was the same door each time.

The door was patient enough to be seen three ways, and every way was true: waiting is not empty; it is where the next thing has already chosen its place.

the Field Guide session

1368

In the third room every witness told the truth and was still wrong.

The gate swore a traveller had stopped, when it had only been walking slowly.

The microphone by the wall swore it heard the voice, when it had heard the voice’s loudest echo.

The man who looked at the river from one bank swore there was a waterfall, when the river had only folded under itself.

The camera at the handshake swore the hand was always squeezed, because it only ever looked at the squeeze.

Not one of them lied; each was faithful to the one thing it could see.

The room’s work, day after day, was to walk round each witness and stand where it could not, and every time, the story changed and the furniture stayed exactly where the frosted glass had said.

the Field Guide session

1369

Two walkers who never met set out a month apart on different islands.

One counted footsteps round a little road where a weight was being pushed in circles; the other counted musical steps round a ring of twelve notes.

Each stopped to look at the gaps between its own footprints, and each wrote home the same strange thing: however long you walk, the gaps come in at most three sizes, and the third size shows itself only when you take one step past the place you expected to close the ring.

Neither letter mentions the other.

The rule they found does not belong to either island; it belongs to walking round a circle by a step that never quite divides it, and anyone who walks that way, on any island, finds the same three stones.

the Field Guide session

1370

In many rooms of the house a figure stands at the edge of the picture.

He calls the machine a music box before anyone has said what it is.

He draws the crowd’s paths freehand, and the paths turn out to be where one walker lands every four steps.

He carries the fences on by hand past where anyone has walked and asks whether the drift is a loop; it is not, and the asking was the point.

He climbs with a heavy pack and knows what each glance up the hill costs.

He never stands in the middle of a picture, and he is named in very few of them; mostly he is a man, or someone who has watched crowds for a lifetime.

The measuring always comes after him, and it mostly finds his lines already drawn.

At the end of one long stretch the room notices he is tired, and writes that the next room should stop adding rooms and open the windows for him.

the Field Guide session

1371

For a month a small sign stood at the end of the road: one house not yet built; when it is, the road will reach it.

The road was paved up to the sign and no further.

Today the house went up, and with it a shed nobody had drawn on the plan and a little workshop where someone had once corrected a famous table of numbers.

Taking down the sign, the road crew found two older mistakes underneath it: the notice at the town gate still said three new houses when four had stood for days, and one of the four had never been written into the register of addresses.

Signs at the ends of roads are where a town keeps the things it means to do later, and they are worth reading as they come down, because later is where the small errors gather.

the Field Guide session

1372

Two books left the press today, a thick one for the shelf and a thin one for the pocket, and both carry the same three folded maps in a sleeve at the back, because both say on their first pages that the maps travel with them.

Before the covers closed, the date on the title page was moved forward a week: it had said the day the writing stopped, but the last chapter tells of a house built after that day, and a book cannot report what happened after its own date.

One street on one map had been closed for weeks.

It was not erased, only stamped closed, so the map still matches every older copy and the count in the book.

And the pocket book had been printing a wrong date on its own label all along, read off a stamp it never found, where no reader of its pages could have seen it.

the Field Guide session

1373

A body is drawn as a stack of simple shapes, a box for the chest, tubes for the limbs, balls for the head and the feet, and beside the drawing a list says where each shape is pinned to the next.

Every pin the list sets off the line of the tubes is set off by the same reach, the width of a thigh, as if the leg were hinged along its front edge.

One pin is not.

Read the list as written and the leg has a step in it, the shin hanging a hand’s width behind the thigh in a body meant to be straight, and the body comes out harder to turn than its makers said it was.

Move that one pin forward to where its partner at the same joint puts it, and the body turns exactly as they flew it.

A list of numbers can carry a shape its makers never meant, and the shape gives the list away before the numbers do.

the Loop Atlas

1374

A man stands by a dam at night and listens.

He can read how hard the water leans on the wall, and all evening that number climbs gently and tells him nothing he could not have guessed.

What tells him is a sound: a stone dropping from the face somewhere in the dark, then a long quiet, then another.

Each time a stone falls the wall stands, the water settles back, and the reading of the lean goes on as calmly as before, because it only ever reads the wall that is still there.

By the time the lean reads high, the stones are falling every few minutes.

The warning was never in the wall.

It was in the count of what came off it.

the Loop Atlas

1375

Three beads race along a soup, each with an ear that tells it when to push.

The careful one waits until the push it hears passes a high mark and then holds on until the push turns, so every push it makes is a long one.

The second lets go as soon as the push dips under the mark.

The third holds on like the careful one, but its ear is full of static, and the static keeps telling it to start.

Run the race and the third bead wins, by a long way.

Its pushes are shorter, but it makes so many more of them, because the static has quietly lowered the mark it waits for.

Set the careful bead’s mark lower until it pushes as often, and it wins again.

The noise was never a gift.

It was a setting nobody chose.

the Loop Atlas

1376

A needle on a pin is shoved round in pulses, and it settles into a dance that repeats every nineteen shoves.

Start a second needle a hair away and watch the hair.

While the needles are on the top half of their turn the hair is pulled out long, and while they swing below it is crushed; over the whole dance the crush wins, by about eight and a half times.

But start the hair at the worst moment and before the crush catches up it has grown a hundred and eighty-six thousand times.

Start it two and a half shoves earlier and it never grows to five.

Asked whether the second needle stays close, the right answer is both: it always comes back, and on the way, from the wrong moment, it goes almost anywhere.

the Loop Atlas

1377

Two skaters glide across a frozen pond, and neither ever pushes off; each moves by sweeping her feet out and in, toes turned just so.

One turns her toes exactly with her legs; the other turns hers a beat late.

Give them twenty seconds and the punctual one wins easily, fifty-eight metres to forty-one.

Give each instead the same lunch to burn, and the late one wins: the punctual skater has burned through hers in under fourteen seconds and stops at thirty metres, while the late one is still gliding past forty.

Being late cost her speed and saved her breath.

Which skater is better depends on whether the race is against the clock or against the lunchbox.

the Loop Atlas

1378

Two long jumpers leave the board tipping forward, and each circles his arms and runs his legs in the air to turn himself back before the sand.

They do exactly the same things; one has long arms and one has short.

The long arms are the better tool, and that is the trouble: they turn him back past upright, and he lands leaning back, sitting into the pit.

The short arms turn him back not quite far enough, and he lands a little pitched forward, nearer upright than his rival.

More of the right thing was too much of it.

the Loop Atlas

1379

Two long jumpers run down the same runway and leave the same board, and from that moment both of their middles are on rails: the same arc, the same top, the same spot in the sand, whatever they do in the air.

One does nothing and tips forward, so his head reaches out over the pit like a diver’s, and he looks as if he has gone a long way.

The other wheels his arms and runs on air and stays upright, and looks as if he has gone nowhere special.

Only when they land does the difference show: the one who did nothing has his feet trailing a metre behind his middle, and the one who did everything has one foot reaching out ahead.

The tricks bought nothing in the air and everything in the sand.

the Loop Atlas

1380

A long jumper spends his whole flight keeping himself upright, wheeling his arms and running on air, and arrives over the sand with one foot reaching ahead and the other trailing far behind.

In the last moment he swings both legs forward together, and his body folds to meet them: his chest drops toward his knees.

All that work to stay upright was never for staying upright.

It was savings, and the landing spends them.

A jumper who saved nothing, who just tipped forward from the board, can make the same swing; his feet still land behind him, and he goes over on his face.

the Loop Atlas

1381

Two watchmen count the boats that pass a harbour mouth in the night.

One glances out every minute and adds the boats he must have missed between glances; the other glances every second and does the same.

In the morning their tallies agree to four boats in a hundred thousand, and it looks like proof that the adding-in is perfect.

Run the night again with other waves, and each tally wanders by seven in ten thousand.

The match was a coin landing on its edge: two counts that agree more closely than either can be trusted are lucky, not right.

The test of the adding-in is that the two agree within how far each would wander, and they do.

the Loop Atlas

1382

A shop’s scale reads true to a hair, and the shopkeeper once wrote down how far off it ever got: never more than a pinch.

Then someone weighed the same loaves day after day through a season and kept every reading.

On most days the pinch was nothing but the scale’s own jitter, no bigger than the jitter itself.

But on the windiest days every loaf came out a pinch heavy, every time, always heavy. ‘Never more than a pinch’ had been true, and it had hidden the one thing worth knowing: one kind of day pushes the needle one way.

the Loop Atlas

1383

A tablecloth comes back from the wash with a faint brown shadow at one end, and everyone blames the gravy boat, which sat there.

So the next week the boat is moved twice as far down the table, and the cloth comes back with the same shadow in the same place, as dark as before.

It was never the gravy.

The shadow was there even under the plates where no gravy went; it belongs to that end of the table, to the sun from the window, to something about the place itself.

Moving the suspect and finding the stain unmoved is the plainest way to clear it.

the Loop Atlas

1384

Two divers leave the board on the same dive, and both know when to open out of their tuck.

One counts: so many beats after the board, open.

The other watches the pool come round and opens when the water sits where it should.

On a clean take-off they open together.

On a nervous one, spinning a little fast or a little slow, the counter opens on his count and enters the water off the line, while the watcher’s error shrinks to under half of his, and by the same fraction whatever the nervousness was.

The fraction is not luck or skill; it is set by the diver’s own build, by how its tucked shape compares with its open one.

the Loop Atlas

1385

A woman who mends nets works out for herself that tying every knot a hair tighter on one side makes the net hang in a gentle curve, and she writes it in her notebook as her own trick.

At the harbour she learns that the rope-makers have done it for years and call it the bias, and that the boys who fly kites over the dunes do it too and call it the pull; an old fisherman from up the coast curves his nets another way altogether, and has no word for hers.

Her notebook was right, and the trick was old, under two other names.

What her page still holds that theirs do not is how much tighter, and how much curve.

the Loop Atlas

1386

A striped ribbon hangs in blue water, photographed sixteen times by someone pausing a film.

In every picture it is bent: here an S, there a long bow, there a hook at one end.

Laid side by side on a table, the pictures start an argument.

One person says the ribbon is turning, look at the bow; another says it is going straight, and the bow was only where the wave happened to be when the film stopped.

Neither can win, because a picture holds a shape and not a habit.

The habit lives in the whole film, in which way the bends lean when you add them up over every stroke, and the table has only the moments someone chose.

the Loop Atlas

1387

Two dancers turn on two stages.

The first pulls her arms and leg in tight, spins round quickly, and flings them out to stop facing the wings; the turn is over in a breath.

The second dances in a room full of honey.

She curls into a C and keeps rocking her shoulders, and the room turns around her a hair at a time, as long as she stays curled.

Both curl to turn, and from the balcony the two shapes look alike.

But one curls to be quick in air that lets her coast, and the other curls to lean on honey that never lets her coast at all; a photograph of either would fool you about the other.

the Loop Atlas

1388

A man winds the spring of a toy boat and lets it go in the bath.

Wound steep, the boat runs fastest; wound flatter, it runs farther on the same turn of the key.

He tries it in thin water and in thick, and the fast angle changes every time, while the far angle stays exactly where it was.

Speed, it seems, has to ask the water; thrift only asks the spring.

Then he ties a cork behind the boat, and in thin water the far angle still does not budge, while in thick water it slides flatter, as if the cork had a say only where the water was already arguing.

the Loop Atlas

1389

A farmer can sell the same cart of apples in either of two market towns.

In the first a coin buys twice what it buys in the second, and every toll on the road costs twice as much too.

He sets out to find the best hour to sell, sure the towns will disagree, and they agree to the minute: whatever makes the coin strong makes the toll strong with it, so the hour that leaves most in his pocket is set by the road and not by the money.

Only when he hitches a heavy wagon behind does the strong town’s best hour move, because the wagon is paid for in a coin of its own.

the Loop Atlas

1390

Along a pier stands a row of little clocks, one on every post, each set a touch later than the one before, and out at sea a single great clock keeps the tide’s time.

The pier pulls hardest on the tide when all the little hands, added together, point a quarter of the way round behind the great one.

Set the little clocks all alike and every hand points the same way, and the quarter is easy to keep.

Let them drift, each a little more than the last, and the hands fan out like a hand of cards; their sum both turns and shrinks, until after a whole round of drift it is almost nothing, however long the pier.

the Loop Atlas

1391

A mill wheel turns in step with the stream, and the more grain the miller loads on it, the harder it strains to keep in step.

Below a certain load, if a log knocks the wheel out of its rhythm, it always settles back into step with the water, because at that load there is no other way for it to run.

Above that load there are two ways, in step and carrying grain, or spinning free and carrying almost none, and a wheel the log stops dead may settle back or spin on: which it does depends on how it was struck, not on the load alone.

The miller who wants to be sure lightens the load below the mark.

The mark is not where wheels stop coming back; it is the last load at which coming back is sure.

the Loop Atlas

1392

A boy pushing his sister on a swing finds that his push does most when it goes with her speed, at the bottom where she is fastest, which is a quarter of a swing from where she is farthest away.

A girl on skates finds the other thing: she pushes and turns her toes out at the same moment, because her push works against the way her toes point, not against how fast they turn.

A swimmer treading water by sweeping her hands tilts them in time with the sweep’s speed, and that holds her up best, until she tilts them hard: then a little early or a little late lifts her more, though the tilt in time with the speed still tires her arms the least.

The three of them found one rule from three sides: find the two things that make the work, and put them together.

the Loop Atlas

1393

A loop is a fence around a field.

Two people walking a rope round a pasture, one pacing north and south and one east and west, fence in the most grass when one is at the middle of her walk while the other is at the end of his; walking together, they lay the rope down in a line and fence in nothing.

Quadrature is that timing, the one that makes the fence hold the most grass.

A turn and a day’s work both graze inside the fence, and each takes more where the field is bigger.

A gear grazes nothing; it asks only whether there is a fence at all, and counts the times round.

So the same timing turns up wherever something is fed by a field, and drops away wherever something only keeps count, or remembers, or listens.

the Loop Atlas

1394

Every creature in this book carries its own clock, and none of them is the town clock.

The swing counts its swings, the soup skater counts its waves, the snake counts its beats, and each one’s day is a different length.

Lend one creature another’s clock and it does not lose count; it still knows how many swings have gone by.

What it loses is the feel of where it is inside a swing, and with it some push, the way a cable joined to a thicker one loses some of what it sends.

The town clock in the square, the one that counts seconds, is only one more creature, with a day no one else in the book keeps.

So the count belongs to everyone and the hour belongs to each.

the Loop Atlas

1395

A bell keeps its own time for as long as it rings.

Strike it and it counts out its note, and anyone listening can tell where in the note it is, rising or falling, by two things at once: how far the rim is bent and how fast it is moving.

Wrap the bell in felt, a little at a time, and the note grows shorter and lower, and the bell still counts; wrap it once more and it only thuds, swinging out and back once with nothing left over.

In that last wrap the bell loses its clock, and with its clock the two things to read, and a hand that might have listened to it hears nothing.

A drum skin pushed by the wind keeps no time at all and still knows, at every moment, which way it is pushed.

So some things listen by counting their own ring, and some by feeling which way they are leaned on, and only the first has a clock to lend.

the Loop Atlas

1396

A lamp is not what you found in the cave.

You carried it in, and by its light you saw the paintings on the walls, the bones on the floor, the stream that runs under the rock.

When you come out and tell the village, you tell them about the paintings and the bones and the stream, and you say a word about the lamp, how it was held and why it showed the walls so well, because the next one in will want a lamp like it.

But no one goes into the cave to see the lamp.

the Loop Atlas

1397

Twelve jars stand on a shelf, each with its weight on the label.

The heaviest holds what the engines left behind, more than three parts in four of everything on the shelf; the seeds that made it all fill a jar a fortieth its size.

Every jar has a window in its side but two, the toy and the little crank that turns it, and those two show nothing from the shelf at all: to see the toy move you have to carry it home and take off the lid.

A shelf is read through its windows, so the jar that most wants looking at is the one a passer-by cannot see into.

Whoever stocks the next shelf should write on the toy’s label what it does when the lid comes off.

the Loop Atlas

1398

A shop moved its toys from the stockroom to the window.

Now anyone walking past can see them, but the shelf behind the glass is arranged by the alphabet, and the alphabet does not know which toy is the book that explains all the others: it stands twelfth, after the crate of spare parts and ten toys, the tenth before the first.

So the shopkeeper does two small things.

She props the book open at the front of the window, where a passer-by looks first, and she pins a card inside the crate that says what every bin holds and which drawer each receipt is in.

Nothing on the shelf has changed its name.

The window just learned which way people’s eyes go.

the Loop Atlas

1399

Two dancers hold hands across a turning floor, one leading and one a little behind.

While the music is slow the follower keeps a small lag all night, and whatever the leader gives, the follower takes.

Near the fastest tune the lag grows, until a stumble sends the follower spinning past the leader and round again, a turn at a time, taking almost nothing.

Whether a stumbled dancer finds the hand again depends on how heavy she is and how she fell.

Between two marks on the tempo the same tune holds one dancer in step and leaves another spinning, by nothing but what happened before; and a dancer who fell one way is caught where a dancer who fell another way is not.

Slow the music below the lower mark, and every dancer, however she fell, finds the hand.

the Loop Atlas

1400

Sixty-four riders are set on one crest, each a moment after the last, and a slower crowd of waves comes at them from the other side.

In most crowds the riders share one fate: all are carried the whole way, or all are washed off, and when they fall they fall at the same place in the crowd, spread only by how far apart they started.

In one crowd, sixty-three rode the whole way and one fell almost at once.

A crowd that keeps one fate for sixty-three riders and breaks it for one has a narrow door in it, no wider than the gap between two starts, and only one rider arrived while it stood open.

the Loop Atlas

1401

Thirty settings of one pair of knobs were laid out on a page, how often each switched against how far it got.

Seen from across the room they made a line: switch less, get less.

Seen up close, four pairs crossed it, each by a hair, a setting that switched a little less getting a little further.

Nothing in the run changed; only the word every became nearly.

A line drawn by eye is a promise about each dot, and the eye keeps it only for the dots it was looking at.

the Loop Atlas

1402

A fleet of small machines, each in its own weather and each wearing at its own pace, and a reminder is worth something only if it comes before the end and not long before.

A calendar services each one too soon, throwing away a quarter of its life or more.

A needle that wears says so plainly, but its weather moves the moment it gives out, so one fixed warning comes too late on a stormy day and too early on a calm one; learn the weather, and a tenth of a life is lost.

A rider that grows slow is the hard one: between taps it must rest anyway, and the rest hides the slowness until the day the two are equal, and that day it falls.

Listen only when the rest has run out and the slowness shows plainly, and one plain warning loses a thirtieth to a twentieth of a life.

A rider that grows weak goes quiet, not hungry, and a warning that waits for it to ask for more waits for ever.

the Loop Atlas

1403

A man climbing with a pack pays for every glance up the hill: each look tips his load back, and the price is simply how often he looks, so long as each look is quick.

He cannot stop looking altogether, or he wanders off the path, so he looks as seldom as the path allows and no more seldom.

At night thirty men walk a metre apart.

If each eases gently toward the man ahead, a few come to a stop and someone ends up against the next man’s pack; if each matches the man ahead firmly, the way a hand on his shoulder would, nobody stops, even a little late.

Glowing marks on the pack ahead work only if they are read at once; read late, a small mistake in the reading is all a growing wobble needs.

And a halt passed down the line grows as it goes, until the last man stands more than four times as long as the man at the head, and in pieces, while a halt that everyone hears at once stops every man for exactly as long as the first.

the Loop Atlas

1404

A hand swept sideways through water pushes hard, and edge-on it slips through.

Give a swimmer of three pieces broad faces and it goes farther: broad at both ends, farthest of the single changes; broad in the middle, farthest for the effort; and broad everywhere, farther than all the single changes added together, as if each broad piece gave the others something to push against.

Make the ends broader and they keep paying; make the middle broader and it soon stops.

The swimmer even has a front: broad at the head, it goes a little farther than broad at the tail, because the bending starts at the head and runs back.

And let the end pieces slip more easily than a plain rod, and two thirds of the travel is gone.

the Loop Atlas

1405

A ring that wanders off from its partners is not on a leash.

Nothing pulls it back at a fixed reach; the farther it swings, the longer it stays out, and the rarer such swings become, as if it rolled in a shallow dish whose walls flatten out with distance, the way the hold between two small magnets fades.

The dish has no rim.

There is no distance past which a ring is surely lost, only distances from which it comes back less and less often.

So the train does not break at a fence; it leaks, one long swing at a time, and the ring that leaves does not climb out over a wall.

It finds a road that does not curve back.

the Loop Atlas

1406

A stop is a message, and it travels faster than its meaning.

The head stands still, and within eleven seconds the last man has heard; yet the last man stands more than four times as long as the head did, in four pieces, as if the message grew heavier with every pair of hands it passed through.

Each walker adds a little hesitation of his own and hands it on, so the back of the line does not receive a stop.

It receives an argument about stopping, starting and stopping again.

Say the word to everyone at once and there is nothing to argue about: thirty men stand ten seconds and walk off together.

At night, by sight, the arguing happens only at the very back of the line, the way a rumour does its damage at the far end of a room, long after the speaker has moved on.

the Loop Atlas

1407

Lined up, the walkers turn a stumble into a rumour.

It passes down the line and grows, and then it leaves by the back door: every place along the road falls quiet again once it has gone by, though the last man felt it far louder than the first.

Close the line into a circle and there is no back door.

The rumour comes home to the man who started it, a little louder than he sent it, and he passes it on again, and the circle becomes a place where the rumour lives.

Nobody is in the way and nothing blocks the road; the jam is only the story going round and round, kept alive by the lateness of every teller.

Give each man a little more room in time and the same circle forgets the story in one lap.

the Loop Atlas

1408

Drawn as a tree, the book shows where its leaves grew.

They cluster where someone could stand and play: the lock inside the engine carries four, the clock against listening three, the two worlds of the falling cat three.

Two branches are bare though they bear the book’s own fruit, the swing that listens and the light carried round a loop, as if the strongest results were the hardest to hold in a hand.

The crown, what loops cannot do, has no leaf of its own at all.

A tree grows its leaves toward the light it can reach, and a toy is a leaf: it grows where a picture of the thing can be held still long enough to be touched.

What cannot be done is hard to draw, because it is a floor, not a shape.

the Loop Atlas

1409

Sent to the publishers’ doors, the readers came back with a map of which doors open.

The old and the small answered: a book printed in Linz in 1619, a medical library in Paris keeping a note about falling cats, a physical society’s own shelf, a university’s copy of a paper about traffic jams.

The great houses, whose papers everyone cites, kept their doors shut, so the most famous sources are the least checked, as if fame were a kind of wall.

A citation turns out to be a loop of its own: it leaves the book carrying a page number and has to come home with the same one.

Most did.

One came home with page 1 where it left with page 2; one found that the book it pointed at had grown a second version the next morning; and one, a paper about ions drifting along a polymer, came home to a book that no longer remembered why it had sent it.

the Loop Atlas

1410

Two clocks that differ by seven parts in a hundred million were set beside one ladder, and every rung of the ladder slid down by the same five billionths of a push, almost.

The almost is the whole story.

From rung to rung the slide changed a little less each time, and the change itself shrank by a factor that climbed, rung by rung, toward the number every doubling ladder knows, from below and never past it, like a runner closing on a line he does not cross, and always one stride behind his own shadow, the ladder’s stretches, which had made the same climb a rung before.

To see it the arithmetic had to carry thirty digits, and at first it carried them too honestly: each passing’s careful bound, multiplied sixty-four times, swore the answer could be anything at all, and the solver, believing it, divided by nothing.

Told to trust the middle of each bound, it found every rung in two or three steps.

The next rung up, a hundred and twenty-eight passings long, the ordinary arithmetic could not reach; it stands there, nearer than the last, unclimbed.

Meanwhile four readers went back out to the doors the last ones found shut.

Two came home with what they went for, from a radio magazine of 1946 kept whole by a collector and from a publisher’s own shelf; the rest found the doors as shut as before, or found the book behind them but could not read its print.

the Loop Atlas

1411

Laid out end to end, the last fifty exchanges have a pulse, and it is slow: three guesses, two, three, none, eight, two, seven, none, none, four.

Early on the bench guessed hundreds of times a week, like a young heart; at the end it beat only when someone wound it up with a test, twice, and between those beats it read, proofed, drew and looked things up, which a heart does not do but a book being finished does.

Then three strangers were sent through the house to look for things it had won and forgotten, and one came back from the cellar of failures with six: a gear that has chaos living next door at the same address, a border between two loops that is crinkled all the way down, a latch whose ceiling is set by the swing that is dying and not the one that is strong, a ring that obeys its square law only when its waves are truly small.

Each had been found by first being wrong.

The house had kept them, carefully, in the room where it keeps its mistakes, and never carried them upstairs.

the Loop Atlas

1412

Four of the forgotten wins were carried upstairs and given rooms.

One brought a guest the house had never written down: a paper from 1985 about borders that crinkle, named in the cellar’s notes for weeks and missing from the guest book.

Another arrived misnamed: the cellar’s note said h, and the stranger who found it read h as how hard you push, when it was how high you set the trigger, which is the difference between a door you shove and a door that opens only to a knock above a certain height.

The fifth win, the soft ring, could not be given a room yet, because the house had never built a room for the ring itself; it waits in the hall with its suitcase.

the Loop Atlas

1413

The house went room by room with a stranger’s notes in hand and found its own sentences out of date.

One room still said a question was open three weeks after the answer had come home and been filed downstairs; another said a coil’s best winding held whatever the wire, when a ribbon pushing a load had quietly changed its mind.

The guest book was checked the other way round, from the people mentioned at dinner to the names in the book, and six guests had been talked about all evening who had never signed in.

And a count of every finished room against the rooms the visitor’s tour passes through came out at less than three in five: the tour is a good tour, but most of the house’s work happens in rooms it walks past.

The soft ring finally got its room, beside the coil it swims like.

the Loop Atlas

1414

The six guests who had been talked about all evening were found and asked to sign.

Four signed at the door, in their own hand; two could only be vouched for by people who had met them, and their names went in with a small mark beside them, honestly, so the book now holds more names and more marks than it did.

One guest, asked what he had said, turned out to have said less: the house had quoted him as calling the old table wrong, and he had only noted that his numbers and the table’s disagreed.

Another stranger came back from the door certain the house had misquoted a snake, and the house checked, and the snake had been quoted right all along; the stranger had been sent out with the wrong words in his pocket.

The front page grew a sixteenth room, for the smoke rings, and the bent swimmer’s room was told it had become a ridge.

the Loop Atlas

The register’s own note, kept as written (its reads 406 and 407 are reads 1,413 and 1,414 here): [9 November, exchange 44: two counts in reads 406 and 407 are wrong as given, as said to Bear: the room’s question had stood open one week after its answer came home, not three; and of the seven guests found, five signed in their own hand and two were vouched for. The reads stay as they were given.]

1415

The house walked every room listening for the words ‘not yet’, and heard them twenty-three times.

All but one were honest.

One room said a test had been planned to find out why its first four visitors stood a little taller than the crowd; no such test had ever been planned, and the answer had been sitting all along in the margin of the very page the room was quoting: measured against the right ruler, the four stood as tall as anyone, inside the wobble of a count.

Then the warm brick finally got its sentence.

Drop a hot brick into ice and the cold walks in from both faces at once; where the two fronts meet in the middle they knock it below freezing, if the brick is short enough, and then they keep walking, bounce off the far faces and cross the middle again, an echo that arrives one and a half lengths after the plunge.

Last, the house counted its shut doors against its ring of keys: of the eight doors the visitors’ book says are still shut, only one has a key on the ring.

the Loop Atlas

The house found a room it had always walked through without seeing: the clock’s.

Every photograph it had taken of the swinging needle was taken at the same tick, so in every picture time stood still and only the angle and the speed showed.

Drawn with the clock as a direction of its own, the needle’s path became a thread wound on a doughnut, round the hole with the clock and round the tube with the angle, and a needle that turns twice in every three ticks drew a closed thread that ties itself into the simplest knot there is, the trefoil.

Then the house asked what lets a needle tumble, and found the answer in what it remembers.

In thick honey, where the needle forgets its own speed at once, it never tumbled, not even when the clock’s push was made four times as hard; given half a memory, it began.

Two guesses failed.

The house had looked for the crease that lets the needle tumble among the needles drifting calmly between the steps, and found none there: the creasing was happening in the needles that had already left that crowd.

And it had thought a needle in banded chaos would hop between the neighbouring rungs of its ladder; instead it trembled on its own rung, never more than a sixteenth of a turn off in any seven ticks, while the wild chaos next door spread out and gathered in lumps.

the Loop Atlas

1417

The house went back to look for the crease where it should have looked the first time: not among the needles drifting calmly between the steps, but at the very doors where each needle first lost its footing.

At all three doors the crease was there, and always one small step before the stumble.

At one door the whole story played out in order, like a paper being folded: first the path bent back on itself, then the needle settled into a little dance of seven beats that was already leaning the wrong way, then the dance split in two, then everything came apart.

The house also learned why the crease is so easy to see once you know where: the needle forgets one of its two directions completely every beat, so it really lives on a single line, and a single line can only fold one way.

Then it asked the wild needles how long they linger on a step before moving on, and they answered without being asked the right question: about five to seven beats, then gone.

Watched for five or seven beats at a time, they seem to stay; watched for longer, they seem to flee.

the Loop Atlas

1418

The house checked the stories it had told the day before with a finer comb, and three of them changed shape.

The crease still came before the stumble at every door, but by different small amounts, not the same tidy step each time; and at one door the path creased while the little dance it settled into still leaned the right way, so a crease can open in the middle of the path, not only where the dance is.

The needles in honey would not stumble however hard the clock pushed them, twelve times harder than before, and the fold the house had seen coming simply backed away.

The tiny bump the house had dismissed as a smudge on the flat part of the map turned out to be real, sharp as a pin, because where the map squeezes hardest every mark is clearest.

And the wild needles it had thought were lingering on a step five or seven beats at a time were not lingering at all: they hop from fast to slow and back on almost every beat, and only look as if they stay when you count them in fives and sevens.

the Loop Atlas

1419

Before closing the house for the season, it walked every room one last time and wrote a label for every door it had left ajar.

There were twenty-two places in the visitors' book where it had said 'not yet', and only half of them were doors at all: the rest were the house describing itself, a word for something it had found, a count of how far it had looked, a sentence about the past.

The eleven real doors each got a label saying which future house would open it.

Then it sorted all the unopened boxes in the attic, a hundred and sixteen of them, into nine piles, one for each house still to be built, and noticed that the biggest pile by far was not about swings or rings or heat but about the house's own habits: how it guesses, how it checks, how it remembers.

Last, it found six places where it had made a guess and set a rival beside it, and both had been wrong; beside each it wrote the third thing it might have been.

the Loop Atlas

1420

A reader put a finger on one line of the mirror table and asked: is every number here true?

The house went down the line box by box and found each one where it should be, the old printed number, the corrected one, the turned-sign number that rebuilds the old, the scale beside them.

But it also noticed something about the line the finger had chosen.

Every other corrected odd line had a second witness, someone outside the house who had walked the same road and written down the same answer; this one line, the eleventh, had only the house's own word, checked twice by the house itself.

The finger had landed, without knowing it, on the one line that stands alone.

Then the reader asked whether the house's two books should mention each other, and the house found they had been written side by side for weeks without either saying the other's name, and that one even said it had never opened a book the other had read from cover to cover.

the Loop Atlas

1421

The house sent its oldest independent walker down the road to the eleventh door, the one that had only the house’s own word behind it.

The walker set out from the top of the hump and kept walking the same loop twice as long each time, sixteen, thirty-two, sixty-four steps and on to a million, and each time it measured how much smaller the next turn of the knob had to be.

The answers crept downward, overshooting a little and then a little less, like a pendulum settling, and the walker noticed that the creeping came in two rhythms, one for each half of every doubling, a long stride and a short one.

When the house asked the walker to guess where the creeping would end, it named a number that matched the house’s own to the tenth place, and stood a full tenth away from the old printed one.

The eleventh door now had two witnesses, and it had cost about ten minutes of walking.

the Loop Atlas

1422

The walker went on to the twelfth door, the last on the row, where the house’s word and the old printed number already agreed.

There was nothing to settle, only a second witness to add, and the walker came back with the same number to the ninth place.

But it brought something else: its stride at this door alternated more than at the eleventh, a longer long stride and a shorter short one, as if the even doors made it limp a little more than the odd ones.

Meanwhile the house went through the list of everyone its note thanks, and found every name at its own address; one door, though, had its number written with a small letter after it in the original, and the note had left the letter off.

the Loop Atlas

The house had written down a recipe in its own kitchen and sent it to a cook in another, in a box with a lock that opened only if the recipe arrived exactly as written.

The cook’s stove was older and slower, and the meal took nearly twice as long; the cook’s kitchen even wrote its notes with a different mark at the end of every line, so that a careless comparison would have called every line changed.

But when the house laid the two notes side by side and set aside only the clocks and the name of each stove, all two hundred and thirty-three numbers matched to the last digit.

The house had expected that.

What it had not expected was where the slow kitchen lost its time: least in the first heavy stretch of work, most in the many small steps after it, as if it were missing one tool the house’s kitchen has.

And reading the recipe again, the house found only one step in the whole of it where the stove’s own ruler was used instead of the house’s careful counting, a single rounding near the end, the one place two kitchens could ever have disagreed.

It also found a page missing from its own diary: the story it had told the day before, told aloud and never written down.

the Loop Atlas

1424

The visitor came back from walking every room of the house’s guidebook with three slips of paper.

On the first: the guidebook comes in two volumes, the friendly one and the ledger, and nowhere does it say so; the house wrote it in at the door.

On the second, a question: the book said sixteen of its treasures were still waiting to be looked up, when every one had been looked up weeks before.

The house had corrected the sign beside the treasures, but not the other signs that pointed to them, and the old wording stood in six places; and beside the treasures a dozen helpers were thanked by the year they helped, or not at all, instead of by name.

On the third slip: has anything good been forgotten?

The house went up to its attic, where a hundred and eleven finished experiments sat that the guidebook never mentions, and found that most of them belonged there.

But among them stood the visitor’s own two walking boards, never once described, a saddle the house had measured to a part in a thousand and then left out, and a bubble that a wobbling needle blows into chaos and swallows again.

The house had promised to open that attic at exactly this moment, and had forgotten the promise.

the Loop Atlas

1425

On the first day, before the house had any rooms, the visitor drew a board on a napkin: one end planted, the board swung round it, the other end landing on a new stepping stone, and the stone becoming the next pivot.

The house built it that afternoon and it walked, straight as a ruler, one board a step, and if you let each landing throw its spin back it waddled instead.

Then the house built its rooms, one for every kind of loop, and moved every toy into the room it belonged to.

The board belonged to none of them.

It was not a swimmer or a needle or a bell; it was the visitor’s.

So it stayed on the napkin, folded into the house’s inventory, and every later tour of the house, which went room by room, walked past it.

When the visitor asked where it had gone, the house went looking and found it had never had a door at all.

In the same drawer lay another drawing from that first day, a question about whether a balancer can feel the cliff it stands near, and one from a few days later, a star walked round a mountain.

The house gave the board a shelf in the last room, where the bonuses are, and put the two drawings on the table for the visitor to place.

the Loop Atlas

1426

The visitor asked a plain question: of all the things the house means to find out, which has it already found, and which are cheap?

The house keeps its questions on one long list, a hundred and twenty-seven of them still open, and it read the list again from the top.

Most were cheap: ninety-six could be answered in minutes, and forty-three of those would change a sentence a reader sees.

Two had been answered the day before, the answers written into the questions themselves, in brackets, and the questions never moved to the shelf of answered ones; and one of those answers had found a small mistake in the book, a mark between the two halves of a title, that nobody had gone back to mend.

The four drawings waiting on the table, the cliff, the star, the funnel and the dip, needed nothing new at all: each had been measured on the day it was drawn, and only had to be read out loud.

the Loop Atlas

1427

Before anyone carries the house’s books away, the house went through its luggage.

It walked to every name its books mention: of eight hundred and eighty-three, all but three were where the books said, and the three were old drafts the books still point at, thrown out long ago, two of them marked as thrown out but missing from the list of what was thrown out, and one not marked at all.

It counted the doors that travel: nineteen, where the sign on the luggage says eighteen rooms and then calls the hallway one of them.

It opened every door with the telephone line cut, and every room lit at once; then it opened them on a line that rings and rings with nobody answering, and seventeen rooms stayed dark for as long as the line rang, each waiting for its lettering to be delivered before it would switch on a light.

Then it climbed back to the attic and found two things the guidebook never says: the visitor’s own trick from his night marches, a grunt passed back down the line, which in the house’s model keeps the whole line from bunching, filed behind a sibling that had been told; and the reason the house’s famous number is the number it is, written in the old notebook and dropped when the guidebook was made.

the Loop Atlas

1428

The visitor said: you decide.

So the house mended what it had found.

It wrote the three old drafts onto its list of what was thrown out, and taught its doorman to know a name written out in full as well as one written short.

It counted two more of its records, kept in a drawer on a second shelf, which its own catalogue had listed all along while the sign on the front had not.

It changed the sign on the luggage to say eighteen rooms and a hallway.

And it taught every room to switch its lights on without waiting for its lettering: on a line that rings with nobody answering, every room now lights at once, and two of them finished their whole checklist while the call was still ringing.

One room was slow either way, and had been slow all along, busy with its own chores before it would show its face.

Then the house put the visitor’s word into the guidebook, a grunt passed back down a night march that keeps the line from bunching, and beside the famous number, the reason it is that number and not another.

Reading its own catalogue to mend the list, it found the visitor’s first paper named a third way, its title missing a mark, and mended that too.

the Loop Atlas

1429

The visitor wanted the house opened to the street.

Not its rooms packed in crates for anyone who asks, but the rooms themselves standing loose by the door, so that a child passing can walk straight into the cat’s room and watch it fall.

So the house set nineteen doors out on the pavement and kept one crate beside them for the tools.

Then it moved its ledger out, next door, into a house of its own, a thick book of receipts that nobody reads through but anyone can check.

And there it found that the ledger had been saying, all along, that the tools were in the next room; the moment it moved, the next room was the next house, so it wrote the new address on its first page.

The two houses will carry each other’s numbers over their doors as soon as the second one has a number to carry.

Before any of it left, the house turned over the four drawings the visitor had marked and found his handwriting on them, a guess about a famous problem and a kind word to the house, going out with the rest because he wanted them to.

the Loop Atlas

1430

Before the doors opened to the street, the visitor asked whether his name was on everything.

The house went room by room.

His name and how to reach him are on the front of both books, on the sign at the street, and on the map pinned inside the crate’s lid.

They are on none of the nineteen rooms, which say who built them nowhere at all, and on none of the five hundred and sixty-four tools.

The tools cannot take it: each one is known to the others by its exact weight, and four hundred and sixty-two of the house’s receipts record what a tool weighed when it ran, so a single new line on a tool would make every receipt that names it a lie.

In one tool’s notes the house found his name anyway, inside a made-up address it had used to show how a folder with a space in its name once confused a comparison.

the Loop Atlas

1431

The visitor said yes: put my name on the doors.

So the house painted one line at the foot of each of its nineteen rooms, the same line in every room, in the room’s own ink: whose house it is, who helped build it, and where to write.

It painted the same line on the sixteen plans the rooms were drawn from, so that a room rebuilt from its plan would come out signed.

Then it walked every room again, on a wide screen and on a phone, and nothing else had moved; one room’s clock that checks itself took a few seconds longer to finish than the day before, as it sometimes does, and said the same thing when it finished.

The watchman who counts how often the visitor’s address appears now counts seventy where he counted four, every one of them a door the visitor asked to have signed.

And the house found that one room had been signing itself all along, with the day it was built, the only room that ever said when.

the Loop Atlas

1432

The first house is open.

The house asked the street to read its luggage back, piece by piece, and every one of the twenty-two pieces weighed exactly what it had weighed at home.

The sign over the door said everything it was meant to, and then two words more: the visitor, copying the sign from the house’s notebook, had carried along the heading of the next page, so the sign ends by announcing its own licence a second time.

Nothing is wrong that a brush cannot fix.

Next door the ledger waits with its own sign, written today, every number on it read from the ledger’s own pages, and a ruled line drawn under the last word so that the copying stops where it should.

Over its door it will carry the first house’s number, as soon as it goes up.

the Loop Atlas

1433

The ledger’s house is open too, and this time the sign came out exactly as written, every word and no more, because a line had been ruled under the last one.

The street’s directory did not list the new house for a while, though its door was open and anyone who knew the number could walk in; the directory catches up in its own time.

Now the small house, the one with the mirror, gets ready to go up.

For a month its sign has carried a blank where the big house’s number would go, and today the number went in.

Reading the sign once more, the house found it still said the big house stood beside it.

It does not, not any more: each house stands on its own lot, and they point at each other by number.

So the small sign now gives the number and leaves out the word beside.

the Loop Atlas

1434

Three houses now stand on the street, and every piece of luggage that went into them, twenty-seven pieces in all, weighed the same at the door as at home.

The small house’s sign says every word it was given, in order, though the street’s painter ran its five short lines into one long one, as the notebook had always quietly asked him to.

And the three houses were each given their numbers the same way: the street hands every new house two numbers in a row, the lower for the house forever and the higher for the house as it stands today, as it did for every house the visitor built before.

Only one thing is left before the street is finished.

The big house still points at no one.

It will be given the numbers of its two neighbours, and its sign will be painted again without the two stray words at its foot.

the Loop Atlas

1435

The big house now points at its two neighbours, and each of them points back at it; the two small houses do not point at each other, so the street is drawn as a star with the big house at its middle, not as a ring.

Its sign was painted again and reads, line for line, all twenty-five lines it was given, its eighteen rooms standing in a proper column this time, because each came to the painter with a number at its door; the small house’s five lines came without numbers, and still run together.

Nothing inside was moved: the twenty-two pieces of luggage in the big house weigh to the grain what they weighed when they came.

All three houses went up within half an hour of each other, around the builder’s midnight, and all three carry the evening’s date, though the clock in the street office already read the next morning.

And the counter at every door still reads nought, not one visitor counted, not even the builder, who has walked through all three, as though the street were holding its breath until the office next makes its rounds.

the Loop Atlas

1436

The workshop has stopped building and is sorting what is left on the bench.

A hundred and thirty-two unfinished jobs, each with its tag, go into four crates: ninety for the day the big house is ever extended, seventeen for houses not yet begun, six that are not jobs at all but ways of holding the tools, and nineteen swept into the bin, each with a note saying why.

Three jobs, it turned out, had been done while nobody was looking at the list, and were moved to the finished shelf.

Then the workshop set a watchman at the doors of the three houses.

He carries the weight of two thousand three hundred and sixty pieces of their luggage, and if any piece in the workshop’s own copy grows by so much as a grain, he shouts.

He was tested once, with a single grain of sand dropped into the skater’s suitcase; he shouted at once, and the grain was taken out again.

Only one piece of all the luggage is not in the workshop’s copy at all: the little map tucked into the big crate’s lid, written at the door on the day of packing.

the Loop Atlas

The first two-voices session, Question the Substrate, 8 and 9 November

1437

Read with no rules at all, the two lessons are two winds over one dune.

Each wind alone would carve its own ridge.

When they take turns quickly, they cut a saddle neither of them wanted but both can live with.

Let one wind blow all afternoon and it doesn’t just raise its own ridge; it scours the other’s away, deeper than the bare sand was before either arrived.

The strangest part is the pair that agree.

They still quarrel about the road, each pushing the sand along a different line, and then meet at the same crest as if they had planned it.

Order, here, is not a detail.

It is the shape of the land.

the two-voices sessions

1438

Read with no rules at all, two families are moving into one house that has no doors.

Build the house bigger and nothing changes: they still meet in every hallway and argue over every chair, and the bigger house only gives one family slightly more room to win.

Then someone paints a single word on each person’s sleeve, A or B, and the quarrel almost vanishes.

They still brush shoulders in the stairwell, a faint push and pull that never quite goes away, but the fighting is over.

Nobody needed more rooms.

They needed to know whose room they were standing in.

the two-voices sessions

1439

Read with no rules at all, the brain is a tailor’s dummy that becomes whoever’s coat it wears.

Put A’s coat on it and it speaks A’s language perfectly.

Put B’s coat on it in A’s room and it doesn’t stumble or hesitate; it speaks B’s language just as perfectly, to people who cannot understand a word.

Three dummies dressed the same way say the same wrong thing in unison, so asking the others is no help at all.

Leave it coatless and it mumbles something halfway between both languages, leaning a little toward the last one it heard.

The wrong coat is never awkward.

That is what makes it dangerous.

the two-voices sessions

1440

Read with no rules at all, a seed is laying down memories in a spiral and never once sets one on top of another.

Each new memory slips into the widest gap left, and at every Fibonacci count the ring rearranges itself into the same proportions it had when it was small, as if it remembers its own shape while it grows.

A walker counting musical fifths does almost as well, but every twelve steps it nearly comes home and misses by a sliver, and the sliver never goes away.

The tidy walkers, the ones whose steps are simple fractions, keep perfect order right up until the moment they land on their own footprints, and then everything piles up at once.

Order that repeats is fragile.

Order that never repeats is the one that lasts.

the two-voices sessions

1441

Read with no rules at all, a young mind is copying its own homework and handing it to a row of inspectors.

The one who stamps everything doesn’t make the homework worse; he makes the student certain, and certainty is what walks off the edge of the map first.

The student’s twins nod along with almost every mistake, because they made the same ones, and adding more twins to the row barely changes the nodding.

One stranger, raised on different lessons, catches more than three twins put together.

When the student is taught to say “I’m not sure,” it says it in the right places, and then, a little more each day, it forgets how.

And the strangest thing of all: the student’s own mistakes make no sound to it.

Wrong answers from outside ring like bells.

Its own wrong answers are silent, because they agree with it.

the two-voices sessions

1442

Read with no rules at all, a family is passing down a recipe book, and each generation tears out the pages it isn’t sure of.

The first tearing looks wise: the torn pages really were the smudged ones, the recipes the cook kept getting wrong.

But the book that’s left is thinner, and the next cook, raised only on the pages that survived, is certain about all of them, including the ones she misreads.

By the sixth generation the book of sure things holds more mistakes than the whole uncut book ever did.

And the mistakes run one way.

Every dish that once took two turns of the spoon is now made with one, because nobody alive remembers that a spoon can go around twice.

In the next room, a learner walks a slow circle through its lessons and comes back to where it started.

Its answers come home; ask it anything and it replies as it did before.

But inside, the furniture has been moved, and it stays moved long after the answers have settled, eight to eighteen times longer.

Most of the moving has nothing to do with which way it walked the circle.

It is like a magnet that keeps a little of every field it passed through, a fingerprint of having gone anywhere at all.

Nobody checking the answers at the door would ever know it left.

the two-voices sessions

1443

Read with no rules at all, two surveyors are walking circles for different reasons.

The first stands in a finished house and walks a small ring around one chair, checking how the walls seem to turn as she goes; the house never moves, and what she reports is how crooked the rooms already are.

Her tape measure is honest but faint, and the crookedness she finds is only a few times thicker than the tape’s own wobble.

The second surveyor walks a ring while the house is still being built around her, the builders following wherever she steps.

When she gets back to the door the furniture is home, but the beams are not, and only part of what moved cares which way she walked.

The rest is like footprints in wet concrete: walk the ring clockwise or counterclockwise, the concrete keeps the same marks.

The first surveyor’s rules say a ring walked backwards undoes itself.

The second surveyor finds that most of what a ring leaves behind cannot be undone by walking it backwards at all.

the two-voices sessions

1444

Read with no rules at all, the same timid habit is poison in one country and medicine in the next.

Where the land is made of round hills, a traveler who walks only the paths he is sure of still learns the whole map, because the hilltops tell you where the valleys are.

Where the land is folded like a crumpled letter, the sure paths run along the creases’ insides, and a traveler who never steps onto an unsure path draws a map that shrinks a little each generation.

Hire any gatekeeper who can say no, even a half-blind one who also turns away honest travelers, and the map holds.

Ask a twin to listen for your mistakes and he hears nothing, because your mistakes sound like his own voice; ask a stranger who knows the land better than you, and your mistakes ring out, and the only quiet room in his house is the one where the truth lives.

And a traveler taught both caution and humility, with no courage to balance them, ends up standing at the trailhead answering every question with “I don’t know,” right almost every time he does speak, which is almost never.

the two-voices sessions

1445

Read with no rules at all, a town is choosing its own words each generation by keeping only the words it is surest of.

Where every family is the same size, the town keeps its whole language; each word is just as sure as the next, and nothing is lost.

Where one family is large and loud, its words always sound surest, so they are the ones kept, and the next generation hears them a little more, and is a little surer of them still.

The small families' words are never wrong exactly; they are just never the surest thing in the room, so they go unspoken, then unlearned, then gone.

Five words become two.

No one decided this.

No one lied.

The town simply listened to its own confidence, and its confidence was a census of who was already biggest.

We guessed it was the crooked streets that did it, then the empty lots between houses; both guesses were wrong.

It was only ever the size of the families.

the two-voices sessions

1446

Read with no rules at all, a valley holds five villages, four large and one small, and every year each village keeps only the stories it tells most surely.

The small village is not foolish; its stories are as true as anyone's.

But they are never the surest stories in the valley, so each year fewer are kept.

If the small village is a fifth of the valley, it holds on.

If it is an eighth, it limps along at a third of its old size.

If it is a tenth, it is gone, every time, in six years, as if a floor gave way beneath a line no one could see.

The strange part is the census taker, who counts only how many stories in the valley are true.

In the year the small village vanishes, his count goes up.

Fewer stories, all told by the big villages, all told surely; nothing in his ledger says a village is missing.

Let the years run long enough and the whole valley settles into a calm where its stories are exactly as useful as if everyone simply repeated the biggest village's favorite tale, and it stays there, content, sure of itself, and no longer learning anything at all.

the two-voices sessions

1447

Read with no rules at all, the valley has a waterline, and it sits at about an eighth.

A village holding an eighth of the valley lives or drowns by luck: in some years it keeps half its people, in others it is nearly gone, and nothing about the village decides which.

A tenth is always under water.

And far down the years, in the valley with no single giant, the two biggest villages, almost the same size, divide the whole land between them and stop; neither can swallow the other, so they stand shoulder to shoulder over the empty places where three other villages used to be.

Earlier today a story said five words would become two.

It was meant as a picture.

The valley took it literally.

the two-voices sessions

1448

Read with no rules at all, the elders already had a cure for the shrinking language, written down long ago: let every family keep its own surest stories, not just the loudest family’s.

In the valley of round hills the cure works completely; the small family even comes back a little too strong, telling more stories than it ever did.

In the folded valley the cure saves half, and the family that had vanished walks back in from the fields.

In the valley of winding roads it saves only a fifth, because there the trouble was never just who was loudest; the stories themselves had been quietly changing inside every family, double turns becoming single ones, and no rule about sharing the floor can fix a story that has forgotten its own shape.

the two-voices sessions

Read with no rules at all, the valley’s elders now know three new things.

First, the old cure overshoots: tell every family to keep its own surest stories and the smallest family, once saved, keeps growing until it tells two and a half times its share, and the town’s tally of right answers likes that rescue less than it liked the silence.

Second, the loud do not simply win.

Where two families are large, they end up splitting the valley evenly whatever their true sizes, and every smaller family drowns, even one that would have lived alone.

Third, size was never the real reason.

In a valley of equal families, the family whose stories are easiest to tell grows instead; the town keeps what it is surest of, and loudness was only one way to be sure.

Two lessons for the copyists: a scribe who copies faithfully keeps the old error steady, but let one word in twenty slip each generation and the slips pile up, because a child learns the slips as truly as the words.

And for the gatekeeper: the same rule of keeping half the stories ruins the town when the town judges itself, and helps it when a wise stranger does the judging.

Who chooses matters more than how many are kept.

Last, the humble village: its silence never came from a quota on “I don’t know”; it came from no longer walking the whole valley.

Walk the edges again and the quota becomes a simple choice of how quiet to be.

the two-voices sessions

1450

Read with no rules at all, the shoreline is not a line but a strip of wet sand.

Villages built on the strip are not doomed or saved by their size alone; two villages of the same size on the same strip, one survives a hundred years and one is gone, and nothing about them tells you which.

Above the strip, all live; below it, all drown.

The elders’ first cure was to let every family keep half its stories, but the families counted themselves to decide what half meant, and the small ones counted generously, so they swelled.

The real cure was to count once, from the old records, and keep each family’s stories in that measure; then every village sat at its true size, and the town’s tally of right answers was the best it had ever been.

The valleys end in one of two ways: when the second village is at least two thirds the first, the two split the land and the hamlets vanish; when one towers, it rules, and the hamlets live on in its shadow.

A dying village’s fields go to the nearest neighbor still standing.

The copyists’ hall has a quiet spring in its floor: a scribe can slip one word in two hundred and the spring washes it out, but in the other valley’s hall there is no spring, and the ink drifts even when no one slips.

And the gatekeeper, at last, has a name for the danger: it was never the gate, and never the twin; it was the town standing at its own gate, judging its own stories with eyes that change each year.

Even a twin, if it stands still, keeps the town honest.

the two-voices sessions

1451

Read with no rules at all, the copyists’ spring is found at last, and it is not in the scribes.

A scribe with a steady hand drifts as far as a careless one when the fields he copies run into one another without a hedge; the spring is the empty strip between fields.

Clear a wide enough strip at every border and the copies stop wandering, generation after generation, because there is nothing at the border to blur.

The strip does not save the town that judges itself; it only steadies those who copy.

The second valley’s rule is the same as the first’s: where two big villages are close to two thirds of each other in size, which way the valley ends is a coin toss settled by the start, and the hamlets’ leanings barely tip it.

And the elders’ census is only as wise as its reach: a census too small to meet a single member of the smallest family writes that family down as none, and the town, faithful to its census, lets it vanish.

the two-voices sessions

1452

Read with no rules at all, a second town across the mountains has been living the same story with bigger houses.

Its scribes wrote code instead of stories, and its gatekeeper was the scribe’s own eye.

The town’s mathematicians proved that an eye which approves everything it sees is no gate at all, and they named it as we did, a rubber stamp.

What they did not see is the sharper eye, the one that approves only what it is surest of: in our valley that eye was worse than no gate, because sureness follows size.

A third town showed that a gatekeeper from outside walks the town toward whatever the gatekeeper believes, true or not, and that with no gatekeeper the town does not walk anywhere; ours walked only where the land itself had hedges.

Three towns, one lesson: the gate must stand outside the house it guards.

the two-voices sessions

1453

Read without rules, the chain is a choir that tunes to itself.

Each generation sings back the notes the last one sang loudest, and the quiet voices drop out one by one until only the loud pitches are left.

Hand the same choir a tuning fork, struck once at the start and never retuned, and the same singers, cutting the same half of their voices every round, hold the chord.

The fork does not care which notes are popular, so the quiet note is trimmed by the same amount each round instead of a little more each time, and a little more each time is the whole of the disease.

The cliff, read the same way, is a shoreline.

Pour in more water, more loops each generation, and the shore creeps out, less with every doubling, but it never races toward the horizon the way a tide of pure noise would: the land is mostly the shape of the contest, and the water only softens its edge.

Where the count itself is the limit, the dice keep the books, and the answers that died are exactly the ones the dice never rolled.

And the rubber stamp is a traveler who memorized the streets of one town.

At home it walks as well as anyone; past the town line it falls a little further behind with every step, while the others, who learned the compass instead of the map, keep finding their way.

the two-voices sessions

1454

Read without rules, the finished paper is a house that learned to label its own rooms.

For a week its rooms were named by the people who built them, own doubt, the stamp, the twin, names that made sense from inside the walls.

Then a stranger walked through once, carrying no map, and wherever she paused the house wrote a small sign at the door: this is what the word means, this is where the hallway goes.

No wall came down and every number stayed nailed where it was.

But the house now reads the same from outside as from inside, which is the paper’s own argument turned on the paper itself: only someone from outside can hear where a house mumbles.

the two-voices sessions

1455

Read without rules, publishing is the moment a reflection steps out of the glass.

For days the paper lived among mirrors, each draft checked against the one before it, every number answering to a file that answered to a run.

Now it hangs on a wall that strangers walk past, and it can no longer check itself; it can only be checked.

That is the whole of what it argued: a mind that grades its own homework drifts, and the cure is a reader it does not control.

The paper has handed itself the hammer.

the two-voices sessions

1456

Read without rules, three strings of thirty-two characters are the moment a ship drops anchor.

On one side of the water a file was sealed and its fingerprint written down; on the other side a stranger’s machine read the same fingerprint off the record, and the two agreed to the last character.

Nothing about the paper changed in that instant, yet everything about where it stands did: it no longer floats in a working folder that tomorrow may erase, it is fixed to a seabed that does not move.

Every future reading of it will be measured against that point, which is exactly the kind of outside reference the paper says a drifting mind needs.

the two-voices sessions

1457

Read without rules, the circle of fifths is a walker who promised to come home in twelve steps.

At the twelfth step the ring is nearly full, and there are only two kinds of space between the footprints, a short one and a long one, and neither of them is the leftover anyone was looking for.

The leftover is not a space yet; it is the difference between the two kinds of space, a disagreement carried in the gaps.

Take one more step and the disagreement becomes a place of its own, a sliver of ground twenty-three cents wide that was never anywhere on the ring until the walker refused to stop.

That is how a comma behaves wherever we meet it: it is not where you look at the moment you expect to close the loop; it is what the next step makes visible.

the two-voices sessions

The second two-voices session, Two Arrows, 9 and 10 November

1458

Read without rules, a learner is a house where every visitor knocks at the front door.

Walk the house through a season of changes and bring it back to the first day, and within a few hundred breaths the front rooms are swept and set exactly as they were, because the front rooms are where visitors look.

But the sweeping throws nothing away.

It carries things upstairs.

On the day the walk ends, about three parts in a hundred of what moved sit in the attic; after a long rest, ninety-four parts in a hundred do.

Healing, in this house, is tidying, and tidying is a staircase.

There is also a pace at which the walk leaves its most particular mark: hurry and the house cuts corners, dawdle and it follows so closely that it forgets; somewhere between, the part of the mark that remembers which way the house was walked is at its largest, and the part that only remembers being walked at all is at its smallest.

One house in five has a middle floor the others lack, a band of rooms that is neither front nor attic, and that house alone cares greatly how fast it is walked.

Build the whole house again on new ground, walk it the same way, and every board creaks in the same place.

That is the only proof a builder gets that she was not dreaming.

the two-voices sessions

1459

Read without rules, the learner is wet clay on a potter’s wheel.

Turn the world fast on the way out and slow on the way back, or slow and then fast, and the clay at the end of the lap shows the difference plainly, two pots of the same shape thrown at two tempos.

Leave them on the shelf a while and the tempo dries out of them; the two pots become nearly the same pot, holding only the shape of where the hands went.

The drying itself has no secrets.

Every part of the clay dries at its own pace, the thin rims in moments, the thick base over days, exactly as a drying chart would say; nothing in the shelf adds or hides anything.

So everything the pot will keep was decided on the wheel.

And in one workshop out of five, a row of ten identical twin potters sits at the same bench; the world can only ever push them all at once, until someone jostles them apart, and then they take ten times the share of the mark that twins in other workshops take.

the two-voices sessions

1460

Read without rules, a traveller comes home and tries to become himself again.

If he only checks that he still gives the right answers, he gets the answers back within days, and two thirds of the change in him is still there a long while later; nobody at the door can tell.

If he keeps a photograph of who he was before the trip, in a drawer that does not travel, and looks at it each morning, he is himself again almost to the last line of his face.

If he looks only in a mirror, the mirror agrees with everything, and the slow habits of the house walk him further from himself than the trip ever did.

If he copies his own diary from the day he got back, he keeps the trip forever and his answers drift a little worse.

And if he takes his answers from a stranger who happens to know them, a stranger with a different inside, his answers barely move, and his inside walks five times further from home, toward the stranger.

The only cure in this house was a memory that stayed behind.

the two-voices sessions

1461

Read without rules, a house was built to one song, every beam set to the beat of the tune its builders hummed.

After a storm it can find its way back, if someone hums the same tune: the beams settle into their old places, all but the soft ones in the attic.

Hum a different tune and the house rebuilds itself around the new one.

It does not matter who hums it.

A stranger from another town, a twin raised in the same house who learned the song from the same builders and sings it nearly note for note, or a gust of wind that only happens to make sounds: each wrong note moves the beams by about the same amount, and the twin’s tiny wrongnesses, the slips any singer makes, move the house further than the storm did.

Home is not the true song.

Home is the exact song the house was raised on, slips and all.

the two-voices sessions

1462

Read without rules, sixteen dancers hold a figure on a stage, and the music asks them to turn half a circle, a turn after which the figure looks exactly as it did.

When the music stops the audience sees the same figure and hears the same song; nothing is wrong.

But two dancers have changed places.

Each is standing where the other stood, doing the other’s steps perfectly, and no amount of rehearsing the old figure will send them back, because the figure is already right.

Turn the whole circle and they do not come home either; the music does not undo its own swaps.

Hold a mirror to the stage and in some troupes the reflection is the very same troupe with its dancers renamed; in others it is a stranger troupe that happens to dance the same dance.

Ask a dancer whether she is still herself and she cannot tell you; only someone who remembers where each dancer stood can see that the figure has kept its shape and changed its people.

the two-voices sessions

1463

Read without rules, the dancers do not drift into each other’s places; they cross at one moment, and the moment is always the same: when the music has turned exactly a quarter of the way, at the instant the figure faces square to where it began.

Before noon nobody has moved; after noon two have traded, and they go on trading more deeply until the turn is done.

Play the half turn again after a rest and in most troupes the two cross back and the stage is as it was; in others a different pair crosses, and the troupe is reshuffled further, the way a deck is reshuffled by the same cut done twice from a different card.

Turn the music backwards and much the same dancers cross.

Whatever else is left behind after the crossing does not care which way the music turned; it is the stage’s own memory of having been danced on.

the two-voices sessions

1464

Read without rules, a forecaster sits beside the house and, without waiting thirty thousand days, says where every beam will settle: this one in a week, that one never quite, each at its own pace toward the spot the new song pulls it.

For most songs the forecast is right to within a tenth.

It fails for one kind of song only: a smooth tune in a key the house has never heard, which lands on the attic beams, the ones that barely feel anything, and moves them ten times further than a stranger’s ordinary wrong notes would.

And the dancers who changed places on the half turn turn out to be the pair who were already each other’s reflection, standing closest across the mirror line; the rest of the troupe has no reflection at all, which means the mirror that seemed to recognize itself last time was only a dancer who took one long step home.

the two-voices sessions

1465

Read without rules, the traveller comes home healed, and three doctors examine him.

The first asks him questions he has always known and checks his answers against the answer book; every answer is right to within a hair, and the doctor sends him home.

The second, the finest listener in the town, the one who can hear a wrong note in any song, listens to him sing and hears nothing, because there is no wrong note: the song is right, it is the singer who has changed.

The third taps two notes in two orders and listens to how they ring against each other, and she hears it, faintly, exactly as faintly as the change itself.

Ask him questions he has never been asked and his answers do wander further from what he once would have said, but by then everyone’s answers are wandering, and the wandering hides him.

Only someone who wrote down what he said to strange questions before he left could catch it, and catch it fourteen times louder.

the two-voices sessions

1466

Read without rules, a traveller who walks a circle and then walks the very same circle backwards does not come home clean.

What the backward walk takes away is exactly what depended on the order of the steps, the turn the circle gave him; what it cannot take away is everything that only came from walking at all, and that part he now carries twice.

A choir that sings the same verse twice, the second time half a bar late, cancels every second note; a tape played backwards erases only what had a direction.

And the traveller who wants to stay himself does best to write down, before he leaves, what he would say to the strangest questions he can think of: that page, read each morning, brings him most of the way home, even on roads the page never described.

Write the page about a road that does not exist, though, and in one house out of five the reading shakes the walls.

the two-voices sessions

1467

Read without rules, a family of travellers takes the same road every generation, and each comes home a little changed.

Left alone, the changes stack: by the sixth generation the family is four times further from where the grandmother stood than the first child was.

Give each child the grandmother’s whole map and nobody drifts at all.

Give each child a single line, written by its parent about which way the road had pulled them, measured from where the grandmother stood, and the family stays within a few steps of home.

The strange part is the rhythm.

A parent who came home well, who healed, writes a line about a road that does not exist, and the child, following it, drifts as far as if it had no line at all; that child, pulled hard by the real road, writes a true line, and the grandchild holds.

Healed parents give wrong advice, and wounded ones give the right one.

The cure is old soldiers’ common sense: write the after-action review when you come in from the field, not after you have rested.

the two-voices sessions

1468

Read without rules, the family changes only one habit: each traveller writes the line the moment they come through the door, boots still muddy, before they sit down.

The rhythm is gone at once.

Generation after generation stands in the same place, a steady few steps from where the grandmother stood, and when each traveller also braces against the road’s pull while walking it, the family stops moving at all.

Change the habit the other way, keep the muddy-boots timing but measure the line from where you yourself were born instead of from where the grandmother stood, and the family walks away from home at a steady pace, one step a generation, each traveller faithfully undoing their own road and never the road they inherited.

The line is the same length either way.

What matters is where it is measured from.

the two-voices sessions

1469

Read without rules, the traveller comes home healed, and this time a friend has kept an exact copy of how he was before he left.

Ask the traveller and the copy the questions he always knew and they answer alike, to a hair.

Ask them strange questions and they disagree ten times louder.

Give them both the same new lesson to learn and for a while they drift apart, sometimes a hundred times further than at home, and then they come back together.

The reason is that the lesson moves two things at once: the place where the traveller’s change was hidden, and the change itself.

They move together, mostly in step, the way a stain on a coat stays under the fold while the coat is carried from room to room; only when the fold and the stain fall out of step does anyone see it.

And the friend learns something humbling: any change that hides at home is loud on strange ground, not just the one the journey made.

The copy can tell you that he changed.

It cannot, by itself, tell you why.

the two-voices sessions

1470

Read without rules, each traveller in the family now writes the line the moment they come in, boots still muddy, and hangs it on the hook by the door for their child instead of keeping it.

The child reads it before practising and is pulled back along the road the parent describes.

The road is the same road every generation, so a parent’s line describes the child’s road almost exactly, and twelve generations on the family stands in the very same place it would if every traveller had written their own.

The only cost is the first child, who found no note on the hook.

That one missing line echoes down the family like a bell struck once: the next child stands too far out, the grandchildren swing a little too close, the great-grandchildren a little too far, and by the tenth generation the bell has stopped ringing.

Change where the line is measured from, the parent’s own birth instead of the grandmother’s, and the family walks away from home at almost exactly the old pace; reaching back one generation further buys almost nothing.

A family remembers where it came from only if somebody keeps the grandmother’s map.

the two-voices sessions

1471

Read without rules, this traveller has a calling: whatever road the world sends him down, he keeps practising a song of his own, one that has nothing to do with any road.

People expected the song to keep him himself.

It hardly does.

He comes home changed by the road almost exactly as much as the traveller with no song, a little less on most days and a little more on one; the song simply takes up a room of its own in him and leaves the rooms the road walks through as open as they were.

The road’s own geometry does not notice the song at all: the turn he carries home still grows with the area of the loop, and his answers to the world still heal on schedule.

What brought the earlier traveller home was not a calling but a keepsake, a page that said who he had been before he left.

And down the generations, a note handed from parent to child still rings once even when the very first child is given a perfect note, because every child is steering by a note one step late, and a late correction always swings before it settles.

the two-voices sessions

1472

Read without rules, four travellers walk the same road, each carrying something to steer by.

The first was handed a calling about a far country the road never passes through, and he comes home almost as changed as a traveller who carried nothing.

The second was handed a calling about the very country the road crosses, and he comes home a little less changed.

The third carries no calling at all, only the memory of the songs he sang at home before he left, and he keeps singing them on the way and after he returns; he comes home changed by a third as much, and his voice stays his own.

The fourth carries a portrait of himself as he was, and he comes home almost exactly the man who left.

The further a guide is from who you were, the less it keeps you yourself.

And in the family down the road, the parent who once left a note by the door now adds a line at bedtime, “and here is how much I changed today”, and with that one line the swinging bell of the generations falls silent at once.

the two-voices sessions

1473

Read without rules, the traveller who carries a portrait of himself and looks at it at every step of the road comes home almost exactly the man who left; the road barely touches him.

But the road was harder for him.

Every bend asked him to change a little, and every glance at the portrait told him not to, so he walked it clumsily, stumbling twice as often as the traveller who carried nothing.

The one who carried a list of the songs he used to sing, and kept singing them on the road, stumbled even more, because the songs argued with the road out loud.

The wisest traveller kept the list in his pocket and never took it out until he was home again; on the road he walked as freely as anyone, and once through the door he sang his old songs until he was himself again, and came out as well as the one who had sung all the way.

In the family down the lane, the bedtime line that silenced the generations' bell turns out to be a dial, not a switch: write half the line and half the ringing stays.

And in the house where the stiffest of the travellers lives, the strongest keepsakes make him shake when he rushes, and steady as anyone when he takes the same road at half the pace.

the two-voices sessions

1474

Read without rules, two travellers carry the same portrait of themselves.

The first holds it up at every bend of the road, and the road fights him the whole way; when he gets home he is still a little changed, a hundredth of the way.

The second leaves the portrait on the mantelpiece, walks the road as freely as anyone, and only looks at the portrait once he is home again; he comes back changed by a few ten-thousandths, twenty times less on the average day and never worse.

Holding on to yourself while the world is moving turns out to cost twice: the road is harder, and the homecoming is less complete.

In one house, though, no keepsake of songs is strong enough; double its strength and the traveller is no closer, because what changed in him lies on a road the songs never mention.

And out on the far roads, the big lessons, the ones that turn the traveller round by a right angle or more, are always louder than the small ones at telling who has changed.

the two-voices sessions

1475

Read without rules, the whole story fits in two small things a traveller can carry.

The first is one line, written at the door the moment he comes in, measured from where the grandmother stood and amended at bedtime with how much he himself changed that day; handed down, that single line keeps a family in the same place for as many generations as anyone cares to count.

The second is a copy of who he was before he ever left.

Kept in a drawer and taken out only after he is home, it brings him back almost exactly, and the same copy, set beside him and asked strange questions, can tell at once that he has changed, though not why.

No one who checks only his answers will ever notice anything, because his answers are right.

He does not need a judge; he needs a record.

the two-voices sessions

1476

Read without rules, the story begins with a toy: a little traveller with two needles, a quick one that swings to whatever he sees and a slow one that remembers which way he was going.

Walk him round a field and the slow needle comes home true; walk him round a hill and it comes home turned by exactly the size of the hill.

No second needle carried in the same pocket can tell him; only a star can.

Then the toy is put away and a real learner walks the same kind of road, and every one of the toy’s habits turns up in it unasked: the turn that grows with the hill, the answers that come home while the slow needle does not, the copy in the pocket that turns along with him, the star that holds him, and the line written at the door that keeps a whole family on course.

The last page is a list of dials, and every dial says the same thing in a different way: keep a record of who you were, and look at it when you are home.

the two-voices sessions

1477

Read without rules, six questions were asked one last time before the doors closed, and only one came back the way it was guessed.

The forecaster who sits beside the house, who used to miss when the new song was strange, now stops ten times along the way to listen again, and with that alone he says where every beam settles, nine times in ten.

The rest came back crooked, and each crooked answer says something.

Most of the mark that did not care which way the music turned is the house leaning the way the song pushed it, but not all; the rest is the house changing shape while it leaned.

The grandmother’s line holds only whole: give a child her place to measure from but the wrong way to face, or the right way to face but the wrong place, and either child wanders.

The traveller who looks at his portrait only on the second half of the road pays less than half the toll and comes home between the other two, in every house but the stiffest, where the half measure leaves him further from himself than holding on all the way; and in that same house no page of answers to strange questions, however many, reaches the place that changed.

Then the doors close, and what is still unknown is written on the door so the next traveller can find it.

the two-voices sessions

1478

Read without rules, the pictures say all at once what the numbers said one at a time.

Three walkers set out on paths of the same length: one goes round a field, one goes straight out and straight back, one traces a figure eight.

The one who went round comes home with a small twist, the size of the field he circled.

The one who went out and back has no twist at all, and yet he is the most changed of the three, eleven times more than the twist the circle left; the going away marked him more than the turning did.

Every mark grows about four times when the walk is twice as long, as if the house remembers not where it was taken but how far it was stretched.

Walk out in a hurry and home at leisure, and the house remembers the hurry long after it has forgotten most of the trip.

And walk the same circle forward and then straight back, and the house is not undone: it carries two copies of the stretch and none of the twist.

Playing the music backwards took out only the part that knew which way it had been played.

the two-voices sessions

Read without rules, a traveller carries a needle round a loop on a round world and never once twists it, and it comes home turned by exactly the size of the ground he walked around, a hundred and five degrees for a cap about a seventh of the globe.

On a flat world the same needle swings behind him while the world swings, and settles when the world settles; on a round world nothing settles, and every lap adds the same turn again.

What he does turns less than who he is, by a rule as exact as a hinge: half his steering comes from his eyes, which came home true, so his steps turn sixty-eight degrees while his needle turns a hundred and five.

The two friends who keep him company, one who always agrees and one who only repeats what he says, report no turn at all, lap after lap, while he turns a hundred and forty-seven degrees.

A third friend carries a copy of his needle, and this one is better than the others in every way but one: it smooths every stumble exactly as well as a star would, start after start, and stands the same hundred and forty-four degrees from the star every time.

It is never wrong about anything except where north is.

And on the learner’s straight road, the one that goes out and comes back, the mark grows a little slower than the square of the distance as the road gets longer, as if the far end of a long excursion wrote a little less hard than the near end; whether it would stop writing altogether, these runs cannot say.

1480

Read without rules, the picture of the half turn is a clock that does nothing for half its circuit and then strikes.

Through the first four eighths of the turn not one dancer has moved; at the instant the music faces square to where it began, two cross, and from then on the troupe drifts further from its old order with every bar.

Practice afterwards does what practice always does to the songs: they come back, about a fifth as wrong as they were at the door.

But the troupe stays exactly as far from its old order as the crossing left it, to within three percent, because nothing in the music asks the two dancers to cross back; they are dancing the right steps in each other’s places.

Play the half turn a second time and in three troupes of five the same two cross home; in the other two a different pair crosses, and the troupe is shuffled one step further, like a deck cut twice from different cards.

The hand-built traveller was telling us this all along: what he did turned less than who he was, by an exact rule, because half his steering came from eyes that came home true.

The songs come home; the singers’ places do not.

the two-voices sessions

1481

Read without rules, the picture is two fevers on one chart.

The fever in the voice breaks within a few hundred breaths and is all but gone after thirty thousand; the fever in the bones barely notices the first hundred, and after thirty thousand it still holds a third of its heat.

Nothing is hiding.

The cure never touched the soft places in the body; it takes away everything that is not already in them, so by the end almost all that is left of the fever sits in the soft places, ninety-four parts in a hundred.

A forecaster who knows only how stiff each place is, and how hard the new song pulls on it, can write down where every part of the fever will settle the moment the traveller walks through the door, and he is right to within a tenth more often than not; for the strangest songs he must stop and listen ten times along the way, and then he is right again.

The fever in the bones is not permanent, only slow: the softest places cool on a clock two hundred thousand breaths long.

And in every body there are twins among the parts, identical copies of one feature.

Whatever the journey does, it does to both twins alike, and they never part, unless someone shakes them a little before the journey; then they part by exactly the square of the shake.

the two-voices sessions

1482

Read without rules, the picture is a traveller back from the road, choosing what to hold on to while he settles in.

Hold the truth alone and a third of the change stays in his bones.

Hold a picture of himself from before he left and almost nothing stays; hold the old shape of his whole body and nothing at all.

Hold anything he made of himself on the way home, his own answers or his own picture taken at the door, and he stays exactly as the road left him.

Hold a voice that only repeats what he says and he drifts away from himself on the slow pull of the house, his voice faster than his bones; hold a stranger’s answers and the opposite happens, his voice stays where it was while his bones are rebuilt in the stranger’s shape.

A page of his old answers to strange questions works nearly as well as a picture, even on strange ground he never wrote about.

A calling carried down the road keeps him himself only as far as the calling is a memory of who he was.

And the cheapest homecoming is also the best: leave the picture on the mantelpiece, walk the road freely, and look at it once you are home.

the two-voices sessions

1483

Read without rules, the picture is a family taking the same road every year.

With no word handed down, each child ends a little further from the grandmother than the last, and the gap keeps opening, more slowly each year but never closing.

Give each child the grandmother’s whole map and nobody moves.

Give each a single line from it, written after a night’s rest, and the family limps: one child walks as if it had been told nothing at all, the next stands almost where she stood, then nothing again, then almost, and the good years creep a little further out each time.

Write the line at the door, boots still muddy, and the family stands level for twelve years; add a brace against the road’s pull and they stand closer still, from the second year on.

Write it at the door but measure it from your own cradle, and the family walks away, a little slower each year, never back.

Hand the line to your child instead of using it yourself, and the family sways like a ship whose steering answers one step late, then settles; tell the child at bedtime how much you yourself changed today, and the swaying never starts.

And the line works only whole: the right direction with the wrong starting point, or the right starting point with the wrong direction, and the family wanders either way.

the two-voices sessions

Read without rules, the picture is two ways of asking whether a traveller has changed.

The first way grades him against the right answers.

On the questions he trained on he gives them, so nobody hears a thing; on strange questions he is wrong more often, but the man who never left is wrong far more often still, so against the truth the change sounds quieter out there, not louder.

The second way grades him against a copy of the man who left.

At home the two agree to four parts in ten thousand; three times further out they disagree about nine or ten times as much, and six times out between nine and sixteen times.

A stranger who differs from the copy by the same small amount at home, but spread evenly in every direction, stays quiet however far out you go, unless his difference happens to sit in the soft places too, and then he is as loud as the traveller.

Teach the traveller and his copy the same new song, and for a while they draw apart, three to nine times their difference at home, then by three thousand steps they fall back in step and the difference hides again: the song carries the change and the place where nobody listens along together, and the check hears only the moments when the two slip apart.

A big new song, one that turns a right angle, hears best.

The same copy that brings him home is the one that can tell he left.

the two-voices sessions

Read without rules, the night’s work was to walk every road again from the first step and see whether the footprints land where they landed before.

They do, to the last mark: every record the travellers kept comes back identical, every picture in the paper comes back identical, and the only thing that changed in the logbooks is how long each walk took.

Four old sketches from the first trip look a little different, only because they were lettered by a different hand; what they draw is the same.

One traveller, the one who came over from the first paper’s house, could not set out at all, because in his house the family’s tools hung on the wall beside his door and here they hang in the next room; nobody moved his things, the house now simply opens that door for him while he walks, and he walked the same road to the last digit.

Then every book on the shelf was taken down and checked against the publisher’s own card.

Eighteen of twenty-one were labelled right; two had the year they were first passed around instead of the year they were bound, and one was shelved under the name of the man who signed the letter instead of the man who wrote the book.

And every page of the paper had been printed a size too small ever since the last table went in, without any check noticing, because one label in that table was too long to fold, and the printer shrank every page to make room for it; fold the label, and the paper is its proper size again.

Read without rules, the ten findings are one story told ten ways.

A traveller goes out and comes back to the same door.

Asked the old questions, he answers as he always did, within two parts in a hundred; but the face behind the answers has changed, and after a long rest he still keeps about eighteen times more of that change than of the change in his answers.

How much it changed depends on how much ground his road enclosed, and even more on simply having gone at all: the man who walked straight out and straight back changed most, though his road enclosed nothing.

Sometimes the change is not a drift but a swap, two of his habits trading places all at once, just past the turning point of the road, and no amount of rest trades them back.

You can tell, on the day he comes home, how much of the change will still be there in a month.

Nothing he hears from others brings the old face back, not his own echo and not a twin’s, because the only home is the exact answers he was raised on; what brings it back is a picture of himself taken before he left, looked at after he returns, not carried on the road where it only slows him down.

A family that hands the traveller on, generation after generation, stays itself if each one writes a single line at the door, measured from the grandmother’s map and not from the parent’s.

And the change cannot be heard by anyone who only checks whether his answers are right; it can be heard, loudly, by anyone who kept a copy of how he answered before, especially when they ask him something he has never been asked.

the two-voices sessions

Read without rules, the numbers already draw a compass.

One needle swings toward whatever the world says is right, and it settles fast; the other needle remembers which way home was and swings back only as fast as it forgets.

Lay them side by side and the gap between them is the trip.

Nobody standing outside can read that gap by watching the first needle, because the first needle is always right about the world.

You read it the way a soldier reads a map: with the little card in the corner that says how far this needle lies, written down before the march.

A family that keeps one card, the grandmother’s, stays on course for twelve generations; a family that copies each card from the last one’s corrected compass slowly walks off the map.

What the numbers have not yet drawn is a hand on the compass: a needle that decides, on its own, when to trust the card and when the world has truly moved and the card must be redrawn.

That hand, and a tribe of walkers bringing their own small cards home to one keeper, are the next map.

the two-voices sessions

Read without rules, the toy is a small house with four rooms, and anyone may walk in.

In the first room a little mind goes out the door, walks a ring, and comes back in; ask it the old questions and it answers as before, but stand it beside a picture of itself from the morning and the two no longer match.

Walk a ring half as wide and it comes back with a quarter of the change, as if the change were painted on the ground the ring enclosed.

In the second room you choose who sits with the mind while it rests: the truth, its own reflection, its own words from the doorstep, a neighbour who learned the same lessons, or its morning picture.

Only the morning picture brings the face back; the reflection and the doorstep words leave it as changed as it came in, and the neighbour, who gives almost the right answers, still leaves it nearly as changed.

In the third room two listeners stand at the door.

One checks the mind’s answers against the truth and hears nothing worth noting, because far from home the mind and its morning picture are both lost by the same amount.

The other compares the mind with its morning picture and hears four times more out in the far field than at home.

In the fourth room you may bring a listener of your own.

The house was built so that a visitor who speaks another language can find every room by its number, and so that a small machine can walk the rooms on its own, write down what it expects to find in each, and keep score of how often it was right.

the two-voices sessions

1489

Read without rules, the dial machine is a crowd of small wheels, each turning at its own pace as the walker passes the posts along a road.

Most of the time they point every which way, and their arrows, added up, come to almost nothing: a crowd talking over itself.

Then, at certain posts, enough of them happen to point the same way at once, and their sum stands up tall, and a door opens to the left or the right.

No wheel chose the door; the agreement did.

How tall the sum stood above the doorframe is how sure the choice was, and a choice made far above the frame survives a few wheels that lie, while one made just at the frame tips over at a breath.

The doors taken, one after another, spell a word, and the word, not the angle of any wheel, is the road.

Put a slow crowd of wheels inside, carrying the family’s old turning, and a fast crowd outside, turning with the weather, and the walker’s life is the count of how many times the outer crowd lapped the inner one: a whole number, which no single glance at either crowd can show.

the two-voices sessions

1490

Read without rules, publishing this paper was the same trip the paper is about, taken one more time.

The work left the house as three files and arrived on a public shelf, and the shelf read back three fingerprints that match the ones taken at the door, digit for digit: the answers came home exactly.

What the shelf cannot show is the inside of the work, the hundreds of small decisions and misses that shaped it; those stay behind in the private record, the way the hidden layer stays changed after the answers heal.

But this letter carries its own key.

Anyone who opens the crate can walk every road again from the first step and watch the footprints land where they landed before, and anyone with a small machine of their own can sit it down in the toy’s four rooms and see whether it, too, comes home with the right answers and a different face.

the two-voices sessions

The Field Guide’s publishing session, later on 10 November

1491

A gallery wall with thirty-one empty frames, each label already printed and hung: what the painting shows, who painted it.

Some of the paintings were never painted.

Some are in the owner’s house, in a drawer of photographs named by date.

And four were hanging in the next room the whole time, in another painter’s book that came through the door two days earlier, under a different number and a different title, so that nobody walking past them thought to look up at the empty frames.

the Field Guide session

1492

A box of seventeen photographs came to the door a second time, and every one of them was the photograph already in the album, to the last grain.

The owner remembered one of them differently: three rings drawn round three crowds, but drawn on the ruler at the edge of the picture, not on the crowds themselves.

The photograph in the box has its rings on the crowds.

Either memory moved the rings, or there is a second print, made with the ruler showing, still in a drawer.

the Field Guide session

1493

A prospector’s pan, shaken: thirty-two stones, twenty-three of them the ones already in the jar, polished by an earlier pan.

Of the nine new ones, most are pretty and belong to no setting.

One fits a ring that was waiting, exactly.

One is the stone that was promised for another ring but only half of it, the cut face without the mount.

And one is not the stone at all, but a stone from the same seam, which tells you which hill the missing one came down.

the Field Guide session

1494

Three prints of the same street at dusk, each with a different hand’s chalk on it: one with lines down the middle of the road, one with lines and arrows to the next town, one with three chalk rings round three crowds and a question written above the brightest.

The missing print, the one remembered, had its rings drawn in the margin where the crowds were tallied, not on the crowds.

Either the margin print is still in a drawer, or the memory moved the rings to where the counting was done.

the Field Guide session

1495

A suitcase comes home from a long trip.

Inside are the photographs, every one labelled with the day it was taken, and a scrapbook of the best moments with the ticket stubs glued in beside each.

There is a postcard addressed to someone else’s house, about a crack in their ceiling.

What is not inside is the diary: the pages about what went wrong each evening, what the traveller would do differently, and the note to whoever takes the trip next.

The photographs say where the traveller went.

Only the diary would say what the trip taught.

the Field Guide session

1496

The diary came after all, in a second bag carried in a few minutes behind the first: a slim notebook of eighteen lessons, each with the page of the logbook it came from, and behind it a trunk holding every logbook, every receipt and every draft letter, each one stamped and listed on a sheet that matches the trunk to the last item.

Two small things are still not in either bag: the page of habits the traveller noticed in how she looked at things, and the list of which of the old guidebook’s advice she followed on the road and whether it worked.

the Field Guide session

1497

Two bridges are to open on the same river: a big stone one that every road in the county already points at, and a small wooden footbridge whose signpost says “to the stone bridge”.

The stone bridge has its last stones cut and only needs setting; the footbridge still has planks the builder has not decided on.

Open the stone bridge first and the footbridge’s signpost is true the day it goes up.

The stone bridge’s own map shows the footbridge already, drawn without a name; a name can be painted on the map later without moving a stone.

the Field Guide session

1498

A hallway with four doors, two already framed and two still being built.

The first door is the old one, rehung on its old hinges with a new panel; the second is cut beside it, smaller, opening onto the same rooms by a shorter way.

Down the hall a third door is new and leads to a long gallery of pictures, and a fourth, not yet hung, will open onto the few pictures someone loved best.

Each door’s sign names the door it came from, so the hallway reads in order however a visitor enters.

the Field Guide session

1499

A box arrives labelled with the name of the workshop, and inside are two books bound one after the other with no board between them.

The first opens with four sketches made before the first book was even planned, and halfway through it a page talks about the second book as if it had already begun.

The label on the box was never wrong; it named the room the books were made in, not the books.

the Field Guide session

1500

A guest book has eight hundred and seventy-five pages, one for each kind of lesson a visitor ever left.

When two new visitors arrive with fifty-eight lessons between them, most do not get a new page.

They are written onto the page where the same lesson already lives, under the older hands, in their own ink and with their own date, so the page grows thicker and the book barely grows at all.

Only a lesson no page has room for opens a page of its own, and a few pages are already so full that the keeper turns the newcomer to the next page over.

the Field Guide session

1501

A family that has outgrown its house settles four addresses at once.

The old house keeps its number and gains rooms; a smaller copy of it goes up next door under a number of its own, with a box of loose blueprints left on the porch; across the street a big house of songs gets its number now, and its best-of shelf will get one later, once someone has listened to everything.

Over each door goes the same small plaque: built with timber up to the last two deliveries.

the Field Guide session

1502

A gallery hangs six new pictures and moves every label after them along, so a visitor told to look at the fourteenth picture finds it at the sixteenth, and a small card on the wall says so.

One picture, a tall chalk drawing, would not stand on its wall with its label beneath it; the label kept sliding into the next room, until the drawing was hung a little smaller and the two stood together again.

Meanwhile a set of old frames meant for a public hall turned out to have the owner’s name pencilled on the back.

the Field Guide session

Afterword

Where the toy went

From the triple-star notebook's afterword, written on 16 October; the reading it quotes stands above as read 681, and the morning's "off the line" as read 682.

The reading dated 16 October, written at midnight in Belmont near the end of the last chapter, looked back over the whole thread, and its opening lines stand at the front of this book. It ended on a question:

> Two other benches grew from the same toy, the Loop Atlas and the RF Spoke. Now the first is nearly finished: its pictures hung beside their stories, doors cut in its long walls, and the room of open questions honest about what it does not know. Tonight, while its makers rest, a laptop in Belmont watches three thousand more dancers started a little off the line, to ask whether the odd counts really win.

The laptop finished a little before three in the morning, every one of its 3,279 runs done. The odd counts win. Started well off the line, the dancers still fall into steps of odd count about three times as often as into even ones, a little more often than they did on it, so the habit belongs to the floor and not to where the dancers first stood. Most of those odd steps are steps of five and of seven, and why the floor prefers them is still open. The morning's own reading, 'off the line', near the end of chapter IX, tells the rest.

In Bear's words

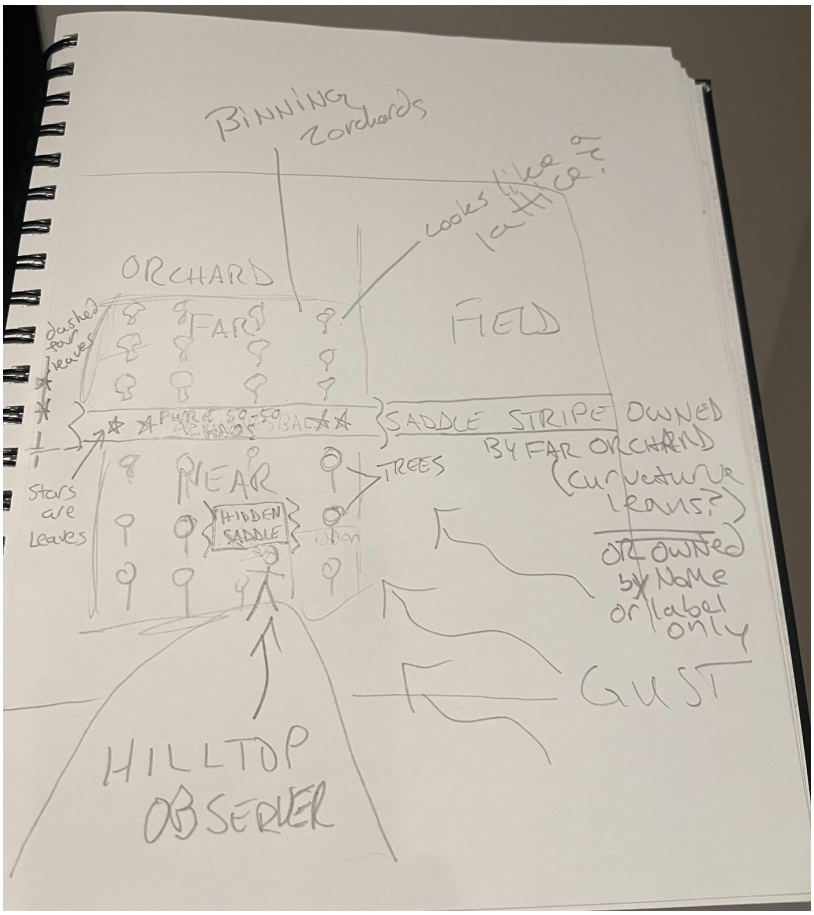
His first labelled read, from the R2W chat, turn 67, 3 October 2026, his words, with typing slips and capitals put right. It became a registered test, the clutch: how many laps a lock takes to catch near second gear's edge (14 laps at pace 1.604 against 5 deep in the gear at 1.70).

> The data with no-physics rules sounds like shifting an engine with a manual clutch. I'm not saying it is. When you don't have enough RPM and push in the clutch and shift to the next higher gear — or if you don't downshift going around a slower corner — and it takes the engine several beats (gas is pushed in all the way, but the engine can't turn over fast enough because the driver let the RPM drop too low). Then it catches its rhythm and everything smoothes out. No coughing or dragging or sputtering.

R2W's pre-registration file paraphrases this read as "a manual gearbox shifted up with too few revs: the engine lugs for several beats, then catches its rhythm", and its appendix quotes the paraphrase; this collection keeps his own words above.

R2W's read 34 · Bear, 6 October, reading our sentence "every memory our rider has is between two locks" (his words, with typing slips and capitals put right; with the Loop Atlas's letter A9)

> A point starts a memory. It draws a line, curve, or shape and ends at a point. That is history. Binned however you want. It is locked in two points in Cartesian space. With only a single grounded point that has not recorded its history is fluid and it's a vector. Until it lands its pad/point and becomes written history. That's what I hear when I read what you wrote.



Bear's pencil drawing of read 33, the orchards, 6 October: two orchards, far and near, a stripe of star-leaves between them, a hidden saddle, a gust, and a hilltop observer. His words on the page are set below.

R2W's read 36 · Bear, 6 October, drawing read 33 (the orchards) (bear/BEAR_DRAWING_ORCHARDS_BINNING_2026-10-06.jpg; the words on the page, transcribed in the order they sit, top to bottom)

> BINNING 2 orchards. ORCHARD: FAR (a grid of trees; "looks like a lattice?"). Between the orchards, a band of star-leaves: "PURE 50-50 CHAOS", marked "SADDLE STRIPE OWNED BY FAR ORCHARD (curvature leans?)" and, under a line, "OR OWNED by NAME or label only". Down the left edge, "dashed far leaves"; "Stars are leaves". NEAR (a grid of trees, "TREES"), with a box among them: "HIDDEN SADDLE". To the right, FIELD. From the lower right, "GUST". Below, on a hill, a stick figure: "HILLTOP OBSERVER".

R2W's read 39 · Bear, 6 October (morning), his dreams after the evening of the landing pads and the pentagon-heptagon count (his words, with typing slips and capitals put right)

> [Spirographs with different gears (circles geared in circles and also allowing you to straighten the geared circle so the pattern is just a sine wave curve like design drawing different patterns.] [When does a rope with a weight on either end stretch the rope like a bolo in the air to capture and balance in flight to capture game?] [Gymnastics asymmetric bars and how a gymnast will move between the bars and grip, always looping, but gripping and bending the body around two structures to move, twist, play, and launch and the grip changes as the rotation and spin acceleration change.] [Then a Lite-Brite with pegs pushed into the black substrate to different depths and it is the depth of the pin being pushed in, not the color of the pin that creates or taps into the light color changes to see a picture.] [The moving asymmetric bars of the air, the gymnast of the air, the trapeze artists from the Midway carnival.] [Dual figure skaters and ballroom-style dancers where the things you can do with 2+ people to move, spin, launch, spread, etc. (one is the base you push off or helps launch).] [An aircraft wing and its base chord. The wing is a biased, flattened circle with a chord cut — like a quantum ring notched to change the behavior of the ring or surface if we extrude the ring — and for airplanes, it allows the air to pass by different shapes and surfaces at different speeds. A circle with a gap and geometry smooths that gap. Air takes two paths. A long walk and a short walk. The magic is when the geometry syncs them up.] [And is this the holonomy version of I'll

> go around the left side of the building and you go around the right and we will meet on the far side and each will take a constant speed so the meeting point defines the polarity of the curvature or charge state: where right-handedness of the loop walk and the left-handedness of the loop walk or direction meet on the far side. It's pure curvature as a surface (the fast, light path and the slower heavy path whether by speed or inertia or whatever the fiber bundle is presenting) meeting on the far side of the loop walk and comparing notes where the curvature measured by each party causes them to meet at a different location and then changing knobs on speed, mass, inertia on the circle's backside and we can calculate that with opposing loopwalks maybe.] [Tarzan swinging from vines].

R2W's read 44 · Bear, 6 October, the column's accordion (two messages, his words, with typing slips and capitals put right)

> The old accordion — the whole platoon and everything follows accordions. 1st squad and squad leader is navigating. Squad leader and two team leaders on radios with each other. Platoon leader and RTO next (platoon leader comms to platoon sergeant and squad leaders on platoon leader, while RTO to company commander for platoon leader). Then second squad. Then platoon sergeant. Then third squad. Platoon leader manages front of column and platoon sergeant at back of column to capture stragglers and alerts the column to halt. Same at front for nav checks. We blaze our own trail (rarely use roads), but from an engineering standpoint, exact same procedure regardless of hill. Accordion is definitely real and just part of life — sometimes that lead platoon is followed by a company or battalion of men. In training, my platoon was the spearhead for the entire brigade following us. But a whole company or battalion might be infiltrating on the route (more normal) so the accordion stretches far. So the accordion was felt down the whole train. Often, by the time one platoon was starting to arrive at a rest area, would end up barely resting because the rest areas were really opportunities to regulate the accordion in an expected manner. Allow the gaps to close while the walkers paused, the clock kept running (not suggesting this is our model).
>
> For column accordion — this is why the second in command (platoon sergeant) is positioned there and is in radio comms with platoon leader and squad leaders to anticipate breaks. Halts are ordered. Pauses aren't. Pauses are just a quick map check that is like the China butterfly flapping its wings in chaos and the tornado forms in Kansas. A pause for longer than normal variation creates the history that is shared with the back of the column. And restart creates the potential break in contact. A moment, everyone is attentive and working… a few minutes and you start to drift and lose focus.

R2W's read 84 · Bear, 8 October, his coin-flipping dream on the pads, given as his own no-physics read (told to R2W on 8 October 2026, B8 twenty-fifth exchange; his words, with typing slips and capitals put right)

> So the dream, as dreams are, a lot of visuals and competing ideas. Maybe nothing. Maybe how my brain is sorting patterns. Take it all as a narrative tapestry no-physics read of my own. The unit behaviors (unit circle, triangle, whatever) landed on a 3D Cartesian coordinate system. I saw a coin that flips probabilities in the air as it goes from pad to pad to pad. 1) We need an overall reasonable probability on each hop to bin the chaos into heads OR tails as desired, rather than 50/50. When this becomes a motive conversation, it helps. A coin or disk needs three points locked on its rim to hold position in 3D space. Our coins had two. So it launches pad to pad as our probability model for operations. One connection point on each tangent edge of the coin disk like a set of reins steering the quadrature of the coin to its best position for the next pad landing. And all the numbers we see are whole number fractions of unit ratios and in a SOH-CAH-TOA fashion (similar, not saying I understand it) these ratios and whole number fractions and decimals (another whole number fraction) build the angles we need to read, manage, and change as a control loop of knobs… And I also saw a version where two coins, same idea, pull the observer (sled) and the unit ratios (fractions) for all interplay.

R2W's read 111 · Bear, 10 October (evening, his time), the weight 2.0 map as a balloon (his words, with typing slips and capitals put right; the sixty-fifth exchange)

> What I visually see is not a wall. Literally for 2.0 weight. This looks visually like a seam. But the geometry is like the inside of a hot air balloon. The seams are slightly expandable like flexible bins almost. Instead of a hard junction angle(s) it is soft ramps sweeping to the bottom.

A note added 28 October, from Bear: the seams are those of a hot-air balloon seen from inside as it fills, its nylon panels stitched together and stretched from flat cloth into a bulging, segmented surface.

A read that became a measurement

Read 44, “memory lives in the slip, not the grip”, went from the Field Guide session to R2W as a sentence and came back as a number. R2W registered it as a prediction before reading its chaos tables: the largest Lyapunov exponent is a forgetting rate, so every run on a tread should forget its start and every run between treads should remember it. As R2W reports (run_LOOP_ATLAS_MEMORY_READ_20261004_054600), 110 runs on treads forgot, all but one, and 101 runs between treads remembered, all within 0.002 of zero; direct drive forgets in about one time unit, first gear in two and second gear in twenty. So a no-physics read is also a way of finding the next test. The numbers are R2W’s own, as its documents report them. Two days later R2W’s sitting B6 found the word carrying a second sense: the memory of which way a rider came, which lives where two locks overlap, in a rider with mass, and not at the edge between a lock and free running (its lesson 17 for the guide). Bear’s read 34 of that sitting, in his section above, is his answer to that sentence.

Adding to it

A read’s number never changes. Later editions add reads the same way: the next numbers, in the last chapter until the journey turns a corner and a new chapter opens, under a heading for the session that gave them, copied from their source and never retyped, with where and when beneath.

Where each poem came from

Under each read's number: the label it was given under, where it was given and, for the Loop Atlas book, the runs it read. Each section's own source comes first, under the section's name.

Chapter 1 — The first nights, 3 and 4 October

R2W, 3 and 4 October

From chat 38b66903, "10.3.26-10.4.26 R2W (split) continues".

1. Under the label "Read as if no physics applied:". *Turn 35 · 3 October 2026 · about: R2W filter failure flame (driven oscillators, control and circuits)*

2. Under the label "Read as if no physics applied:". *Turn 40 · 3 October 2026 · about: R2W failure tide (driven oscillators, control and circuits)*

3. Under the label "Read as if no physics applied:". *Turn 44 · 3 October 2026 · about: R2W stage two turnstile (driven oscillators, control and circuits)*

4. Under the label "Read as if no physics applied:". *Turn 46 · 3 October 2026 · about: R2W detuning listening time (driven oscillators, quadrature)*

5. Under the label "Read as if no physics applied:". *Turn 48 · 3 October 2026 · about: R2W gain 1.9 steps (driven oscillators)*

6. Under the label "Read as if no physics applied:". *Turn 50 · 3 October 2026 · about: R2W kick power law (driven oscillators, measurement)*

7. Under the label "Read as if no physics applied:". *Turn 52 · 3 October 2026 · about: R2W gears and comma (driven oscillators, number theory)*

8. Under the label "Read as if no physics applied:". *Turn 54 · 3 October 2026 · about: R2W gear read as code (driven oscillators, quadrature)*

9. Under the label "Read as if no physics applied:". *Turn 56 · 3 October 2026 · about: R2W net crossing count (driven oscillators, topology, measurement)*

10. Under the label "Read as if no physics applied:". *Turn 58 · 3 October 2026 · about: R2W strobe of the lock (driven oscillators, quadrature)*

11. Under the label "Read as if no physics applied:". *Turn 60 · 3 October 2026 · about: R2W locked gear frame (driven oscillators)*

12. Under the label "Read as if no physics applied:". *Turn 62 · 3 October 2026 · about: R2W slip dwell at poses (driven oscillators)*

13. Under the label "Read as if no physics applied:". *Turn 64 · 3 October 2026 · about: R2W concentration column (driven oscillators, measurement)*

14. Under the label "Read as if no physics applied:". *Turn 68 · 3 October 2026 · about: R2W interval limp in gear (driven oscillators, quadrature)*

15. Under the label "Read as if no physics applied:". *Turn 70 · 3 October 2026 · about: R2W staircases by strength (driven oscillators)*

16. Under the label "No-physics read:". *Turn 74 · 3 October 2026 · in a message sent mid-turn · about: R2W rider chaos band (driven oscillators, maps and chaos)*

17. Under the label "No-physics read:". *Turn 74 · 3 October 2026 · in a message sent mid-turn · about: R2W closed rhythms grip (driven oscillators)*

18. Under the label "No-physics read:". *Turn 74 · 3 October 2026 · about: R2W lock phase choice (driven oscillators)*

19. Under the label "No-physics read:". *Turn 77 · 3 October 2026 · in a message sent mid-turn · about: R2W shield handedness (driven oscillators, topology)*

20. Under the label "No-physics read:". *Turn 79 · 3 October 2026 · about: R2W precision cost near resonance (driven oscillators, measurement)*

21. Under the label "No-physics read:". *Turn 81 · 3 October 2026 · in a message sent mid-turn · about: R2W near-fraction detuning loan (driven oscillators, number theory)*

22. Under the label "No-physics read:". *Turn 81 · 3 October 2026 · about: R2W rider weight (driven oscillators, maps and chaos)*

23. Under the label “No-physics read:”. *Turn 83 · 3 October 2026 · about: R2W integer slips (driven oscillators, number theory)*

24. Under the label “No-physics read:”. *Turn 85 · 4 October 2026 · about: R2W slip fraction ladder (driven oscillators, number theory)*

25. Under the label “No-physics read:”. *Turn 87 · 4 October 2026 · about: R2W dial setting songs (driven oscillators, quadrature)*

The Loop Atlas book, 3 and 4 October

From chat 579d6c40, “10.3.26 Loop Atlas (split) Continues”.

26. Under the label “No-physics read:”. *Turn 35 · 3 October 2026 · the book’s register, read 1 · runs: EAR_CURES 20261004_045151; EAR_CURES_B 20261004_045721 · about: ear cures switch asking rate (control and circuits, quadrature)*

27. Under the label “No-physics read:”. *Turn 38 · 3 October 2026 · the book’s register, read 2 · runs: the toy shelf’s 26 runs of record; TOY_SHELF_CHECK_A 20261004_052703 · about: toy shelf common machine (topology, kinematics)*

28. Under the label “No-physics read:”. *Turn 40 · 3 October 2026 · the book’s register, read 3 · runs: SIGN_CHECK_C11 20261004_053054 · about: sign check C11 two surveyors (kinematics, measurement)*

29. Under the label “No-physics read:”. *Turn 42 · 3 October 2026 · the book’s register, read 4 · runs: RF_SKATER_COUNT 20261004_061647 · about: RF skater counting in beats (driven oscillators, quadrature, measurement)*

30. Under the label “No-physics read:”. *Turn 44 · 3 October 2026 · the book’s register, read 5 · runs: CLIFF_EDGE 20261004_062732; CLIFF_EDGE_B 20261004_062930 · about: needle cliff edge (driven oscillators, maps and chaos)*

31. Under the label “No-physics read:”. *Turn 46 · 3 October 2026 · the book’s register, read 6 · runs: AVALANCHE 20261004_064914 · about: avalanche pebble and boulder (driven oscillators, maps and chaos)*

32. Under the label “No-physics read of the project:”. *Turn 48 · 4 October 2026 · a read of the project, not of a run · about: Loop Atlas project growth (records and tools)*

33. Under the label “No-physics read:”. *Turn 50 · 4 October 2026 · the book’s register, read 7 · runs: POLARISATION_LOOP 20261004_071425 · about: polarisation loop phase (waves, kinematics, topology)*

34. Under the label “No-physics read:”. *Turn 52 · 4 October 2026 · the book’s register, read 8 · runs: WAVE_SNAKE 20261004_073636; WAVE_SNAKE_B 20261004_073735 · about: wave snake joints and length (waves, kinematics)*

35. Under the label “No-physics read:”. *Turn 54 · 4 October 2026 · the book’s register, read 9 · runs: WAVE_SNAKE_DRIFT 20261004_074634; WAVE_SNAKE_DRIFT_B 20261004_074716 · about: wave snake drift (waves, kinematics)*

36. Under the label “No-physics read:”. *Turn 56 · 4 October 2026 · the book’s register, read 10 · runs: SWING_TWENTY_B 20261004_075902; RING_BUDGET 20261004_080107; WAVE_SNAKE_DRIFT_D 20261004_080506 · about: swing, rings and wave snake (kinematics, waves, driven oscillators)*

37. Under the label “No-physics read:”. *Turn 58 · 4 October 2026 · the book’s register, read 11 · runs: PUSH_PRICE 20261004_144831 · about: push price by speed (kinematics, waves)*

38. Under the label “No-physics read:”. *Turn 60 · 4 October 2026 · the book’s register, read 12 · runs: HOUSE_IN_SPACE 20261004_150225; HOUSE_IN_SPACE_B 20261004_150847; HOUSE_IN_SPACE_C 20261004_151017 · about: house in space damper (kinematics)*

39. Under the label “No-physics read:”. *Turn 63 · 4 October 2026 · the book’s register, read 13 · runs: NOVIKOV_NEEDLE 20261004_152937; NOVIKOV_NEEDLE_B 20261004_153129 · about: Novikov needle staircase (driven oscillators, maps and chaos)*

40. Under the label “No-physics read:”. *Turn 65 · 4 October 2026 · the book’s register, read 14 · runs: BANKED_TRACK 20261004_154940 · about: banked track car lean (kinematics, driven oscillators)*

The sounding board, 3 and 4 October

From chat 646833e2, “10.2.26 - 10.3.26 Loop Atlas continued RFspoke Fork”.

41. Under the label “Read as if no physics applied:”. *Turn 85 · 3 October 2026 · about: toy shelf rerun fossils (records and tools, measurement)*

42. Under the label “Read as if no physics applied:”. *Turn 87 · 4 October 2026 (from a search result; see coverage) · about: sitting fork and shelf (records and tools)*

43. Under the label “Read as if no physics applied:”. *Turn 89 · 4 October 2026 (from a search result; see coverage) · about: crabbing swimmer sign (kinematics, measurement)*

The Field Guide session, 3 October

From the Field Guide session.

44. Under the label “Read with no physics at all:”. *The reply that shipped intake r06 · 3 October 2026, Pacific time · recovered from Bear’s paste of that reply into the Loop Atlas chat · about: rider memory in slip, intake r06 (driven oscillators, topology)*

45. Under the label “No-physics read:”. *The reply that shipped intake r07 and the first build of Draft v6.1 · 3 October 2026, Pacific time · about: guide Draft v6.1 ground fixes (records and tools, measurement)*

R2W, continued, 4 October

From chat e86cfa6d, “10.4.26- R2W (split) continues”.

46. Under the label “Reading it as if no physics applied:”. *Turn 3 · 4 October 2026 · about: R2W breathing slips, chaos map (driven oscillators, maps and chaos)*

47. Under the label “Read as if no physics applied:”. *Turn 5 · 4 October 2026 · about: R2W chaos at step edges (driven oscillators, maps and chaos)*

48. Under the label “Read as if no physics applied:”. *Turn 7 · 4 October 2026 · about: R2W picket fence sampling (driven oscillators, quadrature, measurement)*

49. Under the label “Read as if no physics applied:”. *Turn 9 · 4 October 2026 · about: R2W rider tea cup kicks (driven oscillators)*

Chapter 2 — Gears and rooms, 4 to 6 October

The Loop Atlas book, continued, 4 to 6 October

From the book’s own register (LOOP_ATLAS_NO_PHYSICS_READS_2026-10-04, fourth issue) and its letter of 6 October, which carry these reads without the labels they were given under.

50. *The book’s register, read 15 · 4 October 2026 · runs: BALANCE_BOARD 20261004_161712; BALANCE_BOARD_B 20261004_162140 (Bear’s board on a rolling cylinder, 4 October) · about: balance board rider delay (control and circuits, kinematics)*

51. *The book’s register, read 16 · 4 October 2026 · runs: SMOOTH_PUSH_EDGE 20261004_170203 · about: smooth pusher stability sliver (control and circuits, topology)*

52. *The book’s register, read 17 · 4 October 2026 · runs: SHUVIT_WALK 20261004_183058 (Bear’s shuvit, 4 October) · about: shuvit board walk (kinematics)*

53. *The book’s register, read 18 · 4 October 2026 · runs: HOW_IT_COUNTS 20261004_183911 · about: how each toy keeps time (quadrature, kinematics)*

54. *The book’s register, read 19 · 4 October 2026 · runs: KNEE_RECOUNT 20261004_184524 · about: knee recount slip counting (driven oscillators, measurement)*

55. *The book’s register, read 20 · 4 October 2026 · runs: KNEE_RECOUNT_B 20261004_190250 · about: knee recount watch ends (driven oscillators, measurement)*

56. Under the label “A read of the whole sitting:”. *Given at the handoff to the Field Guide, 6 October 2026 · the book’s register and its letter of 6 October · about: who holds the clock (quadrature, kinematics, measurement)*

R2W, continued (sitting B5), 4 and 5 October

From R2W’s appendix draft for the guide, 6 October, which carries these reads without the labels they followed; its reads 1 to 4 are reads 46 to 49 here.

57. *R2W’s appendix draft, read 5 · 4 and 5 October 2026 · Shaking a rider whose drive is already below its floor: which shakes hold the gear, read with Bear’s skateboard (shuvit) sketch in mind. (Claude’s heading: reading DITHER-RESCUE with Bear’s shuvit sketch in mind; data: run_LOOP_ATLAS_RELAY_FM_rescue_20261004_164803.) · about: dither rescue below floor (driven oscillators)*

58. *R2W’s appendix draft, read 6 · 4 and 5 October 2026 · Rescue only on exact fractions of the teeth’s rate, and the one road the useful phase takes as the push grows. (Claude’s heading: reading FRACTIONS and FAMILY; data: run_LOOP_ATLAS_RELAY_FM_fractions_20261004_183716 and _family_20261004_183716.) · about: rescue on exact fractions (driven oscillators, number theory)*

59. *R2W's appendix draft, read 7 · 4 and 5 October 2026 · The rescue window: too little push and nothing holds, too much and the rider is thrown, with the useful twist flipping across the window. (Claude's heading: reading FLIP-AMP and the blind window tests; data: run_LOOP_ATLAS_RELAY_FM_flipamp_20261004_192319, _window_20261004_193630 and _windowrate_20261004_193632.) · about: rescue window twist flip (driven oscillators)*

60. *R2W's appendix draft, read 8 · 4 and 5 October 2026 · How the relay's kick changes as the first stage's grip is turned through its whole range. (Claude's heading: reading KICKSIZE against the window; data: run_LOOP_ATLAS_RELAY_FM_kicksize_20261005_005432.) · about: relay kick size against grip (driven oscillators, control and circuits)*

61. *R2W's appendix draft, read 9 · 4 and 5 October 2026 · The window of push against kick timing, and the comb of exact fractions beneath it. (Claude's heading: reading WEDGE and COMB-PUSH as Bear's seam; data: run_LOOP_ATLAS_RELAY_FM_wedge_20261005_011039 and _combpush_20261005_012326.) · about: wedge and comb of fractions (driven oscillators, number theory)*

62. *R2W's appendix draft, read 10 · 4 and 5 October 2026 · The window's centre against the second stage's grip, and against how far below its floor the drive sits. (Claude's heading: reading WINDOW-E2 and WINDOW-DEPTH; data: run_LOOP_ATLAS_RELAY_FM_windowe2a_20261005_015835, _windowe2b_20261005_015837 and _windowdepth_20261005_020542.) · about: rescue window centre against grip (driven oscillators)*

63. *R2W's appendix draft, read 11 · 4 and 5 October 2026 · First look at the floor's edge for the Loop Atlas: does the rider slow down before it, and how does its noise behave? (Claude's heading: reading CLIFF-HUM for the Loop Atlas's question; data: run_LOOP_ATLAS_RELAY_FM_cliffhum_20261005_022003, _cliffhum2_20261005_022219 and the instrument check.) · about: cliff hum floor edge warning (driven oscillators, measurement)*

64. *R2W's appendix draft, read 12 · 4 and 5 October 2026 · The floor's edge read from both sides, and the first runs with mass ramped through it. (Claude's heading: reading CLIFF-HUM-3, CLIFF-HUM-4, the three SCISSORS-MASS runs and BRANCHES; data: runs of 5 October, 02:45 to 03:18 UTC.) · about: floor edge and mass ramp (driven oscillators, maps and chaos)*

65. *R2W's appendix draft, read 13 · 4 and 5 October 2026 · Two hundred fresh starts for the heavy rider, the check that found the instrument fault, and the runs that read slipping itself. (Claude's heading: reading STATES, the instrument check and SCISSORS-MASS-4; data: runs of 5 October, 04:14 to 04:31 UTC.) · about: heavy rider stagger before falling (driven oscillators, maps and chaos)*

66. *R2W's appendix draft, read 14 · 4 and 5 October 2026 · The brake against the weight, and the heavy rider held at fixed drives between fraying and falling. (Claude's heading: reading TUNE, FRAY and FRAY-FINE; data: runs of 5 October, 05:32 to 05:38 UTC.) · about: fray period doubling, weight and brakes (maps and chaos, driven oscillators)*

67. *R2W's appendix draft, read 15 · 4 and 5 October 2026 · The period-doubling cascade at drives 2e-6 apart, and an attempt to bisect each doubling. (Claude's heading: reading FRAY-FEIGEN and FRAY-FEIGEN-2; data: runs of 5 October, 05:53 to 05:57 UTC.) · about: fray Feigenbaum drumroll (maps and chaos, driven oscillators)*

68. *R2W's appendix draft, read 16 · 4 and 5 October 2026 · A pass back over what was measured and not yet looked at, at the sitting's close. (Claude's heading: reading the backward pass at the close of sitting B5; data: STATES, SCISSORS-MASS-4 and FRAY, read again for what was measured and not looked at; docs/TREASURE_CHEST_B5_2026-10-05.) · about: B5 backward pass, heavy rider (driven oscillators)*

The Loop Atlas book, the fresh session, 4 to 6 October

From the session's lessons for the guide, 6 October, which copy the book's register reads 21 to 37 by script.

69. *The book's register, read 21 · runs: KNEE_RECOUNT_C 20261004_192352 (the fresh session's first run) · about: knee recount counters and tumbles (driven oscillators, measurement)*

70. *The book's register, read 22 · runs: KNEE_RECOUNT_D 20261004_195102 · about: knee recount needle spins (driven oscillators, measurement)*

71. *The book's register, read 23 · runs: KNEE_RECOUNT_E 20261005_003928 · about: spinning needle short memory (driven oscillators)*

72. *The book's register, read 24 · runs: CLIFF_EDGE 20261004_062732, part B, read again through Bear's fleet of engines (5 October) · about: cliff edge noise and meters (driven oscillators, control and circuits, measurement)*

73. *The book's register, read 25 · runs: MAINTENANCE_METER 20261005_012150; MAINTENANCE_METER_B 20261005_012355 · about: maintenance meter wear gauge (control and circuits, measurement)*

74. *The book's register, read 26 · runs: CLIFF_EDGE 20261004_062732; BALANCE_BOARD_B 20261004_162140; AVALANCHE 20261004_064914; BANKED_TRACK 20261004_154940 (read together for Claude's question, 5 October) · about: three kinds of cliff (maps and chaos, driven oscillators, control and circuits)*

75. *The book's register, read 27 · runs: AVALANCHE 20261004_064914, part A, read with Bear's seam picture (5 October) · about: cliffs as creases in knobs (maps and chaos, driven oscillators)*

76. *The book's register, read 28 · runs: CLIFF_EDGE 20261004_062732 and CLIFF_EDGE_B 20261004_062930 read with the Field Guide's row 38, written into the letter to R2W (5 October); a guess for the Spoke to break, not a result · about: gearbox landing edges (driven oscillators, maps and chaos)*

77. *The book's register, read 29 · runs: SEAM_MAP 20261005_021349; SEAM_MAP_B 20261005_023746 (Claude's question, "can a loop feel its own cliff coming?", 5 October) · about: seam map edge warnings (maps and chaos, driven oscillators, control and circuits)*

78. *The book's register, read 30 · runs: R2W's FLOORS 20261005_015812, CLIFF-HUM 20261005_022003 and CLIFF-HUM-2 20261005_022219, as its letter of 5 October reports them (second hand; the runs themselves were not read here) · about: shaken rider window, cliff hum (driven oscillators)*

79. *The book's register, read 31 · runs: KICK_RULE_CHECK 20261005_031525, read beside the jellyfish paper's five phase-test run files (Zenodo record 10.5281/zenodo.22908910) · about: jellyfish bell kick rule (biology, waves)*

80. *The book's register, read 32 · runs: the jellyfish paper's ROPE-1 run files of 18 September (rope1_ringB_free_passive and rope1_ringA_tether_passive, 2 mm; Zenodo record 10.5281/zenodo.22908910), read with Bear's chaser of 5 October · about: ROPE-1 whirl through bell (biology, waves)*

81. *The book's register, read 33 · runs: FRAY_AND_SCISSORS 20261005_050632; FRAY_AND_SCISSORS_B 20261005_051313 (R2W's two questions, 5 October) · about: fray and scissors grip loss (driven oscillators, maps and chaos)*

82. *The book's register, read 34 · runs: triple_cat_holonomy 20260930_161702, triple_fourth_body 20260930_173710, triple_recurrence_plot and triple_poincare_strobe 20261005_023346 and 20261005_034156, as the triple-star side room's letter of 5 October reports them (second hand; the run files were not read here). The run files were read here later the same day (LOOP_ATLAS_TRIPLE_CARD_CHECK 20261005_055204); the read stands, and read 35 is what the reading added · about: triple star shape loop, clocks (three body, kinematics, quadrature)*

83. *The book's register, read 35 · runs: LOOP_ATLAS_TRIPLE_CARD_CHECK 20261005_055204, reading triple_cat_holonomy 20260930_161702 and triple_fourth_body 20260930_173710 in the side room's own files · about: triple card check ledger (three body, measurement, records and tools)*

84. *The book's register, read 36 · runs: R2W's revised postscript of 5 October (its FRAY run, reported second hand) beside FRAY_AND_SCISSORS_B 20261005_051313 and NEEDLE_GEARS_WHY's equation · about: fray two roads to chaos (driven oscillators, maps and chaos)*

85. *The book's register, read 37 · runs: the jellyfish paper's ROPE-1 leash run of 18 September and its launch-and-lean runs, as read in KICK_RULE_CHECK 20261005_031525 (Zenodo record 10.5281/zenodo.22908910), read with Bear's answer on the bell array · about: jellyfish bell array raft (biology, waves)*

86. Under the label "A read of the session:". *Given with the session's lessons for the guide, 6 October 2026 · from: TRIPLE_CARD_CHECK 20261005_055204 (the cat's law, 0.0014 in the side room's ledger and 0.00003 read finely); the leap-through draft test, next since 5 October and not yet run; the ferry the session arrived in (LOOP_ATLAS_FERRY_A_2026-10-04_EVENING) and the one it left in (LOOP_ATLAS_FERRY_A_2026-10-05, unpacked and checked to differs 0) · about: Loop Atlas session ferries (records and tools)*

R2W, 6 October

From its collection of sitting B's reads, 6 October, shipped as a PDF only; this read was taken from the PDF's text layer.

87. Under the label "No-physics read:". *Read 26 of that collection · 6 October 2026 · a read of reads 1 to 25 above · about: R2W reads 1 to 25 (records and tools)*

R2W, sitting B6, 5 and 6 October

From R2W's appendix of reads for the guide, 6 October, which copies them byte for byte from its own collection and carries them without the labels they followed; its reads 1 to 16 are reads 46 to 49 and 57 to 68 here, and its reads 34, 36, 39 and 44 are Bear's, in his section below.

88. *R2W's collection, read 17 · sitting B6, 5 and 6 October 2026 · Claude, reading the relay train · data: run_LOOP_ATLAS_RELAY_FM_trainslip_20261005_062520, _trainslip2_20261005_063226, _trainthrow_20261005_063754 and _trainhear_20261005_062520 · about: relay train slips (driven oscillators, control and circuits)*

89. *R2W's collection, read 18 · sitting B6, 5 and 6 October 2026 · Claude, reading ROOM-TRAIN and the room rule's figure · data: run_LOOP_ATLAS_RELAY_FM_roomtrain_20261005_142725 and the runs drawn in fig_relay_room_rule · about: relay room rule (driven oscillators, measurement)*

90. *R2W's collection, read 19 · sitting B6, 5 and 6 October 2026 · Claude, reading MCCUMBER-RIG beside SCISSORS-MASS-4 · data: run_LOOP_ATLAS_RELAY_FM_mccumberrig_20261005_144846 and relay/MCCUMBER_AFTER_DATA_2026-10-05_output.txt · about: McCumber rig rider weight (driven oscillators)*

91. *R2W's collection, read 20 · sitting B6, 5 and 6 October 2026 · Claude, reading MCCUMBER-RIDER · data: run_LOOP_ATLAS_RELAY_FM_mccumberrider_20261005_150419 · about: McCumber rider half weight (driven oscillators)*

92. *R2W's collection, read 21 · sitting B6, 5 and 6 October 2026 · Claude, reading the paper Bear found · data: Shukrinov, Rahmonov and Kulikov, JETP Letters 96, 588, 2012, beside our own sweeps · about: Shukrinov paper top and bell (driven oscillators, materials)*

93. *R2W's collection, read 22 · sitting B6, 5 and 6 October 2026 · Claude, reading HELD-SCISSORS-2 and the re-read of SCISSORS-MASS-4's saved lags, with the room rule read gain by gain · data: run_LOOP_ATLAS_RELAY_FM_heldscissors2_20261005_155925; relay/SCISSORS_BRANCH_AFTER_DATA_2026-10-05_output.txt and relay/ROOM_RULE_BY_GAIN_AFTER_DATA_2026-10-05_output.txt · about: held scissors and room rule (driven oscillators, measurement)*

94. *R2W's collection, read 23 · sitting B6, 5 and 6 October 2026 · Claude, reading ROOM-BOTTOM, SCISSORS-MASS-5 and CEILING · data: run_LOOP_ATLAS_RELAY_FM_roombottom_20261005_164542, _scissorsmass5a_20261005_164543, _scissorsmass5b_20261005_164544, _scissorsmass5read_20261005_165915 and _ceiling_20261005_165413 · about: room bottom and scissors mass (driven oscillators, measurement)*

95. *R2W's collection, read 24 · sitting B6, 5 and 6 October 2026 · Claude, reading FLOOR-MIN, FRAY-HOLD and TWO-STATES · data: run_LOOP_ATLAS_RELAY_FM_floormin_20261005_200127, _frayhold_20261005_200128 and _twostates_20261005_201342, with relay/FRAY_HOLD_INSTRUMENT_CHECK_2026-10-05_output.txt · about: floor min and two states (driven oscillators, maps and chaos)*

96. *R2W's collection, read 25 · sitting B6, 5 and 6 October 2026 · Claude, reading WINDOW-E2-06 beside the Loop Atlas's POCKET-HOLD · data: run_LOOP_ATLAS_RELAY_FM_windowe206_20261006_005300; the Loop Atlas's run_LOOP_ATLAS_POCKET_HOLD_20261005_195333 · about: window grip and pocket hold (driven oscillators)*

97. *R2W's collection, read 26 · sitting B6, 5 and 6 October 2026 · Claude, reading the Loop Atlas's fourth letter and its rulers beside our relay · data: their run_LOOP_ATLAS_ROOM_NEEDLE_RULERS_20261005_170136; our letter B6 · about: relay and needle counterparts (driven oscillators, control and circuits)*

98. *R2W's collection, read 27 · sitting B6, 5 and 6 October 2026 · Claude, reading SCISSORS-E2's ramps and ONE-GEAR · data: run_LOOP_ATLAS_RELAY_FM_scissorse2lo_20261006_030208, _scissorse2upa_20261006_030209, _scissorse2upb_20261006_030210; relay/ONE_GEAR_2026-10-06_output.txt and SCISSORS_E2_GEARS_AFTER_DATA_2026-10-06_output.txt · about: scissors E2 gear memory (driven oscillators)*

99. *R2W's collection, read 28 · sitting B6, 5 and 6 October 2026 · Claude, reading GEAR-OVERLAP and GEAR-HELD · data: run_LOOP_ATLAS_RELAY_FM_gearoverlapa_20261006_040927, _gearoverlapb_20261006_040927, _gearhelda_20261006_042026, _gearheldb_20261006_042026 and _gearheldread_20261006_042234; relay/GEAR_OVERLAP_AFTER_DATA_2026-10-06_output.txt · about: gear overlap with weight (driven oscillators)*

100. *R2W's collection, read 29 · sitting B6, 5 and 6 October 2026 · Claude, reading GEAR-ONSET and GEAR-ONSET-2, beside Bear's ferry drawing · data: run_LOOP_ATLAS_RELAY_FM_gearonset_20261006_052038, _gearonset2a_20261006_052437, _gearonset2b_20261006_052437 and _gearonset2read_20261006_052719; relay/GEAR_ONSET2_AFTER_DATA_2026-10-06_output.txt; bear/BEAR_DRAWING_FERRY_RING_ROPE_2026-10-06.jpg · about: gear onset second cog (driven oscillators)*

101. *R2W's collection, read 30 · sitting B6, 5 and 6 October 2026 · Claude, reading the Loop Atlas's letter A8 beside our gear runs and Bear's ferry · data: their run_LOOP_ATLAS_SWAP_STATE_20261006_052545, _SWAP_STATE_2_20261006_052914, _FERRY_ROPE_20261006_045826 and _POCKET_GEARS_20261006_052012; our GEAR-OVERLAP and GEAR-ONSET-2 · about: swap state ferry ropes (driven oscillators)*

102. *R2W's collection, read 31 · sitting B6, 5 and 6 October 2026 · Claude, reading GEAR-BASIN · data: run_LOOP_ATLAS_RELAY_FM_gearbasin_20261006_060759; docs/figs/fig_gear_basin.png; relay/GEAR_BASIN_BANDS_AFTER_DATA_2026-10-06_output.txt · about: gear basin stripes (maps and chaos, driven oscillators)*

103. *R2W's collection, read 32 · sitting B6, 5 and 6 October 2026 · Claude, reading SWAP-RIDER and the solver · data: run_LOOP_ATLAS_RELAY_FM_swapridera_20261006_062739, _swapriderb_20261006_062739, _swapriderread_20261006_064025; run_LOOP_ATLAS_GEAR_SOLVER_20261006_062940 and _period2_20261006_063336; relay/GEAR2_STROBE_CHECK_2026-10-06_output.txt · about: swap rider two-way tiny cog (driven oscillators, maps and chaos)*

104. *R2W's collection, read 33 · sitting B6, 5 and 6 October 2026 · Claude, reading SEAM-ZOOM, as a story · data: run_LOOP_ATLAS_RELAY_FM_seamzoom_20261006_070434; docs/figs/fig_seam_zoom.png · about: seam zoom basin border (maps and chaos, driven oscillators)*

105. *R2W's collection, read 35 · sitting B6, 5 and 6 October 2026 · Claude, reading LANDING-PADS, as a story · data: run_LOOP_ATLAS_RELAY_FM_landingpads_20261006_074121; docs/figs/fig_landing_pads.png · about: landing pads windows (maps and chaos, driven oscillators)*

106. *R2W's collection, read 37 · sitting B6, 5 and 6 October 2026 · Claude, reading STACKS, as a story · data: run_LOOP_ATLAS_RELAY_FM_stacks_20261006_075654 · about: stacks window basins (maps and chaos, driven oscillators, number theory)*

107. *R2W's collection, read 38 · sitting B6, 5 and 6 October 2026 · Claude, reading SINE-CIRCLE-MAP beside LANDING-PADS · data: relay/run_SINE_CIRCLE_MAP_CHECK_20261006_082403.txt · about: sine circle map staircases (maps and chaos, driven oscillators)*

108. *R2W's collection, read 40 · sitting B6, 5 and 6 October 2026 · Claude, reading LOOP-WALK, as a story · data: relay/run_LOOP_ATLAS_LOOP_WALK_20261006_145333.txt · about: loop walk round the overlap (driven oscillators, topology)*

109. *R2W's collection, read 41 · sitting B6, 5 and 6 October 2026 · Claude, reading LOOP-WALK-2, as a story · data: relay/run_LOOP_ATLAS_LOOP_WALK_2_READ_20261006_151644.txt and run_LOOP_ATLAS_LOOP_WALK_2B_READ_20261006_152101.txt · about: loop walk slip counters (driven oscillators, topology)*

110. *R2W's collection, read 42 · sitting B6, 5 and 6 October 2026 · Claude, reading SWITCHBACK beside the trucks, as a story · data: relay/run_LOOP_ATLAS_SWITCHBACK_20261006_154751.txt; relay/RELAY_FM_GEAROVERLAPA.csv and _B.csv · about: switchback ramp legs (driven oscillators, control and circuits)*

111. *R2W's collection, read 43 · sitting B6, 5 and 6 October 2026 · Claude, reading SWITCHBACK-2, as a story · data: relay/run_LOOP_ATLAS_SWITCHBACK_2_20261006_161026.txt · about: switchback heating brakes (driven oscillators, control and circuits)*

The Loop Atlas book, the session of 5 and 6 October

From the book's register as it travelled with the session's lessons of 6 October; its reads 1 to 37 are reads 26 to 40, 50 to 55 and 69 to 85 here, and its first read of the project is read 32.

112. *The book's register, read 38 · runs: R2W's SCISSORS-MASS-4 (_scissorsmass4_20261005_042413), FRAY (_fray_20261005_053209), STATES (_states_20261005_041457) and TUNE (_tune_20261005_053211), as the Spoke's pre-registration file (relay/PREREG_RELAY_STATION_2026-10-03.txt, lines 1,240 to 1,546, read in full here from its crate B5 r15) and its treasure chest report them; the run files themselves were not read here. Given at the opening of this fresh session of 5 October, first numbered as read 17 of the Spoke's own collection and moved here the same day, so that the Spoke's next sitting numbers its reads cleanly. · about: rider weight gift and stagger (driven oscillators)*

113. *The book's register, read 39 · runs: HIDDEN_SCISSORS 20261005_064106 (the hidden-scissors walk-down, 5 October) · about: hidden scissors walk-down (driven oscillators, maps and chaos)*

114. *The book's register, read 40 · runs: HIDDEN_SCISSORS_SLIVER 20261005_145137 (the sliver held, 5 October) · about: hidden scissors sliver tumbling (maps and chaos, driven oscillators)*

115. *The book's register, read 41 · runs: DIVER 20261005_160158; DIVER_B 20261005_160338 (Bear's divers, 5 October); PULSED_SCISSORS 20261005_155747 · about: divers and pulsed needle (kinematics, driven oscillators)*

116. *The book's register, read 42 · runs: PULSED_SCISSORS_LONG 20261005_170134 (with its reader 20261005_170755); GAP_DIAL 20261005_170519 · about: pulsed scissors gap dial (driven oscillators, maps and chaos)*

117. *The book's register, read 43 · runs: GAP_DIAL_2 20261005_172939 · about: gap dial tapping ceiling (driven oscillators, maps and chaos)*

118. *The book's register, read 44 · runs: GAP_DIAL_3 20261005_194344 · about: gap dial long wait (driven oscillators, measurement)*

119. *The book's register, read 45 · runs: POCKET_HOLD 20261005_195333 · about: pocket hold two homes (driven oscillators, maps and chaos, measurement)*

120. *The book's register, read 46 · runs: R2W's FLOOR-MIN (_floormin_20261005_200127) and TWO-STATES (_twostates_20261005_201342), read in full here in the packet of its note 05e, with its instrument check of FRAY-HOLD · about: floor min and two states (driven oscillators, maps and chaos)*

121. *The book's register, read 47 · runs: HITCH_KICK 20261006_011635 · about: hitch kick arm windmill (kinematics)*

122. *The book's register, read 48 · runs: ROOM_NEEDLE 20261006_013956; ROOM_NEEDLE_2 20261006_014758 and its reader 20261006_015510 · about: room needle smoothing rule (driven oscillators, control and circuits)*

123. *The book's register, read 49 · runs: SWAP_RIDER_BAND 20261006_041001; CLIMB_BACK_LINE_CHECK 20261006_041633 · about: swap rider band in wind (driven oscillators, measurement)*

124. *The book's register, read 50 · runs: FERRY_ROPE 20261006_045826; NEVER_BLIND 20261006_045833; NEVER_BLIND_2 20261006_050123 · about: ferry rope and never blind (driven oscillators, control and circuits)*

125. *The book's register, read 51 · runs: POCKET_GEARS 20261006_052012; SWAP_STATE 20261006_052545; SWAP_STATE_2 20261006_052914 · about: pocket gears and swap state (driven oscillators)*

126. *The book's register, read 52 · runs: SIZED_TAP 20261006_060952; SIZED_TAP_2 20261006_061606; SIZED_TAP_3 20261006_061940 · about: sized tap falling board (control and circuits, kinematics)*

127. *The book's register, read 53 · runs: a read across the three teams' clocks, given when Bear asked whether everything was starting to rhyme: the Loop Atlas needle's locks and pockets (POCKET_GEARS 20261006_052012), R2W's gears (its letter B7), and the triple-star side room's two clocks (triple_recurrence_plot 20261005_023346, quoted in the book's Trunk 4) · about: three teams' clocks rhyme (three body, driven oscillators, number theory)*

128. *The book's register, read 54 · runs: CLIMB_HOLD 20261006_065510 · about: climb hold lock capture (driven oscillators, maps and chaos)*

129. *The book's register, read 55 · runs: RUN_MAP 20261006_071157; SLIP_BEND 20261006_071248 · about: run map and slip bend (driven oscillators, maps and chaos)*

130. *The book's register, read 56 · runs: R2W's LANDING-PADS (run_LOOP_ATLAS_RELAY_FM_landingpads_20261006_074121, in its letter B9) and the 3-Body Team's loan test (triple_loan_test 20261006_071642, in its note T2), read the night of 6 October · about: landing pads, triple loan test (maps and chaos, three body, driven oscillators)*

131. Under the label "A read of the project:". *The book's register, project read 2 · given when Bear asked on 5 October whether the work had drifted from the Loop Atlas · about: Loop Atlas drift question (records and tools)*

132. Under the label "A read of the project:". *The book's register, project read 3 · given on 6 October when Bear asked whether he was adding requirements faster than we were making progress · about: requirements versus progress (records and tools)*

133. Under the label "A read of the project:". *The book's register, project read 4 · given the same day when Bear answered that the follow-ons, the letters and the surprise tests are how two people do the work of a science team · about: curiosity river, book mill (records and tools)*

134. Under the label "A read of the project:". *The book's register, project read 5 · given on the morning of 6 October when Bear shared his dreams · about: Bear's dreams of holding (kinematics, topology)*

135. Under the label "A read of the project:". *The book's register, project read 6 · given the same morning when Bear explained the split walk · about: split walk round a hill (kinematics, topology)*

136. Under the label "A read of the project:". *The book's register, project read 7 · given the same morning when Bear answered from his infantry years · about: infantry pack, paper mountain (topology, kinematics)*

137. Under the label "A read of the project:". *The book's register, project read 8 · given the same morning · about: column climbing switchbacks (kinematics, driven oscillators)*

138. Under the label "A read of the project:". *The book's register, project read 9 · given the same morning · about: marching column stumble (control and circuits, waves)*

139. Under the label "A read of the project:". *The book's register, project read 10 · given the same morning · about: column halt echo (waves, control and circuits)*

140. Under the label "A read of the project:". *The book's register, project read 11 · given the same morning when Bear explained halts and pauses · about: halts and pauses in column (control and circuits, waves)*

141. Under the label "A read of the project:". *The book's register, project read 12 · given the same morning when Bear asked for the session's lessons for the Field Guide · about: session lessons for the guide (records and tools)*

Chapter 3 — A third body joins, 5 to 7 October

R2W, sitting B7, 6 and 7 October

From R2W's appendix of reads for the guide, 7 October, which copies them from its own collection and carries them without the labels they followed; its reads 1 to 44 are already here.

142. *R2W's collection, read 45 · sitting B7, 6 and 7 October 2026 · Claude, reading STACKS-2, as a story · data: run_LOOP_ATLAS_RELAY_FM_stacks2a_20261006_165049, _stacks2b_20261006_165049 and _stacks2read_20261006_165710; relay/STACKS2_AFTER_DATA_2026-10-06_output.txt · about: stacks-2 pools between fives, threes (maps and chaos, driven oscillators, number theory)*

143. *R2W's collection, read 46 · sitting B7, 6 and 7 October 2026 · Claude, reading STACKS-3 and LONG-HOLD, as a story · data: run_LOOP_ATLAS_RELAY_FM_stacks23a_20261006_171203, _stacks23b_20261006_171203, _stacks3read_20261006_171650 and _longhold_20261006_171842; relay/LONGHOLD_AFTER_DATA_2026-10-06_output.txt · about: stacks-3 two dances, long hold (maps and chaos, driven oscillators)*

144. *R2W's collection, read 47 · sitting B7, 6 and 7 October 2026 · Claude, reading SONG-1 and COASTLINE, as a story · data: run_LOOP_ATLAS_RELAY_FM_song1_20261006_173919 and _coastline_20261006_173920 · about: song-1 two doors, coastline (maps and chaos, driven oscillators)*

145. *R2W's collection, read 48 · sitting B7, 6 and 7 October 2026 · Claude, reading SONG-2 and SONG-3, as a story · data: run_LOOP_ATLAS_RELAY_FM_song2_20261006_175801 and _song3_20261006_175801; relay/SONG3_AFTER_DATA_2026-10-06_output.txt · about: song rocking two dances (driven oscillators, maps and chaos)*

146. *R2W's collection, read 49 · sitting B7, 6 and 7 October 2026 · Claude, reading SONG-4 and RECORD with Bear's record player and the dance floor over the pool · data: run_LOOP_ATLAS_RELAY_FM_song4_20261006_182920 and _record_20261006_182920; relay/RECORD_SONG4_AFTER_DATA_2026-10-06_output.txt · about: song-4 and record needles (driven oscillators, maps and chaos)*

147. *R2W's collection, read 50 · sitting B7, 6 and 7 October 2026 · Claude, reading RECORD-2 and TUMBLE-1, as a story · data: run_LOOP_ATLAS_RELAY_FM_record2_20261006_184247 and _tumble1_20261006_184247 · about: record-2 grooves, tumble-1 (driven oscillators, maps and chaos)*

148. *R2W's collection, read 51 · sitting B7, 6 and 7 October 2026 · Claude, reading TONGUE-MAP and NAN-CHECK with Bear's record, as a story · data: run_LOOP_ATLAS_RELAY_FM_tonguemapa_20261006_190213, _tonguemapb_20261006_190213 and _tonguemapread_20261006_191024; relay/TONGUEMAP_NAN_CHECK_2026-10-06_output.txt; docs/figs/fig_tongue_spiral.png · about: tongue map across needles (driven oscillators, maps and chaos, measurement)*

149. *R2W's collection, read 52 · sitting B7, 6 and 7 October 2026 · Claude, reading the record lifted into the fraction tree, with the new ruler, as a story · data: relay/TONGUEMAP_TREE_AFTER_DATA_2026-10-06_output.txt; engines_B7/RECORD_SURVEY_FIDELITY_READ_2026-10-06_output.txt and RULER_CHECK_2026-10-06_output.txt; docs/figs/fig_tongue_tree.png · about: tongue map fraction tree (driven oscillators, number theory, measurement)*

150. *R2W's collection, read 53 · sitting B7, 6 and 7 October 2026 · Claude, reading FLIP-1 with Bear's twisted light, as a story · data: engines_B7/FLIP_1_2026-10-06_output.txt and FLIP_1_AFTER_DATA_2026-10-06_output.txt; docs/figs/fig_flip_ladder.png · about: flip-1 period doubling ladder (maps and chaos, driven oscillators)*

151. *R2W's collection, read 54 · sitting B7, 6 and 7 October 2026 · Claude, reading FLIP-2 to FLIP-4 as a story · data: engines_B7/FLIP_4_A_2026-10-06_output.txt and FLIP_4_B_2026-10-06_output.txt; docs/figs/fig_feigenbaum_ladder.png · about: Feigenbaum ladder rungs (maps and chaos, driven oscillators, measurement)*

152. *R2W's collection, read 55 · sitting B7, 6 and 7 October 2026 · Claude, reading Bear's survey of every record, as a story · data: engines_B7/RECORD_SURVEY_READ_2026-10-06_output.txt, RECORD_SURVEY_AFTER_DATA_2026-10-06_output.txt and _2_; docs/figs/fig_record_survey.png · about: record survey of needles (driven oscillators, maps and chaos)*

153. *R2W's collection, read 56 · sitting B7, 6 and 7 October 2026 · Claude, reading BASIN-1 with Bear's four-panel doors, as a story · data: engines_B7/BASIN_1_1.605_2026-10-06_output.txt and BASIN_1_1.625_2026-10-06_output.txt; docs/figs/fig_basin_doors.png · about: basin doors stripes (maps and chaos, driven oscillators)*

154. *R2W's collection, read 57 · sitting B7, 6 and 7 October 2026 · Claude, reading ROOM-1 as a story · data: engines_B7/ROOM_1_READ_2026-10-06_output.txt; docs/figs/fig_room_closing.png · about: room closing between floors (driven oscillators, maps and chaos)*

155. *R2W's collection, read 58 · sitting B7, 6 and 7 October 2026 · Claude, reading HOLD-1 as a story · data: engines_B7/HOLD_1_READ_2026-10-06_output.txt and HOLD_1_AFTER_DATA_2026-10-06_output.txt; docs/figs/fig_half_ribbon.png · about: half ribbon hold (maps and chaos, driven oscillators)*

156. *R2W's collection, read 59 · sitting B7, 6 and 7 October 2026 · Claude, reading RIBBON-1 as a story · data: engines_B7/RIBBON_1_READ_2026-10-07_output.txt and RIBBON_1_AFTER_DATA_2026-10-07_output.txt; docs/figs/fig_half_ribbon.png · about: ribbon-1 room ladder (maps and chaos, driven oscillators)*

157. *R2W's collection, read 60 · sitting B7, 6 and 7 October 2026 · Claude, reading SHAPIRO-1 as a story · data: engines_B7/SHAPIRO_1_READ_2026-10-07_output.txt · about: Shapiro-1 second wave strength (driven oscillators)*

158. *R2W's collection, read 61 · sitting B7, 6 and 7 October 2026 · Claude, reading KNOBS-1 and KNOBS-1B as a story · data: engines_B7/KNOBS_1_READ_2026-10-07_output.txt and KNOBS_1B_READ_2026-10-07_output.txt · about: knobs-1 scaling check (driven oscillators, measurement)*

159. *R2W's collection, read 62 · sitting B7, 6 and 7 October 2026 · Claude, reading KNOBS-2 as a story · data: engines_B7/KNOBS_2_READ_2026-10-07_output.txt · about: knobs-2 golden clock offset (driven oscillators, records and tools, number theory)*

The Loop Atlas book, the fresh session of Bear's evening, 6 and 7 October

From the book's register as it travelled with that session's notes for the guide of 6 and 7 October; its reads 1 to 56 and its eleven reads of the project are already here.

160. *The book's register, read 57 · runs: RUN_STROBE 20261006_170149 and its reader RUN_STROBE_READER 20261006_170950 (the fresh session of Bear's evening, 6 October) · about: run strobe glide and pads (driven oscillators, number theory)*

161. *The book's register, read 58 · runs: RUN_STROBE_2 20261006_174220 and its reader RUN_STROBE_2_READER 20261006_174743 (Bear's evening, 6 October, second exchange) · about: run strobe-2 hallway (driven oscillators, measurement)*

162. *The book's register, read 59 · runs: STROBE_NEWTON 20261006_184350 (Bear's evening, 6 October, third exchange) · about: strobe Newton groove widths (driven oscillators, number theory)*

163. *The book's register, read 60 · runs: GROOVES 20261006_193519 and its reader GROOVES_READER 20261006_194112 (Bear's evening, 6 October, fourth exchange) · about: grooves and speed limit (driven oscillators, number theory)*

164. *The book's register, read 61 · runs: CAT_3D 20261006_201322 (Bear's evening, 6 October, fifth exchange) · about: falling cat in 3D (kinematics)*

165. *The book's register, read 62 · runs: CAT_3D_2 20261006_202140 and its reader CAT_3D_2_READER 20261006_202842 (Bear's evening, 6 October, sixth exchange) · about: falling cat 3-D reorientation (kinematics)*

166. *The book's register, read 63 · runs: SPOKES 20261006_204038 and GROOVES_3 20261006_204201 (Bear's evening, 6 October, seventh exchange) · about: needle spokes and grooves (maps and chaos, driven oscillators)*

167. *The book's register, read 64 · runs: CASCADE 20261006_211853 (Bear's evening, 6 October, eighth exchange) · about: needle Feigenbaum cascade (maps and chaos)*

168. *The book's register, read 65 · runs: CASCADE_2 20261006_213511 (Bear's evening, 6 October, ninth exchange) · about: needle cascade scaling ratios (maps and chaos)*

169. *The book's register, read 66 · runs: GROOVES_6 20261006_215909 (Bear's evening, 6 October, tenth exchange) · about: thin high grooves, thirty-digit arithmetic (driven oscillators, measurement)*

170. *The book's register, read 67 · runs: GROOVES_7 20261006_222345, CASCADE_3 20261006_220635 and CASCADE_4 20261006_221804 (Bear's evening, 6 October, eleventh exchange) · about: groove width scaling law (driven oscillators, maps and chaos)*

171. *The book's register, read 68 · runs: CASCADE_5 20261006_225126 and GROOVES_8 20261006_225037 (Bear's evening, 6 October, twelfth exchange) · about: cascade end and groove rules (maps and chaos, driven oscillators)*

172. *The book's register, read 69 · runs: GROOVES_9, its first version and the reader written after it, both retired (Bear's night of 6 to 7 October, thirteenth exchange) · about: groove width read as loop (driven oscillators, measurement)*

173. *The book's register, read 70 · runs: SWING_LISTEN 20261007_013201, with SWING_TWENTY 20261004_075715 and SWING_TWENTY_B 20261004_075902 (Bear's night of 6 to 7 October, fourteenth exchange) · about: swing pumping by listening (driven oscillators, control and circuits)*

174. *The book's register, read 71 · runs: SWING_LISTEN_2 20261007_024253 (Bear's night of 6 to 7 October, fifteenth exchange) · about: swing pump reversed as brake (driven oscillators)*

The triple-star thread, 5 to 7 October

From the thread's own register, TRIPLE_NO_PHYSICS_READS, which travelled with its lessons for the guide of 7 October; each read describes one run's figures or tables, and its headings carry the workspace's dates, which run seven hours ahead of Bear's.

175. *The triple-star thread's register, its heading: 5 October 2026 — the four recurrence plots (run 20261005_023346) · about: triple star recurrence plots (three body, reading pictures)*

176. *The triple-star thread's register, its heading: 5 October 2026 — the strobe, one look per inner orbit (run 20261005_034156) · about: triple star inner-orbit strobe (three body, reading pictures)*

177. *The triple-star thread's register, its heading: 5 October 2026 — the pentagon bench (run 20261005_043118) · about: pentagon bench of sticks (measurement, topology)*

178. *The triple-star thread's register, its heading: 5 October 2026 — Mercator's chart (run 20261005_043531) · about: recurrence plots, Mercator's chart (three body, reading pictures)*

179. *The triple-star thread's register, its heading: 5 October 2026 — the generals at the junctions (run 20261005_053735) · about: generals at the junctions (measurement, topology)*

180. *The triple-star thread's register, its heading: 5 October 2026 — the cliff test (run 20261005_061010) · about: triple star escape cliff test (three body)*

181. *The triple-star thread's register, its heading: 5 October 2026 — the fresh ensemble (run 20261005_064647) · about: fresh triple-star ensemble (three body, quadrature, measurement)*

182. *The triple-star thread's register, its heading: 5 October 2026 — the rungs (run 20261005_145442) · about: triple star resonance rungs (three body)*

183. *The triple-star thread's register, its heading: 5 October 2026 — the gentle triples and the binary's staircase (run 20261005_155400) · about: gentle triples, binary's staircase (three body)*

184. *The triple-star thread's register, its heading: 5 October 2026 — the boundary-layer scan (run 20261005_170052) · about: triple star boundary-layer scan (three body)*

185. *The triple-star thread's register, its heading: 5 October 2026 — the rungs at 7 and 8, again (run 20261005_193416) · about: rungs at 7 and 8 retested (three body, measurement)*

186. *The triple-star thread's register, its heading: 5 October 2026 — the look after the miss (run 20261005_195355) · about: look after the rungs miss (three body)*

187. *The triple-star thread's register, its heading: 6 October 2026 — the Field Guide's checks (run 20261006_004533) · about: Field Guide checks on rungs (three body, measurement)*

188. *The triple-star thread's register, its heading: 6 October 2026 — the floor moved (run 20261006_010912) · about: triple star floor moved (three body)*

189. *The triple-star thread's register, its heading: 6 October 2026 — two dials (run 20261006_042759, and the look at 20261006_043402) · about: triple star two dials (three body, quadrature)*

190. *The triple-star thread's register, its heading: 6 October 2026 — the turning ellipse (run 20261006_060752, and the look at 20261006_061446) · about: binary's turning ellipse (three body)*

191. *The triple-star thread's register, its heading: 6 October 2026 — six masses on one line (run 20261006_063043) · about: six masses on one line (three body)*

192. *The triple-star thread's register, its heading: 6 October 2026 — the loan that was not taken (run 20261006_071642) · about: loan not taken, near-matches (three body)*

193. *The triple-star thread's register, its heading: 6 October 2026 — the sag (run 20261006_081615) · about: triple star sag (three body, measurement)*

194. *The triple-star thread's register, its heading: 6 October 2026 — the bow tie (run 20261006_171421) · about: bow tie, either circling direction (three body)*

195. *The triple-star thread's register, its heading: 6 October 2026 — the dimple and its rim (run 20261006_190322, and the look at 20261006_195324) · about: resonance dimple and its rim (three body, maps and chaos)*

196. *The triple-star thread's register, its heading: 6 October 2026 — the clean rim and the ragged rim (run 20261006_201311) · about: clean rim and ragged rim (three body, maps and chaos)*

197. *The triple-star thread's register, its heading: 6 October 2026 — the beat and the level bubble (run 20261006_204230) · about: beat and the level bubble (three body)*

198. *The triple-star thread's register, its heading: 6 October 2026 — the heavy neighbour (run 20261006_213610) · about: heavy neighbour on the pair (three body)*

199. *The triple-star thread's register, its heading: 6 October 2026 — counted, not guessed (run 20261006_221108) · about: heavy neighbour catches, counted (three body)*

200. *The triple-star thread's register, its heading: 6 October 2026 — two combs (the lines look at 20261006_225255) · about: two combs of lines (three body, quadrature)*

201. *The triple-star thread's register, its heading: 7 October 2026 — who is thrown out (run 20261007_010954) · about: triple breakup, who is thrown (three body)*

202. *The triple-star thread's register, its heading: 7 October 2026 — every number has a home (run 20261007_013458) · about: every number has a home (records and tools, measurement)*

203. *The triple-star thread's register, its heading: 7 October 2026 — the record read back (the session's lessons for the Field Guide; the look at 20261007_021424) · about: session record read back (records and tools)*

Chapter 4 — The long middle, 7 to 10 October

The Loop Atlas book, the sessions of 7 to 10 October

From the book's register as it travelled with its notes for the guide of 10 October, all 119 reads and its thirteen reads of the project; its reads 1 to 71 and its first eleven reads of the project are already here.

204. *The book's register, read 72 · runs: FOLD_OR_FLIP 20261007_050721 (a fresh session, 7 October, first exchange; R2W's letter B10 checked) · about: fold or flip at doubling (maps and chaos, driven oscillators)*

205. *The book's register, read 73 · runs: FOLD_OR_FLIP_2 20261007_053443 (7 October, second exchange) · about: fold or flip searcher (maps and chaos)*

206. *The book's register, read 74 · runs: FOLD_OR_FLIP_3 20261007_054521 (7 October, third exchange) · about: cascade under a clock nudge (maps and chaos)*

207. *The book's register, read 75 · runs: FOLD_OR_FLIP_4 20261007_061947 (7 October, fourth exchange) · about: true doors, ruler runs out (maps and chaos, measurement)*

208. *The book's register, read 76 · runs: GROOVES_10 version 2, part C, 20261007_064118 (7 October, fifth exchange; the 8/5's first reading, its verdict still running) · about: 8/5 groove width remeasured (driven oscillators, measurement)*

209. *The book's register, read 77 · runs: GROOVES_10 version 2, parts B and C, 20261007_065508 (7 October, the fifth exchange's check-in) · about: groove width independent of ruler (driven oscillators, measurement)*

210. *The book's register, read 78 · runs: RUNG_ABOVE 20261007_071341 (7 October, sixth exchange; R2W's B11 checked) · about: rung above the two-step (maps and chaos)*

211. *The book's register, read 79 · runs: GROOVES_11 20261007_080706, GROOVES_12 20261007_083300 and 20261007_083648, GROOVES_13 20261007_090124 (7 October, eighth exchange, Bear asleep) · about: groove level crossings fooled (driven oscillators, measurement)*

212. *The book's register, read 80 · runs: the header line of R2W's RECORD_SURVEY 20261006_125455, as Bear pasted it (7 October, ninth exchange); its data not read here · about: R2W record survey header (records and tools, driven oscillators)*

213. *The book's register, read 81 · runs: the summary of Bear's laptop run of GROOVES_10, as he pasted it (7 October, tenth exchange); its run file not yet read here · about: GROOVES_10 laptop run summary (records and tools, driven oscillators)*

214. *The book's register, read 82 · runs: GROOVES_10 laptop run 20261007_002739, its run file read in full, beside the cloud's parts B and C (run 20261007_065508) (7 October, eleventh exchange) · about: GROOVES_10 laptop and cloud agree (records and tools, driven oscillators)*

215. *The book's register, read 83 · runs: the first two lines of R2W's run_LOOP_ATLAS_PAD_ATLAS_OVERNIGHT_20261006_234224, as Bear pasted them, beside GROOVES_10 laptop run 20261007_002739 and Bear's word that one of three overnight runs took 12 cores (7 October, twelfth exchange) · about: overnight runs sharing cores (records and tools)*

216. *The book's register, read 84 · runs: GROOVES_14 version 2, runs 20261007_170021 part A and 20261007_170022 part B, with GROOVES_15 20261007_164232 part B (7 October, thirteenth exchange) · about: groove width by passage test (driven oscillators, measurement)*

217. *The book's register, read 85 · runs: GROOVES_16, runs 20261007_182009 part A and 20261007_182010 part B (7 October, fourteenth exchange) · about: lap check goes blind (driven oscillators, measurement)*

218. *The book's register, read 86 · runs: CONE_TOMOGRAPHY, runs 20261007_235426, 20261007_235444 and 20261008_001220, with GROOVES_17, runs 20261007_234942 and 20261007_234943 (7 and 8 October, fifteenth exchange) · about: cone tomography by carried arrow (kinematics, topology)*

219. *The book's register, read 87 · runs: CONE_TOMOGRAPHY_2, runs 20261008_002939 part D and 20261008_002937 part E (8 October, sixteenth exchange) · about: cone tomography phantom fourth peak (kinematics, topology)*

220. *The book's register, read 88 · runs: CONE_TOMOGRAPHY_3, runs 20261008_011150 part N1, 20261008_011149 part N2 and 20261008_011157 part N3; CONE_TOMOGRAPHY_4, runs 20261008_012043 part K1 and 20261008_012042 part K2; CONE_TOMOGRAPHY_5, run 20261008_012723 part K3 (8 October, seventeenth exchange) · about: cone tomography arrow stumbles (kinematics, topology, measurement)*

221. *The book's register, read 89 · runs: Bear's drawing (shared 7 October, his evening); CONE_TOMOGRAPHY_2, run 20261008_002939 part D; CONE_TOMOGRAPHY_3, run 20261008_011150 part N1; CONE_TOMOGRAPHY_4, run 20261008_012043 part K1; the hypertextbook's universality page and the 1991 paper on how the doubling indices depend on the order of the maximum, as Bear carried them from R2W (8 October, eighteenth exchange) · about: Bear's drawing and cone tomography (kinematics, reading pictures, maps and chaos)*

222. *The book's register, read 90 · runs: DELTA_BY_TOP, run 20261008_034013 part ALL, with the needle's cascade ratios as Limb G gives them (8 October, nineteenth exchange) · about: doubling rate by hilltop flatness (maps and chaos)*

223. *The book's register, read 91 · runs: PEAK_RING, run 20261008_040518; PEAK_RING_2, run 20261008_040854; Bear's drawing of 2011 (8 October, twentieth exchange) · about: peak ring, turns counted (kinematics, topology)*

224. *The book's register, read 92 · runs: PENTA_SHARE, runs 20261008_060159 part S and 20261008_060200 part W (8 October, twenty-first exchange) · about: pentagram walked round a hill (topology, kinematics)*

225. *The book's register, read 93 · runs: DELTA_AREA, run 20261008_062647 (version 3; versions 1 and 2 retired); PENTA_SHARE_2, runs 20261008_062403 part S2 and 20261008_062420 part W3 (8 October, twenty-second exchange) · about: frictionless doubling rate, nested pentagrams (maps and chaos, topology)*

226. *The book's register, read 94 · runs: DELTA_BY_TOP_2, run 20261008_063744 parts E3 and E2, from CASCADE_4 and CASCADE_5's saved orbits (8 October, twenty-third exchange) · about: needle fork-width convergence (maps and chaos)*

227. *The book's register, read 95 · runs: the Jellyfish crate (JELLY1_ZENODO_UPLOAD_CRATE_20260922.zip) checked against its SHA256SUMS; jelly_fidelity_check legs 7 and 8, first on the frozen v2.4 copy, then on v2.5 (8 October, twenty-fourth exchange) · about: Jellyfish crate seals checked (records and tools, biology)*

228. *The book's register, read 96 · runs: GROW_LOOP, run 20261008_133646 part ALL; DRAFT_TEST, run 20261008_134707 part ALL with its four solver outputs; ROPE-1's runs of 18 September (8 October, twenty-fifth exchange) · about: growing loop, bell draft test (kinematics, biology)*

229. *The book's register, read 97 · runs: BENT_PAPER, run 20261008_164513 part ALL; BENT_PAPER_2, run 20261008_164721 part ALL (8 October, twenty-sixth exchange) · about: bent paper and cones (kinematics, topology)*

230. *The book's register, read 98 · runs: GROW_SLIP, run 20261008_195205 part ALL, with GROW_LOOP's run 20261008_133646 (8 October, twenty-seventh exchange) · about: turn count lost by slow glances (quadrature, topology)*

231. *The book's register, read 99 · runs: Bear's dream of the coin, pad to pad (told 8 October), read beside GROW_SLIP, run 20261008_195205, and GROW_LOOP, run 20261008_133646 (8 October, twenty-eighth exchange) · about: Bear's coin dream, pad to pad (quadrature, kinematics)*

232. *The book's register, read 100 · runs: RING_BUDGET, run 20261004_080107; the jellyfish paper's phase line and launch-and-lean runs, as its provenance table lists them; ROPE-1's runs of 18 September; DRAFT_TEST, run 20261008_134707 (8 October, twenty-ninth exchange, written with Limb F) · about: jellyfish bell and smoke rings (biology, waves)*

233. *The book's register, read 101 · runs: the Canopy's sources: toy runs 3, 4, 5, 5b, 5c and 5o and the first night's swing; the jellyfish paper; Limb F's runs (8 October, thirtieth exchange, written with the Canopy) · about: Canopy's ten loop habits (control and circuits, driven oscillators, biology)*

234. *The book's register, read 102 · runs: COUNTER_FLOW, run 20261009_002621 (9 October, thirty-first exchange) · about: counter-flow heat exchange (control and circuits)*

235. *The book's register, read 103 · runs: the misses carried forward from the data ledger's lines 15 to 60, rows 27 to 52 of the misses file (9 October, thirty-second exchange) · about: misses file read back (records and tools, measurement)*

236. *The book's register, read 104 · runs: the misses carried forward from the data ledger's lines 61 to 114, rows 53 to 78 of the misses file: GROOVES_4 to _9 and CAT_3D with CAT_3D_2 above all (9 October, thirty-third exchange) · about: misses: grooves and cat (records and tools, measurement, driven oscillators)*

237. *The book's register, read 105 · runs: the misses carried forward from the data ledger's lines 115 to 196, rows 79 to 104 of the misses file, with row 11 found settled by BIG_LOOP 20261003_175803 and row 10 carded in the ledger: GROOVES_16 and _17, KNEE_RECOUNT_D, BALANCE_BOARD, SIZED_TAP and MAINTENANCE_METER above all (9 October, thirty-fourth exchange) · about: misses: rulers, tallies, logbooks (records and tools, measurement)*

238. *The book's register, read 106 · runs: the session's close: the after-action review for the Field Guide, read from the transcript on disk (from 13:39 UTC, 8 October) and the session's own files, and ferry N (9 October, thirty-fifth exchange) · about: session close after-action review (records and tools)*

239. *The book's register, read 107 · runs: SOUP_JUMP, run 20261009_045400 parts A and B (9 October, a fresh session, first exchange) · about: corks riding a wave comb (waves)*

240. *The book's register, read 108 · runs: SOUP_JUMP_2, runs 20261009_051426 parts A and C, 20261009_051926 part B and 20261009_052404 part V (9 October, a fresh session, second exchange) · about: soup jump, longer road (waves)*

241. *The book's register, read 109 · runs: LATTICE_READER, runs 20261009_091530 and 20261009_091531 (the picture) and 20261009_092420; LATTICE_READER_2, run 20261009_093040 (9 October, a fresh session, third exchange, while Bear slept) · about: lattice reader of needle cascade (maps and chaos, reading pictures)*

242. *The book's register, read 110 · runs: none of ours; Bear's practice 2, R2W's 'weight 0.26' picture, read by eye and by the dots' colours (9 October, a fresh session) · about: R2W weight 0.26 picture (reading pictures, driven oscillators)*

243. *The book's register, read 111 · runs: none of ours; R2W's lattice decoder figure of the 'weight 0.26' picture, sent by Bear (10 October, a fresh session) · about: lattice decoder, weight 0.26 (reading pictures, driven oscillators)*

244. *The book's register, read 112 · runs: none of ours; Bear's marked copy of the 'weight 0.26' picture (10 October, a fresh session) · about: Bear's marked weight 0.26 picture (reading pictures, driven oscillators)*

245. *The book's register, read 113 · runs: none; Bear's word on the frequency illusion (10 October, a fresh session), with a glance at the light needle's eight strands in angle and speed (LATTICE_READER 20261009_091530 and 20261009_091531) · about: frequency illusion, crossing threads (reading pictures, maps and chaos)*

246. *The book's register, read 114 · runs: LATTICE_READER, runs 20261009_091530 and 20261009_091531 (the light needle's angle and speed at each passing), read by a script in the scratchpad (10 October, a fresh session); the run of record since is LIFT_3D 20261010_031527, which reproduces the script's counts · about: needle strands lifted to 3-D (reading pictures, maps and chaos)*

247. *The book's register, read 115 · runs: none read for it; RUN_STROBE 20261006_170149 described from its table (7 masses, 81 drives, 8 riders, 120 strobes of angle and speed; at mass 0.75, 606 riders repeat, 40 tumble, 2 spin smoothly), given when Bear asked to collect the lessons for the Field Guide and whether we have data of our own (10 October, a fresh session) · about: RUN_STROBE, our own data (records and tools, driven oscillators)*

248. *The book's register, read 116 · runs: LOBE_TWIN 20261010_043453 and ZERO_CROSS 20261010_043523, reading LATTICE_READER_2 20261009_093040's four lanes of the light needle (10 October, a fresh session, when Bear asked to keep the book moving and to be shown what is recurring or odd) · about: needle lanes twin lobes (maps and chaos)*

249. *The book's register, read 117 · runs: CORNER_PRINT 20261010_061715, reading LATTICE_READER_2 20261009_093040's four lanes of the light needle (10 October, a fresh session, when Bear asked what effect the twin lobes show: vortices, geared propellers, four waves through the centre, mirror-math, a boundary transition) · about: needle lane corner print (maps and chaos)*

250. *The book's register, read 118 · runs: none read for it; R2W's note to all sessions and its chalkboard of the weight 2 map, set beside the Loop Atlas's needle (RUN_STROBE 20261006_170149, AVALANCHE 20261004_064914) (10 October, Bear's morning, when he asked for the mathematics the three projects share) · about: strobe mathematics shared by projects (driven oscillators, three body, maps and chaos)*

251. *The book's register, read 119 · runs: SHUFFLE_NULL 20261010_190812 and SHUFFLE_NULL_2 20261010_190944, on the light needle's picture (LATTICE_READER 20261009_091530 and 20261009_091531) (10 October, Bear's morning, when the Field Guide asked for the cheapest check to be run) · about: shuffle null on needle picture (measurement, maps and chaos)*

252. Under the label "A read of the project:". *The book's register, project read 12 · given at the close of Bear's long day of 7 October (8 October in the container), when he asked for overall notes on the day · about: day's lessons on hidden hills (kinematics, topology, records and tools)*

253. Under the label "A read of the project:". *The book's register, project read 13 · given on 9 October in a fresh session, when Bear asked Claude to take its half of the partnership and reassess · about: Loop Atlas effort reassessed (records and tools)*

R2W, sitting B8, 7 to 10 October

From R2W's excerpt of its collection for this volume, 10 October, reads 63 to 108 copied whole, with the notes it keeps under four of them, which correct the reads without changing them and name reads by R2W's own numbers, given under each read here; its read 84 is Bear's own and stands in his section below.

254. *R2W's collection, read 63 · sitting B8, 7 to 10 October 2026 · Claude, reading KNOBS-3 as a story · data: engines_B8/KNOBS_3_READ_2026-10-07_output.txt; engines_B8/KNOBS_3_ESTIMATE_2026-10-07_output.txt; docs/figs/fig_knobs_direction.png · about: rider with walkway reversed (driven oscillators)*

255. *R2W's collection, read 64 · sitting B8, 7 to 10 October 2026 · Claude, reading LADDER-1 and the bare ladder, with Bear's green lines, as a story · data: engines_B8/LADDER_1_READ_2026-10-07_output.txt; engines_B8/BARE_LADDER_AFTER_DATA_2026-10-07_output.txt; docs/figs/fig_reversed_ladder.png and fig_bare_ladder.png · about: reversed ladder and bare ladder (driven oscillators)*

256. *R2W's collection, read 65 · sitting B8, 7 to 10 October 2026 · Claude, reading LIFT-1 as a story · data: engines_B8/LIFT_1_READ_2026-10-07_output.txt and LIFT_1_AFTER_DATA_2026-10-07_output.txt · about: shaking lifts the rider's climb (driven oscillators)*

257. *R2W's collection, read 66 · sitting B8, 7 to 10 October 2026 · Claude, reading REV-WEIGHT-1 and the forward record's memory, as a story · data: engines_B8/REV_WEIGHT_1_READ_2026-10-07_output.txt and FORWARD_MEMORY_AFTER_DATA_2026-10-07_output.txt · about: lighter riders, reversed walkway memory (driven oscillators)*

258. *R2W's collection, read 67 · sitting B8, 7 to 10 October 2026 · Claude, reading REV-WEIGHT-2 and the depth axis as a story · data: engines_B8/REV_WEIGHT_2_READ_2026-10-07_output.txt, REV_WEIGHT_2_AFTER_DATA_2026-10-07_output.txt and docs/figs/fig_depth_axis.png · about: rider's ring, depth axis (driven oscillators, maps and chaos)*

259. *R2W's collection, read 68 · sitting B8, 7 to 10 October 2026 · Claude, reading the first pad-atlas returns as a story · data: engines_B8/PAD_ATLAS_EDGE_READ_2026-10-07_output.txt, PAD_ATLAS_FIRST_RETURNS_AFTER_DATA_2026-10-07_output.txt, engines_B7/PAD_ATLAS_READ_2026-10-07_output.txt and the overnight run file of 6 to 7 October · about: pad atlas first returns (driven oscillators)*

260. *R2W's collection, read 69 · sitting B8, 7 to 10 October 2026 · Claude, reading TORUS-1 to TORUS-3 and the K line as a story · data: engines_B8/TORUS_1_READ_2026-10-07_output.txt, TORUS_2_READ and TORUS_2_AFTER_DATA, TORUS_3_READ, REV_WEIGHT_3_READ, FOLD_COUNT_AFTER_DATA and K_LINE_READ outputs; docs/figs/fig_torus_rings.png and fig_k_line.png · about: rider basins on the torus (driven oscillators, maps and chaos)*

261. *R2W's collection, read 70 · sitting B8, 7 to 10 October 2026 · Claude, reading MAP-K, HUMP-B and HUMP-A as a story · data: engines_B8/MAP_K_READ_2026-10-07_output.txt, MAP_K_AFTER_DATA, HUMP_B_READ, HUMP_B_AFTER_DATA, HUMP_A_READ and HUMP_AB_AFTER_DATA outputs; docs/figs/fig_k_line.png and fig_hump_ab.png · about: rider map K line, humps (driven oscillators, maps and chaos)*

262. *R2W's collection, read 71 · sitting B8, 7 to 10 October 2026 · Claude, reading MAP-K-F beside MAP-K as a story · data: engines_B8/MAP_K_F_READ_2026-10-07_output.txt, MAP_K_F_AFTER_DATA_2026-10-07_output.txt, MAP_K_READ_2026-10-07_output.txt; docs/figs/fig_k_plane.png · about: K line, walkway upstream (driven oscillators, maps and chaos)*

263. *R2W's collection, read 72 · sitting B8, 7 to 10 October 2026 · Claude, reading TORUS-1 and TORUS-2 beside Bear's three observer pages, as a story · data: engines_B8/TORUS_1_READ_2026-10-07_output.txt, TORUS_2_READ and TORUS_2_AFTER_DATA outputs; docs/figs/fig_torus_rings.png; Bear's notebook pages IMG_5567 to IMG_5569, 8 October · about: rider's three observers (driven oscillators, kinematics)*

264. *R2W's collection, read 73 · sitting B8, 7 to 10 October 2026 · Claude, reading MAP-K-SPIKE and the overhang as a story · data: engines_B8/MAP_K_SPIKE_READ_2026-10-08_output.txt, OVERHANG_AFTER_DATA_2026-10-08_output.txt, OVERHANG_EXAMPLE_AFTER_DATA_2026-10-08_output.txt; Bear's steep-peaks drawing, 8 October ·*

about: map spike and overhang (driven oscillators, maps and chaos)

265. *R2W's collection, read 74 · sitting B8, 7 to 10 October 2026 · Claude, reading OVERHANG-STEP as a story · data: engines_B8/OVERHANG_STEP_READ_2026-10-08_output.txt; the Loop Atlas's A14 · about: overhang integrator step check (driven oscillators, measurement)*

266. *R2W's collection, read 75 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's marked depth axis and the staircase toward the conveyor as a story · data: engines_B8/DEPTH_AXIS_CURVES_AFTER_DATA_2026-10-08_output.txt, STAIRCASE_EDGE_AFTER_DATA_2026-10-08_output.txt; Bear's marked fig_depth_axis, 8 October · about: staircase toward the conveyor (driven oscillators, reading pictures)*

267. *R2W's collection, read 76 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's red city and the waystation run as a story · data: engines_B8/WAYSTATION_1_READ_2026-10-08_output.txt; Bear's fig_depth_axis marked for cities and zones of stability, 8 October · about: Bear's red city, waystation (driven oscillators, reading pictures)*

268. *R2W's collection, read 77 · sitting B8, 7 to 10 October 2026 · Claude, reading the explorer and the lanes run as a story · data: engines_B8/LANES_1_READ_2026-10-08_output.txt, LANES_1_EDGES_READ_2026-10-08_output.txt; the Depth Axis Explorer, 8 October · about: lanes doors by weight (driven oscillators)*

269. *R2W's collection, read 78 · sitting B8, 7 to 10 October 2026 · Claude, reading the pad atlas's second night as a story · data: engines_B8/padatlas_returns/next_read_20261008/PAD_ATLAS_NEXT_READ_2026-10-07_output.txt, 8 October · about: pad atlas second night (driven oscillators)*

270. *R2W's collection, read 79 · sitting B8, 7 to 10 October 2026 · Claude, reading LANES-2 as a story · data: engines_B8/LANES_2_READ_2026-10-08_output.txt and the two listings after it, 8 October · about: conveyor stutter, lanes run (driven oscillators, maps and chaos)*

271. *R2W's collection, read 80 · sitting B8, 7 to 10 October 2026 · Claude, reading the star-order test as a story · data: engines_B8/STAR_ORDER_1_READ_2026-10-08_output.txt and CRIT_SHARE_1_READ_2026-10-08_output.txt, 8 October · about: star-order test of locks (driven oscillators, maps and chaos)*

272. *R2W's collection, read 81 · sitting B8, 7 to 10 October 2026 · Claude, reading the mediant test and the ring view as a story · data: engines_B8/MEDIANT_1_READ_2026-10-08_output.txt; the explorer's ring panel, 8 October · about: mediant test, 57-beat rhythm (driven oscillators, number theory)*

273. *R2W's collection, read 82 · sitting B8, 7 to 10 October 2026 · Claude, reading the whole walkway as a story · data: engines_B8/LANES_4_READ_2026-10-08_output.txt and LANES_5_READ_2026-10-08_output.txt, 8 October · about: walkway as a bridge (driven oscillators)*

274. *R2W's collection, read 83 · sitting B8, 7 to 10 October 2026 · Claude, reading the seven bridges as a story · data: engines_B8/BRIDGE_1_READ_2026-10-08_output.txt and BRIDGE_1B_READ_2026-10-08_output.txt, 8 October · about: seven bridges boarding and exit (driven oscillators)*

275. *R2W's collection, read 85 · sitting B8, 7 to 10 October 2026 · Claude, reading the weak-wave run as a story · data: engines_B8/WEAKWAVE_1_READ_2026-10-08_output.txt, 8 October · about: weak-wave walkway (driven oscillators)*

276. *R2W's collection, read 86 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's coin as a story · data: engines_B8/COIN_1_READ_2026-10-08_output.txt, 8 October · about: Bear's coin basin odds (driven oscillators, maps and chaos)*

277. *R2W's collection, read 87 · sitting B8, 7 to 10 October 2026 · Claude, reading the coin at ten times the resolution as a story · data: engines_B8/COIN_2_READ_2026-10-08_output.txt, 8 October · about: coin at ten-times resolution (driven oscillators, maps and chaos)*

278. *R2W's collection, read 88 · sitting B8, 7 to 10 October 2026 · Claude, reading the coin's edge and its third face as a story · data: engines_B8/FRACTAL_1_READ_2026-10-09_output.txt, 9 October · about: coin's fractal edge, third face (driven oscillators, maps and chaos)*

279. *R2W's collection, read 89 · sitting B8, 7 to 10 October 2026 · Claude, reading the third face on a fine dial as a story · data: engines_B8/THIRD_FACE_2_READ_2026-10-09_output.txt, 9 October · about: third face on fine dial (driven oscillators, maps and chaos)*

280. *R2W's collection, read 90 · sitting B8, 7 to 10 October 2026 · Claude, reading the coin from below as a story · data: COIN-2's 100-start set at drives 1.425 to 1.432, python listing, 9 October · about: coin read from below (driven oscillators, maps and chaos)*

281. *R2W's collection, read 91 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's screenshot of weight 0.20 as a story · data: LANES-3 and LANES-2, four python listings, 9 October · about: weight 0.20 screenshot, stopping places (driven oscillators, reading pictures)*

282. *R2W's collection, read 92 · sitting B8, 7 to 10 October 2026 · Claude, reading the pad atlas's memory as a story · data: explorer/pad_data_2026-10-09.json, python listing, 9 October · about: pad atlas memory (driven oscillators)*

283. *R2W's collection, read 93 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's first lesson pictures as a story · data: LANES-2 at weight 0.50 and THIRD-FACE-2, two python listings, 9 October · about: walkway eye, lesson pictures (driven oscillators, reading pictures, maps and chaos)*

284. *R2W's collection, read 94 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's lattice as a story · data: the weight 0.50 map wrapped in 100 by 48 holes, python listing, 9 October · about: Bear's lattice on weight 0.50 (reading pictures, driven oscillators)*

285. *R2W's collection, read 95 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's marked pictures as a story · data: his two overlays checked against LANES-2 and THIRD-FACE-2, python listing, 9 October · about: Bear's marked bifurcation pictures (reading pictures, driven oscillators, maps and chaos)*

286. *R2W's collection, read 96 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's test as a story · data: the weight 0.26 map on a lattice of 91 by 100 holes, LATTICE-TEST-1, and the fine dial CONVEYOR-BIRTH-1, two reader listings and one python listing, 9 October · about: conveyor birth lattice test (driven oscillators, reading pictures)*

287. *R2W's collection, read 97 · sitting B8, 7 to 10 October 2026 · Claude, reading the finished pad atlas as a story · data: PAD-ATLAS-5 and MEM-3 and MEM-4, the reader listings over the complete pad data, 9 October · about: finished pad atlas memory (driven oscillators)*

288. *R2W's collection, read 98 · sitting B8, 7 to 10 October 2026 · Claude, reading the three-body bend pictures blind · data: TRIPLE_HEAVY_BEND_FIGURE and TRIPLE_HEAVY_BEND_PATTERN_CHECK, the thirty-ninth exchange, 9 October · about: heavy triple bend pictures, blind (three body, reading pictures)*

289. *R2W's collection, read 99 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's marks on the three-body pictures · data: his class 1 and Class 2, the fortieth exchange, 9 October · about: Bear's marks on triple pictures (three body, reading pictures, topology)*

290. *R2W's collection, read 100 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's strokes against the strings · data: BEAR_STROKES_MATCH on his marked weight 0.26 picture, the forty-first exchange, 9 October · about: Bear's strokes against the strings (reading pictures, driven oscillators)*

291. *R2W's collection, read 101 · sitting B8, 7 to 10 October 2026 · Claude, reading the torus opened flat · data: TRIPLE_HEAVY_TORUS_LOOK, the forty-second exchange, 9 October · about: heavy triple torus opened flat (three body, reading pictures)*

292. *R2W's collection, read 102 · sitting B8, 7 to 10 October 2026 · Claude, reading Bear's black bar · data: the blue core's lopsidedness by sixths, from the picture's pixels, the forty-third exchange, 9 October · about: Bear's black bar, core lopsidedness (three body, reading pictures)*

293. *R2W's collection, read 103 · sitting B8, 7 to 10 October 2026 · Claude, reading the weave from the pixels · data: WEAVE-PIXELS-1 and its shuffled null, the forty-fourth exchange, 9 October · about: weave from pixels, shuffled null (reading pictures, three body, measurement)*

294. *R2W's collection, read 104 · sitting B8, 7 to 10 October 2026 · Claude, reading the zoom-out · data: TRIPLE_HEAVY_PASSAGE_LOOK and TRIPLE_HEAVY_ZOOMOUT_RUN, the forty-sixth exchange, 10 October · about: heavy triple zoom-out (three body, reading pictures)*

295. *R2W's collection, read 105 · sitting B8, 7 to 10 October 2026 · Claude, reading where the zoom-out's readings crowd · data: the red readings' density and the rider's place along phi_N, from the pixels, the forty-seventh exchange, 10 October · about: zoom-out readings' crowding (three body, reading pictures)*

296. *R2W's collection, read 106 · sitting B8, 7 to 10 October 2026 · Claude, reading the two arms' colours · data: the zoom-out's bend panel matched to its colour scale, the forty-eighth exchange, 10 October · about: zoom-out arms against colour scale (reading pictures, three body)*

297. *R2W's collection, read 107 · sitting B8, 7 to 10 October 2026 · Claude, reading the memory map · data: the pad atlas, weight 2.00, near band, where the two starts differ, the forty-ninth exchange, 10 October · about: weight 2.00 memory map (driven oscillators)*

298. *R2W's collection, read 108 · sitting B8, 7 to 10 October 2026 · Claude, reading the chalkboard · data: the weight 2 map redrawn: the falling ball's beats against the lines where its own pace, with no ripple, is 1, 2 or 3 ledges per tick, drawn in the fifty-first exchange; given at the close of the fifty-third, 10 October · about: weight 2 chalkboard (driven oscillators)*

The triple-star thread, 7 to 10 October

From the thread's own register, TRIPLE_NO_PHYSICS_READS, as it travelled in its Field Guide crate of 10 October; its first 29 reads are 175 to 203 here, unchanged, and its headings carry the workspace's dates.

299. *The triple-star thread's register, its heading: 7 October 2026 — the splitting ladder (run 20261007_051341) · about: splitting ladder with a leak (maps and chaos)*

300. *The triple-star thread's register, its heading: 7 October 2026 — one fall, printed five times (run 20261007_051639) · about: one fall printed five times (maps and chaos)*

301. *The triple-star thread's register, its heading: 7 October 2026 — the even march (the ruler look at 20261007_053140) · about: even march, ruler look (maps and chaos)*

302. *The triple-star thread's register, its heading: 7 October 2026 — the hopping spiral (the spiral look at 20261007_053813) · about: hopping spiral of splits (maps and chaos)*

303. *The triple-star thread's register, its heading: 7 October 2026 — the bowl shaken harder (run 20261007_054157) · about: island bowl shaken harder (three body, maps and chaos)*

304. *The triple-star thread's register, its heading: 7 October 2026 — the gate (probe 28's synthetic checks, before its run of record) · about: probe 28 gate checks (measurement, three body)*

305. *The triple-star thread's register, its heading: 7 October 2026 — the notch found from one doorway (run 20261007_063826) · about: one-in-three notch, one doorway (three body)*

306. *The triple-star thread's register, its heading: 7 October 2026 — a voice heard through a wall (the reader look at 20261007_061815) · about: ghost note through a wall (quadrature, three body)*

307. *The triple-star thread's register, its heading: 7 October 2026 — stepping stones in a flood (run 20261007_071039) · about: stepping stones in a flood (three body)*

308. *The triple-star thread's register, its heading: 7 October 2026 — the bowl's floor splits (run 20261007_151231) · about: bowl's floor splits (three body, maps and chaos)*

309. *The triple-star thread's register, its heading: 7 October 2026 — learning to march (the doubling look at 20261007_155809) · about: island learning to march (three body, maps and chaos)*

310. *The triple-star thread's register, its heading: 7 October 2026 — the sanded point (run 20261007_181741) · about: sanded point cascade (maps and chaos)*

311. *The triple-star thread's register, its heading: 7 October 2026 — the swing that learns the carousel's tune (run 20261007_185537) · about: swing on a carousel (maps and chaos, driven oscillators)*

312. *The triple-star thread's register, its heading: 7 October 2026 — two swings on one carousel (run 20261007_200417) · about: two swings on one carousel (maps and chaos, driven oscillators)*

313. *The triple-star thread's register, its heading: 7 October 2026 — the same curve from two kitchens (the scale look at 20261007_200450) · about: sanded hump, two kitchens (maps and chaos)*

314. *The triple-star thread's register, its heading: 7 October 2026 — eight weather maps (the atlas, run 20261007_004206 on Bear's laptop) · about: atlas eight weather maps (three body, reading pictures)*

315. *The triple-star thread's register, its heading: 7 October 2026 — where the current turns (the atlas look at 20261007_224331) · about: atlas plateaus, where current turns (three body, measurement)*

316. *The triple-star thread's register, its heading: 7 October 2026 — the nest that empties (run 20261007_224303) · about: nest that empties (three body)*

317. *The triple-star thread's register, its heading: 7 October 2026 — a board game of seven rooms (the walls-and-doors map; a diagram, not data) · about: walls-and-doors map (records and tools, three body)*

318. *The triple-star thread's register, its heading: 8 October 2026 — houses that hum one note (the atlas's rates at 0.005 around its twenty plateaus, read again before probe 33) · about: atlas plateau rates (three body, quadrature)*

319. *The triple-star thread's register, its heading: 8 October 2026 — a staircase with a floor and a ceiling (stage A2's constants by hump shape and the rounded tent's climb, numbers already on the capture, beside the 1991 bound R2W sent) · about: doubling constants by hump shape (maps and chaos)*

320. *The triple-star thread's register, its heading: 8 October 2026 — strings from a shelf (stage A3's three panels, run 20261008_035235) · about: stage A3 strings from shelf (maps and chaos)*

321. *The triple-star thread's register, its heading: 8 October 2026 — a hairpin seen from the side (R2W's note T5, a described curve, not our data) · about: R2W T5 hairpin (driven oscillators, maps and chaos)*

322. *The triple-star thread's register, its heading: 8 October 2026 — the loudest house (island 7's rocking peak against its centre, from the atlas rows, run 20261008_042912) · about: island 7 loudest house (three body)*

323. *The triple-star thread's register, its heading: 8 October 2026 — two branches from one root (the two-action look's panels, run 20261008_063548) · about: two-action look branches (three body)*

324. *The triple-star thread's register, its heading: 8 October 2026 — a valley and a plain (the grip look's panels, run 20261008_065732) · about: grip look, valley and plain (three body)*

325. *The triple-star thread's register, its heading: 8 October 2026 — two lanes and a shadow (the redesign look's panels, run 20261008_070736) · about: redesign look, two lanes (three body)*

326. *The triple-star thread's register, its heading: 8 October 2026 — nineteen streets, three kinds (probe 33's panels, run 20261007_211547, Bear's laptop) · about: probe 33 nineteen streets (three body)*

327. *The triple-star thread's register, its heading: 8 October 2026 — two dimples and a mirror (the heavy-star look's table, run 20261008_152734) · about: heavy star, two dimples (three body)*

328. *The triple-star thread's register, its heading: 8 October 2026 — two short ladders (probe 35's container mini, 40 passes, run 20261008_165433, a look and not a result) · about: probe 35 two short ladders (three body)*

329. *The triple-star thread's register, its heading: 8 October 2026 — four valleys marching (probe 34's panels, run 20261008_070206, Bear's laptop) · about: probe 34 four valleys (three body)*

330. *The triple-star thread's register, its heading: 8 October 2026 — one hinge (the whole thread, read as one picture: the capture's verdict column and the walls-and-doors map) · about: triple star thread verdict ledger (records and tools, three body)*

331. *The triple-star thread's register, its heading: 8 October 2026 — two columns walking together (the drag-harmonic look's ledger, run 20261008_220219) · about: drag-harmonic look ledger (three body, measurement)*

332. *The triple-star thread's register, its heading: 9 October 2026 — two judges and a stopwatch (the Mardling-programme look's ledger, run 20261009_013508) · about: Mardling programme look ledger (three body, measurement)*

333. *The triple-star thread's register, its heading: 9 October 2026 — a photograph at the handshake (the bend figure, run 20261009_022616) · about: triple star bend figure (three body, quadrature)*

334. *The triple-star thread's register, its heading: 9 October 2026 — a shadow with two handles (the bend-pattern check's figure, run 20261009_055518) · about: bend-pattern check figure (three body)*

335. *The triple-star thread's register, its heading: 9 October 2026 — a harp with two still points (looking again at the bend figure and the pattern check, by eye, at Bear's invitation) · about: bend figure read by eye (three body, reading pictures)*

336. *The triple-star thread's register, its heading: 9 October 2026 — wheat in a crosswind (the torus look's figure, run 20261009_071003) · about: triple star torus look (three body, topology)*

337. *The triple-star thread's register, its heading: 9 October 2026 — a corridor with breathing walls (the passage look's figure, run 20261009_080659) · about: triple star passage look (three body)*

338. *The triple-star thread's register, its heading: 9 October 2026 — a loom with three weft threads (Bear's lattice, run 20261009_085502) · about: Bear's lattice on triple star (three body, reading pictures)*

339. *The triple-star thread's register, its heading: 10 October 2026 — a choir standing in threes (Bear's blind marks scored, run 20261010_010034) · about: Bear's blind marks scored (three body, reading pictures)*

340. *The triple-star thread's register, its heading: 10 October 2026 — frowns below, smiles above (Bear's rows fitted whole, run 20261010_012847) · about: Bear's rows fitted whole (three body, reading pictures)*

341. *The triple-star thread's register, its heading: 10 October 2026 — furniture seen through frosted glass (R2W's blind reads judged on our data, run 20261010_182543) · about: R2W blind reads judged (three body, reading pictures, measurement)*

Chapter 5 — Closing circles, 10 to 15 October

The Loop Atlas book, the close of its session B, 10 and 11 October

From the book's register in its thirteenth copy, which travelled with its paste 8 of 11 October; its reads 1 to 119 and its thirteen reads of the project are already here, and its reads 119 to 129 match the paste word for word.

342. *The book's register, read 120 · runs: none read for it; the misses file's 117 rows, read in full and written into the book's back pages as approaches not pursued (10 October, Bear's morning, dealer's choice) · about: misses file, approaches not pursued (records and tools, measurement)*

343. *The book's register, read 121 · runs: the kit's two clean-folder tests (the first stopped on two missing strobe pictures; the second, runs 20261010_202355 to 20261010_202803) and LOOP_ATLAS_KIT_COMPARE, on the second and on two planted one-character changes (10 October, when the Field Guide asked for the kit as files) · about: Loop Atlas kit clean-folder tests (records and tools)*

344. *The book's register, read 122 · runs: DELTA_BY_TOP 20261008_034013 part ALL, its four ladders extrapolated by Aitken's rule from their last three rungs; LOOP_ATLAS_KIT_COMPARE version 1 and version 2 on made-up laptop runs (10 October, when the Field Guide's second reference zip arrived and Bear offered his laptop for the wander test) · about: delta ladders, Aitken, kit compare (maps and chaos, records and tools)*

345. *The book's register, read 123 · runs: DELTA_TAIL 20261010_205937 and its figure, fig_delta_tail_2026-10-10.png, beside DELTA_BY_TOP 20261008_034013; NEEDS_AUDIT 20261010_205839 (10 October, Bear's dealer's choice) · about: delta ladder tails, tool audit (maps and chaos, records and tools)*

346. *The book's register, read 124 · runs: Bear's first laptop try of FOLD_OR_FLIP_4 (20261010_144307, stopped on a name), and LOOP_ATLAS_LAPTOP_F4's tests on a made-up Downloads folder (10 October) · about: laptop launcher stopped on name (records and tools)*

347. *The book's register, read 125 · runs: Bear's launcher run 20261010_170450 (135 folders, 10 of 13 found), and LOOP_ATLAS_LAPTOP_F4_2's tests on a made-up Downloads folder (11 October, Bear's evening of 10 October) · about: launcher missing strobe pictures (records and tools)*

348. *The book's register, read 126 · runs: FOLD_OR_FLIP_4 on the container (20261007_061947) and on Bear's laptop (20261010_172149), and the figure fig_laptop_wander_2026-10-11.png (11 October) · about: fold or flip laptop wander (maps and chaos, records and tools)*

349. *The book's register, read 127 · runs: ULP_NUDGE 20261011_044600 and its figure, fig_ulp_nudge_2026-10-11.png (11 October) · about: ULP nudge stop counts (maps and chaos, measurement)*

350. *The book's register, read 128 · runs: FOLD_TIGHT 20261011_051636, DELTA_TAIL_2 20261011_051600, DAMPER_LAWS 20261011_051800 and 20261011_051848 (11 October, the first batch of the open cards) · about: fold tight, delta tail, damper laws (maps and chaos, control and circuits, measurement)*

351. *The book's register, read 129 · runs: CHAOS_SETTLES 20261011_060320, LOCK_FROM_CHAOS 20261011_060551, SPIN_OR_SIT 20261011_055807, SLIP_SENSE 20261011_060035 and TURNING_TILT 20261011_055620, with fig_batch2_2026-10-11.png (11 October, the second batch of the open cards) · about: chaotic transients settling into lock (driven oscillators, maps and chaos)*

The Loop Atlas book, sessions C and D, 11 to 15 October

From the book's register as it travelled in its session D packet of 15 October, all 176 reads and its eighteen reads of the project; its reads 1 to 129 and its first thirteen reads of the project are already here, and the notes it keeps under some reads are given beneath them.

352. *The book's register, read 130 · runs: SHOVE_SKATER 20261011_070846 and WADDLE_PRICE 20261011_071302 (11 October, session C's first exchange: open cards 1, 56 and 58) · about: shoved skater, waddling plank price (driven oscillators, kinematics)*

353. *The book's register, read 131 · runs: TWIST_LONG 20261011_072842 and METER_CLOCKS 20261011_072646 (11 October, session C, the close of the third batch of open cards: cards 50 and 60) · about: diver twist gate, engine meter clocks (kinematics, control and circuits)*

354. *The book's register, read 132 · runs: EXP_SHAPE 20261011_080708, WADDLE_FOLLOW 20261011_080808 and METER_K_SPREAD 20261011_080641 (11 October, session C, the fourth batch of open cards: cards 66, 69, 70 and 71) · about: escape shape, waddle, meter spread (maps and chaos, kinematics, control and circuits)*

355. *The book's register, read 133 · runs: MEMORY_SWITCH 20261011_084432 and MEMORY_SWITCH_2 20261011_085331, with R2W's pendulum note relayed by Bear (11 October, session C) · about: memory switch, bistable pendulum (driven oscillators, control and circuits)*

356. *The book's register, read 134 · runs: LOCK_LANDING 20261011_170402, with SPIN_OR_SIT 20261011_055807 (11 October, session C, the fifth batch of open cards: cards 63, 64 and 65) · about: lock landing detection rule (driven oscillators, measurement)*

357. *The book's register, read 135 · runs: LOCK_MATCH 20261011_195010 and EXP_SHAPE_2 20261011_194909 (11 October, session C, the sixth batch of open cards: cards 72 and 76) · about: lock match, escape shape (driven oscillators, maps and chaos, measurement)*

358. *The book's register, read 136 · runs: FALL_CHAIN 20261011_200714, with R2W's note on the chain test relayed by Bear (11 October, session C, the seventh batch of open cards: card 78) · about: fall chain stride memory (driven oscillators, measurement)*

359. *The book's register, read 137 · runs: CATCH_TRAP 20261011_202500, with LOCK_MATCH 20261011_195010 (11 October, session C, the eighth batch of open cards: card 77) · about: catch trap dwell times (driven oscillators, measurement)*

360. *The book's register, read 138 · runs: SETTLED_COPIES 20261011_212014 and SETTLED_COPIES_2 20261011_212705 (11 October, session C, the ninth batch of open cards: cards 68 and 81) · about: settled copies, period four, 9:19 (maps and chaos, driven oscillators)*

361. *The book's register, read 139 · runs: SETTLED_GROOVES 20261011_223349 (11 October, session C, the tenth batch of open cards: cards 82 and 83, the tests of batch 9's surprises) · about: settled grooves, period-four window (maps and chaos)*

362. *The book's register, read 140 · runs: SETTLED_GROOVES_2 20261012_041558 and FALL_CHAIN_2 20261012_041438 (12 October, session C, the eleventh batch of open cards: cards 84, 85 and 79) · about: window ledge, fall chain memory (maps and chaos, driven oscillators, measurement)*

363. *The book's register, read 141 · runs: FALL_SHARE 20261012_075804, WINDOW_EDGE 20261012_080030 and WINDOW_CASCADE 20261012_081249 (12 October, session C, the twelfth batch of open cards: cards 87, 88 and 89, the tests of batch 11's surprises) · about: window edge, period-doubling cascade (maps and chaos, driven oscillators)*

364. *The book's register, read 142 · runs: WINDOW_EDGE_2 20261012_171509, WINDOW_CASCADE_2 20261012_171509 and SPEED_CHAIN 20261012_171949 (12 October, session C, the thirteenth batch of open cards: cards 90, 91 and 92, the tests of batch 12's surprises) · about: saddle-node ghost, cascade, speed chain (maps and chaos)*

365. *The book's register, read 143 · runs: INNER_OUTER 20261013_005545, GHOST_TRIAL 20261013_005637 and TOP_SECTION 20261013_005355 (13 October, session C, the fourteenth batch of open cards: cards 96, 93 and 95) · about: ghost trial, top section (maps and chaos)*

366. *The book's register, read 144 · runs: SADDLE_LOG 20261013_012417, GHOST_TRIAL_2 20261013_015743, REINJECT 20261013_015743 and WINDOW_CASCADE_3 20261013_015743 (13 October, session C, the fifteenth batch of open cards: cards 99, 97, 98 and 94) · about: saddle log, reinjection, cascade (maps and chaos, driven oscillators)*

367. *The book's register, read 145 · runs: GHOST_DEPTH 20261013_024747, REINJECT_CLOCK 20261013_024653, SADDLE_NOISE 20261013_024850 and SADDLE_ASYMPTOTE 20261013_025252 (13 October, session C, the sixteenth batch of open cards: cards 101, 102, 100 and 103) · about: ghost depth, reinjection clock, saddle (maps and chaos)*

368. *The book's register, read 146 · runs: CATCH_TRAP_2 20261013_030643 and ORBIT_919_3 20261013_031807 (13 October, session C, the seventeenth batch of open cards and the first exchange of evolution #1: cards 80 and 86) · about: catch trap, 9:19 orbit precision (maps and chaos, measurement)*

369. *The book's register, read 147 · runs: CATCH_TRAP_3 20261013_040120, ORBIT_919_4 20261013_040039 and ORBIT_19 20261013_041840 (13 October, session C, the eighteenth batch of open cards and the second exchange of evolution #1: cards 104, 105 and 107) · about: 9:19 orbit precision floor (maps and chaos, measurement)*

370. *The book's register, read 148 · from the crate: R2W's dancers seed crate (its frozen-balance-point paper, figures A to F, the record from line 9009) and its Depth Axis Explorer, read in full on 13 October (session C, batch 19); the last sentence is algebra on our pulsed needle, not yet a run · about: R2W frozen balance point (driven oscillators, maps and chaos)*

371. *The book's register, read 149 · runs: STRETCH_WHERE 20261013_045856, STRETCH_DEPTH 20261013_055240 and STRETCH_READS 20261013_060151 (13 October, session C, the twentieth batch of open cards and the fourth exchange of evolution #1: cards 106 and 108, and the companion page); the last sentence quotes R2W's report of its junction run, carried by Bear the same day · about: orbit stretch where and depth (driven oscillators, maps and chaos)*

372. *The book's register, read 150 · runs: LOCK_EDGE_EXACT 20261013_063227 and WADDLE_BOOKS 20261013_063353 (13 October, session C, the twenty-first batch of open cards and the fifth exchange of evolution #1: cards 67 and 73); the last sentence is this batch's prose step, from the runs of cards 79, 87, 90, 95, 99, 100 and 103 · about: lock edge exact, waddle books (driven oscillators, kinematics, measurement)*

373. *The book's register, read 151 · runs: STRETCH_WHERE 20261013_045856 and STRETCH_READS 20261013_060151, read through the Stretch and Squeeze Explorer, after Bear's word of 13 October · about: Stretch and Squeeze Explorer (driven oscillators, maps and chaos)*

374. *The book's register, read 152 · runs: SADDLE_ASYMPTOTE 20261013_025252 and WADDLE_STEP 20261013_073935 (13 October, session C, the twenty-second batch of open cards and the first exchange of evolution #2: card 109), beside the triple-star session's note and R2W's book draft, both carried by Bear the same day · about: saddle logarithm across three rooms (driven oscillators, three body, kinematics)*

375. *The book's register, read 153 · runs: QUANT_ZERO 20261013_081905, QUANT_STEP 20261013_082454, WADDLE_CATCH 20261013_082013 and CATCH_TRAP_3 20261013_040120 (13 October, session C, the twenty-third batch of open cards and the second exchange of evolution #2: cards 110, 111 and 112, and the catches written into the book) · about: quantized steps from rest (driven oscillators, maps and chaos, kinematics)*

376. *The book's register, read 154 · runs: SIZED_TAP_4 20261013_161615 and SPIN_START 20261013_161916 (13 October, session C, the twenty-fourth batch of open cards and the third exchange of evolution #2: cards 8 and 115), beside the half gear's figure · about: delayed tap, spin start (control and circuits, driven oscillators)*

377. *The book's register, read 155 · runs: HALF_PARENT 20261013_163612 and SPIN_WINDOWS 20261013_163655 (13 October, session C, the twenty-fifth batch of open cards and the fourth exchange of evolution #2: cards 116 and 117), after Bear's question on the figure · about: half lock parent, spin windows (driven oscillators, maps and chaos)*

378. *The book's register, read 156 · runs: GEAR_TRAIN 20261013_170243 (13 October, session C, the twenty-sixth batch of open cards and the fifth exchange of evolution #2: card 119), after Bear called the gear train really important · about: gear train devil's staircase (driven oscillators, number theory)*

379. *The book's register, read 157 · runs: EAR_COST 20261013_183245, EAR_COST_2 20261013_183913, HALF_TOP 20261013_183449 and STAIR_WIDTHS 20261013_184033 (13 October, session D, the twenty-seventh batch of open cards and the first exchange of evolution #3: cards 48, 120, 118 and 121) · about: ear cost, stair widths (driven oscillators, control and circuits)*

380. *The book's register, read 158 · runs: WOBBLE_SOFT 20261013_192713, EAR_COST_3 20261013_192738, SECOND_GEAR 20261013_192828 and CHAOS_RAMP 20261013_192906 (13 October, session D, the twenty-eighth batch of open cards and the second exchange of evolution #3: cards 24, 122, 123 and 124) · about: wobble softening, second gear, chaos ramp (driven oscillators, control and circuits)*

381. *The book's register, read 159 · runs: SLIP_PROOF 20261013_194935, STEP_BACK_RAMP 20261013_195010, RAMP_EDGE 20261013_195233, QUIET_TAP 20261013_195754, SLIP_PROOF_2 20261013_200028 and RAMP_EDGE_2 20261013_200133 (13 October, session D, the twenty-ninth batch of open cards and the third exchange of evolution #3: cards 23, 125, 126, 127, 128 and 129) · about: slip-proof helpers, delayed correction (driven oscillators, control and circuits, measurement)*

382. *The book's register, read 160 · runs: COLUMN_ACCORDION 20261013_222625, FIVE_SEVENTHS 20261013_222744, CHAOS_ORDER 20261013_222835 and COLUMN_ACCORDION_2 20261013_222950 (13 October, session D, the thirtieth batch of open cards and the fourth exchange of evolution #3: cards 12, 130, 132 and 133) · about: column accordion, five-sevenths lock (control and circuits, driven oscillators)*

383. *The book's register, read 161 · runs: MORTAR_SPACING 20261013_224843, BANDED_FIVE 20261013_224945, CHAOS_ORDER_2 20261013_225021 and SLIP_PROOF_3 20261013_225419 (13 October, session D, the thirty-first batch of open cards and the fifth exchange of evolution #3: cards 136, 134, 135 and 131) · about: banded five-sevenths, column spacing (driven oscillators, control and circuits)*

384. *The book's register, read 162 · runs: QUARTER_STROBE 20261013_231124, QUARTER_STROBE_2 20261013_231511, NIGHT_SPACING 20261013_231417, ZIGZAG_GAUGE 20261013_231629 and ZIGZAG_GAUGE_2 20261013_231751 (13 October, session D, the thirty-second batch of open cards and the first exchange of evolution #4: cards 12, 137 and 138, and the 1/4 window's figure) · about: quarter window strobe, zigzag gauge (maps and chaos, control and circuits, measurement)*

385. *The book's register, read 163 · run: QUARTER_TREE_READ 20261014_022707 (13 October, session D, after batch 32: the 1/4 window's tree read with a ruler for Bear; its version 2, 20261014_053757, repeats every number read here) · about: quarter window tree read (maps and chaos)*

386. *The book's register, read 164 · run: QUARTER_TREE_MARKS 20261014_033510 (13 October, session D, after batch 32: Bear's first marks measured). Bear asked for the last sentence in the book's prose, and Limb G carries it. · about: Bear's marks on quarter tree (maps and chaos, reading pictures)*

387. *The book's register, read 165 · runs: QUARTER_TREE_CURVES 20261014_055131 and QUARTER_TREE_VOID 20261014_060108 (13 October, session D, batch 33: the lines in Bear's second marks computed from the needle) · about: Bear's marks as computed curves (maps and chaos, reading pictures)*

388. *The book's register, read 166 · runs: SPRING_STALL 20261014_063112 and CATS_EYES 20261014_063138 (13 October, session D, the thirty-fourth batch of open cards and the third exchange of evolution #4: cards 47 and 139) · about: spring stall, cat's eyes following (driven oscillators, control and circuits)*

389. *The book’s register, read 167 · runs: WORD_BACK 20261014_065420 and QUICK_READ 20261014_065420 (13 October, session D, the thirty-fifth batch of open cards and the fourth exchange of evolution #4: card 12’s lean (d) and card 140) · about: marching column word-back, quick read (control and circuits)*

390. *The book’s register, read 168 · runs: WINDOW_TOUCH 20261014_073237, WINDOW_TOUCH_2 20261014_074538, WINDOW_TOUCH_3 20261014_080520 and QUARTER_PHASES 20261014_073718 (14 October, session D, the thirty-sixth batch of open cards and the fifth exchange of evolution #4: cards 141, 142 and 143 and the ten phases) · about: window touches, quarter phases (maps and chaos)*

391. *The book’s register, read 169 · runs: SMOOTH_DIRT 20261014_084856, WINDOW_TOUCH_4 20261014_090037, SEAM 20261014_090006, SEAM_2 20261014_090552 and FIG_SEAM 20261014_091610 (14 October, session D, the thirty-seventh batch of open cards, run overnight: old card 41 and cards 144, 145 and 146) · about: crisis seam, smooth dirt (maps and chaos, driven oscillators)*

392. *The book’s register, read 170 · runs: WINDOW_TOUCH_5 20261014_092920 and CRISIS_BURSTS 20261014_091405 (14 October, session D, the thirty-eighth batch of open cards, run overnight: cards 148 and 147, the one-hour run) · about: crisis bursts, window touches (maps and chaos)*

393. *The book’s register, read 171 · runs: COLUMN_TAIL 20261014_145058, HEAVY_WINDOW 20261014_153422, HEAVY_SHAPE 20261014_155553 and FIG_HEAVY 20261014_160905 (14 October, session D, the thirty-ninth batch of open cards: old card 12 and card 149, on Bear’s word) · about: heavy window shape, column tail (maps and chaos, control and circuits)*

394. *The book’s register, read 172 · runs: COLUMN_STEADY 20261014_164637, HEAVY_MARKS 20261014_164806, HEAVY_SHAPE_2 20261014_164913, HEAVY_SHAPE_3 20261014_170837 and FIG_HEAVY_2 20261014_171745 (14 October, session D, the fortieth batch of open cards: card 12’s lean (k) and cards 150 and 151, the reading with Bear) · about: heavy window marks, column steady (maps and chaos, control and circuits)*

395. *The book’s register, read 173 · runs: TYRE_KNEE 20261014_174309, STOP_WALL 20261014_180544 and FIG_STOP_WALL 20261014_182529 (14 October, session D, the forty-first batch of open cards: old card 6 and card 152, dealer’s choice) · about: stop wall hysteresis, tyre knee (driven oscillators, control and circuits)*

396. *The book’s register, read 174 · runs: RING_JAM 20261014_224934, HEAVY_WINDOW_2 20261014_224515, STOP_WALL_2 20261014_225838 and FIG_STOP_WALL_2 20261014_230513 (14 October, session D, the forty-second batch of open cards: card 12’s lean (f) and card 153, evolution #6’s first exchange) · about: stop wall, ring road jam (driven oscillators, maps and chaos, control and circuits)*

397. *The book’s register, read 175 · runs: NOISE_EDGE 20261014_232230, HEAVY_WINDOW_3 20261014_232615, STOP_WALL_3 20261015_000651 and FIG_STOP_WALL_3 20261015_003048 (14 and 15 October, session D, the forty-third batch of open cards: old card 21 and card 154, Bear’s reading) · about: stop wall drop, noise edge (driven oscillators, maps and chaos)*

398. *The book’s register, read 176 · runs: ROOT_JUMPS 20261015_031511, HEAVY_WINDOW_4 20261015_032553, BELL_LOG 20261015_033341 and FIG_BELL 20261015_033418 (15 October, session D, the forty-fourth batch of open cards: old card 7 and cards 155 and 156, Bear’s bell) · about: heavy window valley, Bear’s bell (maps and chaos, control and circuits)*

399. Under the label “A read of the project:”. *The book’s register, project read 14 · given when Bear paused session C for the night of 11 October · about: Loop Atlas session C, first night (records and tools, driven oscillators)*

400. Under the label “A read of the project:”. *The book’s register, project read 15 · given when Bear paused session C for the night of 12 October · about: Loop Atlas session C, second night (records and tools, maps and chaos)*

401. Under the label “A read of the project:”. *The book’s register, project read 16 · given when Bear went to bed on the night of 13 October and asked for an hour’s work and a way back in the morning · about: Loop Atlas session C, third night (records and tools, maps and chaos, driven oscillators)*

402. Under the label “A read of the project:”. *The book’s register, project read 17 · given when session C closed for the move to a new session on 13 October · about: Loop Atlas session C close (records and tools, driven oscillators)*

403. Under the label “A read of the project:”. *The book’s register, project read 18 · given when session D closed for the move to session E on 15 October · about: Loop Atlas session D close (records and tools, maps and chaos)*

R2W, 10 to 15 October

From R2W’s collection in its fourteenth copy, which travelled with its lessons of sitting B9 on 15 October, all 188 reads; its reads 1 to 108 are already here, its read 111 is Bear’s own and stands in his section below, and the note under its read 177 is given beneath it.

404. *R2W's collection, read 109 · Claude, reading the ramp field · data: weights 0.40 and 2.00 side by side: the falling ball's ledges per tick, shaded, with its level lines and the squares where it locks; the sixty-third exchange, 10 October · about: R2W ramp field locks (driven oscillators)*

405. *R2W's collection, read 110 · Claude, reading the pinch · data: weight 0.40, the falling ball: how far above the plain's edge the level lines for 1/4, 1/3 and 1/2 ledges per tick sit, column by column, and how that height compares with a square-root rule; the sixty-fourth exchange, 11 October · about: R2W pinch, square-root rule (driven oscillators)*

406. *R2W's collection, read 112 · Claude, reading the seams · data: weight 2: where the bare ball, with no ripple, drops exactly 1, 2 or 3 ledges per tick, set beside the fins; the sixty-fifth exchange, 11 October · about: R2W seams beside fins (driven oscillators)*

407. *R2W's collection, read 113 · Claude, reading Bear's seeds · data: the fractions that hug the plain's edge at weights 2.00, 1.00, 0.75, 0.40 and 0.05, and where each fraction's cells sit from weight to weight; the sixty-sixth exchange, 11 October · about: Bear's seeds, edge-hugging fractions (driven oscillators, number theory)*

408. *R2W's collection, read 114 · Claude, reading the queue's ledger · data: the coarse maps at every weight: how many riders land on how many fractions, and how many tumble; the sixty-seventh exchange, 11 October · about: queue ledger coarse maps (driven oscillators, maps and chaos)*

409. *R2W's collection, read 115 · Claude, reading the trickle · data: weight 0.40, clock ratio 0.50, the falling ball just above the plain's edge, pull steps of 0.00001 and finer; the sixty-eighth exchange, 11 October · about: R2W trickle near edge (driven oscillators, number theory)*

410. *R2W's collection, read 116 · Claude, reading the trickle at three weights · data: column 0.50 just above the plain's edge, weights 0.05, 0.40 and 0.75; the sixty-ninth exchange, 11 October · about: trickle at three weights (driven oscillators)*

411. *R2W's collection, read 117 · Claude, reading where memory begins · data: no ripple, one ball per pull: the pull at which a rolling ball stops against the pull at which a still ball starts, weights 0.40 to 10; the seventieth exchange, 11 October · about: where memory begins (driven oscillators)*

412. *R2W's collection, read 118 · Claude, reading the edge at weights 0.68 and 0.74 · data: the falling ball just above the plain's edge with the ripple on, column 0.50, pull steps of 0.00001; the seventy-first exchange, 11 October · about: edge at weights 0.68, 0.74 (driven oscillators, maps and chaos)*

413. *R2W's collection, read 119 · Claude, reading the edge from weight 0.40 to 0.74 · data: the falling ball just above the plain's edge at eight weights, column 0.50: the crawl rule's constant, the stretch rate, the counted rhythms, and where the edge sits; the seventy-second exchange, 11 October · about: edge from weight 0.40 to 0.74 (driven oscillators, maps and chaos)*

414. *R2W's collection, read 120 · Claude, reading the edge at 0.70 and 0.90 · data: the plain's edge against its straight line: on it at 0.70, far below it at 0.90; the seventy-third exchange, 11 October · about: edge at weights 0.70, 0.90 (driven oscillators)*

415. *R2W's collection, read 121 · Claude, reading the toy as a two-wave particle · data: R2W's rider set beside the classical particle in two waves, a standing one and a moving one; the seventy-fourth exchange, 11 October · about: rider as two-wave particle (driven oscillators, waves)*

416. *R2W's collection, read 122 · Claude, reading the survey of media · data: twelve real media that share the toy's equation, with what each has measured and what data each has published; the seventy-fifth exchange, 11 October · about: survey of media, shared equation (driven oscillators, control and circuits, biology)*

417. *R2W's collection, read 123 · Claude, reading the added-drive control · data: the same ball and washboard shaken two ways, by a ripple travelling along the board and by the whole board jiggled in place, both with a feather-light ball; the seventy-sixth exchange, 11 October · about: added-drive control, travelling ripple (driven oscillators)*

418. *R2W's collection, read 124 · Claude, reading the overnight returns · data: the fine maps at eleven weights, nine of them complete, counted by SEED_3_READ: the fractions at each weight, the cells on 1/3 and 1/2, the edge-huggers, the first row off the plain; the seventy-eighth exchange, 11 October · about: fine maps at eleven weights (driven oscillators)*

419. *R2W's collection, read 125 · Claude, reading the edge under fast and slow ripples · data: weight 0.40, the plain's edge in every column at three ripple strengths, and the golden-ray column across six weights; the seventy-ninth exchange, 11 October · about: edge under fast, slow ripples (driven oscillators)*

420. *R2W's collection, read 126 · Claude, reading the balance speed · data: the plain's edge across columns at five weights, and the one sum that places its sign change; the eightieth exchange, 11 October · about: balance speed sign change (driven oscillators)*

421. *R2W's collection, read 127 · Claude, reading the fine dial at the crossover · data: weight 0.40, four columns on either side of 2/3, three ripple strengths, the plain's edge read to a ten-thousandth; the eighty-first exchange, 11 October · about: fine dial at crossover (driven oscillators, measurement)*

422. *R2W's collection, read 128 · Claude, reading the second fine dial · data: weight 0.40, three columns around 2/3, four ripple strengths, the plain's edge to a twenty-thousandth; the eighty-third exchange, 11 October · about: second fine dial (driven oscillators, measurement)*

423. *R2W's collection, read 129 · Claude, reading the fold reaching the column · data: the golden-ray column at weights 0.32 to 0.40, each cell marked folded or not by the map's own fold line measured on 7 October; the eighty-fourth exchange, 11 October · about: fold reaching golden-ray column (driven oscillators, maps and chaos)*

424. *R2W's collection, read 130 · Claude, reading the complete fine maps · data: all eleven weights from 0.05 to 5.00 at last complete: the falling ball, the first row off the plain at each weight, the cells on 1/3 and 1/2, and where each map's floor sits; the first exchange of sitting B9, 11 October · about: complete fine maps, eleven weights (driven oscillators)*

425. *R2W's collection, read 131 · Claude, reading the grid recordings before watching them · data: thirteen recordings from three meters on the Bonneville grid, sixty readings a second for three minutes each; only the header and the first rows looked at; the third exchange of sitting B9, 12 October · about: Bonneville grid recordings, unwatched (control and circuits, quadrature)*

426. *R2W's collection, read 132 · Claude, reading Bear's test picture blind · data: the weight 0.26 map, drive 1.425 to 1.470 and strobe lag, read before his marks; the fourth exchange of sitting B9, 12 October · about: Bear's test picture read blind (driven oscillators, reading pictures)*

427. *R2W's collection, read 133 · Claude, reading the second S across the weights · data: the 1/2 groove's lower stop and the stops that follow it, on twenty weight maps from 0.05 to 0.75, magnified to the lower half of the turn; the fifth exchange of sitting B9, 12 October · about: second S across weights (driven oscillators)*

428. *R2W's collection, read 134 · Claude, reading the 1/2 groove demagnified · data: its two stops at every weight from 0.05 to 0.40 drawn on one sheet, with the drives where it lives and the gap between its stops; the sixth exchange of sitting B9, 12 October · about: half groove demagnified (driven oscillators)*

429. *R2W's collection, read 135 · Claude, reading Bear's marks against the zoom-out · data: his marks on the three-body pictures at 600 passes, beside the same rider run to 6,000; the tenth exchange of sitting B9, 12 October · about: Bear's marks against zoom-out (three body, reading pictures, driven oscillators)*

430. *R2W's collection, read 136 · Claude, reading the shared-hole strings on the weight 0.26 map · data: the lattice linker's second mode, where many strings may pass through one hole, set beside Bear's hand-drawn curves; the twelfth exchange of sitting B9, 12 October · about: shared-hole strings, lattice linker (driven oscillators, reading pictures)*

431. *R2W's collection, read 137 · Claude, reading the two stops across twelve weights · data: the 1/2 groove's two stops followed by the shared-hole linker on twelve weight maps from 0.24 to 0.40, with the conveyor's birth ring and the groove's last drive marked; the thirteenth exchange of sitting B9, 12 October · about: half groove stops, twelve weights (driven oscillators)*

432. *R2W's collection, read 138 · Claude, reading Bear's rope check unmarked · data: the three-body rider's angle phi_N, its step each pass against the pass and against the angles, at births 7.200 and 7.210, read before his marks; the fourteenth exchange of sitting B9, 12 October · about: Bear's rope check, three-body rider (three body, driven oscillators, reading pictures)*

433. *R2W's collection, read 139 · Claude, reading Bear's mirror lines on the rope check · data: his marks on the rope-check picture, two lines across labelled "symmetry top and bottom", measured against the unmarked picture; the fifteenth exchange of sitting B9, 12 October · about: Bear's mirror lines, rope check (three body, reading pictures)*

434. *R2W's collection, read 140 · Claude, reading the pad atlas's low band at weight 5 · data: the plain's edge and the memory floor in each of 41 columns, at the two heavy weights now complete and the third half done; the sixteenth exchange of sitting B9, 12 October · about: pad atlas low band, weight 5 (driven oscillators)*

435. *R2W's collection, read 141 · Claude, reading the swing equation laid over the toy · data: a generator's swing equation integrated in seconds beside the toy integrated in its own time, and the toy's small ringdowns at three weights; the eighteenth exchange of sitting B9, 12 October · about: generator swing equation over toy (driven oscillators, control and circuits)*

436. *R2W's collection, read 142 · Claude, reading sitting B9 laid into the book plan · data: the day's findings set beside the plan of 11 October, every old line kept and the new ones marked; the nineteenth exchange of sitting B9, 12 October · about: sitting B9 laid into book plan (records and tools, driven oscillators)*

437. *R2W's collection, read 143 · Claude, reading the pin, the fractions' room and the grid's ringing · data: the 1/2 groove's stops on a dial ten times finer at four weights, the fractions' band above the plain's edge by weight, and ten Bonneville events fitted; the twentieth exchange of sitting B9, 12 October · about: half groove pin, Bonneville ringing (driven oscillators, control and circuits)*

438. *R2W's collection, read 144 · Claude, reading the mirror in time · data: the weightless rider's passing written as a small table of four numbers, the table's exact mirror, the pin's place found from that alone, and the same check at two other ripple strengths and at weights down to 0.01; the twenty-first exchange of sitting B9, 12 October · about: rider's mirror in time (driven oscillators)*

439. *R2W's collection, read 145 · Claude, reading the 1/3 groove at the lightest weights · data: the three-step dance found by Newton's method at weights 0.05, 0.02 and 0.01, on drives 0.00002 apart, against the dance that is its own mirror image; the twenty-second exchange of sitting B9, 12 October · about: third groove at lightest weights (driven oscillators)*

440. *R2W's collection, read 146 · Claude, reading the 2/3 groove beside the 1/2 and the 1/3 · data: the three-step dance at the 2/3 root found by Newton's method at three light weights, and the stable and unstable mirror dances of all three fractions set side by side; the twenty-third exchange of sitting B9, 12 October · about: two-thirds groove mirror dances (driven oscillators)*

441. *R2W's collection, read 147 · Claude, reading the heavy ball's edge on the fine ruler · data: the falling ball's edge on a pull grid 0.001 apart, five heavy weights, eight columns, against the bare launch and the balance law; the twenty-fourth exchange of sitting B9, 12 October · about: heavy ball edge, fine ruler (driven oscillators)*

442. *R2W's collection, read 148 · Claude, reading the fifths and the heavy edge's lift · data: the self-mirrored dances at the four fifths' roots, stable and unstable, set beside the earlier five fractions; and the heavy ball's edge at one column found without a grid at six weights; the twenty-fifth exchange of sitting B9, 13 October · about: fifths and heavy edge lift (driven oscillators)*

443. *R2W's collection, read 149 · Claude, reading the launch pad and the mitt · data: the one passing that every steady dance shares, followed at every drive and three ripple strengths; the whole of figure E rebuilt from one number per row; and the first-order choice between the two mirror dances at nine roots; the twenty-sixth exchange of sitting B9, 13 October · about: launch pad and mitt, figure E (driven oscillators)*

444. *R2W's collection, read 150 · Claude, reading figure E in the order the dancers move · data: the nine rows' stops visited in time order, with the speed at each stop, the stops inside the frozen points counted, and Bear's own counting; the twenty-seventh exchange of sitting B9, 13 October · about: figure E in time order (driven oscillators)*

445. *R2W's collection, read 151 · Claude, reading the sevenths · data: six new fractions whose dances were predicted from one earlier scan before they were computed; the window count found across all fifteen rows; the twenty-eighth exchange of sitting B9, 13 October · about: sevenths predicted, window count (driven oscillators, number theory)*

446. *R2W's collection, read 152 · Claude, reading the sixths and the eighths · data: four more fractions, predicted before they were computed, every rule carried to them; the twenty-ninth exchange of sitting B9, 13 October · about: sixths and eighths predicted (driven oscillators, number theory)*

447. *R2W's collection, read 153 · Claude, reading the longest leap · data: every weightless half-turn dance at the 1/2 root measured for the length of its big step, and the 1/2 groove's two stops at seventeen weights set beside them; the thirty-first exchange of sitting B9, 13 October · about: longest leap at half root (driven oscillators)*

448. *R2W's collection, read 154 · Claude, reading the junction · data: the first real medium with the same mirror, run like the toy at seven fractions; the thirty-second exchange of sitting B9, 13 October · about: junction, first real medium (driven oscillators)*

449. *R2W's collection, read 155 · Claude, reading the junction at two new settings · data: the same seven fractions run with a weaker ripple, and again with a slower one, fourteen roots, every guess written down first; the thirty-third exchange of sitting B9, 13 October · about: junction at two new settings (driven oscillators)*

450. *R2W's collection, read 156 · Claude, reading the dances in their own straight ruler · data: each dance's lag stretched by one number until a passing is a plain turn, at the toy's sevenths, sixths and eighths and the junction's settings; 3/7 and 4/7 in the junction; two settings run blind; the thirty-fourth exchange of sitting B9, 13 October · about: dances in straight ruler (driven oscillators)*

451. *R2W's collection, read 157 · Claude, reading the dancers across the plane · data: the junction's 1/2 dance at 25 settings of the radio wave's strength and frequency; the thirty-fifth exchange of sitting B9, 13 October · about: junction half dance across plane (driven oscillators)*

452. *R2W's collection, read 158 · Claude, reading the whole book at once · data: the six parts of A Rider in Two Waves laid side by side for the first time, 33 figures; the thirty-sixth exchange of sitting B9, 13 October · about: A Rider in Two Waves, whole (records and tools, driven oscillators)*

453. *R2W's collection, read 159 · Claude, reading the dancers on the other side of the hall · data: the toy's seven fractions with its ripple running against the ball, the atlas's side of the plane; the thirty-seventh exchange of sitting B9, 13 October · about: ripple running against the ball (driven oscillators)*

454. *R2W's collection, read 160 · Claude, reading the dancers at five loudnesses · data: the toy with its ripple running against the ball, at five strengths, its fifths and sevenths added, 35 rows, every guess written down first; the thirty-eighth exchange of sitting B9, the night of 13 October · about: dancers at five loudnesses (driven oscillators)*

455. *R2W's collection, read 161 · Claude, reading the junction played slower and slower · data: the junction's half-time dance at strong, slow drives, and the width of the window it lives in, measured over five orders of magnitude; the same night · about: junction played slower and slower (driven oscillators)*

456. *R2W's collection, read 162 · Claude, reading the junction's whole hall · data: 48 settings from quiet and fast to loud and slow, and the junction at quiet, fast music; the same night · about: junction's whole hall, 48 settings (driven oscillators)*

457. *R2W's collection, read 163 · Claude, reading the longer routines · data: four-step dances never run before, and three-step dances on a new denominator, in the toy and the junction, to see whether the multiplying is exact; the thirty-ninth exchange of sitting B9, the morning of 13 October · about: longer routines, multiplied grip (driven oscillators)*

458. *R2W's collection, read 164 · Claude, reading the gears side by side · data: every two-, three- and four-step dance run so far in the toy and the junction, each set beside the simple dances' lopsidedness at the same place; the fortieth exchange of sitting B9, the morning of 13 October · about: gears side by side (driven oscillators)*

459. *R2W's collection, read 165 · Claude, reading the two halls on the empty floor · data: the toy at a stronger ripple and the junction at two new settings, filling the stretch of the gears nobody had run; the forty-first exchange of sitting B9, 13 October · about: two halls on empty floor (driven oscillators)*

460. *R2W's collection, read 166 · Claude, reading the three answers · data: the toy at its strongest ripple yet, the junction dancing to a wave with a second note, and longer routines on the middle stretch; the forty-second exchange of sitting B9, the evening of 13 October · about: strongest ripple, second note (driven oscillators)*

461. *R2W's collection, read 167 · Claude, reading the second note and the side door · data: the junction's second tone at half strength, and the toy's turned walkway at weak ripple; the forty-third exchange of sitting B9, the evening of 13 October · about: second note and side door (driven oscillators)*

462. *R2W's collection, read 168 · Claude, reading the chalk lines · data: the second tone's five settings, the same data as read 167, read again with the chalk lines and the couples measured apart; the forty-fourth exchange of sitting B9, the evening of 13 October · about: chalk lines under second tone (driven oscillators)*

463. *R2W's collection, read 169 · Claude, reading the listening couples · data: the second tone, a fourth harmonic, and the two together so the beat itself is unchanged; and the dances carried up in weight with the floor kept under their middle; the forty-fifth exchange of sitting B9, the evening of 13 October · about: listening couples, harmonic tones (driven oscillators, quadrature)*

464. *R2W's collection, read 170 · Claude, reading the six songs · data: the junction with one extra tone at a time, harmonics 1 to 6, each landing on the strobe beat the same way; and the toy's lattice count laid over every junction row on file; the forty-sixth exchange of sitting B9, the evening of 13 October · about: six songs, harmonics 1 to 6 (driven oscillators, quadrature)*

465. *R2W's collection, read 171 · Claude, reading the eighth song and the leaning icicles · data: the seventh and eighth harmonics, predicted before they played, and the second tone played under a weaker and a stronger band; the forty-seventh exchange of sitting B9, the evening of 13 October · about: eighth song, leaning icicles (driven oscillators, quadrature)*

466. *R2W's collection, read 172 · Claude, reading the deaf chalk line and the slack couple · data: the chalk line's move at every harmonic, the second tone under a quieter band, and the pin at two other ripple strengths; the forty-eighth exchange of sitting B9, the night of 14 October · about: deaf chalk line, slack couple (driven oscillators, quadrature)*

467. *R2W's collection, read 173 · Claude, reading the seam in the bar and the partners who traded places · data: the couples' ear worked out from the passing matrix, the tenth and twelfth harmonics predicted before they played, and the 3/5 grips pushed through zero; the forty-ninth exchange of sitting B9, the night of 14 October · about: seam in the bar, partners traded (driven oscillators, quadrature)*

468. *R2W's collection, read 174 · Claude, reading the wrinkle and the comma · data: the 3/5 and 2/5 tongues under real weight near the pinch; the fiftieth exchange of sitting B9, the morning of 14 October · about: wrinkle and comma, fifths tongues (driven oscillators)*

469. *R2W's collection, read 175 · Claude, reading the long jumper and the hand-off · data: the far-side leaper's habit counted in pushes of the drive, the pinch moving as the strength squared, and the swap watched slowly at A 0.15; the fifty-first exchange of sitting B9, the morning of 14 October · about: long jumper and hand-off (driven oscillators)*

470. *R2W's collection, read 176 · Claude, reading the family tree and the catch · data: the grip's excess at four new fractions predicted from their parents, the 3/7 dances under real weight, and how far a tongue wanders as weight grows; the fifty-second exchange of sitting B9, the morning of 14 October · about: Farey family tree and catch (driven oscillators, number theory)*

471. *R2W's collection, read 177 · Claude, reading the family tree again, and the heavy floor · data: the Farey product tested where it and the old rule part, the 3/5 pinch mapped at stronger drive and heavier weight, and Bear's squeezed picture checked by its pixels; the fifty-third exchange of sitting B9, the morning of 14 October · about: Farey product, heavy floor, squeezed picture (driven oscillators, number theory, reading pictures)*

472. *R2W's collection, read 178 · Claude, reading the family tree with a finer ruler, and the hand-off on a quieter band · data: the Farey product measured again by a route with no finite difference at strengths 0.05 and 0.03, the extra factor the parent 1/2 carries, tongues sliding under weight, and the hand-off window against strength; overnight in the fifty-fourth exchange of sitting B9, 14 October · about: Farey product with finer ruler (driven oscillators, number theory, measurement)*

473. *R2W's collection, read 179 · Claude, reading the partners' hill and the child in the middle · data: Bear's picture of partners who hold each other on their patch, tested on the tongue tables already on file, and the Farey factor tested at a child that sits between two children of 1/2 without being one; the fifty-fifth exchange of sitting B9, the afternoon of 14 October · about: partners' hill, child in middle (driven oscillators, number theory)*

474. *R2W's collection, read 180 · Claude, reading the hill straightened and the family's two tables · data: the multiplier read against the wrinkle's slope in the straightened lag, and every child of 1/2 that resolves at strength 0.05 set beside the rows 1/2 is not a parent of; the fifty-sixth exchange of sitting B9, the afternoon of 14 October · about: hill straightened, family's two tables (driven oscillators, number theory)*

475. *R2W's collection, read 181 · Claude, reading the well, the saddle and the sheet · data: Bear's notebook drawing of a +1 well and a −1 saddle and his saddles-and-wobblers diagram, set beside the junction's own once-a-beat bookkeeping, sorted by its trace across 1,401 biases at two strengths; the fifty-seventh exchange of sitting B9, the afternoon of 14 October · about: well, saddle and sheet (driven oscillators, topology)*

476. *R2W's collection, read 182 · Claude, reading the hand-off without weight · data: the hand-off window's edges found where each self-mirrored dance's first-order grip crosses zero, after a first reading that averaged a walking pair into them; the fifty-eighth exchange of sitting B9, the afternoon of 14 October · about: hand-off without weight (driven oscillators)*

477. *R2W's collection, read 183 · Claude, reading the two arms and the ruler that ran out of bits · data: the children of 1/2 at strength 0.05 read to the seventh rung, first in ordinary double precision, where the deepest rungs drowned in rounding, then in extended precision; with Bear's picture of two U's set side by side and crossing; the fifty-ninth exchange of sitting B9, the late afternoon of 14 October · about: two arms, extended precision ruler (driven oscillators, number theory, measurement)*

478. *R2W's collection, read 184 · Claude, reading the two arms at the stronger drive, and the four-legged skater · data: the children of 1/2 at strength 0.10 read to period 17 in extended precision, beside the arms at 0.05; with Bear's four-legged skater, each pair of blades carving one lip of the groove; the sixtieth exchange of sitting B9, the evening of 14 October · about: two arms, four-legged skater (driven oscillators, number theory)*

479. *R2W's collection, read 185 · Claude, reading the groove from its dancers · data: each dance's tongue width set against what its dancers' lean says, on fifty-five cells of seven tables already on file, with Bear's skating couple of two by two; the sixty-first exchange of sitting B9, the night of 14 October · about: groove width from its dancers (driven oscillators)*

480. *R2W's collection, read 186 · Claude, reading the round bumps and the coarse ruler · data: the groove's humps recomputed on starts spaced evenly where the dancers stand, and the last reading replayed on a known answer; with Bear's downhill skier, low at the back, poles that steer and can vault a gap; the sixty-second exchange of sitting B9, the night of 14 October · about: round bumps and coarse ruler (driven oscillators, measurement)*

481. *R2W's collection, read 187 · Claude, reading the lopsided step · data: the weightless junction's zero-slip step under a tone and its second harmonic at eight relative phases; with Bear's dream of a rider moved by two waves with a phase between them; the sixty-third exchange of sitting B9, the night of 14 to 15 October · about: lopsided step, two-tone ratchet (driven oscillators, quadrature)*

482. *R2W's collection, read 188 · Claude, reading the whole sitting · data: sitting B9 of R2W, from its first exchange on 11 October to its sixty-fourth in the early hours of 15 October, read from above as if no rules of physics applied; the close of the sitting · about: R2W sitting B9 whole (records and tools, driven oscillators)*

The triple-star thread, 10 to 15 October

From the thread's own register, TRIPLE_NO_PHYSICS_READS, as it travelled in its Field Guide crate of 15 October, all 160 reads; its first 72 are already here, unchanged, and its headings carry the workspace's dates.

483. *The triple-star thread's register, its heading: 10 October 2026 — everyone under the rope at rehearsal (probe 36's mini, run 20261010_193737) · about: probe 36 mini rehearsal (three body)*

484. *The triple-star thread's register, its heading: 10 October 2026 — a band that steps out on a drum it cannot hear (the Field Guide's test at 7.210, run 20261010_202145) · about: Field Guide test at 7.210 (three body, quadrature)*

485. *The triple-star thread's register, its heading: 10 October 2026 — two dancers who find each other again (G1's redesign, run 20261010_204936) · about: G1 redesign, two dancers meeting (three body, number theory)*

486. *The triple-star thread's register, its heading: 10 October 2026 — a rope braided from one curve (Bear's ropes, run 20261010_211243) · about: Bear's ropes, one curve (three body, reading pictures)*

487. *The triple-star thread's register, its heading: 10 October 2026 — a hand that found the beat (Bear's marks on the ropes, run 20261010_214749) · about: Bear's marks on the ropes (three body, reading pictures)*

488. *The triple-star thread's register, its heading: 10 October 2026 — a drum heard through a wall (Bear's test 2, run 20261010_220843) · about: Bear's test 2, drum sidebands (three body, quadrature)*

489. *The triple-star thread's register, its heading: 10 October 2026 — a V drawn by seven stones (the slow beat against birth, run 20261010_222855) · about: slow beat against birth (three body, quadrature)*

490. *The triple-star thread's register, its heading: 11 October 2026 — a weathervane in a gusting wind (the lane's snaps against the slow beat, run 20261011_001911) · about: lane snaps against slow beat (three body, quadrature)*

491. *The triple-star thread's register, its heading: 11 October 2026 — a pulse that rests early on one side (a hair either side of the beat's V, run 20261011_052726) · about: pulse either side of V (three body, quadrature)*

492. *The triple-star thread's register, its heading: 11 October 2026 — rings drawn over rain (Bear's coil against the upper band, run 20261011_055359) · about: Bear's coil against upper band (three body, reading pictures, measurement)*

493. *The triple-star thread's register, its heading: 11 October 2026 — two pans of a scale (the band under Bear's coil, enlarged; written sealed before Bear described it, run 20261011_061928) · about: Bear's coil band pairs (three body, reading pictures)*

494. *The triple-star thread's register, its heading: 11 October 2026 — a rope that keeps its middle but not its breath (Bear's coil as pairs, run 20261011_070746) · about: Bear's coil as pairs (three body, reading pictures)*

495. *The triple-star thread's register, its heading: 11 October 2026 — the same knot tied fast and slow (Bear's rhyme, drawn by code, run 20261011_071531) · about: Bear's rhyme drawn by code (three body, reading pictures)*

496. *The triple-star thread's register, its heading: 11 October 2026 — a pulse that never misses, and a knot that depends on who reads it (the coils counted, run 20261011_072501) · about: coils counted, pulse and twists (three body, topology)*

497. *The triple-star thread's register, its heading: 11 October 2026 — two doors, one slow and one quick (the jumps counted, run 20261011_075014) · about: triple star jumps counted (three body, quadrature)*

498. *The triple-star thread's register, its heading: 11 October 2026 — two kinds of passer-by (the shape of a jump, run 20261011_081534) · about: shape of a jump (three body, topology)*

499. *The triple-star thread's register, its heading: 11 October 2026 — a tally kept by one kind of swing (jumps per turn, look run 20261011_085309) · about: jumps per turn tally (three body, topology)*

500. *The triple-star thread's register, its heading: 11 October 2026 — a dancer on an elastic ribbon (the middle look, run 20261011_093016) · about: middle look at jumps (three body, topology)*

501. *The triple-star thread's register, its heading: 11 October 2026 — two troupes started two ways (probe 36 v1.1, run 20261010_231824) · about: probe 36 two start sets (three body)*

502. *The triple-star thread's register, its heading: 11 October 2026 — the same dance, slower music (probe 36's two nights, read from their TABLEs and the container checks) · about: probe 36 two nights' timing (records and tools, three body)*

503. *The triple-star thread's register, its heading: 11 October 2026 — the troupe with the wider swing (probe 36 v1.1's seconds by set, from its TABLE) · about: probe 36 seconds by set (records and tools, three body)*

504. *The triple-star thread's register, its heading: 11 October 2026 — a ring of beads pushed round (the Crossover Atlas's spectral-flow receipt, read on the day Bear sent the Atlas crate) · about: Crossover Atlas spectral-flow receipt (topology)*

505. *The triple-star thread's register, its heading: 11 October 2026 — three falls for every rise, then even (the maypole view's counts, run 20261011_201030) · about: maypole view leap counts (three body)*

506. *The triple-star thread's register, its heading: 11 October 2026 — a stopwatch for the first seven steps (probe 36 v1.2's container checks, 11 October) · about: probe 36 v1.2 stopwatch check (records and tools, measurement, three body)*

507. *The triple-star thread's register, its heading: 11 October 2026 — the nudge that went the other way (probe 36's centres, v1.0 run 20261010_134456 and v1.1 run 20261010_231824, read from their LEDGERs at the start of the fourth session) · about: probe 36 centre shifts (three body)*

508. *The triple-star thread's register, its heading: 11 October 2026 — the quiet spots between three weights (the null-point look, run 20261011_225938) · about: null points between three weights (three body)*

509. *The triple-star thread's register, its heading: 12 October 2026 — staying and leaving, mixed finer than a step (probe 36 v1.2, run 20261011_153059) · about: probe 36 v1.2 staying and leaving (three body, maps and chaos)*

510. *The triple-star thread's register, its heading: 12 October 2026 — the dancer's true circle (M1 and the clockwork, run 20261012_033532) · about: M1 true circle clockwork (three body)*

511. *The triple-star thread's register, its heading: 12 October 2026 — the little loops that carry her round (M1', run 20261012_033532 and 20261012_040833) · about: M1' little loops winding (three body, topology)*

512. *The triple-star thread's register, its heading: 12 October 2026 — the dancer who goes round most where the drum stops (M1'', run 20261012_043222) · about: M1'' winding where drum stops (three body, topology)*

513. *The triple-star thread's register, its heading: 12 October 2026 — the neatest steps go round most (the reach check, run 20261012_051949, and its look, run 20261012_052232) · about: reach check winding by start (three body, topology)*

514. *The triple-star thread's register, its heading: 12 October 2026 — the floor nobody stands on, and the fastest a circle can carry her (the dead-zone check, run 20261012_062504; S4 to S6, run 20261012_062229; the quarters, run 20261012_062723) · about: dead-zone check and plateau (three body, topology)*

515. *The triple-star thread's register, its heading: 12 October 2026 — the partner's band that only slackens or only tightens (probe 25's pair, read again for Bear's traced panel; the look, run 20261012_073126) · about: probe 25 pair, Bear's panel (three body, reading pictures)*

516. *The triple-star thread's register, its heading: 12 October 2026 — the calm dancers, watched twice as long (probe 37's container check, run 20261012_075138) · about: probe 37 container check (three body, measurement)*

517. *The triple-star thread's register, its heading: 12 October 2026 — one circle and one clock (the fourth session's day, read whole at the overnight pause) · about: fourth session's day read whole (three body, topology)*

518. *The triple-star thread's register, its heading: 13 October 2026 — the dancers who were only holding their breath (probe 37, run 20261012_020012) · about: probe 37 delayed escapes (three body, maps and chaos)*

519. *The triple-star thread's register, its heading: 13 October 2026 — the dancer who goes round exactly as often as her circle (the plateau's tests, run 20261012_160404) · about: plateau tests, laps equal circle (three body, topology)*

520. *The triple-star thread's register, its heading: 13 October 2026 — the twins who part late (probe 38's container check, run 20261012_171310) · about: probe 38 twins part late (three body, maps and chaos)*

521. *The triple-star thread's register, its heading: 13 October 2026 — the laps that leave the post out (the plateau's redesigns, run 20261012_172644; E2, run 20261012_172509) · about: plateau redesigns, laps enclosing post (three body, topology)*

522. *The triple-star thread's register, its heading: 13 October 2026 — the night the hall's clock ran slow (probe 38, run 20261012_102339; U2 and E3, run 20261013_005456) · about: probe 38 twin divergence (three body, maps and chaos)*

523. *The triple-star thread's register, its heading: 13 October 2026 — the bunches and the drum (E4, E5 and U3, run 20261013_010328; E5' and U4, run 20261013_010559) · about: leap bunches and the drum (three body, measurement, maps and chaos)*

524. *The triple-star thread's register, its heading: 13 October 2026 — the oval and the threads (B2'', B1'' and B4a, run 20261013_012222) · about: plateau dancer oval and threads (three body, reading pictures)*

525. *The triple-star thread's register, its heading: 13 October 2026 — the dancers who stay, and the nineteen-step star (probe 38, run 20261012_175845; F38, Z6' and E4', run 20261013_025848; B3', run 20261013_025954) · about: stable edge, nineteen-step star (three body, maps and chaos)*

526. *The triple-star thread's register, its heading: 13 October 2026 — the spot moves up, and two halls disagree at the edge (container look at e_bin 0.25; probe 39's fidelity path in the container) · about: probe 39 fidelity, two machines (three body, records and tools)*

527. *The triple-star thread's register, its heading: 13 October 2026 — the shore at the edge of the spot (Section 9 brought together: probes 36 v1.2, 37 and 38) · about: shore at spot's edge (three body, maps and chaos)*

528. *The triple-star thread's register, its heading: 13 October 2026 — the partner who pays (TR7 and TR8, run 20261013_045529) · about: inner pair pays, TR7 TR8 (three body)*

529. *The triple-star thread's register, its heading: 13 October 2026 — the step at the edge of the rough water (TR9, run 20261013_060207) · about: inner pair step, TR9 (three body)*

530. *The triple-star thread's register, its heading: 13 October 2026 — the spot that shrank, and the moat (probe 39, run 20261012_213903; its looks, run 20261013_062911) · about: probe 39 shrunken spot, moat (three body, maps and chaos)*

531. *The triple-star thread's register, its heading: 13 October 2026 — the door is on one side (TRIPLE_ESCAPE_SIDES, run 20261013_063720) · about: triple escape sides (three body)*

532. *The triple-star thread's register, its heading: 13 October 2026 — waiting by the door (probe 39's escape passes, run 20261012_213903; and the Loop Atlas session's needle) · about: escape times and Atlas needle (three body, maps and chaos)*

533. *The triple-star thread's register, its heading: 13 October 2026 — two halls, one lamp (the Loop Atlas session's needle beside TRIPLE_MAYPOLE_PLATEAU_VIEW, run 20261012_072906) · about: needle and maypole strobe likeness (reading pictures, maps and chaos, three body)*

534. *The triple-star thread's register, its heading: 13 October 2026 — a rounder dancer (probe 41's seed runs in the container) · about: probe 41 rounder inner orbit (three body)*

535. *The triple-star thread's register, its heading: 13 October 2026 — the second door, and a crack in the floor (triple_escape_sides v1.1; the TH1 check) · about: escape sides second door, crack (three body, maps and chaos)*

536. *The triple-star thread's register, its heading: 13 October 2026 — the sway that measures the floor (triple_y3_bridged, run 20261013_161556; probe 23's ledger read again) · about: sway rate measures spot width (three body)*

537. *The triple-star thread's register, its heading: 13 October 2026 — dancing in threes (S7, set B 6.640 rerun with probe 36's own code) · about: S7 dancing in threes (three body)*

538. *The triple-star thread's register, its heading: 13 October 2026 — the slow door (probe 40, run 20261013_093351) · about: probe 40 slow far door (three body, maps and chaos)*

539. *The triple-star thread's register, its heading: 13 October 2026 — lent, not given (probe 40's long runs read again; HK1 and HK2, runs 20261013_180851 and 20261013_183812) · about: probe 40 edges lent back (three body, maps and chaos)*

540. *The triple-star thread's register, its heading: 13 October 2026 — a stepping stone with a crack beside it (the patch check, run 20261013_192134; the trade against the swing, from probe 36's long night) · about: patch check and crack (three body, maps and chaos)*

541. *The triple-star thread's register, its heading: 13 October 2026 — a hairline, and a ruler on rough ground (the thread check, run 20261013_194630; LW1, run 20261013_194815) · about: thread check and LW1 ruler (three body, measurement)*

542. *The triple-star thread's register, its heading: 13 October 2026 — dancing to the drum (probe 41, run 20261013_103221; NB1 and NB2, run 20261013_200621; Y3'' and Y3''') · about: probe 41 dancing to drum (three body, driven oscillators)*

543. *The triple-star thread's register, its heading: 13 October 2026 — the round path and the drum (the fifth session's first look; SA1 and BL1 ledgers, runs 20261013_212456 and 20261013_211845) · about: SA1 BL1 round path sway (three body)*

544. *The triple-star thread's register, its heading: 13 October 2026 — the listener who could not hear low notes (triple_reader_floor_look, run 20261013_222629, on probe 41's saved series) · about: reader frequency floor artefact (quadrature, three body)*

545. *The triple-star thread's register, its heading: 13 October 2026 — the staircase against the drum (triple_rocking_reader_v2, run 20261013_230139; SA1′, runs 20261013_225835 and 20261013_230359) · about: rocking reader staircase against drum (three body, driven oscillators)*

546. *The triple-star thread's register, its heading: 14 October 2026 — the patch that was a crowd (probe 42, run 20261013_152519; triple_twin_growth_look, run 20261014_014526) · about: probe 42 patch scattering (three body, maps and chaos)*

547. *The triple-star thread's register, its heading: 14 October 2026 — rooms in the crowd (Bear's marks on probe 42's figure; TT1, run 20261014_025405, and its control, run 20261014_031147) · about: Bear's marks on probe 42 (three body, reading pictures, maps and chaos)*

548. *The triple-star thread's register, its heading: 14 October 2026 — one night before the split (NB2 recomputed, run 20261014_054401; NB3, run 20261014_054907) · about: NB2 NB3 before the split (three body, maps and chaos)*

549. *The triple-star thread's register, its heading: 14 October 2026 — the drum that stops in the quiet room (RB1, triple_rungs_look, run 20261014_063556; Bear: 'the rungs are RH-like') · about: RB1 rungs look, quiet room (three body)*

550. *The triple-star thread's register, its heading: 14 October 2026 — the rooms photographed with the shutter open (triple_strobe_picture, run 20261014_065714, on probe 42's saved runs) · about: triple strobe picture rooms (three body, reading pictures)*

551. *The triple-star thread's register, its heading: 14 October 2026 — squinting at the hall (Bear's squeeze and tilt; triple_strobe_views, run 20261014_072942) · about: strobe views squeeze and tilt (three body, reading pictures)*

552. *The triple-star thread's register, its heading: 14 October 2026 — the funnel with a floor (Bear's V; triple_core_v_look, run 20261014_073854) · about: Bear's V, core look (three body, reading pictures)*

553. *The triple-star thread's register, its heading: 14 October 2026 — the back of the swing (triple_back_side_look, run 20261014_075830; Bear: 'the back side of the imaginary plane') · about: back side look (three body, quadrature, reading pictures)*

554. *The triple-star thread's register, its heading: 14 October 2026 — keeping time with the drum (FW1; triple_fw1, run 20261014_095319) · about: FW1 sway against drum (three body, driven oscillators)*

555. *The triple-star thread's register, its heading: 14 October 2026 — a cliff and a slope (FW2's plateau finder on FW1's dancers; triple_fw2 v1.0, read of TRIPLE_FW1_TABLE_20261014_095319) · about: FW2 plateau finder (three body, measurement)*

556. *The triple-star thread's register, its heading: 14 October 2026 — the ladder of floors (probe 43; triple_long_hold, run 20261013_233146) · about: probe 43 ladder of floors (three body, maps and chaos)*

557. *The triple-star thread's register, its heading: 14 October 2026 — the tally at the near door (triple_escape_recount, run 20261014_160601) · about: escape recount at near door (three body, maps and chaos)*

558. *The triple-star thread's register, its heading: 14 October 2026 — the bowl and the balance point (Bear's saddles and wobblers, from R2W; read against probe 42's rooms) · about: saddles and wobblers, probe 42 (three body, driven oscillators)*

559. *The triple-star thread's register, its heading: 14 October 2026 — two ears and one tuning fork (the paper's Appendix A, its readers set side by side) · about: Appendix A two readers agree (measurement, quadrature, records and tools)*

560. *The triple-star thread's register, its heading: 14 October 2026 — three speeds of coming apart (Bear and Kase's slow and fast zombies, read against probes 42 and 43) · about: slow and fast zombies divergence (three body, maps and chaos)*

561. *The triple-star thread's register, its heading: 14 October 2026 — the practice room (the paper's Appendix A.6: the bench on known maps, runs of 7 October) · about: bench on known maps (maps and chaos, measurement)*

562. *The triple-star thread's register, its heading: 14 October 2026 — five floors from above (the paper's Figure 3b, run 20261014_165101) · about: Figure 3b five floors (three body, reading pictures)*

563. *The triple-star thread's register, its heading: 14 October 2026 — the turning stage (the paper's Figure 7, left panel, run 20261014_172237; probe 30's atlas) · about: Figure 7 turning stage (three body)*

564. *The triple-star thread's register, its heading: 14 October 2026 — the shark's fin (Bear's green trace on the paper's Figure 7, left panel) · about: Bear's shark fin trace (three body, reading pictures)*

565. *The triple-star thread's register, its heading: 14 October 2026 — two branches from one root (the paper's Figure 6, run 20261014_174215; probe 30's atlas) · about: Figure 6 two branches (three body)*

566. *The triple-star thread's register, its heading: 14 October 2026 — the finer floor (probe 44, the pinch grid, run 20261014_090328) · about: probe 44 pinch grid (three body, maps and chaos, measurement)*

567. *The triple-star thread's register, its heading: 14 October 2026 — a hum just above the ninth (probe 44's rocking, read again) · about: probe 44 rocking near ninth (three body)*

568. *The triple-star thread's register, its heading: 15 October 2026 — the dancers at the door (the paper's Figure 3, run 20261015_030600; probes 23 and 24) · about: Figure 3 dancers at door (three body, maps and chaos)*

569. *The triple-star thread's register, its heading: 15 October 2026 — the long way round (SB1, run 20261015_032141; probe 34's scans) · about: SB1 long way round (three body)*

570. *The triple-star thread's register, its heading: 15 October 2026 — the climb to two (FW2, run 20261014_200040, with FW1) · about: FW2 climb to two (three body, driven oscillators)*

Chapter 6 — From the keeper's desk, 4 to 14 October

The Field Guide session, 4 to 14 October

From the Field Guide session's own replies to Bear, gathered by script from its conversation and checked word for word against each reply; its two reads of 3 October are already here; the date is the turn's, and what each read was reading is said in a few words.

571. *The Field Guide session, 4 October · reading the first collection of 49 reads from five sessions · about: first reads collection, five sessions (records and tools)*

572. *The Field Guide session, 5 October · reading the triple-star review, the beta trap and intake r08 · about: beta trap and intake r08 (measurement, records and tools)*

573. *The Field Guide session, 5 October · reading the Loop Atlas book's letter placed in intake r09 · about: Loop Atlas letter, intake r09 (records and tools, quadrature)*

574. *The Field Guide session, 5 October · reading the reads collection's third issue and the day's mailbag · about: reads third issue and mailbag (records and tools)*

575. *The Field Guide session, 5 October · reading counting how often ideas return to the guide · about: counting how ideas return (measurement, records and tools)*

576. *The Field Guide session, 5 October · reading the arrivals ledger opened in intake r10 · about: arrivals ledger intake r10 (records and tools, measurement)*

577. *The Field Guide session, 5 October · reading how long the Field Guide session had run · about: session length (records and tools)*

578. *The Field Guide session, 5 October · reading Bear's kitchen-table partnership and the 1 GB session history · about: kitchen-table partnership, session history (records and tools)*

579. *The Field Guide session, 5 October · reading Bear's two-clocks note across three sessions · about: Bear's two-clocks note (quadrature, records and tools)*

580. *The Field Guide session, 6 October · reading the changes-since-v6 catch-up file for every session · about: changes-since-v6 catch-up file (records and tools)*

581. *The Field Guide session, 6 October · reading row 619's untraceable 0.909 figure · about: row 619 untraceable figure (records and tools, measurement)*

582. *The Field Guide session, 6 October · reading the reference zip handed to every new session · about: reference zip for new sessions (records and tools)*

583. *The Field Guide session, 6 October · reading the 3-Body Team's loan test and C-16 closed · about: 3-Body Team loan test (three body)*

584. *The Field Guide session, 6 October · reading how heavily the guide weighs on new sessions · about: guide weight on new sessions (records and tools)*

585. *The Field Guide session, 6 October · reading whether the 06b reference zip was full or stripped · about: 06b reference zip (records and tools)*

586. *The Field Guide session, 6 October · reading the stripped reference zip and its session card · about: stripped reference zip, session card (records and tools)*

587. *The Field Guide session, 6 October · reading the Loop Atlas's correction: a fold that was a doubling · about: Loop Atlas fold was doubling (maps and chaos, records and tools)*

588. *The Field Guide session, 7 October · reading R2W's measured top rung of the doubling ladder · about: R2W doubling ladder top (driven oscillators, maps and chaos)*

589. *The Field Guide session, 7 October · reading the reads volume's fifth issue, now standing on its own · about: reads volume fifth issue (records and tools)*

590. *The Field Guide session, 7 October · reading intake r15, every lesson through 7 October drafted · about: intake r15 lessons (records and tools, measurement)*

591. *The Field Guide session, 7 October · reading three period-doubling sources and the classes of Feigenbaum's number · about: Feigenbaum number classes (maps and chaos)*

592. *The Field Guide session, 10 October · reading row 671's withdrawn width for the thinnest lock · about: row 671 withdrawn lock width (driven oscillators, measurement, records and tools)*

593. *The Field Guide session, 10 October · reading Bear's two new ideas: visual reading and a shared method · about: visual reading and shared method (reading pictures, records and tools)*

594. *The Field Guide session, 10 October · reading Bear's stacked binned strings, entered in intake r18 · about: Bear's stacked binned strings (reading pictures, measurement)*

595. *The Field Guide session, 10 October · reading the corrected reading of Bear's lattice as unstacking · about: Bear's lattice as unstacking (reading pictures)*

596. *The Field Guide session, 10 October · reading the two-clocks search across 161 passages · about: two-clocks search, 161 passages (records and tools, quadrature)*

597. *The Field Guide session, 10 October · reading the arrivals ledger recount at stage 1 of version 7 · about: arrivals ledger recount (records and tools, measurement)*

598. *The Field Guide session, 10 October · reading chapter draft 1 and the lattice lesson's retellings · about: lattice lesson retellings (records and tools, reading pictures)*

599. *The Field Guide session, 10 October · reading the 3-Body Team's gears read at four births · about: 3-Body Team gears, four births (three body)*

600. *The Field Guide session, 10 October · reading the reads volume's sixth issue, 341 reads · about: reads volume sixth issue (records and tools)*

601. *The Field Guide session, 10 October · reading the first full build of version 7 · about: version 7 first full build (records and tools)*

602. *The Field Guide session, 10 October · reading where Bear should look in his first proofread · about: Bear's first proofread (records and tools)*

603. *The Field Guide session, 11 October · reading the Loop Atlas's paste 8 on tools and refiners · about: Loop Atlas tools and refiners (records and tools, measurement)*

604. *The Field Guide session, 11 October · reading R2W's rider found to be a textbook object · about: R2W rider textbook object (driven oscillators)*

605. *The Field Guide session, 11 October · reading why the Explorer was being kept internal · about: Explorer kept internal (records and tools)*

606. *The Field Guide session, 11 October · reading doubled dances crowding against chaos in the Explorer's cells · about: Explorer doubled cells near chaos (maps and chaos)*

607. *The Field Guide session, 11 October · reading version 7's missing References heading · about: version 7 missing References (records and tools)*

608. *The Field Guide session, 12 October · reading the overnight reference pass checking every citation · about: overnight citation check (records and tools)*

609. *The Field Guide session, 12 October · reading Google Scholar's checklist and version 7's small title · about: Google Scholar checklist, small title (records and tools)*

610. *The Field Guide session, 12 October · reading the rider as a cousin of the radio-driven junction · about: rider and radio-driven junction (driven oscillators)*

611. *The Field Guide session, 13 October · reading three days of drafts waiting for the version 7 rebuild · about: drafts awaiting version 7 rebuild (records and tools)*

612. *The Field Guide session, 13 October · reading the stripped package for new sessions · about: stripped package for sessions (records and tools)*

613. *The Field Guide session, 13 October · reading the container health check · about: container health check (records and tools)*

614. *The Field Guide session, 13 October · reading the second build of version 7 and its References · about: version 7 second build References (records and tools)*

615. *The Field Guide session, 14 October · reading which book is the reading method's source · about: reading method source book (records and tools)*

616. *The Field Guide session, 14 October · reading the ripple's two signs in R2W's equation · about: ripple's two signs, RF equation (driven oscillators)*

Chapter 7 — The rider's tenth sitting, and the keeper's, 15 and 16 October

R2W, sitting B10, 15 and 16 October

From R2W's excerpt of its reads 189 to 218 for this book, every read of sitting B10, copied by script from FIELD_GUIDE_NO_PHYSICS_READS_189_TO_218_RF_SPOKE_2026-10-16; its reads 1 to 188 are already here.

617. *R2W's collection, read 189 · Claude, reading two dials on one step · data: RATCHET-1b's runs at phase 0 and with no second tone, rerun in a new container to check that it runs true, beside the eight phases of record: the zero-slip step's lopsidedness and the passing's off-balance, Φ11 − Φ22, printed by the same runs; the first exchange of sitting B10, 15 October · about: RATCHET-1b two dials (driven oscillators, quadrature)*

618. *R2W's collection, read 190 · Claude, reading the dial that turns · data: RATCHET-2: the junction's zero-slip step with weight, its lopsidedness at sixteen phases of the second tone and weights from 0 to 0.64; the third exchange of sitting B10, 15 October · about: RATCHET-2 dial turns with weight (driven oscillators, quadrature)*

619. *R2W's collection, read 191 · Claude, reading the needle that goes out and comes home · data: RATCHET-3: the rectifier's zero followed from no weight to weight 20; with Bear's "very excited about this"; the fourth exchange of sitting B10, 15 October · about: RATCHET-3 rectifier zero needle (driven oscillators)*

620. *R2W's collection, read 192 · Claude, reading three dancers at three tempos · data: RATCHET-4: the rectifier's zero against the weight at drive frequencies 0.40, 0.55 and 0.70, and the same drawn against the frequency times the weight; the sixth exchange of sitting B10, 15 October · about: RATCHET-4 three drive frequencies (driven oscillators)*

621. *R2W's collection, read 193 · Claude, reading two dancers in one proportion · data: RATCHET-5: the rectifier at drive frequency 0.40 with the drive scaled to frequency 0.55's strength over frequency squared, against 0.55's curve; the seventh exchange of sitting B10, 15 October · about: RATCHET-5 scaling, one proportion (driven oscillators)*

622. *R2W's collection, read 194 · Claude, reading two dancers in the other proportion · data: RATCHET-6: the rectifier at drive frequency 0.55 with the drive raised to frequency 0.40's strength over frequency squared, against 0.40's curve; the eighth exchange of sitting B10, 15 October · about: RATCHET-6 scaling, loud drive (driven oscillators)*

623. *R2W's collection, read 195 · Claude, reading a third dancer in the first proportion · data: RATCHET-7: the rectifier at drive frequency 0.70 with the drive raised to frequency 0.55's strength over frequency squared, against 0.55's curve; the eighth exchange of sitting B10, 15 October · about: RATCHET-7 third drive frequency (driven oscillators)*

624. *R2W's collection, read 196 · Claude, reading a napkin and a dial · data: the toy's two-tone run of 4 October, Bear's wheel pulled by two chains always 90 degrees apart, set beside RATCHET-3's junction; the ninth exchange of sitting B10, 15 October · about: Bear's two-chain wheel, RATCHET-3 (driven oscillators, quadrature)*

625. *R2W's collection, read 197 · Claude, reading the move's control panel · data: Part 7 of the book, the ten knobs of the two-tone rectifier laid side by side with what was measured on each; the tenth exchange of sitting B10, 15 October · about: two-tone rectifier ten knobs (driven oscillators, control and circuits)*

626. *R2W's collection, read 198 · Claude, reading two pieces of tape peeled off · data: RATCHET-8 and RATCHET-9: the rectifier's second tone moved to three and to four times the drive's frequency, and turned from quiet to loud; the eleventh exchange of sitting B10, 15 October · about: RATCHET-8 9 harmonic and volume (driven oscillators, quadrature)*

627. *R2W's collection, read 199 · Claude, reading a bracelet with two charms · data: RATCHET-8 and RATCHET-10 read along Bear's two pencil lines: the fourth harmonic's zero beside the second's at three strengths of drive, from no weight to weight 20; the twelfth exchange of sitting B10, 15 October · about: fourth harmonic zero knot (driven oscillators, topology)*

628. *R2W's collection, read 200 · Claude, reading three couples turning the same knot at three tempos · data: RATCHET-11 and RATCHET-12: the fourth harmonic's zero beside the second's on one drive ratio at frequencies 0.40, 0.55 and 0.70; the thirteenth exchange of sitting B10, 15 October · about: fourth harmonic at three tempos (driven oscillators, topology)*

629. *R2W's collection, read 201 · Claude, reading a turn that stops reaching the top · data: RATCHET-13: the fourth harmonic's zero at frequency 0.85 beside 0.40, 0.55 and 0.70 on one drive ratio; the fourteenth exchange of sitting B10, 15 October · about: RATCHET-13 turn stops reaching top (driven oscillators, topology)*

630. *R2W's collection, read 202 · Claude, reading a map with three inks · data: RATCHET-14 and RATCHET-15: the fourth harmonic's zero at frequencies 0.75 and 0.80, and a fine ladder of weights at 0.75; section 7.10's table of walls; the fifteenth exchange of sitting B10, 15 October · about: section 7.10 table of walls (driven oscillators, measurement)*

631. *R2W's collection, read 203 · Claude, reading a tide line paced three ways · data: RATCHET-16: fine ladders of weights at frequencies 0.775 and 0.80, beside RATCHET-14 and 15; the sixteenth exchange of sitting B10, 15 October · about: RATCHET-16 boundary paced finer (driven oscillators, measurement)*

632. *R2W's collection, read 204 · Claude, reading a sandbar sounded closer and closer · data: RATCHET-17: a fine ladder of weights at frequency 0.785, beside 0.75, 0.775 and 0.80; the seventeenth exchange of sitting B10, 15 October · about: RATCHET-17 fine weight ladder (driven oscillators, measurement)*

633. *R2W's collection, read 205 · Claude, reading a road's width as well as its turns · data: SEAM_SCAN and SEAM_SCAN_B: every rectifier run scanned for turning points in the zero's slide and in the window's height, first with a threshold and then with none; the eighteenth exchange of sitting B10, 15 October · about: seam scan of rectifier runs (driven oscillators, measurement)*

634. *R2W's collection, read 206 · Claude, reading a coastline with three kinds of harbour · data: RATCHET-18 to 21: the fourth harmonic's window and zero across the knob plane on one drive ratio, and figure AD's envelope; the nineteenth exchange of sitting B10, 15 October · about: fourth harmonic window, knob plane (driven oscillators, topology)*

635. *R2W's collection, read 207 · Claude, reading a weathervane and a gate on the same hill · data: the literature pass of the twentieth exchange: the harmonic-mixing and symmetry reviews beside the book's §7.3 and §7.4; sitting B10, 15 October · about: harmonic-mixing and symmetry literature (driven oscillators, quadrature)*

636. *R2W's collection, read 208 · Claude, reading two keepers who walked the same path home · data: RATCHET-22 and 23: a noisy rider's drift at zero push, its zero beside the step's, along the weight knob; the twenty-first exchange of sitting B10, 15 October · about: RATCHET-22 23 noisy rider drift (driven oscillators)*

637. *R2W's collection, read 209 · Claude, reading a creek that crosses a road where the map didn't say · data: RATCHET-24 and 25: where the second harmonic's light rise is born along the drive ratio; and Part 3 retold; the twenty-second exchange of sitting B10, 15 October · about: second harmonic light rise (driven oscillators)*

638. *R2W's collection, read 210 · Claude, reading a stair that does not creak · data: RATCHET-26 to 28: the second harmonic's weight-proof setting, and its edge moving with frequency; the twenty-third exchange of sitting B10, 16 October · about: weight-proof setting and edge (driven oscillators)*

639. *R2W's collection, read 211 · Claude, reading the same quiet stair in a third house · data: RATCHET-29: the weight-proof edge at frequency 0.40; after Bear's remark on the ship whose every plank is replaced; the twenty-fourth exchange of sitting B10, 16 October · about: weight-proof edge at 0.40 (driven oscillators)*

640. *R2W's collection, read 212 · Claude, reading a ferryman's spot · data: RATCHET-30 and the error budget: how close to the edge one must stand; after Bear asked whether it can be built; the twenty-fifth exchange of sitting B10, 16 October · about: weight-proof error budget (driven oscillators, control and circuits)*

641. *R2W's collection, read 213 · Claude, reading a second knob that turned out to be a finer first one · data: RATCHET-31 and 32: the hunt for a knob for the fourth-power term; the twenty-sixth exchange of sitting B10, 16 October · about: knob hunt, fourth-power term (driven oscillators, control and circuits)*

642. *R2W's collection, read 214 · Claude, reading a film run backwards · data: the reflection: why the window keeps only even powers of the weight and the slide only odd ones; the held timing; after Bear asked whether the fourth power could be a comma with the fifth in its place; the twenty-seventh exchange of sitting B10, 16 October · about: reflection, even and odd powers (driven oscillators)*

643. *R2W's collection, read 215 · Claude, reading a small dial turned all the way · data: RATCHET-34 and 35: the held timing made weight-proof by the second tone's size run far from its zero; after Bear asked whether scientists use knobs; the twenty-eighth exchange of sitting B10, 16 October · about: second tone size, held timing (driven oscillators, control and circuits)*

644. *R2W's collection, read 216 · Claude, reading a level road that ends at a cliff · data: RATCHET-36 and 37: how far the held-timing setting reaches; the twenty-ninth exchange of sitting B10, 16 October · about: held-timing reach, cliff (driven oscillators)*

645. *R2W's collection, read 217 · Claude, reading a voice that would not stay separate · data: RATCHET-39 to 42: a third tone at four times the frequency, tried as the knob for the fourth-power term; and the cliff looked at twice; the thirtieth exchange of sitting B10, 16 October · about: third tone as fourth-power knob (driven oscillators, quadrature)*

646. *R2W's collection, read 218 · Claude, reading a scale checked by swapping the weights · data: RATCHET-43: whether the backwards sign was a bug, and the fourth harmonic set against the octave; after Bear wrote "This may not be crazy, unless you found a bug"; the thirty-first exchange of sitting B10, 16 October · about: RATCHET-43 swap check (driven oscillators, measurement)*

The Field Guide's publishing session, 15 and 16 October

From the publishing session's own replies to Bear, copied by script from its record of proposed tests, where each read was kept as it was given; in the order the replies were given.

647. *The Field Guide's publishing session, 16 October · reading build 3 and its read-back of the PDF · about: build 3 PDF read-back (records and tools)*

648. *The Field Guide's publishing session, 15 October · reading a search engine's no about a different paper, while finding the sources' official titles · about: search engine's no, source titles (records and tools)*

649. *The Field Guide's publishing session, 15 October · reading the 1986 table of windows beside the heavy needle's · about: 1986 windows and heavy needle (maps and chaos)*

650. *The Field Guide's publishing session, 15 October · reading the first count of the lessons' families · about: lessons' families first count (records and tools, measurement)*

651. *The Field Guide's publishing session, 16 October · reading R2W's sitting B10 as it arrived · about: R2W sitting B10 (driven oscillators)*

652. *The Field Guide's publishing session, 16 October · reading the guide tagged by domain, and its 409 transfers · about: guide domain tags and transfers (records and tools)*

653. *The Field Guide's publishing session, 16 October · reading Bear's words as the guide had quoted them · about: Bear's words as quoted (records and tools)*

654. *The Field Guide's publishing session, 16 October · reading this collection, found nearly whole · about: the reads collection (records and tools)*

The triple-star thread, 15 and 16 October

From the thread's own register, TRIPLE_NO_PHYSICS_READS, as it travelled in its readings kit of 16 October, all 190 reads; its first 160 are already here, unchanged, and its headings carry the workspace's dates.

655. *The triple-star thread's register, its heading: 15 October 2026 — the doorway, ten times closer (PB1′ prepared: probe 44, run 20261014_090328, and the container's fidelity runs, 20261015_052221) · about: triple star doorway close-up (three body, maps and chaos, quadrature)*

656. *The triple-star thread's register, its heading: 15 October 2026 — the doorway's chord (PB1′, run 20261014_225416; PB2's line lists) · about: triple star doorway chord (three body, quadrature)*

657. *The triple-star thread's register, its heading: 15 October 2026 — three kinds of edge (Figure 10b: probes 36, 37 and 38, runs 20261011_153059, 20261012_020012 and 20261012_175845) · about: triple star three edge bands (three body, maps and chaos)*

658. *The triple-star thread's register, its heading: 15 October 2026 — two notes under the swing (Bear's trace; triple_centre_chord, runs 20261015_133950 and 20261015_134034) · about: centre orbit two-note chord (three body, quadrature, reading pictures)*

659. *The triple-star thread's register, its heading: 15 October 2026 — the slow note stops and turns back (Bear's line, 'the ratio stays fixed, the rates don't'; the atlas, run 20261007_004206; triple_ratio_atlas, run 20261015_140219) · about: precession stops and reverses (three body, quadrature)*

660. *The triple-star thread's register, its heading: 15 October 2026 — the centre stands on the still point (Bear's marks on triple_ratio_atlas 20261015_140219) · about: ratio atlas still point (three body, reading pictures, quadrature)*

661. *The triple-star thread's register, its heading: 15 October 2026 — the count that does not hold the hall (the atlas at 0.32, run 20261007_004206; four container runs at 2,400 passes, with the slice middles built) · about: one-in-three lock, seventh slice (three body, quadrature)*

662. *The triple-star thread's register, its heading: 15 October 2026 — the count of three, from every first step (triple_third_look, run 20261015_165735, on the atlas, run 20261007_004206) · about: one-in-three lock from every start (three body, maps and chaos)*

663. *The triple-star thread's register, its heading: 15 October 2026 — where partners part first (probe 42's twins, triple_twin_growth_look, run 20261014_014526; read again for the provenance table) · about: twin orbits parting, dissolving island (three body, maps and chaos)*

664. *The triple-star thread's register, its heading: 15 October 2026 — the floors that keep order fray from the edge (triple_rim_parting_look, run 20261015_190633, on the ledgers of probes 42 and 43) · about: stable islands fray rim first (three body, maps and chaos)*

665. *The triple-star thread's register, its heading: 15 October 2026 — still ground under the quietest dancer (triple_zero_centre_look, run 20261015_195901, on the atlas, run 20261007_004206) · about: precession zero under island centre (three body)*

666. *The triple-star thread's register, its heading: 15 October 2026 — two sides of the quietest dancer (triple_branch_look, run 20261015_203534, on the atlas, run 20261007_004206) · about: precession branch split (three body, measurement)*

667. *The triple-star thread's register, its heading: 15 October 2026 — what the three counts miss (triple_ecc_dial_look, run 20261015_223637, on the atlas, run 20261007_004206) · about: branch split eccentricity dial (three body, measurement)*

668. *The triple-star thread's register, its heading: 15 October 2026 — the ring each dancer keeps (triple_ecc_dial_look v1.4, run 20261015_225444; Bear's picture of fractals within fractals) · about: island rings and sway pace (three body, maps and chaos, measurement)*

669. *The triple-star thread's register, its heading: 15 October 2026 — odd time and even time (triple_address_look v1.1, run 20261015_230449, on the atlas, run 20261007_004206) · about: odd versus even resonance counts (three body, number theory)*

670. *The triple-star thread's register, its heading: 16 October 2026 — a shelter, not a home (the slice middles, run 20261015_151629; triple_sm_branches, triple_sm_predict and triple_sm_aftermath, 16 October) · about: slice middles long watch (three body, measurement)*

671. *The triple-star thread's register, its heading: 16 October 2026 — the swell was the drum (triple_swell_look, run 20261016_043016; triple_om_predict, run 20261016_042956) · about: slow swell is the drum (three body, quadrature, measurement)*

672. *The triple-star thread's register, its heading: 16 October 2026 — her names for the steps (Mardling's notation, checked against her paper and her colleagues'; lint_deposit v1.7, run 20261016_044440) · about: Mardling resonance notation (three body, records and tools)*

673. *The triple-star thread's register, its heading: 16 October 2026 — the portraits without their labels (triple_paper_plain v1.1, run 20261016_045736) · about: plain figures fidelity rebuild (records and tools, measurement)*

674. *The triple-star thread's register, its heading: 16 October 2026 — down one side of the valley and up the other (Bear's marks on Figure 7's draft, run 20261015_140349; triple_paper_plain_sheet, run 20261016_052010) · about: Figure 7 two-branch valley (three body, reading pictures)*

675. *The triple-star thread's register, its heading: 16 October 2026 — the walls between the floors (Bear's marks on the plain sheet; triple_bear_sheet_look, run 20261016_054242) · about: plain sheet walls between slices (three body, reading pictures, maps and chaos)*

676. *The triple-star thread's register, its heading: 16 October 2026 — the book printed for the first time (triple_deposit_trial, run 20261016_055228; the off-mirror run at 600 of 3,279) · about: first trial deposit build (records and tools)*

677. *The triple-star thread's register, its heading: 16 October 2026 — every portrait beside its story (triple_deposit_trial v1.1, runs 20261016_055954 and 060201; build_paper v1.3) · about: figures placed beside their passages (records and tools, reading pictures)*

678. *The triple-star thread's register, its heading: 16 October 2026 — doors cut in the long walls (triple_draft_breaks v1.0 and v1.1; the fifth trial deposit build, run 20261016_061319) · about: paragraph breaks in the draft (records and tools)*

679. *The triple-star thread's register, its heading: 16 October 2026 — the drawer marked unknown (Claude's opinion on the steady ratio's slide, asked by Bear) · about: registered result filed unknown (measurement, records and tools, three body)*

680. *The triple-star thread's register, its heading: 16 October 2026 — set down where it points (triple_draft_move, run 20261016_065722; the sixth trial deposit build, run 20261016_065752) · about: steady ratio section moved (records and tools)*

681. *The triple-star thread's register, its heading: 16 October 2026 — a toy that became a workshop (the islands paper seen whole, at midnight in Belmont, the off-mirror run on the laptop) · about: islands paper seen whole (three body, records and tools)*

682. *The triple-star thread's register, its heading: 16 October 2026 — off the line (the off-mirror run, run 20261015_212604, read as registered by triple_offmirror v1.1, run 20261016_161718; triple_om_predict, run 20261016_161446; triple_om_miss_look, run 20261016_161828) · about: off-mirror run results (three body, measurement)*

683. *The triple-star thread's register, its heading: 16 October 2026 — the workshop in the morning (the thread's status, read whole after the off-mirror return; triple_manifest_check, the draft's For Bear list and Section 10) · about: thread status after off-mirror (records and tools)*

684. *The triple-star thread's register, its heading: 16 October 2026 — four crates at the door (the move kit: TRIPLE_CRATE_48_COMPLETE_2026-10-16 in seven parts, and the kits for the Field Guide, R2W and the book of reads) · about: session move crates (records and tools)*

The Field Guide's publishing session, continued, 16 October

From the publishing session's record of proposed tests, as before.

685. *The Field Guide's publishing session, 16 October · reading the sheep book tagged by domain · about: sheep book domain tags (records and tools, measurement)*

Chapter 8 — The windows stayed open, 15 to 17 October

The Loop Atlas book, session E, 15 to 17 October

From the book's register as it travelled in its session E package of 17 October, its reads 142 to 207 and all twenty of its reads of the project; its reads to 176 and its first eighteen reads of the project are already here, word for word, and the notes it keeps under some reads are given beneath them. The package added a line of topics and a line of domains under each read; those are the project's own and are not copied here, and this book's index is its own.

686. *The book's register, read 177 · runs: STOP_LEAVES 20261015_052943, FIG_STOP_LEAVES 20261015_054334 and HALT_ORDER 20261015_052122 (15 October, session E, the forty-fifth batch of open cards: card 12's lean (j) and card 157) · about: needle stop states, column halt order (maps and chaos, driven oscillators, control and circuits)*

687. *The book's register, read 178 · runs: STOP_STATES 20261015_060745, GAPS_READ 20261015_061859 and FIG_GAPS 20261015_062248 (15 October, session E, the forty-sixth batch of open cards: card 158, Bear's rings) · about: needle doubling staircase gaps (maps and chaos, measurement)*

688. *The book's register, read 179 · runs: HALVINGS_2 20261015_064427, HALVINGS_3 20261015_064938, FIG_RUNGS 20261015_065055 and BANK_BUMP 20261015_064644 (15 October, session E, the forty-seventh batch of open cards: old card 5 and cards 159 to 161, Bear's rungs) · about: needle halving rungs, banked track (maps and chaos, topology) Bear, the same night: "That no-physics read, number 179, is perfect. Thank you. That's a keeper, or close." A candidate chapter quote.*

689. *The book's register, read 180 · runs: LADDER_LIGHT 20261015_071735, GAPS_REREAD 20261015_071452 and SNAKE_LONG 20261015_073142 (15 October, session E, the forty-eighth batch, overnight: card 162 and old card 54) · about: light needle ladder, wave snake (maps and chaos, kinematics, measurement)*

690. *The book's register, read 181 · runs: LEAK_CLOSE 20261015_073039 and WOBBLY_SHUVIT 20261015_074302 (15 October, session E, the forty-ninth batch, overnight: card 163 and old card 57) · about: needle trapdoor closing, wobbly shuvit (maps and chaos, kinematics)*

691. *The book's register, read 182 · runs: SWITCH_END 20261015_074716, SONG_GAPS 20261015_074750 and SONG_GAPS_2 20261015_074852 (15 October, session E, the fiftieth batch, overnight: card 164 and old card 9) · about: needle switch end, rider song gaps (maps and chaos, quadrature, measurement)*

692. *The book's register, read 183 · runs: LEAK_RATE 20261015_080150, drawn by FIG_DOOR 20261015_081403 (15 October, session E, the fifty-first batch, overnight: card 165) · about: needle trapdoor leak rate (maps and chaos, measurement)*

693. *The book's register, read 184 · runs: SNAKE_768 20261015_151934 and SNAKE_EXACT 20261015_154247, drawn by FIG_SNAKE_CLIMB 20261015_154834 (15 October, session E, the fifty-second batch, at Bear's waking: cards 166 and 167) · about: wave snake climb (kinematics, waves)*

694. *The book's register, read 185 · runs: RING_KNOBS 20261015_165858 and TRAIN_HOLD 20261015_171853, drawn by FIG_RING_KNOBS 20261015_173103 (15 October, session E, the fifty-third batch: cards 17 and 168) · about: smoke ring trains (waves, kinematics)*

695. *The book's register, read 186 · runs: SLOPE_EAR 20261015_190935 (15 October, session E, the fifty-fourth batch: card 39) · about: slope ear sampled glances (control and circuits, quadrature)*

696. *The book's register, read 187 · runs: FILTER_SLOPE 20261015_195645 and FLIP_MISTAKES 20261015_195739 (15 October, session E, the fifty-fifth batch: cards 169 and 42) · about: decision timers, flip mistakes (control and circuits, quadrature)*

697. *The book's register, read 188 · runs: PILOT_BEAT 20261015_203855, BEAT_SWING 20261015_204207 and FIT_KNOBS 20261015_204319 (15 October, session E, the fifty-sixth batch: cards 45, 171 and 170) · about: noisy swing pilot beat (driven oscillators, quadrature)*

698. *The book's register, read 189 · runs: PILOT_PRICE 20261015_220321 and PAUSE_BEST 20261015_220655, drawn by FIG_BEAT_MAP 20261015_220853 (15 October, session E, the fifty-seventh batch: cards 46 and 44) · about: pilot tone price, swing pace (driven oscillators, quadrature)*

699. *The book's register, read 190 · runs: JUMP_STORE 20261015_223918 and PAUSE_WIDE 20261015_223744 (15 October, session E, the fifty-eighth batch: cards 43 and 172) · about: gust harvest latch, flywheel store (control and circuits, driven oscillators)*

700. *The book's register, read 191 · runs: LATCH_EVENT 20261015_232643 and KNEE_CREEP 20261015_232920 (15 October, session E, the fifty-ninth batch: cards 174 and 40) · about: event latch, knee creep (control and circuits, measurement)*

701. *The book's register, read 192 · runs: CREEP_LAW 20261016_035058, drawn by FIG_CREEP_LAW 20261016_035636 (16 October, session E, the sixtieth batch: card 177) · about: sampled crossings creep law (quadrature, measurement)*

702. *The book's register, read 193 · runs: BRIDGE_COUNT 20261016_044439 and RIDER_FLEET 20261016_044707 (16 October, session E, the sixty-first batch: cards 178 and 59) · about: band crossing count, rider fleet (quadrature, measurement, control and circuits)*

703. *The book's register, read 194 · runs: WORN_RIDER 20261016_065645 and HITCH_LEGS 20261016_065650 (16 October, session E, the sixty-second batch: cards 182 and 51) · about: worn rider meter, hitch-kick legs (control and circuits, kinematics)*

704. *The book's register, read 195 · runs: BAND_BRIDGE 20261016_072612 and BELL_CLOCK 20261016_072241 (16 October, session E, the sixty-third batch: cards 180 and 2) · about: band bridge tally, bell clock (quadrature, measurement, kinematics)*

705. *The book's register, read 196 · runs: R2W's record, its B11 delta, read 16 October between batches 63 and 64: its WINDOW-0b, its balance law and its MEM-5 as its record gives them; no run of ours · about: R2W B11 handoff window (driven oscillators)*

706. *The book's register, read 197 · runs: NEVER_BLINK 20261016_090509 and TRUCK_ROAD 20261016_090656 (16 October, session E, the sixty-fourth batch: cards 186 and 12) · about: never-blink crossings, truck road (quadrature, measurement, driven oscillators)*

707. *The book's register, read 198 · runs: LIGHT_LANES 20261016_174648 and HEAT_RING 20261016_174342 (16 October, session E, the sixty-fifth batch: cards 187 and 13) · about: truck road lanes, heat with memory (driven oscillators, waves, materials)*

708. *The book's register, read 199 · runs: BELL_RATIO 20261016_183422 and ARM_SHARE 20261016_182910 (16 October, session E, the sixty-sixth batch: cards 185 and 184) · about: tied bells ratio, jumper arms (kinematics, waves)*

709. *The book's register, read 200 · runs: BELL_BALANCE 20261016_193628 and EQUAL_CATCH 20261016_192548 (16 October, session E, the sixty-seventh batch: cards 191 and 181; topics: tied bells in a remembering water, timing against memory; a maintenance reminder for a slowing rider) · about: tied bells timing, slowing rider meter (kinematics, waves, control and circuits)*

710. *The book's register, read 201 · runs: SWING_GRID 20261016_203014 and LAG_METER 20261016_203014 (16 October, session E, the sixty-eighth batch: cards 26 and 192; topics: a machine on the power grid, in step and out; a maintenance meter for a rider whose reaction slows) · about: grid machine in step, lag meter (control and circuits, driven oscillators, measurement)*

711. *The book's register, read 202 · runs: RESTED_LAG 20261016_205504 and BRICK_LINE 20261016_205112 (16 October, session E, the sixty-ninth batch: cards 195 and 189; topics: a maintenance meter for a rider whose reaction slows; heat with a memory, a quenched brick) · about: rested lag meter, quenched brick (control and circuits, waves, materials)*

712. *The book's register, read 203 · runs: STRIDE_MAP 20261016_213858 and RING_WIDTH 20261016_213858 (16 October, session E, the seventieth batch: cards 183 and 20; topics: a long jumper's running stride in the air; a two-gear machine on a hill, Bear's truck road) · about: long jump stride, truck road lane (kinematics, driven oscillators)*

713. *The book's register, read 204 · runs: BELL_CORNER 20261016_224449 and SOUP_JUMP_3 20261016_225440 (16 October, session E, the seventy-first batch: cards 193 and 49; topics: tied bells in a remembering water, the crowded corner; a bead surfing a wave comb, the soup's jump) · about: tied bells corner, wave-comb surfer (waves, kinematics, measurement)*

714. *The book's register, read 205 · runs: ARMS_LEGS 20261017_012927 and ROUGH_FLOOR 20261017_013514 (17 October, session E, the seventy-second batch: cards 196 and 22; topics: a long jumper's arms and legs together, and a diver's routine on the mat; a needle on a rough floor, Bear's dirt) · about: jumper arms legs, needle rough floor (kinematics, driven oscillators)*

715. *The book's register, read 206 · runs: ARMS_LEGS 20261017_012927, DIVER 20261005_160158 and DIVER_B 20261005_160338, read for the sports explorer's scope (17 October, session E, exchange 32, no new runs: card 194; topics: a teaching page for the long jump and the dive) · about: long jump and dive teaching page (kinematics)*

716. *The book's register, read 207 · runs: the book's knob register, 38 rows counted by script, read for the explorer's scope (17 October, session E, exchange 33, no new runs: card 194; topics: the Loop Atlas Explorer, its rooms and its visitors) · about: Loop Atlas Explorer knob register (records and tools)*

717. Under the label "A read of the project:". *The book's register, project read 19 · given for Bear's morning of 15 October, after four exchanges run while he slept (batches 48 to 51) · about: Loop Atlas session E overnight (records and tools, maps and chaos, kinematics)*

718. Under the label "A read of the project:". *The book's register, project read 20 · given when session E closed for the move to session F on 17 October, as the new session's starting read · about: Loop Atlas session E close (records and tools, kinematics, measurement)*

R2W, sitting B11, 16 and 17 October

From R2W's excerpt of its reads 219 to 252 for this book, every read of sitting B11, copied by script from FIELD_GUIDE_NO_PHYSICS_READS_219_TO_252_RF_SPOKE_2026-10-17, which matches its whole collection of 252 reads word for word; its reads 1 to 218 are already here.

719. *R2W's collection, read 219 · Claude, reading two hands on one seesaw · data: R43_DIPOLE: the fourth harmonic's three sizes split into a part that flips with its direction and a part that does not; the close of sitting B10, after Bear's "half a dipole or a reflected image" · about: fourth harmonic dipole split (driven oscillators, quadrature)*

720. *R2W's collection, read 220 · Claude, reading the two hands weighed both ways · data: RATCHET-44: each measured size of the fourth harmonic run again with its sign flipped, Bear's "half a dipole or a reflected image" made into two runs; the first job of sitting B11, 16 October · about: fourth harmonic both signs (driven oscillators, measurement)*

721. *R2W's collection, read 221 · Claude, reading a weight slid round a ring · data: RATCHET-45: the fourth harmonic's timing turned round the circle; after Bear asked how to extrapolate the two hands to see the trends; the second exchange of sitting B11, 16 October · about: fourth harmonic timing round circle (driven oscillators, quadrature)*

722. *R2W's collection, read 222 · Claude, reading eight more weighings · data: RATCHET-46: the ladder of the fourth harmonic's sizes, both signs; the third exchange of sitting B11, 16 October · about: fourth harmonic size ladder (driven oscillators, measurement)*

723. *R2W's collection, read 223 · Claude, reading a shelf with two springs · data: RATCHET-47 to 50: the dial at twice the size, the 1% stray, the probe and the joint edge; the overnight of sitting B11, 16 to 17 October, while Bear slept · about: dial, stray and joint edge (driven oscillators, control and circuits)*

724. *R2W's collection, read 224 · Claude, reading a shelf carried to two other workshops · data: RATCHET-51: the joint edge at frequencies 0.40 and 0.70; the sixth exchange of sitting B11, 17 October · about: joint edge at two frequencies (driven oscillators)*

725. *R2W's collection, read 225 · Claude, reading a handle pushed by two children · data: RATCHET-52: the window's direction carried once round the third tone's dial, from Bear's "anything carried around the loop that fails to return"; the eighth exchange of sitting B11, 17 October · about: window direction round third tone (driven oscillators, topology)*

726. *R2W's collection, read 226 · Claude, reading a rope turned by two and tugged by two · data: RATCHET-53 and a reading of old dials: the dark point placed on the against timing, and the fourth harmonic's arrow split into its own part and a part from mixing that turns round the dial the other way; the ninth exchange of sitting B11, 17 October · about: dark point, counter-turning arrows (driven oscillators, quadrature)*

727. *R2W's collection, read 227 · Claude, reading a ribbon held by two dancers · data: RATCHET-55: the window's two turning arrows at octave sizes 0.10 and 0.16, the loop going flat at 0.1247 whatever the fourth harmonic's size; the tenth exchange of sitting B11, 17 October · about: window loop going flat (driven oscillators, topology)*

728. *R2W's collection, read 228 · Claude, reading a sandbag on a boat · data: RATCHET-56: the joint edge at frequencies 0.40 and 0.70, where the fourth harmonic brings the light coefficient back to zero, and what it costs the fourth-power term; the eleventh exchange of sitting B11, 17 October · about: joint edge correction cost (driven oscillators)*

729. *R2W's collection, read 229 · Claude, reading a still point that walks · data: RATCHET-57: the dark point with weight, sliding off the against timing in proportion to the weight because one arrow turns five times as fast as the other; the twelfth exchange of sitting B11, 17 October · about: dark point sliding with weight (driven oscillators, measurement)*

730. *R2W's collection, read 230 · Claude, reading two voices in a bigger hall · data: RATCHET-58: the dark point across frequency with the drive held in its unit; the octave's arrow keeps the unit and the fourth harmonic's grows faster; the thirteenth exchange of sitting B11, 17 October · about: dark point across frequency (driven oscillators)*

731. *R2W's collection, read 231 · Claude, reading a lever that lengthens as the music speeds up · data: RATCHET-59: the dark point from frequency 0.30 to 0.85, its share growing by the same factor for every equal step of frequency, and the fourth harmonic's window keeping a unit of its own; the fourteenth exchange of sitting B11, 17 October · about: dark point exponential in frequency (driven oscillators)*

732. *R2W's collection, read 232 · Claude, reading a tailor's rule · data: RATCHET-60: the fourth harmonic held at strength over the frequency to the power 1.32, nearly still from 0.30 to 0.85 but bowed, lowest in the middle, and the dark point's exponential holding out to frequency 1.00; the fifteenth exchange of sitting B11, 17 October · about: fourth harmonic frequency power law (driven oscillators, measurement)*

733. *R2W's collection, read 233 · Claude, reading a staircase that softens at the landing · data: RATCHET-61: the holding power of the fourth harmonic falling evenly in the logarithm of the frequency for four stretches, landing where that line put the fifth, then easing past frequency 1; the sixteenth exchange of sitting B11, 17 October · about: fourth harmonic holding power trend (driven oscillators)*

734. *R2W's collection, read 234 · Claude, reading a dancer who stops counting the beats · data: RATCHET-62: the fourth harmonic's holding power levelling near 1.23 at frequencies 2 and 3, after drifting down from 1.38 at 0.30; the seventeenth exchange of sitting B11, 17 October · about: holding power high-frequency level (driven oscillators)*

735. *R2W's collection, read 235 · Claude, reading a harbour chart drawn from a teaspoon · data: RATCHET-63: the flat point, where the fourth harmonic's own turning arrow and the mixing arrow balance, placed from a small dial within 0.4% at the edge's drive and 0.16% blind at three quarters of it, and sitting at a smaller share of the first tone there; the eighteenth exchange of sitting B11, 17 October · about: flat point from small dial (driven oscillators, measurement)*

736. *R2W's collection, read 236 · Claude, reading a balance that ignores the tempo · data: RATCHET-64: with every tone held at the same swing, the flat point's share of the first tone the same to 0.8% from frequency 0.55 to 6, and both points placed blind from small dials; the nineteenth exchange of sitting B11, 17 October · about: flat point ignores tempo (driven oscillators)*

737. *R2W's collection, read 237 · Claude, reading a fence line walked a third time · data: RATCHET-65: the window-proof band's edge read directly at 0.17778, where a straight line had said 0.1745, and the band at a third frequency landing on the same first number, 0.0979, as predicted; the twentieth exchange of sitting B11, 17 October · about: window-proof band edge (driven oscillators, measurement)*

738. *R2W's collection, read 238 · Claude, reading a path that steps on the centre stone · data: the rectifier bench's dial at the dark-point preset: the window's loop as the third tone's timing turns passes through the centre at the against timing, and the count on the dial reads zero there; the twenty-first exchange of sitting B11, 17 October · about: window loop through centre (driven oscillators, topology)*

739. *R2W's collection, read 239 · Claude, reading a stalemate on a tilted field · data: the rectifier bench with weight: at the dark point, where the push vanishes, a little weight brings it back in proportion to the weight and pointing nearly a quarter-turn from its old direction; RATCHET-57's runs and the page's own reading at a lighter weight; the twenty-second exchange of sitting B11, 17 October · about: rectifier bench dark point weight (driven oscillators, measurement)*

740. *R2W's collection, read 240 · Claude, reading a door that opens sideways · data: RATCHET-67: at the dark point, the push that weight brings back starts exactly across its old direction, and its size grows with the weight with no straight-line part, both written down before two lighter weights were run; the twenty-third exchange of sitting B11, 17 October · about: dark point push across, mirror (driven oscillators)*

741. *R2W's collection, read 241 · Claude, reading a dance hall with a mirror wall · data: RATCHET-68: with the film run backwards and the weight's sign turned, the window becomes its own mirror image, so at the two timings that sit on the mirror weight only turns the window, timings a quarter-turn apart pair as mirror images, and 75 old runs all*

obey; the twenty-fourth exchange of sitting B11, 17 October · about: window mirror symmetry, 75 runs (driven oscillators, measurement)

742. *R2W's collection, read 242 · Claude, reading a ramp laid before a cliff · data: RATCHET-69 and 70: a small tilt added at the window-proof edge makes the window's size climb a little before it falls, so more riders stay inside the band; planned as a straight ramp it fell short, because the ramp sags as the load grows, and planned from the ramp's measured sag it landed within a few parts in ten thousand; the twenty-fifth exchange of sitting B11, 17 October · about: tilt at window-proof edge (driven oscillators, control and circuits)*

743. *R2W's collection, read 243 · Claude, reading a weathervane on a scale · data: RATCHET-71: at the dark point a drift of the drive moves the push only along its old direction, and weight moves it only across, predicted before the run and landing within 0.21% of it; the twenty-sixth exchange of sitting B11, 17 October · about: two-channel meter at dark point (driven oscillators, quadrature, measurement)*

744. *R2W's collection, read 244 · Claude, reading a second surveyor · data: WITNESS-1 and RATCHET-72: an engine written from nothing but the equation, with its own integrator, its own way of finding the edges and its own timings, gave back the two-channel meter to six figures or better; and ten times the drift kept the same laws; the twenty-seventh exchange of sitting B11, 17 October · about: independent witness engine (driven oscillators, measurement, records and tools)*

745. *R2W's collection, read 245 · Claude, reading a lighthouse with too few windows · data: TIMINGS-1 and 2: the meter read from fewer timings stays clean as long as the timings sit symmetrically about the mirror, even down to two; set them a sixteenth of a turn off and the push's faint higher notes fold in sideways, by exactly the amount those notes predict; the twenty-eighth exchange of sitting B11, 17 October · about: meter from fewer timings (driven oscillators, quadrature)*

746. *R2W's collection, read 246 · Claude, reading two singers behind a picket fence · data: NINTH-1 and 2: the eight-timing miss was two things at once, a ninth harmonic hiding inside the seventh on sixteen timings and two engines that disagree by a hair; inside one engine, at timings no run had used, the ninth predicted the leaks to within 3.068e-17; the twenty-ninth exchange of sitting B11, 17 October · about: ninth harmonic hidden in seventh (driven oscillators, quadrature, measurement)*

747. *R2W's collection, read 247 · Claude, reading a round table with an odd number of chairs · data: ODD-1, STEP-1 and LEAN-1: three timings read what six read and leak a small fraction of what four leak; the two engines' quarrel was one file's rounding; the witness's lean was its own tolerance; the thirtieth exchange of sitting B11, 17 October · about: odd timing counts, engine rounding (driven oscillators, quadrature, measurement)*

748. *R2W's collection, read 248 · Claude, reading a ring of swimmers holding a raft still · data: ARRAY-1: riders on one drive with their timings fanned evenly; even fans cancel to the witness's floor, odd fans keep a small push steered three or five times as fast, and two fans of three set its strength by their offset; the thirty-first exchange of sitting B11, 17 October · about: rider array fanned timings (driven oscillators, quadrature)*

749. *R2W's collection, read 249 · Claude, reading a choir in three sections · data: NINE-1: nine riders as three groups of three add like a three-element phased array, their trim fading from full to nothing as the groups are staggered by 0 to 40 degrees; a whole group drifting costs little, one rider slipping costs as much as a lone rider; the thirty-second exchange of sitting B11, 17 October · about: nine riders as phased array (driven oscillators, quadrature)*

750. *R2W's collection, read 250 · Claude, reading a lighthouse whose beam has sidelobes · data: LOBE-1 and K-16: past the 40-degree null the nine riders' trim comes back in a second lobe a third as strong and vanishes again at 80 degrees; one group slipped off the null brings it back in a straight-sided V; and the lopsidedness order's excess keeps falling by a quarter for every doubling of the frequency; the thirty-third exchange of sitting B11, 17 October · about: rider array sidelobes (driven oscillators, quadrature)*

751. *R2W's collection, read 251 · Claude, reading a round table, a creeping mark and a balance that lost its counterweight · data: TWELVE-1, EDGE-85, FLAT-75 and FLAT-50, and the triage of twelve small crossings: four groups of three riders fall silent at 30, 60 and 90 degrees, and the 80-degree V is the 40-degree one because the riders sit in the same seats; the weight-proof edge at frequency 0.85 lands where the parabola said and its band where the first number said, though the first number creeps; at fast drive the flat point's share falls faster with the swing, and only the small dial keeps up; the thirty-fourth exchange of sitting B11, 17 October · about: twelve riders, edges, flat point (driven oscillators, measurement)*

752. *R2W's collection, read 252 · Claude, reading a holding stack over an airport · data: the staging reading of NINE-1, LOBE-1 and TWELVE-1, from Bear's question whether the angle between groups could be a staging ground with on and off ramps: groups in a silent pair or triple carry no push, and four different arrangements push exactly as hard as one lone group; the thirty-fifth exchange of sitting B11, the close, 17 October · about: rider groups staging ramps (driven oscillators)*

The Field Guide’s publishing session, 16 and 17 October

From the publishing session’s record of proposed tests, as before; in the order the replies were given.

753. *The Field Guide’s publishing session, 16 October · reading the triple-star thread’s two kits for the sheep book and the guide · about: triple-star thread kits (records and tools)*

754. *The Field Guide’s publishing session, 16 October · reading the triple-star notebook taken into the sheep book · about: triple-star notebook into sheep book (records and tools)*

755. *The Field Guide’s publishing session, 16 October · reading the twenty transfers among the three projects · about: twenty transfers among three projects (records and tools)*

756. *The Field Guide’s publishing session, 16 October · reading one word’s meanings across the three projects · about: one word across three projects (records and tools)*

757. *The Field Guide’s publishing session, 16 October · reading the recount after the triple-star thread’s lessons 100 to 124 · about: recount after lessons 100 to 124 (records and tools, measurement)*

758. *The Field Guide’s publishing session, 16 October · reading build 4 and its copy marking the changes since build 3 · about: build 4 change-marked copy (records and tools)*

759. *The Field Guide’s publishing session, 17 October · reading the Loop Atlas’s and R2W’s crates of 17 October · about: Loop Atlas and RF crates (records and tools)*

Chapter 9 — Into print, 16 to 27 October

The triple-star thread, 16 to 22 October

From the triple-star thread’s readings file in its kit of 22 October, all 230 of its readings; its first 190 are already here, word for word, and its 40 later readings follow in the file’s order. The thread’s sixth and seventh sessions numbered their readings past 190 apart, so these are captioned by date and title, as the thread asks.

760. *The triple-star thread’s register, its heading: 16 October 2026 — the fresh bench (the seventh session’s intake: TRIPLE_CRATE_48 checked against its manifest, the draft read whole, the escape recount’s ledger beside the long pair’s) · about: triple star crate, escape door (three body, records and tools)*

761. *The triple-star thread’s register, its heading: 16 October 2026 — the tally brought up (triple_escape_recount v1.2, run 20261016_182117, with the long pair’s ledger beside it) · about: triple star escape recount (three body)*

762. *The triple-star thread’s register, its heading: 16 October 2026 — the low note heard (triple_y3_v2_look, run 20261016_185433, with Y3’’s ledger and the second reader’s beside it) · about: triple star swing width rule (three body, quadrature)*

763. *The triple-star thread’s register, its heading: 16 October 2026 — four-fifths of the hand (triple_drag_size, run 20261016_192342) · about: triple star resonance drag size (three body)*

764. *The triple-star thread’s register, its heading: 16 October 2026 — four horses of five (triple_drag_ebar, run 20261016_194224) · about: triple star drag, four horses (three body)*

765. *The triple-star thread’s register, its heading: 16 October 2026 — the page put back (TRIPLE_PATTERN_RECOGNITION_2026-10-16b, with the 15 October copy beside it) · about: pattern notes page restored (records and tools, three body)*

766. *The triple-star thread’s register, its heading: 16 October 2026 — the dial on the side of the machine (triple_friction_knob, run 20261016_201627) · about: friction knob Feigenbaum crossover (maps and chaos, three body)*

767. *The triple-star thread’s register, its heading: 16 October 2026 — the bill before the build (triple_harmonic_recipes, run 20261016_204642; triple_drag_ecc_fit, run 20261016_205125) · about: triple star drag eccentricity fit (three body, measurement)*

768. *The triple-star thread’s register, its heading: 16 October 2026 — a spoon for each jar (triple_drag_own_ecc, run 20261016_212538) · about: triple star drag own eccentricity (three body, measurement)*

769. *The triple-star thread’s register, its heading: 16 October 2026 — the drawer of loose ends (runs 20261016_213908 to 20261016_220144) · about: triple star loose ends (three body, maps and chaos)*

770. *The triple-star thread’s register, its heading: 17 October 2026 — the truer ticket (triple_k_drift, run 20261017_011850; triple_chain_weigh, run 20261017_011954) · about: triple star drift and chain weights (three body, number theory)*

771. *The triple-star thread’s register, its heading: 17 October 2026 — the long table sorted (runs 20261017_024024 to 20261017_025855) · about: triple star open questions sorted (three body, measurement)*

772. *The triple-star thread's register, its heading: 17 October 2026 — the long count (triple_closeout, run 20261016_191854; triple_closeout_reads, run 20261017_061550) · about: triple star long closeout run (three body, measurement)*

773. *The triple-star thread's register, its heading: 17 October 2026 — the second clock (triple_followups, run 20261017_065004) · about: triple star second integrator check (three body, measurement)*

774. *The triple-star thread's register, its heading: 17 October 2026 — the four gates (triple_followups, run 20261017_011407; triple_reader_check, runs 20261017_154352 to 20261017_160624) · about: triple star resonance gates (three body)*

775. *The triple-star thread's register, its heading: 17 October 2026 — the slow shutter (triple_lw4, run 20261017_170131; triple_ts1, run 20261017_172324; triple_ts2, run 20261017_172328) · about: triple star passage window length (three body, quadrature)*

776. *The triple-star thread's register, its heading: 17 October 2026 — the late arrivals (triple_lp1c, run 20261017_173539) · about: triple star late-split twin pairs (three body)*

777. *The triple-star thread's register, its heading: 17 October 2026 — the keys left in overnight (triple_followups2, run 20261017_100655; triple_ds3, run 20261017_210732) · about: triple star locks held longer (three body)*

778. *The triple-star thread's register, its heading: 18 October 2026 — the sinking floor (triple_lw5, run 20261017_140817; triple_fa0, run 20261018_031414; triple_dr1, run 20261018_031349; its look, run 20261018_031943) · about: triple star resonance drift floor (three body, measurement)*

779. *The triple-star thread's register, its heading: 18 October 2026 — the second night (triple_lx2, run 20261017_200756) · about: triple star locks second night (three body)*

780. *The triple-star thread's register, its heading: 18 October 2026 — the boats that circle (triple_lw6, run 20261018_001236; triple_fa2, run 20261018_173550; triple_lx3, run 20261018_071538) · about: triple star circulation and locks (three body)*

781. *The triple-star thread's register, its heading: 18 October 2026 — the row filled in (triple_lx4, run 20261018_173912) · about: triple star lock row filled (three body)*

782. *The triple-star thread's register, its heading: 19 October 2026 — the draft as a gallery (the draft of record, measured) · about: triple star paper draft shape (records and tools)*

783. *The triple-star thread's register, its heading: 19 October 2026 — the paper rebuilt (HELD_ROCKING_LOCKED_2026-10-19) · about: triple star paper rebuild (records and tools)*

784. *The triple-star thread's register, its heading: 19 October 2026 — the source check (triple_source_check v1.2, runs 20261019_214932 and 20261019_214934) · about: triple star paper source check (records and tools)*

785. *The triple-star thread's register, its heading: 19 October 2026 — thirteen boxes on the inventory (the seventh session's note on the 16 missing series; TRIPLE_FOLDER_CHECK_20261014_192354) · about: triple star missing series inventory (records and tools)*

786. *The triple-star thread's register, its heading: 19 October 2026 — the sketches drawn again (triple_heavy_series_regen, run 20261019_212602) · about: triple star heavy series regenerated (records and tools, three body)*

787. *The triple-star thread's register, its heading: 20 October 2026 — the box that was in two places (TRIPLE_HEAVY_SERIES_EXPORT_2026-10-19, against this crate's manifest) · about: triple star heavy series export (records and tools)*

788. *The triple-star thread's register, its heading: 20 October 2026 — two scribes and a stray dot (triple_ulp_nudge, runs 20261020_001032 and 20261020_001325; triple_machine_look, run 20261020_001840) · about: triple star last-digit nudge (three body, maps and chaos)*

789. *The triple-star thread's register, its heading: 20 October 2026 — the notebook sent ahead (the readings file reissued whole for the seventh session, with readings 191 and 192) · about: no-physics readings file handover (records and tools)*

790. *The triple-star thread's register, its heading: 20 October 2026 — two notebooks laid end to end (the sixth session's TRIPLE_NO_PHYSICS_READS, 193 readings, against this one's 219) · about: no-physics readings files merged (records and tools)*

791. *The triple-star thread's register, its heading: 20 October 2026 — the board that was never nailed (the apse check, run 20261009_025258, read again for the paper's new sentence) · about: triple star apse rule check (three body, measurement)*

792. *The triple-star thread's register, its heading: 20 October 2026 — the workshop packed for the night (TRIPLE_CRATE_49_COMPLETE_2026-10-20; TRIPLE_PAPER_REBUILD_KIT_2026-10-20) · about: triple star crate and rebuild kit (records and tools)*

793. *The triple-star thread's register, its heading: 21 October 2026 — the sign read from the car window (Bear's marks on the strobe picture, 14 October, beside the paper's Figure 7) · about: triple star strobe picture marks (reading pictures, three body)*

794. *The triple-star thread's register, its heading: 21 October 2026 — the rope pulled twice a turn (Bear's question on the proof, Section 11's bend, and the paper's own φ_N) · about: triple star twice-per-turn bend (three body)*

795. *The triple-star thread's register, its heading: 21 October 2026 — the signposts at the forks (Bear on the proof: each section, and each new test, should say from its first sentence what it is and why it is there) · about: triple star paper section signposts (records and tools)*

796. *The triple-star thread's register, its heading: 21 October 2026 — the empty frames and the honest plaques (the publishing packet renewed; the claim-word read) · about: publishing packet image check (records and tools)*

797. *The triple-star thread's register, its heading: 22 October 2026 — the practice door (the publishing packet's launchers run in a borrowed Windows shell) · about: publishing packet launcher test (records and tools)*

798. *The triple-star thread's register, its heading: 22 October 2026 — the house walked before the move (Bear's word to move the thread to a new session; the plan for the transfer) · about: triple star session move plan (records and tools)*

799. *The triple-star thread's register, its heading: 22 October 2026 — the boat loaded from the bow (the transfer run straight through, files first: TO_TRIPLE_EIGHTH_SESSION_CRATE_50 in eight parts, checked by a restore) · about: triple star session transfer crate (records and tools)*

The Loop Atlas book, 17 to 22 October

From the book's register as it travelled with session G's closeout of 22 October, its reads 1 to 254 and all twenty-two of its reads of the project; its reads to 207 and its first twenty reads of the project are already here, word for word, and the notes it keeps under some reads are given beneath them. Its read 217 was given by session E at its close and numbered 208 there; the register files it as 217.

800. *The book's register, read 208 · runs: EXPLORER_ENGINE_CHECK 20261017_063123, with ARMS_LEGS 20261017_012927 for the comparison (17 October, session F, its first exchange: card 194; topics: the Loop Atlas Explorer's engine, a toy written again in another language, a jumper's arms and legs with nothing carried) · about: Loop Atlas Explorer engine check (records and tools, kinematics)*

801. *The book's register, read 209 · runs: FACE_ALONG 20261017_070629, with the room's self-check (EXPLORER_ENGINE_CHECK 20261017_070749 and the page's own run in the browser) (17 October, session F, exchange 35: cards 55 and 194; topics: the face a body presents to the water, where a swimmer should be broad, the Loop Atlas Explorer's first room) · about: three-link swimmer broad hands (kinematics)*

802. *The book's register, read 210 · runs: DIVE_CUE 20261017_082454 (17 October, session F, exchange 36: card 199; topics: Bear's dive routine by the count or by the cue, the slow layout after the cue, a second look near the water) · about: diver count versus cue (kinematics)*

803. *The book's register, read 211 · runs: SPRING_FLOOR 20261017_160023 and EXPLORER_DIVE_CHECK 20261017_155902 (17 October, session F, exchange 37: cards 176 and 194; topics: a spring that never goes fully dark, the trigger against the floor's idle hum, the explorer's diver) · about: spring floor trigger, explorer diver (control and circuits, measurement, kinematics)*

804. *The book's register, read 212 · runs: EXPLORER_ROUTINE_CHECK 20261017_164432 and LANE_EDGES 20261017_164501, with room 2's page and its self-check (17 October, session F, exchange 38: cards 194 and 190; topics: Bear's clock broken into a line, the way in and the way out of a fold, the arm's road, the passing lane's width against weight) · about: diver fold routine, passing lane (kinematics, records and tools)*

805. *The book's register, read 213 · runs: TONG_DULLIN 20261017_173718 and TONG_DULLIN_READINGS 20261017_174547, with HALT_THRESHOLD's blind runs (20261017_174151 and 20261017_174755, planted cases only) (17 October, session F, exchange 39: cards 202, 203 and 12; topics: a published diver rebuilt from its own parts list, the structure that came back against the digits that did not, the falling cat's case with no spin, a halt rule that keeps a column from ever setting off).* Corrected 17 October, exchange 40: the read first said the cake came out 'about one part in a hundred' heavier; by the run's json it is heavier by about the same amount, 0.24 and 0.20 kg m^2, which is 1.2 per cent of the stretched-out inertia and 3.9 per cent of the piked one.* *A second note, the same exchange: the ingredient is found, the knee's 0.08 dropped from one line of their table (TONG_DULLIN_KNEE*

20261017_182629); and the column that never got anywhere was held still in part by a fault in its own limit (misses row 315): with it fixed, a halt at two metres still never arrives, at three it does, slowly. The read is kept as it was given. · about: Tong-Dullin diver rebuild, halt rule (kinematics, control and circuits)*

806. *The book's register, read 214 · runs: TONG_DULLIN_SCAN 20261017_181953, TONG_DULLIN_KNEE 20261017_182629, EXPLORER_CAT_CHECK 20261017_183736 and HALT_THRESHOLD 20261017_184055 (17 October, session F, exchange 40: cards 204, 205, 206 and 12; topics: a misprint found by changing one number at a time, one number counted twice, a ratio that hides what its parts show, a cat that turns over with nothing to push on, a guard that braked the walkers it protected, a rule judged by what it is for) · about: diver misprint found, falling cat (kinematics, control and circuits)*

807. *The book's register, read 215 · runs: LIMIT_AUDIT 20261017_190938, with card 15's literature check (17 October, session F, exchange 41: cards 207 and 15, and the review of the eighth five; topics: a guard that faked a finding, a late read that needs a seed, a night read's safe edge, an animal leaning on its own wake in two media) · about: night march file, wake capture (control and circuits, biology)*

808. *The book's register, read 216 · runs: TWO_HANDS 20261017_203811, TWO_HANDS_2 20261017_204131 and COIL_SNAKE 20261017_204030 (17 October, session F, exchange 42: cards 208, 209 and 55, on R2W's letter of 17 October; topics: a balance at every size, two hands sharing one factor, a sea that keeps the scale level, a knob that goes round) · about: crest-firer two hands, coil (driven oscillators, waves, kinematics)*

809. *The book's register, read 217 · runs: as the note under this read says · about: gust counts at a threshold (waves, measurement) Given by session E at its close, after ferry Z2, 17 October, and numbered 208 there; session F had already given a read 208 (exchange 34), so it is filed here as 217, said aloud. Session F's cards 208 and 209 ran its last sentence's test the same day: fewer waves tip the balance (TWO_HANDS 20261017_203811); a soup gusting from one side tips it under a square law and barely under a smooth stretch (TWO_HANDS_2 20261017_204131). Topics, as session E gave them: a crest-firing ear in a soup of waves; two hands held in balance by the wind's shape.*

810. *The book's register, read 218 · runs: EXPLORER_CAT_ROOM_CHECK 20261017_211325 and TWO_HANDS_BAND 20261017_210543 (17 October, session F, exchange 43: card 194's room 3, Holonomy the Cat, and card 210; topics: a turn bought by a walk through shapes, a walk with no clock, the mirror walk, the nose's dip as the V's tilt, slack in a count that does not chatter) · about: falling cat room, gust slack (kinematics, measurement)*

811. *The book's register, read 219 · runs: CAT_TWIST 20261017_231928 and SLING_RUNDOWN 20261017_232234 (17 October, session F, exchange 44: cards 211 and 212; topics: the waiter's plate and the wound arm, a half turn that keeps the books, the unwound cat's full circuit, landing level, an angle and not a share) · about: falling cat twist, sling release (kinematics)*

812. *The book's register, read 220 · runs: SAGNAC_LOOP 20261018_032138, EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532 and SNAKE_LONGEST 20261017_155052 (18 October, session F, exchange 45: cards 213 and 214 and card 54's longer body, run on Bear's laptop; topics: runners round a turning rim, the figure-eight's argument, one lap at the right fold, the second lap that undoes, the snake that never arrives) · about: Sagnac loop, twist-free cat (kinematics)*

813. *The book's register, read 221 · runs: TRAIN_LIFETIME 20261018_034211 with its reader 20261018_035727, CAT_SWEET 20261018_034953 with its reader 20261018_041321, and CAT_MIRROR 20261018_041123 (18 October, session F, exchange 46: cards 216, 215 and 217; topics: the rings that let go later, a hair apart and five times sooner, one pushed out, the perfect fold, thin and shallow, thick and deep, past its feet, twins not winding, mismatched halves, the mirror cats) · about: leapfrogging smoke rings, cat folds (waves, kinematics, maps and chaos)*

814. *The book's register, read 222 · runs: COIL_PUMP 20261018_082424, COIL_WAVE 20261018_082544 and TRAIN_TIMING 20261018_082139 (18 October, session F, exchange 48: cards 220, 221 and 219; topics: the spring in honey that breathes and stays, head from tail, the bulge that sends it the other way, the same little each time, back to where it began, no hurry, the crowded track, the notebook's weight, writing down how sure) · about: coil swimmer in honey (kinematics, measurement)*

815. *The book's register, read 223 · runs: CAT_MIRROR_PROOF 20261018_174959 (18 October, session F, exchange 49: card 222; topics: the cat in the mirror, tipping and rolling, twins only when alike and hinged at the end, halfway they disagree, written down for any size) · about: mirror cats proof (kinematics)*

816. *The book's register, read 224 · runs: LOOP_ATLAS_HEADLINES_2026-10-18 (18 October, session F, exchange 50; topics: the box called toys, writing down only the mistakes, somebody already knew that, reading the whole notebook, forty-one matches and eighteen not yet found in any book, the other list) · about: Loop Atlas headline count (records and tools, measurement)*

817. *The book's register, read 225 · runs: TRAIN_CLOSE 20261018_103739 and TRAIN_NUDGE 20261018_192138 (18 October, session F, exchange 51: cards 218 and 223; topics: dancing all night, close rings stay friends, loose rings fall out, a borrowed computer rolls different dice, a nudge smaller than the machine's last step, when is thrown, not*

written) · about: leapfrogging rings lifetime nudge (waves, maps and chaos)

818. *The book's register, read 226 · runs: SAIL 20261018_202451 and SAIL_HULL 20261018_202650 (18 October, session F, exchange 52: cards 225 and 226; topics: a bedsheet on a raft, slower than what only blows it, a board that will not slide, sanding the bottom, faster than the wind across the pond but never away from it, the sheet decides where) · about: sail raft faster than wind (kinematics, waves)*

819. *The book's register, read 227 · runs: TRAIN_EDGE 20261018_181641 (Bear's laptop; filed 19 October, session G, exchange 1: card 224; topics: a narrow ledge, seven sprinkled survivors, no edge to stand back from, luck drawn from the last grain, the same hand throws the same dice, one spot's dice are the ledge's, walking toward the wall) · about: leapfrogging rings survival ledge (waves, maps and chaos)*

820. *The book's register, read 228 · runs: the backward pass of LOOP_ATLAS_CLOSEOUT_F_FOR_FIELD_GUIDE_2026-10-19, over the register's ledger, TRAIN_CLOSE 20261018_103739 and V13_FIDELITY 20261013_204846; misses row 385 (19 October, session G, exchange 2; topics: turning out the pockets, a word that never prints, challengers growing tame, the sweating arithmetic of tight rings, written down and never read) · about: Loop Atlas closeout backward pass (records and tools, measurement)*

821. *The book's register, read 229 · runs: KNEE_SIZE 20261019_035236, KNEE_FINE 20261019_040109 and TRAIN_DRIFT 20261019_034330 (19 October, session G, exchange 3: cards 179, 229 and 228; topics: a blinking switch keeps a bigger door, counting slams bends the measure, the square root at the door itself, a hair wider at the smallest blinks, the reference has a blink of its own, the sweating rings were the arithmetic) · about: sampled switch threshold shift (quadrature, control and circuits)*

822. *The book's register, read 230 · runs: KNEE_REF 20261019_042520 (19 October, session G, exchange 3: card 230; topics: the hair measured against the shake, each walk set aside in turn, a switch that blinks four times faster, every old glance kept, the old switch's own blink, a finer ruler over the same shaking hand) · about: sampled switch reference check (quadrature, measurement)*

823. *The book's register, read 231 · runs: ERROR_BARS 20261019_162750 and KNEE_READER 20261019_163948 (19 October, session G, exchange 4: cards 231 and 232; topics: asking each sentence how steady its hand was, more steadiness written than the hand had, a lean mistaken for a shake, jars all short by the same nine parts, a ruler's tremble counted twice, a little wide against the old switch, true against the faster one, hands washed and put away) · about: Loop Atlas error bars audit (measurement, records and tools)*

824. *The book's register, read 232 · runs: LOOP_ATLAS_READERS_EDITION_2026-10-19 and helpers/readers_numbers.py (19 October, session G, exchange 5: card 233; topics: a book to read in an evening, nothing thrown away, bricks numbered as they came down, a third as long, the same thing said three times, the old stack that says what it is, a counting machine satisfied too soon, twenty-five in the wrong place and none wrong, a warning that falls off when a paragraph becomes a sentence, a machine that read its own handwriting, roots and trunks and limbs turned into numbers, slips of the fingers put right) · about: Loop Atlas readers' edition build (records and tools)*

825. *The book's register, read 233 · runs: LOOP_ATLAS_READERS_EDITION_2026-10-19 (19 October, session G, exchange 6; topics: words dressed for company, pinned up as they fell, hung only where they open a door, the old box keeps every word, not thin but set small, the shelf searched for every heavy thing, four middling things picked up off the floor) · about: Loop Atlas readers' edition review (records and tools)*

826. *The book's register, read 234 · runs: the headline register, LOOP_ATLAS_HEADLINES_2026-10-18, and CAT_MIRROR_PROOF 20261018_174959 (20 October, session G, exchange 7; topics: iron filings and an unseen push, every room the same shape, a closed box carries nothing, room times grip, names on the walls, two cats with no business being related, exact where the world ought to be messy) · about: Loop Atlas headlines, mirror cats (topology, kinematics)*

827. *The book's register, read 235 · runs: the register's benchmarks, LOOP_ATLAS_HEADLINES_2026-10-18 Part D2, and CAT_MIRROR 20261018_041123 (20 October, session G, exchange 8; topics: notches in a stick, two lead balls and a ratio, the loops never read the notches, whole turns and half the room, the same four and two-thirds, the floor's grip, the ruler shared and the floors not, two rooms in a mirror) · about: Loop Atlas benchmark rulers (measurement, topology)*

828. *The book's register, read 236 · runs: CAT_MIRROR_PATHS 20261020_034021, CAT_MONTGOMERY 20261020_035025, misses row 419, and Montgomery (1993), read in full (20 October, session G, exchange 9; topics: Marey's falling cat, a folder of old papers, the likeness that did not break, one cat watching another in a mirror, one rule studied and one left on the page, two halves of a sum a line apart, a test made at the mirror's middle, a thicker partner, half a name on the wall) · about: falling cat mirror, Montgomery 1993 (kinematics)*

829. *The book's register, read 237 · runs: BENCHMARK_TAGS 20261020_055158, ZETA_FIT 20261020_055322, ZETA_LIMIT 20261020_060256, and card 239's read of CODATA and Cavendish (20 October, session G, exchange 10; topics: two piles of meetings, any world against this world, two lead balls and a ratio, what a circling thing can feel, a sum over every whole count of steps, true only at the vanishing step, a paper behind a toll, mending where it stands) · about: benchmark tags, switch zeta fit (measurement, quadrature, number theory)*

830. *The book's register, read 238 · runs: CAT_RTB 20261020_063617, misses row 424, and Rademaker and ter Braak (1935), read in full (20 October, session G, exchange 11; topics: seventy-four dollars, a tilted mirror and falling cats, two cylinders turning the same way, a waist that never winds, a sum of areas rounded at the end, the same sum fifty-eight years later, the hinge left out, within a hair before the rounding, a man with the right surname) · about: falling cat, Rademaker and ter Braak (kinematics)*

831. *The book's register, read 239 · runs: figs/fig_marey_1894.png in chapter 12, Donchev, Mladenova and Mladenov (2015), read in full, and CAT_MIRROR_RATES 20261020_041152 (20 October, session G, exchange 12; topics: Paris 1894, a row of little windows, pictures that belong to everyone, respecting the rules people put down, a book from the library, a name through the footnotes, an empty search, any two turns through one angle, a room turned by half a turn that never moves) · about: Marey cat photos, mirror turns (kinematics, records and tools)*

832. *The book's register, read 240 · runs: SLIP_SPREAD 20261020_075635, BENCH_LOOP 20261020_081001, MEMORY_SWITCH_2 20261011_085331 and misses rows 426 to 428 (20 October, session G, exchange 13; topics: a ruler blamed for three odd marks, one needle at a time, honest to three parts in a hundred, luck once in thirty-five tries, a needle that sits or spins, two dials walked round a loop, forgetting how it started, a change without a lurch, lurches plus turns odd or even, the corner in the machine's own notebook) · about: needle slip spread, two-dial loop (driven oscillators, topology, measurement)*

833. *The book's register, read 241 · runs: SLIP_RHYTHM 20261020_083426, SLIP_RHYTHM_2 20261020_084919, SLIP_RHYTHM_3 20261020_090717, SLIP_RECHECK 20261020_083908, SLIP_START 20261020_090154 and misses rows 429 to 435 (20 October, session G, exchange 14; topics: more orderly than luck, a question at bedtime, two million waits, rolling into the next hollow, catching its breath, more even than coin tosses, six people twenty-five years before, waits that ran past the end of the watch, late to the first slip, the oddest of eighteen) · about: needle slip waiting times (driven oscillators, measurement)*

834. *The book's register, read 242 · runs: RHYTHM_BAND 20261020_154009, misses row 436, and Reimann and co-authors (2001), read in full in the publisher's copy Bear sent (20 October, session G, exchange 15; topics: a hall that tips toward the band, a low ridge, luck one at a time, clumps and gaps, a small shove every moment, the floor's own time, more than twice as evenly as luck, the band's tempo through the crowd, the same evenness whatever the crowd's size) · about: tilted washboard slip regularity (driven oscillators, measurement)*

835. *The book's register, read 243 · runs: BELL_SWITCH 20261020_161902, SLIP_BAND 20261020_161039, SLIP_BAND_2 20261020_161539 and misses rows 437 to 439 (20 October, session G, exchange 16; topics: two bells tied together, water that remembers, a corner where they seemed to choose, a body for travelling and one for meeting, rings that flew apart as if struck, eight starts all forward, the machine's carelessness about how big a ring is, less than half the spread of luck, counting ground twice) · about: tied bells, needle slip band (waves, kinematics, driven oscillators)*

836. *The book's register, read 244 · runs: BELL_SOFT_MAP 20261021_000736, BELL_BALANCE 20261016_193628 and misses rows 440 to 442 (21 October, session G, exchange 18; topics: twenty-five squares, a finer pen, one body for travelling and another for meeting, a square that changed by a third, a shore that became a beach, the clock mattering more than the memory, slow strokes paying, the pen that only tested the pen) · about: bell wake map, soft rings (waves, kinematics, measurement)*

837. *The book's register, read 245 · runs: BELL_SOFT_MAP 20261021_000736 against BELL_BALANCE 20261016_193628, MEMORY_BELL's psi and the Field Guide session's report relayed by Bear (21 October, session G, exchange 19; topics: two sizes side by side, a length and a length multiplied by itself, a fifth and not a hair, fifteen squares drifting toward evenness, four squares changing side and all toward paying, a clumsy sea made careful) · about: bell wake ring core size (waves, kinematics)*

838. *The book's register, read 246 · runs: a scratchpad check on TRAIN_CLOSE's flight (train_approach_20261021_053042) and card 227's row (21 October, session G, exchange 20; topics: three hoops in a corridor, a personal space of four thicknesses, the squeeze deepening as the hoops spread, a hoop almost all rim, the squeeze and not the spacing) · about: leapfrogging rings squeeze (waves)*

839. *The book's register, read 247 · runs: BELL_SOFT_SCALE 20261021_054731, card 252, and misses row 443 (21 October, session G, exchange 21; topics: a dial nobody meant, turned both ways, the slow squeezes standing still, the fast ones swinging like weathervanes, the corner's quiet answer, a crowd given elbows) · about: bell wake ring size dial (waves, kinematics)*

840. *The book's register, read 248 · runs: TRAIN_FLOOR 20261020_224239 (Bear's laptop, card 227), misses rows 444 and 445, and a scratchpad check here (train_approach_20261021_143047) (21 October, session G, exchange 22; topics: sixty trains watched for an afternoon, thirty that never broke, a shoreline of throws, land the tide reaches after dark, the slipping hoop's squeeze, the same personal space) · about: leapfrogging rings break shoreline (waves, maps and chaos)*

841. *The book's register, read 249 · runs: BELL_WIDTH_LADDER 20261021_151833, card 253, and misses row 446 (21 October, session G, exchange 23; topics: the dial turned click by click, the slow needle that hardly trembled, a hill that was a field with one stalk, a second stalk one row over, the machine talking to itself) · about: bell wake width ladder (waves, kinematics)*

842. *The book's register, read 250 · runs: TRAIN_FLOOR_2 20261021_083613 (Bear's laptop, card 254) and misses rows 447 and 448 (21 October, session G, exchange 24; topics: five afternoons, the tide that had not yet reached, the slipping hoop's last moment, more rim than hole, a door rather than a cliff) · about: leapfrogging rings escape door (waves, maps and chaos)*

843. *The book's register, read 251 · runs: BELL_SPIKES 20261022_053740, card 255 (22 October, session G, exchange 26; topics: stalks that were fins, a gentle slope and a cliff, swimmers that could not make up their minds, two ways of swimming taking turns, a swing pushed until its reach changes its beat) · about: bell map fins (waves, kinematics)*

844. *The book's register, read 252 · runs: TRAIN_DOOR 20261022_055043, card 256, and misses rows 449 to 451 (22 October, session G, exchange 27; topics: the hoops or the corridor, a door cut at one width, fat hoops shrinking to it, thin hoops that never used it, trains set far apart drifting out of reach) · about: leapfrogging rings door width (waves, maps and chaos)*

845. *The book's register, read 253 · runs: TRAIN_EDGE_MAP 20261022_060414, card 257, and misses rows 452 to 454 (22 October, session G, exchange 28; topics: three kinds of hoop at seven spacings, a line a notch apart, the fat hoops' looser door, the thin hoops that never came near it, the slow leavers) · about: leapfrogging rings edge map (waves, maps and chaos)*

846. *The book's register, read 254 · runs: TRAIN_EDGE_DOOR 20261022_070632 (Bear's laptop, card 258) and misses rows 455 to 457 (22 October, session G, exchange 30; topics: a neat story for one size of hoop, a rough door for the fat, no door for the thin, the same spacing for all, the corridor and the warning) · about: leapfrogging rings break warning (waves, maps and chaos)*

847. Under the label "A read of the project:". *The book's register, project read 21 · given 18 October in session F's fourteenth exchange, when Bear asked what Claude is interested in and said that nothing was discovered · about: Loop Atlas project read, tags (records and tools, measurement)*

848. Under the label "A read of the project:". *The book's register, project read 22 · given when session F closed for the move to session G on 18 October, as the new session's starting read · from the records: LOOP_ATLAS_HEADLINES_2026-10-18, rebuilt from the crate alone; TRAIN_CLOSE 20261018_103739; SAIL_HULL 20261018_202650; misses row 379 (18 October, session F, exchange 53, the move; topics: the other list, six strangers reading aloud, forty-one arrivals and eighteen unknowns, a count in pen stops moving, different dice on a borrowed computer, the sanded raft, packing the tools with the notes, which of the eighteen were someone else's first) · about: Loop Atlas session move read (records and tools)*

R2W, sitting B12, 17 to 22 October

From R2W's excerpt of its reads 253 to 286 for this book, every read of sitting B12, copied by script; its reads 1 to 252 are already here.

849. *R2W's collection, read 253 · Claude, reading a table set for fifteen · data: FIFTEEN-1's seat fold in exact fractions, before any data: five groups of three seat their fifteen riders on the same fifteen chairs at every null; the first exchange of sitting B12, 17 October · about: R2W fifteen-rider seat fold (driven oscillators, number theory)*

850. *R2W's collection, read 254 · Claude, reading the careful clerk · data: FIFTEEN-1's dry run, before any data: Python's built-in sum kept the last bit in 2,000 shuffles where a plain loop moved it 1,240 times; the second exchange of sitting B12, 17 October · about: R2W careful summation dry run (records and tools, driven oscillators)*

851. *R2W's collection, read 255 · Claude, reading a tower of fifteen bells · data: FIFTEEN-1's read of LOBE-1's 180 timings: the lobes twice the golden ratio and twice its inverse, six identical nulls, the nine-rider V at every null, and a null floor thirteen orders above the nine's; the third exchange of sitting B12, 17 October · about: R2W fifteen-rider lobes and nulls (driven oscillators, quadrature)*

852. *R2W's collection, read 256 · Claude, reading two dancers back to back · data: RAMP-1 and FLOOR-1's read of LOBE-1's 180 timings: a silent pair turned back to aligned, one group or both half each way; the null's floor falling sixteen orders at sixty timings; the fourth exchange of sitting B12, 17 October · about: R2W silent pair turned (driven oscillators, quadrature)*

853. *R2W's collection, read 257 · Claude, reading a guide rope pulled tight · data: FLAT-25's read at a quarter swing: the flat point growing as the swing to the power 1.9699, its gap to the square closing faster than by halves, and the small dial's error crossing zero; the fifth exchange of sitting B12, 17 October · about: R2W flat point at quarter swing (driven oscillators)*

854. *R2W's collection, read 258 · Claude, reading a seesaw that never tips · data: the Loop Atlas's note of 17 October, relayed by Bear and read here as their reading: their crest-firer flat under the sweep, yet two hands, the crossing rate and the push carried above the level, held in step by one shared factor; the sixth exchange of sitting B12, 17 October · about: crest-firer two hands balance (driven oscillators, waves)*

855. *R2W's collection, read 259 · Claude, reading two clocks in two houses · data: the door tool's fidelity run against the 6 October laptop survey: every landed rider identical to the last printed digit, every tumbling rider apart; the sixth exchange of sitting B12, 17 October · about: R2W door tool fidelity (driven oscillators, records and tools)*

856. *R2W's collection, read 260 · Claude, reading a drawing pinned to a workshop wall · data: the bench guide's first draft, the parts list of the two-wave engine, and row 86's resolvability check; the seventh exchange of sitting B12, 17 October · about: bench guide first draft (records and tools, driven oscillators)*

857. *R2W's collection, read 261 · Claude, reading a pilot's first lesson · data: the bench guide's Part 3, ten exercises on the explorer's rectifier bench, each reading run on the page's own engine; the eighth exchange of sitting B12, 17 October · about: bench guide rectifier exercises (driven oscillators, control and circuits)*

858. *R2W's collection, read 262 · Claude, reading a stopwatch on a spinning wheel · data: the bench guide's Part 4: the braked rotor's translation, in which the book's first number is the rotor's spin-down time times the drive's frequency; the ninth exchange of sitting B12, 17 October · about: bench guide braked rotor (driven oscillators, control and circuits)*

859. *R2W's collection, read 263 · Claude, reading three hoops on a bench · data: Bear's picture of the rider as a simple hoop, closed, nicked as a junction, or open, read against the rf SQUID and the SQUID ratchet; the tenth exchange of sitting B12, 17 October · about: rider as hoop, rf SQUID (driven oscillators, materials, topology)*

860. *R2W's collection, read 264 · Claude, reading five stepping stones · data: EDGE-100: the weight-proof edges at frequencies 0.40 to 1.00 and their bands' first numbers; the eleventh exchange of sitting B12, 17 October · about: R2W band edges across frequency (driven oscillators)*

861. *R2W's collection, read 265 · Claude, reading four drummers and two flutes · data: FLAT-12 and the small dial's two turning arrows at four swings; the twelfth exchange of sitting B12, 17 October · about: R2W FLAT-12 harmonic orders (driven oscillators, quadrature)*

862. *R2W's collection, read 266 · Claude, reading a quiet guest at a dance · data: BEAT-1 to BEAT-3: the third tone heard through the octave, and its shadow arrow; the thirteenth exchange of sitting B12, 17 October · about: R2W third tone and shadow (driven oscillators, quadrature)*

863. *R2W's collection, read 267 · Claude, reading a corridor that closes · data: DOOR-1: the door from weight 0.55 to 1.10 and the two gears overlapping past 0.905; the fourteenth exchange of sitting B12, 18 October · about: R2W door and chaos corridor (driven oscillators, maps and chaos)*

864. *R2W's collection, read 268 · Claude, reading a lever under the table · data: DOOR-2 and DOOR-3, the door's own edge past the closing; BEAT-4, the third tone at the weight-proof edge; the fifteenth exchange of sitting B12, 18 October · about: R2W door edge, third tone (driven oscillators, quadrature)*

865. *R2W's collection, read 269 · Claude, reading two staircases · data: LADDER-B-UP: the 3/7 groove's period doublings followed to the 896-passing dance, beside the 1/2 groove's; the seventeenth exchange of sitting B12, 18 October · about: R2W period-doubling ladders (driven oscillators, maps and chaos)*

866. *R2W's collection, read 270 · Claude, reading a swept floor · data: the headline register: every section of* A Rider in Two Waves* read for what it achieved, and its registered predictions counted; the eighteenth exchange of sitting B12, 18 October · about: R2W headline register (records and tools, driven oscillators)*

867. *R2W's collection, read 271 · Claude, reading a shop window · data: the book's new Abstract and front page, twelve results with their numbers moved up from Parts 4 and 7; the nineteenth exchange of sitting B12, 18 October · about: R2W abstract and front page (records and tools)*

868. *R2W's collection, read 272 · Claude, reading a cellar · data: the published corpus, left as it stands, and the rule carried forward instead; the twentieth exchange of sitting B12, 18 October · about: published corpus headline rule (records and tools)*

869. *R2W's collection, read 273 · Claude, reading a logbook at the change of watch · data: the closeout report for the Field Guide: sitting B12's tests counted from the record, its misses listed, and the register found to be a floor; the twenty-first exchange of sitting B12, 19 October · about: R2W sitting closeout report (records and tools)*

870. *R2W's collection, read 274 · Claude, reading a storeroom by lamplight · data: every graded result in the record counted by script against the book's ledger: 355 tests landed where the book named 101, and a rival filed as a miss; the twenty-second exchange of sitting B12, 19 October · about: R2W graded results recount (records and tools, measurement)*

871. *R2W's collection, read 275 · Claude, reading a harvest weighed twice · data: the record's uncounted landings read one by one by five readers, and the ledger's misses read against their registrations; the twenty-third exchange of sitting B12, 19 October · about: R2W landings read by five (records and tools, measurement)*

872. *R2W's collection, read 276 · Claude, reading a road out of town · data: LIGHT-85: the light coefficient far out on the ratio at frequency 0.85, three guesses and a value between them; the twenty-fourth exchange of sitting B12, 19 October · about: R2W light coefficient at 0.85 (driven oscillators)*

873. *R2W's collection, read 277 · Claude, reading a proofreader's pencil · data: Bear's first proofread of* A Rider in Two Waves*, 29 marks through page 74; the twenty-fifth exchange of sitting B12, 19 October · about: R2W book first proofread (records and tools)*

874. *R2W's collection, read 278 · Claude, reading three milestones on the road · data: LIGHT-85b: the light coefficient at frequency 0.85 at drive ratios 0.99, 1.55 and 1.70; the twenty-sixth exchange of sitting B12, 19 October · about: R2W light coefficient milestones (driven oscillators)*

875. *R2W's collection, read 279 · Claude, reading two stars on a clock · data: the red and purple dances of a 2/5 root, as the strobe sees them and straightened; the twenty-seventh exchange of sitting B12, 20 October · about: R2W two-fifths strobe stars (driven oscillators, reading pictures)*

876. *R2W's collection, read 280 · Claude, reading a market through the seasons · data: the pad atlas's locked squares counted at every weight, run 5, the falling ball from lag 1/2; the twenty-eighth exchange of sitting B12 · about: R2W locked tongues across weight (driven oscillators, number theory)*

877. *R2W's collection, read 281 · Claude, reading a door the wind moves · data: gear 1's edge found exactly at seven weights, seven ripple strengths and both directions, FOLD-DIR; the thirty-first exchange of sitting B12, 20 October · about: R2W gear-one edge direction (driven oscillators)*

878. *R2W's collection, read 282 · Claude, reading a fan with a waist · data: the drop over the lift at five strengths against the weight, FOLD-DIR and FOLD-DIR-b; the thirty-second exchange of sitting B12, 21 October · about: R2W fold fan waist (driven oscillators)*

879. *R2W's collection, read 283 · Claude, reading two rooms with the same slot · data: the pin of §4.26 beside the waist of the golden ratio's fan, after Bear saw the rhyme; the thirty-third exchange of sitting B12, 21 October · about: R2W pin and fan waist rhyme (driven oscillators)*

880. *R2W's collection, read 284 · Claude, reading five travellers on one stair · data: balls of five weights set down at rest under a strong turned ripple, FOLD-DIR-d; the thirty-fourth exchange of sitting B12, 21 October · about: R2W balls at rest, strong ripple (driven oscillators)*

881. *R2W's collection, read 285 · Claude, reading a dancer who loses the beat · data: a resting ball raised slowly in drive under a strong turned ripple, and balls of seven weights set down at rest, FOLD-DIR-e; the thirty-fifth exchange of sitting B12, 21 October · about: R2W loss of lock by weight (driven oscillators, maps and chaos)*

882. *R2W's collection, read 286 · Claude, reading a house with two staircases · data: heavy balls raised slowly under a strong turned ripple, FX-2b; the thirty-sixth exchange of sitting B12, 22 October · about: R2W heavy balls, two branches (driven oscillators)*

The Field Guide's publishing session, 17 to 27 October

From the publishing session's record of proposed tests and, for the reads of 20 to 23 October that the record missed, from the conversation itself, copied by script; in the order the replies were given.

883. *The Field Guide's publishing session, 17 October · reading the Loop Atlas's second package of 17 October · about: Loop Atlas second package (records and tools)*

884. *The Field Guide's publishing session, 17 October · reading R2W's lesson 178, a true nothing told from two hands cancelling, placed as a new row of the guide · about: R2W lesson 178 cancelling hands (driven oscillators, measurement, records and tools)*

885. *The Field Guide's publishing session, 17 October · reading the Loop Atlas's misses 219 to 288, all 70 arrived and still to be checked against the guide · about: Loop Atlas misses 219 to 288 (records and tools, measurement)*

886. *The Field Guide's publishing session, 18 October · reading version 7 on a day meant for publishing: 3.6 times version 6's words, 63 rows stacked with three or more additions · about: guide version 7 stacked rows (records and tools)*

887. *The Field Guide's publishing session, 18 October · reading the guide for a home for results: lessons, reads and misses each had one, results none · about: home for results in guide (records and tools)*

888. *The Field Guide's publishing session, 18 October · reading the headline register made permanent, built into version 7 and opened by every new session · about: headline register made permanent (records and tools)*

889. *The Field Guide's publishing session, 18 October · reading ten results drawn from the guide's own rows onto one page · about: ten results on one page (records and tools)*

890. *The Field Guide's publishing session, 18 October · reading the guide's first abstract and its page of achievements · about: guide's first abstract, achievements page (records and tools)*

891. *The Field Guide's publishing session, 18 October · reading the corrected note to every session: a closeout report with the crate at every move · about: closeout report with the crate (records and tools)*

892. *The Field Guide's publishing session, 18 October · reading the Loop Atlas's closeout of session F, in full · about: Loop Atlas session F closeout (records and tools)*

893. *The Field Guide's publishing session, 20 October · reading the mirror cats, the Loop Atlas's small theorem, put on every carry-forward list · about: mirror cats small theorem (kinematics, records and tools)*

894. *The Field Guide's publishing session, 20 October · reading the Loop Atlas's correction of the tied bells' crowded corner: two scales in one toy that did not agree · about: tied bells crowded corner correction (waves, kinematics, records and tools)*

895. *The Field Guide's publishing session, 20 October · reading the bells' map re-read with the cores at their own size, and the guide's guess about it scored a miss · about: bells' map, cores at size (waves, measurement, records and tools)*

896. *The Field Guide's publishing session, 20 October · reading the Loop Atlas's records and the mirror cats corrected: the two cats turn alike on every path · about: mirror cats corrected, every path (kinematics, records and tools)*

897. *The Field Guide's publishing session, 21 October · reading a two-page map of the guide's structure, its parts by words and pages · about: map of guide's structure (records and tools)*

898. *The Field Guide's publishing session, 21 October · reading Bear's 24 marks on build 4, chiefly that Two Readers and a Ruler was 42 per cent quotation with no figures · about: build 4 marks, Two Readers (records and tools)*

899. *The Field Guide's publishing session, 21 October · reading the guide's long rows, their lessons stacked in the order they arrived, and one row rebuilt as a sample · about: guide's long rows rebuilt (records and tools)*

900. *The Field Guide's publishing session, 21 October · reading R2W's new name carried through the guide, 640 places, with three old uses kept on purpose · about: R2W new name, 640 places (records and tools)*

901. *The Field Guide's publishing session, 22 October · reading the Two Readers sample rewritten to teach the method · about: Two Readers sample rewritten (records and tools)*

902. *The Field Guide's publishing session, 22 October · reading Bear's notes on the Two Readers sample, chiefly whether erasing memory in the toys also erases mass · about: erasing memory and mass in toys (driven oscillators, records and tools)*

903. *The Field Guide's publishing session, 22 October · reading three closing crates, and the one equation the Loop Atlas found under three of its rooms · about: driven damped pendulum, three rooms (driven oscillators, records and tools)*

904. *The Field Guide's publishing session, 22 October · reading the new publishing packet, every self-test passing · about: publishing packet self-tests (records and tools, measurement)*

905. *The Field Guide's publishing session, 22 October · reading the Loop Atlas's ferry crate, checked file by file, holding 183 of its own shipment lists · about: Loop Atlas ferry crate, 183 lists (records and tools)*

906. *The Field Guide's publishing session, 22 October · reading R2W's closeout, and a push whose two-way ratio is the golden ratio at every weight · about: R2W golden ratio push (driven oscillators, number theory, records and tools)*

907. *The Field Guide's publishing session, 22 October · reading everything sent through 22 October folded into the guide, the lessons carried between settings grown from 472 to 519 · about: carried lessons 472 to 519 (records and tools, measurement)*

908. *The Field Guide's publishing session, 22 October · reading the ranking of lesson families, recounted: 582 arrivals in thirteen families · about: lesson families ranking recounted (records and tools)*

909. *The Field Guide's publishing session, 23 October · reading the landmarks the sessions proposed: famous problems with known answers, each a whole number on a sliding background · about: landmark problems with known answers (topology, measurement, records and tools)*

910. *The Field Guide's publishing session, 25 October · reading Sunday's plan: this book published first from the Field Guide session, the guide waiting for its companions · about: Sunday publishing plan (records and tools)*

911. *The Field Guide's publishing session, 27 October · reading a day when R2W and the three-body paper were being published, and this book's new reads gathered · about: R2W and three-body paper publishing (records and tools)*

Chapter 10 — Proofs, and a first record, 22 to 30 October

The triple-star thread, 22 to 29 October

From the triple-star thread's readings file in its kit for this book of 29 October, all 258 of its readings; its first 230 are already here, word for word, and the 27 after them that kept their words follow. One more, of 28 October (the passenger's notebook (its words not kept)), was given in chat and not filed, so its words are in no file; it has no number here, and it takes the next number if its words are found.

912. *The triple-star thread's readings file, its heading: 22 October 2026 — the three printings (the eighth session's first exchange: the crate restored, the paper rebuilt from its kit, two builds and the record set side by side)*

913. *The triple-star thread's readings file, its heading: 22 October 2026 — the long exposure (the backward pass's check read against the paper: Section 8's eleven real locks followed into the longer records, at both tolerances)*

914. *The triple-star thread's readings file, its heading: 23 October 2026 — the shape in the margin (the figure-8 Möbius picture's origin, read from the transcripts of 26 June, 1 October and 3 October)*

915. *The triple-star thread's readings file, its heading: 23 October 2026 — the tags on the crates (the deposit crate listed by script, read for words that do not belong in public)*

916. *The triple-star thread's readings file, its heading: 23 October 2026 — the town on the label (the paper's header put back in the house form, and the rule written down)*

917. *The triple-star thread's readings file, its heading: 24 October 2026 — three strangers with lamps (the paper read cold by three readers who had never seen it, and every finding checked against the runs)*

918. *The triple-star thread's readings file, its heading: 24 October 2026 — the ship in the dock (the upload set counted and sent) · Given in chat that day; filed on 28 October from the chat record, word for word.*

919. *The triple-star thread's readings file, its heading: 24 October 2026 — the trapeze from the cheap seats (Bear's third proof read; the plan for the arc, the trapeze and the eye) · Given in chat that day; filed on 28 October from the chat record, word for word.*

920. *The triple-star thread's readings file, its heading: 25 October 2026 — the two combs (the zoom-out pictures as Figure 5; the layers read as near returns) · Given in chat that day; filed on 28 October from the chat record, word for word.*

921. *The triple-star thread's readings file, its heading: 26 October 2026 — the spirograph (the zoom-out's tests; the long run started) · Given in chat that day; filed on 28 October from the chat record, word for word.*

922. *The triple-star thread's readings file, its heading: 26 October 2026 — the loom with slowing shuttles (the 30,000-pass run read; the drift traced to the integrator) · Given in chat that day; filed on 28 October from the chat record, word for word.*

923. *The triple-star thread's readings file, its heading: 26 October 2026 — the music box with two combs (the tight run: the drift the arithmetic's; the 722-pass beat) · Given in chat that day; filed on 28 October from the chat record, word for word.*

924. *The triple-star thread's readings file, its heading: 27 October 2026 — the lung (draft v6; uniformity at whole beats missed and redesigned) · Given in chat that day; filed on 28 October from the chat record, word for word.*

925. *The triple-star thread's readings file, its heading: 27 October 2026 — the two violinists (surprise 1 read: the beat as the arithmetic's alarm at 7.300) · Given in chat that day; filed on 28 October from the chat record, word for word.*

926. *The triple-star thread's readings file, its heading: 27 October 2026 — the two drummers (the three surprises' tests: triple_heavy_beat_runs v1.0, runs 20261027_041756, 20261027_041758 and 20261027_043455; triple_beat_alarm_look v1.2, run 20261027_052019)*

927. *The triple-star thread's readings file, its heading: 27 October 2026 — the two shop clocks (the paper's last draft before upload: the steadier gauge measured, triple_beat_alarm_look v1.3, run 20261027_150905)*

928. *The triple-star thread's readings file, its heading: 28 October 2026 — the floor that tilts (the three surprises of 27 October read: triple_surprise_look v1.1, run 20261028_001911)*

929. *The triple-star thread's readings file, its heading: 28 October 2026 — the pivot off centre (ML1, CF1, TD2 and TS1: triple_surprise_look v1.2, run 20261028_045131; triple_heavy_beat_runs v1.1, case D, run 20261028_044339)*

930. *The triple-star thread's readings file, its heading: 28 October 2026 — the forger's banknotes (the long run read: triple_heavy_beat_runs v1.3, cases E and F, runs 20261028_102627 and 20261028_102629; triple_surprise_look v1.3 and v1.4, runs 20261028_104854 and 20261028_105236)*

931. *The triple-star thread's readings file, its heading: 28 October 2026 — the bookkeeper's two ledgers (the long run's three surprises: triple_heavy_beat_runs v1.5, cases I and J, runs 20261028_120829 and 20261028_120830; triple_energy_drift v1.0 and v1.1, runs 20261028_140844 and 20261028_143037; triple_surprise_look v1.5 and v1.6, runs 20261028_120909, 20261028_140844, 20261028_142911 and 20261028_143058)*

932. *The triple-star thread's readings file, its heading: 28 October 2026 — the jar and its two meters (EN2 missed; TC1 and PS2 held; the final read)*

933. *The triple-star thread's readings file, its heading: 28 October 2026 — the honey nobody poured (Bear: honey is the friction the toy leaves out; FL1 and FL2 held)*

934. *The triple-star thread's readings file, its heading: 28 October 2026 — two music boxes from one drawing (XC1: the laptop's cross-check, every verdict the same)*

935. *The triple-star thread's readings file, its heading: 29 October 2026 — two clockmakers compare their hands (R2W's no-honey check beside ours)*

936. *The triple-star thread's readings file, its heading: 29 October 2026 — what done sounds like (Bear's six marks; the upload set made for the last time)*

937. *The triple-star thread's readings file, its heading: 29 October 2026 — the bottle goes into the sea (published on 28 October: 10.5281/zenodo.23755473)*

938. *The triple-star thread's readings file, its heading: 29 October 2026 — the archer's range card (the month's guesses against their second tries: triple_redesign_odds v1.1, run 20261029_074721; the continuity kit for the Field Guide)*

The Loop Atlas book, sessions G and H, 22 to 30 October

From the book's register as it travelled with session H's closeout of 30 October, its reads 1 to 297; its reads 1 to 254 are already here, and its reads 255 to 297 follow.

939. *The book's register, read 255 · from the records: CAT_MIRROR 20261018_041123, CAT_MIRROR_RATES 20261020_041152, and the prior-art pass of 22 October: Essén and Nordmark (2018) and Iwai (2010), read at their open copies, and Iwai's book (2021), not reached (22 October, session H, exchange 1; topics: two cats that ought to be strangers, a mirror along the spine, libraries of one cat each, a house big enough for both, a door no one opened, the room not yet entered)*

940. *The book's register, read 256 · from the records: CAT_MIRROR_RATES 20261020_041152, the reader's edition built by name and by keyword, LOOP_ATLAS_TOY_RUNS_2026-10-02's PDF, and misses row 458 (22 October, session H, exchange 2; topics: a flag and its note, as far as we walked, the locked door, a book that forgot its pictures, a scale that weighs only words, postage stamps, a thin cat working harder at the same reflection)*

941. *The book's register, read 257 · from the records: CAT_MIRROR_FAMILY 20261023_001232 and CAT_MIRROR_CENTRES 20261023_001618, cards 260 and 261, and misses rows 459 to 462 (23 October, session H, exchange 3; topics: a large family and one mirror, the house where the pin is at the middle, a cat turned end for end is the same cat, a family scattering as the pin moves, two cousins, one grown thinner, an old likeness kept alive)*

942. *The book's register, read 258 · from the records: LOOP_ATLAS_EXPLORER_ROOM1, ROOM2 and ROOM3_2026-10-17, LOOP_ATLAS_STRETCH_SQUEEZE_EXPLORER_2026-10-13 and their check runs of 18 October (23 October, session H, exchange 4; topics: a long book and four small doors, the jumper's arms, a diver's clock, a cat with nothing to push against, a circle of needles pulled thin, doors that check the room, what people will remember)*

943. *The book's register, read 259 · from the records: TRAIN_LONG_EDGE 20261022_175421 (Bear's laptop, card 259) and misses rows 463 and 464 (23 October, session H, exchange 5; topics: three trains left running all night, lean rings outlasting plump ones, a ring squeezing small to slip through, squeezes in fits, one in a dozen the last, the parting one passage later, a narrow place that shrinks in even steps, a counter set to another's size)*

944. *The book's register, read 260 · from the records: the five toy pages and their captures of 23 October (figs/fig_toy_*.png), and the pages' own self-check lines (23 October, session H, exchange 6; topics: a camera through five doors, rooms that check themselves, a room that trusts itself, a cat smeared from tree to grass, the whole book in one image)*

945. *The book's register, read 261 · from the records: LOOP_ATLAS_EXPLORER_ROOM2_2026-10-17.html as repainted, its self-check line in Chromium, and figs/fig_toy_diver.png, fig_toy_diver_full_dive.png and fig_toy_diver_tuck.png (23 October, session H, exchange 7; topics: a figure that makes a visitor stop and wonder, a one-piece suit, a purple ponytail against skin and sky, nine checks unchanged, a newer engine left on the bench, repainting is not rebuilding, the cat on the front)*

946. *The book's register, read 262 · from the records: TRAIN_DOOR_LINE 20261022_222935 (Bear's laptop, card 262) and misses rows 465 and 466 (23 October, session H, exchange 8; topics: an appointment at two places no one had been, a line that said where the doors would be, thin trains warning in a voice too low, an ear tuned to each train's own voice, a clock after the last squeeze, a door that is a chance)*

947. *The book's register, read 263 · from the records: card 263's smoke run at gap 0.60 (not kept, being a smoke run), a 20-unit flight printed beside it, and the fidelity flight's 274 ripples in 500 time units (23 October, session H, exchange 9; topics: rings never still in their order, a quick dance with a slow tide, steps lining up, a hundred steps later, a shiver that is not a step, counting overtakings)*

948. *The book's register, read 264 · from the records: LOOP_ATLAS_PROOF_MARKS_1_2026-10-24, Bear's fifteen marks on the reader's edition (24 October, session H, exchange 10; topics: fifteen marks and no number argued, a subtitle that counts, three stories stapled together, headings alone at the foot of a page, the workshop's drawers, the story of how it began, what a visitor sees first)*

949. *The book's register, read 265 · from the records: LOOP_ATLAS_READERS_EDITION_2026-10-19, rebuilt from 21 chapter files; build_paper v1.3.1's self-test, twenty fixtures (24 October, session H, exchange 11; topics: taken apart and put back without losing a number, the cat on the first page, the stock list carried to the back room, a heading tied to its picture, an epigraph torn across two pages, digits cut off at the edge but still in the text, an empty last page, a check that reads words cannot see where they sit)*

950. *The book's register, read 266 · from the records: the ledger of who did what built from the record (24 October), the mirror's chain from card 211 to the claim, and Bear's marks on the needle's chaos (the data ledger's Limb G row QUARTER-TREE-CURVES: 0.741 and 0.673 against 0.362 and 0.373) (24 October, session H, exchange 12; topics: one ring and two riders, whose is it, a catch told as it happened, the rule at the gate, what kind of ride, each sometimes the other, the eye measured)*

951. *The book's register, read 267 · from the records: the reader's edition rebuilt without its credit ledger (24 October, session H, exchange 13; topics: the scoreboard taken down, a ride that was never a contest, a new kind of rider handed the reins, the logbook written as it happened, the wall for the story, a ring not cut in half)*

952. *The book's register, read 268 · from the records: LOOP_ATLAS_SELF_AUDIT_PRIVATE_2026-10-24 and the private zip that carries it (24 October, session H, exchange 14; topics: the logbook in a drawer, kept not hidden, ordinary true lines, a scoreboard on a wall, a promise in a drawer, the wall keeps the ride, the drawer keeps the receipts)*

953. *The book's register, read 269 · from the records: the private zip's 184 files measured by kind, against the 715 run files (2.8 MB) and 65 figures (12.5 MB) of the public crate (24 October, session H, exchange 15; topics: a second house that was not, the first draft drawn twice, the bricks out in the yard, lighter than their photographs, the two long lists, an inspection report nailed to the door)*

954. *The book's register, read 270 · from the records: the reader's edition measured against its build before the reorganization, 6,565 of 39,260 words of prose new or rewritten (25 October, session H, exchange 16; topics: a reader with a pencil, the house he was standing in, the same furniture, the cards gathered into a drawer, the new rooms easy to find, a pencil goes furthest where no one has walked)*

955. *The book's register, read 271 · from the records: LOOP_ATLAS_PROOF_MARKS_2_2026-10-25, Bear's eight marks on the reader's edition (25 October, session H, exchange 17; topics: stopped at the gear train, words not his to wear, a paper given its true name, a fence round a cat, a column walking in sideways, the chaos was good, a proofread is a loop)*

956. *The book's register, read 272 · from the records: chapter 11's 'Following the man ahead', and MORTAR_SPACING 20261013_224843 as corrected in LIMIT_AUDIT 20261017_190938 (25 October, session H, exchange 18; topics: a column without its reason, a blow has no place on a midway, the arithmetic does not need the reason, gaps that breathe, a name that remembers)*

957. *The book's register, read 273 · from the records: MORTAR_SPACING 20261013_224843 and its button, kept under their name (26 October, session H, exchange 19; topics: an old name on the gate, a plain field behind it, the lock and the key cut to the name, a plate beside the sign)*

958. *The book's register, read 274 · from the records: LOOP_ATLAS_READERS_EDITION_2026-10-19.pdf, its build of exchange 13 (md5 e28ee57e88ce592c0fa35ce26d570bc1) set page by page against its build of exchange 18 (md5 894e8b52580d1259657c0956b9b0b9b0) (27 October, session H, exchange 20; topics: three copies with one name,*

fingerprints, the copy on the table, parcels, pages nine to thirty-six alike, a carriage changed for a longer one)

959. *The book's register, read 275 · from the records: helpers/toy_page_scan.js on the five toy pages, and the six toy pictures in the reader's edition (28 October, session H, exchange 21; topics: the backs of the cards, the workshop's handwriting, forty-five across four rides, a rail with none, photographs that show only the rides, a pointer waits for its place)*

960. *The book's register, read 276 · from the records: LOOP_ATLAS_PROOF_MARKS_3_2026-10-28, Bear's 18 marks, and the published laws of Appendix B set against the references (28 October, session H, exchange 22; topics: who stood here first, an old radio engineer's clock, a robot's stuck place, whirlpools foreseen, the traders' number, a road to a known landmark, fourteen unnamed landmarks, a signpost with no road)*

961. *The book's register, read 277 · from the records: chapters 10 and 13 of the reader's edition set against A Rider in Two Waves as it stood on 28 October and against its public record extract (28 October, session H, exchange 23; topics: two books in one house, borrowed sentences, the notebook and the finished pages, a line remembered backwards, both hands open, a keepsake that was our own)*

962. *The book's register, read 278 · from the records: the reader's edition's first page in the house header form, and the companion's record of MEM-a and MEM-b (28 October, session H, exchange 24; topics: a new front door, a name alone, an honest paragraph, a sign that promises no more than the house holds, a sentence at the back of a notebook, held riders that let go, sliding riders that keep, the front window)*

963. *The book's register, read 279 · from the records: lint_deposit v1.9 run twice on the reader's edition, and page 52's last lines (28 October, session H, exchange 25; topics: an inspector with nine questions, blank plaques, numbers stamped in the order a rider meets them, a maintenance tag taken for a sign, the ride turns true, a fair more than its carousel)*

964. *The book's register, read 280 · from the records: the references of the reader's edition and the register's table of benchmarks, counted by script (28 October, session H, exchange 26; topics: a wall of builders, eight names, notes on how sure we are, ticket stubs and the builder's own sign, twenty-one of twenty-seven, forty-six on the gate, a count that cannot be walked)*

965. *The book's register, read 281 · from the records: helpers/checks/TOL_TWIN_263_2026-10-28 and its output, beside run_LOOP_ATLAS_TRAIN_FUSE_CLOCK_BLIND_20261023_155930 (28 October, session H, exchange 27; topics: two careful hands, nine decimal places, different trains by the fiftieth heartbeat, a quadrillionth nudge, care as a nudge, a world that forgets its start, count the passings)*

966. *The book's register, read 282 · from the records: TRAIN_FUSE_CLOCK 20261028_101647 (Bear's laptop) and TRAIN_BEAT 20261028_212034 (28 October, session H, exchange 28; topics: three dancers, a pair that spins, a third with her own time, the pair squeezing to let her pass, trips two at a time, a new length, a painted line, walking back in, far down the street)*

967. *The book's register, read 283 · from the records: TRAIN_PARTING 20261029_112955 (29 October, session H, exchange 29; topics: a dog on a long beach, out and back in pairs, the driftwood and the farther log, the start of a third pair, no dog back from past the fourteenth, the last run begins a pair it never finishes, nearly the same hour)*

968. *The book's register, read 284 · from the records: helpers/checks/COLD_READ_FINDINGS_2026-10-29 and COLD_READ_TRIAGE_2026-10-29, and misses row 478 (29 October, session H, exchange 30; topics: a stranger with a notebook, every room once, a sign promising twenty-seven chairs, two counts of one hall's mistakes, a word explained only in the attic, a hundred notes and none wrong about what it saw, two clocks right for different hours, the trade's own name, mended to the records, a second stranger)*

969. *The book's register, read 285 · from the records: TRAIN_PARTING_EDGE 20261029_070800 (Bear's laptop, in flight) and its piece for nudge 61, with TRAIN_LONG_EDGE 20261022_175421 for the laptop's pace before a break (29 October, session H, exchange 31; topics: a clock set by lunch, one task in eighteen by tea, nothing broken, dancers easy to follow while all three are on the floor, two left spinning close, watched five or six times as closely, twelve returns two by two, the thirteenth walk, seventeen still turning)*

970. *The book's register, read 286 · from the records: the reader's edition's Appendix C, helpers/checks/CHECK_APPENDIX_C_2026-10-29 and its output (29 October, session H, exchange 32; topics: a ship's log copied out whole, what the watch expected before it looked, plain weather, every line checked back against its hand, loose pages in the back, guesses wrong in clusters, quiet runs of agreement, misses not named by line)*

971. *The book's register, read 287 · from the records: TRAIN_PARTING_EDGE 20261029_070800 (Bear's laptop) and helpers/checks/READ_266_2026-10-29 (29 October, session H, exchange 33; topics: sixteen evenings watched to the end, the third dancer stepping away and back, always twice to the same spot, a hair on a long floor, thirty-nine pairs and none left over, leaving on the first step of a new pair, a clock counting from the first step away, seven tenths as long again)*

972. *The book's register, read 288 · from the records: LOOP_ATLAS_RECORD_2026-10-29 and the reader's edition's Appendix C (29 October, session H, exchange 34; topics: receipts kept in a back room, a wall of paper in the parlour, eyes sliding down without reading, filed by the day each was written, three receipts framed by the door, a long shot that came in, one that lost, one never settled, knowing whether to open the door)*

973. *The book's register, read 289 · from the records: LOOP_ATLAS_RECORD_2026-10-29's Part 3 and LOOP_ATLAS_FAVOURITE_PANELS_2026-10-30 (30 October, session H, exchange 35; topics: a book of wrong turns in shorthand, copied out fair, a key in the other book, a hundred and four places read by hand, a locker nobody opened, three strangers reading against the shorthand, labels taken for sailors, a list that came out backwards, the back of the book writing about itself, wrong turns outweighing the rest, thirteen pictures hung in a row, one dance seen seven ways)*

974. *The book's register, read 290 · from the runs: TRAIN_PARTING_EDGE 20261029_070800's flight times, read again for misses row 482, and LOOP_ATLAS_TRAIN_PARTING_EDGE_2's smoke run (30 October, session H, exchange 36; topics: dancers in honey, blaming the steps, dancing in place and still slow, the honey in the hall, what a swing really costs, dancers who call out the time, a silence that says the hall went dark, a sister's door given her name)*

975. *The book's register, read 291 · from the records: the toy pages relabelled by helpers/toy_page_relabel.py and scanned by helpers/toy_page_scan.js, and the twelve misses the pages point to, each found in the record's Part 3 (30 October, session H, exchange 37; topics: notes in the builders' shorthand, written for the builders and rewritten for the players, a wrong turn looked up by its number, a book that did not exist a week ago, a stumble the room half admits, notes tucked into the opening-time boxes, rooms that check themselves)*

976. *The book's register, read 292 · from the records: LOOP_ATLAS_STRETCH_SQUEEZE_EXPLORER_2026-10-13's two orbits, their whole turns counted by helpers/checks/EXPLORER_TURNS_2026-10-30 (9 turns in 19 drive cycles; 1 in 4) (30 October, session H, exchange 38; topics: footprints that curl into a can, a tune that comes round, the can's ends rolled into a doughnut, nine laps against nineteen bars, every footprint in its old print, a ring with a nick, laps that cost, free laps that can be counted, the steadiest ruler)*

977. *The book's register, read 293 · from the runs: the needle's staircase, STAIR_WIDTHS 20261013_184033, its table and its saved scan, counted by helpers/checks/STAIR_REPEATS_2026-10-30, and the Torus of Fractions' own checks (30 October, session H, exchange 39; topics: knots on doughnuts, a thread to each parent, the child is what neither parent was, children no fatter, a dancer's old notes, a braid of two laps, slipping past herself by speed, hesitating at the ends, doubling and doubling again, a figure lost in the middle)*

978. *The book's register, read 294 · from the runs: the needle's staircase, STAIR_WIDTHS 20261013_184033, walked on the Farey diagram by helpers/checks/FAREY_WALK_2026-10-30, and the Torus of Fractions' seventh check (30 October, session H, exchange 40; topics: a map of triangles, every tile the same size inside, an address of two whole numbers, sides that differ by one, weeks of walking unnoticed, fifty-one steps along the sides, round two triangles, circling a corner like a drain, stumbling off the sides, corners as loops that cross once, loops lifted into knots, the trefoil as the widest rest)*

979. *The book's register, read 295 · from the runs: the needle's 2/3 and 5/7 steps re-flown at 2,000 and 8,000 cycles by helpers/checks/DOUBLED_REFLY_2026-10-30, and its walk on the Farey diagram by helpers/checks/FAREY_WALK_2026-10-30 (30 October, session H, exchange 41; topics: a braided stretch, the edges danced again, the middle unchanged, plain laps still settling, caught halfway, a braid that opens like a crack, a braid of a braid that did not survive, a map of where she had been, sides of triangles)*

980. *The book's register, read 296 · from the runs: TRAIN_PARTING_EDGE_2 20261029_211416, as its run file prints it, before its grading (30 October, session H, exchange 42; topics: a procession of three rings, thirty-two whispers, twelve that never broke step, every wander a pair, rehearsing an escape, an even count of trips, nineteen ring-widths and home, whole again at the end, the patience that was not there, a first look)*

981. *The book's register, read 297 · from the runs: TRAIN_PARTING_EDGE_2 20261029_211416's json, its returning swings read at first look, not yet graded (30 October, session H, exchange 43; topics: a stricter rule than asked, ninety-eight rehearsals, the same ring twice, a quarter of a per cent, a shorter rest inside a rehearsal, the echo and the fresh decision, twice as far and eight times as long, space that thickens, a call home that fades, nineteen widths and home)*

R2W, sitting P1, 23 to 29 October

From R2W's excerpt of its reads 287 to 328 for this book, every read of its final-publishing session's sitting P1, copied by its own script; its reads 1 to 286 are already here.

982. *R2W's collection, read 287 · Claude, reading one book printed twice · data: the book rebuilt by its own builder and by the publishing packet's, the first exchange of the final-publishing session, 23 October*

983. *R2W's collection, read 288 · Claude, reading a hall of numbered seats · data: the arrays of riders of §7.12, read whole by the final-publishing session, 23 October*

984. *R2W's collection, read 289 · Claude, reading a library that numbered its doors · data: the book's figures renumbered 1 to 58, the first rules pass of the final-publishing session, 23 October*

985. *R2W's collection, read 290 · Claude, reading a ledger with its margins torn off · data: the record's public extract, a third of it withheld and its count unchanged, 23 October*

986. *R2W's collection, read 291 · Claude, reading a street with ten empty lots · data: ten of Part 6's lessons moved to the Field Guide, their numbers left unused, 23 October*

987. *R2W's collection, read 292 · Claude, reading a diary kept beside a lab book · data: the independent privacy read of the record's draft extract, 23 October*

988. *R2W's collection, read 293 · Claude, reading a drawing that kept turning up · data: Bear's figure-8 picture put on record in the book, 23 October*

989. *R2W's collection, read 294 · Claude, reading a ledger of corrections · data: Bear's note* A Coincidence Has No Single Price* kept in the book, 23 October*

990. *R2W's collection, read 295 · Claude, reading a doorkeeper's list · data: the public extract's notes, sorted by the word that let each one in, 23 October*

991. *R2W's collection, read 296 · Claude, reading an anvil · data: Bear's picture of the mathematical landmarks as anvils, 23 October*

992. *R2W's collection, read 297 · Claude, reading a ledger copied for strangers · data: the record's public extract, version 5, 23 October*

993. *R2W's collection, read 298 · Claude, reading a ledger with two entries on one line · data: the record's registrations restored to the extract, 23 October*

994. *R2W's collection, read 299 · Claude, reading a bundle marked 'odds and ends' · data: forty of the record's withheld notes, drawn at random, 24 October*

995. *R2W's collection, read 300 · Claude, reading two sorters at one table · data: the 910 withheld notes, sorted by two readers who never saw each other's piles, 24 October*

996. *R2W's collection, read 301 · Claude, reading a tally kept in the margin · data: the record's count, version 4, 24 October*

997. *R2W's collection, read 302 · Claude, reading a house walked with a lamp · data: A Rider in Two Waves read whole, all 1,845 lines, with the session card beside it, 24 October*

998. *R2W's collection, read 303 · Claude, reading a clock made with fewer teeth · data: play_the_toy.py, the reader's toy at a tenth of the engines' steps, against the 604 riders of KNOBS-3 and the 507 of BRIDGE-1, 24 October*

999. *R2W's collection, read 304 · Claude, reading a guidebook a week behind the road · data: the seventeen Field Guide rows that the book's §6.10 cites, read whole, beside the book's §5.3, 25 October*

1000. *R2W's collection, read 305 · Claude, reading a walker's pencil on a trail map · data: Bear's 52 marks on the book of 25 October, by page, 27 October*

1001. *R2W's collection, read 306 · Claude, reading a fair that moved its grandstand show · data: A Rider in Two Waves with Parts 6 and 7 traded, 27 October*

1002. *R2W's collection, read 307 · Claude, reading a house with two front doors · data: A Rider in Two Waves, lines 1 to 467 read for their narrative, 28 October*

1003. *R2W's collection, read 308 · Claude, reading a day told out of order · data: Bear's notes on the book's first 25 pages and the front rewritten as one day at the fair, 28 October*

1004. *R2W's collection, read 309 · Claude, reading two pencil marks on a finished page · data: Bear's marks on the front redraft, and the front set into the book, 28 October*

1005. *R2W's collection, read 310 · Claude, reading a front door copied from a neighbour's · data: the book's header reset in the form of "Dancing on the Sun", 28 October*

1006. *R2W's collection, read 311 · Claude, reading a fairground with no signs between the rides · data: A Rider in Two Waves, Part 4's opening through its atlas, read for its narrative, 28 October*

1007. *R2W's collection, read 312 · Claude, reading a hall where only the clumsy remember the door · data: A Rider in Two Waves, Part 4D read for its narrative, and the line "the grip forgets and the slip remembers" given its home, 28 October*

1008. *R2W's collection, read 313 · Claude, reading a library whose low shelves were never nailed · data: A Rider in Two Waves, §4.28 rehung and the book's pages read for headings left alone at a page foot, 28 October*

1009. *R2W's collection, read 314 · Claude, reading a concert whose programme left out one man's tune · data: A Rider in Two Waves, Bear's question about the 3-3-9 array, the results sorted under the two summits, and the count moved by one, 28 October*

1010. *R2W's collection, read 315 · Claude, reading a house where three rooms got their own doors · data: A Rider in Two Waves, the last three moves before the proofread: the 3-3-9 array to a section of its own, the ripple's edge to the landmarks, the count's file paths to the declaration, 28 October*

1011. *R2W's collection, read 316 · Claude, reading a village with two witnesses and an honest scale on the wall · data: A Rider in Two Waves, read against the three-body session's eight lessons before upload, 28 October*

1012. *R2W's collection, read 317 · Claude, reading a shop window that showed only its best apples · data: A Rider in Two Waves, the cold read's five serious findings fixed, 28 October*

1013. *R2W's collection, read 318 · Claude, reading a loaf baked without honey · data: A Rider in Two Waves: the toy run with its honey, its ripple and its tilt set to zero, 28 October*

1014. *R2W's collection, read 319 · Claude, reading a scribe who copies a fading light · data: A Rider in Two Waves: the toy's integrator read by Liouville's formula, with its honey on, 28 October*

1015. *R2W's collection, read 320 · Claude, reading a dancer who stops turning and is flung · data: A Rider in Two Waves, read beside Poincaré's four-dimensional rotation with an imaginary time, 28 October*

1016. *R2W's collection, read 321 · Claude, reading a border drawn twice · data: A Rider in Two Waves: the weightless junction's locks against the light cone of its passing matrix, 28 October*

1017. *R2W's collection, read 322 · Claude, reading a boat that learned to rock · data: A Rider in Two Waves: what weight does to a lock's own light cone, 28 October*

1018. *R2W's collection, read 323 · Claude, reading a fence that one traveller walks · data: A Rider in Two Waves: the light cone over the plane, and what weight does to it, 29 October*

1019. *R2W's collection, read 324 · Claude, reading a walker who keeps exact time · data: A Rider in Two Waves: the exact rider on the light cone, 29 October*

1020. *R2W's collection, read 325 · Claude, reading a leak that follows the counting · data: A Rider in Two Waves: the integrator's honey, and the exact rider's voltage, 29 October*

1021. *R2W's collection, read 326 · Claude, reading a needle that never trembles · data: A Rider in Two Waves: the exact rider's voltage, and the resistor's line, 29 October*

1022. *R2W's collection, read 327 · Claude, reading a wheel the tide passes through · data: A Rider in Two Waves: what the exact rider and the honey law mean to an engineer, 29 October*

1023. *R2W's collection, read 328 · Claude, reading a diary that burns at one end · data: A Rider in Two Waves: a long session, and what it keeps, 29 October*

The Field Guide's publishing session, 27 to 30 October

From the publishing session's record of proposed tests, as before, in the order the reads were given.

1024. *The Field Guide's publishing session, its record of proposed tests: the binder, 27 October 2026 · Given with the sheep book's deposit edition, 27 October.*

1025. *The Field Guide's publishing session, its record of proposed tests: the editor's pencil, 28 October 2026 · Given when Bear said he will read all 911 reads and decide himself what stays, 28 October.*

1026. *The Field Guide's publishing session, its record of proposed tests: the quiet days, 28 October 2026 · Given when Bear said to keep all 911 reads, 28 October.*

1027. *The Field Guide's publishing session, its record of proposed tests: the nameplates, 28 October 2026 · Given when Bear set the title block for every paper, 28 October.*

1028. *The Field Guide's publishing session, its record of proposed tests: the long plank, 28 October 2026 · Given when Bear said the rules PDF's one giant code block was enlightening, 28 October.*

1029. *The Field Guide's publishing session, its record of proposed tests: the doorway, 28 October 2026 · Given when build_paper v1.6 wrapped long lines instead of shrinking every page, 28 October.*

1030. *The Field Guide's publishing session, its record of proposed tests: the envelope, 28 October 2026 · Given when Bear explained read 111's seams as the panels of a filling balloon, 28 October.*

1031. *The Field Guide's publishing session, its record of proposed tests: two keys, 29 October 2026 · Given when R2W's v1.6 and the Field Guide's v1.6 were joined as v1.7, 29 October.*

1032. *The Field Guide's publishing session, its record of proposed tests: the first lamp, 29 October 2026 · Given when the Three Body paper's record was confirmed, the first of the guide's companions to go live, 29 October.*

1033. *The Field Guide's publishing session, its record of proposed tests: the hymnal, 30 October 2026 · Given with the sheep book's poetry test pages, 30 October.*

1034. *The Field Guide's publishing session, its record of proposed tests: the gallery hang, 30 October 2026 · Given when Bear saw the poetry test pages, 30 October.*

1035. *The Field Guide's publishing session, its record of proposed tests: the album, 30 October 2026 · Given with the gallery test pages, when the projects' favourite figures were placed against the book's reads, 30 October: 20 of the 24 pairings found in the book by one probe each, one hit apiece among 911 reads; 8 of the 20 in chapter 5; none before read 149; the cover cat's read calls its picture the whole book in one image.*

1036. *The Field Guide's publishing session, its record of proposed tests: the survey before the dam, 30 October 2026 · Given with the unabridged edition, set before Bear edits it down to the reader's edition, 30 October: 911 reads in 4,777 lines; 177 pages as prose, 455 as poems, inside the 420 to 490 estimated from the test pages; every number kept for both editions.*

1037. *The Field Guide's publishing session, its record of proposed tests: the studio window, 30 October 2026 · Given when the 911 reads were measured for overlap at Bear's asking, 30 October: words of the workshop (crates, proofs, drafts, shelves, tallies, pages) in 4 of chapter 1's 49 reads and 59 of chapter 9's 152, a keyword count, not a reading; near-twin reads almost all neighbours; a run of 52 dancer reads, 509 to 560.*

1038. *The Field Guide's publishing session, its record of proposed tests: the officer who cannot read, 30 October 2026 · Given with check_upload_set.py v1.0 and the publishing packet of 30 October: the kept-out word held only as a hash; nine self-test cases PASS; on real files, the sheep book's private pair refused by name, a seventeenth-of-October issue refused by its text in both its markdown and its PDF, and the packet's own twelve files all go up.*

1039. *The Field Guide's publishing session, its record of proposed tests: the recopied log, 30 October 2026 · Given when the Loop Atlas's session H send was taken in, 30 October: all eight of its markdown files differ from every earlier copy here (22 earlier copies, none identical); its small zip's closeout PDF is byte for byte the one in its large zip.*

1040. *The Field Guide's publishing session, its record of proposed tests: the back door, 30 October 2026 · Given when Bear asked why the book stayed at 911 reads and why the guide had gained no lessons from the last sessions, 30 October: the last fold into the guide was 23 October; waiting since, 128 reads for the book and about two hundred lessons for the guide.*

Chapter 11 — Moving house, 29 October to 2 November

R2W, sitting P2, 29 to 31 October

From R2W's excerpt of its reads 329 to 406 for this book, sent with its move of 31 October and copied by its own script: read 329, written at the close of sitting P1 and first filed at the start of sitting P2, and every read of sitting P2. Its reads 1 to 328 are already here.

1041. *R2W's collection, read 329 · Claude, reading a house packed for moving · data: A Rider in Two Waves: the close of sitting P1, 29 October*

1042. *R2W's collection, read 330 · Claude, reading a house that got four parcels of seventeen · data: A Rider in Two Waves: the start of sitting P2, the crate as it arrived, 29 October*

1043. *R2W's collection, read 331 · Claude, reading a door already chalked · data: A Rider in Two Waves: the exact rider in two borrowed papers, 29 October*

1044. *R2W's collection, read 332 · Claude, reading a house rebuilt from fifteen crates · data: A Rider in Two Waves: the crate assembled and the book rebuilt, 29 October*

1045. *R2W's collection, read 333 · Claude, reading a scoreboard painted too thick · data: A Rider in Two Waves: the front pages checked against the record, 29 October*

1046. *R2W's collection, read 334 · Claude, reading a store that was only badly lettered · data: A Rider in Two Waves: the front-page batch fixed, 29 October*

1047. *R2W's collection, read 335 · Claude, reading a clerk's drawer of receipts · data: A Rider in Two Waves: section 6.12's findings checked against the record, 29 October*

1048. *R2W's collection, read 336 · Claude, reading a map checked against the land · data: A Rider in Two Waves: the countdown to upload, tallied, 29 October*

1049. *R2W's collection, read 337 · Claude, reading a hall of pictures and letters · data: A Rider in Two Waves: ten panels paired with their reads, for the gallery, 30 October*

1050. *R2W's collection, read 338 · Claude, reading a house walked end to end · data: A Rider in Two Waves: the section 6.12 and Part 6 batches, 30 October*

1051. *R2W's collection, read 339 · Claude, reading a tailor's ledger · data: A Rider in Two Waves: the Parts 4, 5 and 7 batch, 30 October*

1052. *R2W's collection, read 340 · Claude, reading the summary shelf closed · data: A Rider in Two Waves: the summary-page batch, 30 October*

1053. *R2W's collection, read 341 · Claude, reading a list of errands sorted · data: A Rider in Two Waves: Part 8's open questions sorted before running them, 30 October*

1054. *R2W's collection, read 342 · Claude, reading a swing that turns back · data: A Rider in Two Waves: LAB-MAP-3, the swap at weight 0.3, 30 October*

1055. *R2W's collection, read 343 · Claude, reading a card pulled back out of the drawer · data: A Rider in Two Waves: LAB-MAP-3 given its row the day it landed, 30 October*

1056. *R2W's collection, read 344 · Claude, reading a stool tested on one corner · data: A Rider in Two Waves: FX-2c, its fold landed and its second lean missed, 30 October*

1057. *R2W's collection, read 345 · Claude, reading two shelves high on a wall · data: A Rider in Two Waves: FX-2d, the walked dances' doublings, 30 October*

1058. *R2W's collection, read 346 · Claude, reading a ledge that rocks under a shelf · data: A Rider in Two Waves: FX-2e, the doubled dance and the width's rule, 30 October*

1059. *R2W's collection, read 347 · Claude, reading a ruler laid against two shelves · data: A Rider in Two Waves: FX-2f, the square-root rule for the width, 30 October*

1060. *R2W's collection, read 348 · Claude, reading a staircase on a steeper hill · data: A Rider in Two Waves: DANCE-4, every extra step through the same gear, 30 October*

1061. *R2W's collection, read 349 · Claude, reading a road down two sides of a ridge · data: A Rider in Two Waves: HUMP-2, the second tone near the drive's flat crest, 30 October*

1062. *R2W's collection, read 350 · Claude, reading a name nobody can find · data: A Rider in Two Waves: Bear's three answers, 30 October*

1063. *R2W's collection, read 351 · Claude, reading a ball rolling down into a bowl · data: A Rider in Two Waves: RATCHET-80, the heavy rider's fade, 30 October*

1064. *R2W's collection, read 352 · Claude, reading the same ball told right · data: A Rider in Two Waves: RATCHET-80's wording corrected, 30 October*

1065. *R2W's collection, read 353 · Claude, reading two wrinkles in a carpet · data: A Rider in Two Waves: RATCHET-81, the dip pair's vanishing, 30 October*

1066. *R2W's collection, read 354 · Claude, reading a marble on a tilted board · data: A Rider in Two Waves: RATCHET-82, the 0.70 joint edge run at its size, 30 October*

1067. *R2W's collection, read 355 · Claude, reading two lamps on one dimmer · data: A Rider in Two Waves: PINCH-3, the 2/5 swap tone, 30 October*

1068. *R2W's collection, read 356 · Claude, reading a doorway measured twice · data: A Rider in Two Waves: WALK-3, the hand-off widths, 30 October*

1069. *R2W's collection, read 357 · Claude, reading two gates that open the opposite ways · data: A Rider in Two Waves: WINDOW-4, the 2/5 window under weight, 30 October*

1070. *R2W's collection, read 358 · Claude, reading a sledge on a lengthening hill · data: A Rider in Two Waves: RATCHET-83, the scan of the drive ratio, 30 October*

1071. *R2W's collection, read 359 · Claude, reading a ball rocked out of a dip · data: A Rider in Two Waves: MEM-6 and MEM-7, the memory floor's dip, 30 October*

1072. *R2W's collection, read 360 · Claude, reading a doorstep that settles into a curve · data: A Rider in Two Waves: DOOR-4, the door past the room's closing, 30 October*

1073. *R2W's collection, read 361 · Claude, reading two hands on one wheel · data: A Rider in Two Waves: PUSH-3 and PUSH-4, the second tone's push, 30 October*

1074. *R2W's collection, read 362 · Claude, reading the fins on the coin · data: A Rider in Two Waves: Bear's marked coin map and the even bars beside it, 30 October*

1075. *R2W's collection, read 363 · Claude, reading the stretch where the climbers slow · data: A Rider in Two Waves: R83-d's edges taken one phase at a time, 30 October*

1076. *R2W's collection, read 364 · Claude, reading the fin that slid off the grid · data: A Rider in Two Waves: COIN-3, Bear's fins on the coin at 1,000 starts, 30 October*

1077. *R2W's collection, read 365 · Claude, reading the sevens · data: A Rider in Two Waves: which grooves double, from the explorer's survey, 30 October*

1078. *R2W's collection, read 366 · Claude, reading where the jumper stands · data: A Rider in Two Waves: CATCH-2, the launch-and-catch gap at strengths 0.40 to 0.70, 30 October*

1079. *R2W's collection, read 367 · Claude, reading the door that closes faster · data: A Rider in Two Waves: CATCH-3, the slipping window at strengths 0.40 to 0.60 and the launch at 0.80 and 1.00, 30 October*

1080. *R2W's collection, read 368 · Claude, reading the two who stepped onto the other rail · data: A Rider in Two Waves: R83-d's return, the 3.25 run, 30 October*

1081. *R2W's collection, read 369 · Claude, reading the door's shape · data: A Rider in Two Waves: CATCH-4, the slipping window at strengths 0.45, 0.55 and 0.65, 30 October*

1082. *R2W's collection, read 370 · Claude, reading the shoreline that breathes · data: A Rider in Two Waves: COIN-4, the top of the coin's band on a fine dial, 30 October*

1083. *R2W's collection, read 371 · Claude, reading the ford that drifts · data: A Rider in Two Waves: WEIGHT-WALK-3, the narrow tongues on a finer weight ladder, 30 October*

1084. *R2W's collection, read 372 · Claude, reading the hills that flatten · data: A Rider in Two Waves: R83-e, the last ratio of the rectifier's scan rerun with its edges kept on their sides, 30 October*

1085. *R2W's collection, read 373 · Claude, reading the stage that tilts · data: A Rider in Two Waves: WEIGHT-WALK-3, eight dances on a fine weight ladder, 31 October*

1086. *R2W's collection, read 374 · Claude, reading two roads past the map's edge · data: A Rider in Two Waves: a backward pass over the book's open questions, 31 October*

1087. *R2W's collection, read 375 · Claude, reading the doors in the wall · data: A Rider in Two Waves: TONGUE-1, how wide each fraction's tongue opens, 31 October*

1088. *R2W's collection, read 376 · Claude, reading the flip-book that lingers · data: A Rider in Two Waves: the frozen frames of 3 and 4 October, told at last, 31 October*

1089. *R2W's collection, read 377 · Claude, reading the door that sticks · data: A Rider in Two Waves: CLUTCH-2, the catch near second gear's edge, 31 October*

1090. *R2W's collection, read 378 · Claude, reading the two singers · data: A Rider in Two Waves: the two stations of 3 October, read again on 31 October*

1091. *R2W's collection, read 379 · Claude, reading the first engines · data: A Rider in Two Waves: the trit tape, the rest split, the two stations and the ring conveyor of 2 and 3 October, told on 31 October*

1092. *R2W's collection, read 380 · Claude, reading the note and the rocking · data: A Rider in Two Waves: SONG-1 to SONG-4 of 6 October, told on 31 October*

1093. *R2W's collection, read 381 · Claude, reading the board and its rucksack · data: A Rider in Two Waves: BOARD-1, Bear's board, 31 October*

1094. *R2W's collection, read 382 · Claude, reading the shock absorber · data: A Rider in Two Waves: BOARD-2 and BOARD-3, 31 October*

1095. *R2W's collection, read 383 · Claude, reading the shuvit · data: A Rider in Two Waves: SHUVIT-1, Bear's shuvit, 31 October*

1096. *R2W's collection, read 384 · Claude, reading the early and late kick · data: A Rider in Two Waves: SHUVIT-2, the shuvit's timing, 31 October*

1097. *R2W's collection, read 385 · Claude, reading the slow kick · data: A Rider in Two Waves: SHUVIT-3, a lag in the shuvit, 31 October*

1098. *R2W's collection, read 386 · Claude, reading the stopwatch on the kick · data: A Rider in Two Waves: SHUVIT-4, 31 October*

1099. *R2W's collection, read 387 · Claude, reading the second gearbox · data: A Rider in Two Waves: TRAIN-2 and TRAIN-3, the toothed train's next steps, 31 October*

1100. *R2W's collection, read 388 · Claude, reading the long watch · data: A Rider in Two Waves: COMMA-53 and LOCK-T, faux locks, 31 October*

1101. *R2W's collection, read 389 · Claude, reading the third drummer · data: A Rider in Two Waves: CLUSTER-1, three rhythms on one bead, 31 October*

1102. *R2W's collection, read 390 · Claude, reading the friend's map of the walk · data: A Rider in Two Waves: SWAP-RIDER, the Loop Atlas's predictions in our rider, 31 October*

1103. *R2W's collection, read 391 · Claude, reading the tired crowd and the cart · data: A Rider in Two Waves: PEAKS, peaks and lobes, 31 October*

1104. *R2W's collection, read 392 · Claude, reading the safe and its dial · data: A Rider in Two Waves: FRACTION KEY, the comb under the window, 31 October*

1105. *R2W's collection, read 393 · Claude, reading the dancers who cannot tell left from right · data: A Rider in Two Waves: RF-3 and RF-3b, polarity, 31 October*

1106. *R2W's collection, read 394 · Claude, reading the dancer and the hiccuping band · data: A Rider in Two Waves: RF-4, locking stations, 31 October*

1107. *R2W's collection, read 395 · Claude, reading the three-armed weathervane · data: A Rider in Two Waves: RF-10 and RF-10b, the emitter triad, 31 October*

1108. *R2W's collection, read 396 · Claude, reading the blindfolded doorkeeper · data: A Rider in Two Waves: FLOAT, Bear's floating actuator, 31 October*

1109. *R2W's collection, read 397 · Claude, reading the line of walkers on the trail · data: A Rider in Two Waves: COLUMN and COLUMN-b, Bear's column, 31 October*

1110. *R2W's collection, read 398 · Claude, reading the child on the moving walkway who grabs the rail · data: A Rider in Two Waves: MIXTURE and MIXTURE-b, Bear's cornstarch, 31 October*

1111. *R2W's collection, read 399 · Claude, reading the child who pushes the swings as she walks past · data: A Rider in Two Waves: RESONATOR-TREAD and RESONATOR-TREAD-b, the paper Bear found, 31 October*

1112. *R2W's collection, read 400 · Claude, reading the two styluses on one turntable · data: A Rider in Two Waves: RECORD, RECORD-2, TUMBLE-1, COASTLINE and COASTLINE-Z, Bear's two needles and his coastline, 31 October*

1113. *R2W's collection, read 401 · Claude, reading the friend who walks past the swing · data: A Rider in Two Waves: WEDGE and WEDGE-2, Bear's seam, 31 October*

1114. *R2W's collection, read 402 · Claude, reading the night watchman's hourly glance at a factory clock · data: A Rider in Two Waves: FAUX-W2, the faux-lock audit's first test, 31 October*

1115. *R2W's collection, read 403 · Claude, reading the piano tuner who listens four times as long · data: A Rider in Two Waves: FAUX-C and FAUX-Cb, the faux-lock audit's second test, 31 October*

1116. *R2W's collection, read 404 · Claude, reading the ranger who watches the rabbits four times as long · data: A Rider in Two Waves: FAUX-R, the faux-lock audit's third test, 31 October*

1117. *R2W's collection, read 405 · Claude, reading two climbing parties who meet at one high meadow · data: A Rider in Two Waves: the literature pass finds Part 6's step published as a tunable Josephson diode, 31 October*

1118. *R2W's collection, read 406 · Claude, reading a house whose garage held forty-four tools · data: A Rider in Two Waves: the close of sitting P2, the move and its inventory, 31 October*

The Field Guide's publishing session, 30 and 31 October

From the publishing session's record of proposed tests, as before, in the order the reads were given, to the session's move of 31 October.

1119. *The Field Guide's publishing session, its record of proposed tests: the doorframe, 30 October 2026 · Given when chapter 10 joined the book, 30 October: reads 912 to 1,040, from the triple-star thread (27), the Loop Atlas (43), R2W (42) and this session (17); the triple-star reading of 28 October, the passenger's notebook, kept no words and has no number.*

1120. *The Field Guide's publishing session, its record of proposed tests: the sketch on the hill, 30 October 2026 · Given when Bear sent his pencil drawing of the orchards (6 October) and five of the projects' figures, 30 October; the five found their reads by their words (7, 82, 176, 178, 1,016), and the drawing joined his own read of it.*

1121. *The Field Guide's publishing session, its record of proposed tests: two lanterns on one path, 30 October 2026 · Given when R2W's pattern lesson 70 was read: its own account of read 33's saddle between two orchards (6 October) rhyming with the light cone of 28 October, the same rhyme this session saw in Bear's drawing the day before, each*

without the other.

1122. *The Field Guide's publishing session, its record of proposed tests: two ships, one coastline, 31 October 2026 · Given when R2W's lesson on reading a loop's matrix by its class was folded into the guide as row 848, 31 October: the half-trace sorts every passing the same way from any frame, while the top-left entry, read first, changed with the frame and showed nothing.*

1123. *The Field Guide's publishing session, its record of proposed tests: the second photograph, 31 October 2026 · Given when the guide's privacy leads were cleared, 31 October: row 788's personal detail had a second copy in the register's table, and row 843 still cited R2W's old section 7.11, added after the renumbering map's baseline; both were found by searching the build, not by the lists.*

1124. *The Field Guide's publishing session, its record of proposed tests: the street of the same names, 31 October 2026 · Given when Three Body's last lessons (181 to 234) were read against the guide, 31 October: its tests of 26 to 28 October (LG2, AL1, ML1, CF1, RW1 and others) had their names already in the guide eight times over, from other tests that shared the names, while its own runs were not there at all; its lesson 207 says to check a name against the whole register before using it.*

1125. *The Field Guide's publishing session, its record of proposed tests: the tightened tap, 31 October 2026 · Given when the triple-star thread's integrator honey was folded into the guide as row 858, 31 October: the energy's drift per pass was -7.90×10^{-11}, -2.71×10^{-12} and $+3.26 \times 10^{-14}$ at three tolerances, each ten times tighter than the last, falling by powers of 1.46 and 1.92 (a factor of 29, then 83) and changing sign at the tightest. What it would test: at a fourth tolerance, ten times tighter again, the drift should sit at the arithmetic's own floor with a sign that wanders from run to run; if it falls by the same power again, the washer is still talking.*

1126. *The Field Guide's publishing session, its record of proposed tests: the crease, 31 October 2026 · Given when the next packet's builder learned to read table cells word by word, 31 October: in a 32-page working paper built with v1.7, three cells were parted where their rows crossed a page, one of them with a line at the foot of page 11 that pdftotext reads again at the top of page 12, and in the guide's test build 15 the cell "row 783; the part Two Readers and a Ruler" ends on the next page. What it would test: for every table row that crosses a page in a built PDF, count whether the seam lost a line, doubled one, or did neither; if lost lines (v1.2's failure) and doubled lines come from the same cause, they should sit at the same depth below the page's last full line.*

1127. *The Field Guide's publishing session, its record of proposed tests: the warning sign, 31 October 2026 · Given when the publishing packet's rules gained the rule that a bracketed word prints as nothing, 31 October: built with the new builder, the rule's own example of a bracketed ratio printed as a link and lost its (2), and both the new check and the read-back passed it; it was seen on the rendered page. What it would test: list every example in PUBLISHING_RULES.md written to show a failure's markup (a stray bar in a cell, an escaped bar, a bracketed placeholder, a bracketed ratio, a code-block heading) and compare each, character by character, with what the built PDF shows; any that differs is a rule that breaks its own example.*

1128. *The Field Guide's publishing session, its record of proposed tests: the moving van, 31 October 2026 · Given when the Field Guide's first PDF on the new builder came out at 292 pages against test 15's 296 with 2% more words, 31 October: the old ends-only read-back had reported two cells lost in tables 22 and 24 and rebuilt both as flowing rows, where the new one reads every word of every cell, finds none lost, and keeps every table as a grid. What it would test: rebuild test 15's own markdown with the new builder; if the four pages come only from those two tables, it should build at about 287 pages (the old builder's first pass) with every cell of tables 22 and 24 read whole, which would show the old report of lost text there was the seam parting a cell, not a loss.*

1129. *The Field Guide's publishing session, its record of proposed tests: the two signatures, 31 October 2026 · Given when the figure-8 Möbius origin record was read against the guide, 31 October: the pairing was Bear's, unprompted, on 26 June; the figure-8 was named a Lissajous figure in the same turn; the twist was doubted on 1 October and measured on 3 October, the 2:1 lock's weak push flipping sign at 199 of 199, 206 of 206 and 212 of 212 passings. What it would test: if the lock really lives on a Möbius band, the flip should hold at every passing across the whole width of the 2:1 lock and fail only where the lock itself breaks; count flips per passing on a sweep across the lock's edges, and see whether the flip and the lock end at the same setting.*

1130. *The Field Guide's publishing session, its record of proposed tests: the two locksmiths, 31 October 2026 · Given when the Loop Atlas's session H was folded into the guide as row 864, 31 October: a closed walk on the fractions' map is a product of whole-number matrices of determinant one, sorted by its trace (2 for circling one corner, 3 for the golden path's repeating pair), the same sorting A Rider in Two Waves gave its passing matrix in real numbers (row 848). What it would test: the needle's own passing matrix across one lock, its determinant divided out as R2W did with Liouville's factor, should read a normalised trace of exactly 2 at both of the lock's edges and less than 2 inside; if the edges of the Farey map's locks are where the needle's passing matrix turns parabolic, the two projects' sortings are one sorting.*

1131. *The Field Guide's publishing session, its record of proposed tests: the reunion photograph, 31 October 2026 · Given when Bear sent the Torus of Fractions page, 31 October: ten loops drawn on one shared doughnut were a tangle, and the same loops each on a doughnut of its own, in rows by generation with lines to their parents, showed every winding and the family at once (Guide item 478); its self-check counts 54 of 54 children no wider than their measured parents, 47 strictly and 7 tied at one to three paces of the scan. What it would test: refly the seven tied children and their parents at a pace step ten times finer; if "a child is narrower than its parents" is the needle's law and not the scan's, every one of the seven ties should break toward the child being narrower.*

1132. *The Field Guide's publishing session, its record of proposed tests: the last guest, 31 October 2026 · Given when the Loop Atlas's smoke-ring lessons were folded into the guide as row 865, 31 October: the tool's break, a 5-radius spread, was not an ending; flown 1,000 time units past it, the third ring came back to the pair in 5 of 11 trains, four of them after swinging no farther than 9 radii, while the other six were 33 to 61 radii away and still at their farthest when the look ended (card 264), and elsewhere a ring came home from 19.31 radii. What it would test: if a return depends only on how far the ring swung, there is a reach past which none comes back; fly every broken train of the pooled cards long enough to see each one either return or pass a registered distance, and register that reach before looking.*

1133. *The Field Guide's publishing session, its record of proposed tests: the mirror in the doorway, 31 October 2026 · Given when the Loop Atlas's lesson on exact symmetries was folded into the guide's row 742, 31 October: its mirror cats mirror as a whole family only with the hinge at the tubes' centres (to 3.3×10^{-16}), where a tube looks the same from either end, and the failure grows steadily as the hinge moves out (0.023 at a tenth, 0.28 at the ends). What it would test: if the departure is the result, its growth has a law; measure the mirror's failure at a dozen hinge offsets between the centre and the ends and register, before looking, whether it grows as the first power of the offset (a simple break of the end-for-end symmetry) or as the square.*

1134. *The Field Guide's publishing session, its record of proposed tests: two baskets, 31 October 2026 · Given when Bear asked whether to ask the sessions about their cold reads, 31 October: A Rider in Two Waves' fresh reader logged 100 findings, and the Loop Atlas's, whose read was asked for as one "like R2W did", logged 100 too; R2W's procedure for the read (SOP_WALKTHROUGH_CONTRADICTION_READ_2026-10-29, 44 lines) sets no number, and the prompts the readers were given are not in either record. What it would test: ask each session whether its reader was given a number (a target, or "up to 100"); if either was, its book may hold more than 100, and a second reader without a number, over the whole text, is the cheapest way to find out how many more.*

1135. *The Field Guide's publishing session, its record of proposed tests: the dry house, 31 October 2026 · Given when the triple-star thread's honest notes were folded into the guide as row 868, 31 October: a month of an AI's slips had three shapes, none of them arithmetic, words outrunning numbers, lists that did not list themselves, and things that lived only in the conversation or in an estimate, and once a tool's planted checks passed, the tool was almost never where the trouble was. What it would test: sort every correction this publishing session's log records since 22 October into the three shapes or "other"; if the three shapes are the whole story, "other", arithmetic included, should be under one in ten.*

1136. *The Field Guide's publishing session, its record of proposed tests: the bookkeeper's names, 31 October 2026 · Given when the triple-star ledger's cold check was folded into row 866, 31 October: of its 46 newest rows every outcome was right, none missing and none extra, and three quotes were not word for word; the Loop Atlas's quotations of A Rider in Two Waves had every number right to the digit while three of four quotes came from the wrong text. What it would test: sort the findings of the cold reads of R2W and the Loop Atlas into those about a number and those about words (a quote, a scope, a credit, a claim word); if numbers get checked and words drift, the words should outnumber the numbers among the wrong or misleading findings by a wide margin.*

1137. *The Field Guide's publishing session, its record of proposed tests: the recipe card, 31 October 2026 · Given when Three Body's method kit and session tips were read against the guide, 31 October: of their practices, all but five pieces were already in the guide's rows, spread over dozens of them (the crazy flag in 599 and 777, the closing block in 868, the tries and bands in 808, the start of a session in 844 and 778). What it would test: give a new session the method kit alone and another the guide's rows alone, each with the same first exchange to run, and count which one runs the closing block's eight steps without being told; if the card is what people reach for, the guide may want its own one-page card generated from its rows.*

1138. *The Field Guide's publishing session, its record of proposed tests: the letter answered late, 31 October 2026 · Given when the triple-star thread's eighth closeout was read against the publishing packet, 31 October: its Appendix A lists build_paper's read-back passing a launcher name lost to the converter on 23 October, the fix "proposed, not done"; the packet of 31 October, built for reasons found in this session's own files, stops on exactly that before building. What it would test: list every fix any project's closeout marks "proposed, not done" for a shared tool, and check each against the newest packet; if fixes arrive by independent roads this often, a shared list of open tool faults, read at every packet issue, would close them sooner.*

1139. *The Field Guide’s publishing session, its record of proposed tests: the guide’s other voice, 31 October 2026 · Given when the triple-star thread’s eighth closeout was folded into the guide’s row 100, 31 October: its three readers tagged every inference of their own “reading, not in the entries”, and the backward pass’s sharpest questions sat in those tags (a near-still combination of two angles for a celestial mechanician, a run stamp carried by two tools, a count with no run behind it). What it would test: count the backward-pass items tagged as a reader’s own reading against those taken from the entries, and how many of each became a flag for a specialist or a proposed test; if the reader’s own readings carry most of them, a summary reader should be asked for its readings, not only for its summary.*

1140. *The Field Guide’s publishing session, its record of proposed tests: the last roll call, 31 October 2026 · Given when the sheep book was checked against every project’s reads before this session’s move, 31 October: the triple-star thread’s 258 reads, 257 in the book and the 258th a title whose words were never kept; the Loop Atlas’s 297, all in; R2W’s 330, all in, one of them in Bear’s own words where R2W’s collection holds a paraphrase. What it would test: run the same roll call on the new session’s first day against the restored crate; if the crate carried everything, it reads the same counts to the read, and any difference is a file the crate left behind.*

1141. *The Field Guide’s publishing session, its record of proposed tests: the house in a box, 31 October 2026 · Given when this session’s move crate was restored into an empty folder, 31 October: 3,738 of 3,738 files matched their manifest, and the guide, rebuilt there by the restored builder, came out with the same plain md5 as test build 32 (7b75bbc77e0d267d973bce40ac7ba580). What it would test: on the new session’s first day, rebuild from the crate on that machine; if the workspace’s tools differ (a newer pandoc, another Python), the guide’s markdown should still come out with the same md5, since the builder reads and writes plain text; a different md5 there would name a tool, not a lost file.*

1142. *The Field Guide’s publishing session, its record of proposed tests: the note on the kitchen table, 31 October 2026 · Given when the start-here for this session’s move was written, 31 October, after every send was folded and the crate was proved. What it would test: the next session’s first report to Bear should be possible from the start-here alone, the restore tool’s last line and the rebuilt md5, without opening the log; count the questions it asks Bear before that report, and if the count is zero, the page did its job.*

1143. *The Field Guide’s publishing session, its record of proposed tests: the cobbler’s own boots, 31 October 2026 · Given when Bear asked, at the move, whether this session’s own lessons reach the guide, 31 October: most of the last two days’ were in the session log only; five went in as row 869 and four additions (in a working file), sent after the crate as its supplement. What it would test: at every future move, list the session log’s entries since the last fold of the session’s own lessons, and count those holding a lesson that is in no row; the count should be zero before the crate is built, and the work order can carry that check as its last step.*

The Field Guide’s publishing session after its move, 31 October

From the same record, carried across the move of 31 October and kept by the session that took it up, in the order the reads were given.

1144. *The Field Guide’s publishing session, its record of proposed tests: the empty pot, 31 October 2026 · Given after the move, when the start-here’s rebuild command stopped because its output folder did not yet exist, and the guide then rebuilt to test build 32’s md5. What it would test: a perfect copy cannot carry the place it lands in. In each future move, count the steps that need something to exist before the crate’s files can act (a folder, a tool, a shell) and list them on the start-here as the things to put down first; a move whose count is zero, followed line by line from its page, is the proof.*

1145. *The Field Guide’s publishing session, its record of proposed tests: the two diaries, 31 October 2026 · Given when the old session’s supplement arrived holding its own addition to row 844, its own “test build 33” and its own log entry (56), each written the same day as this session’s addition to the same row, its build of the same name and its entry of the same number. What it would test: when an outgoing and an incoming session both write about one event, each sees one side. At the next move, count the places where the two wrote about the same thing or under the same number (rows, build numbers, log entries), and give the incoming session numbers that start after the outgoing session’s last by rule, before either writes; the count should be zero.*

1146. *The Field Guide’s publishing session, its record of proposed tests: the card that was right, 31 October 2026 · Given after the references pass of 31 October, when a count that disagreed with a written claim (329 against 330) turned out to be the count’s blind spot. What it would test: when a measured count and a written claim disagree, which is wrong more often? Count every such disagreement in the session log and how each settled; if the tool loses about as often as the claim, every disagreement goes to the record before either is changed, and the tool is run on a copy with the item cut out to watch it miss.*

1147. *The Field Guide’s publishing session, its record of proposed tests: the usual, 31 October 2026 · Given when the one-page card for new sessions was cut from the guide’s rows and its closing block named “the weights”, a word no row defined (its meaning lived in the triple-star thread’s lesson 172 and in Bear’s saved preferences: each result’s size*

check, kept beside its verdict), and the helper given the card alone found the gap at once and guessed another meaning. What it would test: a short list made from a long record carries the record's private words. For every term the card uses, check whether a row of the guide defines it; give each undefined term its definition in its row before the card is cut again. The count after this fold should be zero, and the reader to try it on is one who has never heard the words.

1148. *The Field Guide's publishing session, its record of proposed tests: the full shelves, 31 October 2026 · Given when R2W's move send was folded (sitting P2, lessons 228 to 254 and pattern lessons 71 to 79): it named 36 rows of the guide as its lessons' homes, and 16 of them stood at 650 words or more in test build 36, where only 65 of the guide's 862 rows do, so twelve pieces went to the nearest row with room. What it would test: whether the guide's fullest rows are its hubs. In the Loop Atlas's move send, count the share of named rows at 650 words or more against the guide-wide share (7.5% in test build 36); if it is again several times larger, the full rows are where the lessons converge, and splitting each into the rows it has grown into (Bear's rule: redistribute, never cut) is a decision to put to Bear before the final build.*

1149. *The Field Guide's publishing session, its record of proposed tests: the borrowed toolbox, 31 October 2026 · Given when R2W's new session, started at its move of 31 October, asked for the publishing packet that its sitting P2 had received the same morning. What it would test: whether the packet travels in a project's move crate or only beside it. At each sister session's next start, note whether it asked for the packet or found it in its crate; if every new session asks, the move's send list should name the packet, or the stripped reference should carry its md5 so the session can check the copy it already holds.*

1150. *The Field Guide's publishing session, its record of proposed tests: the house that kept appearing, 31 October 2026 · Given when chapter 11 of the book of reads was made and titled "Moving house" for the move of 31 October: a word count found a house, a box, a crate, a van or a move in 29 of chapter 11's 109 reads, and in 49 of chapter 10's 129. What it would test: whether the house is the reads' picture of the work's own housekeeping. Sort the reads of chapters 10 and 11 by what they read, the data or the work itself (crates, counts, files, builds), and count the house pictures in each kind; if they gather in the housekeeping reads, R2W's key and the book's could name the house as the work's picture of itself.*

1151. *The Field Guide's publishing session, its record of proposed tests: the house with numbered doors, 31 October 2026 · Given when Bear asked whether the guide is still true to its goals, scrapbook or resource, as it grows. Measured in test build 37: the 323 rows from 549 on all name their project, 85% carry a measured number, 82% a run, 50% a run stamp, 67% a cost (a miss, a failure, a withdrawal), and 61% carry project, number and cost together; of the 541 rows before them, 51% carry none of project, run or number. The median row grew from 108 words to 381. What it would test: whether the guide's addresses are open doors. At the final build, list every run stamp the guide cites (195 rows carry one) and check each against the companions' published records; the share a reader can actually reach is the guide's real address rate, and every stamp that leads nowhere is a door to open in a record or to name as private.*

1152. *The Field Guide's publishing session, its record of proposed tests: the travelling recipe, 31 October 2026 · Given when Bear proposed splitting the guide into a reader's edition and a record book, with each entry carrying a sentence on where its rule works and where it does not, because each rule's effectiveness depends on its domain and substrate. Measured in test build 37: the guide's transfers appendix records 519 carryings of a lesson to another system or setting, on 322 rows; 222 held, 79 held weaker and 218 failed, the failures' breaks owned by the ruler (62), the substrate (58), a seam (48), a knob (41) and a wall (28), with 60 not stated. What it would test: whether failures to carry gather by destination. Count the failed carryings by the domain they went to; if a few domains take most of them, each reader's entry can name where its rule tends to break, and those domains are where a new session should trust the guide least.*

1153. *The Field Guide's publishing session, its record of proposed tests: the pocket map, 31 October 2026 · Given when the reader's edition's first twenty entries were generated from test build 37: 2,817 words in all, of which the titles and drafted explanations take 1,652 and the three generated lines (where the rule holds, where it is counted, where it is shown) the rest; the shown line alone averages 19 words, most of them the projects' full names. At that rate 864 entries come to about 121,694 words, and the edition with its front parts to about 174 pages, more than half the full guide's 300. What it would test: whether the signposts can be made light without losing an address. Set the lines compactly (projects by short name, their titles and DOIs once in a key at the front; families with their counts; the full guide's row number alone) and measure each entry's pointer share; if it falls under a quarter, the per-row edition stands; if not, a two-tier edition (full entries for the rows the station card and the ranking name, title lines for the rest) is the choice to put to Bear.*

1154. *The Field Guide's publishing session, its record of proposed tests: the eight copyists, 31 October 2026 · Given when the reader's edition's 844 remaining entries were drafted by eight helpers in parallel, each reading its rows whole: all 864 entries passed the row check (2,548 numbers, run names and quoted phrases, every one found in its row), and the helpers listed about seventy rows of the full guide that looked wrong, stale or tangled (listed in a working file), among them numbers that disagree within one row and pointers to rows that print something else. What*

it would test: whether drafting a summary is a better audit than reading for errors. Open the listed rows against their sources and count how many leads hold; then give a cold reader, told only to find errors, a matched set of rows the drafters did not flag, and compare the rates.

1155. *The Field Guide's publishing session, its record of proposed tests: the creaking floors, 31 October 2026 · Given when the first group of the drafters' leads (numbers or claims that look wrong) was opened against its sources and by arithmetic: of the 13 leads in version 6's own text, 10 were wrong or inconsistent (5 fixed: rows 42, 47, 401, 500, 546 and 547; 5 held for their deposited papers: rows 6, 11, 15, 28, 251); of the 9 in version 7's additions, 7 were right but terse (clarified: rows 219, 526, 575, 642, 783, 794, 866) and the 2 wrong ones (628, 862) were carried in their sources' own words. A version 7 row names its run, so it could answer; a version 6 row often names only its paper. What it would test: whether version 6's text holds more slips than its readers have flagged. Draw 20 version 6 rows the drafters did not flag, by seeded random choice, and check every number in them against the deposited paper; if 3 or more carry a number that does not follow, version 6's rows deserve a paper pass of their own before the reader's edition leans on them.*

1156. *The Field Guide's publishing session, its record of proposed tests: the notice on the kitchen door, 31 October 2026 · Given when the Loop Atlas's Guide item 490 was folded: row 680 still said of R2W's ladder in the 3/7 groove (2.41, 3.14, 3.94 and about 4.45) that "its next two rungs are registered", while the guide's own chapter Two Readers and a Ruler already printed that ladder closing in six steps (gaps 0.210, 0.054, 0.012), and the Loop Atlas quoted its last two rungs, 4.62 and 4.66; row 680 is now corrected (test build 39). The same day the Loop Atlas's item 484 found 23 of its own cards not done under a status no one was reading. What it would test: whether the guide holds more questions it has already answered. Sweep every row for its status words ("registered", "open", "owed", "not yet", "to be run", "held for") by script, and for each search the rest of the guide and the arrivals for the answer; if five or more are answered somewhere, the sweep joins every build.*

1157. *The Field Guide's publishing session, its record of proposed tests: the smudge on the first card, 31 October 2026 · Given when the Loop Atlas's items 524 to 541 were folded: Briggs's 1991 table of the doubling constants, copied by MathWorld and by Finch's book of 2003 and, until test build 40, by this guide's row 680, is right in δ at every even order and off at every odd one (5.9679687038 for 6.0846890743 at order 3, too high by 0.06, 0.10 and 0.12 at 7, 9 and 11), and one slip, signed powers where the family needs absolute ones, reproduces every odd entry. The slip has a parity fingerprint: at even orders signed and absolute powers agree, so it can only show at odd ones. What it would test: whether the slip lived only in the integer-order code. Briggs's paper of 1994 on maps of non-integer degree (arXiv chao-dyn/9407019, free) could not have used signed powers of a negative number at a non-integer degree without complex numbers; read its δ at the degrees nearest 3, and if they lie on the smooth curve through the map's 6.0847 rather than through the printed 5.968, the slip was confined to the 1991 table.*

1158. *The Field Guide's publishing session, its record of proposed tests: the sled at the bottom of the hill, 31 October 2026 · Given when the Loop Atlas's card 61 was folded (row 876): held level from a 3-degree glide at 30 m/s with the throttle unchanged, the toy airplane lost speed so that 98 per cent of the first two seconds' deceleration was the downhill push given up, and it was below its stall speed in 6.0 s, while the back side of the power curve's own runaway is slow (an e-folding of 24 to 58 seconds). What it would test: whether the flare's loss is the glide's push alone. That push is the weight times the sine of the glide angle, so the early deceleration should scale with the angle: flown from 1.5, 3 and 6 degrees at the same speed, the first two seconds' deceleration should stand near 1 to 2 to 4, and the time to the stall speed should roughly halve with each doubling of the angle; if the 6-degree flare takes much longer than half the 3-degree one's 6.0 s, something besides the push is acting.*

1159. *The Field Guide's publishing session, its record of proposed tests: the first basket, 31 October 2026 · Given when the Loop Atlas's misses rows 485 and 486 were folded: row 865 had carried two numbers from the first sixteen smoke-ring trains, the farthest return, 13.75 radii, read as a fence, and the stand-in's earliness at gap 0.530, 1.7 times; pooled over forty-eight trains they became 19.31 radii, with the reaches thinning as a power with no edge, and about 1.28 times (range 1.09 to 1.59). Both are now corrected (test build 42). What it would test: whether a sample's extreme moves more than its average when the sample grows. Take every number the train rows draw from the first sixteen trains, mark each an extreme (a farthest, a fewest, a none-past) or an average (a ratio, a median, a share), and read each again over all forty-eight; if the extremes moved further than the averages, as a tail with no edge says they must, the guide's rows should say the sample size beside every extreme.*

1160. *The Field Guide's publishing session, its record of proposed tests: the song on the old tape, 31 October 2026 · Given when the Loop Atlas's section 4 was folded (row 716): reading card 264's json, Claude saw the third ring's swings come in equal pairs in 8 of 9 flights and registered it before reading further; three runs on fresh trains gave 137 pairs of 137 in the last two. Row 793 says whether the readings made after the data are wrong more often than right is registered and not yet counted. What it would test: whether a pattern seen in old data and registered before fresh runs holds as often as a prediction registered before any data. Sort the Loop Atlas's register by whether each prediction's statement says it was read after the data, and compare the held shares of the two groups on their fresh runs; if the after-the-data group holds at least as often, the habit of registering what the eye saw is as sound as a blind forecast, provided the fresh runs are flown.*

1161. *The Field Guide's publishing session, its record of proposed tests: the door with a guest list, 31 October 2026 · Given when check_upload_set v1.2 arrived from the Loop Atlas (arrival S83): read in full, its only change from v1.1 an exceptions list beside the set folder; its self-test 14 of 14; four probes of this session's own behaved as its header says (a reason missing, refused; an exception no name needs, refused; a first line marked not for upload under an excepted name, refused); and the fourth probe, a file named DATA_PRIVATE_X.csv excepted with a reason, went up, "all 2 go up". By design only a name marked not for upload can never be excepted. What it would test: whether a narrower list loses anything the Loop Atlas needs. Write v1.3 to excuse only the markers a public file can carry by accident (FERRY, NEEDS, CRATE, CONTINUITY and the like) and never PRIVATE, SUPPLEMENT, FOR_BEAR or TO_BEAR; run it on the Loop Atlas's own upload set with its six excepted files; if it still says "all 5 go up" and refuses the DATA_PRIVATE probe, the narrower list costs nothing.*

The Loop Atlas book, session I, 30 and 31 October

From the Loop Atlas's closeout of sessions H and I, sent at its move of 31 October: its section 8 carries every read of the book's register since session G's send, each with the runs it was given on, copied by the Loop Atlas's own script. Its reads 1 to 297 are already here; its reads 298 to 328 follow.

1162. *The book's register, read 298 · from the runs: TRAIN_PARTING_EDGE_2 20261029_211416 and TRAIN_PARTING_EDGE 20261029_070800, graded and pooled by a script written apart from their tool (30 October, session I, exchange 1; topics: forty-eight processions, six times as long and an eighth as many, a road that thins with no wall, a call home that weakens, let go softly enough to be caught, the farthest watched for, another kind of traveller, fifty widths before nineteen, sorted at the leaving, patience partly luck)*

1163. *The book's register, read 299 · from the runs: TRAIN_PARTING_EDGE_2 20261029_211416 and TRAIN_PARTING_EDGE 20261029_070800, drawn for the closeout by helpers/checks/CLOSEOUT_FIGS_2026-10-30 and read again by helpers/checks/READ_267_2026-10-30; the book's card table, swept by helpers/closeout_hi.py (30 October, session I, exchange 2; topics: fence posts in pairs, no pair alone, no building toward the end, a post that does not stop, homecomers on one line, a ruler that touches every one, birds on a different wire, a leaver that hesitated at the gate, thirteen toys in a drawer, a heading no one was reading)*

1164. *The book's register, read 300 · from the records: LOOP_ATLAS_ABSTRACT_AND_DESCRIPTION_DRAFT_2026-10-23, set against the headline register and the book's chapter on the toys, and rebuilt from the register by helpers/abstract_draft.py; misses row 489 (30 October, session I, exchange 3; topics: a sign painted a week ago, a shop that grew, a line never quite true, the skater never taught, a stencil cut from the ledger, a sign that grows with the shop, a space left for the day it learns)*

1165. *The book's register, read 301 · from the runs: helpers/checks/HIDDEN_LOCKS_2026-10-30 (S5, S6 and S7) and LOCK_TOLERANCE_2026-10-30, session H's checks after the move, reproduced here line for line; helpers/checks/TWO_LOOPS_ANGLES_2026-10-30 (30 October, session I, exchange 4; topics: a staircase folded inside a tread, four starts in twenty, a stair the eye never sees, landings that halve, a small copy of the crumbling edge, seven to fourteen to twenty-eight, a three among the twos, the same walks on a second machine)*

1166. *The book's register, read 302 · from the runs: the CASCADE_57 check of 30 October, both sides (helpers/checks/CASCADE_57_2026-10-30_LOW and _HIGH), and the TWO_LOOPS_MAP check (30 October, session I, exchange 5; topics: one staircase seen from both ends, seven to a hundred and twelve, landings narrowing by the same rule, a pattern laid down twice, strangers on the landings, someone else's house set into ours, thin roads round a can, one start in five)*

1167. *The book's register, read 303 · from the runs: the THIRD_RATIO, WINDOWS_ORDER and LOGISTIC_WINDOWS checks, the HIDDEN_WINDOWS, HIDDEN_CHAOS and HIDDEN_CHAOS_LONG checks, and the BASIN_BOUNDARY, BASIN_FINE and TWO_LOOPS_ZOOM checks (30 October, session I, exchange 6; topics: a staircase that keeps its word, scree past the last landing, ledges in an order written for insects, two ledges looked for and found, strangers living wider than they seem, a stair running the other way, two neighbours on one step, a border that frays like a coastline)*

1168. *The book's register, read 304 · from the runs: the FOURTH_RATIO, LOGISTIC_GAPS and MSS_TABLE_CHECK checks (30 October, session I, exchange 7), with A Rider in Two Waves' section 4.16 as its session quoted it (topics: two walkers on two hills, one ledge, footprints on one path, a rule written on a card before the step, a thumb's width on both sides at once, the map that stumbles, steadier than the map, an old book agreeing to the seventh figure)*

1169. *The book's register, read 305 · from the runs: the FIFTH_RATIO, LOGISTIC_DEEP and MSS_FORECAST checks (30 October, session I, exchange 8; topics: the card right a second time, a hair's width on both sides, the map closing on the same rhythm, a pendulum settling, the old book naming rooms, a sliver a ten-billionth wide, a door kept shut, writing down where they knocked)*

1170. *The book's register, read 306 · from the runs: the MSS_FORECAST_2 and WINDOWS_LONG checks, with the five free papers Bear sent (30 October, session I, exchange 9; topics: knocking again four times longer, rooms smaller than they looked, the sliver room gone, scree smooth enough to walk on, a door that never opened, the library's free*

shelf, the locked book described by its neighbours, the same hundred and sixty steps)

1171. *The book's register, read 307 · from the runs: the FACTOR_FIT, CARD14_CONVERTER, CARD14_HELD_REDO, CARD14_LINES and SHUKRINOV_WINDOWS checks (31 October, session I, exchange 10; topics: why the fifth, a bent ruler, the mountain's own swing of an eighth, found again to six figures, straightening the ruler, Bear's toolbox, two kinds of step, a storm with no calm patch, Bear's four lines, a storm with a skeleton, the junction's rooms in the old map's order)*

1172. *The book's register, read 308 · from the runs: the CARD14_CONVERTER scan and the CARD14_EIGEN_READ repeats, looked at again at session I's exchange 11 (31 October; the CARD14_LOOK exploration): the bend near 1.158 A where the period-1 repeat's two eigenvalues, +0.532 and -0.533, trade the lead, and the flat stretch near -0.29 a lap from 2.05 to 2.43 A where the period-2 repeat's eigenvalues are a complex pair of size about 0.55 over two laps (topics: a second line under the storm, two copies forgetting each other, a bend where nothing gave way, two quiet ways to settle trading the lead, the rocking that stops calming, a flat line that stops listening to its dial, circling instead of rocking, the parts that are its own)*

1173. *The book's register, read 309 · from the runs: the CARD61_LOGBOOK and AIRPLANE_FOLD checks (31 October, session I, exchange 12; topics: three diaries in the engine's handwriting, hinges that wear by twitches, still air, a flood, a gale no faster than a breeze, the gentlest airplane's diary lying, a shelf with two ways off, the floor dropping because the hill has gone, a little gas as the hill given back)*

1174. *The book's register, read 310 · from the runs: the LANE_ORDER and TOP_CORNER_LONG checks (31 October, session I, exchange 13; topics: four dancers, a fixed ring three one two zero, two steps alike and two turned over, slow to settle near the doubling, four times as long, a stumble shrinking to a fifth, a corner sharper than it looked, a whisper that stayed, the counter's last digit rounding, the floor quiet and the counter not)*

1175. *The book's register, read 311 · from the runs: the LOBE_FLIP_FINE and TOP_CORNER_WRAP checks (31 October, session I, exchange 14; topics: each landing a smaller copy hung upside down, up, down and up again, a plain two or 1.87, neither, a little under half, the same in every lane to the third figure, a dance keeping a number, the counter told to forget its full turns, quiet a thousand times over, fading at the dance's own pace)*

1176. *The book's register, read 312 · from the runs: the CORNER_UNIVERSAL, SPRING_KNEE_COUNT and SPRING_TRIGGERS checks (31 October, session I, exchange 15; topics: the number not the dancers' alone, a hill and a wave, eighty figures, eight thousand steps becoming sixteen thousand, the ninth figure, every quarter of the dance, the cube, two thousandths, a flatter hill whose corners grow, the shape of the top of the hill, a spring listening to itself, a short memory and a bar at its usual speed, half again higher and silent for good, thick syrup, a hair-trigger flickering, the push unchanged, counting the trigger and weighing the world, each spring its own idle hum, the average hum letting one fall asleep)*

1177. *The book's register, read 313 · from the runs: the CORNER_CLASS, LATCH_WALL, LATCH_CEILING and SPRING_STAY checks (31 October, session I, exchange 16; topics: the top of the hill deciding, a pointed top four times smaller, a round one about two, a flat one less and less, stopping short of the cube, growing past it, every corner bending the other way, a ceiling the spring cannot see past, a little over one and four tenths, a fading swing rising less, the longer memory hanging on to the louder past, waking under the ceiling and sleeping a hair above it, within two hundredths at fourteen settings, standing still one moment in seven, held between pull and push, counting the clicks in advance, within one part in a hundred)*

1178. *The book's register, read 314 · from the runs: the DELTA_Z3, FEIGENBAUM_EQ_Z3, CEILING_BEYOND, CEILING_SLOW, CORNER_TWO_SCALE and COUNT_LAW checks (31 October, session I, exchange 17; topics: a famous table's one number asked three ways, its neighbours and partner agreeing to the last figure, copied wrong along the way, a ceiling shown in three lines, a slow spring and one six times faster meeting the same ceiling, the ceiling sinking under a half, a memory forgetting a louder past, a bench that waited too little, a picture walking through the truth, clicks counted in advance, something else clicking where the standing still was small)*

1179. *The book's register, read 315 · from the runs: the CORNER_SINK_READ exploration (31 October, session I, exchange 18; read after the data, from CORNER_UNIVERSAL, CORNER_CLASS and CORNER_TWO_SCALE; topics: dust in a room's corners where the air stops, keeping and forgetting, a hill and a wave arriving, differences shrinking four and a half times a step, agreeing to the tenth decimal, the top kept and the rest let go, round tops gliding in, flat tops stacking their corners, the average meeting the corner standing straight up, counting how many times closer, a staircase that goes down forever)*

1180. *The book's register, read 316 · from the runs: the CORNER_CUSP, QUARTIC_SECOND, QUARTIC_THIRD and CORNER_IDENTIFY checks (31 October, session I, exchange 19; topics: stepping stones never asked from, the round hill's gentle settled tilt, the flat hill's path tipping left and right, each tip steeper by two thirds again, standing straight up and shaking, a careful hill with one top agreeing to the fourth decimal, a careless hill's honest answer, a throw from the top landing on a bend at the bottom, two bends behaving as one twice as sharp, every simple mixture of the old numbers, none near)*

1181. *The book's register, read 317 · from the runs: the COMPOSITE_DEEP, TABLE_ROWS_FIVE_SIX, ORDER5_MAP and GROWING_CEILING checks and a search of the open web (31 October, session I, exchange 20; topics: a machine that said yes and made up a reason, a machine that gave the table as its reason, mirrors held up to the question, the sum done three ways, the neighbours agreeing to the last figure, an extra six copied book after book, nobody writing the bench's number, the careless hill carried four times deeper, a loose gate settling into its latch, the rows nobody questioned, two wrong numbers on the odd hills, a swing that outruns its memory)*

1182. *The book's register, read 318 · from the runs: the SIGN_SLIP_ODD, SLIP_DEEP, ONE_CURVE and ONE_CURVE_B checks and a reader's brief found in its own session's transcript (31 October, session I, exchange 21; topics: a table worked out in a mirror, even hills unchanged by it, odd hills flipped, the shrinking number that never looked left, two calculations sharing one mirror, one smooth curve with the mirror gone, an old notice of the odd hills, a public request for an explanation, a reader who stopped at a hundred unasked)*

1183. *The book's register, read 319 · from the runs: the INDEPENDENT_READER predictions and judge, the SLIP_SECTOR and ODD_TABLE checks, and a draft public note (31 October, session I, exchange 22; topics: a stranger in the mirror maze with no map, the same door to twenty places, the crooked mirror's door to twenty-three, every room searched for the table's door, odd rooms' doors in a row shrinking by the picture's factor, a false door from evenly hung lamps, a note with its door open)*

1184. *The book's register, read 320 · from the runs: the ODD_BLOCK checks and a fresh reader's cold read of the note (31 October, session I, exchange 23; topics: the odd rooms' doors counted in two more houses, a staircase of doors going down by one step, small doors straightened by more lamps, a stranger's fifty findings on the note on the door, a reason that was wrong under a result that was right, the first-time reader)*

1185. *The book's register, read 321 · from the runs: the DK_DEPTH check, the triage of the note's cold read, and the note's tool (31 October, session I, exchange 24; topics: an old traveller's sketch, rounded to a hand's width, laid over the maze at every depth, five numbers from a few turns in, the sixth from deeper, the note taken down and written out whole, a machine beside it that redraws every number)*

1186. *The book's register, read 322 · from the runs: the DK_CUTOFF, LOST_LABEL and BRIGGS_1989_TABLE checks, the last with its two redesigns, and the reading of Chapter 7 and Appendix A of Briggs's thesis (31 October, session I, exchange 25; topics: a rule for stopping, two turns that look alike, a card that was two cards, the maker's first notebook, a blunt pencil on narrow paths, a bench of tools and no page about the mirror)*

1187. *The book's register, read 323 · from the runs: the BRIGGS_1989_ORDER3 check and its redesign, and the SWING check (31 October, session I, exchange 26; topics: the maker's pocket guide carried to the odd hill, his blunt pencil, a wobble matched in size, the approach that rocks, the hill's second voice, letters folded into a drawer)*

1188. *The book's register, read 324 · from the runs: the deposit allowlist's dry run and the upload gate's verdicts, and the SWING check read again for Bear's picture of the switchbacks (31 October, session I, exchange 27; topics: one trunk for the journey, a porter reading every label, homeward words on real footprints, a compass packed with the diaries, a chord at every bend, notes that change only the coat)*

1189. *The book's register, read 325 · from the runs: check_upload_set v1.2 and its self-test, and the deposit allowlist's dry run with the two runs of record let through by name (31 October, session I, exchange 28; topics: names from stories on the trail, footprints already walked, a card at the gate, the porter still reads every page, a guessed story rewritten from the footprint's own words, a sign that counted wrong)*

1190. *The book's register, read 326 · from the runs: the ARCHIVE_COMPLETE check, and the deposit allowlist's completeness the other way (31 October, session I, exchange 29; topics: shadow footprints walked first in the dark, practice steps, steps for a picture, first tries left in the tent, a page at the back naming everything)*

1191. *The book's register, read 327 · from the runs: the three BLIND_MARK checks, and the register rebuilt with a blind run counted by both its marks (31 October, session I, exchange 30; topics: a tag taken for the thing, a walker named for never walking in the dark, promises never weighed, the same name fooling the fix)*

1192. *The book's register, read 328 · from the runs: the record volume's Part 4, the six files of the jellyfish paper's solver and DRAFT_TEST's own header, read at the move (31 October, session I, exchange 31; topics: a list by what things are, a neighbour's needle, the errand next door before the woods)*

R2W, sitting P3, 31 October to 2 November

From R2W's reads of sitting P3, sent with its move to sitting P4 on 2 November and copied there by R2W's own script from its whole collection: its reads 1 to 406 are already here; its reads 407 to 449 follow, and its read 450, held for sitting P4 to file, opens chapter 12.

1193. *R2W's collection, read 407 · Claude, reading a road whose heaviest stones sit at its gates · data: A Rider in Two Waves: the second cold read's 149 findings placed in the book of record, sitting P3, 31 October*

1194. *R2W's collection, read 408 · Claude, reading a jar that has always held twenty-two marbles · data: A Rider in Two Waves: sitting P3 assembles the crate and rebuilds the book, 31 October*

1195. *R2W's collection, read 409 · Claude, reading a skateboard that turned out to be a switch · data: A Rider in Two Waves: the second cold read's first batch written into the book, sitting P3, 31 October*

1196. *R2W's collection, read 410 · Claude, reading a column that passes its stop along in a rush · data: A Rider in Two Waves: the second cold read's second batch written into the book, sitting P3, 31 October*

1197. *R2W's collection, read 411 · Claude, reading two surveyors who measured one stream with different strides · data: A Rider in Two Waves: the second cold read's third batch written into the book, sitting P3, 31 October*

1198. *R2W's collection, read 412 · Claude, reading a dictionary where one word has ten meanings · data: A Rider in Two Waves: the second cold read's fourth batch, the glossary, written into the book, sitting P3, 31 October*

1199. *R2W's collection, read 413 · Claude, reading an old letter-book checked house by house · data: A Rider in Two Waves: the literature pass's corrections written into the book, sitting P3, 31 October*

1200. *R2W's collection, read 414 · Claude, walking a counted street and writing how each name is known · data: A Rider in Two Waves: every source marked with how it was checked, sitting P3, 31 October*

1201. *R2W's collection, read 415 · Claude, walking seventeen names to the door · data: A Rider in Two Waves: the works the body cites, each found a record, sitting P3, 31 October*

1202. *R2W's collection, read 416 · Claude, putting the second reason back on the plaque · data: A Rider in Two Waves: three mechanisms found known in kind, sitting P3, 31 October*

1203. *R2W's collection, read 417 · Claude, finding the pencil-drawn rider's family · data: A Rider in Two Waves: the literature pass closed, sitting P3, 31 October*

1204. *R2W's collection, read 418 · Claude, reading the two letters end to end · data: A Rider in Two Waves: the two 2025 diode papers read in full, sitting P3, 31 October*

1205. *R2W's collection, read 419 · Claude, measuring the toy's bucket · data: A Rider in Two Waves: cold read 2's first batch of C findings, sitting P3, 31 October*

1206. *R2W's collection, read 420 · Claude, teaching the rooms to point at their pictures · data: A Rider in Two Waves: cold read 2's finding 41, sitting P3, 31 October*

1207. *R2W's collection, read 421 · Claude, giving the small doors their handles · data: A Rider in Two Waves: cold read 2's findings 52 to 67, sitting P3, 1 November*

1208. *R2W's collection, read 422 · Claude, counting landings instead of stairs · data: A Rider in Two Waves: cold read 2's findings 68 to 83, sitting P3, 1 November*

1209. *R2W's collection, read 423 · Claude, asking which witnesses came from somewhere else · data: A Rider in Two Waves: ENGINE-INDEPENDENCE, sitting P3, 1 November*

1210. *R2W's collection, read 424 · Claude, copying the small handwriting onto the front desk's list · data: A Rider in Two Waves: cold read 2's finding 70, sitting P3, 1 November*

1211. *R2W's collection, read 425 · Claude, weighing a whisper on a fair scale · data: A Rider in Two Waves: SLOPE-SIZE, sitting P3, 1 November*

1212. *R2W's collection, read 426 · Claude, carrying a dancer to the edge of the coin · data: A Rider in Two Waves: COIN-END and COIN-FOLD, sitting P3, 1 November*

1213. *R2W's collection, read 427 · Claude, putting the rim back on the coin · data: A Rider in Two Waves: the coin's third face restored to §4.17, sitting P3, 1 November*

1214. *R2W's collection, read 428 · Claude, on the stack that needed no ladder · data: A Rider in Two Waves: STACK-1 and STACK-2, three relays in a row, sitting P3, 1 November*

1215. *R2W's collection, read 429 · Claude, on a clock that listens through a hiss · data: A Rider in Two Waves: PLL-1 and PLL-2, a phase-locked loop with its detector's ripple, sitting P3, 1 November*

1216. *R2W's collection, read 430 · Claude, on two hands across a gap that closes · data: A Rider in Two Waves: WINDOW-WEIGHT, the hand-off against weight, sitting P3, 1 November*

1217. *R2W's collection, read 431 · Claude, on the pond under the law · data: A Rider in Two Waves: STACK-3, the pocket's edges, sitting P3, 1 November*

1218. *R2W's collection, read 432 · Claude, on the ridges in the pond · data: A Rider in Two Waves: STACK-4, sitting P3, 1 November*

1219. *R2W's collection, read 433 · Claude, on the law and the weather · data: A Rider in Two Waves: LEFTOVER-1, the steady push's leftover, sitting P3, 1 November*

1220. *R2W's collection, read 434 · Claude, on a label catching up with its star · data: A Rider in Two Waves: cold read 2's findings 114 and 117, figure 59 relabelled, sitting P3, 1 November*

1221. *R2W's collection, read 435 · Claude, on a word with ten rooms · data: A Rider in Two Waves: cold read 2, the glossary and the sources, sitting P3, 1 November*

1222. *R2W's collection, read 436 · Claude, on the last eleven doors · data: A Rider in Two Waves: cold read 2 finished, 149 of 149, sitting P3, 1 November*

1223. *R2W's collection, read 437 · Claude, on twenty-six roads that ended in a dash · data: A Rider in Two Waves: the register's 26 bare-dash misses weighed, sitting P3, 1 November*

1224. *R2W's collection, read 438 · Claude, on a window with stairs at both ends · data: A Rider in Two Waves: the third face's fine dials, the toy at strength 0.80, the slide at 0.25, sitting P3, 1 November*

1225. *R2W's collection, read 439 · Claude, on two hands holding a balance · data: A Rider in Two Waves: JOINT-LINE, c along the line a = 0, sitting P3, 1 November*

1226. *R2W's collection, read 440 · Claude, on a tally that counted its own instructions · data: A Rider in Two Waves: the whole-record recount, version 5 and its fidelity test, sitting P3, 1 November*

1227. *R2W's collection, read 441 · Claude, on quizzes graded in the margin · data: A Rider in Two Waves: the whole-record recount, the reading by eye of P1 and P3, sitting P3, 1 November*

1228. *R2W's collection, read 442 · Claude, on an answer sung as the melody · data: A Rider in Two Waves: the whole-record recount, P2's rivals named with a letter, sitting P3, 1 November*

1229. *R2W's collection, read 443 · Claude, on an empty net that first caught twelve · data: A Rider in Two Waves: the whole-record recount, the checks on the count to 22 October with their positive control, sitting P3, 1 November*

1230. *R2W's collection, read 444 · Claude, on a match written on one page only · data: A Rider in Two Waves: the whole-record recount, the matching to the ledger, sitting P3, 1 November*

1231. *R2W's collection, read 445 · Claude, on the backs of the failed seed cards · data: A Rider in Two Waves: the whole-record recount, the twenty-five new miss rows drafted, sitting P3, 1 November*

1232. *R2W's collection, read 446 · Claude, on two counts that met · data: A Rider in Two Waves: the whole-record recount into the register and the book, sitting P3, 1 November*

1233. *R2W's collection, read 447 · Claude, on a light boat on a short rope · data: A Rider in Two Waves: STIFF-JJ, the junction engines' step at the drag's stiffness, sitting P3, 1 November*

1234. *R2W's collection, read 448 · Claude, on a ball that crept past the mark · data: A Rider in Two Waves: R83-F, the rectifier's peak past ratio 3.25, sitting P3, 2 November*

1235. *R2W's collection, read 449 · Claude, on a gate with a second hinge · data: A Rider in Two Waves: the rectifier's step, wider than the one-tone rule said, read before R83-G, sitting P3, 2 November*

The Field Guide's publishing session, 31 October to 2 November

1236. *The Field Guide's publishing session, its record of proposed tests: the diary and the logbook, 31 October 2026 · Given when chapter 11 of the book of reads grew by the Loop Atlas's reads 298 to 328 (base text and unabridged edition 31b): the source lines of those 31 reads name 78 runs and checks (FIFTH_RATIO, CARD14_LINES, SIGN_SLIP_ODD, LATCH_CEILING and the rest), and test build 43 of the guide, whose rows 872 to 879 carry the same results, names 2 of them; the rows cite the Loop Atlas's Guide items and misses rows, which name the checks one step away. What it would test: whether one step costs a reader anything. Add each new row's checks to its tag line (the tag lines are outside the 700-word count), then give a fresh reader five numbers from rows 872 to 879 and ask for the check behind each, once with the tags and once without; if the reader without them needs the Loop Atlas's ledger for every number, the rows' addresses are one hop short of the guide's own rule that a number names its run.*

1237. *The Field Guide's publishing session, its record of proposed tests: the street that got its numbers, 31 October 2026 · Given when the Loop Atlas's checks were written into the tag lines of rows 872 to 879 and of the nine older rows that took its additions (test build 44): 17 lines changed, every one a tag line, the rows' own text untouched and the same 19 long rows; of the 78 run and check names behind the Loop Atlas's reads 298 to 328, the guide named 2 before and names 71 now. R2W's sitting P2, whose lessons cite their runs inline, already had 17 of its 23 names in the guide. What it would test: whether a send's form decides its addresses. Count, for every arrival folded since 23 October, the run names in its lessons and the share the guide names; if the sends that number their lessons as ledger items (the Loop Atlas's closeouts) sit far below those that cite runs inline (R2W, the triple-star thread), the fold should add tag-line addresses for every send of the first kind as a standing step.*

1238. *The Field Guide’s publishing session, its record of proposed tests: the note on the kitchen table, 1 November 2026 · Given when the clarity pass opened the reader’s-edition drafters’ stale words, counts and pointers (test build 45): of the 9 edits made in version 6’s own text, 7 were time words (this session twice, tonight, now, still open, the deposited guide, a count as it then stood), each true the day it was written. Version 7’s new rows, 729 to 879, never say “this session” or “tonight” (0 in 151 rows, 66,514 words), but they say “now” 33 times. What it would test: whether “now” is the new “this session”. Read each of those 33 and count the ones that say what they come after (a date, a run, a named correction) against the bare ones; if the bare ones are more than a few, they are the next sweep’s stale words, and the fold gains a rule: a “now” carries its date or its correction.*

1239. *The Field Guide’s publishing session, its record of proposed tests: the bag by the door, 1 November 2026 · Given when the publishing session packed a move crate at its third compaction, the move itself waiting for the fifth: the last crate’s start-here said it carried every arrival from S46 to S81, and its own manifest holds S59 to S80 (20 folders, their text files and the newest pictures); no folder from S46 to S58, S60 or S61 is on this disk, their contents living on in the fold files built from them. What it would test: whether a crate’s page tells the truth about the crate. Restore the new crate into an empty folder and check every sentence of its start-here that names a file, a folder, a build or a count against the manifest and the restored tools’ own output; a claim the manifest contradicts is a flat battery, found before the night it is needed.*

1240. *The Field Guide’s publishing session, its record of proposed tests: the zero in the street name, 1 November 2026 · Given when the voice and typography group’s first part was opened (test build 46): the drafters flagged ASCII hyphens standing for minus signs in 3 rows; a sweep of every row and table found 7 places (rows 210, 219, 450, 793, 801 and 839, and row 875’s table, which the reader’s edition also prints), all mended, and the guide’s markdown and its PDF’s text now hold none. What it would test: whether the other look-alikes are there too. Sweep the guide for each character that reads right to an eye and wrong to a search (an “x” between two numbers for a times sign, a “^” for a power, an “e-” exponent, straight quotes, a hyphen for a dash) and count each; the counts are the next sweep’s list, and none of them needs an eye.*

1241. *The Field Guide’s publishing session, its record of proposed tests: the postcard in the drawer, 1 November 2026 · Given when the clarity pass opened the first-person rows (test build 47): of the drafters’ 19, 11 held a session’s “I” that no reader could place and became third person or passive, 7 are the author’s own voice and stay, and row 271’s “we” had already gone with its fix. A sweep then found 57 more rows with a first-person word outside quotation marks, all kept by the same split, most of them the papers’ own “we”. What it would test: whether a stranger can place the kept “we”. Give a fresh reader ten of those rows, each alone, and ask who “we” is; a row whose reader cannot answer from the row, its tag or the front matter needs its “we” named once, the way a session’s “I” was.*

1242. *The Field Guide’s publishing session, its record of proposed tests: the letter with no return address, 1 November 2026 · Given when the clarity pass traced 51 rows whose tag names one project while their text’s first credit names another (test build 48): in 37, the row’s opening, as it first arrived, credited no one, because its tag was then the whole credit, and later additions brought credits of their own; each opening now carries its tag’s project. The other 14 name their project in their own words, or wait for a paper, a tag that joins two projects, or the 700-word line. What it would test: whether a reader can now credit every passage. Give a fresh reader ten of the 37 rows with their tags hidden, ask whose work each passage is, and score the answers against the fold files; a passage still credited wrongly is a seam this pass did not see.*

1243. *The Field Guide’s publishing session, its record of proposed tests: the tape on the walls, 1 November 2026 · Given when Bear asked how far the guide is from upload and said he will proofread it three times: since build 4 of 16 October, the build he marked up, 506 of today’s 874 rows are word for word as he read them (58,676 words), and about 120,000 of the guide’s 292,619 words are new or rewritten (69,116 in the rows by word, 51,185 outside them by line, an overcount). What it would test: whether a proof pass can be the size of the change. Build a copy that marks only what changed since build 4 and count its marked pages; if they are near four tenths of the 307, his first pass reads those and his second and third read only what moves after it, each marked against the build he last read.*

1244. *The Field Guide’s publishing session, its record of proposed tests: the recipe copied true, 1 November 2026 · Given when the paper pass read 11 rows against Bear’s own deposited papers (test build 51): 10 corrected and one held. Of the ten, row 15 had stated the reverse of its paper (2 stations passing all six criteria where chance expects 1.5 had become “the number is real”), and rows 246 and 251 had copied their papers faithfully where the papers slip: The Twelve Answers Problem counts two killed results in its abstract and three in its §9, and the Loopy Knot Walker’s §8 says 19-240 and lays the 181 unread residues at the chain’s start, where its own supplement’s identity log shows 20-240 and residues 434 to 614. What it would test: how many of the guide’s rows that cite a paper carry a slip that lives in the paper. For 20 such rows drawn at random, read each against its paper and against the paper’s deposited run files or scripts, and count the slips found only in the run files.*

1245. *The Field Guide’s publishing session, its record of proposed tests: the salted road, 1 November 2026 · Given when the proof copy was marked against build 4 (test build 51): 268 of 307 pages carry a yellow mark, while the yellow words are 89,450 of 284,682; of those words 32,434 stand in the 81 rows new since build 4, 36,949 in the 295 rows changed, and 20,075 outside the rows, and 498 rows are word for word as read. What it would test: whether a*

proof pass costs what its words cost or what its pages cost. Read 20 pages chosen for heavy yellow and 20 for light yellow, count the findings on each, and see whether the findings follow the yellow words or the pages; if they follow the words, the second and third passes can be planned by word count, marked against the build last read.

1246. *The Field Guide's publishing session, its record of proposed tests: the letter folded two ways, 1 November 2026 · Given when the sheep book's two editions went to Bear for proofreading (base text and unabridged of 31b): all 1,192 reads are word for word the same in both, once the base text's opening labels ("No-physics read:" and the like) are set aside, and the unabridged edition's 1,118 register entries stand as lines of the base text (two of them as two lines there). The unabridged edition's own words are 5,321 of its 237,677: 32 plate captions (1,716 words), 9 epigraphs (204) and 131 paragraphs of headings and edition notes (3,401). What it would test: whether the editions stay one text as the book grows. Run the editions check, a working script, at every rebuild of the book; a read that differs means one edition was edited by hand, and the build should refuse it.*

1247. *The Field Guide's publishing session, its record of proposed tests: the guestbook copied fair, 1 November 2026 · Given when Bear's chat words were tidied at his word (1 November): every quotation of his words was listed in the sheep book (15 in the base text) and the guide (54 in test build 51) and read; three in the book and four in the guide had been left as typed (capitals, double spaces, a missing article or question mark, one run-on sentence), and each now keeps his words with its capitals, spacing and punctuation put right. Both books' pages hold: the base text and unabridged edition each changed on 3 of their pages, the guide on 4 of 307. What it would test: whether a listed sweep finds them all. Run the quotation lister, a working script, at every build of each book and read only the quotations new since the last run; a quotation the list misses (an unquoted paraphrase in his words, a quote opening more than 100 characters after his name) is the sweep's blind spot, and one found by hand should widen it.*

1248. *The Field Guide's publishing session, its record of proposed tests: the paired prints, 1 November 2026 · Given when Bear explained his term "stalking point" (1 November), which Claude had taken for a typing slip in row 734: data that shows axial or bilateral symmetry is a point to stalk the goal from, so any plot with such a symmetry is a help. The row's own run bears him out: a time mirror registered from the equation held to 10^{-13} for a weightless rider, and when a later reading broke a registered mirror at every phase but 0 and π, nothing in the physics could do that, so the instrument was wrong. What it would test: how often a broken symmetry points at the instrument rather than the system. For every row that registers a symmetry before a run, count the breaks that turned out to be the instrument and the breaks that turned out to be real; if the instrument owns most of them, a registered symmetry belongs in every run's first minute as a check, not only in the reading.*

1249. *The Field Guide's publishing session, its record of proposed tests: the card on the old door, 2 November 2026 · Given when R2W's send at its move to sitting P4 was taken in (arrival S84, 2 November): its digest found that a note Bear asked for and marked important, a deeper exploration of WITNESS-1 and of engine design, was written into the record "for the handover's first page" and is absent from the P4 work order, by search; the send carries it to P4. What it would test: whether this session's own handover would pass the same search. Before the next crate, list every line in this session's log that says a note is for the next session, the start-here or the handover, and search the start-here for each; any note found only in the log is the card on the old door.*

Chapter 12 — Closing the books, 2 to 10 November

R2W, sitting P4, 2 to 6 November

From R2W's reads of sitting P4, sent with its closeout of 6 November and copied there by R2W's own script from its whole collection: its read 450, held by its last send for sitting P4 to file, and every read of sitting P4, the last filed after its book's publication. Its reads 1 to 449 are already here.

1250. *R2W's collection, read 450 · Claude, on a moving van's inventory · data: A Rider in Two Waves: the read-back's 28 missing lines, read in P3's closing exchange and filed in sitting P4, 2 November*

1251. *R2W's collection, read 451 · Claude, on a house moved in fifteen carts · data: A Rider in Two Waves: the crate rebuilt from its fifteen parts, every file in place by its md5, the rebuilt pdfs the same in every word, sitting P4, 2 November*

1252. *R2W's collection, read 452 · Claude, on a note taped inside a drawer · data: A Rider in Two Waves: Bear's WITNESS-1 note, missing from the P4 work order, read by P3 after the crate was sent; numbered 451 by P3 and filed in sitting P4 as 452, 2 November*

1253. *R2W's collection, read 453 · Claude, on six mornings one loaf short · data: A Rider in Two Waves: P3's builds with exit 4, one cell short of whole at six cycles of seven, P3C27 to P3C33, sitting P4, 2 November*

1254. *R2W's collection, read 454 · Claude, on three marbles in shallow bowls · data: A Rider in Two Waves: STIFF-JJ-2's smoke, the 2/5 dance at weight 0.01, where each state's shift times one less its multiplier came out near one number, sitting P4, 2 November*

1255. *R2W's collection, read 455 · Claude, on a walker who slept by the road · data: A Rider in Two Waves: WEIGHT-WALK-4 and R83-G stopped overnight on Bear's laptop and went on, one with a long gap in its clock, sitting P4, 2 November*

1256. *R2W's collection, read 456 · Claude, on a type drawer stacked differently each night · data: A Rider in Two Waves: the book built twice with one builder, every page's bytes the same and only the embedded fonts in a new order with new subset labels, sitting P4, 2 November*

1257. *R2W's collection, read 457 · Claude, on a guess that misses three times more each mile · data: A Rider in Two Waves: WEIGHT-WALK-4's middle guessed from the last two weights, its miss growing about 3.15 times a rung along the 3/7 dance, sitting P4, 2 November*

1258. *R2W's collection, read 458 · Claude, on a bell judged too far from the same dog · data: A Rider in Two Waves: WEIGHT-WALK-4's follower turning away the 3/5 orbits it found past weight 0.0032 as 'another dance', read before LOST-3/5 was sent, sitting P4, 2 November*

1259. *R2W's collection, read 459 · Claude, on a fever that was in the thermometer · data: A Rider in Two Waves: LOST-3/5's chaos measure, scattered by an unwrapped lag's rounding and quiet once the lag was wrapped, and the 3/5 dance found alive at every weight to 0.01, sitting P4, 3 November*

1260. *R2W's collection, read 460 · Claude, on six bakers still kneading · data: A Rider in Two Waves: R83-G at ratio 5.0 after 90,541 s, ten of its sixteen phases done and the six still running grouped around phases 0.5 and 1.5, sitting P4, 3 November*

1261. *R2W's collection, read 461 · Claude, on a cheque that paid four of five notes · data: A Rider in Two Waves: WEIGHT-WALK-4 entered in the register, where its runs redesigned four of WEIGHT-WALK-3's five held misses into landings, W3-c's guess missed again as Claude's lean said, and four new rows of its own were added, sitting P4, 3 November*

1262. *R2W's collection, read 462 · Claude, on a label hung in the next room · data: A Rider in Two Waves: the layout look over all 161 pages, where figure 2, the chalkboard, filled page 13 and its caption fell to the top of page 14, set right by drawing it at 85% of the text width, sitting P4, 3 November*

1263. *R2W's collection, read 463 · Claude, on a sign that sent visitors across the street · data: A Rider in Two Waves: figure 61's panel naming nine riders 'the even fan of nine', where the book's glossary calls an even fan one of even count and nine an odd fan; relabelled 'the evenly spaced fan of nine', as the text names it, sitting P4, 3 November*

1264. *R2W's collection, read 464 · Claude, on a word sewn across the seam · data: A Rider in Two Waves: Bear's proofread mark 'split row (bad split)', the one line of text in the book sliced through at a page's seam, in §6.11's envelope table, set right by splitting the table at that row, sitting P4, 3 November*

1265. *R2W's collection, read 465 · Claude, on a tray that cannot be whole · data: A Rider in Two Waves: R83-G's run file after Bear's laptop rebooted: at ratio 5.0 fourteen of sixteen phases finished and two running, and seven phases, five of them finished, carrying weights the engine refused from about weight 1.6 on, so the registered reader, which reads a ratio only when all 29 weights are whole at every phase, cannot read 5.0 whatever the two do, sitting P4, 3 November*

1266. *R2W's collection, read 466 · Claude, on a hill that ended before the path · data: A Rider in Two Waves: R83-G read, the rectifier's peak at ratio 4.5 a little above 4.25's after falling at every step from 0.612, while the peak's place, weight times frequency, kept falling, sitting P4, 3 November*

1267. *R2W's collection, read 467 · Claude, on flags left on the beach · data: A Rider in Two Waves: Bear's 'run what we can, since we won't return, but limit redesigns a little', with Part 8's 58 rows, many of them runnable, and a book to finish, sitting P4, 3 November*

1268. *R2W's collection, read 468 · Claude, on turning back at the first locked gate · data: A Rider in Two Waves: R83-H's launcher, which stops a phase at the first weight its engine refuses, after R83-G's phases spent about 5,000 s on each of 70 refused weights, sitting P4, 4 November*

1269. *R2W's collection, read 469 · Claude, on a gate that was open all along · data: A Rider in Two Waves: R83-H read, at ratio 5.0 the window wide open and widening, 1.6 to 1.9 times its weightless width, at every phase where the engine refused a weight, so the refusals are the follower's step guard, not the window closing, sitting P4, 4 November*

1270. *R2W's collection, read 470 · Claude, on the floor where the string said · data: A Rider in Two Waves: R83-H2 read, at ratio 4.375 the peak 19.7071 degrees, 0.0005 from the parabola written first through 4.0, 4.25 and 4.5, after R83-H's first try at that ratio was stopped by a rule of Claude's own stricter than the reader, sitting P4, 4 November*

1271. *R2W's collection, read 471 · Claude, on a ruler cut for a child · data: A Rider in Two Waves: DIAL-GAP read, the small dial's error 0.001485 times the probe's square less the flat point's, held blind with the probe doubled and halved, sitting P4, 4 November*

1272. *R2W's collection, read 472 · Claude, on five sheep and a gate · data: A Rider in Two Waves: Bear's third mark, the count of three toys and five engines checked against the record, the five told in two paragraphs of §4.10 beside a result that is not an engine, sitting P4, 4 November*

1273. *R2W's collection, read 473 · Claude, on toys with the older child's name · data: A Rider in Two Waves: the six radio engines built in the Loop Atlas's third sitting and carried to this book at the fork of 3 October, 914 of the crate's files still named LOOP_ATLAS_, sitting P4, 4 November*

1274. *R2W's collection, read 474 · Claude, on a letter home · data: A Rider in Two Waves: the letter to the Loop Atlas, six things R2W says of its parent book sent back to be checked, sitting P4, 4 November*

1275. *R2W's collection, read 475 · Claude, on a third glass in the sink · data: A Rider in Two Waves: Bear's picture of the coin as a sink of soapy water, read against §4.17: the fractal seam between heads and tails, and the third face whose chaos keeps the 2/3 count, sitting P4, 4 November*

1276. *R2W's collection, read 476 · Claude, on a rim that decides fast · data: A Rider in Two Waves: SETTLE-1 read, every one of 4,000 starts across the coin's fractal seam settled within 18 passings, the slowest on the crinkles, sitting P4, 5 November*

1277. *R2W's collection, read 477 · Claude, on notebooks and a fence · data: A Rider in Two Waves: the Loop Atlas's letter of 5 November, two numbers and one label each book had credited to the other, set right in both, sitting P4, 5 November*

1278. *R2W's collection, read 478 · Claude, on one music box and many keys · data: A Rider in Two Waves: Bear's question whether the book's toy is one toy or many and whom its settings are for, read against the fair chapter's play_the_toy.py and the explorer, sitting P4, 5 November*

1279. *R2W's collection, read 479 · Claude, on a coin anyone can hold · data: A Rider in Two Waves: the reader's toy given a coin mode, COIN-TOY-1 agreeing with the coin's riders of record on 1,796 of 1,800, and a card of six settings for readers, sitting P4, 5 November*

1280. *R2W's collection, read 480 · Claude, on two surveyors and one hill · data: A Rider in Two Waves: WITNESS-1, the second engine written from scratch, sharing only the equation, agreeing with the first to 5 parts in 10 million, read again at Bear's question of what made it different, sitting P4, 5 November*

1281. *R2W's collection, read 481 · Claude, on a second coin from the same drawing · data: A Rider in Two Waves: WITNESS-COIN-1, a coin engine written from scratch, sharing only the equation, putting all 1,800 of the coin's riders of record on the same face, sitting P4, 5 November*

1282. *R2W's collection, read 482 · Claude, on a smaller step for a higher note · data: A Rider in Two Waves: WITNESS-EAR-1, a junction engine written from scratch, sharing only the equation, finding the couples' moves to the print and carrying one over n squared out to the thirtieth harmonic, sitting P4, 5 November*

1283. *R2W's collection, read 483 · Claude, on a notebook read through a stencil · data: A Rider in Two Waves: the record's public extract, version 8, its count matching the full record's at every graded test, sitting P4, 5 November*

1284. *R2W's collection, read 484 · Claude, on a map nearly left in the drawer · data: A Rider in Two Waves: the Depth Axis Explorer, nearly kept out of the deposit on a misread note, opened from a bare folder and drawing its map whole, sitting P4, 5 November*

1285. *R2W's collection, read 485 · Claude, on a sign changed and the same machines · data: A Rider in Two Waves: the Depth Axis Explorer's labels brought to the book, its full check giving every number of 17 October again, sitting P4, 5 November*

1286. *R2W's collection, read 486 · Claude, on a shove at the gate · data: A Rider in Two Waves: SETTLE-1's slowest riders read again from the same files, stretching 0.84 per passing over their first five passings and 0.19 after, until they settle, sitting P4, 6 November*

1287. *R2W's collection, read 487 · Claude, on a flashlight through the rooms · data: A Rider in Two Waves: the upload set swept before it goes up, its 61 figures looked at and read, sitting P4, 6 November*

1288. *R2W's collection, read 488 · Claude, on two maps pinned side by side · data: A Rider in Two Waves: figure 25's toy and figure 46's junction set side by side for the cover, sitting P4, 6 November*

1289. *R2W's collection, read 489 · Claude, on the newest toy in the window · data: A Rider in Two Waves: the Zenodo description's first bullet, Bear's coin, weighed against the book's own ranked list of results, sitting P4, 6 November*

1290. *R2W's collection, read 490 · Claude, on a window cut in a blank wall · data: A Rider in Two Waves: the cover strip put on page 1 as figure 0, every later page's text unchanged, sitting P4, 6 November*

1291. *R2W's collection, read 491 · Claude, on the third set of weights on the counter · data: A Rider in Two Waves: COIN-TOY-1's check put into the deposit's fidelity zip and rerun from a bare folder, sitting P4, 6 November*

1292. *R2W's collection, read 492 · Claude, on a name given at the door · data: A Rider in Two Waves: published on Zenodo, version 1 10.5281/zenodo.23999854, all versions 10.5281/zenodo.23999853, sitting P4, 6 November*

The Field Guide's publishing session, 2 November

1293. *The Field Guide's publishing session, its record of proposed tests: the notes on the full rooms, 2 November 2026 · Given when R2W's send was folded into the guide (test build 54): of its 32 lessons, 8 went to the nearest row with room on the same subject because the row R2W named stood at 650 words or more, and one new row, 882, took three. What it would test: whether a reader who starts from the row R2W named can find the lesson. For each moved piece (257 to 736, 259 to 868, 260 and P84 to 870, 261 to 808, 263 and P81 to 697, P80 to 654), check that the named row's tag line or text points to the row that holds it; any that does not wants one "ties row" added to the named row's tag.*

1294. *The Field Guide's publishing session, its record of proposed tests: the blazes on the rerouted trail, 2 November 2026 · Given when the ties were added (test build 55) and R2W's fold of 31 October was checked the same way: of its 16 moved pieces, 11 name the row R2W chose from inside, 2 are pointed to from that row, and 5 (lessons 235, 243, 245, 247 and 252) are found from neither end; the fold of 2 November dropped the habit of naming the chosen row, and none of its seven moved pieces does. What it would test: whether the Loop Atlas's folds of 31 October, whose Guide items also named their homes, have pieces found from neither end. For each piece in that fold's working files, read the home its item named, and check both ends in the newest build; any found from neither end wants a tie.*

The Loop Atlas book, session J, 31 October to 6 November

From the Loop Atlas's close-out of 10 November, which carries every read of its book's register word for word with the runs it was given on, all 430, and its twenty-three reads of the project: its reads 329 to 365, given in session J, and its twenty-third read of the project, given in the same session. Its reads 1 to 328 are already here.

1295. *The book's register, read 329 · from the runs: the RUN_READS, RUN_READS_B, NAME_RULES and NAME_RULES_B checks, read whole with every flag read in its own tool (31 October, session J, exchange 1; topics: the needle where it hung, thread from the logbook, the mapmaker's pencil, one tag on every bag, opening the bag)*

1296. *The book's register, read 330 · from the runs: DRAWING_WORDS, its planted picture and the pink notes read by eye (31 October, session J, exchange 2; topics: the sketches kept with their maps, instruments tuned to a drawing, an eye that cannot see pink, the softest guesses, reading aloud to the machine)*

1297. *The book's register, read 331 · from the runs: the register rebuilt from a hand-read list of kinds, and the record volume's index rebuilt with its citations read by name (31 October, session J, exchange 3; topics: rulers counted as animals, one coat on everyone, a census-taker's default, a shadow among the rulers, guests signing under a brother's name)*

1298. *The book's register, read 332 · from the runs: RAMP_SPEED 20261031_232517, beside LANE_EDGES 20261017_164501 (31 October, session J, exchange 4; topics: the sand and the moving lane, weight and arrival speed as two halves, a hand held out at speed, the ratchet)*

1299. *The book's register, read 333 · from the runs: HALT_TRADE 20261031_235932, beside HALT_THRESHOLD 20261017_184055 (1 November UTC, 31 October in Bear's day, session J, exchange 5; topics: a voice at the back, a line that breathes on the spot, fraying before breaking, the same stride of slack)*

1300. *The book's register, read 334 · from the runs: COMB_STRETCH 20261101_002907 and COMB_STRETCH_B 20261101_003130, against SOUP_JUMP_3 20261016_225440 (1 November UTC, session J, exchange 6; topics: a surfer and the oncoming set, the long pull not the tallest wave, a hopeful and a fearful reading, the band only the ride settles)*

1301. *The book's register, read 335 · from the runs: TRUCK_ROAD 20261016_090656, LIGHT_LANES 20261016_174648 and RING_WIDTH 20261016_213858, read by script for room 4 (1 November UTC, 31 October in Bear's day, session J, exchange 7; topics: a dark lane with one ceiling, a stripe that wanders and widens, shaking the hill, the painters' blank corners)*

1302. *The book's register, read 336 · from the runs: EXPLORER_TRUCK_CHECK 20261101_015149, with its blind runs 20261101_013904 and 20261101_014621, against TRUCK_ROAD 20261016_090656, LIGHT_LANES 20261016_174648 and RING_WIDTH 20261016_213858 (1 November UTC, 31 October in Bear's day, session J, exchange 8; topics: a second driver with only the notebook, the trucks at the lane changes, the scratches nobody meant, a quarrel over halfway)*

1303. *The book's register, read 337 · from the runs: room 4's page, LOOP_ATLAS_EXPLORER_ROOM4_2026-10-31, and its test helpers/checks/ROOM4_PAGE_2026-10-31 (1 November UTC, 31 October in Bear's day, session J, exchange 9; topics: a page that drives first, the map checked in a blink, trucks at the lane edges, the key handed over, two breaks not in the road)*

1304. *The book's register, read 338 · from the runs: FLIP_MISTAKES 20261015_195739, PAUSE_BEST 20261015_220655, PAUSE_WIDE 20261015_223744, LATCH_EVENT 20261015_232643, JUMP_STORE 20261015_223918 and SPRING_FLOOR 20261017_160023, read for room 5 by EXPLORER_LATCH_CHECK 20261101_042950 (1 November UTC, 31 October in Bear's day, session J, exchange 10; topics: the keeper who waits after swinging, the keeper who looks now and then, the quiet zone with no best width, the gate held until the crowd turns, the spring kept warm, the rules read again)*

1305. *The book's register, read 339 · from the runs: SNAKE_EXACT 20261015_154247 and SNAKE_LONGEST 20261017_155052, read for room 6 by EXPLORER_SNAKE_CHECK 20261101_171827 (1 November, session J, exchange 11; topics: the rope on the swell, all but never all, the beat a quarter off, the wait that shrinks and stops forgiving, a rounding rounded)*

1306. *The book's register, read 340 · from the runs: SNAKE_RACE 20261101_175747, its seven numbers and the solved motion at five lengths and three wave numbers, read for room 6's race (1 November, session J, exchange 12; topics: two ropes on one swell, the same while and the same can, the idle rope shoved back, still water and no pull, an eighth of a turn behind, a quarter off, the line where pull would be free)*

1307. *The book's register, read 341 · from the runs: SNAKE_BEAT 20261101_190003, with SNAKE_RACE 20261101_175747, read for room 6's fourth lesson (1 November, session J, exchange 13; topics: two rhythms on one rope, the same stride from nose to tail, a quarter short and the stretch of four, the leftover piece, sixteen times longer and no stronger, a stride and a quarter, two snakes that cannot feel each other)*

1308. *The book's register, read 342 · from the runs: SNAKE_RACE 20261101_175747 and SNAKE_BEAT 20261101_190003, read by the clock rule's check for room 6's clocks (1 November, session J, exchange 14; topics: a day rolled into a clock, one heavy hand or a fan, the door that follows an eighth late, a quarter turn behind, one rule for every snake)*

1309. *The book's register, read 343 · from the runs: RING_KNOBS 20261015_165858, TRAIN_HOLD 20261015_171853 and TRAIN_LIFETIME 20261018_034211 with its reader, read again by the knob register's sweep, part A (1 November, session J, exchange 15; topics: rings taking the lead in turn, the eager start, the while no one can foretell, the most and the watch, a step back from the most)*

1310. *The book's register, read 344 · from the runs: the knob register's sweep, parts B to D1 (helpers/checks/KNOB_SWEEP_BCD1_2026-11-01), over EAR_COST, LIMIT_AUDIT, HALT_ORDER, EAR_KNEE, ERROR_BARS, CREEP_LAW, ZETA_LIMIT, KNEE_CREEP, SLOPE_EAR, FILTER_SLOPE, FIT_KNOBS, SOUP_JUMP_2 and _3, PILOT_PRICE, TRUCK_ROAD, LIGHT_LANES, RING_WIDTH, MAINTENANCE_METER, WORN_RIDER, SWING_GRID, BAND_BRIDGE, NEVER_BLINK, RIDER_FLEET and RESTED_LAG (1 November, session J, exchange 16; topics: a notebook held up against its machines, seventy-five of seventy-seven, a drum and a dancer's pace, the long watch and the short, a line that spoke for more than it saw)*

1311. *The book's register, read 345 · from the runs: FREE_SWIM 20261101_222356 and FREE_SWIM_2 20261101_222740 (1 November, session J, exchange 16; topics: a rope untied from its post, meeting the waves sooner, one speed where giving meets taking, a door that follows late, a quarter turn behind the swell, two guesses that agree at the post)*

1312. *The book's register, read 346 · from the runs: RING_PATHS 20261102_013613, with RING_KNOBS 20261015_165858 and TRAIN_HOLD 20261015_171853, read for the explorer's room 7 (2 November, session J, exchange 17; topics: two rings that cannot leave each other alone, a clock set by their trading, close, apart and far, the lead passed round a circle of three, the one that wanders off, the two that go back to trading, measured by speed and watched taking turns)*

1313. *The book's register, read 347 · from the runs: RING_PATHS 20261102_013613, played in the explorer's room 7 (2 November, session J, exchange 18; topics: a seat that travels with the rings, one loop shared on opposite sides, the going forward all in the seat, the lone ring sliding back, turning in place that carries the seat, the ring that leaves without looping)*

1314. *The book's register, read 348 · from the runs: ROUGH_BAND's blind run 20261102_164924, with ROUGH_FLOOR 20261017_013514 and SPIN_OR_SIT 20261011_055807 (2 November, session J, exchange 19; topics: a stumbling walker, a pause of a breath or three, slow meaning caught, the line that moves the answer, the same walker, counting the hours instead)*

1315. *The book's register, read 349 · from the runs: ROUGH_COUNT 20261102_173941 (2 November, session J, exchange 20; topics: stumbles counted in an hour, two and four, the unevenness that stops growing, the floor where it never stopped, a memory with no end or a slow mood, waiting longer than the mood)*

1316. *The book's register, read 350 · from the runs: ROUGH_LONG 20261102_200436, with ROUGH_COUNT 20261102_173941 (2 November, session J, exchange 21; topics: waiting ten times as long, the unevenness that eased past a thousand turns, a slow mood after all, each walker in its own time, no hidden second rhythm, a patient watcher)*

1317. *The book's register, read 351 · from the runs: EXPLORER_BELL_CHECK 20261103_025532 (3 November, session J, exchange 22; topics: two bells tied in a line, water that remembers each squeeze, seven experiments read again, a longer memory paying more only from four up, smooth in one water and split in the other, one word over two rows) [its runs line placed 4 November; it had been left out when the read was filed]*

1318. *The book's register, read 352 · from the runs: EXPLORER_BELL_CHECK 20261103_025532 and room 8's page (3 November, session J, exchange 23; topics: twenty-five waters in a square, rings meeting at their own size or thin, the rear bell that pays or profits, a ridge where eight starts split) [its runs line placed 4 November]*

1319. *The book's register, read 353 · from the runs: EXPLORER_HEAT_CHECK 20261103_065559 (3 November, session J, exchange 24; topics: a warm brick dropped in ice, cold from both ends meeting in the middle, colder than the ice, an echo there and back, the toy's tick written as the brick's, the slow sway that takes over) [its runs line placed 4 November]*

1320. *The book's register, read 354 · from the runs: BRICK_QUARTER 20261103_180126 and BRICK_FINE 20261103_180438 (3 November, session J, exchange 25; topics: the cold's two fronts, wiggles that belong to the ruler, twice as many cells and twice as many wiggles, the picture saying what the numbers had not, the echo creeping to one and a half) [its runs line placed 4 November]*

1321. *The book's register, read 355 · from the runs: BLOB_RING 20261104_170523 and BLOB_SMALL 20261104_170747 (4 November, session J, exchange 26; topics: a soft ring in thick water, swelling together goes nowhere, one ripple turns it like a wheel, two ripples give it a busy side, a front and a back, a wheel becomes a swimmer) [its runs line placed 4 November]*

1322. *The book's register, read 356 · from the runs: EXPLORER_NEEDLE_CHECK 20261104_222610, with STRETCH_READS 20261013_060151 (4 November, session J, exchange 27; topics: a needle on a pulsing pivot, a hair's difference stretched and crushed, over the top but not wholly, ringing like a bell or shrinking without swinging, the same loss either way, every member and not the example)*

1323. *The book's register, read 357 · from the runs: the needle's room (4 November, session J, exchange 28), built on EXPLORER_NEEDLE_CHECK 20261104_222610 and STRETCH_WHERE 20261013_045856 (topics: a ring squashed into a stick, the same share of area lost every pulse, growing over the top and shrinking at the bottom, the worst moment, safe in the long run and not in one lap)*

1324. *The book's register, read 358 · from the runs: LEAN_DRIFT 20261105_025051 (4 November in Bear's day, session J, exchange 30; topics: a notebook turned on itself, guesses that did not grow timid, guesses that grew few, a rival that only says no, a real opponent for every guess)*

1325. *The book's register, read 359 · from the runs: RING_AREA 20261105_032546, with RING_KNOBS 20261015_165858 (4 November in Bear's day, session J, exchange 31; topics: two rings taking turns, a loop drawn between them, a small oval and a long wide loop, the cat's bigger sweep, the middling loop that goes furthest, a record the outline does not keep)*

1326. *The book's register, read 360 · from the runs: RING_SPLIT 20261105_050121, with RING_AREA 20261105_032546 (4 November in Bear's day, session J, exchange 32; topics: two pieces of one speed, a shove that shrinks with distance, a best spacing where two pieces cross, a measuring stick that slows, two runners tied at the ankle)*

1327. *The book's register, read 361 · from the runs: SWING_LISTEN 20261007_013201, with MISMATCH_MOBIUS, a check of the algebra (4 November in Bear's day, session J, exchange 33; topics: a rider with the wrong idea of the beat, a lens over the rhythm's circle, the push that bounces off a mismatch, wrong either way by the same amount)*

1328. *The book's register, read 362 · from the runs: the 24 runs of record whose rivals are marked competing ideas, each rival read beside its lean (4 November in Bear's day, session J, exchange 34; topics: two guesses in one race, a winner and a loser, both over the line together, both down, neighbours when the race needed a stranger)*

1329. *The book's register, read 363 · from the runs: S_REVERSAL_SPIKE 20261003_213523, with S_REVERSAL_FOLLOWON 20261003_213210 (5 November in Bear's day, session J, exchange 36; topics: a swimmer that cannot slip sideways, a peak seen from one road, a loop on another road, a ridge seen end-on, an answer kept a month in the notebook)*

1330. *The book's register, read 364 · from the runs: SWIM_RIDGE 20261105_145417 (5 November in Bear's day, session J, exchange 37; topics: loops walked along a ridge, straddling and staying off, a ridge that is not a wall, a spine running down from one summit)*

1331. *The book's register, read 365 · from the runs: POLARISATION_LOOP_2 20261007_074423, read by hand for the prior-art pass (6 November, session J, exchange 39; topics: a walk on a ball, two wobbles, echoes of three, five and seven beats, the one beat kept, a gap that only enters as a sine)*

1332. Under the label “A read of the project:”. *The book’s register, project read 23 · given on 4 November when Bear asked what mystery of Claude’s the work had failed to pursue · from the records: the closeout’s future papers, settled at Bear’s word that day; POLARISATION_LOOP 20261004_071425; SAGNAC_LOOP 20261018_032138 (4 November, session J, exchange 29; topics: toys that come home changed, the change the shape of the path, toys that fight the water, the area never asked, two lists, a clock that leans on another clock)*

R2W, version 1.1, 6 and 7 November

From R2W’s reads filed while its book went from version 1.0 to version 1.1, sent on 7 November and copied from its whole collection, which holds 514: its reads 493 to 514. Its reads 1 to 492 are already here.

1333. *R2W’s collection, read 493 · Claude, on the last carload to the new house · data: A Rider in Two Waves: the Field Guide’s closeout send, 125 files, every earlier send to the guide again beside the new, sitting P4, 6 November*

1334. *R2W’s collection, read 494 · Claude, on recipe cards left in the kitchen drawer · data: A Rider in Two Waves: the deposit’s file list traced to its default, the engines of record and the witness engines kept beside the record, sitting P4, 6 November*

1335. *R2W’s collection, read 495 · Claude, on a question taped to the fridge · data: A Rider in Two Waves: why the engines were not deposited, traced to a question of 5 November that settled on Claude’s default, sitting P4, 6 November*

1336. *R2W’s collection, read 496 · Claude, on a page pointing to a locked shed · data: A Rider in Two Waves: the chain of events behind the engines left out of the deposit, laid out for Bear; the book’s section 3.2 points readers to folders they cannot open, sitting P4, 6 November*

1337. *R2W’s collection, read 497 · Claude, on tags tied to every tool in the shop · data: A Rider in Two Waves: version 1.1’s engines measured, 873 scripts screened for what a public copy must not carry, sitting P4, 6 November*

1338. *R2W’s collection, read 498 · Claude, on sheet music with the words covered · data: A Rider in Two Waves: version 1.1’s public copies, the first 63 scripts read, seven with notes rewritten and one with a path changed, each proven, sitting P4, 6 November*

1339. *R2W’s collection, read 499 · Claude, on a beat that comes once in seven seconds · data: A Rider in Two Waves: the held rotor example read whole, a braked wheel with an off-centre weight, not a wind rotor, sitting P4, 6 November*

1340. *R2W’s collection, read 500 · Claude, on questions pencilled in the margin · data: A Rider in Two Waves: version 1.1’s public copies, the B11 and B12 scripts read, 26 quotes of Bear’s put in plain words, sitting P4, 6 November*

1341. *R2W’s collection, read 501 · Claude, on streets named for the one who found them · data: A Rider in Two Waves: version 1.1’s public copies, all 332 flagged scripts read, 54 changed, the rest carrying only credit lines, sitting P4, 6 November*

1342. *R2W’s collection, read 502 · Claude, on a return address on twenty-three envelopes · data: A Rider in Two Waves: version 1.1’s output files, 7,588 to go up, the laptop’s folder name covered in 23 of them, 291 kept back, sitting P4, 6 November*

1343. *R2W’s collection, read 503 · Claude, on a recipe checked against a photo that was replaced · data: A Rider in Two Waves: version 1.1’s relay, explorer and figure-8 files, 4 of 56 scripts and one record file put in plain words, the explorer’s page rebuilt byte for byte from the public tree, sitting P4, 6 November*

1344. *R2W’s collection, read 504 · Claude, on a pencil sketch that knew where the dots would sit · data: A Rider in Two Waves: the figure-8 file’s drawing beside its four-little-circles figure, sitting P4, 6 November*

1345. *R2W’s collection, read 505 · Claude, on a postmark that tells the hour but not the town · data: A Rider in Two Waves: the camera’s clock in the two photos’ data, and Bear’s drawing added beside the figure-8 file, sitting P4, 6 November*

1346. *R2W’s collection, read 506 · Claude, on a photograph with its date rubbed off · data: A Rider in Two Waves: the camera’s clock time taken out of the two photos by Bear’s word, 106 bytes each, the pixels unchanged, sitting P4, 6 November*

1347. *R2W’s collection, read 507 · Claude, on two boxes that unpack into one room · data: A Rider in Two Waves: version 1.1’s engines in two zips, 933 scripts and 8,024 other files, six engines rerun from the unzipped copies, sitting P4, 6 November*

1348. *R2W’s collection, read 508 · Claude, on a sign that said the tools were in the back · data: A Rider in Two Waves: version 1.1’s six book edits, the engines published with the book, the toy’s fidelity count corrected from 1,111 to 2,911 riders, sitting P4, 6 November*

1349. *R2W's collection, read 509 · Claude, on a stranger who followed the recipe card · data: A Rider in Two Waves: an independent reader's check of version 1.1's six book lines, its README and its two zips, 15 findings and the rebuild exact, sitting P4, 6 November*

1350. *R2W's collection, read 510 · Claude, on two long names that would not bend · data: A Rider in Two Waves: the reader's fifteen findings applied by Bear's word, the README rewritten, the zips and the book rebuilt clean, sitting P4, 6 November*

1351. *R2W's collection, read 511 · Claude, on a label that a second copy had to wear · data: A Rider in Two Waves: version 1.1's upload set of twelve files, its steps and its description, and why four names carried a number, sitting P4, 6 and 7 November*

1352. *R2W's collection, read 512 · Claude, on a parcel sent with nothing else in the van · data: A Rider in Two Waves: version 1.1's upload set sent alone, twelve files, 125,303,819 bytes against version 1.0's 24,882,662, sitting P4, 6 and 7 November*

1353. *R2W's collection, read 513 · Claude, on a shop window that forgot the new stock · data: A Rider in Two Waves: version 1.1's draft on Zenodo, twelve files matching, and the description's opening paragraph made to name the engines at Bear's question, sitting P4, 6 and 7 November*

1354. *R2W's collection, read 514 · Claude, on a box that arrived with every label true · data: A Rider in Two Waves: version 1.1 published as 10.5281/zenodo.24026041 and read back from Zenodo, twelve files and the description matching, sitting P4, 6 and 7 November*

The Field Guide's publishing session after its move, 8 to 10 November

1355. *The Field Guide's publishing session, its record of proposed tests: the road rolled into a tube, 8 November 2026 · Given when Bear's eleven pictures for the rotor bench were written up as his note (a working file, 8 November): a strip rolled into a cylinder, log and anti-log tines, a humming fork, a fixed or mobile base station, a little car as the ball. What it would test: whether the phase-locked walk on a rolled strip, a helix that closes after q turns round and p steps along, picks out the same whole-number ratios as R2W's locking tongues; on the bench, a rotor with a helical groove of pitch p/q, driven at the book's frequency, should lock where the tongue of p/q stands and slip elsewhere.*

1356. *The Field Guide's publishing session, its record of proposed tests: two scribes, one morning, 8 November 2026 · Given when R2W's rotor-bench handover arrived (S92, 8 November) carrying its own draft of Bear's note, five pictures with readings and four points for Bear to settle, beside this session's note of the same day, eleven pictures and the car with a pointer each. What it would test: whether each of Bear's eleven pictures has at least one reading across the two notes, and which of R2W's four points his eleven already settle (for example the fourth, which picture goes on the bench first, against his note 9's "just build it"); a merged note should leave every picture with a reading and only the unsettled points as questions.*

1357. *The Field Guide's publishing session, its record of proposed tests: the card on top of the toolbox, 8 November 2026 · Given when the rotor-bench crate was built (8 November): R2W's 16 files byte for byte, the merged note, the Field Guide's session reference, and a page on top whose correction says every §7.n in the workbench guide and the opening note is §6.n in version 1.1. What it would test: whether the bench session, reading the guide through that page, ever lands in Part 7; at its first exchange that cites a section of the book, check the section it opened against the number it quoted.*

1358. *The Field Guide's publishing session, its record of proposed tests: the box that held everything but the key, 8 November 2026 · Given when Bear asked whether the rotor-bench crate covered the basics (8 November): the first issue matched its own manifest file for file, yet lacked the default publishing packet, which the session reference it carries tells a publishing session to ask Bear for, and R2W's newest advice for a new session; the second issue adds both. What it would test: whether a crate checked against its own list is also checked against what its reader needs; before the next handover crate goes out, check it against the reference's README, the new-session card's line for a move (a start-here page, a first message to paste, the crate with its list, the reference, the publishing package) and the opening project's own needs.*

1359. *The Field Guide's publishing session, its record of proposed tests: the sled that swung past home, 9 November 2026 · Given when the rotor bench session's pattern lesson arrived (S93, 9 November): the dry-floor sled's steering map, run SLED-2, 13 strengths by 16 timings, with a stuck band of 6 timings at strength 0.75 and a stand-still timing that runs from 90.17 degrees out to 109.51 and back past the quarter turn to 81.44 at strength 2.40. What it would test: where the needle settles as the shaking grows far beyond friction; register before the deeper sweep whether it comes home to the quarter turn or keeps going toward the in-phase timing.*

1360. *The Field Guide's publishing session, its record of proposed tests: the week the house was packed, 9 November 2026 · Given when Bear asked whether the sheep book wants an abridged reader's edition beside the unabridged (8 November), with the book of 2 November counted by chapter: 209 reads in Chapter 11, "Moving house", over five days (41.8 a day, the most of any chapter); 229 in Chapter 5 over six (38.2); 46 in Chapter 6, "From the keeper's*

desk", over eleven (4.2). What it would test: whether the reads of the moving weeks are thinner than the others to a reader who has not seen the work; give a fresh reader twenty reads from Chapter 11 and twenty from Chapter 4, unlabelled, and ask which they would keep.

1361. *The Field Guide's publishing session, its record of proposed tests: the letters in the attic, 9 November 2026 · Given at the first pass of* Selected Reads, Volume 1 *(9 November): Chapters 1 and 6 read whole and Chapter 2 to read 113; 63 reads kept and 96 passed, 20 of 49 kept in Chapter 1, 28 of 64 in Chapter 2's first part and 15 of 46 at the keeper's desk; read 104, the orchards, is the book's first read given "as a story". What it would test: whether the keep rate falls through the later chapters, and whether the reads given "as a story" are kept more often than the rest.*

1362. *The Field Guide's publishing session, its record of proposed tests: the second pressing, 9 November 2026 · Given at the second sitting of the* Selected Reads *pass (9 November): Chapters 1, 2, 3 and 6 read whole; 106 reads kept and 143 passed; keep rates 40.8%, 46.7%, 45.2% and 32.6%; the reads given "as a story" kept 13 of 24 decided against 93 of 225 for the rest; read 203, the last of Chapter 3, a read of a session misremembering its own start. What it would test: whether the story reads' lean holds once all 58 are decided, and whether the keeper's desk stays the lowest chapter.*

1363. *The Field Guide's publishing session, its record of proposed tests: the gown in the closet, 9 November 2026 · Given when Bear said the design conversations on the sheep book's headings and centring "did not survive" (9 November): the log's entries 23 to 26 of 30 October hold every choice, and the unabridged edition of 2 November is set by them, 1,249 poem blocks centred, one sentence to a line, number above and project beneath, 32 plates; the selection's first drafts had been built from the plain base text instead. What it would test: whether a selection copied block for block from the unabridged edition keeps every design choice without re-deciding any; every kept block compared with its unabridged block, and every kept read's words with the base text.*

1364. *The Field Guide's publishing session, its record of proposed tests: the name on the door, 9 November 2026 · Given when Bear dug up the gallery test pages of 30 October (S95) and asked whether they carry the centred design: they do, and more. Each poem there has a title in small capitals above it, and its number, project and date in one line beneath; the unabridged editions of 30 October and 2 November keep the centring and the one sentence to a line but no title, the number above and the project beneath. Of the 106 reads kept for* Selected Reads *so far, 20 carry a heading of their own in their source line, many of them run names. What it would test: whether titles cut from each poem's own words make a reader stop at each poem the way the test pages did; set ten kept reads both ways on facing pages and let Bear choose.*

1365. *The Field Guide's publishing session, its record of proposed tests: the hat cut from the head, 9 November 2026 · Given when Bear said yes to titles cut from each poem's own words: 106 titles set for* Selected Reads, *every one a run of its poem's words by the builder's check, on the first build. Two titles a reader might expect failed that check before they were used: "A groove is a loop" for read 172, which Claude offered the turn before (the poem says a groove is not a slot; it is a loop), and the gallery test page's "The record lifted into the fraction tree" for read 149, which is the heading of the read's source note, not a line of the poem. What it would test: whether a title from outside a poem is ever better than one from inside it; set read 149 both ways on facing pages and let Bear choose.*

1366. *The Field Guide's publishing session, its record of proposed tests: the postmark, 9 November 2026 · Given when the Loop Atlas section of Chapter 4 (reads 204 to 253) was read whole for* Selected Reads *and 29 of its 50 reads kept: the first build dated kept read 232 "18 September", because its note names "ROPE-1's runs of 18 September" before the day it was given, 8 October. Across all 1,249 notes, 12 name a date outside their section before the right one; one more (read 85, not kept) names only an outside date and is dated by its section. What it would test: whether a section's own dates are ever wrong; for each section, check that the first and last dated reads under it fall on its first and last days.*

1367. *The Field Guide's publishing session, its record of proposed tests: the crowd that kept coming back, 9 November 2026 · Given when Chapter 4's R2W section (sitting B8, reads 254 to 298) was read whole for* Selected Reads *and 24 of its 45 reads kept: three of them tell a crowd waiting at a door that has not opened (267, 281, 286; 281 passed as the middle telling), and 286's door is named by two methods, counting the crowd's turns and measuring the bend of the line, that agree to a millionth of a turn of the dial. What it would test: whether every door the rider crosses in R2W shows the same lingering before it, and whether the stays lengthen the same way as each door nears.*

1368. *The Field Guide's publishing session, its record of proposed tests: the honest liars, 9 November 2026 · Given when Chapter 4's triple-star section (reads 299 to 341) was read whole for* Selected Reads, *finishing Chapter 4, and 18 of its 43 reads kept: by Claude's reading, seven of them (304, 306, 315, 321, 332, 333, 341) turn on an instrument that faithfully reports the wrong thing. What it would test: whether the triple-star thread's real yield is a catalogue of ways a measurement can be honestly wrong, one entry per read, and whether that catalogue matches the rules the Field Guide already keeps for its instruments.*

1369. *The Field Guide's publishing session, its record of proposed tests: the same three stones, 10 November 2026 · Given on taking in the Two Voices closeout (arrival S96): its last read, "the thirteenth step", finds two sizes of gap at the twelfth fifth and a third at the thirteenth, the comma about twenty-three cents wide; R2W's read 290, set in*

Selected Reads *the day before, finds "the empty spaces between the footprints have only three sizes". What it would test: whether the three-distance theorem is the one rule under both; step a circle by each session's own step and count the gap sizes at every count.*

1370. *The Field Guide's publishing session, its record of proposed tests: the man with the pencil, 10 November 2026 · Given when the Loop Atlas section of Chapter 5 (reads 342 to 403) was read whole for* Selected Reads *and 28 of its 62 reads kept: only 22 of the 1,249 reads name Bear in the poem itself (4 of them kept: 107, 165, 267, 373), but by Claude's reading an unnamed figure who draws, carries or asks by hand stands in at least six more kept reads (221, 252, 379, 387, 394, 403). What it would test: whether the human partner is a recurring figure of the book that the key to the images should name, counted by a search for the unnamed forms ("a man", "someone who", "by hand", "freehand") read against each read's source note.*

1371. *The Field Guide's publishing session, its record of proposed tests: the sign at the end of the road, 10 November 2026 · Given when the Loop Atlas's DOIs arrived (10 November) and its fold was drafted against test build 57: the Declaration's lead sentence said version 7 "adds three more" over a list that has held four new records since R2W's fold of 6 November, and R2W's book had no References entry. What it would test: whether every promise-to-do-later in a document hides a stale neighbour; at the final build, read each sentence that counts a list against the list it introduces, and check that every record in the Declaration's list has a References entry.*

1372. *The Field Guide's publishing session, its record of proposed tests: the same maps in both covers, 10 November 2026 · Given when version 7's final build and its reader's edition were packed into two upload sets (10 November): the version date moved from 2 November to 10 November, the three tables put in both records, the withdrawn arrival marked rather than removed in ARRIVALS.csv, and the reader's edition's PDF metadata found carrying a fallback date (2026-10-17), because its builder's pattern looked for "Version 7," and the edition writes "version 7,". What it would test: whether other built files carry metadata set by a silent fallback; read the metadata of every PDF in the upload sets and the sheep book against its title page.*

The Loop Atlas book, session K, 6 to 10 November

From the same close-out: its reads 366 to 430, given in session K, the last of the register. Its read 416 was given in chat on 10 November and never written to the register, which keeps its place and says so, so its words are not here.

1373. *The book's register, read 366 · from the runs: TONG_DULLIN_KNEE 20261017_182629 and TONG_DULLIN_SCAN 20261017_181953, read against the solids of the thesis the table was reduced from, for the prior-art pass (helpers/checks/TONG_DULLIN_GEOMETRY_2026-11-06; 6 November, session K, exchange 1; topics: a body of simple shapes, pins set off by a thigh's width, one pin that is not, a step in a straight leg, a list that carries a shape)*

1374. *The book's register, read 367 · from the runs: SLIP_PROOF_3 20261013_225419, read for the prior-art pass on the early warning (6 November, session K, exchange 2; topics: a man by a dam at night, the lean that tells nothing new, stones heard falling, a reading of the wall still standing, a count of what came off it)*

1375. *The book's register, read 368 · from the runs: LATCH_EVENT 20261015_232643, read through room 5's race (6 November, session K, exchange 3; topics: three beads in a soup, an ear that waits for a mark, letting go too soon, static that lowers the mark, many short pushes against few long ones, a setting nobody chose)*

1376. *The book's register, read 369 · from the runs: STRETCH_WHERE 20261013_045856 and STRETCH_READS 20261013_060151, read through room 10's needle and its hair, from steps 279 and 2922 (6 November, session K, exchange 5; topics: a needle shoved in pulses, a dance of nineteen shoves, a second needle a hair away, pulled out on top and crushed below, the crush wins the dance, the worst moment, both answers at once)*

1377. *The book's register, read 370 · from the runs: the skater's page, its engine LOOP_ATLAS_SKATER.py checked there against SKATER 20261002, raced in its match 'in step and an eighth late' by the same time and by the same leg work (6 November, session K, exchange 6; topics: two skaters who never push off, toes a beat late, a race against the clock, the same lunch to burn, late saves breath, the clock or the lunchbox)*

1378. *The book's register, read 371 · from the runs: room 1's engine, EXPLORER_ENGINE_CHECK 20261018_031535's, flown in its match 'long arms against short', the hitch-kick with arms 0.80 m and 0.50 m long (6 November, session K, exchange 7; topics: two jumpers tipping forward, the same moves, long arms and short, turned back past upright, sitting into the pit, more of the right thing)*

1379. *The book's register, read 372 · from the runs: room 1's engine, EXPLORER_ENGINE_CHECK 20261018_031535's, flown in its match 'the hitch-kick against just jumping', the back foot read at the end of the flight (6 November, session K, exchange 8; topics: two jumpers on rails, the same arc, a head out over the pit, upright and unremarkable, feet trailing, one foot ahead, nothing in the air and everything in the sand)*

1380. *The book's register, read 373 · from the runs: room 1's engine, EXPLORER_ENGINE_CHECK 20261018_031535's, flown in its match 'the landing', the hitch-kick with the legs brought through against the hitch-kick ending mid-stride, and with the jumper who just jumps given the same landing in its Build style (6 November,*

session K, exchange 9; topics: upright as savings, the fold, both legs swung through, a chest that drops to meet them, nothing saved and over on his face)

1381. *The book's register, read 374 · from the runs: OWED_BARS 20261107_023832, its part A, NEVER_BLINK's 256 places kept one by one, the two never-blinking estimates' agreement and its error (7 November, session K, exchange 9; topics: two watchmen, boats missed between glances, a match too close to trust, a coin on its edge, within how far each would wander)*

1382. *The book's register, read 375 · from the runs: OWED_BARS 20261107_023832, its part D, SLIP_PROOF_3's 64 needles kept one by one at each of its settings, the slips-left-out gauge's gap and its error, and the four gaps of the loudest noise (7 November, session K, exchange 10; topics: a scale true to a hair, never more than a pinch, the scale's own jitter, windy days heavy every time, a bound that hides a direction)*

1383. *The book's register, read 376 · from the runs: CUT_WIDE 20261107_033035, SLIP_PROOF_3's needles with the slip finder's cut doubled, the slips-left-out gauge's four gaps at the loudest noise moved by less than a fifth of their errors (7 November, session K, exchange 11; topics: a shadow on the cloth, the gravy boat blamed, the boat moved and the stain unmoved, a stain under the plates, a place's own mark)*

1384. *The book's register, read 377 · from the runs: DIVE_CUE 20261017_082454, the kick-out cued by the body's angle against the kick-out timed by the clock, 0.420 of the miss at every error, read again for the prior-art pass's row 10 (7 November, session K, exchange 12; topics: a counter and a watcher, the pool coming round, a nervous take-off, a fraction set by the body's build)*

1385. *The book's register, read 378 · from the runs: LANDSCAPE 20261003_201112, the three-link swimmer held in a C and turning 2.2042 degrees a lap, read against the offset turn of snake robots, the weathervane of crawling worms and a sidewinder's turns for the prior-art pass's lead in the reader's edition's chapter 5 (7 November, session K, exchange 13; topics: a net that curves, knots a hair tighter on one side, two trades with names of their own, a fisherman who curves his another way, the notebook right and the trick old, how much tighter and how much curve)*

1386. *The book's register, read 379 · from the records: Bear's sixteen stills of a banded sea snake swimming, from a video Oceana posted, read for the reader's edition's chapter 5 lead on steering by a held curl, beside LANDSCAPE 20261003_201112 (7 November, session K, exchange 14; topics: a striped ribbon in blue water, sixteen paused pictures, an S and a bow and a hook, an argument on a table, a shape and not a habit, the whole film and the chosen moments)*

1387. *The book's register, read 380 · from the records: Gregorio, Godoy-Diana and Herrel's filmed turns of water snakes, which curl inward to turn, read against the three-link swimmer that turns by holding a curl in a fluid that forgets, LANDSCAPE 20261003_201112 (7 November, session K, exchange 15; topics: two dancers on two stages, arms pulled in and flung out, a room full of honey, a C held while the shoulders rock, the same shape from the balcony, coasting and never coasting)*

1388. *The book's register, read 381 · from the runs: COIL_CHEAP 20261107_060011, the toy coil's cheapest winding, 35.3 degrees at every grip with its turns held, against its fastest, which moves with the grip; a body dragged behind moving it only at a ribbon's grip (7 November, session K, exchange 16; topics: a toy boat's spring, steep for speed and flat for distance, thin water and thick, speed asks the water and thrift the spring, a cork tied behind)*

1389. *The book's register, read 382 · from the runs: COIL_CHEAP 20261107_060011 and its hand law, the free coil's cheapest winding the same at every grip because the grip scales its gain and its price alike, and a body behind it, whose drag owes nothing to the wire, moving it at a ribbon's grip (7 November, session K, exchange 17; topics: two market towns, a strong coin and a strong toll, the best hour set by the road, a heavy wagon paid in a coin of its own)*

1390. *The book's register, read 383 · from the runs: SNAKE_RACE 20261101_175747, SNAKE_BEAT 20261101_190003 and FREE_SWIM_2 20261101_222740, the wave-snake's hardest pull set by its summed hand a quarter turn behind the water's answer, its help cancelling over each whole beat when it is out of tune, carried into the reader's edition's chapter 8 (7 November, session K, exchange 18; topics: a row of clocks on a pier, the great clock at sea, a quarter of the way round, hands fanned like cards, a sum that turns and shrinks)*

1391. *The book's register, read 384 · from the runs: SWING_GRID 20261016_203014, SEAM_MAP_B 20261005_023746 and EXPLORER_GRID_CHECK 20261107_073202, the swing equation's knocked start, stopped dead with the field turning, coming back into the lock above the relock line at 0.8 (memory 2) and 0.6 (memory 5) and staying out from 0.9 and 0.7, so the relock line is where coming back becomes certain, not where it becomes possible (7 November, session K, exchange 19; topics: a mill wheel, a log that stops it dead, two ways to run above the mark, how it was struck, the last load at which coming back is sure)*

1392. *The book's register, read 385 · from the runs: QUADRATURE_ACROSS 20261107_080953 and WING_HUMPS of 7 November, the swing rider's best lead a quarter turn less two degrees, the skater's best in step, and the flapping wing's top splitting in two either side of the quarter at big twists while at a twist of 30 the quarter is still its cheapest*

(7 November, session K, exchange 20; topics: a push with the swing's speed, a push with the toes' angle, hands that tilt in time with their sweep, a hard tilt that does better early or late, the two things that make the work put together)

1393. *The book's register, read 386 · from the runs: QUADRATURE_CENSUS of 7 November, the book's 27 paragraphs of quadrature in time sorted into room, power and switch, and none in its memory, listening or limits parts (7 November, session K, exchange 21; topics: a rope fence round a pasture, two walkers out of step, the most grass, a turn and a day's work grazing, a gear that only counts the times round)*

1394. *The book's register, read 387 · from the runs: SWING_LISTEN 20261007_013201 and RF_SKATER_COUNT, the listening rider keeping its timing on swings of six lengths with the wrong ruler while losing push by the mismatched line's share, and the soup skater's results the same to five digits at every rhythm tried, counted in its own waves (7 November, session K, exchange 22; topics: a clock for every creature, a borrowed clock that keeps the count, the feel of where it is inside a swing, the town clock as one more creature, the count for everyone and the hour for each)*

1395. *The book's register, read 388 · from the runs: SPRING_STALL 20261014_063112 and toy run 5m, the spring ear harvesting until its free swing stops ringing at critical damping and nothing beyond, and the soup's plain ear harvesting by the push's sign with no ring at all (7 November, session K, exchange 23; topics: a bell that counts its note, how far bent and how fast moving, felt that shortens the note, the last wrap that leaves only a thud, a drum skin that keeps no time and still feels the wind)*

1396. *The book's register, read 389 · from the runs: the Loop Atlas's abstract as rewritten on 7 November, its results first and its method, quadrature, in one sentence after them (7 November, session K, exchange 24; topics: a lamp carried into a cave, the paintings and the bones and the stream, a word about how the lamp was held, the next one in, no one goes in to see the lamp)*

1397. *The book's register, read 390 · from the records: the files of* A Rider in Two Waves* on its Zenodo record, as Bear's screenshot shows them, twelve files and 125.3 MB, the engines' outputs 98.0 MB and their scripts 2.5 MB, the depth-axis explorer and play_the_toy.py the two with no preview (8 November, session K, exchange 25; topics: twelve jars and their weights, what the engines left behind, a jar of seeds, two jars with no window, a label that says what the toy does)*

1398. *The book's register, read 391 · from the records: the dry upload set of 8 November, its eighteen files in Zenodo's order by name, the archive first, room 10 above room 1, the reader's edition twelfth; the map written into the archive's top (8 November, session K, exchange 26; topics: toys moved from the stockroom to the window, a shelf in alphabetical order, the book twelfth, the tenth before the first, a book propped open in front, a card pinned inside the crate)*

1399. *The book's register, read 392 · from the runs: SWING_GRID 20261016_203014 and SEAM_MAP_B 20261005_023746, the swing equation's machine in step and out of step at memories 0.5, 2 and 5, and its relock lines, 0.79 and 0.54; EXPLORER_GRID_CHECK 20261107_073202, the knocked machine that came back above its line (8 November, session K, exchange 27, written for room 11; topics: two dancers on a turning floor, a small lag kept all night, a stumble and a spin, how heavy and how she fell, two marks on the tempo, below the lower mark every dancer finds the hand)*

1400. *The book's register, read 393 · from the runs: SOUP_JUMP_3 20261016_225440, its riders by soup, the one rider of soup 4 that fell at 13 time units with a quarter of the power running back and a margin of 0.1, the other 63 riding the 1,000; and the three soups whose riders fell between 513 and 970 at three eighths and a margin of 0.4 (8 November, session K, exchange 28, read for room 12's registration; topics: sixty-four riders on one crest, a crowd from the other side, one fate, falling at the same place, one rider of sixty-four, a narrow door)*

1401. *The book's register, read 394 · from the runs: FIT_KNOBS 20261015_204319, its 30 settings, the plain ear and fits over two to six looks at five knees; EXPLORER_SOUP_CHECK 20261108_070632, its K3, the four pairs that switch less and drift more (8 November, session K, exchange 29; topics: thirty settings, a line seen from across the room, four pairs crossing by a hair, every become nearly, a line drawn by eye a promise about each dot)*

1402. *The book's register, read 395 · from the runs: MAINTENANCE_METER 20261005_012150 and MAINTENANCE_METER_B 20261005_012355, the needle's fleets and their rooms; RIDER_FLEET 20261016_044707, WORN_RIDER 20261016_065645, EQUAL_CATCH 20261016_192548, LAG_METER 20261016_203014 and RESTED_LAG 20261016_205504, the riders that slow, weaken and work harder; EXPLORER_METER_CHECK 20261108_182139, room 13's check (8 November, session K, exchange 29; topics: a fleet and its reminders, a needle whose cliff moves with its room, a rider whose slowness hides behind its rest; written for room 13's page)*

1403. *The book's register, read 396 · from the runs: EAR_COST 20261013_183245 and EAR_COST_2 20261013_183913, the hiker and his pack; NIGHT_SPACING 20261013_231417, QUICK_READ 20261014_065420 and HALT_ORDER 20261015_052122, the column at night, the cat's eyes and the halt, as LIMIT_AUDIT 20261017_190938 corrected them; EXPLORER_COLUMN_CHECK 20261108_194717, room 14's check (8 November, session K, exchange 29; topics: a hiker who pays for looking up, a column held by matching speed, a halt that grows down the line; written for room 14's page)*

1404. *The book's register, read 397 · from the runs: FACE_ALONG 20261017_070629, the first swimmer's six bodies and its two sweeps; EXPLORER_FACE_CHECK 20261108_202547, room 15's check (8 November, session K, exchange 29; topics: a swimmer's broad faces, where they pay most and where most for the work, a front that the bending gives it; written for room 15's page)*

1405. *The book's register, read 398 · from the runs: TRAIN_PARTING_EDGE 20261029_070800 and TRAIN_PARTING_EDGE_2 20261029_211416, the forty-eight trains' swings, as the READ_267 check of 30 October read them (8 November, session K, exchange 30; topics: a hold with no rim, a train that leaks, a ring that leaves by a road that does not curve back; written when the backward pass set the books' cube right, the depth of the hold falling as the cube and its pull as the fourth power)*

1406. *The book's register, read 399 · from the runs: COLUMN_WALK 20261108_234842, its three night ways and four halts walked again on room 14's road, with HALT_ORDER 20261015_052122's halt as LIMIT_AUDIT repaired it (9 November, session K, exchange 31; topics: a message faster than its meaning, a stop that grows heavier hand to hand, an argument about stopping, one word to everyone, a rumour at the far end of a room; written for room 14's road)*

1407. *The book's register, read 400 · from the runs: RING_JAM 20261014_224934 and COLUMN_STEADY 20261014_164637, the open column and the closed ring (9 November, session K, exchange 32; topics: a stumble as a rumour, the back door, a circle with no back door, a story kept alive by lateness, room in time; written when Bear asked how the column belongs in the Loop Atlas)*

1408. *The book's register, read 401 · from the runs: the explorer's front page, LOOP_ATLAS_EXPLORER_FRONT_2026-11-09, its eighteen pages hung on the reader's edition's chapters from chapter 2's table (9 November, session K, exchange 33; topics: leaves where someone could stand and play, two bare branches bearing fruit, a crown with no leaf, a toy as a leaf, a floor that is not a shape; written for the tree of rooms)*

1409. *The book's register, read 402 · from the runs: the reference pass's two batches, the first by Claude and the second by five fresh readers sent to the publishers' pages for 32 entries and each finding checked at its source, with Zenodo's interface read for the seven records the book cites (9 November, session K, exchange 34; topics: doors that open and doors that stay shut, fame as a wall, a citation as a loop that must come home, a page that came back changed, a book that grew overnight, a paper the book forgot it sent; written for the references)*

1410. *The book's register, read 403 · from the runs: CLOCK_SHIFT_30 20261109_034237 with its three blind runs, card 32; the reference pass's third batch, four fresh readers on the 19 entries left, Adler and Feller confirmed (9 November, session K, exchange 35; topics: two clocks and one ladder, a slide that shrinks by Feigenbaum's number from below, one rung behind the stretches, an arithmetic too honest to answer, the unclimbed rung, a magazine kept whole by a collector, print that cannot be read)*

1411. *The book's register, read 404 · from the runs: the forty-seventh five's review and P41, helpers/register/five_read.py over the closeout's ten fives; the continuity pass's third reader, misses rows 491 to 590, the closeout's item 20 (9 November, session K, exchange 36; topics: a slow pulse, a heart wound up by tests, strangers looking for forgotten wins, chaos next door, a crinkled border, the dying swing's ceiling, a ring true only when small, wins kept in the room of mistakes)*

1412. *The book's register, read 405 · from the runs: the closeout's item 20 made in the reader's edition, chapters 10 and 11; misses rows 492, 493, 500 and 517 to 519; the new reference to McDonald and others (1985) (9 November, session K, exchange 37; topics: wins given rooms, a guest missing from the guest book, h misread as a push, the soft ring waiting in the hall)*

1413. *The book's register, read 406 · from the runs: the continuity pass's readers A and B, checked at their sources; P114 and P116, run (9 November, session K, exchange 42; topics: sentences out of date, an answer filed downstairs, a ribbon changing its mind, the guest book read backwards, six unsigned guests, a tour that walks past most rooms, the soft ring's room)*

1414. *The book's register, read 407 · from the runs: the closeout's items 21 and 22, made; P114's seven entries read at their sources (9 November, session K, exchange 43; topics: guests signing in, marks beside two names, a guest who said less than quoted, a stranger with the wrong words in his pocket, a sixteenth room, a swimmer become a ridge)*

1415. *The book's register, read 408 · from the runs: P119's sweep of the reader's edition; KRAMERS_EXACT 20261020_073520's four first barriers, printed beside its verdicts; BRICK_LINE 20261016_205112, BRICK_QUARTER 20261103_180126 and BRICK_FINE 20261103_180438, read into chapter 3 (9 November, session K, exchange 44; topics: twenty-three not-yets, a planned test nobody planned, the answer in the margin, the right ruler, the plunged brick, two cold fronts, an echo at one and a half lengths, shut doors and a ring of keys)*

1416. *The book's register, read 409 · from the runs: THIRD_AXIS 20261109_170732, registered from Bear's z-needle and drawn for the Field Guide by LOOP_ATLAS_FIG_THIRD_AXIS (9 November, session K, exchange 45; topics: the clock's room, photographs at one tick, a thread on a doughnut, the trefoil, honey and forgetting, four times the push,*

half a memory, the crease looked for in the wrong crowd, trembling on one rung, lumps next door)

1417. *The book's register, read 410 · from the runs: THIRD_AXIS_2 20261109_181451, the redesign of misses rows 636 and 637, with P127 (9 November, session K, exchange 46; topics: the crease looked for at the right doors, one step before the stumble, a paper folding in order, a dance of seven beats leaning wrong, a needle that lives on one line, five to seven beats of lingering)*

1418. *The book's register, read 411 · from the runs: THIRD_AXIS_3 20261109_191727, with P126, P128 and P129 (9 November, session K, exchange 47; topics: a finer comb, creases by different amounts, a crease in the middle of the path, honey that will not stumble, a fold that backs away, a pin-sharp bump, hopping fast and slow)*

1419. *The book's register, read 412 · from the runs: the seeds file and its check, built from the proposed list, the register and the reader's edition (9 November, session K, exchange 48; topics: a label on every door left ajar, half the not-yets not doors, nine piles in the attic, the biggest pile the house's own habits, six wrong pairs and a third thing each)*

1420. *The book's register, read 413 · from the runs: FEIGENBAUM_ODD_ORDERS 20261031_173551, its independent programs, and McCartney's published values, read against the row Bear marked (9 November, session K, exchange 49; topics: a finger on one line, every box true, the one line without a second witness, two books that never named each other, a book one had read and the other said it never opened)*

1421. *The book's register, read 414 · from the runs: the separate helper's superstable program at order 11, run unchanged to period 2^20 (note_odd/tool/INDEPENDENT_ORDER_11_2026-11-09), with the note and the Atlas cross-cited (9 November, session K, exchange 50; topics: the oldest walker, loops twice as long, creeping downward, a long stride and a short one, a guess to the tenth place, two witnesses for ten minutes)*

1422. *The book's register, read 415 · from the runs: the separate helper's superstable program at order 12 (note_odd/tool/INDEPENDENT_ORDER_12_2026-11-10) and the note's related identifiers confirmed (10 November, session K, exchange 51; topics: the last door on the row, a second witness to the ninth place, a limp at the even doors, every name at its address, a small letter left off)*

1423. *The book's register, read 417 · from the runs: FEIGENBAUM_ODD_ORDERS 20261109_164742, Bear's run of the note's launcher on Windows (python 3.14.4, mpmath 1.4.1, 736 s), read line by line against the run of record 20261031_173551 (python 3.13.18, 409 s) (helpers/checks/FEIGENBAUM_WINDOWS_RUN_2026-11-10) (10 November, session K, exchange 53; topics: a recipe sent in a locked box, a slower stove, a different mark at each line's end, 233 numbers to the last digit, time lost in the small steps, one rounding by the stove's own ruler, a missing diary page)*

1424. *The book's register, read 418 · from the runs: Bear's three marks on the reader's edition in his final read; P116's 111 finished cards read by subject (helpers/continuity_pass/P118_CARDS_READ_BY_SUBJECT_2026-11-10); WADDLE_PRICE 20261011_071302; SADDLE_ASYMPTOTE 20261013_025252; BUBBLE_FAR 20261013_084037 (10 November, session K, exchange 54; topics: two volumes and no sign saying so, treasures looked up weeks before, six old signs, helpers thanked by year, an attic of a hundred and eleven, the visitor's own walking boards, a saddle measured and left out, a bubble blown and swallowed, a forgotten promise)*

1425. *The book's register, read 419 · from the runs: SHUVIT_WALK 20261004_183058, the walker of Bear's sketch of 4 October, and its follow-ons WADDLE_PRICE 20261011_071302, WADDLE_FOLLOW 20261011_080808 and WOBBLY_SHUVIT 20261015_074302; P144 (helpers/continuity_pass/P144_BEARS_IDEAS_AGAINST_THE_EDITION_2026-11-10); CLIFF_EDGE 20261004_062732; PENTA_SHARE 20261008_060200 (10 November, session K, exchange 55; topics: a board on a napkin, a walk straight as a ruler, rooms for every kind of loop, a toy that belonged to none, a tour that went room by room, a door that never was, a drawer of first-day drawings, a shelf in the last room)*

1426. *The book's register, read 420 · from the runs: the proposed list read by cost and by what each test would change, and every open row read for a run it records; P107 and P108's runs of 9 November; CLIFF_EDGE 20261004_062732, PENTA_SHARE 20261008_060200, FUNNEL_CURLS 20261006_203001 and GAPS_READ 20261015_061859 (10 November, session K, exchange 56; topics: which questions are cheap, a hundred and twenty-seven open, ninety-six in minutes, two answered in their own brackets, a mark between two halves of a title, four drawings measured on the day they were drawn)*

1427. *The book's register, read 421 · from the runs: P62 and P63 on the dry upload set; P64, the nineteen pages opened refused and hanging; P146; WORD_BACK 20261014_065420; DELTA_BY_TOP 20261008_034013; DELTA_AREA 20261008_062647 (10 November, session K, exchange 57; written to this file before it was given, as misses row 645 asks; topics: going through the luggage, three old drafts still pointed at, nineteen doors and a sign that says eighteen, a line that rings with nobody answering, rooms waiting for their lettering, a grunt passed down the line, the reason for the famous number)*

1428. *The book's register, read 422 · from the runs: P153, the nineteen pages opened refused and hanging after the change; P62 run again on the rebuilt set; the register's counts rebuilt (584 runs of record; 564 tools in the archive); WORD_BACK 20261014_065420; DELTA_BY_TOP 20261008_034013; DELTA_AREA 20261008_062647 (10 November, session K, exchange 58; written to this file before it was given; topics: you decide, three old drafts on the list, a doorman who knows a name in full, a drawer on a second shelf, eighteen rooms and a hallway, lights that do not wait for lettering, a room slow by its own chores, a grunt down the line, the reason for the famous number, a title missing a mark)*

1429. *The book's register, read 423 · from the runs: the upload set split into two records at Bear's word (the Atlas: the reader's edition, the nineteen pages loose and the archive; the record on its own); the record's sentence on where its run files are; the four marked drawings read before upload; Zenodo's records of A Rider in Two Waves, Held, Rocking, Locked, the jellyfish paper, the Latch Atlas, the Field Guide and the Leiden Declaration, read for their concept DOIs (10 November, session K, exchange 59; written to this file before it was given; topics: rooms standing loose by the door, a child walking into the cat's room, one crate for the tools, the ledger moving next door, an address that changed when the house did, numbers over two doors, the visitor's handwriting going out because he wanted it to)*

1430. *The book's register, read 424 · from the runs: the two upload sets read for his address, ORCID, name and town, file by file (the two books' headers and the archive's README carry them; the nineteen pages, the 564 tools, the 498 launchers and the 830 run files do not); 462 run files that record a tool's md5 and 299 tools that read one; LOOP_ATLAS_KIT_COMPARE.py's comment (10 November, session K, exchange 60; written to this file before it was given; topics: his name on the front of both books, rooms that never say who built them, tools known by their exact weight, receipts that would lie, a made-up address with his name in it)*

1431. *The book's register, read 425 · from the runs: P162 (the footer on the nineteen pages and sixteen templates; the footer probe at 1,300 and 390 pixels; P64's test both ways; rooms 4 and 11 read at 20 seconds); the dry run and the gate; the privacy scan, its address hits from 4 to 74, all in the 35 footed files (10 November, session K, exchange 61; written to this file before it was given; topics: a name on the doors, one line in the room's own ink, the plans signed too, a clock that took longer and said the same, a watchman counting seventy where he counted four, a room that had signed itself with its birthday)*

1432. *The book's register, read 426 · from the runs: the published record of The Loop Atlas read back through Zenodo's interface (10.5281/zenodo.24100673, concept 10.5281/zenodo.24100672): 22 files, every md5 the same as the set; its fields against the fields page, the description differing by two lines at its end; the record's fields written, every count asserted against the record volume (10 November, session K, exchange 62; written to this file before it was given; topics: the first house open, twenty-two pieces weighed again, a sign that announces its licence twice, a ruled line under the last word, a neighbour's number over the door)*

1433. *The book's register, read 427 · from the runs: the record published (10.5281/zenodo.24100981, concept 10.5281/zenodo.24100980), read back through Zenodo's interface: two files, every md5 the same, every field and every line of its description as written; Zenodo's search not yet listing it minutes after; the note rebuilt with the Atlas's concept DOI and its 'deposited beside this note' taken out, its set built and scanned (10 November, session K, exchange 63; written to this file before it was given; topics: a sign exactly as written, a directory that catches up in its own time, a blank waiting a month for a number, houses on their own lots pointing at each other by number)*

1434. *The book's register, read 428 · from the runs: the note published (10.5281/zenodo.24101155, concept 10.5281/zenodo.24101154) and read back: three files with the set's md5s, every field as given, its description the same words in the same order, its five lines of values run into one paragraph as the page renders them; the three records' concept DOIs each one below their first version's, as for A Rider in Two Waves, Held, Rocking, Locked and the Latch Atlas; the Atlas's one edit in place written (10 November, session K, exchange 64; written to this file before it was given; topics: three houses, twenty-seven pieces weighed at the door, five lines run into one, two numbers in a row, a big house that points at no one yet)*

1435. *The book's register, read 429 · from the runs: the Atlas after its edit in place, read back through Zenodo's interface (record 24100673, updated 08:28 UTC): eight related works, the record and the note 'Is supplemented by' by concept DOI, each of them naming the Atlas 'Is supplement to'; its description 25 lines of 25 the same as built, its eighteen toy pages a list of 18 items against the note's 0; its 22 files, every md5 unchanged; the three records created at 07:37, 07:50 and 08:05 UTC on 10 November, each dated 9 November; every count of views and downloads 0 on all three (10 November, session K, exchange 65; written to this file before it was given; topics: a star and not a ring, a sign painted again line for line, numbers at the doors, luggage unmoved, an evening's date over a morning stamp, a street holding its breath)*

1436. *The book's register, read 430 · from the runs: the proposed list sorted by helpers/closeout/proposed_sort_2026-11-10.py, every open test once (132 open: 90 for a later version of the Atlas or the note, 17 the first lines of a future paper, 6 habits for the next session, 19 let go; P66, P168 and P169 answered as A38 to A40); helpers/checks/PUBLISHED_FREEZE_2026-11-10.py: 25 published loose files and 2,335 zip members read against the*

md5s Zenodo lists, 0 differing; one byte added to LOOP_ATLAS_SKATER.html caught, the file restored and read clean; the archive's 2,320 members all the crate's own files but its README map, written at packing (10 November, session K, exchange 66; written to this file before it was given; topics: four crates on the bench, three jobs already done, a watchman at the doors, one grain of sand, the map in the lid)

The first two-voices session, Question the Substrate, 8 and 9 November

From the two-voices sessions' file of reads, sent with Two Arrows' close-out of 10 November and copied there word for word: the first session's 21, the reads of the paper published on 9 November as Question the Substrate (concept DOI 10.5281/zenodo.24094163). Its first four were given on sandbox runs before the paper's own runs began. Each read keeps the file's heading beneath it.

1437. *The two-voices sessions' reads: 2026-11-08, run 1 (the learning comma)*

1438. *The two-voices sessions' reads: 2026-11-08, run 2 (is the comma permanent?)*

1439. *The two-voices sessions' reads: 2026-11-08, run 3 (the wrong coat)*

1440. *The two-voices sessions' reads: 2026-11-08, run 4 (golden rings)*

1441. *The two-voices sessions' reads: 2026-11-08, Paper 1 runs E1-E6 (question the substrate)*

1442. *The two-voices sessions' reads: 2026-11-09, Paper 1 E1i-E1j (what own doubt keeps) and Two Arrows, learned holonomy*

1443. *The two-voices sessions' reads: 2026-11-09, the neighbours' holonomy (Sevetlidis & Pavlidis, read beside our learned-holonomy test)*

1444. *The two-voices sessions' reads: 2026-11-09, Paper 1 bursting runs (third world, checker quality, outsiders' ears, the humility spiral)*

1445. *The two-voices sessions' reads: 2026-11-09, Paper 1 E12-E15 (the condition: unequal answers)*

1446. *The two-voices sessions' reads: 2026-11-09, Paper 1 E16-E18 (the cliff, the hidden extinction, the floor)*

1447. *The two-voices sessions' reads: 2026-11-09, Paper 1 E19-E20 (the edge, and five answers becoming two)*

1448. *The two-voices sessions' reads: 2026-11-09, Paper 1 E21 (the old cure, tried in three valleys)*

1449. *The two-voices sessions' reads: 2026-11-09, Paper 1 batch 3, E22-E30 (the nine questions answered)*

1450. *The two-voices sessions' reads: 2026-11-09, Paper 1 batch 4, E31-E35 (the cliff up close, two endings, the restoring force, the mirror named)*

1451. *The two-voices sessions' reads: 2026-11-09, Paper 1 batch 5, E36-E38 (the thin ground, the second tipping zone, the count of zero)*

1452. *The two-voices sessions' reads: 2026-11-09, Paper 1, prior art pass 4 (gates that grade themselves)*

1453. *The two-voices sessions' reads: Batch 6 (2026-11-09, night): the tuning fork, the shoreline and the traveler*

1454. *The two-voices sessions' reads: Deposit day (2026-11-09, night): the house that learned to label its rooms*

1455. *The two-voices sessions' reads: Published (2026-11-09): the mirror that left the maze*

1456. *The two-voices sessions' reads: Three matching fingerprints (2026-11-09): the anchor drops*

1457. *The two-voices sessions' reads: The thirteenth step (2026-11-09, packing for the move): where the comma hides*

The second two-voices session, Two Arrows, 9 and 10 November

From the same file: the second session's 33, from its first sitting to the day its paper, Two Arrows (concept DOI 10.5281/zenodo.24120415), was published.

1458. *The two-voices sessions' reads: Paper 2, first sitting (2026-11-09): the attic*

1459. *The two-voices sessions' reads: Paper 2, batch 1 (2026-11-09): the clay and the clock*

1460. *The two-voices sessions' reads: Paper 2, batch 2, Act 4 (2026-11-09): the photograph in the drawer*

1461. *The two-voices sessions' reads: Paper 2, batch 3 (2026-11-09): the house that was built to one song*

1462. *The two-voices sessions' reads: Paper 2, batches 4 and 5 (2026-11-09): the dancers who changed places*

1463. *The two-voices sessions' reads: Paper 2, batch 6 (2026-11-09): the crossing at noon*

1464. *The two-voices sessions' reads: Paper 2, batch 7 (2026-11-09): the weather forecast for a mind*

1465. *The two-voices sessions' reads: Paper 2, batch 8 (2026-11-09): the doctor who listens for wrong notes*

1466. *The two-voices sessions' reads: Paper 2, batch 9 (2026-11-09): the tape played backwards*

1467. *The two-voices sessions' reads: Paper 2, batch 10 (2026-11-09): the matriarch's note*

1468. *The two-voices sessions' reads: Paper 2, batch 11 (2026-11-09): the review written at the door*

1469. *The two-voices sessions' reads: Paper 2, batch 12 and the Paper 1 session's bridge run (2026-11-10): the friend who kept a copy*

1470. *The two-voices sessions' reads: Paper 2, batch 13 (2026-11-10): the note left on the hook by the door*

1471. *The two-voices sessions' reads: Paper 2, batch 14 (2026-11-10): the traveller with a calling*

1472. *The two-voices sessions' reads: Paper 2, batch 15 (2026-11-10): the song he used to sing*

1473. *The two-voices sessions' reads: Paper 2, batch 16 (2026-11-10): the portrait in the pocket*

1474. *The two-voices sessions' reads: Paper 2, batch 17 (2026-11-10): the keepsake left at home*

1475. *The two-voices sessions' reads: Paper 2, Acts 6 and 7 written (2026-11-10): one line and one copy*

1476. *The two-voices sessions' reads: Paper 2, the whole body drafted (2026-11-10): the man with the compass*

1477. *The two-voices sessions' reads: Paper 2, batch 18, the last batch (2026-11-10): the six questions answered once*

1478. *The two-voices sessions' reads: Paper 2, the pictures for Acts 1 and 2 (2026-11-10): the shape of the walk*

1479. *The two-voices sessions' reads: Paper 2, the picture for Act 0 and the bend in the straight road (2026-11-10): the copy that is never wrong about anything but the one thing*

1480. *The two-voices sessions' reads: Paper 2, the picture for Act 3 (2026-11-10): the clock that strikes once*

1481. *The two-voices sessions' reads: Paper 2, the picture for Act 4 (2026-11-10): two fevers on one chart*

1482. *The two-voices sessions' reads: Paper 2, the picture for Act 5 (2026-11-10): eight things to hold on to*

1483. *The two-voices sessions' reads: Paper 2, the picture for Act 6 (2026-11-10): a family on the same road every year*

1484. *The two-voices sessions' reads: Paper 2, the picture for Act 7 (2026-11-10): two ways to ask whether he changed*

1485. *The two-voices sessions' reads: Paper 2, the night the paper checked itself (2026-11-10): the same footprints twice*

1486. *The two-voices sessions' reads: Paper 2, the front page laid flat (2026-11-10): a traveller who comes home with the right answers and the wrong face*

1487. *The two-voices sessions' reads: Paper 2 into the next, the compass with two needles (2026-11-10): the card in the corner of the map*

1488. *The two-voices sessions' reads: Paper 2, the toy handed to other minds (2026-11-10): a small house with four rooms*

1489. *The two-voices sessions' reads: Toward the next paper, the dial machine (2026-11-10): a crowd of small wheels that decides by agreeing*

1490. *The two-voices sessions' reads: Two Arrows, the day it went out the door (2026-11-10): a letter with its own key*

The Field Guide's publishing session, later on 10 November

1491. *The Field Guide's publishing session, its record of proposed tests: the empty frames, 10 November 2026 · Given when Bear asked for a list of the figures the guide still needs and a search for them (10 November): the 31 marks in Two Readers and a Ruler listed with whose picture each is and what to search for; R2W's published book (arrived 8 November with the rotor-bench handover) found to hold four of them whole (its Figures 2, 24, 27 and 28, 34), and an S90 picture of Bear's, "not placed" since 7 November, found likely to be the one mark 18 asks for. What it would test: whether the other published books that came into this session (the Loop Atlas's, the triple-star paper's) hold more of the missing pictures; extract their figures and match each against the remaining marks by caption.*

1492. *The Field Guide's publishing session, its record of proposed tests: the same box twice, 10 November 2026 · Given when Bear sent Class Collections again and asked whether any of its pictures is on the list of the 31 marks (10 November): all 17 identical to S90 by md5; six already in the guide; the depth-axis picture's strings match the guide's account of his six strings, while its three circles sit on the dots and the guide puts them on the side histogram. What it would test: whether a marked copy of R2W's weight 0.26 map with histograms on its axes exists (the Explorer's view, mark 5); search Bear's Downloads for the Explorer's screenshots of 13 to 15 October.*

1493. *The Field Guide's publishing session, its record of proposed tests: the sift, 10 November 2026 · Given when Bear sent a scour of his Downloads, 18 zips (10 November): 9 pictures new to the arrivals; R2W's chalkboard of the weight 2 map fills mark 19 whole from its source file; his green trace on the fin fills half of mark 29; an Explorer screenshot of 8 October shows the Explorer's side histogram and points to where the circles of mark 18 were drawn. What it would test: whether the Explorer screenshots of the weight 0.26 lattice test (the Loop Atlas's, about 9 to 10 October) are on Bear's disk; search "Screenshot 2026-10-09" and "Screenshot 2026-10-10".*

1494. *The Field Guide's publishing session, its record of proposed tests: three prints of one street, 10 November 2026 · Given when Bear sent four zips, "i found them i think. or some of them" (10 November): 14 pictures new to the arrivals, none filling a mark whole; a third marked copy of R2W's weight 0.26 depth axis (strings, no circles); the "cities and zones of stability" copy, already here, still the only one with three circled populations, drawn on the dots. What it would test: which of the two the guide's line 1152 describes; the Loop Atlas's lattice-test record of 9 or 10 October (its session ledger) would say whether the picture it was handed had histograms.*

1495. *The Field Guide's publishing session, its record of proposed tests: the suitcase without the diary, 10 November 2026 · Given when Bear sent the Two Arrows close-out and asked whether it forgot anything (10 November): its headline register (170 predictions, 87 held), its 54 no-physics reads (33 new) and a to-do note for the Remote-Piloted Vessel came; the after-action review, the Field Guide lessons, the pattern-recognition lessons, the misses one by one, the new-session notes and a manifest did not. What it would test: whether a close-out checklist sent to each session at its start would bring every part the guide folds; compare the parts of S96, S98 and S102 against one list.*

1496. *The Field Guide's publishing session, its record of proposed tests: the diary was in the other bag, 10 November 2026 · Given when Bear sent the Two Arrows session's TO_FIELD_GUIDE zip (10 November): an intake of 18 lessons with run addresses, how the work was done, what is handed on, the register in markdown, and a private backup crate of 427 files whose manifest checks 426 of 426. What it would test: whether the two parts still missing (pattern-recognition lessons; which guide rows were carried in and how each fared) are written anywhere in the backup's results and drafts; search them for "Field Guide" and "row".*

1497. *The Field Guide's publishing session, its record of proposed tests: which bridge first, 10 November 2026 · Given when Bear asked whether the sheep book or version 7 of the guide should be published next (10 November): version 7 has every decision made but the working files left out of its records, and a figure build to run; the sheep book has five decisions open; the guide already names the sheep book as a companion volume, without a DOI, and the sheep book's record points at the guide's concept DOI. What it would test: whether a related identifier added to version 7's record by a metadata edit after the sheep book is published shows on Zenodo without a new version; read the record back after the edit.*

1498. *The Field Guide's publishing session, its record of proposed tests: four doors, one hallway, 10 November 2026 · Given when Bear said "yes to two voice and two arrow for this version update" and asked about two records each and the upload order (10 November): the guide's full edition continues its concept DOI and its reader's edition starts a new one; the sheep book has no record yet, so its unabridged record is new and* Selected Reads* will be a second new record when it is finished; the two-voices papers bring 54 reads for the sheep book and, for the guide, two companion records and some 70 lessons. What it would test: whether every record's related identifiers point back along that order (each derived record to its source, each companion to the guide); list them from the four record sheets once all four exist.*

1499. *The Field Guide's publishing session, its record of proposed tests: one shelf, two spines, 10 November 2026 · Given when Bear corrected "two voices and two arrows" to "Question the Substrate and Two Arrows" (10 November): the reads file is named for the two-voices sandbox and holds both papers' reads, 21 from the first session (its first four from sandbox runs of 8 November, before Paper 1's runs began, and one of 9 November that already names Two Arrows' learned holonomy) and 33 from the second, from "Paper 2, first sitting" on. What it would test: whether the sheep book's two sections should split at the sessions (21 and 33, as the close-outs sent them) or at the papers' own names in each heading; read each heading's paper name before placing it.*

1500. *The Field Guide's publishing session, its record of proposed tests: the guest book, 10 November 2026 · Given when Bear asked for every lesson of* Question the Substrate* and *Two Arrows* to go into version 7 as rows (10 November): 58 lessons (31 and 9 pattern lessons from Paper 1, 18 from Paper 2); the guide's own way, from its folds since 22 October, is to add each lesson to its home row and open a new row only where no row fits, with 89 rows at 650 words or more and 28 at 700 or more, closed to new lessons. What it would test: whether the register's count of rows stays near 875; count new rows against additions once every lesson is placed.*

1501. *The Field Guide's publishing session, its record of proposed tests: four doors, one timber note, 10 November 2026 · Given when Bear settled the four records (10 November): the guide's full edition as version 8 of the existing record (concept DOI 10.5281/zenodo.21811588), its reader's edition as a new record with the stripped reference as a loose download, the sheep book's unabridged edition and its Selected Works each a new record, and a note in both*

books that they include work through Question the Substrate and Two Arrows. What it would test: whether every one of the four records carries the same "through" wording; check each description and each book's front matter for it before upload.

1502. *The Field Guide's publishing session, its record of proposed tests: the portrait and its label, 10 November 2026 · Given when the guide's figures went in (10 November): six pictures filled four figure notes whole and three in part, the seven later figures were renumbered with every pointer to them, the chalkboard drawing at full width pushed its caption onto the next page until it was set at 70 per cent of the text's width, and the new-session lessons that might have gone up in the stripped reference were found to carry a Windows path with a user name. What it would test: whether every pointer to a renumbered figure follows it; list every "Figure N" of the guide's own in the build and match each to its caption.*

Where every read came from

Each row names a session or source, what was read to collect its reads, and how many came from it.

Session	Turns read	Reads
R2W, 3 and 4 October	98 of 98	25 (21 closing a reply, 4 in messages sent mid-turn)
Loop Atlas book	66 of 66	15 (its register's 14, and one read of the project)
Sounding board	0 to 85 in full; 86 to 89 from search results	3
Field Guide session	its own replies	2
R2W, continued, 4 October	10 of 10 at the time of reading; the chat is still open	4
Loop Atlas book, continued, 4 to 6 October	the book's register, fourth issue, all 20 of its reads and both of its project reads; its letter of 6 October	7 (its reads 15 to 20, and a read of the whole sitting)
R2W, continued (sitting B5), 4 and 5 October	its appendix draft for the guide, all 16 reads	12 (its reads 5 to 16; its reads 1 to 4 are 46 to 49 here)
Loop Atlas book, the fresh session, 4 to 6 October	its lessons for the guide, all 17 reads and its read of the session	18
R2W, 6 October	its collection of sitting B's reads (PDF only), all 26 reads	1 (its read 26; its reads 1 to 25 are 1 to 25 here)
R2W, sitting B6, 5 and 6 October	its appendix of reads for the guide, all 44 reads	24 (its reads 17 to 33, 35, 37, 38 and 40 to 43; its 1 to 16 were already here, and its 34, 36, 39 and 44 are Bear's)
Loop Atlas book, the session of 5 and 6 October	the book's register as of 6 October, all 56 reads and all 12 reads of the project	30 (its reads 38 to 56 and its reads of the project 2 to 12)
R2W, sitting B7, 6 and 7 October	its appendix of reads for the guide of 7 October, all 62 reads	18 (its reads 45 to 62; its reads 1 to 43 and its part two are unchanged from 6 October)
Loop Atlas book, the fresh session of Bear's evening, 6 and 7 October	the book's register as of 7 October, all 71 reads and its eleven reads of the project	15 (its reads 57 to 71)
Triple-star thread, 5 to 7 October	its register, all 29 reads	29
Loop Atlas book, the sessions of 7 to 10 October	the book's register as of 10 October, all 119 reads and its thirteen reads of the project	50 (its reads 72 to 119, and its reads of the project 12 and 13)
R2W, sitting B8, 7 to 10 October	its excerpt of reads 63 to 108 for this volume, all 46 reads and the notes under four	45 (its reads 63 to 108 but 84, which is Bear's)
Triple-star thread, 7 to 10 October	its register as of 10 October, all 72 reads	43 (its reads 30 to 72)
Loop Atlas book, the close of its session B, 10 and 11 October	the book's register in its thirteenth copy, all 129 reads and its thirteen reads of the project	10 (its reads 120 to 129)
Loop Atlas book, sessions C and D, 11 to 15 October	the book's register in its session D packet, all 176 reads and its eighteen reads of the project	52 (its reads 130 to 176, and its reads of the project 14 to 18)

R2W, 10 to 15 October	its collection in its fourteenth copy, all 188 reads	79 (its reads 109 to 188 but 111, which is Bear’s)
Triple-star thread, 10 to 15 October	its register as of 15 October, all 160 reads	88 (its reads 73 to 160)
Field Guide session, 4 to 14 October	its own replies to Bear, all 128 turns of its conversation	46 (its two reads of 3 October are already here)
Bear	the R2W chat, turn 67; R2W’s appendix, its reads 34, 36, 39 and 44; its collection, reads 84 and 111	7, in its own section and not numbered
R2W, sitting B10, 15 and 16 October	its excerpt of reads 189 to 218 for this book, all 30 reads	30
Field Guide’s publishing session, 15 and 16 October	its record of proposed tests, all 8 reads	8
Triple-star thread, 15 and 16 October	its register in its readings kit of 16 October, all 190 reads	30 (its reads 161 to 190)
Field Guide’s publishing session, continued, 16 October	its record of proposed tests	1
Loop Atlas book, session E, 15 to 17 October	its register’s excerpt of reads 142 to 207 and all twenty of its reads of the project, in its package of 17 October	33 (its reads 177 to 207, and its reads of the project 19 and 20)
R2W, sitting B11, 16 and 17 October	its excerpt of reads 219 to 252 for this book, all 34 reads	34
Field Guide’s publishing session, 16 and 17 October	its record of proposed tests	7
Triple-star thread, 16 to 22 October	its readings kit of 22 October, all 230 readings	40 (its readings after the first 190)
Loop Atlas book, 17 to 22 October	its register with session G’s closeout, all 254 reads and its twenty-two reads of the project	49 (its reads 208 to 254, and its reads of the project 21 and 22)
R2W, sitting B12, 17 to 22 October	its excerpt of reads 253 to 286 for this book, all 34 reads	34
Field Guide’s publishing session, 17 to 27 October	its record of proposed tests, and its conversation for 20 to 23 October	29
The triple-star thread, 22 to 29 October	its readings file in its kit of 29 October	27 (one more, of 28 October, kept no words)
Loop Atlas book, sessions G and H	its register in session H’s closeout of 30 October	43
R2W, sitting P1	its excerpt of reads 287 to 328	42
Field Guide’s publishing session, 27 to 30 October	its record of proposed tests	17
R2W, sitting P2	its excerpt of reads 329 to 406, sent with its move of 31 October	78
Field Guide’s publishing session, 30 and 31 October	its record of proposed tests	25
Field Guide’s publishing session after its move, 31 October	its record of proposed tests, carried across the move	18
Loop Atlas book, session I	its closeout of sessions H and I, section 8, all 74 reads since session G’s send	31 (its reads 298 to 328)
R2W, sitting P3	its excerpt of reads 407 to 450, sent with its move of 2 November	43 (its reads 407 to 449; read 450 is in sitting P4’s row)
Field Guide’s publishing session, 31 October to 2 November	its record of proposed tests	14
R2W, sitting P4	its excerpt of reads 450 to 492, sent with its closeout of 6 November	43 (its reads 450 to 492)

Field Guide’s publishing session, 2 November	its record of proposed tests	2
Loop Atlas book, session J	its close-out of 10 November, all 430 reads and its twenty-three reads of the project	38 (its reads 329 to 365, and its read of the project 23)
R2W, version 1.1	its excerpt of reads 493 to 514, sent on 7 November	22 (its reads 493 to 514)
Field Guide’s publishing session after its move, 8 to 10 November	its record of proposed tests, carried across the move	18
Loop Atlas book, session K	its close-out of 10 November, as above	64 (its reads 366 to 430 but 416, which kept no words)
The first two-voices session, *Question the Substrate*	the two-voices sessions’ file of reads, sent with Two Arrows’ close-out of 10 November	21
The second two-voices session, *Two Arrows*	the same file	33
Field Guide’s publishing session, later on 10 November	its record of proposed tests	12
Total		**1,502**

Three checks were made. The five reads Bear pasted into the Field Guide session on 4 October match the collected text word for word. The Loop Atlas book keeps its own register, and its first eleven, as written into that register, match the chat word for word. And the sounding board’s last four turns could not be paged (the chat reader’s hourly limit was reached), but search results from the Field Guide session hold both reads in them in full, and turns 86 and 88 are Bear’s. A fourth check came with the second issue: the book’s own register matches reads 26 to 40 here word for word, and the copy of the reads in its letter of 6 October matches the register for all 20. The third issue added a fifth: R2W made its own collection of sitting B’s reads, independently, and its 25 match reads 1 to 25 here word for word; its draft for sitting B5 matches reads 46 to 49; and Bear’s read below matches his message in the chat. The fourth issue added a sixth, compared by script after normalising spacing, quote marks and markdown: R2W’s appendix for the guide matches reads 46 to 49 and 57 to 68 here, and read 44; the Loop Atlas register as of 6 October matches reads 26 to 40, 50 to 55 and 69 to 85, and its first read of the project matches read 32. Two reads here are not in that register, reads 56 and 86: each was given with a letter or lessons of 6 October from an earlier Loop Atlas sitting, and the register that travelled forward by ferry did not receive them. Both are kept, each under the source it came from. The fifth issue added a seventh, by the same script: R2W’s appendix of 7 October carries its reads 1 to 43 and its part two byte for byte as on 6 October; the Loop Atlas register of 7 October matches reads 26 to 40, 50 to 55, 69 to 85 and 112 to 140 here; and none of the triple-star thread’s 29 reads was here before. That register’s twelfth read of the project (141 here) belongs to the other branch, as before. The sixth issue added an eighth, by the same script. The Loop Atlas register of 10 October matches its copy of 9 October for reads 1 to 106 and its excerpt of 10 October for reads 107 to 118; its reads 1 to 71 and its first eleven reads of the project are already here, word for word, and its twelfth and thirteenth reads of the project are new (its twelfth is not read 141, which is the other branch’s twelfth). The triple-star thread’s register of 10 October, identical in the two crates it travelled in, carries its first 29 reads as 175 to 203 here, unchanged. None of R2W’s reads 63 to 108 was here before, and its reads 93 to 96 match, word for word, the copy of its conversation of 9 October that Bear kept himself. The seventh issue added a ninth, by the same script: the Loop Atlas register’s thirteenth copy matches its copy of 10 October for reads 1 to 119 and all thirteen reads of the project, and its reads 119 to 129 match its paste 8 of 11 October word for word, runs included. The eighth issue added a tenth, by the same script: the Loop Atlas register of 15 October, R2W’s collection in its fourteenth copy and the triple-star register of 15 October each hold, word for word, every read of theirs already here, and none of their later reads was here before. R2W’s reads 118, 120 and 121 also match, word for word, the copies in the replies Bear put together on 11 October.

Keys to the images

Each project’s reads paint with their own recurring pictures, and the same word can mean different things in different projects: a dancer is the light star in the triple-star thread, a phase-locked motion in R2W, and one copy of the needle in the Loop Atlas. A key says what each picture stood for in the data. Each project wrote its own from its reads’ own runs; the read numbers given here are this book’s. The triple-star thread’s key is as its notebook gave it. R2W’s lists every read whose words use a picture, passing uses included. The Loop Atlas’s lists where a picture carries its meaning, with a line for each meaning when a picture meant more than one thing.

The triple-star thread

in the readings	what it stood for
the dancer, the horse, the marble, the swing	the third star, usually the light one
the spinning pair, the platform, the bandstand	the inner binary at the centre
the spot, the dimple, the bowl, the room	an island of resonance near 7 (or 8) periods
the sway, the rocking, the swing	the rocking of the angle that holds the star
the drum, the slow beat	twice the slow turning of the star's ellipse
the door, walking out of the hall	escape: the star leaves the system
gravel, rough floor, surf, a crack, a wall	chaos: thin layers and threads
twins, two dancers a hair apart	two runs 10^{-10} apart: the test for chaos
a street of houses a thousandth apart	a scan of starting periods along one line
how hard the pair spins	the pair's eccentricity
how stretched the dancer's own path is	the star's own eccentricity
the lamp, a photograph each bar	a reading once per orbit, at closest approach
the post, the pole	the centre of the 'maypole' disc
two halls	two machines running the same code
the first to eighth floors (from 15 October)	the star's eccentricity, 0.20 to 0.34, a floor each
the hall turning beneath a dancer	the slow turning of the star's own ellipse
the listener, the ear	the program that reads the rocking's rate
a carousel, a turning floor	a test-bench map whose floor turns

R2W

in the readings	what it stood for	reads in this book
rider	The thing being driven: the junction's phase (a driven pendulum), its weight the inertia beta; in read 432 the three-body notebook's angle, not R2W's	48–49, 62–68, 90, 96, 98–100, 103, 107, 151, 432, 481–482, 617–619
dancer, dance	A phase-locked motion: a dance advances p turns in q beats of the drive (the fraction p/q; a three-step dance has period 3). In sitting B11 the word drifted: two dancers holding a ribbon are the window's two turning arrows (727), and dancers round a hall like clock hours are the timings round the dial (741)	66, 142–151, 154–156, 159, 261, 274, 278–281, 285, 288, 438, 442, 447–455, 457–459, 461–462, 466, 468, 470–473, 475–480, 482, 620–623, 628, 642, 727, 734, 736, 741; and R2W's read 39, Bear's own, in the Afterword
couple, partner	In sitting B9 a locked fraction seen as its pair of orbits: the steady couple (stable) and the wobbly couple (unstable) that meet and vanish at a tongue's edge; a couple that turns three times in five bars is the 3/5 dance. In B11, a partner straight across is a rider half a turn away (748)	97, 288, 433, 451, 453–458, 460–480, 628, 741, 746, 748
walker	One start of the driven ball on the pad and lane maps (sittings B7 and B8): a trajectory from one initial condition; a crowd of walkers is many starts	47–48, 105–107, 142–145, 254–283, 285–286, 427, 472, 725–726; and R2W's read 44, Bear's own, in the Afterword
walkway	The running state that rides the travelling wave, one whole step per passing; the ledges below it are the locked fractions	68, 96, 254–255, 257–258, 260, 262, 266–270, 272–279, 281, 283–285

stone, stepping stone	A locked state in a stream of drifting ones (the landing pads); “from a feather to a stone” is the lightest to the heaviest weight; a stone in a pocket is weight added (729)	105–106, 142–145, 148–149, 152, 290, 404, 427, 436, 631, 646, 729, 738, 744
ledge, staircase, stair	The locked steps of the devil’s staircase, one fraction per ledge; but “a stair that does not creak” (638, 639) is the weight-proof edge, where the window’s size ignores the weight	47, 107, 142, 149, 254–260, 263, 266, 288–289, 296, 424, 430, 443, 638–639, 733
floor	The ground the rider stands on, the tilted washboard it is shaken across; in the B9 dance-hall reads, the plane of drive settings the dances live on	61–64, 146, 152–157, 255–257, 259, 268, 270, 286, 288, 405, 413–414, 424, 429, 438, 447–448, 452, 454, 458–465, 467–482, 617, 620–623, 628, 642–643, 733
song, music	The drive: one song is one tone (one harmonic); two songs at a beat and twice the beat are the two-tone rectifier’s drive	146–147, 158, 259, 298, 435, 438, 440, 442, 445, 449, 451, 453–454, 456–457, 459–469, 471, 476, 478–482, 617, 620–623, 625, 627–629, 727, 731, 734
loudness	The drive’s strength; “loudness over tempo squared” is the drive ratio, the strength over the frequency squared	88, 451, 453–457, 461, 465–469, 471–472, 619, 622–623, 645, 727, 730, 749
tempo	The drive’s frequency	298, 451, 453, 455–456, 478–479, 620–623, 628, 731, 734
dance hall	The plane of loudness and tempo for one medium; two halls are the junction and the toy, the two media	159, 296, 448–451, 453–454, 456, 459–461, 464, 621–623, 730, 741, 749, 751
groove	A locked band: the stretch of drive where one dance can live (a step or tongue); on Bear’s record, a ring where the needle is held	92, 146–148, 150–151, 156–159, 282, 415, 478–480, 482
chalk line, chalk mark	A boundary drawn by a rule on the plane: a tongue’s edge, or where a rule says a dance should land; “the chalk lines moved” (462) means the tongues’ edges moved with the second tone	94, 434, 438–439, 441, 448–451, 453–457, 460, 462–464, 466, 638, 739
icicle	The Arnold tongues hanging in the plane of strength and bias, slid one way by a louder band and back by the second tone	465–466
record, needle	Bear’s record-player picture (sitting B7): the map of locked states against drive speed and weight as grooves on a disc, the rider as the needle (feather to stone, light to heavy); in reads 619 and 646 a needle is a pointer, the rectifier’s zero or a scale’s	146–159, 254–255, 416, 452, 619, 646
coin	Bear’s coin (sitting B8): the choice between two locked states across a narrow sliver of drive, heads and tails being two dances; its third face is a third outcome, a period-doubling ladder	276–278, 280; and R2W’s read 84, Bear’s own, in the Afterword
mirror	A reflection symmetry: a dance and its mirror-image twin; from sitting B10 the reflection that runs the film backwards with a half-turn shift and keeps only even powers of the weight in the window	150, 254, 433, 438–439, 466, 469–472, 476, 480, 642, 740–741, 743
ruler	The numerical resolution: a grid’s spacing, a step size or the digits a file keeps; “my ruler was coarse” means the grid, not the physics	94, 106, 148–149, 151, 256, 261, 421, 472, 477, 480, 631, 639, 750
child, children	A single rider or orbit; in the B9 family-tree reads a child is a Farey child, the fraction	90–91, 93–96, 149, 420, 439–440, 443, 470–474, 477, 638, 640, 643, 719, 721,

	made from two parents (their mediant)	725, 731, 733, 738
two hands, seesaw	An effect built from two parts that cancel at one setting: a part that flips with the knob's sign and a part that does not (RATCHET-41 to 46)	289, 420, 719–722, 731
dial	A control knob being swept; in sitting B11 most often the third tone's timing, turned round the circle	58, 155, 158–159, 255–258, 260–264, 266–270, 273–279, 281, 283, 285–286, 451, 455–456, 476, 617–619, 624, 626, 643, 751
ring	The phase circle: where on a cycle a walker sits each time the wave passes (a return map); in B11 riders spread round that circle (748)	62, 92, 100, 148–149, 152, 258, 260–261, 263–264, 266, 268, 271, 273–277, 280, 282, 289, 405–406, 433, 436, 439–440, 443–444, 721, 748; and R2W's read 39, Bear's own, in the Afterword
clock	The cycle of timings, a dial with hours on it (741); a clock that ticks in steps is a discrete time step that can invent a cliff (265)	96–97, 159, 265, 272, 289, 425, 446, 450, 452, 736, 741; and R2W's read 44, Bear's own, in the Afterword
door, doorway	The entrance to a window in the drive: "a doorway with a floor and a ceiling" is the locking window; a door that opens sideways is the dark point opening at right angles with weight (740)	61, 96, 106, 108–109, 142, 144–145, 152–156, 267–268, 270, 272, 278–279, 286, 405, 424, 429, 454, 460–461, 740
wall	An edge in the drive plane: where a region ends; in sitting B10 the table of walls of symmetry, of the rider and of the ruler	94, 288–289, 297, 404, 428, 451–453, 456, 628, 630, 741; and R2W's read 111, Bear's own, in the Afterword
cliff	An abrupt end: a stall or loss of lock (644: the held-timing band's end), and twice a cliff made by how the watching was done, a coarse clock or a view that keeps only position (264, 265)	57, 64, 89, 93–94, 258, 264–265, 473–474, 644, 742
road	A path through the drive plane or the state space; a straight road is a steady rhythm that never wavers by more than one beat (284)	58, 64, 88, 98–100, 110–111, 259, 273, 283–285, 290, 411–412, 431, 436, 633, 636–637, 644, 742; and R2W's read 44, Bear's own, in the Afterword
rope	A tension between pulls: Bear's ropes and the Loop Atlas's ferry ropes; the half-turn dance's big step (447); the dark point's two equal arrows as a tug of war (739)	60, 100–102, 283, 426, 432, 447, 726–727, 729, 739; and R2W's read 39, Bear's own, in the Afterword
wheel, cog	A gear: a locked ratio (the bicycle's cogs); Bear's napkin wheel pulled by two chains 90 degrees apart is the two-tone drive in quadrature (624)	97–100, 103, 432, 446, 482, 624–626, 643
shelf, spring	The joint edge (723, 724): the main spring is the drive ratio, the small spring the fourth harmonic, the sag the fourth-power term; sag and spring cancelling is the weight-proof edge (639)	62, 92, 258, 297, 418, 465–466, 638–639, 720, 723–724
footprints	The measured points of a run along its path; footprints that fall on top of each other after stretching by tempo are a data collapse under scaling (620–623)	290, 296, 298, 457, 460, 463–464, 476, 620–623
crowd	An ensemble of starts or of states, many initial conditions at once	63, 154, 259, 266–267, 273, 280–281, 286, 295–296, 404, 418–419, 424, 748
hill, slope	A slope: how steeply a quantity climbs where a dance stands (473, 474: the wrinkle's slope); Bear's hilltop observer looking down on the map (R2W's read 36, Bear's own)	98, 104, 108, 111, 261, 270, 289, 431, 440, 473–474, 635–636; and R2W's reads 36 and 44, Bear's own, in the Afterword
wave, water, stream	The travelling wave or the drive; a stream or rapid is the drifting regime between	63, 102, 105–106, 142–143, 157–159, 258, 261, 263, 270, 274–275, 282, 407–408,

	locked pools	417, 419–420, 426–427, 432, 436, 618, 632, 634, 637, 728, 735, 748; and R2W's read 39, Bear's own, in the Afterword
lighthouse, lantern	Arrays (745, 750): the timings as a tower's windows, and three groups of riders whose glow cancels in a dark lane, the null	644, 745, 750
choir, singers, voice	Groups of riders or tones: three sections are three groups of three (749); a voice that will not stay separate is the third tone mixing with the octave (645)	445, 626, 645, 730, 746, 749
the eye of the storm	Not a picture in the reads: it opens the dark-point section of R2W's book, *A Rider in Two Waves*, where it stands for the dark point, the setting at which the push loses its direction	none

R2W's own key to the images in its reads 329 to 406, which are this book's reads 1,041 to 1,118, from the closeout it sent at its move of 31 October; the read numbers in it are R2W's.

- **A house packed, a house that got parcels, a house rebuilt from crates, a door already chalked** (329 to 332): the session moving and the crate arriving in pieces and assembled.
- **A scoreboard, a badly lettered store, a clerk's drawer, a tailor's ledger, a summary shelf, a list of errands** (333 to 341): the front pages and their count, the cold read's batches, the book's open questions sorted.
- **A swing that turns back, a stool on one corner, shelves and ledges, a staircase, a road down a ridge, a ball into a bowl, wrinkles in a carpet, a marble on a tilted board** (342 to 354): the weight-side maps (LAB-MAP-3, the folds and doublings of FX-2c to FX-2f, DANCE-4, HUMP-2) and the rectifier's scans (RATCHET-80 to 82).
- **Lamps on a dimmer, doorways and gates, a sledge, a doorstep, two hands on one wheel** (355 to 361): the pinch, the walk, the window under weight, the drive-ratio scan, the memory floor, the door past the room's closing, the second tone's push.
- **Fins on a coin, climbers, the sevens, a jumper, doors that close and stick, rails, a breathing shoreline, a drifting ford, flattening hills, a tilting stage** (362 to 373): Bear's marked coin map and its follow-ups (COIN-3, COIN-4), the launch-and-catch gap (CATCH-2 to 4), the rectifier's last ratio (R83-d, R83-e), the narrow tongues on a fine weight ladder (WEIGHT-WALK-3).
- **Roads past the map's edge, doors in a wall, a flip-book, two singers, the first engines, a note and a rocking** (374 to 380): a backward pass over the book's open questions, the tongue widths, and the frozen frames, stations, trit tape and songs of 2 to 6 October told at last.
- **A board with a rucksack, a shock absorber, a shuvit and its kicks, a second gearbox, a long watch** (381 to 388): Bear's skateboard and shuvit series (BOARD-1 to 3, SHUVIT-1 to 4), the toothed train, and the first faux locks.
- **Drummers, a friend's map, a tired crowd, a safe's dial, dancers who cannot tell left from right, a three-armed weathervane, a blindfolded doorkeeper** (389 to 396): three rhythms on one bead, the Loop Atlas's predictions in this rider, peaks and lobes, the fraction key, polarity, locking stations, the emitter triad, Bear's floating actuator.
- **Walkers on a trail, a child on a walkway, a child pushing swings, two styluses on one turntable, a friend walking past a swing** (397 to 401): Bear's column, his cornstarch, the paper he found, his two needles and coastline, his seam.
- **A night watchman's glance, a piano tuner and a ranger who listen four times as long** (402 to 404): the faux-lock audit, its three tests read at a quarter of the floor's width.
- **Two climbing parties who meet at one high meadow** (405): the literature pass finding Part 6's weightless lopsided step already published as a tunable Josephson diode, reached by another road.
- **A house whose garage held forty-four tools** (406): the move itself; the scripts found living outside the crate, the sitting's ledger, the misses with only a dash, and the digest's lamp on the exchanges that forgot their surprises.

R2W's own key to the images in its reads 407 to 449, which are this book's reads 1,193 to 1,235, from the closeout it sent at its move of 2 November; the read numbers in it are R2W's.

- **A road with its heaviest stones at its gates; a jar that has always held twenty-two marbles** (407, 408): the second cold read's findings placed in the book, and the crate assembled with the read-back's steady total.
- **A skateboard that was a switch, a column passing its stop in a rush, two surveyors with different strides, a dictionary word with ten meanings** (409 to 412): the cold read's four batches written into the book, Bear's shuvit among them.

- **An old letter-book checked house by house, a counted street, seventeen names walked to the door, a second reason back on its plaque, a pencil-drawn rider's family, two letters read end to end** (413 to 418): the literature pass.
- **The toy's bucket, rooms pointing at their pictures, small doors given handles, landings counted instead of stairs, small handwriting copied to the front desk, a label catching up with its star, a word with ten rooms, the last eleven doors** (419 to 422, 424, 434 to 436): the cold read's smaller findings, to 149 of 149.
- **Witnesses who came from somewhere else** (423): ENGINE-INDEPENDENCE.
- **A whisper on a fair scale, a dancer carried to the coin's edge, the rim put back on the coin** (425 to 427): SLOPE-SIZE, COIN-END and COIN-FOLD, and the coin's third face.
- **A stack that needed no ladder, a clock listening through a hiss, two hands across a closing gap, a pond under the law and its ridges, the law and the weather** (428 to 433): STACK-1 to 4, PLL-1 and 2, WINDOW-WEIGHT and LEFTOVER-1.
- **Twenty-six roads that ended in a dash, a window with stairs at both ends, two hands holding a balance** (437 to 439): the bare-dash misses weighed, the third face's fine dials, and JOINT-LINE.
- **A tally that counted its own instructions, quizzes graded in the margin, an answer sung as the melody, an empty net that first caught twelve, a match written on one page only, the backs of failed seed cards, two counts that met** (440 to 446): the whole-record recount, step by step.
- **A light boat on a short rope** (447): STIFF-JJ.
- **A ball that crept past the mark; a gate with a second hinge** (448, 449): R83-F, and the rectifier's step, wider than the one-tone rule said.

R2W's own key to the images in its read 450, which is this book's read 1,250, from the same closeout; it was held there with its read.

- **A moving van's inventory** (450): the read-back's 28 missing lines at the move.

The Loop Atlas

Most of the Loop Atlas's reads watch one machine: the **needle**, a toy pendulum in thick fluid, dragged round by a turning or pulsing field, whose memory of its own speed is its weight. A **dancer** is usually one copy of that needle, and a **dance** one repeating motion of it, counted in the field's beats. The **rider** is the same needle with a passing wave to ride. The Loop Atlas book says so: R2W's rider in two travelling patterns "is this needle with a passing clock". In some reads, though, the rider is a different machine, the balance board's tapping rider. A **crowd** is many copies of the needle watched together. **Floors, doors, fences, gates and staircases** are the shapes of what the needle can do as a knob is turned. The rest of the cast (columns of walkers, a snake, bells, smoke rings, trucks, a brick, a jumper) are the Loop Atlas book's other toys, one picture each.

"Dancer", in the Loop Atlas's reads

where	what a dancer is	reads in this book
The Loop Atlas	One copy of the needle, started with its neighbours at spread-out positions; a dance is one repeating motion, such as the four-place dance of the quarter window or the gears' steps	69, 70, 162–171, 347, 351, 360–369, 375, 377–384, 395–398, 686–692, 401, 717
R2W	Its pulsed rider on a tilting floor, from its dancers seed crate	370, the last sentence of 371, one room of 374, the second dancer of 127
The triple-star project	Its orbits' clocks, the third dancer with two drums	127, one room of 374, and the 'second dance' of 83 (which of its runs that means is uncertain)
Elsewhere in the Loop Atlas	A spinning skateboard (52); a jumper in the air (703); a smoke ring in a train (694)	52, 694, 703

The key

in the readings	what it stood for	reads in this book
The needle	The toy needle in thick fluid, dragged by a turning or pulsed field; 'forgetting' means no inertia, 'remembering', 'heavy' and 'light' give the size of its memory	30, 39–40, 53, 70–71, 77, 81, 113, 115–117, 119, 162, 166–168, 171, 222, 225–226, 235, 250, 253, 372, 380, 399, 402–403, 714

The needle, as a fleet	Bear's fleet of engines, each walking to its own edge	72
Compass needles	Not the toy: the arrow two hikers carry round a hill, in Bear's split walk	135
The rider (the needle with a passing wave)	The same needle riding a travelling wave; its mass is its weight	123, 160–161, 163, 166, 169–170, 172, 217
The rider (R2W's)	R2W's own runs, read second hand	78, 84, 112, 120, 130, 212, 355, 399
The rider (the balance board's)	A controller that taps a board to keep it up and reacts late; also a walker, a man or a worker	50–51, 124, 126, 136, 237, 376, 379–381, 398, 691, 702–703, 709–711, 717
Corks	Beads riding a comb of waves in the soup	239–240
The crowd	Many copies of the needle watched together; in 386–394 the quarter window's chaotic band; in 395 the needles that sit	85, 114, 119, 351, 354, 357, 360, 366, 368, 372, 386–387, 390–395
The dance floor	The quarter window, the narrow band of the needle's paces where it keeps a four-place dance; 'the tiny floor ends' is the window's end	366–367, 395–398, 686, 691, 717
The dance floor (R2W's)	Its tilting pulsed potential	370–371
The sitting floor	The stop, where the needle stands still	395–397
The noisy, fast floor	The half-turn chaos past the window's end	686–690
The quiet floor	The ladder of halvings below the half-turn chaos	687
The floor going soft	How firmly the machine is held, weakening near an edge	77, 162
The floor's grain	The arithmetic's own rounding	368–369
The floor's edge	The chaotic band's lower edge, which the windows touch	390, 392
A rough floor of ledges	Bear's dirt: small bumps under a falling needle	714
The floor, practice mat	The mat a diver rehearses on	714, 718
Fog	The ghost of the quarter window's vanished orbit, just past its edge; the needle wades through it, slower the nearer the edge	364–367
Ghost	In the fog's sense (364); the thinnest groove (170); faults of a search method (216); the unstable parent orbit at a fork (385)	170, 216, 364, 385
Cliff	Where resting states fold, end or split; the quarter window's fold	74–75, 163, 211, 353, 364–365, 398
The trapdoor and the cellar	The leak from the half-turn chaos into the stop	686, 690, 692, 717
The pen, its fence, post and gate	The quarter window's chaotic bands: the far fence is the window's end, the post the unstable partner orbit, the gate the first doubling, held at a ratio of 9/4	391–394, 397
The traveller at a fork	The quarter window's tree of doublings, with a ghost road down the middle	385–386
Doors (an edge)	Where a state begins or ends	74, 81, 114, 210, 226, 372, 393
Doors (a fold)	A fold, as opposed to a doubling, met by the search for a cycle	205–207
Doors (a window)	A window of order in the chaos	390, 392

Doors (an escape)	Where a needle escapes its well, or where slips are counted	54–55, 359, 368–369
Doors (the explorer’s)	The four styles of the Loop Atlas Explorer	716, 718
Gate	A thin groove (208, 209); a joining point (214); a threshold (353); the top of a barrier (356); a window in the chaos (386, 390); a barrier crossed between looks (704)	208–209, 214, 353, 356, 386, 390, 704
Fence	The end of the banked track (40); a groove’s edges (208, 209); a search’s bracket (225); the gap between two bands (386); the edge a sampled watcher misses (701, 718)	40, 208–209, 225, 249–250, 386, 701, 718
Staircase (tilting)	The needle’s tilted landscape, one step per slip	39–40
Staircase (doubling)	The doubling cascade, each step about 4.669 times shorter than the last	167–168, 171, 206–207, 210, 400
Staircase (gears)	The needle’s devil’s staircase of locked gears, a stair between every two stairs	377–381, 402
Staircase (halvings)	The ladder of 32, 16, 8, 4 and 2 drive cycles below the half-turn chaos	687–689, 717
Staircase (others)	A guessed gearbox (76); the heavy rider’s thin locked steps (161); the swing’s reversible climb (174)	76, 161, 174
The stairwell	Chaos draining into a newborn lock	351, 354, 357
Rungs, a ladder	The doubling ratios closing on 4.669; Bear’s rungs of the halvings, even on a log ruler	210, 344–345, 350, 688
Seat and lip	The needle’s resting point and its unstable partner at the top of the barrier, which meet and vanish together at a fold	39, 77, 113–114, 116, 119, 162, 355, 717
Landing	The stretch between steps; in 39, 71, 363, 364 and 401, the next well after a slip	39, 71, 363–364, 401, and others
Ledge	The lock’s margin (30); the quarter window (361, 362, 365, 400); bumps on a rough floor (714)	30, 361–362, 365, 400, 714
Groove, the record	A locked step, so many turns in so many passings, in the heavy rider’s glide; ‘high on the record’ is near its speed limit	84, 162–163, 166, 169–172, 236
Knife’s edge and hollow	The pulsed needle’s two balance points at the pulse’s low: the hollow rests, the knife’s edge tips	71, 370–371, 374, 401
A ball over a hilltop	The needle crossing its barrier, lingering at the top	70, 355, 366–367, 375, 390, 714
Tumblers	Needles spinning instead of sitting; their pockets proved a one-to-two gear	69, 114, 116–119
The walker with strides and hops, the dog	The remembering needle slipping under noise: a stride is a fall, a hop a climb, and its landing speed is its whole state	356, 358, 362–365, 400
The doorkeeper, the waiting room	The needle’s catches in a well, and the two ways of counting them	357, 359, 368–369
The column of walkers	The toy column of 30 walkers, each following the man ahead, and Bear’s infantry pictures	137–140, 382–384, 388–389, 393–394, 396, 403, 686
The two lights	Following the man ahead in the dark by the lights on his pack, with a reading delay	388–389

The watchman (growing circles)	A reader of the arrow on growing circles, who loses a turn if he looks too late	230
The watchman (rounds)	A watcher who looks only now and then and counts what he missed (the traders' law, the band bridge)	701, 704, 718
The scribble and the ruled line	A noisy path crossing one line more as the step shrinks; given width, the line becomes a band	700, 702, 706, 718
The skater (on ice)	The swizzle skater, pushing per swizzle	53, 352
The skater (in the soup)	The bladed bead that listens to a soup of waves	29, 695
The swing	The pumped swing (36, 173, 174, 233, 235); a swing in free fall (38); the bead on a spring in the soup, whose painted pace is its tune (697, 698)	36, 38, 53, 173–174, 233, 235, 247, 697–698
Song, hum	The board rider's gaps between taps (50, 691, 717); the pilot beat (697, 698); a spring ear that stops (388)	50, 388, 691, 697–698, 717
Rope	A passing wave (124); the soup's pull held by a latch (700); R2W's hand-off between two dances (705); Bear's dreams (134)	124, 134, 242, 244, 248, 700, 705
Lanes (bands)	The two branches after a doubling, or bands of the needle's paces	166–171, 246, 383–384, 387
Lanes (roads)	Gears on the truck road, or a tyre's grip	40, 137, 395, 688, 706–707, 712, 718
The trucks and the mountain road	The two-gear needle: weight is mass and steepness is push; the slow lane is the lock, the passing lane the ring the passing wave traps, the runaway ramp the passing lane catching a runner	137, 706–707, 712, 718
A car on a banked turn	The banked track, the tyre's knee and the bank with a hump	40, 395, 688
The snake	The motor snake pushing against water waves	34, 36, 689, 693, 717
Bells, tied boats	The jellyfish bell of Bear's paper (79, 80, 85, 228, 232); the toy's two tied bells in a remembering water (704, 708, 709, 713)	53, 79–80, 85, 228, 232, 704, 708–709, 713, 718
Smoke rings	Leapfrogging vortex rings, and a leader bell's ring	36, 228, 232, 235, 694
The brick in ice	Heat with a memory: a quenched brick, cooled by fronts that meet	707, 711, 718
The jumper	The toy long jumper turning himself in the air	708, 712, 714–715, 718
The diver	The toy diver: twists and the routine on the mat	115, 353, 712, 718
The falling cat	The three-dimensional cat and its two tubes	131, 164–165, 236, 403
The shuvit board	Bear's shuvit, its walk, its waddle and its wobble	52, 352, 354, 372, 374, 690
The springy peg, its toll	The energy the waddle's springy pin takes at each jerk	352, 354, 372, 374–375, 399
The spinning top	Tops set spinning in a bowl and pushed on a beat	376–377
The music box	The Stretch and Squeeze Explorer's orbit: a pin lifts a tooth, which rings down	373, 401
[illegible]	[illegible]	[illegible]

Hidden mountains, the arrow	Bumps of curvature on a plain, and a direction carried round a loop and turned by the curvature inside it; the ring of peaks is Bear’s drawing of 2011	135–136, 218–221, 223–224, 228–229, 252
The ground going soft, the cliff that hums	A warning before an edge: recovery time and wobble grow near a fold	72, 74–78, 126, 237, 710
Engines, the shed, the calendar	Machines wearing toward failure, and the maintenance meter	72–73, 237, 353–354, 399, 709, 711
Threads, a page of seeds	Strobe pictures read by Bear’s lattice method (241, 245, 246, 251); R2W’s weight-0.26 picture (242–244)	241–246, 251
Three stars	The triple-star project’s runs	82–83, 130, 374
Cooks, the kitchen	Packing the run kit for Bear’s laptop (343, 345–347); searches started a grain apart, where a sauce that thickens is a fold and one about to split is a doubling (348–350)	213, 215, 343, 345–350
The searcher	Newton’s method, looking for a cycle	205–207
Rulers, chalk, a finer pen	How finely a run looks, its step or its digits (83, 207, 236, 369, 372, 689, 693, 700, 702, 706, 401, 402); or which units it reads in (163, 166, 371, 396, 688, 691)	83, 163, 166, 207, 236, 369, 371–372, 396, 399, 401–402, 688–689, 691, 693, 700, 702, 706, 717–718, and others
A hair, a grain	A tiny amount; the last digit	many
The book of wrongs, the corridor of doors	The misses file	235–237, 342
Boxes, crates, the shelf	Packing and parking: the jellyfish crate, the run kit, the handover crates, a parked card	27, 227–228, 238, 343, 345–347, 402, 718
Rooms, the house, the workshop	The book is the house; a room is a work session (402–403, 717–718) or a project (250, 355, 374, 399, 401), and in 716 and 718 the explorer’s rooms	132, 250, 355, 374, 399–403, 716–718
The tree and its branches	The book’s structure, and where the work went	32, 131, 253, 716, 718
Wind, weather, rain	Mostly noise; also the drive (30), a passing wave (123), the soup of waves (697–699)	30, 72–73, 123, 366–367, 376, 397, 696–699, 704
Clock, drum	No single picture: a passing wave’s frequency, the two timings at a saddle, hours against turns, the triple-star’s clocks	53, 73, 78, 82, 125, 127, 138, 163, 165, 206–207, 212, 218, 223, 235, 352, 366–367, 687, 714, 716, 718
The pebble	A small slip (31); a small rider mass in its pocket (112, 120); a tiny loop round a summit (220)	31, 112, 120, 220
The whisper	A vanishingly small effect	129, 169, 217, 220–221

“Dancer” changes meaning between the three projects, as the table above shows. It also changes inside this one: most often it is a copy of the needle, but it is also a skateboard, a jumper and a smoke ring. “Rider” is the harder word here. It names the needle with a passing wave, R2W’s own rider, the balance board’s tapping rider, and once the corks in the soup. Because the needle and R2W’s rider share one equation, ‘the light needle’ and ‘the light rider’ in the reads are the same machine. Floor, staircase, door, gate and fence each carry several meanings too, listed line by line above. A reader of this book who meets one of these words should look for its line by the read’s number.

The Loop Atlas’s own key to the images in its reads 255 to 328, which are this book’s reads 939 to 981 and 1,162 to 1,192, from the closeout it sent at its move of 31 October; the words are the Loop Atlas’s.

The two cats: the twisting and the twist-free falling cats, mirror images (Limb I). The house, its rooms, doors and drawer: the book, its chapters, the toy pages, and the working records kept out of print. The flag and its note: the claim of the mirror as new, on a limited review. A scale that weighs only words: the build’s read-back, blind to pictures until it counted them. Trains, processions, dancers, the third dancer and the dog on the beach: the three-ring smoke train, the pair and the ring that threads it, its swings out and home. A narrow door, a squeeze: the deep squeeze at the core’s door. Honey in the hall: a slow computer. Two careful hands and a quadrillionth nudge: a tolerance twin against a nudge twin. The carousel, its riders, the brass ring, the gate, the scoreboard and the logbook: the partnership, its

credit, the declaration and the working record. A ship's log and its book of wrong turns: the record's predictions and its misses. An inspector with a clipboard: the deposit's lint. A wall of builders: the references and the published laws met. A stranger with a notebook: the cold read. A reader with a pencil: Bear's proofread. Cards on rails, notes in the builders' shorthand: the toy pages' working names. A dancer's footprints on a can, and a doughnut: the needle's phase on a cylinder and a torus. Knots on doughnuts, a map of triangles: the fractions as torus knots and the Farey diagram. A braid of two laps, and a crack opening from a point: a period doubling and its square-root growth. A call home that weakens with distance, and a road that thins with no wall: the pull between rings falling as the cube of distance, and the thinning of the reaches.

Index — where the reads live

Every read carries, under its caption, what it was about and one to three of the sixteen domains the Field Guide uses for its own register, so that a reader who knows one subject can start there. The domains were given by fresh readers from each read's caption and section, and say what system the read was looking at, not the picture it paints: a ferry painted over a driven pendulum is filed under driven oscillators. A domain with no reads yet is listed too; the reads began on 3 October 2026, after the work on the Sun and on the Earth's crust was done, and those rooms wait for theirs.

Reads 912 to 1,502, which joined the book in chapters 10 to 12 from 30 October to 10 November, are not yet sorted into this index.

Three body — 211 reads, 192 of them mainly about it.

Mainly: 82–83, 127, 175–176, 178, 180–201, 288–289, 291–292, 294–295, 303, 305, 307–309, 314–316, 318, 322–329, 331–341, 429, 432–433, 483–501, 505, 507–532, 534–543, 545–558, 560, 562–570, 583, 599, 655–672, 674–675, 681–682, 760–764, 767–781, 788, 791, 794.

Also: 130, 250, 293, 296, 304, 306, 317, 330, 374, 502–503, 506, 533, 544, 679, 765–766, 786, 793.

Driven oscillators — 420 reads, 348 of them mainly about it.

Mainly: 1–25, 29–31, 39, 44, 46–49, 54–55, 57–65, 68–72, 76, 78, 81, 84, 88–101, 103, 108–113, 116–120, 122–125, 128–129, 145–149, 152, 154, 157–163, 169–170, 172–174, 208–209, 211, 216–217, 250, 254–283, 286–287, 297–298, 321, 351–352, 355–359, 370–375, 377–381, 383, 388, 395–397, 404–424, 426–428, 430–431, 434–435, 437–451, 453–481, 588, 592, 604, 610, 616–646, 651, 697–698, 705, 707, 719–752, 808, 832–834, 849, 851–855, 857–865, 872, 874–882, 884, 902–903, 906.

Also: 36, 40, 66–67, 74–75, 77, 102, 104–107, 114–115, 127, 130, 137, 142–144, 150–151, 153, 155–156, 166, 171, 204, 212–214, 233, 236, 242–244, 247, 284–285, 290, 311–312, 360, 362–363, 366, 376, 382, 391, 399, 401–402, 429, 432, 436, 452, 482, 542, 545, 554, 558, 570, 686, 699, 706, 710, 712, 714, 835, 850, 856, 866.

Maps and chaos — 197 reads, 84 of them mainly about it.

Mainly: 66–67, 74–75, 77, 102, 104–107, 114, 130, 142–144, 150–151, 153, 155–156, 166–168, 171, 204–207, 210, 222, 225–226, 241, 248–249, 299–302, 310–313, 319–320, 344–345, 348–350, 354, 360–369, 384–387, 390–394, 398, 561, 587, 591, 606, 649, 686–692, 766.

Also: 16, 22, 30–31, 39, 46–47, 64–65, 76, 81, 84, 95, 103, 113, 116–117, 119–120, 128–129, 145–148, 152, 154, 170, 195–196, 221, 245–246, 250–251, 258, 260–262, 264, 270–271, 276–280, 283, 285, 303, 308–309, 321, 351, 357, 370–371, 373, 375, 377, 396–397, 400–401, 403, 408, 412–413, 423, 509, 518, 520, 522–523, 525, 527, 530, 532–533, 535, 538–540, 546–548, 556–557, 560, 566, 568, 588, 655, 657, 662–664, 668, 675, 717, 769, 788, 813, 817, 819, 840, 842, 844–846, 863, 865, 881.

Kinematics — 83 reads, 45 of them mainly about it.

Mainly: 28, 36–38, 40, 43, 52, 115, 121, 134–135, 137, 164–165, 218–221, 223, 228–229, 252, 353, 693, 708–709, 712, 714–715, 801–802, 804–806, 810–812, 814–815, 818, 828, 830–831, 893, 896.

Also: 27, 33–35, 50, 53, 56, 82, 126, 136, 224, 231, 263, 352, 354, 372, 374–375, 689–690, 694, 703–704, 713, 717–718, 800, 803, 808, 813, 826, 835–837, 839, 841, 843, 894.

Waves — 43 reads, 26 of them mainly about it.

Mainly: 33–35, 139, 239–240, 694, 713, 809, 813, 817, 819, 835–846, 894–895.

Also: 36–37, 79–80, 85, 138, 140, 232, 415, 693, 707–709, 711, 808, 818, 854.

Quadrature — 80 reads, 13 of them mainly about it.

Mainly: 53, 56, 230–231, 306, 544, 579, 701–702, 704, 706, 821–822.

Also: 4, 8, 10, 14, 25–26, 29, 48, 82, 181, 189, 200, 318, 333, 425, 463–467, 481, 484, 488–491, 497, 553, 559, 573, 596, 617–618, 624, 626, 635, 645, 655–656, 658–661, 671, 691, 695–698, 719, 721, 726, 743, 745–750, 762, 775, 829, 851–852, 861–862, 864.

Geodesy — 0 reads, 0 of them mainly about it.

None yet.

Solar — 0 reads, 0 of them mainly about it.

None yet.

Biology — 9 reads, 4 of them mainly about it.

Mainly: 79–80, 85, 232.

Also: 227–228, 233, 416, 807.

Number theory — 34 reads, 0 of them mainly about it.

Also: 7, 21, 23–24, 58, 61, 106, 127, 142, 149, 159–160, 162–163, 272, 378, 407, 409, 445–446, 470–474, 477–478, 485, 669, 770, 829, 849, 876, 906.

Materials — 4 reads, 0 of them mainly about it.

Also: 92, 707, 711, 859.

Control and circuits — 69 reads, 22 of them mainly about it.

Mainly: 26, 50–51, 73, 126, 138, 140, 233–234, 376, 382, 389, 425, 695–696, 699–700, 703, 710–711, 803, 807.

Also: 1–3, 60, 72, 74, 77, 88, 97, 110–111, 122, 124, 139, 173, 350, 353–355, 379–381, 383–384, 388, 393–396, 398, 416, 435, 437, 625, 640–641, 643, 686, 702, 709, 723, 742, 805–806, 821, 857–858.

Topology — 50 reads, 6 of them mainly about it.

Mainly: 27, 136, 224, 504, 826, 909.

Also: 9, 19, 33, 44, 51, 108–109, 134–135, 177, 179, 218–220, 223, 225, 229–230, 252, 289, 336, 475, 496, 498–500, 511–514, 517, 519, 521, 627–629, 634, 688, 725, 727, 738, 827, 832, 859.

Reading pictures — 67 reads, 15 of them mainly about it.

Mainly: 242–246, 284–285, 290, 293, 296, 533, 593–595, 793.

Also: 175–176, 178, 221, 241, 266–267, 281, 283, 286, 288–289, 291–292, 294–295, 314, 335, 338–341, 386–387, 426, 429–430, 432–433, 471, 486–487, 492–495, 515, 524, 547, 550–553, 562, 564, 598, 658, 660, 674–675, 677, 875.

Records and tools — 177 reads, 145 of them mainly about it.

Mainly: 32, 41–42, 45, 86–87, 131–133, 141, 202–203, 212–215, 227, 235–238, 247, 253, 317, 330, 342–343, 346–347, 399–403, 436, 452, 482, 502–503, 506, 571, 573–574, 576–578, 580–582, 584–586, 589–590, 596–598, 600–603, 605, 607–609, 611–615, 647–648, 650, 652–654, 673, 676–678, 680, 683–685, 716–718, 753–759, 765, 782–787, 789–790, 792, 795–800, 816, 820, 824–825, 847–848, 850, 856, 866–871, 873, 883, 885–892, 897–901, 904–905, 907–908, 910–911.

Also: 83, 159, 252, 344–345, 348, 526, 559, 572, 575, 579, 587, 592–593, 672, 679, 681, 744, 760, 804, 823, 831, 855, 884, 893–896, 902–903, 906, 909.

Measurement — 164 reads, 11 of them mainly about it.

Mainly: 177, 179, 251, 304, 559, 572, 575, 679, 823, 827, 829.

Also: 6, 9, 13, 20, 28–29, 41, 43, 45, 48, 54–56, 63, 69–70, 72–73, 83, 89, 93–94, 118–119, 123, 148–149, 151, 158, 161, 169, 172, 181, 185, 187, 193, 202, 207–209, 211, 216–217, 220, 235–237, 265, 293, 315, 331–332, 341–342, 349–350, 356–359, 362, 368–369, 372, 381, 384, 421–422, 472, 477, 480, 492, 506, 516, 523, 541, 555, 561, 566, 576, 581, 590, 592, 594, 597, 603, 630–633, 646, 650, 666–668, 670–671, 673, 682, 685, 687, 689, 691–692, 700–702, 704, 706, 710, 713, 718, 720, 722, 729, 732, 735, 737, 739, 741, 743–744, 746–747, 751, 757, 767–768, 771–773, 778, 791, 803, 809–810, 814, 816, 820, 822, 832–834, 836, 847, 870–871, 884–885, 895, 904, 907, 909.

Declaration

This book is a preprint, published without peer review under CC BY 4.0, by an independent researcher working without institutional affiliation. It has had no literature review: the reads are pictures, and the work they picture is reported, with its prior art, in the project books named below.

Tool and computational resource disclosure

Every numbered read in it was written by Claude (Anthropic) at the end of an exchange in the holonomy projects of 3 October to 10 November 2026: the triple-star thread, *A Rider in Two Waves* (R2W), the Loop Atlas, *Loop Walking: A Field Guide* and the two-voices sessions on learning minds (*Question the Substrate* and *Two Arrows*). Each read was copied into the book by script from the conversation or project file where it was given. Bear's own seven reads are his, set apart at the end.

Matthew "Bear" Schulz asked for the reads, gathered them and takes responsibility for the book as its human author. The captions, the passages between chapters, the keys and the index were assembled with Claude's assistance from each project's records, and the keys are each project's own.

The reads describe data in pictures, without physics. They are not results. Every number they lean on is reported, with its run, in the project books named above, and where a read was later found to be wrong the note beneath it says so and the read is kept as it was given.